

0902.0431: formula evidence

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Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_F00001	1	■ 1.000	$A_{\{n\}}, B_{\{n\}}, C_{\{n\}}, D_{\{n\}}$	A_n, B_n, C_n, D_n	Lie algebras, called A_n, B_n, C_n, D_n (classical type) and G_2, F_4, E_6, E_7, E_8 (exceptional
0902.0431_F00002	1	■ 1.000	$G_{\{2\}}, F_{\{4\}}, E_{\{6\}}, E_{\{7\}}, E_{\{8\}}$	G_2, F_4, E_6, E_7, E_8	Lie algebras, called A_n, B_n, C_n, D_n (classical type) and G_2, F_4, E_6, E_7, E_8 (exceptional
0902.0431_F00003	1	■ 1.000	G	G	G concretely. Next, we find all involutive automorphisms σ of G , and determine the
0902.0431_F00004	1	■ 1.000	σ	σ	G concretely. Next, we find all involutive automorphisms σ of G , and determine the
0902.0431_F00005	1	■ 1.000	G^{σ}	G^{σ}	group structures of the fixed points subgroup G^{σ} by σ . Note that they correspond to
0902.0431_F00006	1	■ 1.000	G / G^{σ}	G / G^{σ}	classification of all irreducible compact symmetric spaces G / G^{σ} of exceptional type,
0902.0431_F00007	7	■ 1.000	$G_{\{2\}}$	G_2	Definition. The group G_2 is defined to be the automorphism group of the Cayley
0902.0431_F00008	6	■ 1.000	$\mathfrak{C}$	$\mathfrak{C}$	We denote the division Cayley algebra by $\mathfrak{C}$. We now explain this algebra. Con-
0902.0431_F00009	44	■ 1.000	$\mathfrak{d}_4$	$\mathfrak{d}_4$	Proposition 2.3.7. The Lie subalgebra $\mathfrak{d}_4$ of $\mathfrak{f}_4$:
0902.0431_F00010	8	■ 1.000	$\mathfrak{g}_2$	$\mathfrak{g}_2$	In order to study the Lie algebra $\mathfrak{g}_2$ of the group G_2 , we consider the Lie algebra
0902.0431_F00011	16	■ 1.000	$\mathfrak{su}(3)$	$\mathfrak{su}(3)$	that we identify $\mathfrak{su}(3)$ and $\varphi_*(\mathfrak{su}(3))$. We set
0902.0431_F00012	17	■ 1.000	$\mathfrak{g}_2^C$	$\mathfrak{g}_2^C$	Theorem 1.6.3. The Lie algebra $\mathfrak{g}_2^C$ is simple and so $\mathfrak{g}_2$ is also simple.
0902.0431_F00013	21	■ 1.000	w	w	an automorphism w of order 3 of $\mathfrak{g}_2$.
0902.0431_F00014	23	■ 0.942	$SU(3)$	$SU(3)$	isotropy subgroup $(G_2)_{e_1}$ of G_2 at e_1 is $SU(3)$ (Theorem 1.9.1). Thus we have the
0902.0431_F00015	21	■ 1.000	γ	γ	an involution γ of $\mathfrak{g}_2$.
0902.0431_F00017	26	■ 1.000	$z(G_2)$	$z(G_2)$	1.11. Center $z(G_2)$ of G_2
0902.0431_F00018	26	■ 1.000	G_2^C	G_2^C	1.12. Complex exceptional Lie group G_2^C
0902.0431_F00019	27	■ 1.000	$G_{2(2)}$	$G_{2(2)}$	$\mathfrak{C}^C \mid \tau \gamma x = x$ as algebras. Now, the group $G_{2(2)}$ is defined to be the automorphism
0902.0431_F00020	8	■ 1.000	$SO(8)$	$SO(8)$	of the Lie group $SO(8)$.
0902.0431_F00021	33	■ 0.867	$\operatorname{Spin}(7)$	$\operatorname{Spin}(7)$	Therefore $\tilde{B}_3$ is isomorphic to $\operatorname{Spin}(7)$ as the universal covering group of $SO(7)$.
0902.0431_F00022	33	■ 0.996	$\operatorname{Spin}(8)$	$\operatorname{Spin}(8)$	1.16. Spinor group $\operatorname{Spin}(8)$
0902.0431_F00023	38	■ 1.000	F_4	F_4	Definition. The group F_4 is defined to be the automorphism group of the Jordan
0902.0431_F00024	36	■ 1.000	$\mathfrak{J}$	$\mathfrak{J}$	$\mathfrak{J}$ is a 27 dimensional $\mathbf{R}$ -vector space. In $\mathfrak{J}$, the multiplication $X \circ Y$, called the Jordan
0902.0431_F00025	41	■ 1.000	$\mathfrak{f}_4$	$\mathfrak{f}_4$	In order to investigate the Lie algebra $\mathfrak{f}_4$ of the group F_4 , it will be helpful to
0902.0431_F00026	46	■ 1.000	$\mathfrak{f}_4^C$	$\mathfrak{f}_4^C$	2.4. Simplicity of $\mathfrak{f}_4^C$
0902.0431_F00027	56	■ 0.990	$\operatorname{Spin}(9)$	$\operatorname{Spin}(9)$	2.7. Subgroup $\operatorname{Spin}(9)$ of F_4
0902.0431_F00028	64	■ 1.000	$z(F_4)$	$z(F_4)$	2.10. Center $z(F_4)$ of F_4
0902.0431_F00029	64	■ 0.965	$(Sp(1) \times Sp(3)) / \mathbf{Z}_2$	$(Sp(1) \times Sp(3)) / \mathbf{Z}_2$	2.11. Involution γ and subgroup $(Sp(1) \times Sp(3)) / \mathbf{Z}_2$ of F_4
0902.0431_F00030	68	■ 1.000	$(SU(3) \times SU(3)) / \mathbf{Z}_3$	$(SU(3) \times SU(3)) / \mathbf{Z}_3$	2.12. Automorphism w of order 3 and subgroup $(SU(3) \times SU(3)) / \mathbf{Z}_3$ of
0902.0431_F00031	71	■ 1.000	F_4^C	F_4^C	Definition. The group F_4^C is defined to be the automorphism group of the
0902.0431_F00032	72	■ 1.000	$F_{4(4)}$	$F_{4(4)}$	2.14. Non-compact exceptional Lie groups $F_{4(4)}$ and $F_{4(-20)}$ of type F_4
0902.0431_F00033	72	■ 1.000	$F_{4(-20)}$	$F_{4(-20)}$	2.14. Non-compact exceptional Lie groups $F_{4(4)}$ and $F_{4(-20)}$ of type F_4
0902.0431_F00034	73	■ 0.999	E_6	E_6	Definition. We define the groups E_6^C and E_6 respectively by
0902.0431_F00035	73	■ 1.000	$\mathfrak{e}_6$	$\mathfrak{e}_6$	3.2. Lie algebra $\mathfrak{e}_6$ of E_6
0902.0431_F00036	46	■ 1.000	$\mathfrak{e}_6^C$	$\mathfrak{e}_6^C$	We define Lie algebras $\mathfrak{e}_6^C$ and $\mathfrak{f}_4^C$ respectively by
0902.0431_F00037	76	■ 1.000	$A \vee B$	$A \vee B$	3.4. Element $A \vee B$ of $\mathfrak{e}_6^C$
0902.0431_F00038	5	■ 1.000	τ	τ	The complex conjugation in V^C is denoted by τ :
0902.0431_F00039	90	■ 1.000	$z(E_6)$	$z(E_6)$	3.9. Center $z(E_6)$ of E_6
0902.0431_F00040	91	■ 0.909	$(U(1) \times \operatorname{Spin}(10)) / \mathbf{Z}_4$	$(U(1) \times \operatorname{Spin}(10)) / \mathbf{Z}_4$	3.10. Involution σ and subgroup $(U(1) \times \operatorname{Spin}(10)) / \mathbf{Z}_4$ of E_6
0902.0431_F00041	95	■ 0.935	$(Sp(1) \times SU(6)) / \mathbf{Z}_2$	$(Sp(1) \times SU(6)) / \mathbf{Z}_2$	3.11. Involution γ and subgroup $(Sp(1) \times SU(6)) / \mathbf{Z}_2$ of E_6

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0902.0431_F00042	87	■ 1.000	$\tau \gamma$	$\tau \gamma$	points of $\tau \gamma$ is type C_4 .
0902.0431_F00045	71	■ 0.993	$SU(3) / \boldsymbol{Z}_3$	$SU(3)/\boldsymbol{Z}_3$	$SU(3)/\boldsymbol{Z}_3$ and $\boldsymbol{Z}_2$ (the action of $\boldsymbol{Z}_2 = \{1, \gamma_1\}$ on the group $(SU(3) \times SU(3))$ is
0902.0431_F00046	73	■ 0.999	E_6^C	E_6^C	Definition. We define the groups E_6^C and E_6 respectively by
0902.0431_F00047	110	■ 1.000	$E_{6(6)}, E_{6(2)}, E_{6(-14)}$	$E_{6(6)}, E_{6(2)}, E_{6(-14)}$	3.15. Non-compact exceptional Lie groups $E_{6(6)}, E_{6(2)}, E_{6(-14)}$ and $E_{6(-26)}$
0902.0431_F00048	42	■ 1.000	$E_{6(-26)}$	$E_{6(-26)}$	Lemma 2.3.1. The Lie algebra $\mathfrak{e}_{6(-26)}$ of the group $E_{6(-26)}$ is given by
0902.0431_F00049	113	■ 0.961	E_7	E_7	Definition. We define the groups E_7^C and E_7 respectively by
0902.0431_F00050	111	■ 1.000	$\mathfrak{P}^C$	$\mathfrak{P}^C$	Definition. We define a C -vector space $\mathfrak{P}^C$, called the Freudenthal C -vector
0902.0431_F00051	113	■ 1.000	$\mathfrak{e}_7$	$\mathfrak{e}_7$	Before we investigate the Lie algebra $\mathfrak{e}_7$ of the group E_7 , we shall first study the
0902.0431_F00052	113	■ 0.881	$\mathfrak{e}_7^C$	$\mathfrak{e}_7^C$	Lie algebra $\mathfrak{e}_7^C$ of the group E_7^C .
0902.0431_F00053	94	■ 0.999	$U(1)$	$U(1)$	Note that $U(1)$ is a subgroup of $(E_6)^\sigma$. From now on, we identify these two groups
0902.0431_F00054	134	■ 1.000	$z(E_7)$	$z(E_7)$	Theorem 4.9.1. The center $z(E_7)$ of the group E_7 is the cyclic group of order 2:
0902.0431_F00055	129	■ 1.000	ι	ι	points of an involution ι of $\mathfrak{e}_7$.
0902.0431_F00057	208	■ 0.986	$(SU(2) \times Spin(12)) / \boldsymbol{Z}_2$	$(SU(2) \times Spin(12))/\boldsymbol{Z}_2$	[56] I. Yokota and O. Yasukura, Subgroup $(SU(2) \times Spin(12))/\boldsymbol{Z}_2$ of compact simple
0902.0431_F00059	151	■ 1.000	$(SU(3) \times SU(6)) / \boldsymbol{Z}_3$	$(SU(3) \times SU(6))/\boldsymbol{Z}_3$	4.13. Automorphism w of order 3 and subgroup $(SU(3) \times SU(6))/\boldsymbol{Z}_3$ of
0902.0431_F00060	113	■ 0.961	E_7^C	E_7^C	Definition. We define the groups E_7^C and E_7 respectively by
0902.0431_F00061	160	■ 1.000	$E_{7(7)}, E_{7(-5)}$	$E_{7(7)}, E_{7(-5)}$	4.15. Non-compact exceptional Lie groups $E_{7(7)}, E_{7(-5)}$ and $E_{7(-25)}$ of
0902.0431_F00062	160	■ 1.000	$E_{7(-25)}$	$E_{7(-25)}$	4.15. Non-compact exceptional Lie groups $E_{7(7)}, E_{7(-5)}$ and $E_{7(-25)}$ of
0902.0431_F00063	166	■ 0.988	E_8	E_8	$\mathfrak{e}_8^C$. It is known that $\text{Aut}(\mathfrak{e}_8^C)/\text{Inn}(\mathfrak{e}_8^C) = \{1\}$ holds for the C -algebra of E_8 type
0902.0431_F00064	162	■ 1.000	$\mathfrak{e}_8^C$	$\mathfrak{e}_8^C$	5.1. Lie algebra $\mathfrak{e}_8^C$
0902.0431_F00065	165	■ 1.000	E_8^C	E_8^C	Definition. The group E_8^C is defined to be the automorphism group of the Lie
0902.0431_F00066	177	■ 1.000	v	v	points of an involution v of $\mathfrak{e}_8$.
0902.0431_F00067	189	■ 0.834	$(SU(2) \times E_7) / \boldsymbol{Z}_2$	$(SU(2) \times E_7)/\boldsymbol{Z}_2$	$(SU(2) \times E_7)/\boldsymbol{Z}_2$ (Theorem 5.7.6), and so $\alpha \in z(\varphi(SU(2) \times E_7))$. Therefore by
0902.0431_F00068	177	■ 0.955	$\tilde{\lambda} \gamma$	$\tilde{\lambda} \gamma$	of an involution $\tilde{\lambda} \gamma$ of $\mathfrak{e}_8$.
0902.0431_F00069	181	■ 0.854	$Ss(16)$	$Ss(16)$	5.8. Involution $\tilde{\lambda} \gamma$ and subgroup $Ss(16)$ of E_8
0902.0431_F00070	189	■ 1.000	$z(E_8)$	$z(E_8)$	5.9. Center $z(E_8)$ of E_8
0902.0431_F00072	177	■ 1.000	w_3	w_3	of an automorphism w_3 of order 3 of $\mathfrak{e}_8$.
0902.0431_F00074	177	■ 1.000	z_5	z_5	points of an automorphism z_5 of order 5 of $\mathfrak{e}_8$.
0902.0431_F00075	197	■ 0.999	$(SU(5) \times SU(5)) / \boldsymbol{Z}_5$	$(SU(5) \times SU(5))/\boldsymbol{Z}_5$	5.12. Automorphism z_5 of order 5 and subgroup $(SU(5) \times SU(5))/\boldsymbol{Z}_5$ of
0902.0431_F00076	204	■ 1.000	$E_{8(8)}$	$E_{8(8)}$	Section 5.1. Now, we define groups $E_{8(8)}$ and $E_{8(-24)}$ by
0902.0431_F00077	204	■ 1.000	$E_{8(-24)}$	$E_{8(-24)}$	Section 5.1. Now, we define groups $E_{8(8)}$ and $E_{8(-24)}$ by
0902.0431_F00078	5	■ 1.000	$\boldsymbol{R}, \boldsymbol{C} = \boldsymbol{R} \oplus \boldsymbol{R}e_1, \boldsymbol{H} = \boldsymbol{R} \oplus \boldsymbol{R}e_1 \oplus \boldsymbol{R}e_2 \oplus \boldsymbol{R}e_3$	$\boldsymbol{R}, \boldsymbol{C} = \boldsymbol{R} \oplus \boldsymbol{R}e_1, \boldsymbol{H} = \boldsymbol{R} \oplus \boldsymbol{R}e_1 \oplus \boldsymbol{R}e_2 \oplus \boldsymbol{R}e_3$	$\boldsymbol{R}, \boldsymbol{C} = \boldsymbol{R} \oplus \boldsymbol{R}e_1, \boldsymbol{H} = \boldsymbol{R} \oplus \boldsymbol{R}e_1 \oplus \boldsymbol{R}e_2 \oplus \boldsymbol{R}e_3$ denote the fields of real, complex,
0902.0431_F00079	5	■ 1.000	$\boldsymbol{R}$	$\boldsymbol{R}$	For $\boldsymbol{R}$ -vector space V , its complexification $\{u + iv \mid u, v \in V\}$ is denoted by V^C .
0902.0431_F00080	5	■ 1.000	V	V	For $\boldsymbol{R}$ -vector space V , its complexification $\{u + iv \mid u, v \in V\}$ is denoted by V^C .
0902.0431_F00081	5	■ 1.000	$\{u + iv \mid u, v \in V\}$	$\{u + iv \mid u, v \in V\}$	For $\boldsymbol{R}$ -vector space V , its complexification $\{u + iv \mid u, v \in V\}$ is denoted by V^C .
0902.0431_F00082	5	■ 1.000	V^C	V^C	For $\boldsymbol{R}$ -vector space V , its complexification $\{u + iv \mid u, v \in V\}$ is denoted by V^C .
0902.0431_F00083	5	■ 1.000	$\boldsymbol{R}^C$	$\boldsymbol{R}^C$	$\boldsymbol{R}^C$ is briefly denoted by C .
0902.0431_F00084	5	■ 1.000	C	C	$\boldsymbol{R}^C$ is briefly denoted by C .
0902.0431_F00085	5	■ 1.000	K	K	For K -vector space V ($K = \boldsymbol{R}, \boldsymbol{C}, C$), $\text{Iso}_K(V)$ denotes all K -linear isomorphisms
0902.0431_F00086	5	■ 1.000	$V(K = \boldsymbol{R}, \boldsymbol{C}, C), \text{Iso}_K(V)$	$V(K = \boldsymbol{R}, \boldsymbol{C}, C), \text{Iso}_K(V)$	For K -vector space V ($K = \boldsymbol{R}, \boldsymbol{C}, C$), $\text{Iso}_K(V)$ denotes all K -linear isomorphisms
0902.0431_F00087	5	■ 0.963	f	f	of V . For a K -linear mapping f of V , V_f denotes $\{v \in V \mid f(v) = v\}$.
0902.0431_F00088	5	■ 0.963	V, V_f	V, V_f	of V . For a K -linear mapping f of V , V_f denotes $\{v \in V \mid f(v) = v\}$.
0902.0431_F00089	5	■ 0.963	$\{v \in V \mid f(v) = v\}$	$\{v \in V \mid f(v) = v\}$	of V . For a K -linear mapping f of V , V_f denotes $\{v \in V \mid f(v) = v\}$.
0902.0431_F00090	5	■ 1.000	$V, W(K = \boldsymbol{R}, \boldsymbol{C}), \text{Hom}_K(V, W)$	$V, W(K = \boldsymbol{R}, \boldsymbol{C}), \text{Hom}_K(V, W)$	For K -vector spaces V, W ($K = \boldsymbol{R}, \boldsymbol{C}$), $\text{Hom}_K(V, W)$ denotes all K -homomorph-
0902.0431_F00091	5	■ 0.915	$f: V \rightarrow W, \text{Hom}_K(V, V)$	$f: V \rightarrow W, \text{Hom}_K(V, V)$	ism $f: V \rightarrow W$. $\text{Hom}_K(V, V)$ is briefly denoted by $\text{Hom}_K(V)$.
0902.0431_F00092	5	■ 0.915	$\text{Hom}_K(V)$	$\text{Hom}_K(V)$	ism $f: V \rightarrow W$. $\text{Hom}_K(V, V)$ is briefly denoted by $\text{Hom}_K(V)$.

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0902.0431_F00093	5	■1.000	$\{g \in G \mid \sigma(g) =$	$\{g \in G \mid \sigma(g) =$	Let G be a group and σ an automorphism of G . Then G^σ denotes $\boxed{g \in G \mid \sigma(g) =$
0902.0431_F00094	5	■1.000	$g\}$	$g\}$	$g\}$. For $s \in G$, G^s denotes $\{g \in G \mid sgs^{-1} = g\}$.
0902.0431_F00095	5	■1.000	$s \in G, G^{\{s\}}$	$s \in G, G^s$	$g\}$. For $s \in G$, G^s denotes $\{g \in G \mid sgs^{-1} = g\}$.
0902.0431_F00096	5	■1.000	$\left\{g \in G \mid sgs^{-1} = g\right\}$	$\{g \in G \mid sgs^{-1} = g\}$	$g\}$. For $s \in G$, G^s denotes $\boxed{\{g \in G \mid sgs^{-1} = g\}}$.
0902.0431_F00097	5	■1.000	$X, Y, X \simeq Y$	$X, Y, X \simeq Y$	For topological spaces X, Y , $X \simeq Y$ denotes that X and Y are homeomorphic.
0902.0431_F00098	5	■1.000	X	X	For topological spaces X, Y , $X \simeq Y$ denotes that X and Y are homeomorphic.
0902.0431_F00099	5	■1.000	Y	Y	For topological spaces X, Y , $X \simeq Y$ denotes that X and Y are homeomorphic.
0902.0431_F00100	5	■1.000	$G, G^{\{\prime\}}, G \cong G^{\{\prime\}}$	$G, G', G \cong G'$	For groups G, G' , $G \cong G'$ denotes that G and G' are isomorphic as groups. Iso-
0902.0431_F00101	5	■1.000	$G^{\{\prime\}}$	G'	For groups G, G' , $G \cong G'$ denotes that G and G' are isomorphic as groups. Iso-
0902.0431_F00102	5	■1.000	$G, G^{\{\prime\}}$	G, G'	morphic two groups G, G' are often identified: $G = G'$.
0902.0431_F00103	5	■1.000	$G = G^{\{\prime\}}$	$G = G'$	morphic two groups G, G' are often identified: $G = G'$.
0902.0431_F00104	5	■1.000	$M(n, K)$	$M(n, K)$	$M(n, K)$ denotes all $n \times n$ matrices with entries in K .
0902.0431_F00105	5	■1.000	$n \times n$	$n \times n$	$M(n, K)$ denotes all $\boxed{n \times n}$ matrices with entries in K .
0902.0431_F00106	5	■1.000	$E = \operatorname{diag}(1, \cdots, 1) \in M(n, K)$	$E = \operatorname{diag}(1, \cdots, 1) \in M(n, K)$	$\boxed{E = \operatorname{diag}(1, \cdots, 1) \in M(n, K)}$ is the unit matrix.
0902.0431_F00107	5	■1.000	$A \in M(n, K), \{ \}^{\{t\}} A$	$A \in M(n, K), {}^t A$	For $A \in M(n, K)$, ${}^t A$ denotes the transposed matrix of A and A^* denotes the
0902.0431_F00108	5	■1.000	A	A	For $A \in M(n, K)$, ${}^t A$ denotes the transposed matrix of A and A^* denotes the
0902.0431_F00109	5	■1.000	$A^{\{*\}}$	A^*	For $A \in M(n, K)$, ${}^t A$ denotes the transposed matrix of A and A^* denotes the
0902.0431_F00110	5	■0.999	$A: A^{\{*\}} = \{ \}^{\{t\}} \bar{A}$	$A: A^* = {}^t \bar{A}$	conjugate transposed matrix of A : $A^* = {}^t \bar{A}$.
0902.0431_F00111	5	■0.977	$O(n) = \left\{ A \in M(n, \mathbf{R}) \mid A^t A = E \right\}$	$O(n) = \left\{ A \in M(n, \mathbf{R}) \mid A^t A = E \right\}$	$\boxed{O(n) = \{A \in M(n, \mathbf{R}) \mid A^t A = E\}}$ (orthogonal group),
0902.0431_F00112	5	■0.982	$SO(n) = \{A \in O(n) \mid \det A = 1\}$	$SO(n) = \{A \in O(n) \mid \det A = 1\}$	$\boxed{SO(n) = \{A \in O(n) \mid \det A = 1\}}$ (spcecial orthogonal group),
0902.0431_F00113	5	■1.000	$U(n) = \left\{ A \in M(n, \mathbf{C}) \mid A^* A = E \right\}$	$U(n) = \{A \in M(n, \mathbf{C}) \mid A^* A = E\}$	$U(n) = \{A \in M(n, \mathbf{C}) \mid A^* A = E\}$ or $\{A \in M(n, \mathbf{C}) \mid \tau({}^t A)A = E\}$ (unitary
0902.0431_F00114	5	■1.000	$\left\{ A \in M(n, \mathbf{C}) \mid \tau({}^t A)A = E \right\}$	$\{A \in M(n, \mathbf{C}) \mid \tau({}^t A)A = E\}$	$U(n) = \{A \in M(n, \mathbf{C}) \mid A^* A = E\}$ or $\{A \in M(n, \mathbf{C}) \mid \tau({}^t A)A = E\}$ (unitary
0902.0431_F00115	5	■0.534	$SU(n) = \{A \in U(n) \mid \det A = 1\}$	$SU(n) = \{A \in U(n) \mid \det A = 1\}$	$\boxed{SU(n) = \{A \in U(n) \mid \det A = 1\}}$ (speccial unitary group),
0902.0431_F00116	5	■0.936	$\operatorname{Sp}(n) = \left\{ A \in M(n, \mathbf{H}) \mid A^* A = E \right\}$	$\operatorname{Sp}(n) = \{A \in M(n, \mathbf{H}) \mid A^* A = E\}$	$\boxed{\operatorname{Sp}(n) = \{A \in M(n, \mathbf{H}) \mid A^* A = E\}}$ (symplectic group).
0902.0431_F00117	5	■0.942	$\mathfrak{g}$	$\mathfrak{g}$	letter $\mathfrak{g}$. For example, $\mathfrak{su}(n)$ is the Lie algebra of the special unitary group $SU(n)$.
0902.0431_F00118	5	■0.942	$\mathfrak{su}(n)$	$\mathfrak{su}(n)$	letter $\mathfrak{g}$. For example, $\mathfrak{su}(n)$ is the Lie algebra of the special unitary group $SU(n)$.
0902.0431_F00119	5	■0.942	$\operatorname{SU}(n)$	$SU(n)$	letter $\mathfrak{g}$. For example, $\mathfrak{su}(n)$ is the Lie algebra of the special unitary group $SU(n)$.
0902.0431_F00120	6	■1.000	$\{e_0 = 1, e_1, e_2, e_3, e_4, e_5, e_6, e_7\}$	$\{e_0 = 1, e_1, e_2, e_3, e_4, e_5, e_6, e_7\}$	sider an 8 dimensional $\mathbf{R}$ -vector space with basis $\boxed{e_0 = 1, e_1, e_2, e_3, e_4, e_5, e_6, e_7}$ and
0902.0431_F00121	6	■1.000	e_1, e_2, e_3	e_1, e_2, e_3	between e_1, e_2, e_3 is defined as
0902.0431_F00122	6	■1.000	$e_1 e_6 = e_7, e_4 e_7 = e_3$	$e_1 e_6 = e_7, e_4 e_7 = e_3$	and defined similarly on the other lines. For example, $\boxed{e_1 e_6 = e_7, e_4 e_7 = e_3}$ etc. We
0902.0431_F00123	6	■1.000	e_2, e_5, e_7	e_2, e_5, e_7	regard that $\boxed{e_2, e_5, e_7}$ are also collinear, for example, $e_5 e_7 = e_2$. $e_0 = 1$ is the unit of
0902.0431_F00124	6	■1.000	$e_5 e_7 = e_2, e_0 = 1$	$e_5 e_7 = e_2, e_0 = 1$	regard that e_2, e_5, e_7 are also collinear, for example, $e_5 e_7 = e_2$. $e_0 = 1$ is the unit of
0902.0431_F00125	6	■1.000	$x1, x \in \mathbf{R}$	$x1, x \in \mathbf{R}$	and the distributive law. Thus $\mathfrak{C}$ has a multiplication. $x1, x \in \mathbf{R}$ is briefly denoted
0902.0431_F00126	6	■1.000	x	x	by x . In $\mathfrak{C}$, the conjugate $\bar{x}$, an inner product (x, y) , the length $ x $ and the real part
0902.0431_F00127	6	■1.000	$\bar{x}$	$\bar{x}$	by x . In $\mathfrak{C}$, the conjugate $\bar{x}$, an inner product (x, y) , the length $ x $ and the real part
0902.0431_F00128	6	■1.000	(x, y)	(x, y)	by x . In $\mathfrak{C}$, the conjugate $\bar{x}$, an inner product (x, y) , the length $ x $ and the real part
0902.0431_F00129	6	■1.000	$ x $	$ x $	by x . In $\mathfrak{C}$, the conjugate $\bar{x}$, an inner product (x, y) , the length $ x $ and the real part
0902.0431_F00130	6	■1.000	$R(x)$	$R(x)$	$R(x)$ are defined respectively by
0902.0431_F00131	6	■1.000	$x \in \mathfrak{C}, x \neq 0$	$x \in \mathfrak{C}, x \neq 0$	For $x \in \mathfrak{C}$, $x \neq 0$, we denote $\frac{\bar{x}}{ x ^2}$ by x^{-1} , then we have $xx^{-1} = x^{-1}x = 1$, and $\mathfrak{C}$
0902.0431_F00132	6	■1.000	$\frac{\bar{x}}{ x ^2}$	$\frac{\bar{x}}{ x ^2}$	For $x \in \mathfrak{C}$, $x \neq 0$, we denote $\frac{\bar{x}}{ x ^2}$ by x^{-1} , then we have $xx^{-1} = x^{-1}x = 1$, and $\mathfrak{C}$
0902.0431_F00133	6	■1.000	x^{-1}	x^{-1}	For $x \in \mathfrak{C}$, $x \neq 0$, we denote $\frac{\bar{x}}{ x ^2}$ by x^{-1} , then we have $xx^{-1} = x^{-1}x = 1$, and $\mathfrak{C}$
0902.0431_F00134	6	■1.000	$xx^{-1} = x^{-1}x = 1$	$xx^{-1} = x^{-1}x = 1$	For $x \in \mathfrak{C}$, $x \neq 0$, we denote $\frac{\bar{x}}{ x ^2}$ by x^{-1} , then we have $\boxed{xx^{-1} = x^{-1}x = 1}$, and $\mathfrak{C}$
0902.0431_F00135	6	■1.000	$x(yz) = (xy)z$	$x(yz) = (xy)z$	satisfies all axioms of a field except the associative law $x(yz) = (xy)z$. Of course

Identifier	Page	Conf.	LaTeX source	Rendered	Image
9902.0431_F00136	6	■ 1.000	$x\ y=y\ x$	$xy = yx$	the commutative law $xy = yx$ does not hold. Since the associative law does not
9902.0431_F00137	7	■ 1.000	$a, b, x, y \in \mathfrak{C}$	$a, b, x, y \in \mathfrak{C}$	Freudenthal [6] or Yokota [58]). For $\underline{a, b, x, y} \in \mathfrak{C}$, we have
9902.0431_F00138	7	■ 0.987	$1\ \text{\quad}(x\ y,\ x\ y)=(x,\ x)(y,\ y),\ \text{\quad} x\ y = x y $	$1\ \text{\quad}(xy, xy) = (x, x)(y, y),\ \text{\quad} xy = x y $	1 $\text{\quad}(xy, xy) = (x, x)(y, y),\ \text{\quad} xy = x y $.
9902.0431_F00139	7	■ 0.644	$2\ \text{\quad}(a\ x,\ a\ y)=(a,\ a)(x,\ y)=(x\ a,\ y\ a)$	$2\ \text{\quad}(ax, ay) = (a, a)(x, y) = (xa, ya)$	2 $\text{\quad}(ax, ay) = (a, a)(x, y) = (xa, ya)$.
9902.0431_F00140	7	■ 0.998	$3(a\ x,\ b\ y)+(b\ x,\ a\ y)=2(a,\ b)(x,\ y)$	$3(ax, by) + (bx, ay) = 2(a, b)(x, y)$	3 $\text{\quad}(ax, by) + (bx, ay) = 2(a, b)(x, y)$,
9902.0431_F00141	7	■ 1.000	$4\ \text{\quad}(a\ x,\ y)=(x,\ \bar{a}\ y),\ \text{\quad}(x\ a,\ y)=(x,\ y\ \bar{a})$	$4\ \text{\quad}(ax, y) = (x, \bar{a}y),\ \text{\quad}(xa, y) = (x, y\bar{a})$	4 $\text{\quad}(ax, y) = (x, \bar{a}y),\ \text{\quad}(xa, y) = (x, y\bar{a})$.
9902.0431_F00142	7	■ 0.424	$5\ \text{\quad}\overline{\bar{x}}=x,\ \text{\quad}\overline{x+y}=\bar{x}+\bar{y},\ \text{\quad}\overline{xy}=\bar{y}\bar{x}$	$5\ \text{\quad}\bar{\bar{x}} = x,\ \text{\quad}\overline{x+y} = \bar{x} + \bar{y},\ \text{\quad}\overline{xy} = \bar{y}\bar{x}$	5 $\text{\quad}\bar{\bar{x}} = x.\ \text{\quad}x + y = \bar{x} + \bar{y}.\ \text{\quad}\overline{xy} = \bar{y}\bar{x}$.
9902.0431_F00143	7	■ 0.982	$6\ \text{\quad}(x,\ y)=(y,\ x)=\frac{1}{2}(\bar{x}\ y+y\ \bar{x})=\frac{1}{2}(x\ \bar{y}+y\ \bar{x}),\ \text{\quad}\bar{x}\ x=x\ \bar{x}= x ^2$	$6\ \text{\quad}(x, y) = (y, x) = \frac{1}{2}(\bar{x}y + \bar{y}x) = \frac{1}{2}(x\bar{y} + y\bar{x}),\ \text{\quad}\bar{x}x = x\bar{x} = x ^2$	6 $\text{\quad}(x, y) = (y, x) = \frac{1}{2}(\bar{x}y + \bar{y}x) = \frac{1}{2}(x\bar{y} + y\bar{x}),\ \text{\quad}\bar{x}x = x\bar{x} = x ^2$.
9902.0431_F00144	7	■ 0.988	$7\ \text{\quad}a(\bar{a}\ x)=(a\ \bar{a})\ x,\ \text{\quad}a(x\ \bar{a})=(a\ x)\ \bar{a},\ \text{\quad}x(a\ \bar{a})=(x\ a)\ \bar{a}$	$7\ \text{\quad}a(\bar{a}x) = (a\bar{a})x,\ \text{\quad}a(x\bar{a}) = (ax)\bar{a},\ \text{\quad}x(a\bar{a}) = (xa)\bar{a}$	7 $\text{\quad}a(\bar{a}x) = (a\bar{a})x,\ \text{\quad}a(x\bar{a}) = (ax)\bar{a},\ \text{\quad}x(a\bar{a}) = (xa)\bar{a}$.
9902.0431_F00145	7	■ 1.000	$a(a\ x)=(a\ a)\ x,\ \text{\quad}a(x\ a)=(a\ x)\ a,\ \text{\quad}x(a\ a)=(x\ a)\ a$	$a(ax) = (aa)x,\ \text{\quad}a(xa) = (ax)a,\ \text{\quad}x(aa) = (xa)a$	8 $\text{\quad}a(ax) = (aa)x,\ \text{\quad}a(xa) = (ax)a,\ \text{\quad}x(aa) = (xa)a$,
9902.0431_F00146	7	■ 1.000	$8\ \text{\quad}\bar{b}(ax) + \bar{a}(bx) = 2(a, b)x = (xa)\bar{b} + (xb)\bar{a}$	$8\bar{b}(ax) + \bar{a}(bx) = 2(a, b)x = (xa)\bar{b} + (xb)\bar{a}$	8 $\text{\quad}\bar{b}(ax) + \bar{a}(bx) = 2(a, b)x = (xa)\bar{b} + (xb)\bar{a}$.
9902.0431_F00147	7	■ 1.000	$\{x,\ y,\ z\}=(x\ y)\ z-x(y\ z)$	$\{x, y, z\} = (xy)z - x(yz)$	9 We use a notation $\underline{\{x, y, z\} = (xy)z - x(yz)}$, called the associator of x, y, z .
9902.0431_F00148	7	■ 1.000	$x,\ y,\ z$	x, y, z	9 We use a notation $\{x, y, z\} = (xy)z - x(yz)$, called the associator of x, y, z .
9902.0431_F00149	7	■ 0.774	$10\ \text{\quad}(a\ x)(y\ a)=a(x\ y)\ a\ \text{\quad}\text{\quad}$	$10\ \text{\quad}(ax)(ya) = a(xy)a$	10 $\text{\quad}(ax)(ya) = a(xy)a$ (Moufang's formula).
9902.0431_F00150	7	■ 0.957	$11\ R(x\ y)=R(y\ x),\ \text{\quad}R(x(y\ z))=R(y(z\ x))=R(z(x\ y)) (=R(x\ y\ z))$	$11R(xy) = R(yx),\ \text{\quad}R(x(yz)) = R(y(zx)) = R(z(xy)) (= R(xyz))$	11 $\text{\quad}R(xy) = R(yx),\ \text{\quad}R(x(yz)) = R(y(zx)) = R(z(xy)) (= R(xyz))$.
9902.0431_F00151	7	■ 0.999	$\left\{1,\ a_{\{1\}},\ a_{\{2\}},\ \cdots,\ a_{\{7\}}\right\}$	$\{1, a_1, a_2, \cdots, a_7\}$	For an orthonormal basis $\underline{\{1, a_1, a_2, \cdots, a_7\}}$ of $\mathfrak{C}$ with respect to the inner product
9902.0431_F00152	7	■ 0.801	$12.1\ \text{\quad}a_{\{i\}}\left(a_{\{j\}}\ x\right)=-a_{\{j\}}\left(a_{\{i\}}\ x\right),\ \text{\quad}$	$12.1\ \text{\quad}a_i(a_jx) = -a_j(a_ix),$	12.1 $\text{\quad}a_i(a_jx) = -a_j(a_ix)$ in particular, $a_ia_j = -a_ja_i,\ i \neq j$.
9902.0431_F00153	7	■ 0.801	$\text{\quad}a_{\{i\}}\ a_{\{j\}}=-a_{\{j\}}\ a_{\{i\}},\ \text{\quad}i\ \neq\ j$	$\text{\quad}a_ia_j = -a_ja_i,\ i \neq j$	12.1 $\text{\quad}a_i(a_jx) = -a_j(a_ix)$, in particular, $\underline{a_ia_j = -a_ja_i,\ i \neq j}$.
9902.0431_F00154	7	■ 0.985	$12.2\ \text{\quad}a_{\{i\}}\left(a_{\{i\}}\ x\right)=-x$	$12.2\ \text{\quad}a_i(a_ix) = -x$	12.2 $\text{\quad}a_i(a_ix) = -x$, in particular, $\underline{a_i^2 = -1}$.
9902.0431_F00155	7	■ 0.985	$\text{\quad}a_{\{i\}}^2=-1$	$\text{\quad}a_i^2 = -1$	12.2 $\text{\quad}a_i(a_ix) = -x$, in particular, $\underline{a_i^2 = -1}$.
9902.0431_F00156	7	■ 1.000	$12.3\ a_{\{i\}}\left(a_{\{j\}}\ a_{\{k\}}\right)=a_{\{j\}}\left(a_{\{k\}}\ a_{\{i\}}\right)\ \text{\quad}\text{\quad}$	$12.3a_i(a_ja_k) = a_j(a_ka_i) = a_k(a_ia_j),\ i, j, k$	12.3 $\text{\quad}a_i(a_ja_k) = a_j(a_ka_i) = a_k(a_ia_j),\ \underline{i, j, k}$ are distinct.
9902.0431_F00157	7	■ 0.999	$\alpha\ \in\ G_2$	$\alpha \in G_2$	Lemma 1.2.1. For $\alpha \in G_2$, we have
9902.0431_F00158	7	■ 1.000	$x\ \in\ \mathfrak{C}$	$x \in \mathfrak{C}$	Proof. For $\alpha \in G_2$ and $x \in \mathfrak{C}$, we have
9902.0431_F00159	8	■ 1.000	$(\alpha 1)(\alpha 1)=\alpha(1\ \cdot\ 1)=\alpha 1$	$(\alpha 1)(\alpha 1) = \alpha(1 \cdot 1) = \alpha 1$	From $\underline{(\alpha 1)(\alpha 1) = \alpha(1 \cdot 1) = \alpha 1}$ and $\alpha 1 \neq 0$, we have $\alpha 1 = 1$, while from $(\alpha e_i)(\alpha e_i) =$
9902.0431_F00160	8	■ 1.000	$\alpha 1\ \neq\ 0$	$\alpha 1 \neq 0$	From $(\alpha 1)(\alpha 1) = \alpha(1 \cdot 1) = \alpha 1$ and $\alpha 1 \neq 0$, we have $\alpha 1 = 1$, while from $(\alpha e_i)(\alpha e_i) =$
9902.0431_F00161	8	■ 1.000	$\alpha 1=1$	$\alpha 1 = 1$	From $(\alpha 1)(\alpha 1) = \alpha(1 \cdot 1) = \alpha 1$ and $\alpha 1 \neq 0$, we have $\alpha 1 = 1$, while from $(\alpha e_i)(\alpha e_i) =$
9902.0431_F00162	8	■ 1.000	$\left(\alpha e_i\right)\left(\alpha e_i\right)=$	$(\alpha e_i)(\alpha e_i) =$	From $(\alpha 1)(\alpha 1) = \alpha(1 \cdot 1) = \alpha 1$ and $\alpha 1 \neq 0$, we have $\alpha 1 = 1$, while from $(\alpha e_i)(\alpha e_i) =$
9902.0431_F00163	8	■ 1.000	$\alpha\left(e_i e_i\right)=\alpha(-1)=-\alpha 1=-1$	$\alpha(e_ie_i) = \alpha(-1) = -\alpha 1 = -1$	$\underline{\alpha(e_ie_i) = \alpha(-1) = -\alpha 1 = -1}$ for $i \neq 0$, we get $\overline{\alpha e_i} = -\alpha e_i$. Now,
9902.0431_F00164	8	■ 1.000	$i\ \neq\ 0$	$i \neq 0$	$\alpha(e_ie_i) = \alpha(-1) = -\alpha 1 = -1$ for $i \neq 0$, we get $\overline{\alpha e_i} = -\alpha e_i$. Now,
9902.0431_F00165	8	■ 1.000	$\overline{\alpha e_i}=-\alpha e_i$	$\overline{\alpha e_i} = -\alpha e_i$	$\alpha(e_ie_i) = \alpha(-1) = -\alpha 1 = -1$ for $i \neq 0$, we get $\overline{\alpha e_i} = -\alpha e_i$. Now,
9902.0431_F00166	8	■ 1.000	$\alpha\ \in\ G_2,\ G_2$	$\alpha \in G_2, G_2$	Remark. Since $\alpha 1 = 1$ for $\alpha \in G_2$, G_2 is a subgroup of the orthogonal group
9902.0431_F00167	8	■ 1.000	$O(7)=\{\alpha\ \in\ O(\mathfrak{C})\ \ \alpha 1=1\}$	$O(7) = \{\alpha \in O(\mathfrak{C}) \mid \alpha 1 = 1\}$	$\underline{O(7) = \{\alpha \in O(\mathfrak{C}) \mid \alpha 1 = 1\}}$, that is, $G_2 \subset O(7)$.
9902.0431_F00168	8	■ 1.000	$G_2\ \subset\ O(7)$	$G_2 \subset O(7)$	$O(7) = \{\alpha \in O(\mathfrak{C}) \mid \alpha 1 = 1\}$, that is, $\underline{G_2 \subset O(7)}$.
9902.0431_F00169	8	■ 1.000	$\mathfrak{D}_4$	$\mathfrak{D}_4$	Then $G_{ij} \in \mathfrak{D}_4$ and $\{G_{ij} \mid 0 \leq i < j \leq 7\}$ forms an R -basis of $\mathfrak{D}_4$. Furthermore we
9902.0431_F00170	8	■ 1.000	$G_{\{i\ j\}}:\mathfrak{C}\rightarrow\mathfrak{C},\ i,\ j=0,1,\ \cdots,\ 7,\ i\ \neq\ j$	$G_{ij}:\mathfrak{C} \rightarrow \mathfrak{C}, i, j = 0, 1, \cdots, 7, i \neq j$	We define R -linear mappings $\underline{G_{ij}:\mathfrak{C} \rightarrow \mathfrak{C}, i, j = 0, 1, \cdots, 7, i \neq j}$ satisfying
9902.0431_F00171	8	■ 1.000	$G_{\{i\ j\}}\ \in\ \mathfrak{D}_4$	$G_{ij} \in \mathfrak{D}_4$	Then $\underline{G_{ij} \in \mathfrak{D}_4}$ and $\{G_{ij} \mid 0 \leq i < j \leq 7\}$ forms an R -basis of $\mathfrak{D}_4$. Furthermore we
9902.0431_F00172	8	■ 1.000	$\left\{G_{\{i\ j\}}\ \mid\ 0\ \leq\ i\ <\ j\ \leq\ 7\right\}$	$\{G_{ij} \mid 0 \leq i < j \leq 7\}$	Then $G_{ij} \in \mathfrak{D}_4$ and $\{G_{ij} \mid 0 \leq i < j \leq 7\}$ forms an R -basis of $\mathfrak{D}_4$. Furthermore we
9902.0431_F00173	8	■ 1.000	$F_{\{i\ j\}}:\mathfrak{C}\rightarrow\mathfrak{C},\ i,\ j=0,1,\ \cdots,\ 7,\ i\ \neq\ j$	$F_{ij}:\mathfrak{C} \rightarrow \mathfrak{C}, i, j = 0, 1, \cdots, 7, i \neq j$	define R -linear mappings $\underline{F_{ij}:\mathfrak{C} \rightarrow \mathfrak{C}, i, j = 0, 1, \cdots, 7, i \neq j}$ by
9902.0431_F00174	8	■ 0.999	$i,\ j=0,1,\ \cdots,\ 7,\ i\ \neq\ j$	$i, j = 0, 1, \cdots, 7, i \neq j$	Lemma 1.3.1. For $i, j = 0, 1, \cdots, 7, i \neq j$, we have $F_{ij} \in \mathfrak{D}_4$, and when $i < j$,

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0902.0431_F00175	8	■ 0.999	$F_{\{i\ j\}} \in \mathfrak{D}_{\{4\}}$	$F_{ij} \in \mathfrak{D}_4$	Lemma 1.3.1. For $i, j = 0, 1, \dots, 7, i \neq j$, we have $F_{ij} \in \mathfrak{D}_4$, and when $i < j$,
0902.0431_F00176	8	■ 0.999	$i < j$	$i < j$	Lemma 1.3.1. For $i, j = 0, 1, \dots, 7, i \neq j$, we have $F_{ij} \in \mathfrak{D}_4$, and when $i < j$,
0902.0431_F00177	8	■ 1.000	$F_{\{i\ j\}}$	F_{ij}	F_{ij} is expressed in terms of G_{ij} as follows.
0902.0431_F00178	8	■ 1.000	$G_{\{i\ j\}}$	G_{ij}	F_{ij} is expressed in terms of G_{ij} as follows.
0902.0431_F00179	9	■ 1.000	$\left\{F_{\{i\ j\}} \mid 0 \leq i < j \leq 7\right\}$	$\{F_{ij} \mid 0 \leq i < j \leq 7\}$	In particular, $\{F_{ij} \mid 0 \leq i < j \leq 7\}$ forms an <i>R</i> -basis of $\mathfrak{D}_4$.
0902.0431_F00180	9	■ 1.000	$\kappa, \pi, \nu: \mathfrak{D}_{\{4\}} \rightarrow \mathfrak{D}_{\{4\}}$	$\kappa, \pi, \nu: \mathfrak{D}_4 \rightarrow \mathfrak{D}_4$	Definition. We define <i>R</i> -linear mappings $\kappa, \pi, \nu: \mathfrak{D}_4 \rightarrow \mathfrak{D}_4$ respectively by
0902.0431_F00181	9	■ 1.000	κ, π, ν	κ, π, ν	Lemma 1.3.2. The mappings κ, π, ν are automorphisms of the Lie algebra $\mathfrak{D}_4$:
0902.0431_F00182	9	■ 1.000	$\kappa^2=1, \kappa$	$\kappa^2 = 1, \kappa$	Proof. Since $\kappa^2 = 1$, κ is an <i>R</i> -linear isomorphism of $\mathfrak{D}_4$. We have
0902.0431_F00183	9	■ 1.000	$\kappa \in \operatorname{Aut}(\mathfrak{D}_{\{4\}})$	$\kappa \in \operatorname{Aut}(\mathfrak{D}_4)$	Hence $\kappa \in \operatorname{Aut}(\mathfrak{D}_4)$. Since π maps the <i>R</i> -basis $\{G_{ij} \mid 0 \leq i < j \leq 7\}$ to the <i>R</i> -basis
0902.0431_F00184	9	■ 1.000	π	π	Hence $\kappa \in \operatorname{Aut}(\mathfrak{D}_4)$. Since π maps the <i>R</i> -basis $\{G_{ij} \mid 0 \leq i < j \leq 7\}$ to the <i>R</i> -basis
0902.0431_F00185	9	■ 1.000	$\mathfrak{D}_{\{4\}}, \pi$	$\mathfrak{D}_4, \pi$	$\{F_{ij} \mid 0 \leq i < j \leq 7\}$ of $\mathfrak{D}_4$, π is an <i>R</i> -linear isomorphism. To show that π is an
0902.0431_F00186	9	■ 1.000	i, j, k, l	i, j, k, l	In the following calculations, let i, j, k, l be all distinct and $i, j, k, l \neq 0$. For $x \in \mathfrak{C}$,
0902.0431_F00187	9	■ 1.000	$i, j, k, l \neq 0$	$i, j, k, l \neq 0$	In the following calculations, let i, j, k, l be all distinct and $i, j, k, l \neq 0$. For $x \in \mathfrak{C}$,
0902.0431_F00188	10	■ 1.000	$\pi \in \operatorname{Aut}(\mathfrak{D}_{\{4\}})$	$\pi \in \operatorname{Aut}(\mathfrak{D}_4)$	Hence $\pi \in \operatorname{Aut}(\mathfrak{D}_4)$. Finally, since $\nu = \pi\kappa$, we have $\nu \in \operatorname{Aut}(\mathfrak{D}_4)$.
0902.0431_F00189	10	■ 1.000	$\nu=\pi \kappa$	$\nu = \pi\kappa$	Hence $\pi \in \operatorname{Aut}(\mathfrak{D}_4)$. Finally, since $\nu = \pi\kappa$, we have $\nu \in \operatorname{Aut}(\mathfrak{D}_4)$.
0902.0431_F00190	10	■ 1.000	$\nu \in \operatorname{Aut}(\mathfrak{D}_{\{4\}})$	$\nu \in \operatorname{Aut}(\mathfrak{D}_4)$	Hence $\pi \in \operatorname{Aut}(\mathfrak{D}_4)$. Finally, since $\nu = \pi\kappa$, we have $\nu \in \operatorname{Aut}(\mathfrak{D}_4)$.
0902.0431_F00191	10	■ 1.000	$a \in \mathfrak{C}$	$a \in \mathfrak{C}$	Definition. For $a \in \mathfrak{C}$, we define <i>R</i> -linear mappings $L_a, R_a, T_a: \mathfrak{C} \rightarrow \mathfrak{C}$ respec-
0902.0431_F00192	10	■ 1.000	$L_{\{a\}}, R_{\{a\}}, T_{\{a\}}: \mathfrak{C} \rightarrow \mathfrak{C}$	$L_a, R_a, T_a: \mathfrak{C} \rightarrow \mathfrak{C}$	Definition. For $a \in \mathfrak{C}$, we define <i>R</i> -linear mappings $L_a, R_a, T_a: \mathfrak{C} \rightarrow \mathfrak{C}$ respec-
0902.0431_F00193	10	■ 1.000	$\mathfrak{C}_{\{0\}}$	$\mathfrak{C}_0$	Hereafter, we denote by $\mathfrak{C}_0$ the subset $\{a \in \mathfrak{C} \mid \bar{a} = -a\}$ of $\mathfrak{C}$.
0902.0431_F00194	10	■ 1.000	$\{a \in \mathfrak{C} \mid \bar{a} = -a\}$	$\{a \in \mathfrak{C} \mid \bar{a} = -a\}$	Hereafter, we denote by $\mathfrak{C}_0$ the subset $\{a \in \mathfrak{C} \mid \bar{a} = -a\}$ of $\mathfrak{C}$.
0902.0431_F00195	10	■ 1.000	$a \in \mathfrak{C}_0$	$a \in \mathfrak{C}_0$	Lemma 1.3.3. For $a \in \mathfrak{C}_0$, we have
0902.0431_F00196	10	■ 1.000	$\left(L_{\{a\}} x, y\right)=(a x, y)=(x, \bar{a} y)=-\left(x, a y\right)=-\left(x, L_{a y}\right), x, y \in \mathfrak{C}$	$(L_a x, y) = (a x, y) = (x, \bar{a} y) = -(x, a y) = -(x, L_a y), x, y \in \mathfrak{C}$	Proof. (1) $(L_a x, y) = (a x, y) = (x, \bar{a} y) = -(x, a y) = -(x, L_a y), x, y \in \mathfrak{C}$. Hence
0902.0431_F00197	10	■ 1.000	$L_{\{a\}} \in \mathfrak{D}_{\{4\}}$	$L_a \in \mathfrak{D}_4$	$L_a \in \mathfrak{D}_4$. Similarly, $R_a \in \mathfrak{D}_4$ and $T_a = L_a + R_a \in \mathfrak{D}_4$.
0902.0431_F00198	10	■ 1.000	$R_{\{a\}} \in \mathfrak{D}_{\{4\}}$	$R_a \in \mathfrak{D}_4$	$L_a \in \mathfrak{D}_4$. Similarly, $R_a \in \mathfrak{D}_4$ and $T_a = L_a + R_a \in \mathfrak{D}_4$.
0902.0431_F00199	10	■ 1.000	$T_{\{a\}}=L_{\{a\}}+R_{\{a\}} \in \mathfrak{D}_{\{4\}}$	$T_a = L_a + R_a \in \mathfrak{D}_4$	$L_a \in \mathfrak{D}_4$. Similarly, $R_a \in \mathfrak{D}_4$ and $T_a = L_a + R_a \in \mathfrak{D}_4$.
0902.0431_F00200	10	■ 1.000	$\left(\kappa L_{\{a\}}\right) x=\overline{L_{\{a\}} \bar{x}}=\overline{a \bar{x}}=x \bar{a}=-x a=-R_{\{a\}} x, x \in \mathfrak{C}$	$(\kappa L_a) x = \overline{L_a \bar{x}} = \overline{a \bar{x}} = x \bar{a} = -x a = -R_a x, x \in \mathfrak{C}$	(2) $(\kappa L_a) x = \overline{L_a \bar{x}} = \overline{a \bar{x}} = x \bar{a} = -x a = -R_a x, x \in \mathfrak{C}$. Hence $\kappa L_a = -R_a$. The
0902.0431_F00201	10	■ 1.000	$\kappa L_{\{a\}}=-R_{\{a\}}$	$\kappa L_a = -R_a$	(2) $(\kappa L_a) x = \overline{L_a \bar{x}} = \overline{a \bar{x}} = x \bar{a} = -x a = -R_a x, x \in \mathfrak{C}$. Hence $\kappa L_a = -R_a$. The
0902.0431_F00202	10	■ 1.000	$a=e_{\{i\}}, i=1, \cdots, 7$	$a = e_i, i = 1, \dots, 7$	(3) It is sufficient to show that these relations hold for $a = e_i, i = 1, \dots, 7$. We
0902.0431_F00203	11	■ 1.000	$\left\{L_{\{a\}} \mid a \in \mathfrak{C}_{\{0\}}\right\}$	$\{L_a \mid a \in \mathfrak{C}_0\}$	Lemma 1.3.4. The Lie algebra $\mathfrak{D}_4$ is generated by $\{L_a \mid a \in \mathfrak{C}_0\}$ (by taking at
0902.0431_F00204	11	■ 1.000	$D \in \mathfrak{D}_{\{4\}}$	$D \in \mathfrak{D}_4$	most one time the Lie bracket $[\quad, \quad]$), that is, any $D \in \mathfrak{D}_4$ is expressed by
0902.0431_F00205	11	■ 1.000	$\mathfrak{D}^{\prime}$	$\mathfrak{D}'$	Proof. Let $\mathfrak{D}'$ be the Lie subalgebra of $\mathfrak{D}_4$ generated by $\{L_a \mid a \in \mathfrak{C}_0\}$. Since
0902.0431_F00206	11	■ 0.996	$L_{\{e_{\{i\}}\}}=2 F_{\{i\ 0\}}$	$L_{e_i} = 2F_{i0}$	$L_{e_i} = 2F_{i0}$ and $[L_{e_i}, L_{e_j}] = 4[F_{i0}, F_{j0}] = -4F_{ij}, i \neq 0, j \neq 0, i \neq j$, we see that $\mathfrak{D}'$
0902.0431_F00207	11	■ 0.996	$\left[L_{\{e_{\{i\}}\}}, L_{\{e_{\{j\}}\}}\right]=4\left[F_{\{i\ 0\}}, F_{\{j\ 0\}}\right]=-4 F_{\{i\ j\}}, i \neq 0, j \neq 0, i \neq j$	$[L_{e_i}, L_{e_j}] = 4[F_{i0}, F_{j0}] = -4F_{ij}, i \neq 0, j \neq 0, i \neq j$	$L_{e_i} = 2F_{i0}$ and $[L_{e_i}, L_{e_j}] = 4[F_{i0}, F_{j0}] = -4F_{ij}, i \neq 0, j \neq 0, i \neq j$, we see that $\mathfrak{D}'$
0902.0431_F00208	11	■ 1.000	$F_{\{i\ j\}}, i, j=0,1, \cdots, 7, i \neq j$	$F_{ij}, i, j = 0, 1, \dots, 7, i \neq j$	contains $F_{ij}, i, j = 0, 1, \dots, 7, i \neq j$. Since $\{F_{ij}, i < j\}$ is an <i>R</i> -basis of $\mathfrak{D}_4$ (Lemma
0902.0431_F00209	11	■ 1.000	$\left\{F_{\{i\ j\}}, i < j\right\}$	$\{F_{ij}, i < j\}$	contains $F_{ij}, i, j = 0, 1, \dots, 7, i \neq j$. Since $\{F_{ij}, i < j\}$ is an <i>R</i> -basis of $\mathfrak{D}_4$ (Lemma
0902.0431_F00210	11	■ 1.000	$\operatorname{Aut}(\mathfrak{D}_{\{4\}})$	$\operatorname{Aut}(\mathfrak{D}_4)$	Theorem 1.3.5. In the automorphism group $\operatorname{Aut}(\mathfrak{D}_4)$ of $\mathfrak{D}_4$, the subgroup $\mathfrak{S}_3$
0902.0431_F00211	11	■ 1.000	$\mathfrak{S}_{\{3\}}$	$\mathfrak{S}_3$	Theorem 1.3.5. In the automorphism group $\operatorname{Aut}(\mathfrak{D}_4)$ of $\mathfrak{D}_4$, the subgroup $\mathfrak{S}_3$
0902.0431_F00212	11	■ 0.989	κ	κ	generated by κ and π is isomorphic to the symmetric group S_3 of degree 3. Further-
0902.0431_F00213	11	■ 0.989	$S_{\{3\}}$	S_3	generated by κ and π is isomorphic to the symmetric group S_3 of degree 3. Further-
0902.0431_F00214	11	■ 1.000	$\kappa^2=1, \pi^2=1, \nu^3=1$	$\kappa^2 = 1, \pi^2 = 1, \nu^3 = 1$	$\kappa^2 = 1, \pi^2 = 1, \nu^3 = 1$ etc. for L_a . However, these relations follow from Lemma
0902.0431_F00215	11	■ 1.000	$L_{\{a\}}$	L_a	$\kappa^2 = 1, \pi^2 = 1, \nu^3 = 1$ etc. for L_a . However, these relations follow from Lemma
0902.0431_F00216	11	■ 1.000	$f: \mathfrak{S}_{\{3\}} \rightarrow S_{\{3\}}$	$f: \mathfrak{S}_3 \rightarrow S_3$	1.3.3.(2),(3),(4). The mapping $f: \mathfrak{S}_3 \rightarrow S_3$ defined by the correspondence
0902.0431_F00217	11	■ 0.999	$D_{\{1\}} \in$	$D_1 \in$	Theorem 1.3.6. (Principle of infinitesimal triality in $\mathfrak{D}_4$). For any $D_1 \in$
0902.0431_F00218	11	■ 1.000	$D_{\{2\}}, D_{\{3\}} \in \mathfrak{D}_{\{4\}}$	$D_2, D_3 \in \mathfrak{D}_4$	$\mathfrak{D}_4$, there exist $D_2, D_3 \in \mathfrak{D}_4$ such that

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0902.0431_F00219	11	■ 1.000	$D_{\{2\}}, D_{\{3\}}$	D_2, D_3	Also such D_2, D_3 are uniquely determined for D_1 and we have
0902.0431_F00220	11	■ 1.000	$D_{\{1\}}$	D_1	Also such D_2, D_3 are uniquely determined for D_1 and we have
0902.0431_F00221	11	■ 1.000	$L_{\{a\}}, R_{\{a\}}, T_{\{a\}} \in \mathfrak{D}_4$	$L_a, R_a, T_a \in \mathfrak{D}_4$	Proof. If $a \in \mathfrak{C}_0$, then $L_a, R_a, T_a \in \mathfrak{D}_4$ (Lemma 1.3.3.(1)) and the equality
0902.0431_F00222	11	■ 1.000	$(a \ x) \ y + x(y \ a) = a(x \ y) + (x \ y) \ a$	$(ax)y + x(ya) = a(xy) + (xy)a$	$(ax)y + x(ya) = a(xy) + (xy)a$ implies that
0902.0431_F00223	12	■ 1.000	$b \in \mathfrak{C}_0$	$b \in \mathfrak{C}_0$	Similarly, for $b \in \mathfrak{C}_0$, we have
0902.0431_F00224	12	■ 1.000	$T_{\{a\}}$	T_a	Applying T_a on (ii) and using (i), we have
0902.0431_F00225	12	■ 1.000	a	a	Exchanging a for b , and subtracting from the above, we get
0902.0431_F00226	12	■ 1.000	b	b	Exchanging a for b , and subtracting from the above, we get
0902.0431_F00227	12	■ 1.000	$D_{\{1\}} \in \mathfrak{D}_4$	$D_1 \in \mathfrak{D}_4$	Since $D_1 \in \mathfrak{D}_4$ is expressed as
0902.0431_F00228	12	■ 1.000	$D_{\{2\}} = R_{\{a\}} + \sum \left[R_{\{b\}}, R_{\{c\}} \right], D_{\{3\}} = T_{\{a\}} + \sum \left[T_{\{b\}}, T_{\{c\}} \right]$	$D_2 = R_a + \sum [R_b, R_c], D_3 = T_a + \sum [T_b, T_c]$	(Lemma 1.3.4), putting $D_2 = R_a + \sum [R_b, R_c], D_3 = T_a + \sum [T_b, T_c]$ and using (i),
0902.0431_F00229	12	■ 1.000	$D_{\{2\}}$	D_2	Next, we shall show that D_2 and D_3 are determined uniquely for D_1 . To prove this,
0902.0431_F00230	12	■ 1.000	$D_{\{3\}}$	D_3	Next, we shall show that D_2 and D_3 are determined uniquely for D_1 . To prove this,
0902.0431_F00231	12	■ 1.000	$D_{\{1\}} = 0$	$D_1 = 0$	it is sufficient to show for $D_1 = 0$ that $D_2 = D_3 = 0$. Now, suppose
0902.0431_F00232	12	■ 1.000	$D_{\{2\}} = D_{\{3\}} = 0$	$D_2 = D_3 = 0$	it is sufficient to show for $D_1 = 0$ that $D_2 = D_3 = 0$. Now, suppose
0902.0431_F00233	12	■ 1.000	$x = 1$	$x = 1$	Putting $x = 1$, we have $D_2 y = D_3 y$, so that $D_2 = D_3 (= D)$. Therefore
0902.0431_F00234	12	■ 1.000	$D_{\{2\}} y = D_{\{3\}} y$	$D_2 y = D_3 y$	Putting $x = 1$, we have $D_2 y = D_3 y$, so that $D_2 = D_3 (= D)$. Therefore
0902.0431_F00235	12	■ 1.000	$D_{\{2\}} = D_{\{3\}} (= D)$	$D_2 = D_3 (= D)$	Putting $x = 1$, we have $D_2 y = D_3 y$, so that $D_2 = D_3 (= D)$. Therefore
0902.0431_F00236	12	■ 1.000	$D_1 = p$	$D_1 = p$	Putting $D_1 = p$, we have $2(p, 1) = (p, 1) + (1, p) = (D_1, 1) + (1, D_1) = 0$, which implies
0902.0431_F00237	12	■ 1.000	$2(p, 1) = (p, 1) + (1, p) = (D_1, 1) + (1, D_1) = 0$	$2(p, 1) = (p, 1) + (1, p) = (D_1, 1) + (1, D_1) = 0$	Putting $D_1 = p$, we have $2(p, 1) = (p, 1) + (1, p) = (D_1, 1) + (1, D_1) = 0$, which implies
0902.0431_F00238	12	■ 0.997	$p \in \mathfrak{C}_0$	$p \in \mathfrak{C}_0$	$p \in \mathfrak{C}_0$. Furthermore, putting $y = 1$ in (iv), we have $xp = Dx$, so (iv) becomes
0902.0431_F00239	12	■ 0.997	$y = 1$	$y = 1$	$p \in \mathfrak{C}_0$. Furthermore, putting $y = 1$ in (iv), we have $xp = Dx$, so (iv) becomes
0902.0431_F00240	12	■ 0.997	$x \ p = D \ x$	$xp = Dx$	$p \in \mathfrak{C}_0$. Furthermore, putting $y = 1$ in (iv), we have $xp = Dx$, so (iv) becomes
0902.0431_F00241	12	■ 0.997	$p \in \mathbf{R}$	$p \in \mathbf{R}$	We therefore see that $p \in \mathbf{R}$, so that $p = 0$ since $p \in \mathfrak{C}_0$. Hence $Dx = xp = 0$,
0902.0431_F00242	12	■ 0.997	$p = 0$	$p = 0$	We therefore see that $p \in \mathbf{R}$, so that $p = 0$ since $p \in \mathfrak{C}_0$. Hence $Dx = xp = 0$,
0902.0431_F00243	12	■ 0.997	$D \ x = x \ p = 0$	$Dx = xp = 0$	We therefore see that $p \in \mathbf{R}$, so that $p = 0$ since $p \in \mathfrak{C}_0$. Hence $Dx = xp = 0$,
0902.0431_F00244	12	■ 0.999	$D = 0$	$D = 0$	and so $D = 0$. This proves the uniqueness. Finally, if we express $D_1 \in \mathfrak{D}_4$ as
0902.0431_F00245	12	■ 1.000	$D_{\{1\}} = L_{\{a\}} + \sum \left[L_{\{b\}}, L_{\{c\}} \right], a, b, c \in \mathfrak{C}_0$	$D_1 = L_a + \sum [L_b, L_c], a, b, c \in \mathfrak{C}_0$	$D_1 = L_a + \sum [L_b, L_c], a, b, c \in \mathfrak{C}_0$, then $D_2 = R_a + \sum [R_b, R_c], D_3 = T_a + \sum [T_b, T_c]$
0902.0431_F00246	12	■ 1.000	$D_{\{2\}} = \nu D_{\{1\}}, D_{\{3\}} = \pi D_{\{1\}}$	$D_2 = \nu D_1, D_3 = \pi D_1$	from the arguments above and the uniqueness. Hence we have $D_2 = \nu D_1, D_3 = \pi D_1$
0902.0431_F00247	12	■ 1.000	$D_{\{1\}}, D_{\{2\}}, D_{\{3\}} \in \mathfrak{D}_4$	$D_1, D_2, D_3 \in \mathfrak{D}_4$	Lemma 1.3.7. For $D_1, D_2, D_3 \in \mathfrak{D}_4$, the relation
0902.0431_F00248	13	■ 1.000	$D_{\{2\}} \in \mathfrak{D}_4$	$D_2 \in \mathfrak{D}_4$	Proof. For $D_2 \in \mathfrak{D}_4$, there exist $D_3', D_1' \in \mathfrak{D}_4$ such that
0902.0431_F00249	13	■ 1.000	$D_{\{3\}} \in \mathfrak{D}_4, D_{\{1\}} \in \mathfrak{D}_4$	$D_3', D_1' \in \mathfrak{D}_4$	Proof. For $D_2 \in \mathfrak{D}_4$, there exist $D_3', D_1' \in \mathfrak{D}_4$ such that
0902.0431_F00250	13	■ 1.000	$D_{\{3\}} \in \mathfrak{D}_4, \kappa D_{\{1\}} \in \mathfrak{D}_4$	$D_3' = \nu D_2, \kappa D_1' = \pi D_2$	and $D_3' = \nu D_2, \kappa D_1' = \pi D_2$ (Theorem 1.3.6). The assumption of the lemma is
0902.0431_F00251	13	■ 1.000	$D_{\{2\}} = \nu D_{\{1\}}, \kappa D_{\{3\}} = \pi D_{\{1\}}$	$D_2 = \nu D_1, \kappa D_3 = \pi D_1$	$D_2 = \nu D_1, \kappa D_3 = \pi D_1$ (Theorem 1.3.6). Hence, using Theorem 1.3.5, we have
0902.0431_F00252	13	■ 1.000	$D \in \text{Hom}_{\mathbf{R}}(\mathfrak{C})$	$D \in \text{Hom}_{\mathbf{R}}(\mathfrak{C})$	Proof. If $D \in \text{Hom}_{\mathbf{R}}(\mathfrak{C})$ satisfies $(\exp tD)(xy) = ((\exp tD)x)((\exp tD)y), t \in \mathbf{R}$,
0902.0431_F00253	13	■ 1.000	$(\exp tD)(xy) = ((\exp tD)x)((\exp tD)y), t \in \mathbf{R}$	$(\exp tD)(xy) = ((\exp tD)x)((\exp tD)y), t \in \mathbf{R}$	Proof. If $D \in \text{Hom}_{\mathbf{R}}(\mathfrak{C})$ satisfies $(\exp tD)(xy) = ((\exp tD)x)((\exp tD)y), t \in \mathbf{R}$,
0902.0431_F00254	13	■ 1.000	t	t	then by differentiating with respect to t and putting $t = 0$, we get $D(xy) = (Dx)y +$
0902.0431_F00255	13	■ 1.000	$t = 0$	$t = 0$	then by differentiating with respect to t and putting $t = 0$, we get $D(xy) = (Dx)y +$
0902.0431_F00256	13	■ 1.000	$D(x \ y) = (D \ x) \ y +$	$D(xy) = (Dx)y +$	then by differentiating with respect to t and putting $t = 0$, we get $D(xy) = (Dx)y +$
0902.0431_F00257	13	■ 1.000	$x(D \ y)$	$x(Dy)$	$x(Dy)$. Conversely, if $D \in \text{Hom}_{\mathbf{R}}(\mathfrak{C})$ satisfies $D(xy) = (Dx)y + x(Dy)$, then it is not
0902.0431_F00258	13	■ 1.000	$D(x \ y) = (D \ x) \ y + x(D \ y)$	$D(xy) = (Dx)y + x(Dy)$	$x(Dy)$. Conversely, if $D \in \text{Hom}_{\mathbf{R}}(\mathfrak{C})$ satisfies $D(xy) = (Dx)y + x(Dy)$, then it is not
0902.0431_F00259	13	■ 1.000	$\alpha = \exp tD$	$\alpha = \exp tD$	difficult to verify that $\alpha = \exp tD$ satisfies $\alpha(xy) = (\alpha x)(\alpha y)$.
0902.0431_F00260	13	■ 1.000	$\alpha(xy) = (\alpha x)(\alpha y)$	$\alpha(xy) = (\alpha x)(\alpha y)$	difficult to verify that $\alpha = \exp tD$ satisfies $\alpha(xy) = (\alpha x)(\alpha y)$.
0902.0431_F00261	13	■ 0.997	$\mathfrak{b}_3 = \mathfrak{so}(7)$	$\mathfrak{b}_3 = \mathfrak{so}(7)$	In order to study the Lie algebra $\mathfrak{g}_2$, we need the following Lie algebra $\mathfrak{b}_3 = \mathfrak{so}(7)$
0902.0431_F00262	13	■ 0.987	$SO(7)$	$SO(7)$	of $SO(7)$:
0902.0431_F00263	13	■ 1.000	$\mathfrak{b}_3$	$\mathfrak{b}_3$	Lemma 1.4.2. $\mathfrak{g}_2$ is a Lie subalgebra of $\mathfrak{b}_3$. Moreover we have
0902.0431_F00264	13	■ 1.000	$D \in \mathfrak{g}_2$	$D \in \mathfrak{g}_2$	Proof. Let $D \in \mathfrak{g}_2$. Putting $x = y = 1$ in $D(xy) = (Dx)y + x(Dy)$, we get
0902.0431_F00265	13	■ 1.000	$x = y = 1$	$x = y = 1$	Proof. Let $D \in \mathfrak{g}_2$. Putting $x = y = 1$ in $D(xy) = (Dx)y + x(Dy)$, we get

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0902.0431_F00266	13	■ 1.000	$D \, 1=0$	$D1 = 0$	$D1 = 0$. We next show that
0902.0431_F00267	13	■ 0.997	$\left(D \, e_{\{i\}}\right) \, e_{\{i\}}+e_{\{i\}}\left(D \, e_{\{i\}}\right)=D \, \left(e_{\{i\}} \, e_{\{i\}}\right)=D(-1)=0$	$(De_i)e_i + e_i(De_i) = D(e_ie_i) = D(-1) = 0$	If $i \neq 0$, then $\boxed{(De_i)e_i + e_i(De_i) = D(e_ie_i) = D(-1) = 0}$, and so $De_i \in \mathfrak{C}_0$. Together
0902.0431_F00268	13	■ 0.997	$D \, e_{\{i\}} \, \in \, \mathfrak{C}_0$	$De_i \in \mathfrak{C}_0$	If $i \neq 0$, then $(De_i)e_i + e_i(De_i) = D(e_ie_i) = D(-1) = 0$, and so $\boxed{De_i \in \mathfrak{C}_0}$. Together
0902.0431_F00269	13	■ 0.999	$D \, x \, \in \, \mathfrak{C}_0, \, x \, \in \, \mathfrak{C}$	$Dx \in \mathfrak{C}_0, x \in \mathfrak{C}$	with $D1 = 0$, we have $\boxed{Dx \in \mathfrak{C}_0, x \in \mathfrak{C}}$. Now, note that
0902.0431_F00270	13	■ 0.998	D	D	Applying D on the relation above, we have $(Dx)y + x(Dy) + (Dy)x + y(Dx) = 0$.
0902.0431_F00271	13	■ 0.998	$(D \, x) \, y+x(D \, y)+(D \, y) \, x+y(D \, x)=0$	$(Dx)y + x(Dy) + (Dy)x + y(Dx) = 0$	Applying D on the relation above, we have $\boxed{(Dx)y + x(Dy) + (Dy)x + y(Dx) = 0}$.
0902.0431_F00272	13	■ 1.000	$D \, x, \, D \, y \, \in \, \mathfrak{C}_0$	$Dx, Dy \in \mathfrak{C}_0$	Since $\boxed{Dx, Dy \in \mathfrak{C}_0}$, we get
0902.0431_F00273	14	■ 1.000	$\mathfrak{g}_2 \subset \mathfrak{b}_3$	$\mathfrak{g}_2 \subset \mathfrak{b}_3$	Hence $\mathfrak{g}_2 \subset \mathfrak{b}_3$. The relation in the lemma follows easily from Theorem 1.3.5 and
0902.0431_F00274	14	■ 0.998	$D \, \in \, \mathfrak{b}_3$	$D \in \mathfrak{b}_3$	Proof. Let $D \in \mathfrak{g}_2$. Since $D \in \mathfrak{b}_3$ (Lemma 1.4.2), $D = \sum_{0 < i < j} \lambda_{ij} G_{ij}$, $\lambda_{ij} \in \mathbf{R}$.
0902.0431_F00275	14	■ 0.998	$D=\sum_{0<i<j} \lambda_{ij} \, G_{ij}, \, \lambda_{ij} \in \mathbf{R}$	$D = \sum_{0 < i < j} \lambda_{ij} G_{ij}, \lambda_{ij} \in \mathbf{R}$	Proof. Let $D \in \mathfrak{g}_2$. Since $D \in \mathfrak{b}_3$ (Lemma 1.4.2), $D = \boxed{\sum_{0 < i < j} \lambda_{ij} G_{ij}, \lambda_{ij} \in \mathbf{R}}$.
0902.0431_F00276	14	■ 0.996	$\pi \, D=D$	$\pi D = D$	The condition $\boxed{\pi D = D}$ (Lemma 1.4.2) implies that
0902.0431_F00277	14	■ 1.000	$1 \, \in \, \mathfrak{C}$	$1 \in \mathfrak{C}$	Applying the above element on $1 \in \mathfrak{C}$, we have
0902.0431_F00278	14	■ 1.000	$e_{\{j\}} \, e_{\{i\}}$	$e_j e_i$	Replacing $e_j e_i$ by e_k and comparing the coefficient of each $e_1, e_2, \dots, e_7$, we have
0902.0431_F00279	14	■ 1.000	$e_{\{k\}}$	e_k	Replacing $e_j e_i$ by e_k and comparing the coefficient of each $e_1, e_2, \dots, e_7$, we have
0902.0431_F00280	14	■ 1.000	$e_{\{1\}}, \, e_{\{2\}}, \, \cdots, \, e_{\{7\}}$	$e_1, e_2, \dots, e_7$	Replacing $e_j e_i$ by e_k and comparing the coefficient of each $\boxed{e_1, e_2, \dots, e_7}$, we have
0902.0431_F00281	14	■ 1.000	$\mathfrak{C}$	$\mathbf{C}$	The Cayley algebra $\mathfrak{C}$ naturally contains the field $\mathbf{C}$ of complex numbers as
0902.0431_F00282	15	■ 1.000	$\mathfrak{C} \oplus \mathfrak{C}^3$	$\mathbf{C} \oplus \mathbf{C}^3$	We associate such element x of $\mathfrak{C}$ with the element of $\mathbf{C} \oplus \mathbf{C}^3$
0902.0431_F00283	15	■ 1.000	$(\mathfrak{m}, \mathfrak{n})$	$(\mathbf{m}, \mathbf{n})$	where the real valued symmetric inner product $(\mathbf{m}, \mathbf{n})$, the Hermitian inner product
0902.0431_F00284	15	■ 1.000	$\langle \mathfrak{m}, \mathfrak{n} \rangle$	$\langle \mathbf{m}, \mathbf{n} \rangle$	$\boxed{\langle \mathbf{m}, \mathbf{n} \rangle}$ and the exterior product $\mathbf{m} \times \mathbf{n}$ are usually defined respectively by
0902.0431_F00285	15	■ 1.000	$\mathfrak{m} \times \mathfrak{n}$	$\mathbf{m} \times \mathbf{n}$	$\langle \mathbf{m}, \mathbf{n} \rangle$ and the exterior product $\mathbf{m} \times \mathbf{n}$ are usually defined respectively by
0902.0431_F00286	15	■ 0.786	$\mathfrak{m}=\left(\begin{array}{c} m_1 \\ m_2 \\ m_3 \end{array}\right), \, \mathfrak{n}=\left(\begin{array}{c} n_1 \\ n_2 \\ n_3 \end{array}\right) \in \mathfrak{C}^3$	$\mathbf{m} = \begin{pmatrix} m_1 \\ m_2 \\ m_3 \end{pmatrix}, \mathbf{n} = \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix} \in \mathbf{C}^3$	for $\mathbf{m} = \begin{pmatrix} m_1 \\ m_2 \\ m_3 \end{pmatrix}, \mathbf{n} = \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix} \in \mathbf{C}^3$. Since these operations correspond to their
0902.0431_F00287	15	■ 1.000	$\left(\mathfrak{g}_2\right)_{e_1}$	$(\mathfrak{g}_2)_{e_1}$	We shall study the following subalgebra $(\mathfrak{g}_2)_{e_1}$ of $\mathfrak{g}_2$:
0902.0431_F00288	15	■ 0.951	$\mathfrak{su}(3)$	$(\mathfrak{g}_2)_{e_1} \cong \mathfrak{su}(3)$	Theorem 1.5.1. $\boxed{(\mathfrak{g}_2)_{e_1} \cong \mathfrak{su}(3)}$.
0902.0431_F00289	15	■ 1.000	$\varphi_*: \mathfrak{su}(3) \rightarrow \mathfrak{g}_2$	$\varphi_*: \mathfrak{su}(3) = \{D \in M(3, \mathbf{C}) \mid D^* = -D, \text{tr}(D) = 0\}$	Proof. We define a mapping $\boxed{\varphi_*: \mathfrak{su}(3) = \{D \in M(3, \mathbf{C}) \mid D^* = -D, \text{tr}(D) = 0\}}$
0902.0431_F00290	15	■ 1.000	$0 \rightarrow \left(\mathfrak{g}_2\right)_{e_1}$	$0 \rightarrow (\mathfrak{g}_2)_{e_1}$	$\boxed{0 \rightarrow (\mathfrak{g}_2)_{e_1}}$ by
0902.0431_F00291	15	■ 1.000	$\varphi_*(D) \in \left(\mathfrak{g}_2\right)_{e_1}$	$\varphi_*(D) \in (\mathfrak{g}_2)_{e_1}$	We first prove that $\boxed{\varphi_*(D) \in (\mathfrak{g}_2)_{e_1}}$. For elements
0902.0431_F00292	16	■ 0.671	$\mathfrak{su}(3)$	$\mathfrak{su}(3)$	of an $\mathbf{R}$ -basis of $\mathfrak{su}(3)$ (where $E_{kl} \in M(3, \mathbf{R})$ is the matrix with the (k, l) -entry is 1
0902.0431_F00293	16	■ 0.671	$E_{kl} \in M(3, \mathbf{R})$	$E_{kl} \in M(3, \mathbf{R})$	of an $\mathbf{R}$ -basis of $\mathfrak{su}(3)$ (where $\boxed{E_{kl} \in M(3, \mathbf{R})}$ is the matrix with the (k, l) -entry is 1
0902.0431_F00294	16	■ 0.671	(k, l)	(k, l)	of an $\mathbf{R}$ -basis of $\mathfrak{su}(3)$ (where $E_{kl} \in M(3, \mathbf{R})$ is the matrix with the (k, l) -entry is 1
0902.0431_F00295	16	■ 1.000	$\varphi_*(D) \subset \mathfrak{g}_2$	$\varphi_*(D) \subset \mathfrak{g}_2$	Hence $\varphi_*(D) \subset \mathfrak{g}_2$ (Theorem 1.4.3). Clearly $\varphi_*(D)e_1 = 0$, so that $\varphi_*(D) \subset (\mathfrak{g}_2)_{e_1}$.
0902.0431_F00296	16	■ 1.000	$\varphi_*(D) \, e_1=0$	$\varphi_*(D)e_1 = 0$	Hence $\varphi_*(D) \subset \mathfrak{g}_2$ (Theorem 1.4.3). Clearly $\boxed{\varphi_*(D)e_1 = 0}$, so that $\varphi_*(D) \subset (\mathfrak{g}_2)_{e_1}$.
0902.0431_F00297	16	■ 1.000	$\varphi_*(D) \subset \left(\mathfrak{g}_2\right)_{e_1}$	$\varphi_*(D) \subset (\mathfrak{g}_2)_{e_1}$	Hence $\varphi_*(D) \subset \mathfrak{g}_2$ (Theorem 1.4.3). Clearly $\varphi_*(D)e_1 = 0$, so that $\varphi_*(D) \subset (\mathfrak{g}_2)_{e_1}$.
0902.0431_F00298	16	■ 0.998	$\varphi_*: \mathfrak{su}(3) \rightarrow \mathfrak{g}_2$	$\varphi_*: \mathfrak{su}(3) \rightarrow \mathfrak{g}_2$	Obviously $\varphi_*: \mathfrak{su}(3) \rightarrow \mathfrak{g}_2$ is a homomorphism as Lie algebras and is injective, so
0902.0431_F00299	16	■ 1.000	$\varphi_*(\mathfrak{su}(3))$	$\varphi_*(\mathfrak{su}(3))$	that we identify $\mathfrak{su}(3)$ and $\varphi_*(\mathfrak{su}(3))$. We set
0902.0431_F00300	16	■ 1.000	$\mathfrak{S}$	$\mathfrak{S}$	and let $\mathfrak{S}$ be the $\mathbf{R}$ -vector subspace of $\mathfrak{g}_2$ spanned by $S_1, \dots, S_6$. Then we have the
0902.0431_F00301	16	■ 1.000	$S_1, \dots, S_6$	$S_1, \dots, S_6$	and let $\mathfrak{S}$ be the $\mathbf{R}$ -vector subspace of $\mathfrak{g}_2$ spanned by $\boxed{S_1, \dots, S_6}$. Then we have the
0902.0431_F00302	16	■ 0.999	$\varphi_*: \mathfrak{su}(3) \rightarrow \left(\mathfrak{g}_2\right)_{e_1}$	$\varphi_*: \mathfrak{su}(3) \rightarrow (\mathfrak{g}_2)_{e_1}$	Now, we shall show that $\boxed{\varphi_*: \mathfrak{su}(3) \rightarrow (\mathfrak{g}_2)_{e_1}}$ is onto. Let $B \in (\mathfrak{g}_2)_{e_1}$. Denote
0902.0431_F00303	16	■ 0.999	$B \in \left(\mathfrak{g}_2\right)_{e_1}$	$B \in (\mathfrak{g}_2)_{e_1}$	Now, we shall show that $\varphi_*: \mathfrak{su}(3) \rightarrow (\mathfrak{g}_2)_{e_1}$ is onto. Let $B \in (\mathfrak{g}_2)_{e_1}$. Denote
0902.0431_F00304	16	■ 1.000	$B \, e_1=0$	$Be_1 = 0$	From the condition $Be_1 = 0$, we have

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0902.0431_F00305	16	■ 1.000	$x_{\{1\}}=\cdots=x_{\{6\}}=0$	$x_1 = \cdots = x_6 = 0$	Hence $x_1 = \cdots = x_6 = 0$, so that $B = D \in \mathfrak{su}(3)$. Thus the proof of Theorem 1.5.1 is
0902.0431_F00306	16	■ 1.000	$B=D \in \mathfrak{su}(3)$	$B = D \in \mathfrak{su}(3)$	Hence $x_1 = \cdots = x_6 = 0$, so that $B = D \in \mathfrak{su}(3)$. Thus the proof of Theorem 1.5.1 is
0902.0431_F00307	16	■ 1.000	$\mathfrak{C}^C=\left\{x_{\{1\}}+i x_{\{2\}} \mid x_{\{1\}}, x_{\{2\}} \in \mathfrak{C}\right\}$	$\mathfrak{C}^C = \{x_1 + ix_2 \mid x_1, x_2 \in \mathfrak{C}\}$	Let $\mathfrak{C}^C = \{x_1 + ix_2 \mid x_1, x_2 \in \mathfrak{C}\}$ be the complexification of the Cayley algebra $\mathfrak{C}$.
0902.0431_F00308	16	■ 1.000	$\mathfrak{C}^C$	$\mathfrak{C}^C$	In the same manner as in $\mathfrak{C}$, we can also define in $\mathfrak{C}^C$ the multiplication xy , the inner
0902.0431_F00309	16	■ 1.000	$x y$	xy	In the same manner as in $\mathfrak{C}$, we can also define in $\mathfrak{C}^C$ the multiplication xy , the inner
0902.0431_F00310	16	■ 0.535	$1 \sim 12.3$	$1 \sim 12.3$	product (x, y) such that they satisfy properties ~ 12.3 of Section 1.1 (except some
0902.0431_F00311	17	■ 1.000	$\mathfrak{su}(3) \subset \mathfrak{g}_2$	$\mathfrak{su}(3) \subset \mathfrak{g}_2$	For a while, in the Lie algebra $\mathfrak{su}(3) \subset \mathfrak{g}_2$, we use the following notations.
0902.0431_F00312	17	■ 1.000	$[D, S], D \in \mathfrak{su}(3), S \in \mathfrak{S}$	$[D, S], D \in \mathfrak{su}(3), S \in \mathfrak{S}$	Lemma 1.6.1. The Lie brackets $[D, S], D \in \mathfrak{su}(3), S \in \mathfrak{S}$ are given as follows.
0902.0431_F00318	17	■ 1.000	$H_{\{1\}}$	H_1	the loss of generality). Applying H_1 on S , we have $x_1 S_2 - x_2 S_1 - x_3 S_4 + x_4 S_3 \in W$.
0902.0431_F00326	17	■ 1.000	$L_{\{13\}}$	L_{13}	Next, applying L_{13} and L_{31} on it, we have
0902.0431_F00328	17	■ 1.000	$L_{\{31\}}$	L_{31}	Next, applying L_{13} and L_{31} on it, we have
0902.0431_F00331	17	■ 0.964	W	W	Proof. Evidently $\mathfrak{S}$ is a $\mathfrak{su}(3)$ - $\mathbf{R}$ -module. Let W be a non-zero $\mathfrak{su}(3)$ -invariant
0902.0431_F00332	17	■ 1.000	$S_{\{k\}}$	S_k	$\mathbf{R}$ -submodule of $\mathfrak{S}$. If W contains some S_k , then, from the table of Lemma 1.6.1,
0902.0431_F00333	17	■ 1.000	$S_{\{k\}}, k=1,2, \cdots, 6$	$S_k, k = 1, 2, \cdots, 6$	we can see that W contains all $S_k, k = 1, 2, \cdots, 6$, and hence $W = \mathfrak{S}$. Now, let
0902.0431_F00334	17	■ 1.000	$W=\mathfrak{S}$	$W = \mathfrak{S}$	we can see that W contains all $S_k, k = 1, 2, \cdots, 6$, and hence $W = \mathfrak{S}$. Now, let
0902.0431_F00335	17	■ 1.000	$S=\sum_{k=1}^6 x_{\{k\}} S_{\{k\}}, x_{\{k\}} \in \mathbf{R}$	$S = \sum_{k=1}^6 x_k S_k, x_k \in \mathbf{R}$	$S = \sum_{k=1}^6 x_k S_k, x_k \in \mathbf{R}$ be a non-zero element of W and assume that $x_1 \neq 0$ (without
0902.0431_F00336	17	■ 1.000	$x_{\{1\}} \neq 0$	$x_1 \neq 0$	$S = \sum_{k=1}^6 x_k S_k, x_k \in \mathbf{R}$ be a non-zero element of W and assume that $x_1 \neq 0$ (without
0902.0431_F00337	17	■ 1.000	S	S	the loss of generality). Applying H_1 on S , we have $x_1 S_2 - x_2 S_1 - x_3 S_4 + x_4 S_3 \in W$.
0902.0431_F00338	17	■ 1.000	$x_{\{1\}} S_{\{2\}}-x_{\{2\}} S_{\{1\}}-x_{\{3\}} S_{\{4\}}+x_{\{4\}} S_{\{3\}} \in W$	$x_1 S_2 - x_2 S_1 - x_3 S_4 + x_4 S_3 \in W$	the loss of generality). Applying H_1 on S , we have $x_1 S_2 - x_2 S_1 - x_3 S_4 + x_4 S_3 \in W$.
0902.0431_F00339	17	■ 0.304	$\cdots ($	$\cdots ($	$\cdots (ii)$
0902.0431_F00340	17	■ 0.304	$)$	$)$	—
0902.0431_F00341	17	■ 0.877	$\times x_{\{2\}}$	$\times x_2$	Taking (ii) $\times x_2 - (i) \times x_1$, we have $(x_1^2 + x_2^2) S_5 \in W$. Since $x_1^2 + x_2^2 \neq 0$, we have
0902.0431_F00342	17	■ 0.877	$\times x_{\{1\}}$	$\times x_1$	Taking (ii) $\times x_2 - (i) \times x_1$, we have $(x_1^2 + x_2^2) S_5 \in W$. Since $x_1^2 + x_2^2 \neq 0$, we have
0902.0431_F00343	17	■ 0.877	$\left(x_{\{1\}}^2+x_{\{2\}}^2\right) S_{\{5\}} \in W$	$(x_1^2 + x_2^2) S_5 \in W$	Taking (ii) $\times x_2 - (i) \times x_1$, we have $(x_1^2 + x_2^2) S_5 \in W$. Since $x_1^2 + x_2^2 \neq 0$, we have
0902.0431_F00344	17	■ 0.877	$x_{\{1\}}^2+x_{\{2\}}^2 \neq 0$	$x_1^2 + x_2^2 \neq 0$	Taking (ii) $\times x_2 - (i) \times x_1$, we have $(x_1^2 + x_2^2) S_5 \in W$. Since $x_1^2 + x_2^2 \neq 0$, we have
0902.0431_F00345	17	■ 1.000	$S_{\{5\}} \in W$	$S_5 \in W$	$S_5 \in W$ and so $W = \mathfrak{S}$. Consequently the irreducibility of $\mathfrak{S}$ is proved. Finally, since
0902.0431_F00346	17	■ 0.986	$[\mathfrak{su}(3), \mathfrak{S}]$	$[\mathfrak{su}(3), \mathfrak{S}]$	$[\mathfrak{su}(3), \mathfrak{S}]$ is a $\mathfrak{su}(3)$ -invariant $\mathbf{R}$ -submodule of $\mathfrak{S}$, from the irreducibility of $\mathfrak{S}$, we have
0902.0431_F00347	17	■ 0.967	$[\mathfrak{su}(3), \mathfrak{S}]=\mathfrak{S}$	$[\mathfrak{su}(3), \mathfrak{S}] = \mathfrak{S}$	$[\mathfrak{su}(3), \mathfrak{S}] = \mathfrak{S}$.
0902.0431_F00348	17	■ 0.995	$p: \mathfrak{g}_2 \rightarrow \mathfrak{su}(3)$	$p: \mathfrak{g}_2 \rightarrow \mathfrak{su}(3)$	Let $p: \mathfrak{g}_2 \rightarrow \mathfrak{su}(3)$ and $q: \mathfrak{g}_2 \rightarrow \mathfrak{S}$ be projections of $\mathfrak{g}_2 = \mathfrak{su}(3) \oplus \mathfrak{S}$. Now, let $\mathfrak{a}$ be
0902.0431_F00349	17	■ 0.995	$q: \mathfrak{g}_2 \rightarrow \mathfrak{S}$	$q: \mathfrak{g}_2 \rightarrow \mathfrak{S}$	Let $p: \mathfrak{g}_2 \rightarrow \mathfrak{su}(3)$ and $q: \mathfrak{g}_2 \rightarrow \mathfrak{S}$ be projections of $\mathfrak{g}_2 = \mathfrak{su}(3) \oplus \mathfrak{S}$. Now, let $\mathfrak{a}$ be
0902.0431_F00350	17	■ 0.995	$\mathfrak{g}_2=\mathfrak{su}(3) \oplus \mathfrak{S}$	$\mathfrak{g}_2 = \mathfrak{su}(3) \oplus \mathfrak{S}$	Let $p: \mathfrak{g}_2 \rightarrow \mathfrak{su}(3)$ and $q: \mathfrak{g}_2 \rightarrow \mathfrak{S}$ be projections of $\mathfrak{g}_2 = \mathfrak{su}(3) \oplus \mathfrak{S}$. Now, let $\mathfrak{a}$ be
0902.0431_F00351	17	■ 0.995	$\mathfrak{a}$	$\mathfrak{a}$	Let $p: \mathfrak{g}_2 \rightarrow \mathfrak{su}(3)$ and $q: \mathfrak{g}_2 \rightarrow \mathfrak{S}$ be projections of $\mathfrak{g}_2 = \mathfrak{su}(3) \oplus \mathfrak{S}$. Now, let $\mathfrak{a}$ be
0902.0431_F00352	17	■ 1.000	$p(\mathfrak{a})$	$p(\mathfrak{a})$	a non-zero ideal of $\mathfrak{g}_2$. Then $p(\mathfrak{a})$ is an ideal of $\mathfrak{g}_2$. Indeed, if $D \in p(\mathfrak{a})$, then there
0902.0431_F00353	17	■ 1.000	$D \in p(\mathfrak{a})$	$D \in p(\mathfrak{a})$	a non-zero ideal of $\mathfrak{g}_2$. Then $p(\mathfrak{a})$ is an ideal of $\mathfrak{g}_2$. Indeed, if $D \in p(\mathfrak{a})$, then there
0902.0431_F00354	17	■ 0.998	$S \in \mathfrak{S}$	$S \in \mathfrak{S}$	exists $S \in \mathfrak{S}$ such that $D + S \in \mathfrak{a}$. For any $D' \in \mathfrak{su}(3)$, we have
0902.0431_F00355	17	■ 0.998	$D+S \in \mathfrak{a}$	$D + S \in \mathfrak{a}$	exists $S \in \mathfrak{S}$ such that $D + S \in \mathfrak{a}$. For any $D' \in \mathfrak{su}(3)$, we have
0902.0431_F00356	17	■ 0.998	$D' \in \mathfrak{su}(3)$	$D' \in \mathfrak{su}(3)$	exists $S \in \mathfrak{S}$ such that $D + S \in \mathfrak{a}$. For any $D' \in \mathfrak{su}(3)$, we have
0902.0431_F00357	18	■ 1.000	$[D', D] \in p(\mathfrak{a})$	$[D', D] \in p(\mathfrak{a})$	(Lemma 1.6.1), hence $[D', D] \in p(\mathfrak{a})$.
0902.0431_F00358	18	■ 0.686	$\mathfrak{su}(3) \cap \mathfrak{a} \neq \{0\}$	$\mathfrak{su}(3) \cap \mathfrak{a} \neq \{0\}$	We show that either $\mathfrak{su}(3) \cap \mathfrak{a} \neq \{0\}$ or $\mathfrak{S} \cap \mathfrak{a} \neq \{0\}$. Assume that $\mathfrak{su}(3) \cap \mathfrak{a} = \{0\}$
0902.0431_F00359	18	■ 0.686	$\mathfrak{S} \cap \mathfrak{a} \neq \{0\}$	$\mathfrak{S} \cap \mathfrak{a} \neq \{0\}$	We show that either $\mathfrak{su}(3) \cap \mathfrak{a} \neq \{0\}$ or $\mathfrak{S} \cap \mathfrak{a} \neq \{0\}$. Assume that $\mathfrak{su}(3) \cap \mathfrak{a} = \{0\}$
0902.0431_F00360	18	■ 0.686	$\mathfrak{su}(3) \cap \mathfrak{a} = \{0\}$	$\mathfrak{su}(3) \cap \mathfrak{a} = \{0\}$	We show that either $\mathfrak{su}(3) \cap \mathfrak{a} \neq \{0\}$ or $\mathfrak{S} \cap \mathfrak{a} \neq \{0\}$. Assume that $\mathfrak{su}(3) \cap \mathfrak{a} = \{0\}$
0902.0431_F00361	18	■ 0.928	$\mathfrak{S} \cap \mathfrak{a} = \{0\}$	$\mathfrak{S} \cap \mathfrak{a} = \{0\}$	and $\mathfrak{S} \cap \mathfrak{a} = \{0\}$. The mapping $p _{\mathfrak{a}}: \mathfrak{a} \rightarrow \mathfrak{su}(3)$ is injective because $\mathfrak{S} \cap \mathfrak{a} = \{0\}$.
0902.0431_F00362	18	■ 0.928	$p _{\mathfrak{a}}: \mathfrak{a} \rightarrow \mathfrak{su}(3)$	$p _{\mathfrak{a}}: \mathfrak{a} \rightarrow \mathfrak{su}(3)$	and $\mathfrak{S} \cap \mathfrak{a} = \{0\}$. The mapping $p _{\mathfrak{a}}: \mathfrak{a} \rightarrow \mathfrak{su}(3)$ is injective because $\mathfrak{S} \cap \mathfrak{a} = \{0\}$.
0902.0431_F00363	18	■ 0.985	$p(\mathfrak{a}) = \mathfrak{su}(3)$	$p(\mathfrak{a}) = \mathfrak{su}(3)$	Since $p(\mathfrak{a})$ is a non-zero ideal of $\mathfrak{su}(3)$ and $\mathfrak{su}(3)$ is simple, we have $p(\mathfrak{a}) = \mathfrak{su}(3)$.
0902.0431_F00364	18	■ 0.994	$\dim \mathfrak{a} = \dim p(\mathfrak{a}) = \dim \mathfrak{su}(3) = 8$	$\dim \mathfrak{a} = \dim p(\mathfrak{a}) = \dim \mathfrak{su}(3) = 8$	Hence $\dim \mathfrak{a} = \dim p(\mathfrak{a}) = \dim \mathfrak{su}(3) = 8$. On the other hand, since $\mathfrak{su}(3) \cap \mathfrak{a} = \{0\}$,

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0902.0431_F00365	18	■ 1.000	$q \mid \mathfrak{a} : \mathfrak{a} \rightarrow \mathfrak{S}$	$q \mid \mathfrak{a} : \mathfrak{a} \rightarrow \mathfrak{S}$	$q \mid \mathfrak{a} : \mathfrak{a} \rightarrow \mathfrak{S}$ is also injective, we have $\dim \mathfrak{a} \leq \dim \mathfrak{S} = 6$. This leads to a contradiction.
0902.0431_F00366	18	■ 1.000	$\operatorname{dim} \mathfrak{a} \leq \operatorname{dim} \mathfrak{S} = 6$	$\dim \mathfrak{a} \leq \dim \mathfrak{S} = 6$	$q \mid \mathfrak{a} : \mathfrak{a} \rightarrow \mathfrak{S}$ is also injective, we have $\dim \mathfrak{a} < \dim \mathfrak{S} = 6$. This leads to a contradiction.
0902.0431_F00367	18	■ 0.919	$\mathfrak{s} \cap \mathfrak{u}(3) \cap \mathfrak{a} = \mathfrak{s} \cap \mathfrak{u}(3)$	$\mathfrak{su}(3) \cap \mathfrak{a} = \mathfrak{su}(3)$	(1) Case $\mathfrak{su}(3) \cap \mathfrak{a} \neq \{0\}$. From the simplicity of $\mathfrak{su}(3)$, we have $\mathfrak{su}(3) \cap \mathfrak{a} = \mathfrak{su}(3)$,
0902.0431_F00368	18	■ 0.984	$\mathfrak{a} \supset \mathfrak{s} \cap \mathfrak{u}(3)$	$\mathfrak{a} \supset \mathfrak{su}(3)$	hence $\mathfrak{a} \supset \mathfrak{su}(3)$. On the other hand, we have
0902.0431_F00369	18	■ 0.977	$\mathfrak{a} \supset \mathfrak{s} \cap \mathfrak{u}(3) \oplus \mathfrak{g}_2$	$\mathfrak{a} \supset \mathfrak{su}(3) \oplus \mathfrak{S} = \mathfrak{g}_2$	Hence $\mathfrak{a} \supset \mathfrak{su}(3) \oplus \mathfrak{S} = \mathfrak{g}_2$.
0902.0431_F00370	18	■ 1.000	$S \in \mathfrak{S} \cap \mathfrak{a} \subset \mathfrak{a}$	$S \in \mathfrak{S} \cap \mathfrak{a} \subset \mathfrak{a}$	(2) Case $\mathfrak{S} \cap \mathfrak{a} \neq \{0\}$. Choose a non-zero element $S \in \mathfrak{S} \cap \mathfrak{a} \subset \mathfrak{a}$. Under the actions
0902.0431_F00371	18	■ 0.910	$S_1 \in \mathfrak{a}$	$S_1 \in \mathfrak{a}$	of $\mathfrak{su}(3)$, we can see that $S_1 \in \mathfrak{a}$ (Lemma 1.6.1). Hence $0 \neq 4H_1 + 2H_2 = [S_1, S_2] \in \mathfrak{a}$.
0902.0431_F00372	18	■ 0.910	$0 \neq 4H_1 + 2H_2 = [S_1, S_2] \in \mathfrak{a}$	$0 \neq 4H_1 + 2H_2 = [S_1, S_2] \in \mathfrak{a}$	of $\mathfrak{su}(3)$, we can see that $S_1 \in \mathfrak{a}$ (Lemma 1.6.1). Hence $0 \neq 4H_1 + 2H_2 = [S_1, S_2] \in \mathfrak{a}$.
0902.0431_F00373	18	■ 1.000	$\mathfrak{a} = \mathfrak{g}_2$	$\mathfrak{a} = \mathfrak{g}_2$	So this case can be reduced to the case (1). Thus we have $\mathfrak{a} = \mathfrak{g}_2$, which proves the
0902.0431_F00374	18	■ 0.895	L_{ij}, L_{ji}	L_{ij}, L_{ji}	Lemma 1.7.1. In $\mathfrak{su}(3) \subset \mathfrak{g}_2$, the Lie brackets between H_1 and L_{ij}, L_{ji} of Section
0902.0431_F00375	18	■ 1.000	B_2	B_2	Theorem 1.7.2. The killing form B_2 of the Lie algebra $\mathfrak{g}_2^C$ is given by
0902.0431_F00376	18	■ 1.000	$\operatorname{tr}(D_1 D_2)$	$\operatorname{tr}(D_1 D_2)$	Proof. Since $\operatorname{tr}(D_1 D_2)$ is a $\mathfrak{g}_2^C$ -adjoint invariant bilinear form of $\mathfrak{g}_2^C$ and $\mathfrak{g}_2^C$ is
0902.0431_F00377	18	■ 0.999	$k \in C$	$k \in C$	simple (Theorem 1.6.3), there exists $k \in C$ such that
0902.0431_F00378	18	■ 1.000	k	k	To determine this k , let $D_1 = D_2 = H_1$. Then from
0902.0431_F00379	18	■ 1.000	$D_1 = D_2 = H_1$	$D_1 = D_2 = H_1$	To determine this k , let $D_1 = D_2 = H_1$. Then from
0902.0431_F00380	19	■ 1.000	$\operatorname{tr}(H_1 H_1) = (-1) \times 4 = -4$	$\operatorname{tr}(H_1 H_1) = (-1) \times 4 = -4$	Hence $\operatorname{tr}(H_1 H_1) = (-1) \times 4 = -4$. Therefore $k = 4$.
0902.0431_F00381	19	■ 1.000	$k=4$	$k = 4$	Hence $\operatorname{tr}(H_1 H_1) = (-1) \times 4 = -4$. Therefore $k = 4$.
0902.0431_F00382	19	■ 0.984	$f_* : \mathfrak{sl}(3, C) \rightarrow \mathfrak{su}(3)^C$	$f_* : \mathfrak{sl}(3, C) \rightarrow \mathfrak{su}(3)^C$	We recall the C -Lie isomorphism $f_* : \mathfrak{sl}(3, C) \rightarrow \mathfrak{su}(3)^C$ and the embedding $\varphi_* :$
0902.0431_F00383	19	■ 0.984	φ_*	φ_*	We recall the C -Lie isomorphism $f_* : \mathfrak{sl}(3, C) \rightarrow \mathfrak{su}(3)^C$ and the embedding $\varphi_* :$
0902.0431_F00384	19	■ 1.000	$\mathfrak{su}(3)^C \rightarrow \mathfrak{g}_2^C$	$\mathfrak{su}(3)^C \rightarrow \mathfrak{g}_2^C$	$\mathfrak{su}(3)^C \rightarrow \mathfrak{g}_2^C$,
0902.0431_F00385	19	■ 1.000	$\mathfrak{sl}(3, C)$	$\mathfrak{sl}(3, C)$	and we regard $\mathfrak{sl}(3, C)$ as a subalgebra of $\mathfrak{g}_2^C$ under the composition of f_* and φ_* .
0902.0431_F00386	19	■ 1.000	f_*	f_*	and we regard $\mathfrak{sl}(3, C)$ as a subalgebra of $\mathfrak{g}_2^C$ under the composition of f_* and φ_* .
0902.0431_F00387	19	■ 1.000	$\pm(\lambda_k - \lambda_l), 1 \leq k < l \leq 3$	$\pm(\lambda_k - \lambda_l), 1 \leq k < l \leq 3$	Further we know that the Lie algebra $\mathfrak{sl}(3, C)$ has roots $\pm(\lambda_k - \lambda_l), 1 \leq k < l \leq 3$
0902.0431_F00388	19	■ 1.000	$\mathfrak{sl}(3, C)$	$\mathfrak{sl}(3, C)$	of $\mathfrak{sl}(3, C)$, and E_{kl} is a root vector associated with the root $\lambda_k - \lambda_l$.
0902.0431_F00389	19	■ 1.000	E_{kl}	E_{kl}	of $\mathfrak{sl}(3, C)$, and E_{kl} is a root vector associated with the root $\lambda_k - \lambda_l$.
0902.0431_F00390	19	■ 1.000	$\lambda_k - \lambda_l$	$\lambda_k - \lambda_l$	of $\mathfrak{sl}(3, C)$, and E_{kl} is a root vector associated with the root $\lambda_k - \lambda_l$.
0902.0431_F00391	19	■ 1.000	$\lambda_1 + \lambda_2 + \lambda_3 = 0$	$\lambda_1 + \lambda_2 + \lambda_3 = 0$	with $\lambda_1 + \lambda_2 + \lambda_3 = 0$.
0902.0431_F00392	19	■ 0.999	$\mathfrak{h} = \{-i\lambda_1 G_{23} - i\lambda_2 G_{45} - i\lambda_3 G_{67} \in \mathfrak{g}_2^C \mid \lambda_k \in C\} \subset \mathfrak{sl}(3, C) \subset \mathfrak{g}_2^C$	$\mathfrak{h} = \{-i\lambda_1 G_{23} - i\lambda_2 G_{45} - i\lambda_3 G_{67} \in \mathfrak{g}_2^C \mid \lambda_k \in C\} \subset \mathfrak{sl}(3, C) \subset \mathfrak{g}_2^C$	Proof. $\mathfrak{h} = \{-i\lambda_1 G_{23} - i\lambda_2 G_{45} - i\lambda_3 G_{67} \in \mathfrak{g}_2^C \mid \lambda_k \in C\} \subset \mathfrak{sl}(3, C) \subset \mathfrak{g}_2^C$
0902.0431_F00393	19	■ 0.777	$\mathfrak{sl}(3, C)$	$\mathfrak{sl}(3, C)$	$\mathfrak{sl}(3, C)$ is also roots of $\mathfrak{g}_2^C$, so we have the table of roots and associated root vectors
0902.0431_F00394	20	■ 1.000	$\Pi = \{\alpha_1, \alpha_2\}$	$\Pi = \{\alpha_1, \alpha_2\}$	Hence $\Pi = \{\alpha_1, \alpha_2\}$ is a fundamental root system of $\mathfrak{g}_2^C$. The real part of $\mathfrak{h}_R$ of $\mathfrak{h}$ is
0902.0431_F00395	20	■ 1.000	$\mathfrak{h}_R$	$\mathfrak{h}_R$	Hence $\Pi = \{\alpha_1, \alpha_2\}$ is a fundamental root system of $\mathfrak{g}_2^C$. The real part of $\mathfrak{h}_R$ of $\mathfrak{h}$ is
0902.0431_F00396	20	■ 1.000	$\mathfrak{h}$	$\mathfrak{h}$	Hence $\Pi = \{\alpha_1, \alpha_2\}$ is a fundamental root system of $\mathfrak{g}_2^C$. The real part of $\mathfrak{h}_R$ of $\mathfrak{h}$ is
0902.0431_F00397	20	■ 1.000	$H = -i\lambda_1 G_{23} - i\lambda_2 G_{45} - i\lambda_3 G_{67}, H' = -i\lambda_1' G_{23} - i\lambda_2' G_{45} - i\lambda_3' G_{67} \in \mathfrak{h}_R$	$H = -i\lambda_1 G_{23} - i\lambda_2 G_{45} - i\lambda_3 G_{67}, H' = -i\lambda_1' G_{23} - i\lambda_2' G_{45} - i\lambda_3' G_{67} \in \mathfrak{h}_R$	for $H = -i\lambda_1 G_{23} - i\lambda_2 G_{45} - i\lambda_3 G_{67}, H' = -i\lambda_1' G_{23} - i\lambda_2' G_{45} - i\lambda_3' G_{67} \in \mathfrak{h}_R$ (The-
0902.0431_F00398	20	■ 0.996	$H_{\alpha_i} \in \mathfrak{h}_R$	$H_{\alpha_i} \in \mathfrak{h}_R$	orem 1.7.2). Now, the canonical element $H_{\alpha_i} \in \mathfrak{h}_R$ associated with α_i ($B_2(H_{\alpha_i}, H) =$
0902.0431_F00399	20	■ 0.996	$\alpha_i(B_2(H_{\alpha_i}, H)) =$	$\alpha_i(B_2(H_{\alpha_i}, H)) =$	orem 1.7.2). Now, the canonical element $H_{\alpha_i} \in \mathfrak{h}_R$ associated with α_i ($B_2(H_{\alpha_i}, H) =$
0902.0431_F00400	20	■ 0.531	$\alpha(H), H \in \mathfrak{h}_R$	$\alpha(H), H \in \mathfrak{h}_R$	$\alpha(H), H \in \mathfrak{h}_R$ are determined as follows.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_F00401	21	■ 1.000	$C_{\{1\}} \oplus C_{\{1\}}$	$C_1 \oplus C_1$	(1) One is a subalgebra of type $C_1 \oplus C_1$ which is obtained as the fixed points by
0902.0431_F00402	21	■ 1.000	$A_{\{2\}}$	A_2	(2) The other is a subalgebra of type A_2 which is obtained as the fixed points by
0902.0431_F00403	21	■ 1.000	$\left(G_{\{2\}}\right)_{\{e_{\{1\}}\}}$	$(G_2)_{e_1}$	We shall study the following subgroup $(G_2)_e$ of G_2 :
0902.0431_F00404	21	■ 0.946	$\quad\quad\left(G_{\{2\}}\right)_{\{e_{\{1\}}\}} \cong S U(3)$	$(G_2)_{e_1} \cong SU(3)$	Theorem 1.9.1. $(G_2)_{e_1} \cong SU(3)$.
0902.0431_F00405	21	■ 0.998	$\varphi\colon S U(3) \rightarrow\left(G_{\{2\}}\right)_{\{e_{\{1\}}\}}$	$\varphi: SU(3) \rightarrow (G_2)_{e_1}$	Proof (cf. Theorem 1.5.1). We define a mapping $\varphi: SU(3) \rightarrow (G_2)_e$ by
0902.0431_F00406	21	■ 1.000	$\varphi(A) \in\left(G_{\{2\}}\right)_{\{e_{\{1\}}\}}$	$\varphi(A) \in (G_2)_{e_1}$	We first prove that $\varphi(A) \in (G_2)_{e_1}$. For $\alpha = \varphi(A), A \in SU(3)$ and $x = a + \boldsymbol{m}, y =$
0902.0431_F00407	21	■ 1.000	$\alpha=\varphi(A), A \in S U(3)$	$\alpha = \varphi(A), A \in SU(3)$	We first prove that $\varphi(A) \in (G_2)_{e_1}$. For $\alpha = \varphi(A), A \in SU(3)$ and $x = a + \boldsymbol{m}, y =$
0902.0431_F00408	21	■ 1.000	$x=a+\boldsymbol{m}, y=$	$x = a + \boldsymbol{m}, y =$	We first prove that $\varphi(A) \in (G_2)_{e_1}$. For $\alpha = \varphi(A), A \in SU(3)$ and $x = a + \boldsymbol{m}, y =$
0902.0431_F00409	21	■ 0.850	$b+\boldsymbol{n} \in \boldsymbol{C} \oplus \boldsymbol{C}^3 \oplus \boldsymbol{C}^3=\mathfrak{C}$	$b + \boldsymbol{n} \in \boldsymbol{C} \oplus \boldsymbol{C}^3 = \mathfrak{C}$	$b + \boldsymbol{n} \in \boldsymbol{C} \oplus \boldsymbol{C}^3 = \mathfrak{C}$, using that if $A \in SU(3)$ then $\tilde{A}$ (which is the adjoint matrix of
0902.0431_F00410	21	■ 0.850	$A \in S U(3)$	$A \in SU(3)$	$b + \boldsymbol{n} \in \boldsymbol{C} \oplus \boldsymbol{C}^3 = \mathfrak{C}$, using that if $A \in SU(3)$ then $\tilde{A}$ (which is the adjoint matrix of
0902.0431_F00411	21	■ 0.850	$\widetilde{A}$	$\tilde{A}$	$b + \boldsymbol{n} \in \boldsymbol{C} \oplus \boldsymbol{C}^3 = \mathfrak{C}$, using that if $A \in SU(3)$ then $\tilde{A}$ (which is the adjoint matrix of
0902.0431_F00412	21	■ 1.000	$A=A^{-1}=A^*$	$A) = A^{-1} = A^*$	$A) = A^{-1} = A^*$, we have
0902.0431_F00413	21	■ 1.000	$\varphi(A) \in G_2$	$\varphi(A) \in G_2$	Hence $\varphi(A) \in G_2$. Clearly $\varphi(A)e_1 = e_1$, so that $\varphi(A) \in (G_2)_{e_1}$. Evidently φ is a
0902.0431_F00414	21	■ 1.000	$\varphi(A) e_{\{1\}}=e_{\{1\}}$	$\varphi(A)e_1 = e_1$	Hence $\varphi(A) \in G_2$. Clearly $\varphi(A)e_1 = e_1$, so that $\varphi(A) \in (G_2)_{e_1}$. Evidently φ is a
0902.0431_F00415	21	■ 1.000	φ	φ	Hence $\varphi(A) \in G_2$. Clearly $\varphi(A)e_1 = e_1$, so that $\varphi(A) \in (G_2)_{e_1}$. Evidently φ is a
0902.0431_F00416	21	■ 1.000	$\alpha \in\left(G_{\{2\}}\right)_{\{e_{\{1\}}\}}$	$\alpha \in (G_2)_{e_1}$	homomorphism. We show that φ is onto. Let $\alpha \in (G_2)_{e_1}$. Note that α induces a
0902.0431_F00417	21	■ 1.000	α	α	homomorphism. We show that φ is onto. Let $\alpha \in (G_2)_{e_1}$. Note that α induces a
0902.0431_F00418	21	■ 1.000	$\boldsymbol{C}^3$	$\boldsymbol{C}^3$	$\boldsymbol{C}$ -linear transformation of $\boldsymbol{C}^3$. Now let
0902.0431_F00419	21	■ 0.996	$A=\left(\boldsymbol{a}_{\{1\}}, \boldsymbol{a}_{\{2\}}, \boldsymbol{a}_{\{3\}}\right) \in M(3, \boldsymbol{C})$	$A = (\boldsymbol{a}_1, \boldsymbol{a}_2, \boldsymbol{a}_3) \in M(3, \boldsymbol{C})$	and construct a matrix $A = (\boldsymbol{a}_1, \boldsymbol{a}_2, \boldsymbol{a}_3) \in M(3, \boldsymbol{C})$. From $(\alpha e_2)(\alpha e_4) = \alpha(e_2 e_4) =$
0902.0431_F00420	21	■ 0.996	$\left(\alpha e_2\right)\left(\alpha e_4\right)=\alpha\left(e_2 e_4\right)=$	$(\alpha e_2)(\alpha e_4) = \alpha(e_2 e_4) =$	and construct a matrix $A = (\boldsymbol{a}_1, \boldsymbol{a}_2, \boldsymbol{a}_3) \in M(3, \boldsymbol{C})$. From $(\alpha e_2)(\alpha e_4) = \alpha(e_2 e_4) =$
0902.0431_F00421	21	■ 1.000	$-\alpha e_6$	$-\alpha e_6$	$-\alpha e_6$, we have $\boldsymbol{a}_1 \boldsymbol{a}_2 = -\boldsymbol{a}_3$, namely, $-\langle \boldsymbol{a}_1, \boldsymbol{a}_2 \rangle - \overline{\boldsymbol{a}_1} \times \boldsymbol{a}_2 = -\boldsymbol{a}_3$, hence we have
0902.0431_F00422	21	■ 1.000	$\boldsymbol{a}_1 \boldsymbol{a}_2=-\boldsymbol{a}_3$	$\boldsymbol{a}_1 \boldsymbol{a}_2 = -\boldsymbol{a}_3$	$-\alpha e_6$, we have $\boldsymbol{a}_1 \boldsymbol{a}_2 = -\boldsymbol{a}_3$, namely, $-\langle \boldsymbol{a}_1, \boldsymbol{a}_2 \rangle - \overline{\boldsymbol{a}_1} \times \boldsymbol{a}_2 = -\boldsymbol{a}_3$, hence we have
0902.0431_F00423	21	■ 1.000	$-\left\langle \boldsymbol{a}_1, \boldsymbol{a}_2\right\rangle -\overline{\boldsymbol{a}_1} \times \boldsymbol{a}_2=-\boldsymbol{a}_3$	$-\langle \boldsymbol{a}_1, \boldsymbol{a}_2 \rangle - \overline{\boldsymbol{a}_1} \times \boldsymbol{a}_2 = -\boldsymbol{a}_3$	$-\alpha e_6$, we have $\boldsymbol{a}_1 \boldsymbol{a}_2 = -\boldsymbol{a}_3$, namely, $-\langle \boldsymbol{a}_1, \boldsymbol{a}_2 \rangle - \overline{\boldsymbol{a}_1} \times \boldsymbol{a}_2 = -\boldsymbol{a}_3$, hence we have
0902.0431_F00424	22	■ 1.000	$\left\langle \boldsymbol{a}_2, \boldsymbol{a}_3\right\rangle =\left\langle \boldsymbol{a}_3, \boldsymbol{a}_1\right\rangle =0$	$\langle \boldsymbol{a}_2, \boldsymbol{a}_3 \rangle = \langle \boldsymbol{a}_3, \boldsymbol{a}_1 \rangle = 0$	Similarly we have $\langle \boldsymbol{a}_2, \boldsymbol{a}_3 \rangle = \langle \boldsymbol{a}_3, \boldsymbol{a}_1 \rangle = 0$. Moreover from $(\alpha e_k)(\alpha e_k) = \alpha(e_k e_k) =$
0902.0431_F00425	22	■ 1.000	$\left(\alpha e_k\right)\left(\alpha e_k\right)=\alpha\left(e_k e_k\right)=$	$(\alpha e_k)(\alpha e_k) = \alpha(e_k e_k) =$	Similarly we have $\langle \boldsymbol{a}_2, \boldsymbol{a}_3 \rangle = \langle \boldsymbol{a}_3, \boldsymbol{a}_1 \rangle = 0$. Moreover from $(\alpha e_k)(\alpha e_k) = \alpha(e_k e_k) =$
0902.0431_F00426	22	■ 1.000	$\alpha(-1)=-1$	$\alpha(-1) = -1$	$\alpha(-1) = -1$, we have $\langle \boldsymbol{a}_k, \boldsymbol{a}_k \rangle = 1$, hence $A \in U(3)$. Finally $\det A = (\boldsymbol{a}_3, \boldsymbol{a}_1 \times$
0902.0431_F00427	22	■ 1.000	$\left\langle \boldsymbol{a}_k, \boldsymbol{a}_k\right\rangle =1$	$\langle \boldsymbol{a}_k, \boldsymbol{a}_k \rangle = 1$	$\alpha(-1) = -1$, we have $\langle \boldsymbol{a}_k, \boldsymbol{a}_k \rangle = 1$, hence $A \in U(3)$. Finally $\det A = (\boldsymbol{a}_3, \boldsymbol{a}_1 \times$
0902.0431_F00428	22	■ 1.000	$A \in U(3)$	$A \in U(3)$	$\alpha(-1) = -1$, we have $\langle \boldsymbol{a}_k, \boldsymbol{a}_k \rangle = 1$, hence $A \in U(3)$. Finally $\det A = (\boldsymbol{a}_3, \boldsymbol{a}_1 \times$
0902.0431_F00429	22	■ 1.000	$\operatorname{det} A=\left(\boldsymbol{a}_{\{3\}}, \boldsymbol{a}_{\{1\}} \times \right.$	$\det A = (\boldsymbol{a}_3, \boldsymbol{a}_1 \times$	$\alpha(-1) = -1$, we have $\langle \boldsymbol{a}_k, \boldsymbol{a}_k \rangle = 1$, hence $A \in U(3)$. Finally $\det A = (\boldsymbol{a}_3, \boldsymbol{a}_1 \times$
0902.0431_F00430	22	■ 0.776	$\left.\boldsymbol{a}_{\{2\}}\right)=\left(\boldsymbol{a}_{\{3\}}, \overline{\boldsymbol{a}_{\{3\}}}\right)=\left\langle \boldsymbol{a}_{\{3\}}, \boldsymbol{a}_{\{3\}}\right\rangle =1$	$\boldsymbol{a}_2) = (\boldsymbol{a}_3, \overline{\boldsymbol{a}_3}) = \langle \boldsymbol{a}_3, \boldsymbol{a}_3 \rangle = 1$	$\boldsymbol{a}_2) = (\boldsymbol{a}_3, \overline{\boldsymbol{a}_3}) = \langle \boldsymbol{a}_3, \boldsymbol{a}_3 \rangle = 1$ (where $(\boldsymbol{a}, \boldsymbol{b})$ is the ordinary inner product in $\boldsymbol{C}^3$:
0902.0431_F00431	22	■ 0.776	$(\boldsymbol{a}, \boldsymbol{b})$	$(\boldsymbol{a}, \boldsymbol{b})$	$\boldsymbol{a}_2) = (\boldsymbol{a}_3, \overline{\boldsymbol{a}_3}) = \langle \boldsymbol{a}_3, \boldsymbol{a}_3 \rangle = 1$ (where $(\boldsymbol{a}, \boldsymbol{b})$ is the ordinary inner product in $\boldsymbol{C}^3$:
0902.0431_F00432	22	■ 0.888	$\left.\boldsymbol{a}, \boldsymbol{b}\right\rangle =\left\langle \boldsymbol{a}, \boldsymbol{b}\right\rangle$	$(\boldsymbol{a}, \boldsymbol{b}) = {}^t \boldsymbol{a} \boldsymbol{b}$	$\langle \boldsymbol{a}, \boldsymbol{b} \rangle = {}^t \boldsymbol{a} \boldsymbol{b}$. Hence we have $A \in SU(3)$ and $\varphi(A) = \alpha$, which shows that φ is onto.
0902.0431_F00433	22	■ 0.888	$\varphi(A)=\alpha$	$\varphi(A) = \alpha$	$(\boldsymbol{a}, \boldsymbol{b}) = {}^t \boldsymbol{a} \boldsymbol{b}$. Hence we have $A \in SU(3)$ and $\varphi(A) = \alpha$, which shows that φ is onto.
0902.0431_F00434	22	■ 0.984	$\operatorname{Ker} \varphi=\{E\}$	$\operatorname{Ker} \varphi = \{E\}$	$\operatorname{Ker} \varphi = \{E\}$ is easily obtained. Thus we have the isomorphism $SU(3) \cong (G_2)_{e_1}$.
0902.0431_F00435	22	■ 0.984	$S U(3) \cong\left(G_{\{2\}}\right)_{\{e_{\{1\}}\}}$	$SU(3) \cong (G_2)_{e_1}$	$\operatorname{Ker} \varphi = \{E\}$ is easily obtained. Thus we have the isomorphism $SU(3) \cong (G_2)_{e_1}$.
0902.0431_F00436	22	■ 0.961	$G_{\{2\}} / S U(3) \simeq S^6$	$G_2 / SU(3) \simeq S^6$	Theorem 1.9.2. $G_2 / SU(3) \simeq S^6$.
0902.0431_F00437	22	■ 1.000	$S^6=\left\{a \in \mathfrak{C} \mid \bar{a}=-a, a =1\right\}$	$S^6 = \{a \in \mathfrak{C} \bar{a} = -a, a = 1\}$	Proof. $S^6 = \{a \in \mathfrak{C} \bar{a} = -a, a = 1\}$ is a 6 dimensional sphere. Since the group
0902.0431_F00438	22	■ 1.000	S^6	S^6	on S^6 . We shall show that this action is transitive. To prove this, it is sufficient to

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0902.0431_F00439	22	■ 1.000	$a \in S^6$	$a \in S^6$	show that any element $a \in S^6$ can be transformed to $e_1 \in S^6$ by some $\alpha \in G_2$. Now,
0902.0431_F00440	22	■ 1.000	$e_{\{1\}} \in S^6$	$e_1 \in S^6$	show that any element $a \in S^6$ can be transformed to $e_1 \in S^6$ by some $\alpha \in G_2$. Now,
0902.0431_F00441	22	■ 1.000	$a_{\{1\}} \in S^6$	$a_1 \in S^6$	for $a_1 \in S^6$, choose any element $a_2 \in S^6$ such that $(a_1, a_2) = 0$. Let
0902.0431_F00442	22	■ 1.000	$a_{\{2\}} \in S^6$	$a_2 \in S^6$	for $a_1 \in S^6$, choose any element $a_2 \in S^6$ such that $(a_1, a_2) = 0$. Let
0902.0431_F00443	22	■ 1.000	$\left(a_{\{1\}}, a_{\{2\}}\right)=0$	$(a_1, a_2) = 0$	for $a_1 \in S^6$, choose any element $a_2 \in S^6$ such that $\boxed{a_1, a_2} = \mathbb{0}$. Let
0902.0431_F00444	22	■ 1.000	$a_{\{3\}} \in S^6$	$a_3 \in S^6$	Then $a_3 \in S^6$ and a_3 satisfies $(a_1, a_3) = (a_2, a_3) = 0$. Choose any element $a_4 \in S^6$
0902.0431_F00445	22	■ 1.000	$a_{\{3\}}$	a_3	Then $a_3 \in S^6$ and a_3 satisfies $(a_1, a_3) = (a_2, a_3) = 0$. Choose any element $a_4 \in S^6$
0902.0431_F00446	22	■ 1.000	$\left(a_{\{1\}}, a_{\{3\}}\right)=\left(a_{\{2\}}, a_{\{3\}}\right)=0$	$(a_1, a_3) = (a_2, a_3) = 0$	Then $a_3 \in S^6$ and a_3 satisfies $\boxed{a_1, a_3} = \boxed{a_2, a_3} = \mathbb{0}$. Choose any element $a_4 \in S^6$
0902.0431_F00447	22	■ 1.000	$a_{\{4\}} \in S^6$	$a_4 \in S^6$	Then $a_3 \in S^6$ and a_3 satisfies $(a_1, a_3) = (a_2, a_3) = 0$. Choose any element $a_4 \in S^6$
0902.0431_F00448	22	■ 1.000	$\left(a_{\{1\}}, a_{\{4\}}\right)=\left(a_{\{2\}}, a_{\{4\}}\right)=\left(a_{\{3\}}, a_{\{4\}}\right)=0$	$(a_1, a_4) = (a_2, a_4) = (a_3, a_4) = 0$	such that $\boxed{a_1, a_4} = \boxed{a_2, a_4} = \boxed{a_3, a_4} = \mathbb{0}$. Let
0902.0431_F00449	22	■ 0.996	$\left\{a_{\{0\}}=1, a_{\{1\}}, a_{\{2\}}, \cdots, a_{\{7\}}\right\}$	$\{a_0 = 1, a_1, a_2, \cdots, a_7\}$	Then the set $\boxed{a_0 = 1, a_1, a_2, \cdots, a_7}$ is an orthonormal $\mathbf{R}$ -basis of $\mathfrak{C}$. Indeed, $ a_i =$
0902.0431_F00450	22	■ 0.996	$\left a_{\{i\}}\right =$	$ a_i =$	Then the set $\{a_0 = 1, a_1, a_2, \cdots, a_7\}$ is an orthonormal $\mathbf{R}$ -basis of $\mathfrak{C}$. Indeed, $ a_i =$
0902.0431_F00451	22	■ 1.000	$1, 0 \leq i \leq 7$	$1, 0 \leq i \leq 7$	$1, 0 \leq i \leq 7$ are trivial, and we need to verify that $(a_i, a_j) = 0, i \neq j$. However this
0902.0431_F00452	22	■ 1.000	$\left(a_{\{i\}}, a_{\{j\}}\right)=0, i \neq j$	$(a_i, a_j) = 0, i \neq j$	$1, 0 \leq i \leq 7$ are trivial, and we need to verify that $\boxed{a_i, a_j} = 0, i \neq j$. However this
0902.0431_F00453	22	■ 0.998	$\left\{e_{\{0\}}=1, e_{\{1\}}, e_{\{2\}}, \cdots, e_{\{7\}}\right\}$	$\{e_0 = 1, e_1, e_2, \cdots, e_7\}$	Now, since $\{e_0 = 1, e_1, e_2, \cdots, e_7\}$ and $\{a_0 = 1, a_1, a_2, \cdots, a_7\}$ are both orthonormal
0902.0431_F00454	22	■ 1.000	$\alpha: \mathfrak{C} \rightarrow \mathfrak{C}$	$\alpha: \mathfrak{C} \rightarrow \mathfrak{C}$	$\mathbf{R}$ -bases, the $\mathbf{R}$ -linear isomorphism $\alpha: \mathfrak{C} \rightarrow \mathfrak{C}$ satisfying
0902.0431_F00455	22	■ 0.844	$O(7)$	$O(7)$	belongs to $O(7)$: $\alpha \in O(7)$. Moreover we claim that $\alpha \in G_2$, that is, α satisfies
0902.0431_F00456	22	■ 0.844	$\alpha \in O(7)$	$\alpha \in O(7)$	belongs to $O(7)$: $\alpha \in O(7)$. Moreover we claim that $\alpha \in G_2$, that is, α satisfies
0902.0431_F00457	23	■ 1.000	$\alpha e_{\{1\}}=a_{\{1\}}$	$\alpha e_1 = a_1$	Hence $\alpha \in G_2$ and $\alpha e_1 = a_1$, and so $\alpha^{-1}a_1 = e_1$. This shows the transitivity. The
0902.0431_F00458	23	■ 1.000	$\alpha^{-1} a_{\{1\}}=e_{\{1\}}$	$\alpha^{-1}a_1 = e_1$	Hence $\alpha \in G_2$ and $\alpha e_1 = a_1$, and so $\alpha^{-1}a_1 = e_1$. This shows the transitivity. The
0902.0431_F00459	23	■ 0.942	$e_{\{1\}}$	e_1	isotropy subgroup $(G_2)_{e_1}$ of G_2 at e_1 is $SU(3)$ (Theorem 1.9.1). Thus we have the
0902.0431_F00460	23	■ 1.000	$G_{\{2\}}=\left\{\alpha \in \operatorname{Iso}_{\mathbf{R}}\left(\mathfrak{C}\right) \mid \alpha(x y)=(\alpha x)(\alpha y)\right\}$	$G_2 = \{\alpha \in \operatorname{Iso}_{\mathbf{R}}(\mathfrak{C}) \mid \alpha(xy) = (\alpha x)(\alpha y)\}$	Theorem 1.9.3. $\boxed{G_2 = \{\alpha \in \operatorname{Iso}_{\mathbf{R}}(\mathfrak{C}) \mid \alpha(xy) = (\alpha x)(\alpha y)\}}$ is a simply connected
0902.0431_F00461	23	■ 1.000	$S O(7)=\left\{\alpha \in O(7) \mid \operatorname{det} \alpha=\right.$	$SO(7) = \{\alpha \in O(7) \mid \det \alpha =$	Remark 1. Since G_2 is connected, G_2 is contained in $\boxed{SO(7) = \{\alpha \in O(7) \mid \det \alpha =$
0902.0431_F00462	23	■ 0.975	$\left.1\right\}: G_{\{2\}} \subset S O(7)$	$1\}: G_2 \subset SO(7)$	$\boxed{1\}: G_2 \subset SO(7)}$.
0902.0431_F00463	23	■ 1.000	$\operatorname{dim} G_{\{2\}}=$	$\dim G_2 =$	Remark 2. Since we know that the dimension of the group G_2 as $\dim G_2 =$
0902.0431_F00464	23	■ 0.999	$\operatorname{dim} \mathfrak{g}_{\{2\}}=14$	$\dim \mathfrak{g}_2 = 14$	$\dim \mathfrak{g}_2 = 14$ (Theorem 1.4.3), $G_2/SU(3) \simeq S^6$ is proved as follows. The group G_2
0902.0431_F00465	23	■ 0.761	$S U(3)$	$SU(3)$	acts on S^6 . The isotropy subgroup $(G_2)_{e_1}$ of G_2 at e_1 is $SU(3)$ (Theorem 1.9.1) and
0902.0431_F00466	23	■ 1.000	$\left(\operatorname{dim} \left(G_{\{2\}} \backslash \left(G_{\{2\}}\right)_{e_{\{1\}}}\right)=\operatorname{dim} G_{\{2\}}-\operatorname{dim} S U(3)=14-8=6=\operatorname{dim} S^6\right)$	$\dim \left(G_2 / \left(G_2\right)_{e_1}\right) = \dim G_2 - \dim SU(3) = 14 - 8 = 6 = \dim S^6$	$\boxed{\dim \left(G_2 / \left(G_2\right)_{e_1}\right) = \dim G_2 - \dim SU(3) = 14 - 8 = 6 = \dim S^6}$. Therefore we have
0902.0431_F00467	23	■ 0.997	$\varphi: S U(3) \rightarrow G_{\{2\}}$	$\varphi: SU(3) \rightarrow G_2$	Using the mapping $\boxed{\varphi: SU(3) \rightarrow G_2}$, we define a mapping $w: \mathfrak{C} \rightarrow \mathfrak{C}$ by
0902.0431_F00468	23	■ 0.997	$w: \mathfrak{C} \rightarrow \mathfrak{C}$	$w: \mathfrak{C} \rightarrow \mathfrak{C}$	Using the mapping $\varphi: SU(3) \rightarrow G_2$, we define a mapping $\boxed{w: \mathfrak{C} \rightarrow \mathfrak{C}}$ by
0902.0431_F00469	23	■ 0.996	$\omega_{\{1\}}=-\frac{1}{2}+\frac{\sqrt{3}}{2} e_{\{1\}} \in \mathbf{C} \subset \mathfrak{C}$	$\omega_1 = -\frac{1}{2} + \frac{\sqrt{3}}{2}e_1 \in \mathbf{C} \subset \mathfrak{C}$	where $\boxed{\omega_1 = -\frac{1}{2} + \frac{\sqrt{3}}{2}e_1 \in \mathbf{C} \subset \mathfrak{C}}$. This w is defined as
0902.0431_F00470	23	■ 1.000	$w \in G_{\{2\}}$	$w \in G_2$	Then $w \in G_2$ and $w^3 = 1$.
0902.0431_F00471	23	■ 1.000	$w^3=1$	$w^3 = 1$	Then $w \in G_2$ and $w^3 = 1$.
0902.0431_F00472	23	■ 1.000	$\left(G_{\{2\}}\right)^w$	$(G_2)^w$	We shall study the following subgroup $\boxed{\left(G_2\right)^w}$ of G_2 :
0902.0431_F00473	23	■ 0.996	$\left(G_{\{2\}}\right)^w=\left(G_{\{2\}}\right)_{e_1} \cong S U(3)$	$(G_2)^w = (G_2)_{e_1} \cong SU(3)$	Theorem 1.9.4. $\boxed{\left(G_2\right)^w = \left(G_2\right)_{e_1} \cong SU(3)}$.
0902.0431_F00474	23	■ 0.997	$\varphi(S U(3)) \in\left(G_{\{2\}}\right)^w$	$\varphi(SU(3)) \in (G_2)^w$	$\boxed{\varphi(SU(3)) \in (G_2)^w}$. Indeed, for $A \in SU(3)$ and $a + \mathbf{m} \in \mathbf{C} \oplus \mathbf{C}^3 = \mathfrak{C}$, we have
0902.0431_F00475	23	■ 0.997	$a+\mathbf{m} \in \mathbf{C} \oplus \mathbf{C}^3=\mathfrak{C}$	$a + \mathbf{m} \in \mathbf{C} \oplus \mathbf{C}^3 = \mathfrak{C}$	$\varphi(SU(3)) \in (G_2)^w$. Indeed, for $A \in SU(3)$ and $\boxed{a + \mathbf{m} \in \mathbf{C} \oplus \mathbf{C}^3 = \mathfrak{C}}$, we have
0902.0431_F00476	23	■ 1.000	$w \varphi(A)=\varphi(A) w$	$w\varphi(A) = \varphi(A)w$	Hence $w\varphi(A) = \varphi(A)w$, so that $\varphi(A) \in (G_2)^w$. Conversely, let $\alpha \in (G_2)^w$. We
0902.0431_F00477	23	■ 1.000	$\varphi(A) \in\left(G_{\{2\}}\right)^w$	$\varphi(A) \in (G_2)^w$	Hence $w\varphi(A) = \varphi(A)w$, so that $\varphi(A) \in (G_2)^w$. Conversely, let $\alpha \in (G_2)^w$. We
0902.0431_F00478	23	■ 1.000	$\alpha \in\left(G_{\{2\}}\right)^w$	$\alpha \in (G_2)^w$	Hence $w\varphi(A) = \varphi(A)w$, so that $\varphi(A) \in (G_2)^w$. Conversely, let $\alpha \in (G_2)^w$. We
0902.0431_F00479	23	■ 1.000	$\mathfrak{C}_w=\left\{x \in \mathfrak{C} \mid w x=x\right\}$	$\mathfrak{C}_w = \{x \in \mathfrak{C} \mid wx = x\}$	consider the $\mathbf{R}$ -vector subspace $\boxed{\mathfrak{C}_w = \{x \in \mathfrak{C} \mid wx = x\}}$ of $\mathfrak{C}$, then $\mathfrak{C}_w = \mathbf{C}$. Since α

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0902.0431_F00480	23	■ 1.000	$\mathfrak{C}_{\{w\}}=\boldsymbol{C}$	$\mathfrak{C}_w = C$	consider the $\boldsymbol{R}$ -vector subspace $\mathfrak{C}_w = \{x \in \mathfrak{C} \mid wx = x\}$ of $\mathfrak{C}$, then $\mathfrak{C}_w = C$. Since α
0902.0431_F00481	23	■ 1.000	$w \alpha=\alpha w, \mathfrak{C}_{\{w\}}$	$w\alpha = \alpha w, \mathfrak{C}_w$	satisfies $w\alpha = \alpha w$, $\mathfrak{C}_w$ is invariant under α . Since the restriction of α to $\mathfrak{C}_w$ induces
0902.0431_F00482	23	■ 1.000	$\mathfrak{C}_{\{w\}}$	$\mathfrak{C}_w$	satisfies $w\alpha = \alpha w$, $\mathfrak{C}_w$ is invariant under α . Since the restriction of α to $\mathfrak{C}_w$ induces
0902.0431_F00483	24	■ 0.999	$\gamma_1\colon \mathfrak{C} \rightarrow \mathfrak{C}$	$\gamma_1 : \mathfrak{C} \rightarrow \mathfrak{C}$	In the latter case, consider the mapping $\gamma_1 : \mathfrak{C} \rightarrow \mathfrak{C}$ defined by $\gamma_1(a + m) = \bar{a} + \overline{m}$.
0902.0431_F00484	24	■ 0.999	$\gamma_1(a+\boldsymbol{m})=\bar{a}+\overline{\boldsymbol{m}}$	$\gamma_1(a + m) = \bar{a} + \overline{m}$	In the latter case, consider the mapping $\gamma_1 : \mathfrak{C} \rightarrow \mathfrak{C}$ defined by $\gamma_1(a + m) = \bar{a} + \overline{m}$.
0902.0431_F00485	24	■ 1.000	$\gamma_1 \in G_2$	$\gamma_1 \in G_2$	Then we have $\gamma_1 \in G_2$ and $\gamma_1 e_1 = -e_1$. Let $\beta = \gamma_1 \alpha$. Since $\beta e_1 = e_1$, we have
0902.0431_F00486	24	■ 1.000	$\gamma_1 e_1=-e_1$	$\gamma_1 e_1 = -e_1$	Then we have $\gamma_1 \in G_2$ and $\gamma_1 e_1 = -e_1$. Let $\beta = \gamma_1 \alpha$. Since $\beta e_1 = e_1$, we have
0902.0431_F00487	24	■ 1.000	$\beta=\gamma_1 \alpha$	$\beta = \gamma_1 \alpha$	Then we have $\gamma_1 \in G_2$ and $\gamma_1 e_1 = -e_1$. Let $\beta = \gamma_1 \alpha$. Since $\beta e_1 = e_1$, we have
0902.0431_F00488	24	■ 1.000	$\beta e_1=e_1$	$\beta e_1 = e_1$	Then we have $\gamma_1 \in G_2$ and $\gamma_1 e_1 = -e_1$. Let $\beta = \gamma_1 \alpha$. Since $\beta e_1 = e_1$, we have
0902.0431_F00489	24	■ 0.940	$\beta \in SU(3) \subset (G_2)_{e_1}$	$\beta \in SU(3) \subset (G_2)_{e_1}$	$\beta \in SU(3) \subset (G_2)_{e_1}$ (Theorem 1.9.1) $\subset (G_2)^w$. Therefore, $\gamma_1 = \beta \alpha^{-1} \in (G_2)^w$,
0902.0431_F00490	24	■ 0.940	$\subset (G_2)^w$	$\subset (G_2)^w$	$\beta \in SU(3) \subset (G_2)_{e_1}$ (Theorem 1.9.1) $\subset (G_2)^w$. Therefore, $\gamma_1 = \beta \alpha^{-1} \in (G_2)^w$,
0902.0431_F00491	24	■ 0.940	$\gamma_1=\beta \alpha^{-1} \in (G_2)^w$	$\gamma_1 = \beta \alpha^{-1} \in (G_2)^w$	$\beta \in SU(3) \subset (G_2)_{e_1}$ (Theorem 1.9.1) $\subset (G_2)^w$. Therefore, $\gamma_1 = \beta \alpha^{-1} \in (G_2)^w$,
0902.0431_F00492	24	■ 0.966	$\alpha e_1=e_1$	$\alpha e_1 = e_1$	which is false. Therefore $\alpha e_1 = e_1$, and so $\alpha \in SU(3)$ (Theorem 1.9.1). Thus we have
0902.0431_F00493	24	■ 0.966	$\alpha \in SU(3)$	$\alpha \in SU(3)$	which is false. Therefore $\alpha e_1 = e_1$, and so $\alpha \in SU(3)$ (Theorem 1.9.1). Thus we have
0902.0431_F00494	24	■ 0.905	$\left(G_2\right)^w=\left(G_2\right)_{e_1}$	$(G_2)^w = (G_2)_{e_1}$	$(G_2)^w = (G_2)_{e_1}$.
0902.0431_F00495	24	■ 1.000	$\boldsymbol{H}$	$\boldsymbol{H}$	The Cayley algebra $\mathfrak{C}$ naturally contains the field $\boldsymbol{H}$ of quaternions as
0902.0431_F00496	24	■ 0.996	$\boldsymbol{H} \oplus \boldsymbol{H} e_4$	$\boldsymbol{H} \oplus \boldsymbol{H} e_4$	In $\boldsymbol{H} \oplus \boldsymbol{H} e_4$, we define a multiplication, an inner product (,), a conjugation —
0902.0431_F00497	24	■ 1.000	$\left(G_2\right)^{\gamma}$	$(G_2)^\gamma$	We shall study the following subgroup $(G_2)^\gamma$ of G_2 :
0902.0431_F00498	24	■ 0.551	$\left(G_2\right)^{\gamma} \cong\left(S p(1) \times S p(1)\right) / \boldsymbol{Z}_2, \boldsymbol{Z}_2=\{(1,1),(-1,-1)\}$	$(G_2)^\gamma \cong (Sp(1) \times Sp(1))/\boldsymbol{Z}_2, \boldsymbol{Z}_2 = \{(1,1), (-1, -1)\}$	Theorem 1.10.1. $(G_2)^\gamma \cong (Sp(1) \times Sp(1))/\boldsymbol{Z}_2, \boldsymbol{Z}_2 = \{(1,1), (-1, -1)\}$.
0902.0431_F00499	24	■ 0.999	$\varphi: \operatorname{Sp}(1) \times \operatorname{Sp}(1) \rightarrow\left(G_2\right)^{\gamma}$	$\varphi : \operatorname{Sp}(1) \times \operatorname{Sp}(1) \rightarrow (G_2)^\gamma$	Proof. We define a mapping $\varphi : Sp(1) \times Sp(1) \rightarrow (G_2)^\gamma$ by
0902.0431_F00500	25	■ 0.973	$\varphi(p, q) \in\left(G_2\right)^{\gamma}$	$\varphi(p, q) \in (G_2)^\gamma$	We first show that $\varphi(p, q) \in (G_2)^\gamma$. For $\alpha = \varphi(p, q)$, $p, q \in Sp(1)$ and $x = m + ae_4$,
0902.0431_F00501	25	■ 0.973	$\alpha=\varphi(p, q), p, q \in S p(1)$	$\alpha = \varphi(p, q), p, q \in Sp(1)$	We first show that $\varphi(p, q) \in (G_2)^\gamma$. For $\alpha = \varphi(p, q)$, $p, q \in Sp(1)$ and $x = m + ae_4$,
0902.0431_F00502	25	■ 0.973	$x=m+a e_4$	$x = m + ae_4$	We first show that $\varphi(p, q) \in (G_2)^\gamma$. For $\alpha = \varphi(p, q)$, $p, q \in Sp(1)$ and $x = m + ae_4$,
0902.0431_F00503	25	■ 0.999	$y=n+b e_4 \in \boldsymbol{H} \oplus \boldsymbol{H} e_4=\mathfrak{C}$	$y = n + be_4 \in \boldsymbol{H} \oplus \boldsymbol{H} e_4 = \mathfrak{C}$	$y = n + be_4 \in \boldsymbol{H} \oplus \boldsymbol{H} e_4 = \mathfrak{C}$, we have
0902.0431_F00504	25	■ 1.000	$\varphi(p, q) \in G_2$	$\varphi(p, q) \in G_2$	Hence $\varphi(p, q) \in G_2$. Clearly $\gamma \varphi(p, q) = \varphi(p, q) \gamma$, so that $\varphi(p, q) \in (G_2)^\gamma$. Evidently φ
0902.0431_F00505	25	■ 1.000	$\gamma \varphi(p, q)=\varphi(p, q) \gamma$	$\gamma \varphi(p, q) = \varphi(p, q) \gamma$	Hence $\varphi(p, q) \in G_2$. Clearly $\gamma \varphi(p, q) = \varphi(p, q) \gamma$, so that $\varphi(p, q) \in (G_2)^\gamma$. Evidently φ
0902.0431_F00506	25	■ 1.000	$\alpha \in\left(G_2\right)^{\gamma}$	$\alpha \in (G_2)^\gamma$	is a homomorphism. We shall show that φ is onto. Let $\alpha \in (G_2)^\gamma$. Since α satisfies
0902.0431_F00507	25	■ 1.000	$\gamma \alpha=\alpha \gamma, \mathfrak{C}_{\gamma}=\{x \in \mathfrak{C} \mid \gamma x=x\}=\boldsymbol{H}$	$\gamma \alpha = \alpha \gamma, \mathfrak{C}_\gamma = \{x \in \mathfrak{C} \mid \gamma x = x\} = \boldsymbol{H}$	$\gamma \alpha = \alpha \gamma, \mathfrak{C}_\gamma = \{x \in \mathfrak{C} \mid \gamma x = x\} = \boldsymbol{H}$ is invariant under α , so that the restriction of
0902.0431_F00508	25	■ 1.000	$\mathfrak{C}_{\gamma}$	$\mathfrak{C}_\gamma$	α to $\mathfrak{C}_\gamma$ induces an automorphism of $\boldsymbol{H}$. Hence there exists $q \in Sp(1)$ satisfying
0902.0431_F00509	25	■ 1.000	$q \in \operatorname{Sp}(1)$	$q \in \operatorname{Sp}(1)$	α to $\mathfrak{C}_\gamma$ induces an automorphism of $\boldsymbol{H}$. Hence there exists $q \in Sp(1)$ satisfying
0902.0431_F00510	25	■ 1.000	$\beta=\varphi(1, q)^{-1} \alpha$	$\beta = \varphi(1, q)^{-1} \alpha$	(Proposition 108 of Yokota [58]). By putting $\beta = \varphi(1, q)^{-1} \alpha$, we have $\beta \in (G_2)^\gamma$ and
0902.0431_F00511	25	■ 1.000	$\beta \in\left(G_2\right)^{\gamma}$	$\beta \in (G_2)^\gamma$	(Proposition 108 of Yokota [58]). By putting $\beta = \varphi(1, q)^{-1} \alpha$, we have $\beta \in (G_2)^\gamma$ and
0902.0431_F00512	25	■ 0.994	$\beta \mid \boldsymbol{H}=1$	$\beta \mid \boldsymbol{H} = 1$	$\beta \mid \boldsymbol{H} = 1$. Therefore β induces a transformation of $\mathfrak{C}_{-\gamma} = \{x \in \mathfrak{C} \mid \gamma x = -x\} = \boldsymbol{H} e_4$.
0902.0431_F00513	25	■ 0.994	β	β	$\beta \mid \boldsymbol{H} = 1$. Therefore β induces a transformation of $\mathfrak{C}_{-\gamma} = \{x \in \mathfrak{C} \mid \gamma x = -x\} = \boldsymbol{H} e_4$.
0902.0431_F00514	25	■ 0.994	$\mathfrak{C}_{-\gamma}=\{x \in \mathfrak{C} \mid \gamma x=-x\}=\boldsymbol{H} e_4$	$\mathfrak{C}_{-\gamma} = \{x \in \mathfrak{C} \mid \gamma x = -x\} = \boldsymbol{H} e_4$	$\beta \mid \boldsymbol{H} = 1$. Therefore β induces a transformation of $\mathfrak{C}_{-\gamma} = \{x \in \mathfrak{C} \mid \gamma x = -x\} = \boldsymbol{H} e_4$.
0902.0431_F00515	25	■ 1.000	$\beta e_4=p e_4$	$\beta e_4 = pe_4$	By putting $\beta e_4 = pe_4$, where $p \in \boldsymbol{H}$, we have $ p = pe_4 = \beta e_4 = e_4 = 1$, which
0902.0431_F00516	25	■ 1.000	$p \in \boldsymbol{H}$	$p \in \boldsymbol{H}$	By putting $\beta e_4 = pe_4$, where $p \in \boldsymbol{H}$, we have $ p = pe_4 = \beta e_4 = e_4 = 1$, which
0902.0431_F00517	25	■ 1.000	$ p = \left p e_4\right =\left \beta e_4\right =\left e_4\right =1$	$ p = pe_4 = \beta e_4 = e_4 = 1$	By putting $\beta e_4 = pe_4$, where $p \in \boldsymbol{H}$, we have $ p = pe_4 = \beta e_4 = e_4 = 1$, which
0902.0431_F00518	25	■ 0.983	$p \in \operatorname{Sp}(1)$	$p \in \operatorname{Sp}(1)$	implies $p \in Sp(1)$. From
0902.0431_F00519	25	■ 1.000	$\beta=\varphi(p, 1)$	$\beta = \varphi(p, 1)$	we have $\beta = \varphi(p, 1)$. Therefore we have
0902.0431_F00520	25	■ 0.999	$\operatorname{Ker} \varphi=\{(1,1),(-1,-1)\}=\boldsymbol{Z}_2$	$\operatorname{Ker} \varphi = \{(1,1), (-1, -1)\} = \boldsymbol{Z}_2$	which shows that φ is onto. $\operatorname{Ker} \varphi = \{(1,1), (-1, -1)\} = \boldsymbol{Z}_2$ is easily obtained. Thus
0902.0431_F00521	25	■ 0.802	$\left(S p(1) \times S p(1)\right) / \boldsymbol{Z}_2 \cong\left(G_2\right)^{\gamma}$	$(Sp(1) \times Sp(1))/\boldsymbol{Z}_2 \cong (G_2)^\gamma$	we have the isomorphism $(Sp(1) \times Sp(1))/\boldsymbol{Z}_2 \cong (G_2)^\gamma$.
0902.0431_F00522	25	■ 0.829	$\left(S p(1) \times S p(1)\right) / \boldsymbol{Z}_2 \cong S O(4)$	$(Sp(1) \times Sp(1))/\boldsymbol{Z}_2 \cong SO(4)$	Remark 1. $(Sp(1) \times Sp(1))/\boldsymbol{Z}_2 \cong SO(4)$.

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0902.0431_F00523	25	0.868	f: \operatorname{Sp}(1) \times \operatorname{Sp}(1) \rightarrow \operatorname{SO}(4)=\operatorname{SO}(\boldsymbol{H})	$f : \operatorname{Sp}(1) \times \operatorname{Sp}(1) \rightarrow \operatorname{SO}(4) = \operatorname{SO}(\boldsymbol{H})$	Indeed, a mapping $f : \operatorname{Sp}(1) \times \operatorname{Sp}(1) \rightarrow \operatorname{SO}(4) = \operatorname{SO}(\boldsymbol{H})$ defined by
0902.0431_F00524	25	1.000	\left(\mathfrak{g}_2\right)^{\gamma}	$(\mathfrak{g}_2)^\gamma$	forms an $\boldsymbol{R}$ -basis of $(\mathfrak{g}_2)^\gamma$. So $\dim(\mathfrak{g}_2)^\gamma = 6 = 3 + 3 = \dim(\mathfrak{sp}(1) \oplus \mathfrak{sp}(1))$. Hence φ
0902.0431_F00525	25	1.000	\operatorname{dim}\left(\mathfrak{g}_2\right)^{\gamma}=6=3+3=\operatorname{dim}\left(\mathfrak{s p}(1) \oplus \mathfrak{s p}(1)\right)	$\dim(\mathfrak{g}_2)^\gamma = 6 = 3 + 3 = \dim(\mathfrak{sp}(1) \oplus \mathfrak{sp}(1))$	forms an $\boldsymbol{R}$ -basis of $(\mathfrak{g}_2)^\gamma$. So $\dim(\mathfrak{g}_2)^\gamma = 6 = 3 + 3 = \dim(\mathfrak{sp}(1) \oplus \mathfrak{sp}(1))$. Hence φ
0902.0431_F00526	26	1.000	\alpha \in z\left(G_2\right)	$\alpha \in z(G_2)$	Proof. Let $\alpha \in z(G_2)$. From the commutativity with γ : $\gamma\alpha = \alpha\gamma$, we have
0902.0431_F00527	26	1.000	\gamma: \gamma \alpha=\alpha \gamma	$\gamma : \gamma\alpha = \alpha\gamma$	Proof. Let $\alpha \in z(G_2)$. From the commutativity with γ : $\gamma\alpha = \alpha\gamma$, we have
0902.0431_F00528	26	0.913	\alpha \in S O(4)	$\alpha \in \operatorname{SO}(4)$	$\alpha \in \operatorname{SO}(4)$ (Remark 1 of Theorem 1.10.1) and so $\alpha \in z(\operatorname{SO}(4))$. Since the center
0902.0431_F00529	26	0.913	\alpha \in z(S O(4))	$\alpha \in z(\operatorname{SO}(4))$	$\alpha \in \operatorname{SO}(4)$ (Remark 1 of Theorem 1.10.1) and so $\alpha \in z(\operatorname{SO}(4))$. Since the center
0902.0431_F00530	26	1.000	z(S O(4))	$z(\operatorname{SO}(4))$	$z(\operatorname{SO}(4))$ of $\operatorname{SO}(4)$ is the group of order 2, we have
0902.0431_F00531	26	1.000	S O(4)	$\operatorname{SO}(4)$	$z(\operatorname{SO}(4))$ of $\operatorname{SO}(4)$ is the group of order 2, we have
0902.0431_F00532	26	1.000	\gamma \notin z\left(G_2\right)	$\gamma \notin z(G_2)$	However $\gamma \notin z(G_2)$ (Theorem 1.10.1). Hence $\alpha = 1$.
0902.0431_F00533	26	1.000	\alpha=1	$\alpha = 1$	However $\gamma \notin z(G_2)$ (Theorem 1.10.1). Hence $\alpha = 1$.
0902.0431_F00534	26	1.000	\alpha \in G_2^C	$\alpha \in G_2^C$	Lemma 1.12.1. For $\alpha \in G_2^C$, we have
0902.0431_F00535	26	1.000	\alpha x=(\alpha 1)(\alpha x)	$\alpha x = (\alpha 1)(\alpha x)$	Proof. The equality $\alpha x = (\alpha 1)(\alpha x)$ holds for all $x \in \mathfrak{C}^C$, and so we have $\alpha 1 = 1$.
0902.0431_F00536	26	1.000	x \in \mathfrak{C}^C	$x \in \mathfrak{C}^C$	Proof. The equality $\alpha x = (\alpha 1)(\alpha x)$ holds for all $x \in \mathfrak{C}^C$, and so we have $\alpha 1 = 1$.
0902.0431_F00537	26	1.000	\left(\alpha e_k\right)\left(\alpha e_k\right)=\alpha\left(e_k e_k\right)=\alpha(-1)=-1	$(\alpha e_k)(\alpha e_k) = \alpha(e_k e_k) = \alpha(-1) = -1$	Note that $(\alpha e_k)(\alpha e_k) = \alpha(e_k e_k) = \alpha(-1) = -1$. Let $x = \alpha e_k$ and $N(x) = x\bar{x} \in C$.
0902.0431_F00538	26	1.000	x=\alpha e_k	$x = \alpha e_k$	Note that $(\alpha e_k)(\alpha e_k) = \alpha(e_k e_k) = \alpha(-1) = -1$. Let $x = \alpha e_k$ and $N(x) = x\bar{x} \in C$.
0902.0431_F00539	26	1.000	N(x)=x \bar{x} \in C	$N(x) = x\bar{x} \in C$	Note that $(\alpha e_k)(\alpha e_k) = \alpha(e_k e_k) = \alpha(-1) = -1$. Let $x = \alpha e_k$ and $N(x) = x\bar{x} \in C$.
0902.0431_F00540	26	1.000	x x=-1	$xx = -1$	Then $xx = -1$ and $N(x)N(x) = 1$, which shows that $N(x) = \pm 1$. If $N(x) = -1$,
0902.0431_F00541	26	1.000	N(x) N(x)=1	$N(x)N(x) = 1$	Then $xx = -1$ and $N(x)N(x) = 1$, which shows that $N(x) = \pm 1$. If $N(x) = -1$,
0902.0431_F00542	26	1.000	N(x)=\pm 1	$N(x) = \pm 1$	Then $xx = -1$ and $N(x)N(x) = 1$, which shows that $N(x) = \pm 1$. If $N(x) = -1$,
0902.0431_F00543	26	1.000	N(x)=-1	$N(x) = -1$	Then $xx = -1$ and $N(x)N(x) = 1$, which shows that $N(x) = \pm 1$. If $N(x) = -1$,
0902.0431_F00544	26	1.000	\bar{x}=-x x \bar{x}=-x N(x)=x	$\bar{x} = -xx\bar{x} = -xN(x) = x$	then $\bar{x} = -xx\bar{x} = -xN(x) = x$, so $x \in C$. From $xx = -1$, we have $x = \pm i$. Then
0902.0431_F00545	26	1.000	x \in C	$x \in C$	then $\bar{x} = -xx\bar{x} = -xN(x) = x$, so $x \in C$. From $xx = -1$, we have $x = \pm i$. Then
0902.0431_F00546	26	1.000	x=\pm i	$x = \pm i$	then $\bar{x} = -xx\bar{x} = -xN(x) = x$, so $x \in C$. From $xx = -1$, we have $x = \pm i$. Then
0902.0431_F00547	26	1.000	\alpha\left(e_k\right)=\pm \alpha(i)	$\alpha(e_k) = \pm\alpha(i)$	$\alpha(e_k) = \pm\alpha(i)$, and so $e_k = \pm i$, which is a contradiction. Hence $N(x) = 1$, that is,
0902.0431_F00548	26	1.000	e_k=\pm i	$e_k = \pm i$	$\alpha(e_k) = \pm\alpha(i)$, and so $e_k = \pm i$, which is a contradiction. Hence $N(x) = 1$, that is,
0902.0431_F00549	26	1.000	N(x)=1	$N(x) = 1$	$\alpha(e_k) = \pm\alpha(i)$, and so $e_k = \pm i$, which is a contradiction. Hence $N(x) = 1$, that is,
0902.0431_F00550	26	1.000	x \bar{x}=1	$x\bar{x} = 1$	$x\bar{x} = 1$ and $\bar{x} = -xx\bar{x} = -xN(x) = -x$. Thus we have
0902.0431_F00551	26	1.000	\bar{x}=-x x \bar{x}=-x N(x)=-x	$\bar{x} = -xx\bar{x} = -xN(x) = -x$	$x\bar{x} = 1$ and $\bar{x} = -xx\bar{x} = -xN(x) = -x$. Thus we have
0902.0431_F00552	26	1.000	\langle x, y\rangle	$\langle x, y \rangle$	We define a positive definite Hermitian inner product $\langle x, y \rangle$ in $\mathfrak{C}^C$ by
0902.0431_F00553	27	1.000	\alpha \in \operatorname{Hom}_C\left(\mathfrak{C}^C\right)	$\alpha \in \operatorname{Hom}_C(\mathfrak{C}^C)$	For $\alpha \in \operatorname{Hom}_C(\mathfrak{C}^C)$, we denote the complex conjugate transpose of α with respect to
0902.0431_F00554	27	1.000	\alpha^*: \langle \alpha^* x, y\rangle=\langle x, \alpha y\rangle	$\alpha^* : \langle \alpha^* x, y \rangle = \langle x, \alpha y \rangle$	the inner product $\langle x, y \rangle$ by $\alpha^* : \langle \alpha^* x, y \rangle = \langle x, \alpha y \rangle$.
0902.0431_F00555	27	1.000	\alpha^*=\tau \alpha^{-1} \tau \in G_2^C	$\alpha^* = \tau\alpha^{-1}\tau \in G_2^C$	Lemma 1.12.2. (1) For $\alpha \in G_2^C$, we have $\alpha^* = \tau\alpha^{-1}\tau \in G_2^C$.
0902.0431_F00556	27	1.000	\alpha^C: \mathfrak{C}^C \rightarrow \mathfrak{C}^C	$\alpha^C : \mathfrak{C}^C \rightarrow \mathfrak{C}^C$	(2) For any $\alpha \in G_2$, its complexified mapping $\alpha^C : \mathfrak{C}^C \rightarrow \mathfrak{C}^C$ belongs to G_2^C :
0902.0431_F00557	27	0.998	\alpha^C \in G_2^C	$\alpha^C \in G_2^C$	$\alpha^C \in G_2^C$. Identifying α with α^C , we regard G_2 as a subgroup of G_2^C : $G_2 \subset G_2^C$.
0902.0431_F00558	27	0.998	\alpha^C	α^C	$\alpha^C \in G_2^C$. Identifying α with α^C , we regard G_2 as a subgroup of G_2^C : $G_2 \subset G_2^C$.
0902.0431_F00559	27	0.998	G_2^C: G_2 \subset G_2^C	$G_2^C : G_2 \subset G_2^C$	$\alpha^C \in G_2^C$. Identifying α with α^C , we regard G_2 as a subgroup of G_2^C : $G_2 \subset G_2^C$.
0902.0431_F00560	27	1.000	\tau \alpha=\alpha \tau	$\tau\alpha = \alpha\tau$	Now, for $\alpha \in G_2^C$, we have, $\alpha \in G_2$ if and only if $\alpha = \alpha\tau$, that is,
0902.0431_F00561	27	1.000	\left\langle \alpha^* x, y\right\rangle=\langle x, \alpha y\rangle=\left(\tau x, \alpha y\right)=\left\langle \tau \alpha^{-1} \tau x, y\right\rangle	$\langle \alpha^* x, y \rangle = \langle x, \alpha y \rangle = (\tau x, \alpha y) = \langle \tau\alpha^{-1}\tau x, y \rangle$	Proof. (1) $\langle \alpha^* x, y \rangle = \langle x, \alpha y \rangle = (\tau x, \alpha y) = \langle \alpha^{-1}\tau x, y \rangle = \langle \tau\alpha^{-1}\tau x, y \rangle$ for all
0902.0431_F00562	27	1.000	x, y \in \mathfrak{C}^C	$x, y \in \mathfrak{C}^C$	$x, y \in \mathfrak{C}^C$. Hence $\alpha^* = \tau\alpha^{-1}\tau \in G_2^C$.

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0902.0431_F00563	27	■ 1.000	$\tau \alpha x = \alpha \tau x = \alpha x$	$\tau \alpha x = \alpha \tau x = \alpha x$	(2) Let $\alpha \in G_2^C$ satisfy $\tau \alpha = \alpha \tau$. Then since $\tau \alpha x = \alpha \tau x = \alpha x$, we have $\alpha x \in \mathfrak{C}$
0902.0431_F00564	27	■ 1.000	$\alpha x \in \mathfrak{C}$	$\alpha x \in \mathfrak{C}$	(2) Let $\alpha \in G_2^C$ satisfy $\tau \alpha = \alpha \tau$. Then since $\tau \alpha x = \alpha \tau x = \alpha x$, we have $\alpha x \in \mathfrak{C}$
0902.0431_F00565	27	■ 1.000	$\alpha^{\prime}$	$\alpha^{\prime}$	for $x \in \mathfrak{C}$. Hence α induces an <i>R</i> -transformation $\alpha^{\prime}$ of $\mathfrak{C}$ and $\alpha^{\prime} \in G_2$, further we
0902.0431_F00566	27	■ 1.000	$\alpha^{\prime} \in G_2$	$\alpha^{\prime} \in G_2$	for $x \in \mathfrak{C}$. Hence α induces an <i>R</i> -transformation $\alpha^{\prime}$ of $\mathfrak{C}$ and $\alpha^{\prime} \in G_2$, further we
0902.0431_F00567	27	■ 1.000	$\alpha = (\alpha^{\prime})^C$	$\alpha = (\alpha^{\prime})^C$	have $\alpha = (\alpha^{\prime})^C$.
0902.0431_F00568	27	■ 1.000	$\operatorname{Iso}_{\mathfrak{C}}(\mathfrak{C}) = GL(8, C)$	$\operatorname{Iso}_C(\mathfrak{C}^C) = GL(8, C)$	Proof. Evidently G_2^C is an algebraic subgroup of $\operatorname{Iso}_C(\mathfrak{C}^C) = GL(8, C)$. If
0902.0431_F00569	27	■ 1.000	$\alpha^* \in G_2^C$	$\alpha^* \in G_2^C$	$\alpha \in G_2^C$, then $\alpha^* \in G_2^C$ (Lemma 1.12.2.(1)). Hence, from Chevalley's lemma
0902.0431_F00570	27	■ 1.000	$U(\mathfrak{C}^C) = \{\alpha \in \operatorname{Iso}_C(\mathfrak{C}^C) \mid \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\}$	$U(\mathfrak{C}^C) = \{\alpha \in \operatorname{Iso}_C(\mathfrak{C}^C) \mid \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\}$	where $U(\mathfrak{C}^C) = \{\alpha \in \operatorname{Iso}_C(\mathfrak{C}^C) \mid \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\}$ and $d = \dim G_2^C - \dim G_2 =$
0902.0431_F00571	27	■ 1.000	$d = \operatorname{dim} G_2^C - \operatorname{dim} G_2 =$	$d = \dim G_2^C - \dim G_2 =$	where $U(\mathfrak{C}^C) = \{\alpha \in \operatorname{Iso}_C(\mathfrak{C}^C) \mid \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\}$ and $d = \dim G_2^C - \dim G_2 =$
0902.0431_F00572	27	■ 1.000	$2 \times 14 - 14 = 14$	$2 \times 14 - 14 = 14$	$2 \times 14 - 14 = 14$. Since G_2 is simply connected (Theorem 1.9.3), G_2^C is also simply
0902.0431_F00573	27	■ 1.000	$\mathfrak{C}^{\prime} = H \oplus H e_4^{\prime}$	$\mathfrak{C}^{\prime} = H \oplus H e_4^{\prime}$	In $\mathfrak{C}^{\prime} = H \oplus H e_4^{\prime}$, we define a multiplication by
0902.0431_F00574	27	■ 1.000	$\mathfrak{C}^{\prime}$	$\mathfrak{C}^{\prime}$	This algebra $\mathfrak{C}^{\prime}$ is called the split Cayley algebra, and is isomorphic to $(\mathfrak{C}^C)_{\tau\gamma} = \{x \in$
0902.0431_F00575	27	■ 1.000	$(\mathfrak{C}^C)_{\tau\gamma} = \{x \in$	$(\mathfrak{C}^C)_{\tau\gamma} = \{x \in$	This algebra $\mathfrak{C}^{\prime}$ is called the split Cayley algebra, and is isomorphic to $(\mathfrak{C}^C)_{\tau\gamma} = \{x \in$
0902.0431_F00576	27	■ 1.000	$\mathfrak{C}^C \mid \tau\gamma x = x\}$	$\mathfrak{C}^C \mid \tau\gamma x = x\}$	$\mathfrak{C}^C \mid \tau\gamma x = x\}$ as algebras. Now, the group $G_{2(2)}$ is defined to be the automorphism
0902.0431_F00577	28	■ 1.000	$(x, y)^{\prime}$	$(x, y)^{\prime}$	Proof. In the split Cayley algebra $\mathfrak{C}^{\prime}$, the inner product $(x, y)^{\prime}$ is defined as
0902.0431_F00578	28	■ 0.998	$(x, y)^{\prime} = \frac{1}{2}(x\bar{y} + y\bar{x})$	$(x, y)^{\prime} = \frac{1}{2}(x\bar{y} + y\bar{x})$	$(x, y)^{\prime} = \frac{1}{2}(x\bar{y} + y\bar{x})$. If we define an inner product (x, y) of $\mathfrak{C}^{\prime}$ by $(\gamma x, y)^{\prime}$, then
0902.0431_F00579	28	■ 0.998	$(\gamma x, y)^{\prime}$	$(\gamma x, y)^{\prime}$	$(x, y)^{\prime} = \frac{1}{2}(x\bar{y} + y\bar{x})$. If we define an inner product (x, y) of $\mathfrak{C}^{\prime}$ by $(\gamma x, y)^{\prime}$, then
0902.0431_F00580	28	■ 0.986	$\alpha \in G_{2(2)}$	$\alpha \in G_{2(2)}$	(x, y) is a positive definite inner product. For $\alpha \in G_{2(2)}$, the transpose ${}^t\alpha$ of α with
0902.0431_F00581	28	■ 0.986	${}^t\alpha$	${}^t\alpha$	(x, y) is a positive definite inner product. For $\alpha \in G_{2(2)}$, the transpose ${}^t\alpha$ of α with
0902.0431_F00582	28	■ 0.858	${}^t\alpha = \gamma \alpha^{-1} \gamma \in G_{2(2)}$	${}^t\alpha = \gamma \alpha^{-1} \gamma \in G_{2(2)}$	respect to the inner product (x, y) is ${}^t\alpha = \gamma \alpha^{-1} \gamma \in G_{2(2)}$. Since $G_{2(2)}$ is an algebraic
0902.0431_F00583	28	■ 1.000	$\operatorname{Iso}_{\mathbf{R}}(\mathfrak{C}^{\prime}) = GL(8, \mathbf{R})$	$\operatorname{Iso}_{\mathbf{R}}(\mathfrak{C}^{\prime}) = GL(8, \mathbf{R})$	subgroup of $\operatorname{Iso}_{\mathbf{R}}(\mathfrak{C}^{\prime}) = GL(8, \mathbf{R})$, from Chevalley's lemma, we have
0902.0431_F00584	28	■ 1.000	$O(8) = \{\alpha \in \operatorname{Iso}_{\mathbf{R}}(\mathfrak{C}^{\prime}) \mid (\alpha x, \alpha y) = (x, y)\}$	$O(8) = \{\alpha \in \operatorname{Iso}_{\mathbf{R}}(\mathfrak{C}^{\prime}) \mid (\alpha x, \alpha y) = (x, y)\}$	where $O(8) = \{\alpha \in \operatorname{Iso}_{\mathbf{R}}(\mathfrak{C}^{\prime}) \mid (\alpha x, \alpha y) = (x, y)\}$.
0902.0431_F00585	28	■ 0.999	$z(G_{2(2)})$	$z(G_{2(2)})$	Theorem 1.13.2. <i>The center $z(G_{2(2)})$ of the group $G_{2(2)}$ is trivial:</i>
0902.0431_F00586	28	■ 1.000	$a \in S^{n-1} = \{a \in \mathbf{R}^n \mid (a, a) = 1\}$	$a \in S^{n-1} = \{a \in \mathbf{R}^n \mid (a, a) = 1\}$	For $a \in S^{n-1} = \{a \in \mathbf{R}^n \mid (a, a) = 1\}$, we define an element $D_a \in O(n) =$
0902.0431_F00587	28	■ 1.000	$D_a \in O(n) =$	$D_a \in O(n) =$	For $a \in S^{n-1} = \{a \in \mathbf{R}^n \mid (a, a) = 1\}$, we define an element $D_a \in O(n) =$
0902.0431_F00588	28	■ 0.999	$O(\mathbf{R}^n) = \{A \in \operatorname{Iso}_{\mathbf{R}}(\mathbf{R}^n) \mid (Ax, Ay) = (x, y)\}$	$O(\mathbf{R}^n) = \{A \in \operatorname{Iso}_{\mathbf{R}}(\mathbf{R}^n) \mid (Ax, Ay) = (x, y)\}$	$O(\mathbf{R}^n) = \{A \in \operatorname{Iso}_{\mathbf{R}}(\mathbf{R}^n) \mid (Ax, Ay) = (x, y)\}$ by
0902.0431_F00589	28	■ 1.000	D_a	D_a	D_a is called the reflection with respect to the hyperplane orthogonal to a . Its deter-
0902.0431_F00590	28	■ 0.570	$\det(D_a) = -1$	$\det(D_a) = -1$	minant is -1 : $\det(D_a) = -1$.
0902.0431_F00591	28	■ 1.000	$O(n)$	$O(n)$	Lemma 1.14.1. <i>The group $O(n)$ is generated by reflections, that is, any $A \in O(n)$</i>
0902.0431_F00592	28	■ 1.000	$A \in O(n)$	$A \in O(n)$	Lemma 1.14.1. <i>The group $O(n)$ is generated by reflections, that is, any $A \in O(n)$</i>
0902.0431_F00593	28	■ 1.000	$A \in SO(n)$	$A \in SO(n)$	<i>In particular, $A \in SO(n)$ can be expressed by the product of even number of reflections.</i>
0902.0431_F00594	29	■ 0.999	$\alpha_3 \in SO(8)$	$\alpha_3 \in SO(8)$	Theorem 1.14.2. (Principle of triality in $SO(8)$). <i>For any $\alpha_3 \in SO(8)$, there</i>
0902.0431_F00595	29	■ 1.000	$\alpha_1, \alpha_2 \in SO(8)$	$\alpha_1, \alpha_2 \in SO(8)$	<i>exist $\alpha_1, \alpha_2 \in SO(8)$ such that</i>
0902.0431_F00596	29	■ 1.000	α_1, α_2	α_1, α_2	<i>Moreover, α_1, α_2 are determined uniquely for α_3 up to the sign, that is, for α_3, such</i>
0902.0431_F00597	29	■ 1.000	α_3	α_3	<i>Moreover, α_1, α_2 are determined uniquely for α_3 up to the sign, that is, for α_3, such</i>
0902.0431_F00598	29	■ 1.000	$-\alpha_1, -\alpha_2$	$-\alpha_1, -\alpha_2$	<i>α_1, α_2 have to be α_1, α_2 or $-\alpha_1, -\alpha_2$.</i>

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0902.0431_F00599	29	■ 1.000	$\alpha_3=D_{\mathfrak{C}}\alpha, a, b \in \mathfrak{C}, a = b =1$	$\alpha_3 = D_b D_a, a, b \in \mathfrak{C}, a = b = 1$	$\alpha_3 = D_b D_a, a, b \in \mathfrak{C}, a = b = 1$. Now, since
0902.0431_F00600	29	■ 1.000	$\alpha_3x=D_{\mathfrak{C}}\alpha x=b(\bar{\alpha}x\bar{\alpha})b$	$\alpha_3x = D_b D_a x = b(\bar{a}x\bar{a})b$	we have $\alpha_3x = D_b D_a x = b(\bar{a}x\bar{a})b$. Define mappings $\alpha_1, \alpha_2 : \mathfrak{C} \rightarrow \mathfrak{C}$ respectively by
0902.0431_F00601	29	■ 1.000	$\alpha_1, \alpha_2 : \mathfrak{C} \rightarrow \mathfrak{C}$	$\alpha_1, \alpha_2 : \mathfrak{C} \rightarrow \mathfrak{C}$	we have $\alpha_3x = D_b D_a x = b(\bar{a}x\bar{a})b$. Define mappings $\alpha_1, \alpha_2 : \mathfrak{C} \rightarrow \mathfrak{C}$ respectively by
0902.0431_F00602	29	■ 1.000	$\alpha_1x=b(\bar{\alpha}x), \alpha_2x=(x\bar{\alpha})b$	$\alpha_1x = b(\bar{a}x), \alpha_2x = (x\bar{a})b$	$\alpha_1x = b(\bar{a}x), \alpha_2x = (x\bar{a})b$. We then see that $\alpha_1, \alpha_2 \in SO(8)$ and
0902.0431_F00603	29	■ 1.000	$\alpha_3=1$	$\alpha_3 = 1$	sufficient to show in the case $\alpha_3 = 1$. Now, let
0902.0431_F00604	29	■ 1.000	$\alpha_11=p$	$\alpha_11 = p$	Let $\alpha_11 = p$, then $ p = 1$ and $p(\alpha_2y) = y$, so $\alpha_2y = \bar{p}y$. Similarly we have $\alpha_1x = x\bar{q}$,
0902.0431_F00605	29	■ 1.000	$ p =1$	$ p = 1$	Let $\alpha_11 = p$, then $ p = 1$ and $p(\alpha_2y) = y$, so $\alpha_2y = \bar{p}y$. Similarly we have $\alpha_1x = x\bar{q}$,
0902.0431_F00606	29	■ 1.000	$p\left(\alpha_2y\right)=y$	$p(\alpha_2y) = y$	Let $\alpha_11 = p$, then $ p = 1$ and $p(\alpha_2y) = y$, so $\alpha_2y = \bar{p}y$. Similarly we have $\alpha_1x = x\bar{q}$,
0902.0431_F00607	29	■ 1.000	$\alpha_2y=\bar{p}y$	$\alpha_2y = \bar{p}y$	Let $\alpha_11 = p$, then $ p = 1$ and $p(\alpha_2y) = y$, so $\alpha_2y = \bar{p}y$. Similarly we have $\alpha_1x = x\bar{q}$,
0902.0431_F00608	29	■ 1.000	$\alpha_1x=x\bar{q}$	$\alpha_1x = x\bar{q}$	Let $\alpha_11 = p$, then $ p = 1$ and $p(\alpha_2y) = y$, so $\alpha_2y = \bar{p}y$. Similarly we have $\alpha_1x = x\bar{q}$,
0902.0431_F00609	29	■ 1.000	$q=\alpha_21$	$q = \alpha_21$	where $q = \alpha_21$. Hence $(x\bar{q})(\bar{p}y) = xy$. If we let $x = y = 1$, then $\bar{q}\bar{p} = 1$, so $\bar{q} = p$.
0902.0431_F00610	29	■ 1.000	$(x\bar{q})(\bar{p}y)=xy$	$(x\bar{q})(\bar{p}y) = xy$	where $q = \alpha_21$. Hence $(x\bar{q})(\bar{p}y) = xy$. If we let $x = y = 1$, then $\bar{q}\bar{p} = 1$, so $\bar{q} = p$.
0902.0431_F00611	29	■ 1.000	$\bar{q}\bar{p}=1$	$\bar{q}\bar{p} = 1$	where $q = \alpha_21$. Hence $(x\bar{q})(\bar{p}y) = xy$. If we let $x = y = 1$, then $\bar{q}\bar{p} = 1$, so $\bar{q} = p$.
0902.0431_F00612	29	■ 1.000	$\bar{q}=p$	$\bar{q} = p$	where $q = \alpha_21$. Hence $(x\bar{q})(\bar{p}y) = xy$. If we let $x = y = 1$, then $\bar{q}\bar{p} = 1$, so $\bar{q} = p$.
0902.0431_F00613	29	■ 1.000	py	py	Putting py instead of y , we have
0902.0431_F00614	29	■ 1.000	y	y	Putting py instead of y , we have
0902.0431_F00615	29	■ 1.000	p	p	From this, we see that p is a real number. Hence $p = \pm 1$ because $ p = 1$. Thus we
0902.0431_F00616	29	■ 1.000	$p=\pm 1$	$p = \pm 1$	From this, we see that p is a real number. Hence $p = \pm 1$ because $ p = 1$. Thus we
0902.0431_F00617	29	■ 1.000	$\alpha_1=\alpha_2=1$	$\alpha_1 = \alpha_2 = 1$	have $\alpha_1 = \alpha_2 = 1$ or $\alpha_1 = \alpha_2 = -1$.
0902.0431_F00618	29	■ 1.000	$\alpha_1=\alpha_2=-1$	$\alpha_1 = \alpha_2 = -1$	have $\alpha_1 = \alpha_2 = 1$ or $\alpha_1 = \alpha_2 = -1$.
0902.0431_F00619	29	■ 1.000	$\alpha_1, \alpha_2, \alpha_3 \in O(8)$	$\alpha_1, \alpha_2, \alpha_3 \in O(8)$	Lemma 1.14.3. For $\alpha_1, \alpha_2, \alpha_3 \in O(8)$, the relation
0902.0431_F00620	29	■ 1.000	$x=0$	$x = 0$	Proof. If $x = 0$ or $y = 0$, then the statement is trivially valid, so we may
0902.0431_F00621	29	■ 1.000	$y=0$	$y = 0$	Proof. If $x = 0$ or $y = 0$, then the statement is trivially valid, so we may
0902.0431_F00622	29	■ 1.000	$x, y \neq 0$	$x, y \neq 0$	assume $x, y \neq 0$. Now, multiply $\bar{\alpha}_1\bar{x}$ from left and $\alpha_3(\bar{xy})$ from right on the relation
0902.0431_F00623	29	■ 1.000	$\overline{\alpha_1x}$	$\overline{\alpha_1x}$	assume $x, y \neq 0$. Now, multiply $\bar{\alpha}_1\bar{x}$ from left and $\alpha_3(\bar{xy})$ from right on the relation
0902.0431_F00624	29	■ 1.000	$\alpha_3(\overline{xy})$	$\alpha_3(\overline{xy})$	assume $x, y \neq 0$. Now, multiply $\bar{\alpha}_1\bar{x}$ from left and $\alpha_3(\bar{xy})$ from right on the relation
0902.0431_F00625	30	■ 0.862	$\left(\alpha_1x\right)\left(\alpha_2y\right)=\overline{\alpha_3(\overline{xy})}$	$(\alpha_1x)(\alpha_2y) = \overline{\alpha_3(\overline{xy})}$	$(\alpha_1x)(\alpha_2y) = \overline{\alpha_3(\overline{xy})}$. Then we get $ x ^2(\alpha_2y)(\alpha_3(\overline{xy})) = \overline{\alpha_1\bar{x}} xy ^2$. Therefore
0902.0431_F00626	30	■ 0.862	$ x ^2(\alpha_2y)(\alpha_3(\overline{xy}))=\overline{\alpha_1\bar{x}} xy ^2$	$ x ^2(\alpha_2y)(\alpha_3(\overline{xy})) = \overline{\alpha_1\bar{x}} xy ^2$	$(\alpha_1x)(\alpha_2y) = \overline{\alpha_3(\overline{xy})}$. Then we get $ x ^2(\alpha_2y)(\alpha_3(\overline{xy})) = \overline{\alpha_1\bar{x}} xy ^2$. Therefore
0902.0431_F00627	30	■ 1.000	$\overline{y\bar{z}}$	$\overline{y\bar{z}}$	Putting $\overline{y\bar{z}}$ instead of x , we have $(\alpha_2y)(\alpha_3(\overline{y\bar{z}})) = \overline{\alpha_1(\overline{y\bar{z}})} y ^2$, that is,
0902.0431_F00628	30	■ 1.000	$(\alpha_2y)(\alpha_3(\overline{y\bar{z}}))=\overline{\alpha_1(\overline{y\bar{z}})} y ^2$	$(\alpha_2y)(\alpha_3(\overline{y\bar{z}})) = \overline{\alpha_1(\overline{y\bar{z}})} y ^2$	Putting $\overline{y\bar{z}}$ instead of x , we have $(\alpha_2y)(\alpha_3(\overline{y\bar{z}})) = \overline{\alpha_1(\overline{y\bar{z}})} y ^2$, that is,
0902.0431_F00629	30	■ 1.000	$\alpha_1, \alpha_2, \alpha_3 \in SO(8)$	$\alpha_1, \alpha_2, \alpha_3 \in SO(8)$	then $\alpha_1, \alpha_2, \alpha_3 \in SO(8)$.
0902.0431_F00630	30	■ 1.000	$\alpha_1 \notin SO(8)$	$\alpha_1 \notin SO(8)$	Proof. Suppose $\alpha_1 \notin SO(8)$. Since the mapping $\epsilon : \mathfrak{C} \rightarrow \mathfrak{C}$ defined by $\epsilon x = \bar{x}$
0902.0431_F00631	30	■ 1.000	$\epsilon : \mathfrak{C} \rightarrow \mathfrak{C}$	$\epsilon : \mathfrak{C} \rightarrow \mathfrak{C}$	Proof. Suppose $\alpha_1 \notin SO(8)$. Since the mapping $\epsilon : \mathfrak{C} \rightarrow \mathfrak{C}$ defined by $\epsilon x = \bar{x}$
0902.0431_F00632	30	■ 1.000	$\epsilon x=\bar{x}$	$\epsilon x = \bar{x}$	Proof. Suppose $\alpha_1 \notin SO(8)$. Since the mapping $\epsilon : \mathfrak{C} \rightarrow \mathfrak{C}$ defined by $\epsilon x = \bar{x}$
0902.0431_F00633	30	■ 1.000	$O(8)$	$O(8)$	belongs to $O(8)$, and since $\det \epsilon = -1$, we have $\beta_1 = \epsilon\alpha_1^{-1} \in SO(8)$. Using the
0902.0431_F00634	30	■ 1.000	$\det \epsilon = -1$	$\det \epsilon = -1$	belongs to $O(8)$, and since $\det \epsilon = -1$, we have $\beta_1 = \epsilon\alpha_1^{-1} \in SO(8)$. Using the
0902.0431_F00635	30	■ 1.000	$\beta_1=\epsilon\alpha_1^{-1} \in SO(8)$	$\beta_1 = \epsilon\alpha_1^{-1} \in SO(8)$	belongs to $O(8)$, and since $\det \epsilon = -1$, we have $\beta_1 = \epsilon\alpha_1^{-1} \in SO(8)$. Using the
0902.0431_F00636	30	■ 1.000	β_1	β_1	principle of triality (Theorem 1.14.2) on the element β_1 (cf. Lemma 1.14.3), there
0902.0431_F00637	30	■ 0.995	$\beta_2, \beta_3 \in SO(8)$	$\beta_2, \beta_3 \in SO(8)$	exist $\beta_2, \beta_3 \in SO(8)$ such that
0902.0431_F00638	30	■ 1.000	$\beta_2\alpha_2=\gamma_2$	$\beta_2\alpha_2 = \gamma_2$	Setting $\beta_2\alpha_2 = \gamma_2$ and $\beta_3\alpha_3 = \gamma_3$, the above relation becomes
0902.0431_F00639	30	■ 1.000	$\beta_3\alpha_3=\gamma_3$	$\beta_3\alpha_3 = \gamma_3$	Setting $\beta_2\alpha_2 = \gamma_2$ and $\beta_3\alpha_3 = \gamma_3$, the above relation becomes
0902.0431_F00640	30	■ 1.000	$\gamma_2y=\gamma_3y$	$\gamma_2y = \gamma_3y$	Put $x = 1$, then $\gamma_2y = \gamma_3y$, so $\gamma_2 = \gamma_3$. Hence we have
0902.0431_F00641	30	■ 1.000	$\gamma_2=\gamma_3$	$\gamma_2 = \gamma_3$	Put $x = 1$, then $\gamma_2y = \gamma_3y$, so $\gamma_2 = \gamma_3$. Hence we have
0902.0431_F00642	30	■ 1.000	$\gamma_21=p$	$\gamma_21 = p$	Put $\gamma_21 = p$, then $ p = 1$. Lett $y = 1$ in (i), then $\bar{x}p = \gamma_2x$. Hence (i) becomes
0902.0431_F00643	30	■ 1.000	$\bar{x}p=\gamma_2x$	$\bar{x}p = \gamma_2x$	Put $\gamma_21 = p$, then $ p = 1$. Lett $y = 1$ in (i), then $\bar{x}p = \gamma_2x$. Hence (i) becomes
0902.0431_F00644	30	■ 1.000	$y=p$	$y = p$	Again let $y = p$, then $\bar{x} = (\bar{x}p)p$, and so $\bar{x}\bar{p} = \bar{x}p$. Then

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0902.0431_F00645	30	■ 1.000	$\bar{x}=(\overline{x\ p})\ p$	$\bar{x} = (\overline{xp})p$	Again let $y = p$, then $\bar{x} = (\overline{xp})p$, and so $\bar{x}\ \bar{p} = \overline{xp}$. Then
0902.0431_F00646	30	■ 1.000	$\bar{x}\ \bar{p}=\overline{x\ p}$	$\bar{x}\bar{p} = \overline{xp}$	Again let $y = p$, then $\bar{x} = (\overline{xp})p$, and so $\bar{x}\ \bar{p} = \overline{xp}$. Then
0902.0431_F00647	30	■ 0.999	$\bar{x}\ \bar{y}=\overline{x\ y}$	$\bar{x}\bar{y} = \overline{xy}$	Therefore $p \in \boldsymbol{R}$, hence $p = \pm 1$ since $ p = 1$. Thus (ii) becomes $\bar{x}\ \bar{y} = \overline{xy}$, and so
0902.0431_F00648	30	■ 1.000	$\alpha_1\in S\ 0(8)$	$\alpha_1 \in SO(8)$	But this is a contradiction. Therefore $\alpha_1 \in SO(8)$. As for α_2, α_3 , use Lemma 1.14.3
0902.0431_F00649	30	■ 1.000	α_2, α_3	α_2, α_3	But this is a contradiction. Therefore $\alpha_1 \in SO(8)$. As for α_2, α_3 , use Lemma 1.14.3
0902.0431_F00650	30	■ 1.000	$\alpha_2, \alpha_3\in S\ 0(8)$	$\alpha_2, \alpha_3 \in SO(8)$	and the argument above, to show $\alpha_2, \alpha_3 \in SO(8)$.
0902.0431_F00651	31	■ 0.943	$ a =1$	$ a = 1$	Proposition 1.14.5. For $a \in \mathfrak{C}$ such that $ a = 1$, the mapping $\alpha_a : \mathfrak{C} \rightarrow \mathfrak{C}$
0902.0431_F00652	31	■ 0.943	$\alpha_a:\mathfrak{C}\rightarrow\mathfrak{C}$	$\alpha_a : \mathfrak{C} \rightarrow \mathfrak{C}$	Proposition 1.14.5. For $a \in \mathfrak{C}$ such that $ a = 1$, the mapping $\alpha_a : \mathfrak{C} \rightarrow \mathfrak{C}$
0902.0431_F00653	31	■ 1.000	$a^3=\pm 1$	$a^3 = \pm 1$	belongs to the group G_2 if and only if $a^3 = \pm 1$.
0902.0431_F00654	31	■ 1.000	$(\bar{a}x)(y\bar{a})=\bar{a}(xy)\bar{a}$	$(\bar{a}x)(y\bar{a}) = \bar{a}(xy)\bar{a}$	Proof. If we apply Lemma 1.14.3 to the Moufang formula $(\bar{a}x)(y\bar{a}) = \bar{a}(xy)\bar{a}$,
0902.0431_F00655	31	■ 1.000	$a\ x$	ax	If we replace x by ax and y by ya respectively, then
0902.0431_F00656	31	■ 1.000	$y\ a$	ya	If we replace x by ax and y by ya respectively, then
0902.0431_F00657	31	■ 1.000	α_a	α_a	Now, the mapping α_a belongs to G_2 if and only if
0902.0431_F00658	31	■ 1.000	$a^2=\pm\bar{a}$	$a^2 = \pm\bar{a}$	Therefore $a^2 = \pm\bar{a}$, so that $a^3 = \pm 1$. The converse also holds.
0902.0431_F00659	31	■ 0.357	$\omega_1^3=1$	$\omega_1^3 = 1$	Let $\omega_1 = -\frac{1}{2} + \frac{\sqrt{3}}{2}e_1 \in \boldsymbol{C} \subset \mathfrak{C}$. Then $\omega_1^3 = 1$, so $\alpha_{\bar{\omega}_1} \in G_2$ by Proposition 1.14.5.
0902.0431_F00660	31	■ 0.357	$\alpha_{\bar{\omega}_1}\in G_2$	$\alpha_{\bar{\omega}_1} \in G_2$	Let $\omega_1 = -\frac{1}{2} + \frac{\sqrt{3}}{2}e_1 \in \boldsymbol{C} \subset \mathfrak{C}$. Then $\omega_1^3 = 1$, so $\alpha_{\bar{\omega}_1} \in G_2$ by Proposition 1.14.5.
0902.0431_F00661	31	■ 1.000	$\alpha_{\bar{\omega}_1}$	$\alpha_{\bar{\omega}_1}$	This $\alpha_{\bar{\omega}_1}$ is nothing but w of Section 1.9: $\alpha_{\bar{\omega}_1} = w$, because
0902.0431_F00662	31	■ 1.000	$\alpha_{\bar{\omega}_1}=w$	$\alpha_{\bar{\omega}_1} = w$	This $\alpha_{\bar{\omega}_1}$ is nothing but w of Section 1.9: $\alpha_{\bar{\omega}_1} = w$, because
0902.0431_F00663	31	■ 0.991	$\alpha\in S\ 0(7)$	$\alpha \in SO(7)$	If we use the principle of triality for $\alpha \in SO(7)$, then there exist $\tilde{\alpha}, \alpha' \in SO(8)$
0902.0431_F00664	31	■ 0.991	$\tilde{\alpha}, \alpha'\in S\ 0(8)$	$\tilde{\alpha}, \alpha' \in SO(8)$	If we use the principle of triality for $\alpha \in SO(7)$, then there exist $\tilde{\alpha}, \alpha' \in SO(8)$
0902.0431_F00665	31	■ 1.000	$\tilde{\alpha}y=\alpha'y$	$\tilde{\alpha}y = \alpha'y$	Putting $x = 1$, we have $\tilde{\alpha}y = \alpha'y$, and so $\tilde{\alpha} = \alpha'$. Then (i) becomes
0902.0431_F00666	31	■ 1.000	$\tilde{\alpha}=\alpha'$	$\tilde{\alpha} = \alpha'$	Putting $x = 1$, we have $\tilde{\alpha}y = \alpha'y$, and so $\tilde{\alpha} = \alpha'$. Then (i) becomes
0902.0431_F00667	31	■ 1.000	$\alpha, \tilde{\alpha}\in S\ 0(8)$	$\alpha, \tilde{\alpha} \in SO(8)$	Conversely, suppose that $\alpha, \tilde{\alpha} \in SO(8)$ satisfy (ii). Putting $x = 1$, we have $(\alpha 1)(\tilde{\alpha}y) =$
0902.0431_F00668	31	■ 1.000	$(\alpha 1)(\tilde{\alpha}y)=$	$(\alpha 1)(\tilde{\alpha}y) =$	Conversely, suppose that $\alpha, \tilde{\alpha} \in SO(8)$ satisfy (ii). Putting $x = 1$, we have $(\alpha 1)(\tilde{\alpha}y) =$
0902.0431_F00669	31	■ 0.986	$\tilde{\alpha}y$	$\tilde{\alpha}y$	$\tilde{\alpha}y$, and so we have $\alpha 1 = 1$. Hence $\alpha \in SO(7)$.
0902.0431_F00670	31	■ 1.000	$\tilde{B}_3$	$\tilde{B}_3$	Definitioin. We define a subgroup $\tilde{B}_3$ of $SO(8)$ by
0902.0431_F00671	32	■ 0.671	$\tilde{B}_3/G_2\simeq S^7,\ \ \tilde{B}_3\cap SO(7)=G_2$	$\tilde{B}_3/G_2 \simeq S^7, \quad \tilde{B}_3 \cap SO(7) = G_2$	Theorem 1.15.1. $\tilde{B}_3/G_2 \simeq S^7, \quad \tilde{B}_3 \cap SO(7) = G_2$.
0902.0431_F00672	32	■ 1.000	$S^7=\{a\in\mathfrak{C}\mid a =1\}$	$S^7 = \{a \in \mathfrak{C} \mid a = 1\}$	Proof. $S^7 = \{a \in \mathfrak{C} \mid a = 1\}$ is a 7 dimensional sphere. Since $\tilde{B}_3$ is a subgroup
0902.0431_F00673	32	■ 1.000	S^7	S^7	of $SO(8)$, the group $\tilde{B}_3$ acts on S^7 . We shall show that $\tilde{B}_3$ acts transitively on S^7 .
0902.0431_F00674	32	■ 1.000	$b_0\in S^7$	$b_0 \in S^7$	To prove this, it is sufficient to show that any $b_0 \in S^7$ can be transformed to $1 \in S^7$
0902.0431_F00675	32	■ 1.000	$1\in S^7$	$1 \in S^7$	To prove this, it is sufficient to show that any $b_0 \in S^7$ can be transformed to $1 \in S^7$
0902.0431_F00676	32	■ 0.999	$\alpha\in\tilde{B}_3$	$\alpha \in \tilde{B}_3$	by some $\alpha \in \tilde{B}_3$. Now, we choose any element $a_1 \in S^7$ such that $(1, a_1) = 0$. and
0902.0431_F00677	32	■ 0.999	$a_1\in S^7$	$a_1 \in S^7$	by some $\alpha \in \tilde{B}_3$. Now, we choose any element $a_1 \in S^7$ such that $(1, a_1) = 0$. and
0902.0431_F00678	32	■ 0.999	$(1,a_1)=0$	$(1, a_1) = 0$	by some $\alpha \in \tilde{B}_3$. Now, we choose any element $a_1 \in S^7$ such that $(1, a_1) = 0$. and
0902.0431_F00679	32	■ 1.000	$a_2\in S^7$	$a_2 \in S^7$	choose any element $a_2 \in S^7$ such that $(1, a_2) = (a_1, a_2) = 0$. Let $a_3 \in S^7$ be the
0902.0431_F00680	32	■ 1.000	$(1,a_2)=(a_1,a_2)=0$	$(1, a_2) = (a_1, a_2) = 0$	choose any element $a_2 \in S^7$ such that $(1, a_2) = (a_1, a_2) = 0$. Let $a_3 \in S^7$ be the
0902.0431_F00681	32	■ 1.000	$a_3\in S^7$	$a_3 \in S^7$	choose any element $a_2 \in S^7$ such that $(1, a_2) = (a_1, a_2) = 0$. Let $a_3 \in S^7$ be the
0902.0431_F00682	32	■ 1.000	$a_3b_0=a_2(a_1b_0)$	$a_3b_0 = a_2(a_1b_0)$	element determined by $a_3b_0 = a_2(a_1b_0)$. More precisely, if we let $a_3 = (a_2(a_1b_0))\bar{b}_0$,
0902.0431_F00683	32	■ 1.000	$a_3=(a_2(a_1b_0))\bar{b}_0$	$a_3 = (a_2(a_1b_0))\bar{b}_0$	element determined by $a_3b_0 = a_2(a_1b_0)$. More precisely, if we let $a_3 = (a_2(a_1b_0))\bar{b}_0$,
0902.0431_F00684	32	■ 1.000	$(1,a_3)=(a_1,a_3)=(a_2,a_3)=0$	$(1, a_3) = (a_1, a_3) = (a_2, a_3) = 0$	then $a_3 \in S^7$ and satisfies $(1, a_3) = (a_1, a_3) = (a_2, a_3) = 0$. Choose any element
0902.0431_F00685	32	■ 1.000	$a_4\in S^7$	$a_4 \in S^7$	$a_4 \in S^7$ such that $(1, a_4) = (a_1, a_4) = (a_2, a_4) = (a_3, a_4) = 0$. Let $a_5, a_6, a_7 \in S^7$ be
0902.0431_F00686	32	■ 1.000	$(1,a_4)=(a_1,a_4)=(a_2,a_4)=(a_3,a_4)=0$	$(1, a_4) = (a_1, a_4) = (a_2, a_4) = (a_3, a_4) = 0$	$a_4 \in S^7$ such that $(1, a_4) = (a_1, a_4) = (a_2, a_4) = (a_3, a_4) = 0$. Let $a_5, a_6, a_7 \in S^7$ be
0902.0431_F00687	32	■ 1.000	$a_5,a_6,a_7\in S^7$	$a_5, a_6, a_7 \in S^7$	$a_4 \in S^7$ such that $(1, a_4) = (a_1, a_4) = (a_2, a_4) = (a_3, a_4) = 0$. Let $a_5, a_6, a_7 \in S^7$ be

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0902.0431_F00729	35	■ 0.982	$\backslash kappa, \backslash pi, \backslash nu: \operatorname{Spin}(8) \rightarrow \operatorname{Spin}(8)$	$\kappa, \pi, \nu : \operatorname{Spin}(8) \rightarrow \operatorname{Spin}(8)$	Theorem 1.16.4. We define automorphisms $\kappa, \pi, \nu : \operatorname{Spin}(8) \rightarrow \operatorname{Spin}(8)$ respec-
0902.0431_F00730	35	■ 1.000	$\backslash kappa: S\,0(8) \rightarrow S\,0(8)$	$\kappa : SO(8) \rightarrow SO(8)$	where $\mathfrak{k} : SO(8) \rightarrow SO(8)$ in the right side is defined by $(\kappa\alpha)x = \overline{\alpha\bar{x}}$, $x \in \mathfrak{C}$. Then
0902.0431_F00731	35	■ 1.000	$(\backslash kappa \backslash alpha) x=\overline{\backslash alpha \backslash bar{x}}, x \in \backslash mathfrak{C}$	$(\kappa\alpha)x = \overline{\alpha\bar{x}}, x \in \mathfrak{C}$	where $\kappa : SO(8) \rightarrow SO(8)$ in the right side is defined by $(\kappa\alpha)x = \alpha\bar{x}$, $x \in \mathfrak{C}$. Then
0902.0431_F00732	35	■ 0.884	$\backslash kappa, \backslash pi$	κ, π	The subgroup $\mathfrak{S}_3$ generated by κ, π in the automorphism group $\operatorname{Aut}(\operatorname{Spin}(8))$ of $\operatorname{Spin}(8)$
0902.0431_F00733	35	■ 0.884	$\operatorname{Aut}(\operatorname{Spin}(8))$	$\operatorname{Aut}(\operatorname{Spin}(8))$	The subgroup $\mathfrak{S}_3$ generated by κ, π in the automorphism group $\operatorname{Aut}(\operatorname{Spin}(8))$ of $\operatorname{Spin}(8)$
0902.0431_F00734	35	■ 1.000	$\backslash left(\backslash alpha_{\backslash 1}, \backslash alpha_{\backslash 2}, \backslash alpha_{\backslash 3}\backslash right) \in \operatorname{Spin}(8)$	$(\alpha_1, \alpha_2, \alpha_3) \in \operatorname{Spin}(8)$	If $(\alpha_1, \alpha_2, \alpha_3) \in \operatorname{Spin}(8)$ satisfies $\nu(\alpha_1, \alpha_2, \alpha_3) = (\alpha_1, \alpha_2, \alpha_3)$, then we have $\alpha_1 =$
0902.0431_F00735	35	■ 1.000	$\backslash nu\backslash left(\backslash alpha_{\backslash 1}, \backslash alpha_{\backslash 2}, \backslash alpha_{\backslash 3}\backslash right)=\backslash left(\backslash alpha_{\backslash 1}, \backslash alpha_{\backslash 2}, \backslash alpha_{\backslash 3}\backslash right)$	$\nu(\alpha_1, \alpha_2, \alpha_3) = (\alpha_1, \alpha_2, \alpha_3)$	If $(\alpha_1, \alpha_2, \alpha_3) \in \operatorname{Spin}(8)$ satisfies $\nu(\alpha_1, \alpha_2, \alpha_3) = (\alpha_1, \alpha_2, \alpha_3)$, then we have $\alpha_1 =$
0902.0431_F00736	35	■ 1.000	$\backslash alpha_{\backslash 1}=$	$\alpha_1 =$	If $(\alpha_1, \alpha_2, \alpha_3) \in \operatorname{Spin}(8)$ satisfies $\nu(\alpha_1, \alpha_2, \alpha_3) = (\alpha_1, \alpha_2, \alpha_3)$, then we have $\alpha_1 =$
0902.0431_F00737	35	■ 1.000	$\backslash alpha_{\backslash 2}=\backslash alpha_{\backslash 3}(=\backslash alpha)$	$\alpha_2 = \alpha_3 (= \alpha)$	$\alpha_2 = \alpha_3 (= \alpha)$, that is,
0902.0431_F00738	35	■ 1.000	$a=\backslash alpha\,1$	$a = \alpha 1$	Put $a = \alpha 1$, then $ a = 1$, and put $x = 1$ and $y = 1$ in (i), then we have $a(\alpha y) = \kappa\alpha(y)$
0902.0431_F00739	35	■ 1.000	$a(\backslash alpha\,y)=\backslash kappa \backslash alpha(y)$	$a(\alpha y) = \kappa\alpha(y)$	Put $a = \alpha 1$, then $ a = 1$, and put $x = 1$ and $y = 1$ in (i), then we have $\overline{a(\alpha y)} = \kappa\alpha(y)$
0902.0431_F00740	35	■ 1.000	$(\backslash alpha\,x) \, a=\backslash kappa \backslash alpha(x)$	$(\alpha x)a = \kappa\alpha(x)$	and $(\alpha x)a = \kappa\alpha(x)$, respectively. Hence, we get
0902.0431_F00741	35	■ 1.000	$a \in \backslash boldsymbol{R}$	$a \in \boldsymbol{R}$	hence $a \in \boldsymbol{R}$, and so $a = \pm 1$ from $ a = 1$. In the case $a = -1$, let $x = y = 1$ in
0902.0431_F00742	35	■ 1.000	$a= \backslash pm\,1$	$a = \pm 1$	hence $a \in \boldsymbol{R}$, and so $a = \pm 1$ from $ a = 1$. In the case $a = -1$, let $x = y = 1$ in
0902.0431_F00743	35	■ 1.000	$a=-1$	$a = -1$	hence $a \in \boldsymbol{R}$, and so $a = \pm 1$ from $ a = 1$. In the case $a = -1$, let $x = y = 1$ in
0902.0431_F00744	35	■ 1.000	$(-1)(-1)=-1$	$(-1)(-1) = -1$	(i), then $(-1)(-1) = -1$ which is a contradiction. Hence $a = 1$, so that $\kappa\alpha = \alpha$.
0902.0431_F00745	35	■ 1.000	$a=1$	$a = 1$	(i), then $(-1)(-1) = -1$ which is a contradiction. Hence $a = 1$, so that $\kappa\alpha = \alpha$.
0902.0431_F00746	35	■ 1.000	$\backslash kappa \backslash alpha=\backslash alpha$	$\kappa\alpha = \alpha$	(i), then $(-1)(-1) = -1$ which is a contradiction. Hence $a = 1$, so that $\kappa\alpha = \alpha$.
0902.0431_F00747	35	■ 1.000	$\backslash nu^{\backslash 2}: \operatorname{Spin}(8) \rightarrow \operatorname{Spin}(8)$	$\nu^2 : \operatorname{Spin}(8) \rightarrow \operatorname{Spin}(8)$	Therefore we have $\alpha \in G_2$. Finally, the automorphism $\nu^2 : \operatorname{Spin}(8) \rightarrow \operatorname{Spin}(8)$
0902.0431_F00748	35	■ 0.997	$\backslash nu^{\backslash 2}$	ν^2	hence ν^2 induces the isomorphism $SO(8) \cong Ss(8)$.
0902.0431_F00749	35	■ 0.997	$S\,0(8) \cong S\,s(8)$	$SO(8) \cong Ss(8)$	hence ν^2 induces the isomorphism $\overline{SO(8)} \cong \overline{Ss(8)}$.
0902.0431_F00750	36	■ 0.999	$\backslash mathfrak{J}=\backslash mathfrak{J}(3, \backslash mathfrak{C})$	$\mathfrak{J} = \mathfrak{J}(3, \mathfrak{C})$	Let $\mathfrak{J} = \mathfrak{J}(3, \mathfrak{C})$ denote all 3×3 Hermitian matrices with entries in the Cayley
0902.0431_F00751	36	■ 0.999	3×3	3×3	Let $\mathfrak{J} = \mathfrak{J}(3, \mathfrak{C})$ denote all 3×3 Hermitian matrices with entries in the Cayley
0902.0431_F00752	36	■ 0.998	$X^*=\{ \, \}^{\backslash t}\backslash bar{X}$	$X^* = {}^t\bar{X}$	where $X^* = {}^t\bar{X}$. Any element $X \in \mathfrak{J}$ is of the form
0902.0431_F00753	36	■ 0.998	$X \in \backslash mathfrak{J}$	$X \in \mathfrak{J}$	where $X^* = {}^t\bar{X}$. Any element $X \in \mathfrak{J}$ is of the form
0902.0431_F00754	36	■ 1.000	$X \circ Y$	$X \circ Y$	$\mathfrak{J}$ is a 27 dimensional $\boldsymbol{R}$ -vector space. In $\mathfrak{J}$, the multiplication $X \circ Y$, called the Jordan
0902.0431_F00755	36	■ 0.964	$\operatorname{trace} \operatorname{tr}(X)$	trace $\operatorname{tr}(X)$	In $\mathfrak{J}$, we define the trace $\operatorname{tr}(X)$, an inner product (X, Y) and a trilinear form $\operatorname{tr}(X, Y, Z)$
0902.0431_F00756	36	■ 0.964	$(X, \, Y)$	(X, Y)	In $\mathfrak{J}$, we define the trace $\operatorname{tr}(X)$, an inner product (X, Y) and a trilinear form $\operatorname{tr}(X, Y, Z)$
0902.0431_F00757	36	■ 0.964	$\operatorname{tr}(X, \, Y, \, Z)$	$\operatorname{tr}(X, Y, Z)$	In $\mathfrak{J}$, we define the trace $\operatorname{tr}(X)$, an inner product (X, Y) and a trilinear form $\operatorname{tr}(X, Y, Z)$
0902.0431_F00758	36	■ 1.000	$X \times Y$	$X \times Y$	Moreover, in $\mathfrak{J}$, we define a multiplication $X \times Y$, called the Freudenthal multiplica-
0902.0431_F00759	36	■ 0.839	E	E	(where E is the 3×3 unit matrix) and a trilinear form (X, Y, Z) and the determinant
0902.0431_F00760	36	■ 0.839	$(X, \, Y, \, Z)$	(X, Y, Z)	(where E is the 3×3 unit matrix) and a trilinear form (X, Y, Z) and the determinant
0902.0431_F00761	36	■ 1.000	$\operatorname{det} X$	$\det X$	$\det X$ respectively by
0902.0431_F00762	36	■ 1.000	$X=X(\backslash xi, \, x), \, Y=Y(\backslash eta, \, y)$	$X = X(\xi, x), Y = Y(\eta, y)$	For $\overline{X = X(\xi, x), Y = Y(\eta, y)}$ and $Z = Z(\zeta, z) \in \mathfrak{J}$, the explicit forms in the terms
0902.0431_F00763	36	■ 1.000	$Z=Z(\backslash zeta, \, z) \in \backslash mathfrak{J}$	$Z = Z(\zeta, z) \in \mathfrak{J}$	For $X = X(\xi, x), Y = Y(\eta, y)$ and $Z = Z(\zeta, z) \in \mathfrak{J}$, the explicit forms in the terms
0902.0431_F00764	37	■ 1.000	$X \circ Y=Y \circ X, \quad X \times Y=Y \times X$	$X \circ Y = Y \circ X, \quad X \times Y = Y \times X$	(1) (i) $\overline{X \circ Y = Y \circ X, \quad X \times Y = Y \times X}$.
0902.0431_F00765	37	■ 1.000	$E \circ X=X, \quad E \times X=\frac{1}{2}(\operatorname{tr}(X)E-X), \quad E \times E=E$	$E \circ X = X, \quad E \times X = \frac{1}{2}(\operatorname{tr}(X)E - X), \quad E \times E = E$	(ii) $\overline{E \circ X = X, \quad E \times X = \frac{1}{2}(\operatorname{tr}(X)E - X), \quad E \times E = E}$.
0902.0431_F00766	37	■ 0.990	$\operatorname{tr}(X, \, Y, \, Z)=\operatorname{tr}(Y, \, Z, \, X)=\operatorname{tr}(Z, \, X, \, Y)=\operatorname{tr}(X, \, Z, \, Y)=\operatorname{tr}(Y, \, X, \, Z)$	$\operatorname{tr}(X, Y, Z) = \operatorname{tr}(Y, Z, X) = \operatorname{tr}(Z, X, Y) = \operatorname{tr}(X, Z, Y) = \operatorname{tr}(Y, X, Z)$	(ii) $\overline{\operatorname{tr}(X, Y, Z) = \operatorname{tr}(Y, Z, X) = \operatorname{tr}(Z, X, Y) = \operatorname{tr}(X, Z, Y) = \operatorname{tr}(Y, X, Z)}$
0902.0431_F00767	37	■ 0.971	$=\operatorname{tr}(Z, \, Y, \, X)$	$= \operatorname{tr}(Z, Y, X)$	$\overline{= \operatorname{tr}(Z, Y, X)}$.
0902.0431_F00768	37	■ 0.958	$\quad (X, \, E)=(X, \, E, \, E)=\operatorname{tr}(X, \, E, \, E)=\operatorname{tr}(X), \quad \operatorname{tr}(X, \, Y, \, E)=(X, \, Y)$	$(X, E) = (X, E, E) = \operatorname{tr}(X, E, E) = \operatorname{tr}(X), \quad \operatorname{tr}(X, Y, E) = (X, Y)$	(iii) $\overline{(X, E) = (X, E, E) = \operatorname{tr}(X, E, E) = \operatorname{tr}(X), \quad \operatorname{tr}(X, Y, E) = (X, Y)}$.

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0902.0431_F00769	37	0.772	$\quad \operatorname{tr}(X \times Y) = \frac{1}{2}(\operatorname{tr}(X) \operatorname{tr}(Y) - (X, Y))$	$\operatorname{tr}(X \times Y) = \frac{1}{2}(\operatorname{tr}(X) \operatorname{tr}(Y) - (X, Y))$	(iv) $\operatorname{tr}(X \times Y) = \frac{1}{2}(\operatorname{tr}(X) \operatorname{tr}(Y) - (X, Y))$.
0902.0431_F00770	37	1.000	$(X \times X) \circ X = (\det X) E$	$(X \times X) \circ X = (\det X) E$	(3) (i) $(X \times X) \circ X = (\det X) E$ (Hamilton-Cayley).
0902.0431_F00771	37	1.000	$(X \times X) \times (X \times X) = (\det X) X$	$(X \times X) \times (X \times X) = (\det X) X$	(ii) $(X \times X) \times (X \times X) = (\det X) X$.
0902.0431_F00772	37	0.997	$(X, Y), \operatorname{tr}(X, Y, Z), (X, Y, Z)$	$(X, Y), \operatorname{tr}(X, Y, Z), (X, Y, Z)$	(2) is clear from the explicit forms of $(X, Y), \operatorname{tr}(X, Y, Z), (X, Y, Z)$ etc.
0902.0431_F00773	38	0.673	$\bmod 3$	$\bmod 3$	where the indexes are considered as mod 3.
0902.0431_F00774	38	1.000	$\alpha \in F_4$	$\alpha \in F_4$	Lemma 2.2.1. (1) For $\alpha \in F_4$, we have $\alpha E = E$.
0902.0431_F00775	38	1.000	$\alpha E = E$	$\alpha E = E$	Lemma 2.2.1. (1) For $\alpha \in F_4$, we have $\alpha E = E$.
0902.0431_F00776	38	1.000	$\operatorname{tr}(\alpha X) = \operatorname{tr}(X), X \in \mathfrak{J}$	$\operatorname{tr}(\alpha X) = \operatorname{tr}(X), X \in \mathfrak{J}$	(2) For $\alpha \in F_4$, we have $\operatorname{tr}(\alpha X) = \operatorname{tr}(X), X \in \mathfrak{J}$.
0902.0431_F00777	38	1.000	$E \circ X = X$	$E \circ X = X$	Proof. (1) Applying α on $E \circ X = X$, we have $\alpha E \circ \alpha X = \alpha X$. Let $X = \alpha^{-1} E$,
0902.0431_F00778	38	1.000	$\alpha E \circ \alpha X = \alpha X$	$\alpha E \circ \alpha X = \alpha X$	Proof. (1) Applying α on $E \circ X = X$, we have $\alpha E \circ \alpha X = \alpha X$. Let $X = \alpha^{-1} E$,
0902.0431_F00779	38	1.000	$X = \alpha^{-1} E$	$X = \alpha^{-1} E$	Proof. (1) Applying α on $E \circ X = X$, we have $\alpha E \circ \alpha X = \alpha X$. Let $X = \alpha^{-1} E$,
0902.0431_F00780	38	1.000	$\alpha E \circ E = E$	$\alpha E \circ E = E$	then $\alpha E \circ E = E$, that is, $\alpha E = E$.
0902.0431_F00781	38	1.000	$X \circ (X \times X) = (\det X) E$	$X \circ (X \times X) = (\det X) E$	(2) We use the Hamilton-Cayley identity $X \circ (X \times X) = (\det X) E$ (Lemma
0902.0431_F00782	38	1.000	αX	αX	We put αX in the place of X of (i) and then apply $\alpha^{-1} \in F_4$ on the obtained
0902.0431_F00783	38	1.000	$\alpha^{-1} \in F_4$	$\alpha^{-1} \in F_4$	We put αX in the place of X of (i) and then apply $\alpha^{-1} \in F_4$ on the obtained
0902.0431_F00784	38	1.000	$X = F_i(e_j), i = 1, 2, 3, j = 0, 1, \dots, 7$	$X = F_i(e_j), i = 1, 2, 3, j = 0, 1, \dots, 7$	Let $X = F_i(e_j), i = 1, 2, 3, j = 0, 1, \dots, 7$ then
0902.0431_F00785	38	1.000	$\operatorname{tr}((\alpha F_i(e_j))^2) = 2$	$\operatorname{tr}((\alpha F_i(e_j))^2) = 2$	and $\operatorname{tr}((\alpha F_i(e_j))^2) = 2$, hence
0902.0431_F00786	39	1.000	$i = 1, 2, 3$	$i = 1, 2, 3$	for $i = 1, 2, 3$. Consequently, we have $\operatorname{tr}(\alpha X) = \operatorname{tr}(X)$ for every $X = E_i, F_i(e_j)$ of the
0902.0431_F00787	39	1.000	$\operatorname{tr}(\alpha X) = \operatorname{tr}(X)$	$\operatorname{tr}(\alpha X) = \operatorname{tr}(X)$	for $i = 1, 2, 3$. Consequently, we have $\operatorname{tr}(\alpha X) = \operatorname{tr}(X)$ for every $X = E_i, F_i(e_j)$ of the
0902.0431_F00788	39	1.000	$X = E_i, F_i(e_j)$	$X = E_i, F_i(e_j)$	for $i = 1, 2, 3$. Consequently, we have $\operatorname{tr}(\alpha X) = \operatorname{tr}(X)$ for every $X = E_i, F_i(e_j)$ of the
0902.0431_F00789	39	1.000	$\alpha \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{J})$	$\alpha \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{J})$	For $\alpha \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{J})$, we denote by ${}^t\alpha$ the transpose of α with respect to the inner
0902.0431_F00790	39	1.000	$(X, Y) : ({}^t\alpha X, Y) = (X, \alpha Y)$	$(X, Y) : ({}^t\alpha X, Y) = (X, \alpha Y)$	product $(X, Y) : ({}^t\alpha X, Y) = (X, \alpha Y)$.
0902.0431_F00791	39	1.000	$\alpha \in \operatorname{Iso}_{\mathbf{R}}(\mathfrak{J})$	$\alpha \in \operatorname{Iso}_{\mathbf{R}}(\mathfrak{J})$	Lemma 2.2.2. For $\alpha \in \operatorname{Iso}_{\mathbf{R}}(\mathfrak{J})$, the following four conditions are equivalent.
0902.0431_F00792	39	0.588	$\det(\alpha X) = \det X$	$\det(\alpha X) = \det X$	(1) $\det(\alpha X) = \det X$ for all $X \in \mathfrak{J}$.
0902.0431_F00793	39	0.999	$(\alpha X, \alpha Y, \alpha Z) = (X, Y, Z)$	$(\alpha X, \alpha Y, \alpha Z) = (X, Y, Z)$	(2) $(\alpha X, \alpha Y, \alpha Z) = (X, Y, Z)$ for all $X, Y, Z \in \mathfrak{J}$.
0902.0431_F00794	39	0.999	$X, Y, Z \in \mathfrak{J}$	$X, Y, Z \in \mathfrak{J}$	(2) $(\alpha X, \alpha Y, \alpha Z) = (X, Y, Z)$, for all $X, Y, Z \in \mathfrak{J}$.
0902.0431_F00795	39	1.000	$\alpha X \times \alpha Y = {}^t\alpha^{-1}(X \times Y)$	$\alpha X \times \alpha Y = {}^t\alpha^{-1}(X \times Y)$	(3) $\alpha X \times \alpha Y = {}^t\alpha^{-1}(X \times Y)$ for all $X, Y \in \mathfrak{J}$.
0902.0431_F00796	39	1.000	$X, Y \in \mathfrak{J}$	$X, Y \in \mathfrak{J}$	(3) $\alpha X \times \alpha Y = {}^t\alpha^{-1}(X \times Y)$, for all $X, Y \in \mathfrak{J}$.
0902.0431_F00797	39	0.997	$\alpha X \times \alpha X = {}^t\alpha^{-1}(X \times X)$	$\alpha X \times \alpha X = {}^t\alpha^{-1}(X \times X)$	(4) $\alpha X \times \alpha X = {}^t\alpha^{-1}(X \times X)$ for all $X \in \mathfrak{J}$.
0902.0431_F00798	39	0.992	$\det(\alpha X) = \det X$	$\det(\alpha X) = \det X$	Proof. (1) $\Rightarrow$ (2) $\det(\alpha X) = \det X$ implies that $(\alpha X, \alpha X, \alpha X) = (X, X, X)$.
0902.0431_F00799	39	0.992	$(\alpha X, \alpha X, \alpha X) = (X, X, X)$	$(\alpha X, \alpha X, \alpha X) = (X, X, X)$	Proof. (1) $\Rightarrow$ (2) $\det(\alpha X) = \det X$ implies that $(\alpha X, \alpha X, \alpha X) = (X, X, X)$.
0902.0431_F00800	39	1.000	$\lambda X + \mu Y + \nu Z$	$\lambda X + \mu Y + \nu Z$	Putting $\lambda X + \mu Y + \nu Z$ in place of X and comparing the coefficient of $\lambda\mu\nu$, we obtain
0902.0431_F00801	39	1.000	$\lambda\mu\nu$	$\lambda\mu\nu$	Putting $\lambda X + \mu Y + \nu Z$ in place of X and comparing the coefficient of $\lambda\mu\nu$, we obtain
0902.0431_F00802	39	1.000	$\lambda X + \mu Y$	$\lambda X + \mu Y$	(4) $\Rightarrow$ (3) Putting $\lambda X + \mu Y$ in place of X and comparing the coefficient of $\lambda\mu$, we
0902.0431_F00803	39	1.000	$\lambda\mu$	$\lambda\mu$	(4) $\Rightarrow$ (3) Putting $\lambda X + \mu Y$ in place of X and comparing the coefficient of $\lambda\mu$, we
0902.0431_F00804	39	1.000	$Y = X \times X, X \in \mathfrak{J}$	$Y = X \times X, X \in \mathfrak{J}$	Let $Y = X \times X, X \in \mathfrak{J}$ in the above, then
0902.0431_F00805	39	1.000	$\det X \neq 0$	$\det X \neq 0$	(1) Case $\det X \neq 0$. In this case, we have ${}^t\alpha^{-1}X \times {}^t\alpha^{-1}X = \alpha(X \times X)$. Hence,
0902.0431_F00806	39	1.000	${}^t\alpha^{-1}X \times {}^t\alpha^{-1}X = \alpha(X \times X)$	${}^t\alpha^{-1}X \times {}^t\alpha^{-1}X = \alpha(X \times X)$	(1) Case $\det X \neq 0$. In this case, we have ${}^t\alpha^{-1}X \times {}^t\alpha^{-1}X = \alpha(X \times X)$. Hence,
0902.0431_F00807	39	1.000	$\det({}^t\alpha^{-1}X) = \det X$	$\det({}^t\alpha^{-1}X) = \det X$	hence we have $\det({}^t\alpha^{-1}X) = \det X$. Next, if we use α^{-1} instead of α , we can see
0902.0431_F00808	39	1.000	α^{-1}	α^{-1}	hence we have $\det({}^t\alpha^{-1}X) = \det X$. Next, if we use α^{-1} instead of α , we can see

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0902.0431_F00809	39	■ 1.000	$\operatorname{det}\left(\{ \}^t \alpha X\right)=\operatorname{det} X$	$\det({}^t\alpha X) = \det X$	also that $\det({}^t\alpha X) = \det X$.
0902.0431_F00810	40	■ 1.000	$\operatorname{det} X=0$	$\det X = 0$	(2) Case $\det X = 0$. If $\det({}^t\alpha^{-1}X) \neq 0$, we can use the result of (1). If we put ${}^t\alpha X$
0902.0431_F00811	40	■ 1.000	$\operatorname{det}\left(\{ \}^t \alpha^{-1} X\right) \neq 0$	$\det({}^t\alpha^{-1}X) \neq 0$	(2) Case $\det X = 0$. If $\det({}^t\alpha^{-1}X) \neq 0$, we can use the result of (1). If we put ${}^t\alpha X$
0902.0431_F00812	40	■ 1.000	$\{ \}^t \alpha X$	${}^t\alpha X$	(2) Case $\det X = 0$. If $\det({}^t\alpha^{-1}X) \neq 0$, we can use the result of (1). If we put ${}^t\alpha X$
0902.0431_F00813	40	■ 0.670	$\left.\operatorname{det}\left(\{ \}^t \alpha^{-1}\right)\left(\{ \}^t \alpha X\right)\right)=\operatorname{det}\left(\{ \}^t \alpha X\right)$	$\det{}^t\alpha^{-1}({}^t\alpha X) = \det({}^t\alpha X)$	instead of X in $\det({}^t\alpha^{-1}X) = \det X$ of (1), then $\det{}^t\alpha^{-1}({}^t\alpha X) = \det({}^t\alpha X)$. Hence
0902.0431_F00814	40	■ 0.817	$\left.\operatorname{det} X=\operatorname{det}^t \alpha X\right) \neq 0$	$0 = \det X = \det{}^t\alpha X \neq 0$	$0 = \det X = \det{}^t\alpha X \neq 0$ which is a contradiction. Thus we have $\det({}^t\alpha^{-1}X) =$
0902.0431_F00815	40	■ 0.817	$\operatorname{det}\left(\{ \}^t \alpha^{-1} X\right)=$	$\det({}^t\alpha^{-1}X) =$	$0 = \det X = \det{}^t\alpha X \neq 0$ which is a contradiction. Thus we have $\det({}^t\alpha^{-1}X) =$
0902.0431_F00816	40	■ 1.000	$\operatorname{det}\left(\{ \}^t \alpha X\right)=0$	$\det({}^t\alpha X) = 0$	$\det({}^t\alpha X) = 0$, so $\det({}^t\alpha^{-1}X) = \det({}^t\alpha X) = \det X$ is also valid.
0902.0431_F00817	40	■ 1.000	$\operatorname{det}\left(\{ \}^t \alpha^{-1} X\right)=\operatorname{det}\left(\{ \}^t \alpha X\right)=\operatorname{det} X$	$\det({}^t\alpha^{-1}X) = \det({}^t\alpha X) = \det X$	$\det({}^t\alpha X) = 0$, so $\det({}^t\alpha^{-1}X) = \det({}^t\alpha X) = \det X$ is also valid.
0902.0431_F00818	40	■ 0.924	$\operatorname{tr}\left(\alpha X \circ \alpha Y\right)=\operatorname{tr}\left(\alpha\left(X \circ Y\right)\right)=\operatorname{tr}\left(X \circ Y\right)$	$\Rightarrow (2)(\alpha X, \alpha Y) = \operatorname{tr}(\alpha X \circ \alpha Y) = \operatorname{tr}(\alpha(X \circ Y)) = \operatorname{tr}(X \circ Y)$	Proof. (1) $\Rightarrow (2) (\alpha X, \alpha Y) = \operatorname{tr}(\alpha X \circ \alpha Y) = \operatorname{tr}(\alpha(X \circ Y)) = \operatorname{tr}(X \circ Y)$ (Lemma
0902.0431_F00819	40	■ 0.521	(X, Y)	(X, Y)	2.2.1) (X, Y) . Also $\operatorname{tr}(\alpha X, \alpha Y, \alpha Z) = (\alpha X, \alpha Y \circ \alpha Z) = (\alpha X, \alpha(Y \circ Z)) = (X, Y \circ$
0902.0431_F00820	40	■ 0.521	$\operatorname{tr}\left(\alpha X, \alpha Y, \alpha Z\right)=\left(\alpha X, \alpha Y \circ \alpha Z\right)=\left(\alpha X, \alpha\left(Y \circ Z\right)\right)=\left(X, Y \circ$	$\operatorname{tr}(\alpha X, \alpha Y, \alpha Z) = (\alpha X, \alpha Y \circ \alpha Z) = (\alpha X, \alpha(Y \circ Z)) = (X, Y \circ$	2.2.1) (X, Y) . Also $\operatorname{tr}(\alpha X, \alpha Y, \alpha Z) = (\alpha X, \alpha Y \circ \alpha Z) = (\alpha X, \alpha(Y \circ Z)) = (X, Y \circ$
0902.0431_F00821	40	■ 1.000	$\operatorname{tr}\left(X, Y, Z\right)$	$Z) = \operatorname{tr}(X, Y, Z)$	$Z) = \operatorname{tr}(X, Y, Z)$.
0902.0431_F00822	40	■ 0.699	$\operatorname{tr}\left(\alpha X \circ \alpha Y, \alpha Z\right)=\operatorname{tr}\left(\alpha X, \alpha Y, \alpha Z\right)=\operatorname{tr}\left(X, Y, Z\right)=\left(X \circ Y, Z\right)=\left(\alpha\left(X \circ$	$\Rightarrow (1)(\alpha X \circ \alpha Y, \alpha Z) = \operatorname{tr}(\alpha X, \alpha Y, \alpha Z) = \operatorname{tr}(X, Y, Z) = (X \circ Y, Z) = (\alpha(X \circ$	(2) $\Rightarrow (1) (\alpha X \circ \alpha Y, \alpha Z) = \operatorname{tr}(\alpha X, \alpha Y, \alpha Z) = \operatorname{tr}(X, Y, Z) = (X \circ Y, Z) = (\alpha(X \circ$
0902.0431_F00823	40	■ 0.998	$Y), \alpha Z)$	$Y), \alpha Z)$	$Y), \alpha Z)$ holds for all αZ , so we have $\alpha X \circ \alpha Y = \alpha(X \circ Y)$.
0902.0431_F00824	40	■ 0.998	αZ	αZ	$Y), \alpha Z)$ holds for all αZ , so we have $\alpha X \circ \alpha Y = \alpha(X \circ Y)$.
0902.0431_F00825	40	■ 0.998	$\alpha X \circ \alpha Y=\alpha(X \circ Y)$	$\alpha X \circ \alpha Y = \alpha(X \circ Y)$	$Y), \alpha Z)$ holds for all αZ , so we have $\alpha X \circ \alpha Y = \alpha(X \circ Y)$.
0902.0431_F00826	40	■ 0.946	$\operatorname{tr}\left(\alpha\left(X \times Y\right), \alpha Z\right)=\left(X \times Y, Z\right)=\left(X, Y, Z\right)=\left(\alpha X, \alpha Y, \alpha Z\right)$	$\Rightarrow (5)(\alpha(X \times Y), \alpha Z) = (X \times Y, Z) = (X, Y, Z) = (\alpha X, \alpha Y, \alpha Z)$	(3) $\Rightarrow (5) (\alpha(X \times Y), \alpha Z) = (X \times Y, Z) = (X, Y, Z) = (\alpha X, \alpha Y, \alpha Z)$ (Lemma
0902.0431_F00827	40	■ 1.000	$\left(\alpha X \times \alpha Y, \alpha Z\right)$	$= (\alpha X \times \alpha Y, \alpha Z)$	2.2.2) $\Rightarrow (\alpha X \times \alpha Y, \alpha Z)$ holds for all $\alpha Z \in \mathfrak{J}$, so we have $\alpha X \times \alpha Y = \alpha(X \times Y)$.
0902.0431_F00828	40	■ 1.000	$\alpha Z \in \mathfrak{J}$	$\alpha Z \in \mathfrak{J}$	2.2.2) $= (\alpha X \times \alpha Y, \alpha Z)$ holds for all $\alpha Z \in \mathfrak{J}$, so we have $\alpha X \times \alpha Y = \alpha(X \times Y)$.
0902.0431_F00829	40	■ 1.000	$\alpha X \times \alpha Y=\alpha(X \times Y)$	$\alpha X \times \alpha Y = \alpha(X \times Y)$	2.2.2) $= (\alpha X \times \alpha Y, \alpha Z)$ holds for all $\alpha Z \in \mathfrak{J}$, so we have $\alpha X \times \alpha Y = \alpha(X \times Y)$.
0902.0431_F00830	40	■ 0.923	$(5) \Rightarrow (4)$	$(5) \Rightarrow (4)$	$(5) \Rightarrow (4)$
0902.0431_F00831	40	■ 1.000	$\alpha E=P$	$\alpha E = P$	if we denote $\alpha E = P$, then
0902.0431_F00832	40	■ 1.000	$X=\alpha^{-1} E_1$	$X = \alpha^{-1} E_1$	Let $X = \alpha^{-1} E_1$ and $P = \rho_1 E_1 + \rho_2 E_2 + \rho_3 E_3 + F_1(p_1) + F_2(p_2) + F_3(p_3)$, then
0902.0431_F00833	40	■ 1.000	$P=\rho_{1} E_1+\rho_{2} E_2+\rho_{3} E_3+F_1\left(p_1\right)+F_2\left(p_2\right)+F_3\left(p_3\right)$	$P = \rho_1 E_1 + \rho_2 E_2 + \rho_3 E_3 + F_1(p_1) + F_2(p_2) + F_3(p_3)$	Let $X = \alpha^{-1} E_1$ and $P = \rho_1 E_1 + \rho_2 E_2 + \rho_3 E_3 + F_1(p_1) + F_2(p_2) + F_3(p_3)$, then
0902.0431_F00834	40	■ 1.000	$\lambda=\operatorname{tr}\left(\alpha^{-1} E_1\right)$	$\lambda = \operatorname{tr}(\alpha^{-1} E_1)$	where $\lambda = \operatorname{tr}(\alpha^{-1} E_1)$. By comparing entries of both sides, we have
0902.0431_F00835	41	■ 1.000	$p_2=p_3=0$	$p_2 = p_3 = 0$	Consequently we have $p_2 = p_3 = 0$. Similarly, by letting $X = \alpha^{-1} E_2$, we also have
0902.0431_F00836	41	■ 1.000	$X=\alpha^{-1} E_2$	$X = \alpha^{-1} E_2$	Consequently we have $p_2 = p_3 = 0$. Similarly, by letting $X = \alpha^{-1} E_2$, we also have
0902.0431_F00837	41	■ 1.000	$p_1=0$	$p_1 = 0$	$p_1 = 0$. Again put $X = \alpha^{-1} F_1(1)$, then
0902.0431_F00838	41	■ 1.000	$X=\alpha^{-1} F_1(1)$	$X = \alpha^{-1} F_1(1)$	$p_1 = 0$. Again put $X = \alpha^{-1} F_1(1)$, then
0902.0431_F00839	41	■ 1.000	$\mu=\operatorname{tr}\left(\alpha^{-1} F_1(1)\right)$	$\mu = \operatorname{tr}(\alpha^{-1} F_1(1))$	where $\mu = \operatorname{tr}(\alpha^{-1} F_1(1))$. By Comparing entries of F_1 -parts, we see that $\rho_1 = 1$.
0902.0431_F00840	41	■ 1.000	F_1	F_1	where $\mu = \operatorname{tr}(\alpha^{-1} F_1(1))$. By Comparing entries of F_1 -parts, we see that $\rho_1 = 1$.
0902.0431_F00841	41	■ 1.000	$\rho_1=1$	$\rho_1 = 1$	where $\mu = \operatorname{tr}(\alpha^{-1} F_1(1))$. By Comparing entries of F_1 -parts, we see that $\rho_1 = 1$.
0902.0431_F00842	41	■ 1.000	$\rho_2=\rho_3=1$	$\rho_2 = \rho_3 = 1$	Similarly $\rho_2 = \rho_3 = 1$. Therefore we have $\alpha E = E$.

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0902.0431_F00843	41	■ 1.000	$(X, Y, Z)=\operatorname{tr}(X, Y, Z)-\frac{1}{2}\operatorname{tr}(X)(Y, Z)-\frac{1}{2}\operatorname{tr}(Y)(Z, X)-\frac{1}{2}\operatorname{tr}(Z)(X, Y)+\frac{1}{2}\operatorname{tr}(X)\operatorname{tr}(Y)\operatorname{tr}(Z)$	$(X, Y, Z) = \operatorname{tr}(X, Y, Z) - \frac{1}{2} \operatorname{tr}(X)(Y, Z) - \frac{1}{2} \operatorname{tr}(Y)(Z, X) - \frac{1}{2} \operatorname{tr}(Z)(X, Y) + \frac{1}{2} \operatorname{tr}(X) \operatorname{tr}(Y) \operatorname{tr}(Z)$	Next, using the relation $(X, Y, Z) = \operatorname{tr}(X, Y, Z) - \frac{1}{2} \operatorname{tr}(X)(Y, Z) - \frac{1}{2} \operatorname{tr}(Y)(Z, X) - \frac{1}{2} \operatorname{tr}(Z)(X, Y) + \frac{1}{2} \operatorname{tr}(X) \operatorname{tr}(Y) \operatorname{tr}(Z)$ and $(\alpha X, \alpha Y, \alpha Z) = (X, Y, Z)$, we obtain
0902.0431_F00844	41	■ 0.992	$(\alpha X, \alpha Y, \alpha Z)=(X, Y, Z)$	$(\alpha X, \alpha Y, \alpha Z) = (X, Y, Z)$	$\frac{1}{2} \operatorname{tr}(Z)(X, Y) + \frac{1}{2} \operatorname{tr}(X) \operatorname{tr}(Y) \operatorname{tr}(Z)$ and $(\alpha X, \alpha Y, \alpha Z) = (X, Y, Z)$, we obtain
0902.0431_F00845	41	■ 1.000	$\widetilde{\alpha}: \mathfrak{J} \rightarrow \mathfrak{J}$	$\widetilde{\alpha}: \mathfrak{J} \rightarrow \mathfrak{J}$	consider a mapping $\widetilde{\alpha}: \mathfrak{J} \rightarrow \mathfrak{J}$,
0902.0431_F00846	41	■ 0.941	$\widetilde{\alpha} \in F_4$	$\widetilde{\alpha} \in F_4$	Then $\widetilde{\alpha} \in F_4$. So we identify $\alpha \in G_2$ with $\widetilde{\alpha} \in F_4: G_2 \subset F_4$.
0902.0431_F00847	41	■ 0.941	$\widetilde{\alpha} \in F_4: G_2 \subset F_4$	$\widetilde{\alpha} \in F_4: G_2 \subset F_4$	Then $\widetilde{\alpha} \in F_4$. So we identify $\alpha \in G_2$ with $\widetilde{\alpha} \in F_4: G_2 \subset F_4$.
0902.0431_F00848	41	■ 1.000	$\mathfrak{e}_{6(-26)}$	$\mathfrak{e}_{6(-26)}$	study the Lie algebra $\mathfrak{e}_{6(-26)}$ of the group
0902.0431_F00849	42	■ 1.000	$\phi \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{J})$	$\phi \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{J})$	Lemma 2.2.2). Now, if $\phi \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{J})$ satisfies $((\exp t\phi)X, (\exp t\phi)X, (\exp t\phi)X) =$
0902.0431_F00850	42	■ 1.000	$((\exp t\phi)X, (\exp t\phi)X, (\exp t\phi)X) =$	$((\exp t\phi)X, (\exp t\phi)X, (\exp t\phi)X) =$	Lemma 2.2.2). Now, if $\phi \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{J})$ satisfies $((\exp t\phi)X, (\exp t\phi)X, (\exp t\phi)X) =$
0902.0431_F00851	42	■ 1.000	(X, X, X)	(X, X, X)	(X, X, X) for all $t \in \mathbf{R}$, then we have $(\phi X, X, X) = 0$ by putting $t = 0$ after differen-
0902.0431_F00852	42	■ 1.000	$t \in \mathbf{R}$	$t \in \mathbf{R}$	(X, X, X) for all $t \in \mathbf{R}$, then we have $(\phi X, X, X) = 0$ by putting $t = 0$ after differen-
0902.0431_F00853	42	■ 1.000	$(\phi X, X, X)=0$	$(\phi X, X, X) = 0$	(X, X, X) for all $t \in \mathbf{R}$, then we have $(\phi X, X, X) = 0$ by putting $t = 0$ after differen-
0902.0431_F00854	42	■ 0.998	$-{}^t\phi(X \times Y)=\phi X \times$	$-{}^t\phi(X \times Y) = \phi X \times$	tiating with respect to t . Conversely, if $\phi \in \operatorname{Hom}_{\mathbf{R}}(\mathfrak{J})$ satisfies $-{}^t\phi(X \times Y) = \phi X \times$
0902.0431_F00855	42	■ 1.000	$Y+X \times \phi Y$	$Y + X \times \phi Y$	$Y + X \times \phi Y$, then it is easy to verify that $\alpha = \exp t\phi$ satisfies ${}^t\alpha^{-1}(X \times Y) = \alpha X \times \alpha Y$.
0902.0431_F00856	42	■ 1.000	$\alpha=\exp t\phi$	$\alpha = \exp t\phi$	$Y + X \times \phi Y$, then it is easy to verify that $\alpha = \exp t\phi$ satisfies ${}^t\alpha^{-1}(X \times Y) = \alpha X \times \alpha Y$.
0902.0431_F00857	42	■ 1.000	$\{ \}^t\alpha^{-1}(X \times Y)=\alpha X \times \alpha Y$	${}^t\alpha^{-1}(X \times Y) = \alpha X \times \alpha Y$	$Y + X \times \phi Y$, then it is easy to verify that $\alpha = \exp t\phi$ satisfies $\alpha^{-1}(X \times Y) = \alpha X \times \alpha Y$.
0902.0431_F00858	42	■ 1.000	$\mathfrak{M}^{-}$	$\mathfrak{M}^{-}$	We define an $\mathbf{R}$ -vector space $\mathfrak{M}^{-}$ by
0902.0431_F00859	42	■ 1.000	$X, Y \in M(3, \mathfrak{C})$	$X, Y \in M(3, \mathfrak{C})$	For $X, Y \in M(3, \mathfrak{C})$, we define $[X, Y] \in M(3, \mathfrak{C})$ by
0902.0431_F00860	42	■ 1.000	$[X, Y] \in M(3, \mathfrak{C})$	$[X, Y] \in M(3, \mathfrak{C})$	For $X, Y \in M(3, \mathfrak{C})$, we define $[X, Y] \in M(3, \mathfrak{C})$ by
0902.0431_F00861	42	■ 0.997	$\left[\mathfrak{M}^{-}, \mathfrak{J}\right] \subset \mathfrak{J}, \quad[\mathfrak{J}, \mathfrak{J}] \subset \mathfrak{M}^{-}$	$[\mathfrak{M}^{-}, \mathfrak{J}] \subset \mathfrak{J}, \quad[\mathfrak{J}, \mathfrak{J}] \subset \mathfrak{M}^{-}$	Lemma 2.3.3. $[\mathfrak{M}^{-}, \mathfrak{J}] \subset \mathfrak{J}, \quad[\mathfrak{J}, \mathfrak{J}] \subset \mathfrak{M}^{-}$.
0902.0431_F00862	42	■ 0.998	$\left[\mathfrak{M}^{-}, \mathfrak{J}\right] \subset \mathfrak{J}$	$[\mathfrak{M}^{-}, \mathfrak{J}] \subset \mathfrak{J}$	Since $[\mathfrak{M}^{-}, \mathfrak{J}] \subset \mathfrak{J}$, any element $A \in \mathfrak{M}^{-}$ induces an $\mathbf{R}$ -linear mapping $\widetilde{A}: \mathfrak{J} \rightarrow \mathfrak{J}$
0902.0431_F00863	42	■ 0.998	$A \in \mathfrak{M}^{-}$	$A \in \mathfrak{M}^{-}$	Since $[\mathfrak{M}^{-}, \mathfrak{J}] \subset \mathfrak{J}$, any element $A \in \mathfrak{M}^{-}$ induces an $\mathbf{R}$ -linear mapping $\widetilde{A}: \mathfrak{J} \rightarrow \mathfrak{J}$
0902.0431_F00864	42	■ 0.998	$\widetilde{A}: \mathfrak{J} \rightarrow \mathfrak{J}$	$\widetilde{A}: \mathfrak{J} \rightarrow \mathfrak{J}$	Since $[\mathfrak{M}^{-}, \mathfrak{J}] \subset \mathfrak{J}$, any element $A \in \mathfrak{M}^{-}$ induces an $\mathbf{R}$ -linear mapping $\widetilde{A}: \mathfrak{J} \rightarrow \mathfrak{J}$
0902.0431_F00865	42	■ 0.995	$X=\left(x_{i j}\right), x_{i j} \in \mathfrak{C}, \bar{x}_{i j}=x_{j i}$	$X = (x_{ij}), x_{ij} \in \mathfrak{C}, \bar{x}_{ij} = x_{ji}$	Proof. Let $X = (x_{ij}), x_{ij} \in \mathfrak{C}, \bar{x}_{ij} = x_{ji}$. The (i, j) -entry a_{ij} of $[X, XX] =$
0902.0431_F00866	42	■ 0.995	(i, j)	(i, j)	Proof. Let $X = (x_{ij}), x_{ij} \in \mathfrak{C}, \bar{x}_{ij} = x_{ji}$. The (i, j) -entry a_{ij} of $[X, XX] =$
0902.0431_F00867	42	■ 0.995	$a_{i j}$	a_{ij}	Proof. Let $X = (x_{ij}), x_{ij} \in \mathfrak{C}, \bar{x}_{ij} = x_{ji}$. The (i, j) -entry a_{ij} of $[X, XX] =$
0902.0431_F00868	42	■ 0.995	$[X, X X]=$	$[X, XX] =$	Proof. Let $X = (x_{ij}), x_{ij} \in \mathfrak{C}, \bar{x}_{ij} = x_{ji}$. The (i, j) -entry a_{ij} of $[X, XX] =$
0902.0431_F00869	42	■ 1.000	$X(X X)-(X X) X$	$X(XX) - (XX)X$	$X(XX) - (XX)X$ is given by
0902.0431_F00870	43	■ 0.996	$x_{i i}$	x_{ii}	Since x_{ii} is real, if the bracket $\{ \ , \ , \ }$ contains x_{ii} , then $\{ \ , \ , \ }$ is 0. If $i \neq j$, then
0902.0431_F00871	43	■ 0.996	$i \neq j$	$i \neq j$	Since x_{ii} is real, if the bracket $\{ \ , \ , \ }$ contains x_{ii} , then $\{ \ , \ , \ }$ is 0. If $i \neq j$, then
0902.0431_F00872	43	■ 1.000	$a(x a)-(a x) a, a(\bar{a} x)-(a \bar{a}) x$	$a(xa) - (ax)a, a(\bar{a}x) - (a\bar{a})x$	a_{ij} is a sum of $a(xa) - (ax)a, a(\bar{a}x) - (a\bar{a})x$ and so on, so $a_{ij} = 0$. If $i = j$, then
0902.0431_F00873	43	■ 1.000	$a_{i j}=0$	$a_{ij} = 0$	a_{ij} is a sum of $a(xa) - (ax)a, a(\bar{a}x) - (a\bar{a})x$ and so on, so $a_{ij} = 0$. If $i = j$, then
0902.0431_F00874	43	■ 1.000	$i=j$	$i = j$	a_{ij} is a sum of $a(xa) - (ax)a, a(\bar{a}x) - (a\bar{a})x$ and so on, so $a_{ij} = 0$. If $i = j$, then
0902.0431_F00875	43	■ 1.000	$a_{11}=a_{22}=a_{33}(=a)$	$a_{11} = a_{22} = a_{33}(=a)$	however they are equal, that is, $a_{11} = a_{22} = a_{33}(=a)$. Hence we have $[X, XX] = aE$.
0902.0431_F00876	43	■ 1.000	$[X, X X]=a E$	$[X, XX] = aE$	however they are equal, that is, $a_{11} = a_{22} = a_{33}(=a)$. Hence we have $[X, XX] = aE$.
0902.0431_F00877	43	■ 1.000	$X, X X \in \mathfrak{J}$	$X, XX \in \mathfrak{J}$	Since $[X, XX] \in \mathfrak{J}$, we have $[X, XX] \in \mathfrak{M}^{-}$ (Lemma 2.3.3). Therefore, $(aE)^* = -aE$
0902.0431_F00878	43	■ 1.000	$[X, X X] \in \mathfrak{M}^{-}$	$[X, XX] \in \mathfrak{M}^{-}$	Since $X, XX \in \mathfrak{J}$, we have $[X, XX] \in \mathfrak{M}^{-}$ (Lemma 2.3.3). Therefore, $(aE)^* = -aE$
0902.0431_F00879	43	■ 1.000	$(a E)^{*}=-a E$	$(aE)^* = -aE$	Since $X, XX \in \mathfrak{J}$, we have $[X, XX] \in \mathfrak{M}^{-}$ (Lemma 2.3.3). Therefore, $(aE)^* = -aE$

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0902.0431_F00881	43	■1.000	$\backslash\mathrm{bar}\{a\} \ E=-a \ E$	$\bar{a}E = -aE$	and so $\bar{a}E = -aE$, which imply that $\bar{a} = -a$.
0902.0431_F00882	43	■1.000	$\backslash\mathrm{bar}\{a\}=-a$	$\bar{a} = -a$	and so $\bar{a}E = -aE$, which imply that $\bar{a} = -a$.
0902.0431_F00883	43	■1.000	$M(3, \ \backslash\mathrm{mathfrak{C}})$	$M(3, \mathfrak{C})$	To prove the following Proposition 2.3.6, in $M(3, \mathfrak{C})$, we define a real valued sym-
0902.0431_F00884	43	■1.000	$X=\backslash\mathrm{left}(x_{\{i \ j\}}\backslash\mathrm{right}), \ Y=\backslash\mathrm{left}(y_{\{i \ j\}}\backslash\mathrm{right}), \ Z=\backslash\mathrm{left}(z_{\{i \ j\}}\backslash\mathrm{right})$	$X = (x_{ij}), Y = (y_{ij}), Z = (z_{ij})$	Proof. Let $X = (x_{ij}), Y = (y_{ij}), Z = (z_{ij})$. Then we have
0902.0431_F00885	43	■0.997	$A \ \backslash\mathrm{in} \ \backslash\mathrm{mathfrak{M}}^{\{-\}}, \ \backslash\mathrm{operatorname{tr}}\{A\}=0$	$A \in \mathfrak{M}^-, \mathrm{tr}(A) = 0$	Proposition 2.3.6. For $A \in \mathfrak{M}^-, \mathrm{tr}(A) = 0$, we have $\tilde{A} \in \mathfrak{f}_4$.
0902.0431_F00886	43	■0.997	$\backslash\mathrm{widetilde}\{A\} \ \backslash\mathrm{in} \ \backslash\mathrm{mathfrak{f}}_{\{4\}}$	$\tilde{A} \in \mathfrak{f}_4$	Proposition 2.3.6. For $A \in \mathfrak{M}^-, \mathrm{tr}(A) = 0$, we have $\tilde{A} \in \mathfrak{f}_4$.
0902.0431_F00887	43	■1.000	$= (A \ X, \ Y) - (X \ A, \ Y) + (X, \ A \ Y) - (X, \ Y \ A) = 0$	$= (AX, Y) - (XA, Y) + (X, AY) - (X, YA) = 0$	Now, the left side of the first formula $= (AX, Y) - (XA, Y) + (X, AY) - (X, YA) = 0$
0902.0431_F00888	43	■1.000	$[X, \ X \ X]=a \ E, \ a \ \backslash\mathrm{in} \ \backslash\mathrm{mathfrak{C}}_{\{0\}}$	$[X, XX] = aE, a \in \mathfrak{C}_0$	Certainly, if $[X, XX] = aE, a \in \mathfrak{C}_0$ (Lemma 2.3.4), then
0902.0431_F00889	44	■1.000	$\backslash\mathrm{lambda} \ X + \backslash\mathrm{mu} \ X + \backslash\mathrm{nu} \ Z$	$\lambda X + \mu X + \nu Z$	Now, putting $\lambda X + \mu X + \nu Z$ in the place of X , and comparing the coefficient of $\lambda\mu\nu$,
0902.0431_F00890	44	■1.000	$\backslash\mathrm{widetilde}\{A\}_{\{i\}}(a) \ \backslash\mathrm{in} \ \backslash\mathrm{mathfrak{f}}_{\{4\}}$	$\tilde{A}_i(a) \in \mathfrak{f}_4$	Then $\tilde{A}_i(a) \in \mathfrak{f}_4$ (Proposition 2.3.6) and the operation of $\tilde{A}_i(a)$ on $\mathfrak{J}$ is given by
0902.0431_F00891	44	■1.000	$\backslash\mathrm{widetilde}\{A\}_{\{i\}}(a)$	$\tilde{A}_i(a)$	Then $\tilde{A}_i(a) \in \mathfrak{f}_4$ (Proposition 2.3.6) and the operation of $\tilde{A}_i(a)$ on $\mathfrak{J}$ is given by
0902.0431_F00892	44	■1.000	$\backslash\mathrm{mathfrak{D}}_{\{4\}}=\backslash\mathrm{mathfrak{s o}}(8)$	$\mathfrak{D}_4 = \mathfrak{so}(8)$	is isomorphic to the Lie algebra $\mathfrak{D}_4 = \mathfrak{so}(8)$:
0902.0431_F00893	45	■1.000	$\backslash\mathrm{varphi}_{\{*\}}: \ \backslash\mathrm{mathfrak{D}}_{\{4\}} \ \backslash\mathrm{rightarrow} \ \backslash\mathrm{mathfrak{d}}_{\{4\}}$	$\varphi_* : \mathfrak{D}_4 \rightarrow \mathfrak{d}_4$	Proof. We define a mapping $\varphi_* : \mathfrak{D}_4 \rightarrow \mathfrak{d}_4$ by $\varphi_*(D_1) = \delta$. We first prove that
0902.0431_F00894	45	■1.000	$\backslash\mathrm{varphi}_{\{*\}}\backslash\mathrm{left}(D_{\{1\}}\backslash\mathrm{right})=\backslash\mathrm{delta}$	$\varphi_*(D_1) = \delta$	Proof. We define a mapping $\varphi_* : \mathfrak{D}_4 \rightarrow \mathfrak{d}_4$ by $\varphi_*(D_1) = \delta$. We first prove that
0902.0431_F00895	45	■1.000	$\backslash\mathrm{delta} \ \backslash\mathrm{in} \ \backslash\mathrm{mathfrak{d}}_{\{4\}} \ \backslash\mathrm{subset} \ \backslash\mathrm{mathfrak{f}}_{\{4\}}$	$\delta \in \mathfrak{d}_4 \subset \mathfrak{f}_4$	$\delta \in \mathfrak{d}_4 \subset \mathfrak{f}_4$. Indeed,
0902.0431_F00896	45	■1.000	$\backslash\mathrm{delta} \ \backslash\mathrm{in} \ \backslash\mathrm{mathfrak{f}}_{\{4\}} \ . \ \backslash\mathrm{delta} \ E_{\{i\}}=0, \ i=1,2,3$	$\delta \in \mathfrak{f}_4. \delta E_i = 0, i = 1, 2, 3$	Hence $\delta \in \mathfrak{f}_4$. $\delta E_i = 0, i = 1, 2, 3$ are evident, so that $\delta \in \mathfrak{d}_4$. Clearly φ_* is injective.
0902.0431_F00897	45	■1.000	$\backslash\mathrm{delta} \ \backslash\mathrm{in} \ \backslash\mathrm{mathfrak{d}}_{\{4\}}$	$\delta \in \mathfrak{d}_4$	Hence $\delta \in \mathfrak{f}_4$. $\delta E_i = 0, i = 1, 2, 3$ are evident, so that $\delta \in \mathfrak{d}_4$. Clearly φ_* is injective.
0902.0431_F00898	45	■1.000	$\backslash\mathrm{delta} \ X \ \backslash\mathrm{in} \ \backslash\mathrm{mathfrak{J}}_{\{i\}}$	$\delta X \in \mathfrak{J}_i$	Since $\delta X \in \mathfrak{J}_i$ for $X \in \mathfrak{J}_i$, δ induces <i>R</i> -linear mappings $\delta : \mathfrak{J}_i \rightarrow \mathfrak{J}_i$ and $D_i : \mathfrak{C} \rightarrow \mathfrak{C}$
0902.0431_F00899	45	■1.000	$X \ \backslash\mathrm{in} \ \backslash\mathrm{mathfrak{J}}_{\{i\}}, \ \backslash\mathrm{delta}$	$X \in \mathfrak{J}_i, \delta$	Since $\delta X \in \mathfrak{J}_i$ for $X \in \mathfrak{J}_i$, δ induces <i>R</i> -linear mappings $\delta : \mathfrak{J}_i \rightarrow \mathfrak{J}_i$ and $D_i : \mathfrak{C} \rightarrow \mathfrak{C}$
0902.0431_F00900	45	■1.000	$\backslash\mathrm{delta}: \ \backslash\mathrm{mathfrak{J}}_{\{i\}} \ \backslash\mathrm{rightarrow} \ \backslash\mathrm{mathfrak{J}}_{\{i\}}$	$\delta : \mathfrak{J}_i \rightarrow \mathfrak{J}_i$	Since $\delta X \in \mathfrak{J}_i$ for $X \in \mathfrak{J}_i$, δ induces <i>R</i> -linear mappings $\delta : \mathfrak{J}_i \rightarrow \mathfrak{J}_i$ and $D_i : \mathfrak{C} \rightarrow \mathfrak{C}$
0902.0431_F00901	45	■1.000	$D_{\{i\}}: \ \backslash\mathrm{mathfrak{C}} \ \backslash\mathrm{rightarrow} \ \backslash\mathrm{mathfrak{C}}$	$D_i : \mathfrak{C} \rightarrow \mathfrak{C}$	Since $\delta X \in \mathfrak{J}_i$ for $X \in \mathfrak{J}_i$, δ induces <i>R</i> -linear mappings $\delta : \mathfrak{J}_i \rightarrow \mathfrak{J}_i$ and $\backslash\mathrm{D}_i: \mathfrak{C} \rightarrow \mathfrak{C}$
0902.0431_F00902	45	■1.000	$\backslash\mathrm{delta}$	δ	for $i = 1, 2, 3$. Applying δ on $F_i(x) \circ F_i(y) = (x, y)(E_{i+1} + E_{i+2})$, we have $F_i(D_i x) \circ$
0902.0431_F00903	45	■1.000	$F_{\{i\}}(x) \ \backslash\mathrm{circ} \ F_{\{i\}}(y)=(x, \ y)\backslash\mathrm{left}(E_{\{i+1\}}+E_{\{i+2\}} \ \backslash\mathrm{right})$	$F_i(x) \circ F_i(y) = (x, y)(E_{i+1} + E_{i+2})$	for $i = 1, 2, 3$. Applying δ on $\backslash\mathrm{F}_i(x) \circ F_i(y) = (x, y)(E_{i+1} + E_{i+2})$, we have $F_i(D_i x) \circ$
0902.0431_F00904	45	■1.000	$F_{\{i\}}\backslash\mathrm{left}(D_{\{i\}} \ x\backslash\mathrm{right}) \ \backslash\mathrm{circ}$	$F_i(D_i x) \circ$	for $i = 1, 2, 3$. Applying δ on $F_i(x) \circ F_i(y) = (x, y)(E_{i+1} + E_{i+2})$, we have $F_i(D_i x) \circ$
0902.0431_F00905	45	■1.000	$F_{\{i\}}(y)+F_{\{i\}}(x) \ \backslash\mathrm{circ} \ F_{\{i\}}\backslash\mathrm{left}(D_{\{i\}} \ y\backslash\mathrm{right})=0$	$F_i(y) + F_i(x) \circ F_i(D_i y) = 0$	$\backslash\mathrm{F}_i(y) + F_i(x) \circ F_i(D_i y) = 0$, and hence we have
0902.0431_F00906	45	■1.000	$D_{\{i\}} \ \backslash\mathrm{in} \ \backslash\mathrm{mathfrak{D}}_{\{4\}}, \ i=1,2,3$	$D_i \in \mathfrak{D}_4, i = 1, 2, 3$	Hence $\backslash\mathrm{D}_i \in \mathfrak{D}_4, i = 1, 2, 3$. Moreover, by applying δ on $F_1(x) \circ F_2(y) = \frac{1}{2}F_3(\overline{xy})$, we
0902.0431_F00907	45	■1.000	$F_{\{1\}}(x) \ \backslash\mathrm{circ} \ F_{\{2\}}(y)=\mathrm{frac}\{1\}\{2\} \ F_{\{3\}}(\backslash\mathrm{overline}\{x \ y\})$	$F_1(x) \circ F_2(y) = \frac{1}{2}F_3(\overline{xy})$	Hence $D_i \in \mathfrak{D}_4, i = 1, 2, 3$. Moreover, by applying δ on $\backslash\mathrm{F}_1(x) \circ F_2(y) = \frac{1}{2}F_3(\overline{xy})$, we
0902.0431_F00908	45	■1.000	$\backslash\mathrm{delta} \ \backslash\mathrm{in} \ \backslash\mathrm{mathfrak{f}}_{\{4\}}$	$\delta \in \mathfrak{f}_4$	Theorem 2.3.8. Any element $\backslash\mathrm{D} \in \mathfrak{f}_4$ is uniquely expressed by
0902.0431_F00909	45	■0.988	$\backslash\mathrm{operatorname{diag}} \ A=0$	$\mathrm{diag} \ A = 0$	where $\mathrm{diag} A = 0$ means that all diagonal elements a_{ii} of A are 0. In particular, the
0902.0431_F00910	45	■0.988	$a_{\{i \ i\}}$	a_{ii}	where $\mathrm{diag} A = 0$ means that all diagonal elements a_{ii} of A are 0. In particular, the
0902.0431_F00911	45	■1.000	$E_{\{i\}} \ \backslash\mathrm{circ} \ E_{\{i\}}=E_{\{i\}}$	$E_i \circ E_i = E_i$	Proof. Applying $\delta \in \mathfrak{f}_4$ on $E_i \circ E_i = E_i$ and $E_i \circ E_j = 0, i \neq j$, we have
0902.0431_F00912	45	■1.000	$E_{\{i\}} \ \backslash\mathrm{circ} \ E_{\{j\}}=0, \ i \ \backslash\mathrm{neq} \ j$	$E_i \circ E_j = 0, i \neq j$	Proof. Applying $\delta \in \mathfrak{f}_4$ on $E_i \circ E_i = E_i$ and $\backslash\mathrm{E}_i \circ E_i = 0, i \neq j$, we have
0902.0431_F00913	46	■1.000	$\backslash\mathrm{delta} \ E_{\{i\}}$	δE_i	From these relations, we see that each δE_i is of the form
0902.0431_F00914	46	■1.000	$a_{\{i\}} \ \backslash\mathrm{in} \ \backslash\mathrm{mathfrak{C}}$	$a_i \in \mathfrak{C}$	$a_i \in \mathfrak{C}$. If we construct a matrix $A = 2 \begin{pmatrix} 0 & a_3 & -\bar{a}_2 \\ -\bar{a}_3 & 0 & a_1 \\ a_2 & -\bar{a}_1 & 0 \end{pmatrix}$ using these elements a_i ,
0902.0431_F00915	46	■1.000	$A=2\backslash\mathrm{left}(\backslash\mathrm{begin}\{\mathrm{array}\}\{\mathrm{ccc}\}0 \ \& \ a_{\{3\}} \ \& \ -\backslash\mathrm{bar}\{a\}_{\{2\}} \ \backslash\backslash \ -\backslash\mathrm{bar}\{a\}_{\{3\}} \ \& \ 0 \ \& \ a_{\{1\}} \ \backslash\backslash \ a_{\{2\}} \ \& \ -\backslash\mathrm{bar}\{a\}_{\{1\}} \ \& \ 0 \ \backslash\mathrm{end}\{\mathrm{array}\}\backslash\mathrm{right})$	$A = 2 \begin{pmatrix} 0 & a_3 & -\bar{a}_2 \\ -\bar{a}_3 & 0 & a_1 \\ a_2 & -\bar{a}_1 & 0 \end{pmatrix}$	$a_i \in \mathfrak{C}$. If we construct a matrix $A = 2 \begin{pmatrix} 0 & a_3 & -\bar{a}_2 \\ -\bar{a}_3 & 0 & a_1 \\ a_2 & -\bar{a}_1 & 0 \end{pmatrix}$ using these elements a_i ,
0902.0431_F00916	46	■1.000	$a_{\{i\}}$	a_i	$a_i \in \mathfrak{C}$. If we construct a matrix $A = 2 \begin{pmatrix} 0 & a_3 & -\bar{a}_2 \\ -\bar{a}_3 & 0 & a_1 \\ a_2 & -\bar{a}_1 & 0 \end{pmatrix}$ using these elements a_i ,
0902.0431_F00917	46	■1.000	$D=\backslash\mathrm{delta}-\backslash\mathrm{widetilde}\{A\}$	$D = \delta - \tilde{A}$	If we put $D = \delta - \tilde{A}$, then $DE_i = 0, i = 1, 2, 3$, hence $D \in \mathfrak{d}_4$. Thus, δ can be
0902.0431_F00918	46	■1.000	$D \ E_{\{i\}}=0, \ i=1,2,3$	$DE_i = 0, i = 1, 2, 3$	If we put $D = \delta - \tilde{A}$, then $DE_i = 0, i = 1, 2, 3$, hence $D \in \mathfrak{d}_4$. Thus, δ can be

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0902.0431_F00919	46	■ 1.000	$D \in \mathfrak{d}_4$	$D \in \mathfrak{d}_4$	If we put $D = \delta - \tilde{A}$, then $DE_i = 0$, $i = 1, 2, 3$, hence $D \in \mathfrak{d}_4$. Thus, δ can be
0902.0431_F00920	46	■ 0.546	$\delta = D + \tilde{A}$	$\delta = D + \tilde{A}$	expressed by $\delta = D + \tilde{A}$, where $D \in \mathfrak{d}_4$, $A \in \mathfrak{M}^-$, $\text{diag} A = 0$. To show the uniqueness
0902.0431_F00921	46	■ 0.546	$D \in \mathfrak{d}_4, A \in \mathfrak{M}^-, \text{diag} A = 0$	$D \in \mathfrak{d}_4, A \in \mathfrak{M}^-, \text{diag} A = 0$	expressed by $\delta = D + \tilde{A}$, where $D \in \mathfrak{d}_4, A \in \mathfrak{M}^-, \text{diag} A = 0$. To show the uniqueness
0902.0431_F00922	46	■ 1.000	$E_{\{i\}}$	E_i	Certainly, if we apply it on E_i , then $\tilde{A}E_i = 0$, $i = 1, 2, 3$, hence $A = 0$ and so $D = 0$.
0902.0431_F00923	46	■ 1.000	$\tilde{A}E_{\{i\}} = 0, i = 1, 2, 3$	$\tilde{A}E_i = 0, i = 1, 2, 3$	Certainly, if we apply it on E_i , then $\tilde{A}E_i = 0, i = 1, 2, 3$, hence $A = 0$ and so $D = 0$.
0902.0431_F00924	46	■ 1.000	$A = 0$	$A = 0$	Certainly, if we apply it on E_i , then $\tilde{A}E_i = 0, i = 1, 2, 3$, hence $A = 0$ and so $D = 0$.
0902.0431_F00925	46	■ 0.999	$\dim \mathfrak{f}_4 = 28 + 24 = 52$	$\dim \mathfrak{f}_4 = 28 + 24 = 52$	Finally, we have $\dim \mathfrak{f}_4 = 28 + 24 = 52$ from the expression above. Thus the theorem
0902.0431_F00926	46	■ 1.000	$\mathfrak{J}^C = \{X_1 + iX_2 \mid X_1, X_2 \in \mathfrak{J}\}$	$\mathfrak{J}^C = \{X_1 + iX_2 \mid X_1, X_2 \in \mathfrak{J}\}$	Let $\mathfrak{J}^C = \{X_1 + iX_2 \mid X_1, X_2 \in \mathfrak{J}\}$ be the complexification of the Jordan algebra
0902.0431_F00927	46	■ 1.000	$\mathfrak{J}^C$	$\mathfrak{J}^C$	$\mathfrak{J}$. In the same manner as in $\mathfrak{J}$, in $\mathfrak{J}^C$ we can also define the multiplications $X \circ Y$,
0902.0431_F00928	46	■ 0.991	$\text{tr}(X, Y, Z), (X, Y, Z)$	$\text{tr}(X, Y, Z), (X, Y, Z)$	$X \times Y$, the inner product (X, Y) , the trilinear forms $\text{tr}(X, Y, Z)$, (X, Y, Z) and the
0902.0431_F00929	46	■ 0.996	$\mathfrak{J} \cdot \mathfrak{J}^C$	$\mathfrak{J} \cdot \mathfrak{J}^C$	determinant $\det X$. They have the same properties as those of $\mathfrak{J}$. $\mathfrak{J}^C$ is called the
0902.0431_F00930	46	■ 1.000	$\mathfrak{f}_4$	$\mathfrak{f}_4$	These are the complexification of the Lie algebras $\mathfrak{e}_{6(-26)}$ and $\mathfrak{f}_4$ respectively. Then the
0902.0431_F00931	47	■ 0.982	$A \in \mathfrak{J}^C$	$A \in \mathfrak{J}^C$	For $A \in \mathfrak{J}^C$, we define a C -linear mapping $\tilde{A} : \mathfrak{J}^C \rightarrow \mathfrak{J}^C$ by
0902.0431_F00932	47	■ 0.982	$\tilde{A} : \mathfrak{J}^C \rightarrow \mathfrak{J}^C$	$\tilde{A} : \mathfrak{J}^C \rightarrow \mathfrak{J}^C$	For $A \in \mathfrak{J}^C$, we define a C -linear mapping $\tilde{A} : \mathfrak{J}^C \rightarrow \mathfrak{J}^C$ by
0902.0431_F00933	47	■ 0.996	$A \in \mathfrak{J}^C, \text{tr}(A) = 0$	$A \in \mathfrak{J}^C, \text{tr}(A) = 0$	Proposition 2.4.1. (1) For $A \in \mathfrak{J}^C, \text{tr}(A) = 0$, we have $\tilde{A} \in \mathfrak{e}_6^C$.
0902.0431_F00934	47	■ 0.996	$\tilde{A} \in \mathfrak{e}_6^C$	$\tilde{A} \in \mathfrak{e}_6^C$	Proposition 2.4.1. (1) For $A \in \mathfrak{J}^C, \text{tr}(A) = 0$, we have $\tilde{A} \in \mathfrak{e}_6^C$.
0902.0431_F00935	47	■ 1.000	$A, B \in \mathfrak{J}^C$	$A, B \in \mathfrak{J}^C$	(2) For $A, B \in \mathfrak{J}^C$, we have $[\tilde{A}, \tilde{B}] \in \mathfrak{f}_4^C$.
0902.0431_F00936	47	■ 1.000	$[\tilde{A}, \tilde{B}] \in \mathfrak{f}_4^C$	$[\tilde{A}, \tilde{B}] \in \mathfrak{f}_4^C$	(2) For $A, B \in \mathfrak{J}^C$, we have $[\tilde{A}, \tilde{B}] \in \mathfrak{f}_4^C$.
0902.0431_F00937	47	■ 1.000	$(\tilde{A}X, X, X) = (A \circ X, X \times X) = (A, X \circ (X \times X))$	$(\tilde{A}X, X, X) = (A \circ X, X \times X) = (A, X \circ (X \times X))$	Proof. (1) $(\tilde{A}X, X, X) = (A \circ X, X \times X) = (A, X \circ (X \times X))$
0902.0431_F00938	47	■ 1.000	$\mathfrak{d}_4^C$	$\mathfrak{d}_4^C$	successive actions of $\mathfrak{d}_4^C$.
0902.0431_F00939	47	■ 0.997	$\mathfrak{d}_4^C \mathfrak{e}^C = \left\{ \sum_i D_i a_i \mid D_i \in \mathfrak{d}_4^C, a_i \in \mathfrak{e}^C \right\} = \mathfrak{e}^C$	$\mathfrak{d}_4^C \mathfrak{e}^C = \left\{ \sum_i D_i a_i \mid D_i \in \mathfrak{d}_4^C, a_i \in \mathfrak{e}^C \right\} = \mathfrak{e}^C$	(3) $\mathfrak{d}_4^C \mathfrak{e}^C = \left\{ \sum_i D_i a_i \mid D_i \in \mathfrak{d}_4^C, a_i \in \mathfrak{e}^C \right\} = \mathfrak{e}^C$.
0902.0431_F00940	47	■ 1.000	$0 \neq x = \sum_{i=0}^7 x_i e_i, x_i \in C$	$0 \neq x = \sum_{i=0}^7 x_i e_i, x_i \in C$	Proof. (1) Let $0 \neq x = \sum_{i=0}^7 x_i e_i, x_i \in C$. Suppose $x_i \neq 0$ (moreover we may
0902.0431_F00941	47	■ 1.000	$x_i \neq 0$	$x_i \neq 0$	Proof. (1) Let $0 \neq x = \sum_{i=0}^7 x_i e_i, x_i \in C$. Suppose $x_i \neq 0$ (moreover we may
0902.0431_F00942	47	■ 1.000	$x_0 \neq 0$	$x_0 \neq 0$	The case $x_0 \neq 0$ is reduced to the above by the action of G_{i0} .
0902.0431_F00943	47	■ 1.000	G_{i0}	G_{i0}	The case $x_0 \neq 0$ is reduced to the above by the action of G_{i0} .
0902.0431_F00944	47	■ 1.000	$W \neq \{0\}$	$W \neq \{0\}$	(2) Let $W \neq \{0\}$ be a $\mathfrak{d}_4^C$ -invariant C -submodule of $\mathfrak{e}^C$. If $e_0 = 1 \in W$, then
0902.0431_F00945	47	■ 1.000	$e_0 = 1 \in W$	$e_0 = 1 \in W$	(2) Let $W \neq \{0\}$ be a $\mathfrak{d}_4^C$ -invariant C -submodule of $\mathfrak{e}^C$. If $e_0 = 1 \in W$, then
0902.0431_F00946	47	■ 1.000	$G_{i0} e_0 = e_i$	$G_{i0} e_0 = e_i$	$G_{i0} e_0 = e_i$. Therefore $W = \mathfrak{e}^C$. Now, since a non-zero element x of W can be
0902.0431_F00947	47	■ 1.000	$W = \mathfrak{e}^C$	$W = \mathfrak{e}^C$	$G_{i0} e_0 = e_i$. Therefore $W = \mathfrak{e}^C$. Now, since a non-zero element x of W can be
0902.0431_F00948	48	■ 1.000	$\mathfrak{d}_4^C \mathfrak{e}^C$	$\mathfrak{d}_4^C \mathfrak{e}^C$	(3) Since $\mathfrak{d}_4^C \mathfrak{e}^C$ is a $\mathfrak{d}_4^C$ -invariant C -submodule of $\mathfrak{e}^C$, from the irreducibility of
0902.0431_F00949	48	■ 0.985	$\mathfrak{d}_4^C \mathfrak{e}^C = \mathfrak{e}^C$	$\mathfrak{d}_4^C \mathfrak{e}^C = \mathfrak{e}^C$	$\mathfrak{e}^C$ of (2), we have $\mathfrak{d}_4^C \mathfrak{e}^C = \mathfrak{e}^C$.
0902.0431_F00950	48	■ 1.000	$\delta \in \mathfrak{f}_4^C$	$\delta \in \mathfrak{f}_4^C$	Recall that any element $\delta \in \mathfrak{f}_4^C$ is uniquely expressed by
0902.0431_F00951	48	■ 1.000	$A_i(a_i) \in (\mathfrak{M}^-)^C$	$A_i(a_i) \in (\mathfrak{M}^-)^C$	where $A_i(a_i) \in (\mathfrak{M}^-)^C$ (Theorem 2.3.8).
0902.0431_F00952	48	■ 1.000	$\tilde{\mathfrak{A}}_i^C = \{\tilde{A}_i(a) \mid a \in \mathfrak{e}^C\}$	$\tilde{\mathfrak{A}}_i^C = \{\tilde{A}_i(a) \mid a \in \mathfrak{e}^C\}$	Proof. Let denote $\tilde{\mathfrak{A}}_i^C = \{\tilde{A}_i(a) \mid a \in \mathfrak{e}^C\}$ and $\tilde{\mathfrak{A}}^C = \tilde{\mathfrak{A}}_1^C \oplus \tilde{\mathfrak{A}}_2^C \oplus \tilde{\mathfrak{A}}_3^C$, then

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0902.0431_F00953	48	■ 1.000	$\widetilde{\mathfrak{A}}^C = \widetilde{\mathfrak{A}}_1^C \oplus \widetilde{\mathfrak{A}}_2^C \oplus \widetilde{\mathfrak{A}}_3^C$	$\widetilde{\mathfrak{A}}^C = \widetilde{\mathfrak{A}}_1^C \oplus \widetilde{\mathfrak{A}}_2^C \oplus \widetilde{\mathfrak{A}}_3^C$	Proof. Let denote $\widetilde{\mathfrak{A}}_i^C = \{\tilde{A}_i(a) \mid a \in \mathfrak{C}^C\}$ and $\widetilde{\mathfrak{A}}^C = \widetilde{\mathfrak{A}}_1^C \oplus \widetilde{\mathfrak{A}}_2^C \oplus \widetilde{\mathfrak{A}}_3^C$, then
0902.0431_F00954	48	■ 0.998	$p: \mathfrak{f}_4^C \rightarrow \mathfrak{d}_4^C$	$p: \mathfrak{f}_4^C \rightarrow \mathfrak{d}_4^C$	Let $p: \mathfrak{f}_4^C \rightarrow \mathfrak{d}_4^C$ and $q: \mathfrak{f}_4^C \rightarrow \widetilde{\mathfrak{A}}^C$ be projections of $\mathfrak{f}_4^C = \mathfrak{d}_4^C \oplus \widetilde{\mathfrak{A}}^C$. Now, let $\mathfrak{a}$ be
0902.0431_F00955	48	■ 0.998	$q: \mathfrak{f}_4^C \rightarrow \widetilde{\mathfrak{A}}^C$	$q: \mathfrak{f}_4^C \rightarrow \widetilde{\mathfrak{A}}^C$	Let $p: \mathfrak{f}_4^C \rightarrow \mathfrak{d}_4^C$ and $q: \mathfrak{f}_4^C \rightarrow \widetilde{\mathfrak{A}}^C$ be projections of $\mathfrak{f}_4^C = \mathfrak{d}_4^C \oplus \widetilde{\mathfrak{A}}^C$. Now, let $\mathfrak{a}$ be
0902.0431_F00956	48	■ 0.998	$\mathfrak{f}_4^C = \mathfrak{d}_4^C \oplus \widetilde{\mathfrak{A}}^C$	$\mathfrak{f}_4^C = \mathfrak{d}_4^C \oplus \widetilde{\mathfrak{A}}^C$	Let $p: \mathfrak{f}_4^C \rightarrow \mathfrak{d}_4^C$ and $q: \mathfrak{f}_4^C \rightarrow \widetilde{\mathfrak{A}}^C$ be projections of $\mathfrak{f}_4^C = \mathfrak{d}_4^C \oplus \widetilde{\mathfrak{A}}^C$. Now, let $\mathfrak{a}$ be
0902.0431_F00957	48	■ 1.000	$a_i \in \mathfrak{C}^C, i=1,2,3$	$a_i \in \mathfrak{C}^C, i=1,2,3$	exist $a_i \in \mathfrak{C}^C, i=1,2,3$, such that $D + \sum_{i=1}^3 \tilde{A}_i(a_i) \in \mathfrak{a}$. For any $D' \in \mathfrak{d}_4^C$, we have
0902.0431_F00958	48	■ 1.000	$D + \sum_{i=1}^3 \tilde{A}_i(a_i) \in \mathfrak{a}$	$D + \sum_{i=1}^3 \tilde{A}_i(a_i) \in \mathfrak{a}$	exist $a_i \in \mathfrak{C}^C, i=1,2,3$, such that $D + \sum_{i=1}^3 \tilde{A}_i(a_i) \in \mathfrak{a}$. For any $D' \in \mathfrak{d}_4^C$, we have
0902.0431_F00959	48	■ 1.000	$D' \in \mathfrak{d}_4^C$	$D' \in \mathfrak{d}_4^C$	exist $a_i \in \mathfrak{C}^C, i=1,2,3$, such that $D + \sum_{i=1}^3 \tilde{A}_i(a_i) \in \mathfrak{a}$. For any $D' \in \mathfrak{d}_4^C$, we have
0902.0431_F00960	48	■ 1.000	$\mathfrak{d}_4^C \cap \mathfrak{a} \neq \{0\}$	$\mathfrak{d}_4^C \cap \mathfrak{a} \neq \{0\}$	We show that either $\mathfrak{d}_4^C \cap \mathfrak{a} \neq \{0\}$ or $\widetilde{\mathfrak{A}}^C \cap \mathfrak{a} \neq \{0\}$. Assume that $\mathfrak{d}_4^C \cap \mathfrak{a} = \{0\}$
0902.0431_F00961	48	■ 1.000	$\widetilde{\mathfrak{A}}^C \cap \mathfrak{a} \neq \{0\}$	$\widetilde{\mathfrak{A}}^C \cap \mathfrak{a} \neq \{0\}$	We show that either $\mathfrak{d}_4^C \cap \mathfrak{a} \neq \{0\}$ or $\widetilde{\mathfrak{A}}^C \cap \mathfrak{a} \neq \{0\}$. Assume that $\mathfrak{d}_4^C \cap \mathfrak{a} = \{0\}$
0902.0431_F00962	48	■ 1.000	$\mathfrak{d}_4^C \cap \mathfrak{a} = \{0\}$	$\mathfrak{d}_4^C \cap \mathfrak{a} = \{0\}$	We show that either $\mathfrak{d}_4^C \cap \mathfrak{a} \neq \{0\}$ or $\widetilde{\mathfrak{A}}^C \cap \mathfrak{a} \neq \{0\}$. Assume that $\mathfrak{d}_4^C \cap \mathfrak{a} = \{0\}$
0902.0431_F00963	48	■ 1.000	$\widetilde{\mathfrak{A}}^C \cap \mathfrak{a} = \{0\}$	$\widetilde{\mathfrak{A}}^C \cap \mathfrak{a} = \{0\}$	and $\widetilde{\mathfrak{A}}^C \cap \mathfrak{a} = \{0\}$. The mapping $p _{\mathfrak{a}}: \mathfrak{a} \rightarrow \mathfrak{d}_4^C$ is injective because $\mathfrak{A}^C \cap \mathfrak{a} = \{0\}$.
0902.0431_F00964	48	■ 1.000	$p _{\mathfrak{a}}: \mathfrak{a} \rightarrow \mathfrak{d}_4^C$	$p _{\mathfrak{a}}: \mathfrak{a} \rightarrow \mathfrak{d}_4^C$	and $\widetilde{\mathfrak{A}}^C \cap \mathfrak{a} = \{0\}$. The mapping $p _{\mathfrak{a}}: \mathfrak{a} \rightarrow \mathfrak{d}_4^C$ is injective because $\mathfrak{A}^C \cap \mathfrak{a} = \{0\}$.
0902.0431_F00965	48	■ 1.000	$\mathfrak{A}^C \cap \mathfrak{a} = \{0\}$	$\mathfrak{A}^C \cap \mathfrak{a} = \{0\}$	and $\widetilde{\mathfrak{A}}^C \cap \mathfrak{a} = \{0\}$. The mapping $p _{\mathfrak{a}}: \mathfrak{a} \rightarrow \mathfrak{d}_4^C$ is injective because $\mathfrak{A}^C \cap \mathfrak{a} = \{0\}$.
0902.0431_F00966	48	■ 1.000	$p(\mathfrak{a}) = \mathfrak{d}_4^C$	$p(\mathfrak{a}) = \mathfrak{d}_4^C$	Since $p(\mathfrak{a})$ is a non-zero ideal of $\mathfrak{d}_4^C$ and $\mathfrak{d}_4^C$ is simple, we have $p(\mathfrak{a}) = \mathfrak{d}_4^C$. Hence
0902.0431_F00967	48	■ 1.000	$\dim_C \mathfrak{a} = \dim_C p(\mathfrak{a}) = \dim_C \mathfrak{d}_4^C = 28$	$\dim_C \mathfrak{a} = \dim_C p(\mathfrak{a}) = \dim_C \mathfrak{d}_4^C = 28$	$\dim_C \mathfrak{a} = \dim_C p(\mathfrak{a}) = \dim_C \mathfrak{d}_4^C = 28$. On the other hand, since $\mathfrak{d}_4^C \cap \mathfrak{a} = \{0\}$,
0902.0431_F00968	48	■ 1.000	$q _{\mathfrak{a}}: \mathfrak{a} \rightarrow \widetilde{\mathfrak{A}}^C$	$q _{\mathfrak{a}}: \mathfrak{a} \rightarrow \widetilde{\mathfrak{A}}^C$	$q _{\mathfrak{a}}: \mathfrak{a} \rightarrow \widetilde{\mathfrak{A}}^C$ is also injective, we have $\dim_C \mathfrak{a} \leq \dim_C \widetilde{\mathfrak{A}}^C = 8 \times 3 = 24$. This leads
0902.0431_F00969	48	■ 1.000	$\dim_C \mathfrak{a} \leq \dim_C \widetilde{\mathfrak{A}}^C = 8 \times 3 = 24$	$\dim_C \mathfrak{a} \leq \dim_C \widetilde{\mathfrak{A}}^C = 8 \times 3 = 24$	$q _{\mathfrak{a}}: \mathfrak{a} \rightarrow \widetilde{\mathfrak{A}}^C$ is also injective, we have $\dim_C \mathfrak{a} < \dim_C \widetilde{\mathfrak{A}}^C = 8 \times 3 = 24$. This leads
0902.0431_F00970	48	■ 1.000	$\mathfrak{d}_4^C \cap \mathfrak{a} = \mathfrak{d}_4^C$	$\mathfrak{d}_4^C \cap \mathfrak{a} = \mathfrak{d}_4^C$	(1) Case $\mathfrak{d}_4^C \cap \mathfrak{a} \neq \{0\}$. From the simplicity of $\mathfrak{d}_4^C$, we have $\mathfrak{d}_4^C \cap \mathfrak{a} = \mathfrak{d}_4^C$, hence
0902.0431_F00971	48	■ 1.000	$\mathfrak{a} \supset \mathfrak{d}_4^C$	$\mathfrak{a} \supset \mathfrak{d}_4^C$	$\mathfrak{a} \supset \mathfrak{d}_4^C$. On the other hand, we have
0902.0431_F00972	48	■ 0.991	$[D, \tilde{A}_i(a_i)] = \tilde{A}_i(Da_i)$	$[D, \tilde{A}_i(a_i)] = \tilde{A}_i(Da_i)$	The last equation follows from $[D, \tilde{A}_i(a_i)] = \tilde{A}_i(Da_i)$ (Lemma 2.4.2) and $\mathfrak{d}_4^C \mathfrak{C}^C = \mathfrak{C}^C$
0902.0431_F00973	48	■ 0.874	$\mathfrak{a} \supset \mathfrak{d}_4^C \oplus \widetilde{\mathfrak{A}}_1^C \oplus \widetilde{\mathfrak{A}}_2^C \oplus \widetilde{\mathfrak{A}}_3^C = \mathfrak{f}_4^C$	$\mathfrak{a} \supset \mathfrak{d}_4^C \oplus \widetilde{\mathfrak{A}}_1^C \oplus \widetilde{\mathfrak{A}}_2^C \oplus \widetilde{\mathfrak{A}}_3^C = \mathfrak{f}_4^C$	(Lemma 2.4.3.(3)). Hence $\mathfrak{a} \supset \mathfrak{d}_4^C \oplus \widetilde{\mathfrak{A}}_1^C \oplus \widetilde{\mathfrak{A}}_2^C \oplus \widetilde{\mathfrak{A}}_3^C = \mathfrak{f}_4^C$.
0902.0431_F00974	48	■ 1.000	$a_1 \neq 0$	$a_1 \neq 0$	If $a_1 \neq 0$, under the actions of $\mathfrak{d}_4^C$ of Lemma 2.4.3.(1), we have
0902.0431_F00975	49	■ 1.000	$b=c=0$	$b=c=0$	If $b=c=0$, then $\tilde{A}_1(1) \in \mathfrak{a}$, hence from Lemma 2.4.2,
0902.0431_F00976	49	■ 1.000	$\tilde{A}_1(1) \in \mathfrak{a}$	$\tilde{A}_1(1) \in \mathfrak{a}$	If $b=c=0$, then $\tilde{A}_1(1) \in \mathfrak{a}$, hence from Lemma 2.4.2,
0902.0431_F00977	49	■ 0.645	$[\tilde{A}_1(1), \tilde{A}_1(e_1)] F_2(1) = -2F_2(e_1)$	$[\tilde{A}_1(1), \tilde{A}_1(e_1)] F_2(1) = -2F_2(e_1)$	(as for the first inequality, note that $[\tilde{A}_1(1), \tilde{A}_1(e_1)] F_2(1) = -2F_2(e_1)$), so the case is

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0902.0431_F00978	49	■ 1.000	$b \neq 0$	$b \neq 0$	reduced to the case (1). If $b \neq 0$, then take the Lie bracket with $\tilde{A}_1(1)$, then
0902.0431_F00979	49	■ 1.000	$\widetilde{A}_1(1)$	$\tilde{A}_1(1)$	reduced to the case (1). If $b \neq 0$, then take the Lie bracket with $\tilde{A}_1(1)$, then
0902.0431_F00980	49	■ 1.000	$c^{\prime}=0$	$c' = 0$	under some actions of $\mathfrak{d}_4^C$. If $c' = 0$, then this can be reduced to the case (1) as in
0902.0431_F00981	49	■ 1.000	$c^{\prime} \neq 0$	$c' \neq 0$	the above. The case $c' \neq 0$ can also be reduced to the case (1) by the same argument
0902.0431_F00982	49	■ 1.000	$\mathfrak{a}=\mathfrak{f}_4\{ \}^C$	$\mathfrak{a} = \mathfrak{f}_4^C$	as above. Thus we have $\mathfrak{a} = \mathfrak{f}_4^C$, which proves the simplicity of $\mathfrak{f}_4^C$.
0902.0431_F00983	49	■ 0.821	$\delta=\sum_i\left[\widetilde{A}_i, \widetilde{B}_i\right], A_i, B_i \in \mathfrak{J}^C$	$\delta = \sum_i \left[\tilde{A}_i, \tilde{B}_i \right], A_i, B_i \in \mathfrak{J}^C$	(2) Any element $\delta \in \mathfrak{f}_4^C$ is expressed as $\delta = \sum_i [\tilde{A}_i, \tilde{B}_i], A_i, B_i \in \mathfrak{J}^C$.
0902.0431_F00984	49	■ 1.000	$[\delta, [\widetilde{A}, \widetilde{B}]] X=\delta[\widetilde{A}, \widetilde{B}] X-[\widetilde{A}, \widetilde{B}] \delta X$	$[\delta, [\tilde{A}, \tilde{B}]]X = \delta[\tilde{A}, \tilde{B}]X - [\tilde{A}, \tilde{B}]\delta X$	Proof. (1) $[\delta, [\tilde{A}, \tilde{B}]]X = \delta[\tilde{A}, \tilde{B}]X - [\tilde{A}, \tilde{B}]\delta X$
0902.0431_F00985	49	■ 0.994	$\mathfrak{a}=\left\{\sum_i\left[\widetilde{A}_i, \widetilde{B}_i\right] \mid A_i, B_i \in \mathfrak{J}^C\right\}$	$\mathfrak{a} = \left\{ \sum_i \left[\tilde{A}_i, \tilde{B}_i \right] \mid A_i, B_i \in \mathfrak{J}^C \right\}$	(2) By (1) we see that $\mathfrak{a} = \{ \sum_i [\tilde{A}_i, \tilde{B}_i] \mid A_i, B_i \in \mathfrak{J}^C \}$ is an ideal of $\mathfrak{f}_4^C$. From the
0902.0431_F00986	49	■ 0.999	$\mathfrak{J}_0\{ \}^C=\left\{X \in \mathfrak{J}^C \mid \operatorname{tr}(X)=0\right\}$	$\mathfrak{J}_0^C = \{ X \in \mathfrak{J}^C \mid \operatorname{tr}(X) = 0 \}$	Proposition 2.4.6. (1) $\mathfrak{J}_0^C = \{ X \in \mathfrak{J}^C \mid \operatorname{tr}(X) = 0 \}$ is an $\mathfrak{f}_4^C$ -irreducible C -
0902.0431_F00987	49	■ 0.952	$\mathfrak{f}_4\{ \}^C \mathfrak{J}_0\{ \}^C=\left\{\sum_i \delta_i B_i \mid \delta_i \in \mathfrak{f}_4^C, B_i \in \mathfrak{J}_0^C\right\}=\mathfrak{J}_0^C$	$\mathfrak{f}_4^C \mathfrak{J}_0^C = \left\{ \sum_i \delta_i B_i \mid \delta_i \in \mathfrak{f}_4^C, B_i \in \mathfrak{J}_0^C \right\} = \mathfrak{J}_0^C$	(2) $\mathfrak{f}_4^C \mathfrak{J}_0^C = \left\{ \sum_i \delta_i B_i \mid \delta_i \in \mathfrak{f}_4^C, B_i \in \mathfrak{J}_0^C \right\} = \mathfrak{J}_0^C$.
0902.0431_F00988	49	■ 0.964	$\mathfrak{J}_0\{ \}^C$	$\mathfrak{J}_0^C$	Proof. (1) Evidently $\mathfrak{J}_0^C$ is an $\mathfrak{f}_4^C$ - C -module (Lemma 2.2.1.(2)). Now, let W be
0902.0431_F00989	49	■ 0.964	$\mathfrak{f}_4\{ \}^C-C$	$\mathfrak{f}_4^C - C$	Proof. (1) Evidently $\mathfrak{J}_0^C$ is an $\mathfrak{f}_4^C$ - C -module (Lemma 2.2.1.(2)). Now, let W be
0902.0431_F00990	49	■ 1.000	$F_i(x) \neq 0(i=1,2,3)$	$F_i(x) \neq 0(i = 1, 2, 3)$	Case (i) If W contains some $F_i(x) \neq 0$ ($i = 1, 2, 3$), then $W = \mathfrak{J}_0^C$. Indeed, under
0902.0431_F00991	49	■ 1.000	$W=\mathfrak{J}_0\{ \}^C$	$W = \mathfrak{J}_0^C$	Case (i) If W contains some $F_i(x) \neq 0$ ($i = 1, 2, 3$), then $W = \mathfrak{J}_0^C$. Indeed, under
0902.0431_F00992	49	■ 1.000	$F_i(1) \in W$	$F_i(1) \in W$	some actions of $\mathfrak{d}_4^C$, we can assume $F_i(1) \in W$ (Lemma 2.4.3.(1)). Next, from
0902.0431_F00993	49	■ 1.000	$\widetilde{A}_i(a) F_{i+2}(1)=F_{i+1}(\bar{a})$	$\tilde{A}_i(a)F_{i+2}(1) = F_{i+1}(\bar{a})$	and $\tilde{A}_i(a)F_{i+2}(1) = F_{i+1}(\bar{a})$, etc., we have $E_i - E_{i+1}, F_i(a) \in W$ and hence we have
0902.0431_F00994	49	■ 1.000	$E_i-E_{i+1}, F_i(a) \in W$	$E_i - E_{i+1}, F_i(a) \in W$	and $\tilde{A}_i(a)F_{i+2}(1) = F_{i+1}(\bar{a})$, etc., we have $E_i - E_{i+1}, F_i(a) \in W$ and hence we have
0902.0431_F00995	50	■ 1.000	$y_2=y_3=0$	$y_2 = y_3 = 0$	If $y_2 = y_3 = 0$, then $F_1(1) \in W$, hence we can reduce to the case (i). If $y_2 \neq$
0902.0431_F00996	50	■ 1.000	$F_1(1) \in W$	$F_1(1) \in W$	If $y_2 = y_3 = 0$, then $F_1(1) \in W$, hence we can reduce to the case (i). If $y_2 \neq$
0902.0431_F00997	50	■ 1.000	$y_2 \neq$	$y_2 \neq$	If $y_2 = y_3 = 0$, then $F_1(1) \in W$, hence we can reduce to the case (i). If $y_2 \neq$
0902.0431_F00998	50	■ 0.996	$a \in \mathfrak{C}^C, a \neq 0$	$a \in \mathfrak{C}^C, a \neq 0$	0, choose $a \in \mathfrak{C}^C, a \neq 0$, such that $(y_2, a) = 0$, and apply $\tilde{A}_2(a)$, then we have
0902.0431_F00999	50	■ 0.996	$\left(y_2, a\right)=0$	$(y_2, a) = 0$	0, choose $a \in \mathfrak{C}^C, a \neq 0$, such that $(y_2, a) = 0$, and apply $\tilde{A}_2(a)$, then we have
0902.0431_F01000	50	■ 0.996	$\widetilde{A}_2(a)$	$\tilde{A}_2(a)$	0, choose $a \in \mathfrak{C}^C, a \neq 0$, such that $(y_2, a) = 0$, and apply $\tilde{A}_2(a)$, then we have
0902.0431_F01001	50	■ 1.000	$-F_3(\bar{a})+F_1(\overline{a y_3}) \in W$	$-F_3(\bar{a}) + F_1(\overline{a y_3}) \in W$	$-F_3(\bar{a}) + F_1(\overline{a y_3}) \in W$. Again under the action of $\mathfrak{d}_4^C$, we have
0902.0431_F01002	50	■ 1.000	$z_1=0$	$z_1 = 0$	If $z_1 = 0$, we can reduce to the case (i). If $z_1 \neq 0$, choose $b \in \mathfrak{C}^C, b \neq 0$, such that
0902.0431_F01003	50	■ 1.000	$z_1 \neq 0$	$z_1 \neq 0$	If $z_1 = 0$, we can reduce to the case (i). If $z_1 \neq 0$, choose $b \in \mathfrak{C}^C, b \neq 0$, such that
0902.0431_F01004	50	■ 1.000	$b \in \mathfrak{C}^C, b \neq 0$	$b \in \mathfrak{C}^C, b \neq 0$	If $z_1 = 0$, we can reduce to the case (i). If $z_1 \neq 0$, choose $b \in \mathfrak{C}^C, b \neq 0$, such that
0902.0431_F01005	50	■ 0.941	$\left(z_1, b\right)=0$	$(z_1, b) = 0$	$(z_1, b) = 0$ and apply $\tilde{A}_1(b)$, then $F_2(\bar{b}) \in W$, which is again reduced to the case (i).
0902.0431_F01006	50	■ 0.941	$\widetilde{A}_1(b)$	$\tilde{A}_1(b)$	$(z_1, b) = 0$ and apply $\tilde{A}_1(b)$, then $F_2(\bar{b}) \in W$, which is again reduced to the case (i).
0902.0431_F01007	50	■ 0.941	$F_2(\bar{b}) \in W$	$F_2(\bar{b}) \in W$	$(z_1, b) = 0$ and apply $\tilde{A}_1(b)$, then $F_2(\bar{b}) \in W$, which is again reduced to the case (i).
0902.0431_F01008	50	■ 1.000	$x_2 \neq 0$	$x_2 \neq 0$	Similarly for $x_2 \neq 0$ or $x_3 \neq 0$ the arguments above hold.
0902.0431_F01009	50	■ 1.000	$x_3 \neq 0$	$x_3 \neq 0$	Similarly for $x_2 \neq 0$ or $x_3 \neq 0$ the arguments above hold.
0902.0431_F01010	50	■ 1.000	$x_1=x_2=x_3=0$	$x_1 = x_2 = x_3 = 0$	Case $x_1 = x_2 = x_3 = 0$. Applying $\tilde{A}_1(1)$ on the non-zero element
0902.0431_F01011	50	■ 1.000	$F_1(\xi-2 \eta) \in W$	$F_1(\xi - 2\eta) \in W$	we have $F_1(\xi - 2\eta) \in W$. If $\xi - 2\eta \neq 0$, we can reduce to the case (i). If $\xi - 2\eta = 0$,
0902.0431_F01012	50	■ 1.000	$\xi-2 \eta \neq 0$	$\xi - 2\eta \neq 0$	we have $F_1(\xi - 2\eta) \in W$. If $\xi - 2\eta \neq 0$, we can reduce to the case (i). If $\xi - 2\eta = 0$,
0902.0431_F01013	50	■ 1.000	$\xi-2 \eta=0$	$\xi - 2\eta = 0$	we have $F_1(\xi - 2\eta) \in W$. If $\xi - 2\eta \neq 0$, we can reduce to the case (i). If $\xi - 2\eta = 0$,
0902.0431_F01014	50	■ 1.000	$\widetilde{A}_3(1)$	$\tilde{A}_3(1)$	applying $\tilde{A}_3(1)$, we have $F_3(3\eta) \in W$ which is again the case (i). We have just proved

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0902.0431_F01015	50	■ 1.000	$F_{\{3\}}(3 \eta) \in W$	$F_3(3\eta) \in W$	applying $\tilde{A}_3(1)$, we have $F_3(3\eta) \in W$ which is again the case (i). We have just proved
0902.0431_F01016	50	■ 1.000	$\mathfrak{f}_{\{4\}}^{\{C\}} \mathfrak{J}_{\{0\}}^{\{C\}}$	$\mathfrak{f}_4^C \mathfrak{J}_0^C$	(2) Since $\mathfrak{f}_4^C \mathfrak{J}_0^C$ is an $\mathfrak{f}_4^C$ -invariant C -submodule of $\mathfrak{J}_0^C$, we have $\mathfrak{f}_4^C \mathfrak{J}_0^C = \mathfrak{J}_0^C$
0902.0431_F01017	50	■ 1.000	$\mathfrak{f}_{\{4\}}^{\{C\}} \mathfrak{J}_{\{0\}}^{\{C\}} = \mathfrak{J}_{\{0\}}^{\{C\}}$	$\mathfrak{f}_4^C \mathfrak{J}_0^C = \mathfrak{J}_0^C$	(2) Since $\mathfrak{f}_4^C \mathfrak{J}_0^C$ is an $\mathfrak{f}_4^C$ -invariant C -submodule of $\mathfrak{J}_0^C$, we have $\mathfrak{f}_4^C \mathfrak{J}_0^C = \mathfrak{J}_0^C$
0902.0431_F01018	50	■ 0.997	$A, B, C, D \in \mathfrak{J}^C$	$A, B, C, D \in \mathfrak{J}^C$	Lemma 2.5.1. For $A, B, C, D \in \mathfrak{J}^C$, we have
0902.0431_F01019	50	■ 1.000	$\left(\delta_1, \delta_2\right)_4$	$(\delta_1, \delta_2)_4$	Definition. In $\mathfrak{f}_4^C$, we define an inner product $(\delta_1, \delta_2)_4$ by
0902.0431_F01020	50	■ 1.000	$\delta_1 = \sum_i \left[\widetilde{A}_i, \widetilde{B}_i \right], \delta_2 = \sum_j \left[\widetilde{C}_j, \widetilde{D}_j \right], A_i, B_i, C_j, D_j \in \mathfrak{J}^C$	$\delta_1 = \sum_i \left[\widetilde{A}_i, \widetilde{B}_i \right], \delta_2 = \sum_j \left[\widetilde{C}_j, \widetilde{D}_j \right], A_i, B_i, C_j, D_j \in \mathfrak{J}^C$	for $\delta_1 = \sum_i [A_i, B_i], \delta_2 = \sum_j [C_j, D_j], A_i, B_i, C_j, D_j \in \mathfrak{J}^C$ (Lemma 2.4.5.(2)). Then,
0902.0431_F01021	50	■ 1.000	δ_1, δ_2	δ_1, δ_2	of the choice of the expressions of δ_1, δ_2 , and that $(\delta_1, \delta_2)_4$ is symmetric.
0902.0431_F01022	51	■ 0.999	$\delta_1 = [\widetilde{A}, \widetilde{B}], \delta_2 = [\widetilde{C}, \widetilde{D}], A, B, C, D \in \mathfrak{J}^C$	$\delta_1 = [\widetilde{A}, \widetilde{B}], \delta_2 = [\widetilde{C}, \widetilde{D}], A, B, C, D \in \mathfrak{J}^C$	Proof. It is sufficient to show the lemma for $\delta_1 = [\widetilde{A}, \widetilde{B}], \delta_2 = [\widetilde{C}, \widetilde{D}], A, B, C, D \in \mathfrak{J}^C$
0902.0431_F01023	51	■ 0.658	$(X \circ Y, \delta Z) = -(\delta X \circ Y, Z) - (X \circ \delta Y, Z)$	$(X \circ Y, \delta Z) = -(\delta X \circ Y, Z) - (X \circ \delta Y, Z)$	which is equal to 0, if we use the relation $(X \circ Y, \delta Z) = -(\delta X \circ Y, Z) - (X \circ \delta Y, Z)$
0902.0431_F01024	51	■ 1.000	B_4	B_4	Theorem 2.5.3. The Killing form B_4 of the Lie algebra $\mathfrak{f}_4^C$ is given by
0902.0431_F01025	51	■ 1.000	$k, k' \in C$	$k, k' \in C$	Proof. Since $\mathfrak{f}_4^C$ is simple (Theorem 2.4.4), there exist $k, k' \in C$ such that
0902.0431_F01026	51	■ 1.000	$k, k' \in C$	$k, k' \in C$	To determine these k, k' , we put $\delta = \delta_1 = \delta_2 = \tilde{A}_1(1)$. Since $\tilde{A}_1(1) = -2[\tilde{E}_3, \tilde{F}_1(1)]$
0902.0431_F01027	51	■ 1.000	$\delta = \delta_1 = \delta_2 = \tilde{A}_1(1)$	$\delta = \delta_1 = \delta_2 = \tilde{A}_1(1)$	To determine these k, k' , we put $\delta = \delta_1 = \delta_2 = \tilde{A}_1(1)$. Since $\tilde{A}_1(1) = -2[\tilde{E}_3, \tilde{F}_1(1)]$
0902.0431_F01028	51	■ 1.000	$\tilde{A}_1(1) = -2[\tilde{E}_3, \tilde{F}_1(1)]$	$\tilde{A}_1(1) = -2[\tilde{E}_3, \tilde{F}_1(1)]$	To determine these k, k' , we put $\delta = \delta_1 = \delta_2 = \tilde{A}_1(1)$. Since $\tilde{A}_1(1) = -2[\tilde{E}_3, \tilde{F}_1(1)]$
0902.0431_F01029	51	■ 1.000	$(\operatorname{ad} \delta)^2$	$(\operatorname{ad} \delta)^2$	On the other hand, $(\operatorname{ad} \delta)^2$ is calculated as follows.
0902.0431_F01030	51	■ 1.000	$k=9$	$k=9$	Therefore $k=9$. Next, we will calculate $\operatorname{tr}(\delta\delta)$.
0902.0431_F01031	51	■ 1.000	$\operatorname{tr}(\delta\delta)$	$\operatorname{tr}(\delta\delta)$	Therefore $k=9$. Next, we will calculate $\operatorname{tr}(\delta\delta)$.
0902.0431_F01032	52	■ 1.000	$k'=3$	$k'=3$	Therefore $k'=3$.
0902.0431_F01033	52	■ 1.000	$A \in \mathfrak{J}_0^C$	$A \in \mathfrak{J}_0^C$	Lemma 2.5.4. For a non-zero element $A \in \mathfrak{J}_0^C$, there exists $B \in \mathfrak{J}_0^C$ such that
0902.0431_F01034	52	■ 1.000	$B \in \mathfrak{J}_0^C$	$B \in \mathfrak{J}_0^C$	Lemma 2.5.4. For a non-zero element $A \in \mathfrak{J}_0^C$, there exists $B \in \mathfrak{J}_0^C$ such that
0902.0431_F01035	52	■ 1.000	$[\widetilde{A}, \widetilde{B}] \neq 0$	$[\widetilde{A}, \widetilde{B}] \neq 0$	$[\widetilde{A}, \widetilde{B}] \neq 0$.
0902.0431_F01036	52	■ 1.000	$[\widetilde{A}, \widetilde{B}] = 0$	$[\widetilde{A}, \widetilde{B}] = 0$	Proof. Assume that $[\widetilde{A}, \widetilde{B}] = 0$ for all $B \in \mathfrak{J}_0^C$. Then $0 = (\delta, [\widetilde{A}, \widetilde{B}])_4 =$
0902.0431_F01037	52	■ 1.000	$0 = (\delta, [\widetilde{A}, \widetilde{B}])_4 =$	$0 = (\delta, [\widetilde{A}, \widetilde{B}])_4 =$	Proof. Assume that $[\widetilde{A}, \widetilde{B}] = 0$ for all $B \in \mathfrak{J}_0^C$. Then $0 = (\delta, [\widetilde{A}, \widetilde{B}])_4 =$
0902.0431_F01038	52	■ 0.999	$-(\delta, [\widetilde{B}, \widetilde{A}])_4 = -(\delta B, A)$	$-(\delta, [\widetilde{B}, \widetilde{A}])_4 = -(\delta B, A)$	$-(\delta, [\widetilde{B}, \widetilde{A}])_4 = -(\delta B, A)$ for any $\delta \in \mathfrak{f}_4^C$. Since $\mathfrak{f}_4^C \mathfrak{J}_0^C = \mathfrak{J}_0^C$ (Proposition 2.4.6.(2)),
0902.0431_F01039	52	■ 1.000	$(\mathfrak{J}_0^C, A) = 0$	$(\mathfrak{J}_0^C, A) = 0$	we have $(\mathfrak{J}_0^C, A) = 0$, so that $A = 0$.
0902.0431_F01040	52	■ 1.000	$\mathfrak{D}_4^C$	$\mathfrak{D}_4^C$	algebra $\mathfrak{D}_4^C$:
0902.0431_F01041	52	■ 1.000	$H_k = -iG_{k4+k}$	$H_k = -iG_{k4+k}$	If we let $H_k = -iG_{k4+k}$ for $k = 0, 1, 2, 3$, then
0902.0431_F01042	52	■ 1.000	$k = 0, 1, 2, 3$	$k = 0, 1, 2, 3$	If we let $H_k = -iG_{k4+k}$ for $k = 0, 1, 2, 3$, then
0902.0431_F01043	53	■ 0.963	ν, π	ν, π	(Theorem 1.3.6, Proposition 2.3.7). Outer automorphisms ν, π of $\mathfrak{D}_4^C$ induce C -
0902.0431_F01044	53	■ 0.996	H_0, H_1, H_2, H_3	H_0, H_1, H_2, H_3	H_0, H_1, H_2, H_3 of $\mathfrak{h}$ are respectively given by
0902.0431_F01045	53	■ 1.000	$\mathfrak{h} = \left\{ H = \sum_{k=0}^3 \lambda_k H_k \mid \lambda_k \in C \right\} \subset \mathfrak{d}_4^C \subset \mathfrak{f}_4^C$	$\mathfrak{h} = \left\{ H = \sum_{k=0}^3 \lambda_k H_k \mid \lambda_k \in C \right\} \subset \mathfrak{d}_4^C \subset \mathfrak{f}_4^C$	Let $\mathfrak{h} = \left\{ H = \sum_{k=0}^3 \lambda_k H_k \mid \lambda_k \in C \right\} \subset \mathfrak{d}_4^C \subset \mathfrak{f}_4^C$. Since $\mathfrak{h}$ is a Cartan subalgebra of
0902.0431_F01046	53	■ 0.997	$[H, \tilde{A}_1(a)] = \tilde{A}_1(Ha)$	$[H, \tilde{A}_1(a)] = \tilde{A}_1(Ha)$	are also roots of $\mathfrak{f}_4^C$. Furthermore, from $[H, \tilde{A}_1(a)] = \tilde{A}_1(Ha)$ (Lemma 2.4.2), where
0902.0431_F01047	53	■ 0.996	λ_k	λ_k	we see that λ_k is a root of $\mathfrak{f}_4^C$ and $\tilde{A}_1(e_k + ie_{4+k})$ is an associated root vector for
0902.0431_F01048	53	■ 0.996	$\tilde{A}_1(e_k + ie_{4+k})$	$\tilde{A}_1(e_k + ie_{4+k})$	we see that λ_k is a root of $\mathfrak{f}_4^C$ and $\tilde{A}_1(e_k + ie_{4+k})$ is an associated root vector for
0902.0431_F01049	53	■ 0.996	$0 \leq k \leq 3$	$0 \leq k \leq 3$	$0 \leq k \leq 3$. Similarly, $-\lambda_k$ for $0 \leq k \leq 3$ is root of $\mathfrak{f}_4^C$ and $\tilde{A}_1(e_k - ie_{4+k})$ is
0902.0431_F01050	53	■ 0.996	$-\lambda_k$	$-\lambda_k$	$0 \leq k \leq 3$. Similarly, $-\lambda_k$ for $0 \leq k \leq 3$ is root of $\mathfrak{f}_4^C$ and $\tilde{A}_1(e_k - ie_{4+k})$ is

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0902.0431_F01051	53	■ 0.996	$\widetilde{\mathrm{A}}_1\mathrm{e}_k-i\mathrm{e}_{4+k}$	$\widetilde{A}_1(e_k - ie_{4+k})$	$0 \leq k \leq 3$. Similarly, $-\lambda_k$ for $0 \leq k \leq 3$ is root of $\mathfrak{f}_4^C$ and $\boxed{A_1(e_k - ie_{4+k})}$ is
0902.0431_F01052	54	■ 1.000	$\left[H,\widetilde{\mathrm{A}}_2(a)\right]=\widetilde{\mathrm{A}}_2((\nu H)a)$	$\left[H,\widetilde{A}_2(a)\right]=\widetilde{A}_2((\nu H)a)$	an associated root vector. Next, by Lemma 2.4.2 we have $\boxed{H,A_2(a)}=\boxed{A_2((\nu H)a)}$,
0902.0431_F01053	54	■ 1.000	$\widetilde{\mathrm{A}}_2\mathrm{e}_k+$	$\widetilde{A}_2(e_k+$	and so we see that coefficients of H_0, H_1, H_2, H_3 are roots of $\mathfrak{f}_4^C$ and that $\widetilde{A}_2(e_k+$
0902.0431_F01054	54	■ 0.593	$i+i\mathrm{e}_{4+k}$	$i+ie_{4+k}$	$i+ie_{4+k})$ is associated root vector for $0 \leq k \leq 3$. The roots above with negative sign
0902.0431_F01055	54	■ 1.000	$\widetilde{\mathrm{A}}_2\mathrm{e}_k-i\mathrm{e}_{4+k}$	$\widetilde{A}_2(e_k - ie_{4+k})$	are also roots of $\mathfrak{f}_4^C$ and $\boxed{A_2(e_k - ie_{4+k})}$ are associated root vectors. Finally, from
0902.0431_F01056	54	■ 1.000	$\left[H,\widetilde{\mathrm{A}}_3(a)\right]=\widetilde{\mathrm{A}}_3((\kappa\pi H)a)$	$\left[H,\widetilde{A}_3(a)\right]=\widetilde{A}_3((\kappa\pi H)a)$	$\boxed{H,A_3(a)}=\boxed{A_3((\kappa\pi H)a)}$ (Lemma 2.4.2), where
0902.0431_F01057	54	■ 0.585	$\alpha_1, \alpha_2, \alpha_3, \alpha_4$	$\alpha_1, \alpha_2, \alpha_3, \alpha_4$	Proof. All positive roots of $\mathfrak{f}_4^C$ are expressed by $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ as follows.
0902.0431_F01058	55	■ 1.000	$\Pi=\left\{\alpha_1, \alpha_2, \alpha_3, \alpha_4\right\}$	$\Pi=\left\{\alpha_1, \alpha_2, \alpha_3, \alpha_4\right\}$	Hence $\boxed{I=\left\{\alpha_1, \alpha_2, \alpha_3, \alpha_4\right\}}$ is a fundamental root system of $\mathfrak{f}_4^C$. The real part $\mathfrak{h}_R$ of
0902.0431_F01059	55	■ 1.000	$B_4\left(\delta_1, \delta_2\right)=3 \operatorname{tr}\left(\delta_1 \delta_2\right)$	$B_4\left(\delta_1, \delta_2\right)=3 \operatorname{tr}\left(\delta_1 \delta_2\right)$	The Killing form B_4 of $\mathfrak{f}_4^C$ is $B_4\left(\delta_1, \delta_2\right)=3 \operatorname{tr}\left(\delta_1 \delta_2\right)$ (Theorem 2.5.3), so that
0902.0431_F01060	56	■ 0.648	$\alpha_i\left(B_4\left(H_{\alpha}, H\right)=\alpha(H), H \in\right.$	$\alpha_i\left(B_4\left(H_{\alpha}, H\right)=\alpha(H), H \in\right.$	Now, the canonical elements $H_{\alpha_i} \in \mathfrak{h}_R$ corresponding to $\boxed{d_i\left(B_4\left(H_{\alpha}, H\right)=\alpha(H), H \in\right.}$
0902.0431_F01061	56	■ 1.000	$-\mu$	$-\mu$	induced by the Killing form B_4 between $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ and $-\mu$ are given by
0902.0431_F01062	56	■ 1.000	$C_1 \oplus C_3$	$C_1 \oplus C_3$	(2) The second one is a subalgebra of type $C_1 \oplus C_3$ obtained as the fixed points
0902.0431_F01063	56	■ 1.000	$A_2 \oplus A_2$	$A_2 \oplus A_2$	(3) The third one is a subalgebra of type $A_2 \oplus A_2$ obtained as the fixed points of
0902.0431_F01064	57	■ 0.876	$\left\{\alpha \in F_4 \mid \alpha E_i=E_i, i=1,2,3\right\} \cong \operatorname{Spin}(8)$	$\left\{\alpha \in F_4 \mid \alpha E_i=E_i, i=1,2,3\right\} \cong \operatorname{Spin}(8)$	Theorem 2.7.1. $\boxed{\left\{\alpha \in F_4 \mid \alpha E_i=E_i, i=1,2,3\right\} \cong \operatorname{Spin}(8)}$.
0902.0431_F01065	57	■ 1.000	$\varphi: \operatorname{Spin}(8) \rightarrow D_4=\left\{\alpha \in F_4 \mid \alpha E_i=E_i, i=1,2,3\right\}$	$\varphi: \operatorname{Spin}(8) \rightarrow D_4=\left\{\alpha \in F_4 \mid \alpha E_i=E_i, i=1,2,3\right\}$	$\boxed{\varphi: \operatorname{Spin}(8) \rightarrow D_4=\left\{\alpha \in F_4 \mid \alpha E_i=E_i, i=1,2,3\right\}}$ by
0902.0431_F01066	57	■ 1.000	$\alpha=\varphi\left(\alpha_1, \alpha_2, \alpha_3\right) \in D_4$	$\alpha=\varphi\left(\alpha_1, \alpha_2, \alpha_3\right) \in D_4$	We first prove that $\boxed{\alpha=\varphi\left(\alpha_1, \alpha_2, \alpha_3\right) \in D_4}$. Indeed, the fact that $\det (\alpha X)=\det X$
0902.0431_F01067	57	■ 1.000	$\alpha E_i=E_i, i=1,2,3$	$\alpha E_i=E_i, i=1,2,3$	which together with $\alpha E=E$, shows that $\alpha \in F_4$ and $\boxed{\alpha E_i=E_i, i=1,2,3}$. Therefore
0902.0431_F01068	57	■ 1.000	$\alpha \in D_4$	$\alpha \in D_4$	$\alpha \in D_4$. Certainly φ is a homomorphism. We shall show that φ is onto. Let $\alpha \in D_4$.
0902.0431_F01069	57	■ 1.000	$\alpha X \in \mathfrak{J}_i, X \in \mathfrak{J}_i, \alpha$	$\alpha X \in \mathfrak{J}_i, X \in \mathfrak{J}_i, \alpha$	Since $\alpha X \in \mathfrak{J}_i, X \in \mathfrak{J}_i, \alpha$ induces <i>R</i> -isomorphisms $\alpha: \mathfrak{J}_i \rightarrow \mathfrak{J}_i$ and $\alpha_i: \mathfrak{C} \rightarrow \mathfrak{C}$
0902.0431_F01070	57	■ 1.000	$\alpha: \mathfrak{J}_i \rightarrow \mathfrak{J}_i$	$\alpha: \mathfrak{J}_i \rightarrow \mathfrak{J}_i$	Since $\alpha X \in \mathfrak{J}_i, X \in \mathfrak{J}_i, \alpha$ induces <i>R</i> -isomorphisms $\alpha: \mathfrak{J}_i \rightarrow \mathfrak{J}_i$ and $\alpha_i: \mathfrak{C} \rightarrow \mathfrak{C}$
0902.0431_F01071	57	■ 1.000	$\alpha_i: \mathfrak{C} \rightarrow \mathfrak{C}$	$\alpha_i: \mathfrak{C} \rightarrow \mathfrak{C}$	Since $\alpha X \in \mathfrak{J}_i, X \in \mathfrak{J}_i, \alpha$ induces <i>R</i> -isomorphisms $\alpha: \mathfrak{J}_i \rightarrow \mathfrak{J}_i$ and $\alpha_i: \mathfrak{C} \rightarrow \mathfrak{C}$
0902.0431_F01072	57	■ 1.000	$F_i\left(\alpha_i x\right) \circ F_i\left(\alpha_i y\right)=\left(\alpha_i x, \alpha_i y\right)\left(E_{i+1}+E_{i+2}\right)$	$F_i\left(\alpha_i x\right) \circ F_i\left(\alpha_i y\right)=\left(\alpha_i x, \alpha_i y\right)\left(E_{i+1}+E_{i+2}\right)$	becomes $\boxed{F_i\left(\alpha_i x\right) \circ F_i\left(\alpha_i y\right)=\left(\alpha_i x, \alpha_i y\right)\left(E_{i+1}+E_{i+2}\right)}$, we have
0902.0431_F01073	57	■ 1.000	$\alpha_i \in O(8), i=1,2,3$	$\alpha_i \in O(8), i=1,2,3$	Hence $\alpha_i \in O(8), i=1,2,3$. Moreover, by applying α on $F_1(x) \circ F_2(y)=\frac{1}{2} F_3(\overline{xy})$,
0902.0431_F01074	57	■ 1.000	$\varphi\left(\alpha_1, \alpha_2, \alpha_3\right)=\alpha$	$\varphi\left(\alpha_1, \alpha_2, \alpha_3\right)=\alpha$	$\boxed{\varphi\left(\alpha_1, \alpha_2, \alpha_3\right)=\alpha}$. Therefore φ is onto. Evidently $\operatorname{Ker} \varphi=\{(1,1,1)\}$. Consequently
0902.0431_F01075	57	■ 1.000	$\operatorname{Ker} \varphi=\{(1,1,1)\}$	$\operatorname{Ker} \varphi=\{(1,1,1)\}$	$\varphi\left(\alpha_1, \alpha_2, \alpha_3\right)=\alpha$. Therefore φ is onto. Evidently $\boxed{\operatorname{Ker} \varphi=\{(1,1,1)\}}$. Consequently
0902.0431_F01076	57	■ 1.000	$\left(F_4\right)_{E_1}$	$\left(F_4\right)_{E_1}$	We shall study the following subgroup $\boxed{\left(F_4\right)_E}$ of F_4 :
0902.0431_F01077	57	■ 1.000	$\mathfrak{J}_{01}, \mathfrak{J}_{23}$	$\mathfrak{J}_{01}, \mathfrak{J}_{23}$	We define <i>R</i> -vector subspaces $\mathfrak{J}_{01}, \mathfrak{J}_{23}$ of $\mathfrak{J}$ respectively by
0902.0431_F01078	58	■ 1.000	$A \in \mathfrak{J}_{01}$	$A \in \mathfrak{J}_{01}$	Lemma 2.7.2. Suppose that we are given an element $\boxed{A \in \mathfrak{J}_{01}}$ such that $(A, A)=$
0902.0431_F01079	58	■ 1.000	$(A, A)=$	$(A, A)=$	Lemma 2.7.2. Suppose that we are given an element $A \in \mathfrak{J}_{01}$ such that $\boxed{(A, A)=}$
0902.0431_F01080	58	■ 1.000	$X_0 \in \mathfrak{J}_{01}$	$X_0 \in \mathfrak{J}_{01}$	2. Choose any element $\boxed{X_0 \in \mathfrak{J}_{01}}$ such that $\left(X_0, X_0\right)=2,\left(A, X_0\right)=0$. Choose any
0902.0431_F01081	58	■ 1.000	$\left(X_0, X_0\right)=2,\left(A, X_0\right)=0$	$\left(X_0, X_0\right)=2,\left(A, X_0\right)=0$	2. Choose any element $X_0 \in \mathfrak{J}_{01}$ such that $\boxed{\left(X_0, X_0\right)=2,\left(A, X_0\right)=0}$. Choose any
0902.0431_F01082	58	■ 1.000	$Y_0 \in \mathfrak{J}_{23}$	$Y_0 \in \mathfrak{J}_{23}$	element $\boxed{Y_0 \in \mathfrak{J}_{23}}$ such that $\left(Y_0, Y_0\right)=2,2 A \circ Y_0=-Y_0$. Let
0902.0431_F01083	58	■ 1.000	$\left(Y_0, Y_0\right)=2,2 A \circ Y_0=-Y_0$	$\left(Y_0, Y_0\right)=2,2 A \circ Y_0=-Y_0$	element $Y_0 \in \mathfrak{J}_{23}$ such that $\boxed{\left(Y_0, Y_0\right)=2,2 A \circ Y_0=-Y_0}$. Let
0902.0431_F01084	58	■ 1.000	$X_1 \in \mathfrak{J}_{01}$	$X_1 \in \mathfrak{J}_{01}$	Choose any element $X_1 \in \mathfrak{J}_{01}$ such that $\left(X_1, X_1\right)=2,\left(A, X_1\right)=\left(X_0, X_1\right)=0$.
0902.0431_F01085	58	■ 1.000	$\left(X_1, X_1\right)=2,\left(A, X_1\right)=\left(X_0, X_1\right)=0$	$\left(X_1, X_1\right)=2,\left(A, X_1\right)=\left(X_0, X_1\right)=0$	Choose any element $X_1 \in \mathfrak{J}_{01}$ such that $\left(X_1, X_1\right)=2,\left(A, X_1\right)=\left(X_0, X_1\right)=0$.
0902.0431_F01086	58	■ 1.000	$X_2 \in \mathfrak{J}_{01}$	$X_2 \in \mathfrak{J}_{01}$	Choose any element $X_2 \in \mathfrak{J}_{01}$ such that $\left(X_2, X_2\right)=2,\left(A, X_2\right)=\left(X_0, X_2\right)=$
0902.0431_F01087	58	■ 1.000	$\left(X_2, X_2\right)=2,\left(A, X_2\right)=\left(X_0, X_2\right)=$	$\left(X_2, X_2\right)=2,\left(A, X_2\right)=\left(X_0, X_2\right)=$	Choose any element $X_2 \in \mathfrak{J}_{01}$ such that $\left(X_2, X_2\right)=2,\left(A, X_2\right)=\left(X_0, X_2\right)=$

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0902.0431_F01088	58	■ 1.000	$\left(X_{\{1\}}, X_{\{2\}}\right)=0$	$(X_1, X_2) = 0$	$X_1, X_2) = 0$. Let
0902.0431_F01089	58	■ 1.000	$X_{\{4\}} \in \mathfrak{J}_{01}$	$X_4 \in \mathfrak{J}_{01}$	Finally, choose any element $X_4 \in \mathfrak{J}_{01}$ such that $(X_4, X_4) = 2, (A, X_4) = (X_0, X_4) =$
0902.0431_F01090	58	■ 1.000	$\left(X_{\{4\}}, X_{\{4\}}\right)=2, \left(A, X_{\{4\}}\right)=\left(X_{\{0\}}, X_{\{4\}}\right)=$	$(X_4, X_4) = 2, (A, X_4) = (X_0, X_4) =$	Finally, choose any element $X_4 \in \mathfrak{J}_{01}$ such that $X_4, X_4) = 2, (A, X_4) = (X_0, X_4) =$
0902.0431_F01091	58	■ 1.000	$\left(X_{\{1\}}, X_{\{4\}}\right)=\left(X_{\{2\}}, X_{\{4\}}\right)=\left(X_{\{3\}}, X_{\{4\}}\right)=0$	$(X_1, X_4) = (X_2, X_4) = (X_3, X_4) = 0$	$X_1, X_4) = (X_2, X_4) = (X_3, X_4) = 0$. Let
0902.0431_F01092	58	■ 1.000	$\alpha: \mathfrak{J} \rightarrow \mathfrak{J}$	$\alpha: \mathfrak{J} \rightarrow \mathfrak{J}$	Then, an $\mathbf{R}$ -linear mapping $\alpha: \mathfrak{J} \rightarrow \mathfrak{J}$ satisfying
0902.0431_F01093	58	■ 1.000	$X, Y = E_i$	$X, Y = E_i$	For this purpose, it is sufficient to show that this holds for generators $X, Y = E_i$,
0902.0431_F01094	58	■ 1.000	$F_{\{j\}} \left(e_{\{k\}} \right)$	$F_j(e_k)$	$F_j(e_k)$ of $\mathfrak{J}$. We have to check $27^2 = 729$ times if honestly doing so we reduce the
0902.0431_F01095	58	■ 1.000	$27^2=729$	$27^2 = 729$	$F_j(e_k)$ of $\mathfrak{J}$. We have to check $27^2 = 729$ times if honestly doing so we reduce the
0902.0431_F01096	58	■ 0.995	$\left(F_{\{4\}}\right)_{E_1} / \operatorname{Spin}(8) \simeq S^8$	$(F_4)_{E_1} / \operatorname{Spin}(8) \simeq S^8$	Proposition 2.7.3. $(F_4)_{E_1} / \operatorname{Spin}(8) \simeq S^8$.
0902.0431_F01097	58	■ 1.000	$S^8 = \left\{ X \in \mathfrak{J}_{01} \mid (X, X) = 2 \right\}$	$S^8 = \{X \in \mathfrak{J}_{01} \mid (X, X) = 2\}$	Proof. $S^8 = \{X \in \mathfrak{J}_{01} \mid (X, X) = 2\}$ is an 8 dimensional sphere. For $\alpha \in (F_4)_{E_1}$
0902.0431_F01098	58	■ 1.000	$\alpha \in \left(F_{\{4\}}\right)_{E_1}$	$\alpha \in (F_4)_{E_1}$	Proof. $S^8 = \{X \in \mathfrak{J}_{01} \mid (X, X) = 2\}$ is an 8 dimensional sphere. For $\alpha \in (F_4)_{E_1}$
0902.0431_F01099	58	■ 1.000	$X \in S^8$	$X \in S^8$	and $X \in S^8$, we have $\alpha X \in S^8$. Hence the group $(F_4)_{E_1}$ acts on S^8 . This action is
0902.0431_F01100	58	■ 1.000	$\alpha X \in S^8$	$\alpha X \in S^8$	and $X \in S^8$, we have $\alpha X \in S^8$. Hence the group $(F_4)_{E_1}$ acts on S^8 . This action is
0902.0431_F01101	58	■ 1.000	S^8	S^8	and $X \in S^8$, we have $\alpha X \in S^8$. Hence the group $(F_4)_{E_1}$ acts on S^8 . This action is
0902.0431_F01102	58	■ 0.999	$A \in S^8$	$A \in S^8$	transitive. Indeed, for a given $A \in S^8$, by constructing $\alpha \in (F_4)_{E_1}$ of Lemma 2.7.2,
0902.0431_F01103	59	■ 1.000	$\alpha \left(E_2 - E_3\right) = A$	$\alpha(E_2 - E_3) = A$	we have $\alpha(E_2 - E_3) = A$, and which shows the transitivity. We determine the isotropy
0902.0431_F01104	59	■ 1.000	$E_2 - E_3 \in S^8$	$E_2 - E_3 \in S^8$	subgroup of $(F_4)_{E_1}$ at $E_2 - E_3 \in S^8$. Let $\alpha \in (F_4)_{E_1}$ satisfy $\alpha(E_2 - E_3) = E_2 - E_3$.
0902.0431_F01105	59	■ 1.000	$\alpha \left(E_2 - E_3\right) = E_2 - E_3$	$\alpha(E_2 - E_3) = E_2 - E_3$	subgroup of $(F_4)_{E_1}$ at $E_2 - E_3 \in S^8$. Let $\alpha \in (F_4)_{E_1}$ satisfy $\alpha(E_2 - E_3) = E_2 - E_3$.
0902.0431_F01106	59	■ 1.000	$\alpha E_1 = E_1$	$\alpha E_1 = E_1$	Since $\alpha \in (F_4)_{E_1}$ satisfies $\alpha E_1 = E_1$ and $\alpha E = E$, it also satisfies $\alpha(E_2 + E_3) = E_2 +$
0902.0431_F01107	59	■ 1.000	$\alpha \left(E_2 + E_3\right) = E_2 +$	$\alpha(E_2 + E_3) = E_2 +$	Since $\alpha \in (F_4)_{E_1}$ satisfies $\alpha E_1 = E_1$ and $\alpha E = E$, it also satisfies $\alpha(E_2 + E_3) = E_2 +$
0902.0431_F01108	59	■ 1.000	E_3	E_3	E_3 . Therefore we have $\alpha E_2 = E_2$ and $\alpha E_3 = E_3$, so that $\alpha \in \operatorname{Spin}(8)$. Conversely,
0902.0431_F01109	59	■ 1.000	$\alpha E_2 = E_2$	$\alpha E_2 = E_2$	E_3 . Therefore we have $\alpha E_2 = E_2$ and $\alpha E_3 = E_3$, so that $\alpha \in \operatorname{Spin}(8)$. Conversely,
0902.0431_F01110	59	■ 1.000	$\alpha E_3 = E_3$	$\alpha E_3 = E_3$	E_3 . Therefore we have $\alpha E_2 = E_2$ and $\alpha E_3 = E_3$, so that $\alpha \in \operatorname{Spin}(8)$. Conversely,
0902.0431_F01111	59	■ 1.000	$\alpha \in \operatorname{Spin}(8)$	$\alpha \in \operatorname{Spin}(8)$	E_3 . Therefore we have $\alpha E_2 = E_2$ and $\alpha E_3 = E_3$, so that $\alpha \in \operatorname{Spin}(8)$. Conversely,
0902.0431_F01112	59	■ 0.994	$E_2 - E_3$	$E_2 - E_3$	The isotropy subgroup of $(F_4)_{E_1}$ at $E_2 - E_3$ is $\operatorname{Spin}(8)$ and $\dim((F_4)_{E_1} / \operatorname{Spin}(8)) =$
0902.0431_F01113	59	■ 0.994	$\dim \left(\left(F_{\{4\}}\right)_{E_1} / \operatorname{Spin}(8) \right) =$	$\dim((F_4)_{E_1} / \operatorname{Spin}(8)) =$	The isotropy subgroup of $(F_4)_{E_1}$ at $E_2 - E_3$ is $\operatorname{Spin}(8)$ and $\dim((F_4)_{E_1} / \operatorname{Spin}(8)) =$
0902.0431_F01114	59	■ 0.892	$\dim \left(F_{\{4\}}\right)_{E_1} - \dim \operatorname{Spin}(8) = 36 - 28 = \dim S^8$	$\dim(F_4)_{E_1} - \dim \operatorname{Spin}(8) = 36 - 28 = \dim S^8$	$\dim(F_4)_{E_1} - \dim \operatorname{Spin}(8) = 36 - 28 = \dim S^8$. Therefore, we have $(F_4)_{E_1} / \operatorname{Spin}(8) \simeq$
0902.0431_F01115	59	■ 0.892	$\left(F_{\{4\}}\right)_{E_1} / \operatorname{Spin}(8) \simeq$	$(F_4)_{E_1} / \operatorname{Spin}(8) \simeq$	$\dim(F_4)_{E_1} - \dim \operatorname{Spin}(8) = 36 - 28 = \dim S^8$. Therefore, we have $(F_4)_{E_1} / \operatorname{Spin}(8) \simeq$
0902.0431_F01116	59	■ 1.000	$\left(F_{\{4\}}\right)_{E_1} \cong \operatorname{Spin}(9)$	$(F_4)_{E_1} \cong \operatorname{Spin}(9)$	Theorem 2.7.4. $(F_4)_{E_1} \cong \operatorname{Spin}(9)$.
0902.0431_F01117	59	■ 1.000	$O(9) = O(\mathfrak{J}_{01}) = \left\{ \alpha' \in \operatorname{Iso}_{\mathbf{R}}(\mathfrak{J}_{01}) \mid (\alpha' X, \alpha' Y) = (X, Y) \right\}$	$O(9) = O(\mathfrak{J}_{01}) = \{\alpha' \in \operatorname{Iso}_{\mathbf{R}}(\mathfrak{J}_{01}) \mid (\alpha' X, \alpha' Y) = (X, Y)\}$	Proof. Let $O(9) = O(\mathfrak{J}_{01}) = \{\alpha' \in \operatorname{Iso}_{\mathbf{R}}(\mathfrak{J}_{01}) \mid (\alpha' X, \alpha' Y) = (X, Y)\}$. Consider
0902.0431_F01118	59	■ 1.000	$\alpha' = \alpha \mid \mathfrak{J}_{01}$	$\alpha' = \alpha \mid \mathfrak{J}_{01}$	the restriction $\alpha' = \alpha \mid \mathfrak{J}_{01}$ of $\alpha \in (F_4)_{E_1}$ to $\mathfrak{J}_{01}$, then $\alpha' \in O(9)$. Hence we can define
0902.0431_F01119	59	■ 1.000	$\mathfrak{J}_{01}$	$\mathfrak{J}_{01}$	the restriction $\alpha' = \alpha \mid \mathfrak{J}_{01}$ of $\alpha \in (F_4)_{E_1}$ to $\mathfrak{J}_{01}$, then $\alpha' \in O(9)$. Hence we can define
0902.0431_F01120	59	■ 1.000	$\alpha' \in O(9)$	$\alpha' \in O(9)$	the restriction $\alpha' = \alpha \mid \mathfrak{J}_{01}$ of $\alpha \in (F_4)_{E_1}$ to $\mathfrak{J}_{01}$, then $\alpha' \in O(9)$. Hence we can define
0902.0431_F01121	59	■ 1.000	$p: \left(F_{\{4\}}\right)_{E_1} \rightarrow O(9)$	$p: (F_4)_{E_1} \rightarrow O(9)$	a homomorphism $p: (F_4)_{E_1} \rightarrow O(9)$ by $p(\alpha) = \alpha'$. Since p is continuous and $(F_4)_{E_1}$
0902.0431_F01122	59	■ 1.000	$p(\alpha) = \alpha'$	$p(\alpha) = \alpha'$	a homomorphism $p: (F_4)_{E_1} \rightarrow O(9)$ by $p(\alpha) = \alpha'$. Since p is continuous and $(F_4)_{E_1}$
0902.0431_F01123	59	■ 1.000	$SO(8) = \left\{ \alpha' \in SO(9) \mid \alpha' \left(E_2 - E_3\right) = E_2 - E_3 \right\}$	$SO(8) = \{\alpha' \in SO(9) \mid \alpha'(E_2 - E_3) = E_2 - E_3\}$	We shall show p is onto. Let $SO(8) = \{\alpha' \in SO(9) \mid \alpha'(E_2 - E_3) = E_2 - E_3\}$. The
0902.0431_F01124	59	■ 0.643	p'	p'	restriction p' of $p: (F_4)_{E_1} \rightarrow SO(9)$ to $\operatorname{Spin}(8) = \{\alpha \in (F_4)_{E_1} \mid \alpha(E_2 - E_3) = E_2 - E_3\}$
0902.0431_F01125	59	■ 0.643	$p: \left(F_{\{4\}}\right)_{E_1} \rightarrow SO(9)$	$p: (F_4)_{E_1} \rightarrow SO(9)$	restriction p' of $p: (F_4)_{E_1} \rightarrow SO(9)$ to $\operatorname{Spin}(8) = \{\alpha \in (F_4)_{E_1} \mid \alpha(E_2 - E_3) = E_2 - E_3\}$

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0902.0431_F01126	59	0.643	$\operatorname{Spin}(8)=\left\{\alpha\in(F_4)_{E_1}\mid\alpha(E_2-E_3)=E_2-E_3\right\}$	$\operatorname{Spin}(8)=\{\alpha\in(F_4)_{E_1}\mid\alpha(E_2-E_3)=E_2-E_3\}$	restriction p' of $p:(F_4)_{E_1}\rightarrow SO(9)$ to $\mathit{Spin}(8)=\{\alpha\in(F_4)_{E_1}\mid\alpha(E_2-E_3)=E_2-E_3\}$
0902.0431_F01127	59	0.999	$p:\operatorname{Spin}(9)\rightarrow S_0(8)$	$p:\operatorname{Spin}(9)\rightarrow SO(8)$	coincides with the homomorphism $p:\operatorname{Spin}(9)\rightarrow SO(8)$ in Theorem 1.16.2. In
0902.0431_F01128	59	0.912	$p^{\prime}:\operatorname{Spin}(8)\rightarrow S_0(8)$	$p':\operatorname{Spin}(8)\rightarrow SO(8)$	particular, $p':\operatorname{Spin}(8)\rightarrow SO(8)$ is onto. Hence, from the following commutative
0902.0431_F01129	59	1.000	$\operatorname{Ker}p=\{1,\sigma\}$	$\operatorname{Ker}p=\{1,\sigma\}$	we see that $p:(F_4)_{E_1}\rightarrow SO(9)$ is onto by the five lemma. $\operatorname{Ker}p=\{1,\sigma\}$, where $\sigma=$
0902.0431_F01130	59	1.000	$\sigma=$	$\sigma=$	we see that $p:(F_4)_{E_1}\rightarrow SO(9)$ is onto by the five lemma. $\operatorname{Ker}p=\{1,\sigma\}$, where $\sigma=$
0902.0431_F01131	59	1.000	$\varphi(1,-1,-1)$	$\varphi(1,-1,-1)$	$\varphi(1,-1,-1)$ (which corresponds to σ defined in the following Section 2.9). Indeed,
0902.0431_F01132	59	1.000	$\alpha\in\operatorname{Ker}p$	$\alpha\in\operatorname{Ker}p$	let $\alpha\in\operatorname{Ker}p$, then α satisfies $\alpha X=X$ for all $X\in\mathfrak{J}_{01}$. From $\alpha(E_2-E_3)=E_2-E_3$
0902.0431_F01133	59	1.000	$\alpha X=X$	$\alpha X=X$	let $\alpha\in\operatorname{Ker}p$, then α satisfies $\alpha X=X$ for all $X\in\mathfrak{J}_{01}$. From $\alpha(E_2-E_3)=E_2-E_3$
0902.0431_F01134	59	1.000	$X\in\mathfrak{J}_{01}$	$X\in\mathfrak{J}_{01}$	let $\alpha\in\operatorname{Ker}p$, then α satisfies $\alpha X=X$ for all $X\in\mathfrak{J}_{01}$. From $\alpha(E_2-E_3)=E_2-E_3$
0902.0431_F01135	59	1.000	$\alpha=(\alpha_1,\alpha_2,\alpha_3)\in\operatorname{Spin}(8)$	$\alpha=(\alpha_1,\alpha_2,\alpha_3)\in\operatorname{Spin}(8)$	follows that $\alpha\in\operatorname{Spin}(8)$. Let $\alpha=(\alpha_1,\alpha_2,\alpha_3)\in\operatorname{Spin}(8)$. Since $\alpha F_1(x)=F_1(x)$,
0902.0431_F01136	59	1.000	$\alpha F_1(x)=F_1(x)$	$\alpha F_1(x)=F_1(x)$	follows that $\alpha\in\operatorname{Spin}(8)$. Let $\alpha=(\alpha_1,\alpha_2,\alpha_3)\in\operatorname{Spin}(8)$. Since $\alpha F_1(x)=F_1(x)$,
0902.0431_F01137	59	1.000	$F_1(\alpha_1x)=F_1(x)$	$F_1(\alpha_1x)=F_1(x)$	that is, $F_1(\alpha_1x)=F_1(x)$, so $\alpha_1x=x$ for all $x\in\mathfrak{C}$, hence we have $\alpha_1=1$. From the
0902.0431_F01138	59	1.000	$\alpha_1x=x$	$\alpha_1x=x$	that is, $F_1(\alpha_1x)=F_1(x)$, so $\alpha_1x=x$ for all $x\in\mathfrak{C}$, hence we have $\alpha_1=1$. From the
0902.0431_F01139	59	1.000	$\alpha_1=1$	$\alpha_1=1$	that is, $F_1(\alpha_1x)=F_1(x)$, so $\alpha_1x=x$ for all $x\in\mathfrak{C}$, hence we have $\alpha_1=1$. From the
0902.0431_F01140	59	1.000	$\alpha=(1,1,1)=1$	$\alpha=(1,1,1)=1$	principle of triality, we have $\alpha=(1,1,1)=1$ or $\alpha=(1,-1,-1)=\sigma$. Hence we have
0902.0431_F01141	59	1.000	$\alpha=(1,-1,-1)=\sigma$	$\alpha=(1,-1,-1)=\sigma$	principle of triality, we have $\alpha=(1,1,1)=1$ or $\alpha=(1,-1,-1)=\sigma$. Hence we have
0902.0431_F01142	59	1.000	$SO(9)$	$SO(9)$	of $SO(9)$.
0902.0431_F01143	60	0.970	$\operatorname{Spin}(9)/\operatorname{Spin}'(7)\simeq S^{15}$	$\operatorname{Spin}(9)/\operatorname{Spin}'(7)\simeq S^{15}$	Theorem 2.7.5. $\operatorname{Spin}(9)/\operatorname{Spin}'(7)\simeq S^{15}$.
0902.0431_F01144	60	0.985	$\operatorname{Spin}'(7)=\{\tilde{\alpha}\in SO(8)\mid(\tilde{\alpha}x)(\alpha y)=\tilde{\alpha}(xy),x,y\in\mathfrak{C}\}$	$\operatorname{Spin}'(7)=\{\tilde{\alpha}\in SO(8)\mid(\tilde{\alpha}x)(\alpha y)=\tilde{\alpha}(xy),x,y\in\mathfrak{C}\}$	where $\operatorname{Spin}'(7)=\{\tilde{\alpha}\in SO(8)\mid(\tilde{\alpha}x)(\alpha y)=\tilde{\alpha}(xy),x,y\in\mathfrak{C}\}$ for some $\alpha\in SO(7)\}$.
0902.0431_F01145	60	0.985	$\alpha\in SO(7)$	$\alpha\in SO(7)$	where $\operatorname{Spin}'(7)=\{\tilde{\alpha}\in SO(8)\mid(\tilde{\alpha}x)(\alpha y)=\tilde{\alpha}(xy),x,y\in\mathfrak{C}\}$ for some $\alpha\in SO(7)\}$.
0902.0431_F01146	60	0.999	$S^{15}=\{Y\in\mathfrak{J}_{23}\mid(Y,Y)=2\}$	$S^{15}=\{Y\in\mathfrak{J}_{23}\mid(Y,Y)=2\}$	Proof. Let $S^{15}=\{Y\in\mathfrak{J}_{23}\mid(Y,Y)=2\}$. For $\alpha\in\operatorname{Spin}(9)$ and $Y\in S^{15}$, we have
0902.0431_F01147	60	0.999	$\alpha\in\operatorname{Spin}(9)$	$\alpha\in\operatorname{Spin}(9)$	Proof. Let $S^{15}=\{Y\in\mathfrak{J}_{23}\mid(Y,Y)=2\}$. For $\alpha\in\operatorname{Spin}(9)$ and $Y\in S^{15}$, we have
0902.0431_F01148	60	0.999	$Y\in S^{15}$	$Y\in S^{15}$	Proof. Let $S^{15}=\{Y\in\mathfrak{J}_{23}\mid(Y,Y)=2\}$. For $\alpha\in\operatorname{Spin}(9)$ and $Y\in S^{15}$, we have
0902.0431_F01149	60	1.000	$\alpha Y\in S^{15}$	$\alpha Y\in S^{15}$	$\alpha Y\in S^{15}$. Hence $\operatorname{Spin}(9)$ acts on S^{15} . We shall show that this action is transitive.
0902.0431_F01150	60	1.000	S^{15}	S^{15}	$\alpha Y\in S^{15}$. Hence $\operatorname{Spin}(9)$ acts on S^{15} . We shall show that this action is transitive.
0902.0431_F01151	60	1.000	$Y_0\in S^{15}$	$Y_0\in S^{15}$	Let $Y_0\in S^{15}$. Choose any element $A\in\mathfrak{J}_{01}$ such that $(A,A)=2,2A\circ Y_0=-Y_0$.
0902.0431_F01152	60	1.000	$(A,A)=2,2A\circ Y_0=-Y_0$	$(A,A)=2,2A\circ Y_0=-Y_0$	Let $Y_0\in S^{15}$. Choose any element $A\in\mathfrak{J}_{01}$ such that $(A,A)=2,2A\circ Y_0=-Y_0$.
0902.0431_F01153	60	1.000	Y_0	Y_0	Using these A and Y_0 , construct X_i,Y_i,Z_i and α of Lemma 2.7.2, then $\alpha\in\operatorname{Spin}(9)$
0902.0431_F01154	60	1.000	X_i,Y_i,Z_i	X_i,Y_i,Z_i	Using these A and Y_0 , construct X_i,Y_i,Z_i and α of Lemma 2.7.2, then $\alpha\in\operatorname{Spin}(9)$
0902.0431_F01155	60	1.000	$\alpha F_2(1)=Y_0$	$\alpha F_2(1)=Y_0$	and satisfies $\alpha F_2(1)=Y_0$, which shows the transitivity. We determine the isotropy
0902.0431_F01156	60	0.999	$\operatorname{Spin}(9)_{F_2(1)}$	$\operatorname{Spin}(9)_{F_2(1)}$	subgroup $\operatorname{Spin}(9)_{F_2(1)}$ of $\operatorname{Spin}(9)$ at $F_2(1)\in S^{15}$. Let $\alpha\in\operatorname{Spin}(9)$ satisfies $\alpha F_2(1)=$
0902.0431_F01157	60	0.999	$F_2(1)\in S^{15}$	$F_2(1)\in S^{15}$	subgroup $\operatorname{Spin}(9)_{F_2(1)}$ of $\operatorname{Spin}(9)$ at $F_2(1)\in S^{15}$. Let $\alpha\in\operatorname{Spin}(9)$ satisfies $\alpha F_2(1)=$
0902.0431_F01158	60	0.999	$\alpha F_2(1)=$	$\alpha F_2(1)=$	subgroup $\operatorname{Spin}(9)_{F_2(1)}$ of $\operatorname{Spin}(9)$ at $F_2(1)\in S^{15}$. Let $\alpha\in\operatorname{Spin}(9)$ satisfies $\alpha F_2(1)=$
0902.0431_F01159	60	1.000	$F_2(1)$	$F_2(1)$	$F_2(1)$. Applying α on $F_2(1)\circ F_2(1)=E_1+E_3$ we have $F_2(1)\circ F_2(1)=E_1+\alpha E_3$, so we
0902.0431_F01160	60	1.000	$F_2(1)\circ F_2(1)=E_1+E_3$	$F_2(1)\circ F_2(1)=E_1+E_3$	$F_2(1)$. Applying α on $F_2(1)\circ F_2(1)=E_1+E_3$ we have $F_2(1)\circ F_2(1)=E_1+\alpha E_3$, so we
0902.0431_F01161	60	1.000	$F_2(1)\circ F_2(1)=E_1+\alpha E_3$	$F_2(1)\circ F_2(1)=E_1+\alpha E_3$	$F_2(1)$. Applying α on $F_2(1)\circ F_2(1)=E_1+E_3$ we have $F_2(1)\circ F_2(1)=E_1+\alpha E_3$, so we
0902.0431_F01162	60	1.000	$\alpha(E_2+E_3)=E_2+E_3$	$\alpha(E_2+E_3)=E_2+E_3$	get $\alpha E_3=E_3$. Since α always satisfies $\alpha(E_2+E_3)=E_2+E_3$, we have $\alpha E_2=E_2$, and
0902.0431_F01163	60	0.999	$\alpha=(\alpha_1,\alpha_2,\alpha_3)$	$\alpha=(\alpha_1,\alpha_2,\alpha_3)$	so $\alpha\in\operatorname{Spin}(8)$. Denote $\alpha=(\alpha_1,\alpha_2,\alpha_3)$, then $F_2(\alpha_2(1))=F_2(1)$ implies $\alpha_21=1$,
0902.0431_F01164	60	0.999	$F_2(\alpha_2(1))=F_2(1)$	$F_2(\alpha_2(1))=F_2(1)$	so $\alpha\in\operatorname{Spin}(8)$. Denote $\alpha=(\alpha_1,\alpha_2,\alpha_3)$, then $F_2(\alpha_2(1))=F_2(1)$ implies $\alpha_21=1$,
0902.0431_F01165	60	0.999	$\alpha_21=1$	$\alpha_21=1$	so $\alpha\in\operatorname{Spin}(8)$. Denote $\alpha=(\alpha_1,\alpha_2,\alpha_3)$, then $F_2(\alpha_2(1))=F_2(1)$ implies $\alpha_21=1$,
0902.0431_F01166	60	0.943	$\alpha_2\in SO(7)$	$\alpha_2\in SO(7)$	so that $\alpha_2\in SO(7)$. Hence $\alpha=(\alpha_1,\alpha_2,\alpha_3)\in\operatorname{Spin}'(7)$. Conversely $\alpha\in\operatorname{Spin}(7)$
0902.0431_F01167	60	0.943	$\alpha=(\alpha_1,\alpha_2,\alpha_3)\in\operatorname{Spin}'(7)$	$\alpha=(\alpha_1,\alpha_2,\alpha_3)\in\operatorname{Spin}'(7)$	so that $\alpha_2\in SO(7)$. Hence $\alpha=(\alpha_1,\alpha_2,\alpha_3)\in\operatorname{Spin}'(7)$. Conversely $\alpha\in\operatorname{Spin}(7)$
0902.0431_F01168	60	0.943	$\alpha\in\operatorname{Spin}(7)$	$\alpha\in\operatorname{Spin}(7)$	so that $\alpha_2\in SO(7)$. Hence $\alpha=(\alpha_1,\alpha_2,\alpha_3)\in\operatorname{Spin}'(7)$. Conversely $\alpha\in\operatorname{Spin}(7)$

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0902.0431_F01169	60	■ 1.000	$\alpha F_{\{2\}}(1)=F_{\{2\}}(1)$	$\alpha F_2(1) = F_2(1)$	satisfies $\alpha F_2(1) = F_2(1)$. Therefore $Spin(9)_{F_2(1)} = Spin'(7)$ is proved. Thus we have
0902.0431_F01170	60	■ 1.000	$\operatorname{Spin}(9)_{F_{\{2\}}(1)}=\operatorname{Spin}^{\prime}(7)$	$Spin(9)_{F_2(1)} = Spin'(7)$	satisfies $\alpha F_2(1) = F_2(1)$. Therefore $Spin(9)_{F_2(1)} = Spin'(7)$ is proved. Thus we have
0902.0431_F01171	60	■ 0.991	$\operatorname{Spin}(9) / \operatorname{Spin}^{\prime}(7) \simeq S^{15}$	$Spin(9)/Spin'(7) \simeq S^{15}$	the homeomorphism $Spin(9)/Spin'(7) \simeq S^{15}$.
0902.0431_F01172	60	■ 0.870	$\left(\operatorname{Spin}^{\prime}(7)\right)$	$(Spin'(7))$	$(Spin'(7))$ and $Spin(7)$ are conjugate in the group $O(7)$. Indeed, a mapping $f :$
0902.0431_F01173	60	■ 0.870	$f:$	$f :$	$(Spin'(7))$ and $Spin(7)$ are conjugate in the group $O(7)$. Indeed, a mapping $f :$
0902.0431_F01174	60	■ 0.992	$\operatorname{Spin}^{\prime}(7) \rightarrow \operatorname{Spin}(7)$	$Spin'(7) \rightarrow Spin(7)$	$Spin'(7) \rightarrow Spin(7)$,
0902.0431_F01175	60	■ 0.687	$\left.\operatorname{Spin}^{\prime}(7)\right \cong \operatorname{Spin}(7)$	$Spin'(7) \cong Spin(7)$	gives the conjugation. In particular, $Spin'(7) \cong Spin(7)$
0902.0431_F01176	60	■ 0.856	$\left(F_4\right)_0$	$(F_4)_0$	We denote by $(F_4)_0$ the connected component of F_4 containing the identity 1.
0902.0431_F01177	60	■ 1.000	$\beta_1(a): \mathfrak{J} \rightarrow \mathfrak{J}$	$\beta_1(a): \mathfrak{J} \rightarrow \mathfrak{J}$	Lemma 2.8.1. For $a \in \mathfrak{C}$, we define a mapping $\beta_1(a): \mathfrak{J} \rightarrow \mathfrak{J}$ by $\beta_1(a)X(\xi, x) =$
0902.0431_F01178	60	■ 1.000	$\beta_1(a) X(\xi, x)=$	$\beta_1(a)X(\xi, x) =$	Lemma 2.8.1. For $a \in \mathfrak{C}$, we define a mapping $\beta_1(a): \mathfrak{J} \rightarrow \mathfrak{J}$ by $\beta_1(a)X(\xi, x) =$
0902.0431_F01179	60	■ 1.000	$Y(\eta, y)$	$Y(\eta, y)$	$Y(\eta, y)$, where
0902.0431_F01180	60	■ 0.995	$a=0$	$a = 0$	$\left(\text{if } a=0, \text{ then } \frac{\sin a }{ a } \text{ means } 1\right)$, then $\beta_1(a) \in (F_4)_0$.
0902.0431_F01181	60	■ 0.995	$\frac{\sin a }{ a }$	$\frac{\sin a }{ a }$	$\left(\text{if } a = 0, \text{ then } \frac{\sin a }{ a } \text{ means } 1\right)$, then $\beta_1(a) \in (F_4)_0$.
0902.0431_F01182	60	■ 0.995	$\beta_1(a) \in\left(F_4\right)_0$	$\beta_1(a) \in (F_4)_0$	$\left(\text{if } a = 0, \text{ then } \frac{\sin a }{ a } \text{ means } 1\right)$, then $\beta_1(a) \in (F_4)_0$.
0902.0431_F01183	60	■ 1.000	$A_1(a)=\left(\begin{array}{ccc} 0 & 0 & 0 \\ 0 & 0 & a \\ 0 & -\bar{a} & 0 \end{array}\right)$	$A_1(a)=\left(\begin{array}{ccc} 0 & 0 & 0 \\ 0 & 0 & a \\ 0 & -\bar{a} & 0 \end{array}\right)$	Proof. For $A_1(a)=\left(\begin{array}{ccc} 0 & 0 & 0 \\ 0 & 0 & a \\ 0 & -\bar{a} & 0 \end{array}\right)$, we have $\tilde{A}_1(a) \in \mathfrak{f}_4$ (Proposition 2.3.6) and
0902.0431_F01184	60	■ 1.000	$\tilde{A}_1(a) \in \mathfrak{f}_4$	$\tilde{A}_1(a) \in \mathfrak{f}_4$	Proof. For $A_1(a)=\left(\begin{array}{ccc} 0 & 0 & 0 \\ 0 & 0 & a \\ 0 & -\bar{a} & 0 \end{array}\right)$, we have $\tilde{A}_1(a) \in \mathfrak{f}_4$ (Proposition 2.3.6) and
0902.0431_F01185	61	■ 0.896	$\beta_1(a)=\exp \tilde{A}_1(a)$	$\beta_1(a) = \exp \tilde{A}_1(a)$	$\beta_1(a) = \exp A_1(a)$. Hence $\beta_1(a) \in (F_4)_0$.
0902.0431_F01186	61	■ 1.000	$\alpha \in\left(F_4\right)_0$	$\alpha \in (F_4)_0$	by some element $\alpha \in (F_4)_0$:
0902.0431_F01187	61	■ 1.000	ξ_1, ξ_2, ξ_3	ξ_1, ξ_2, ξ_3	Moreover, ξ_1, ξ_2, ξ_3 are uniquely determined (up to their permutation) independent of
0902.0431_F01188	61	■ 1.000	$\mathfrak{X}=\left\{\alpha X \mid \alpha \in\left(F_4\right)_0\right\}$	$\mathfrak{X} = \{\alpha X \mid \alpha \in (F_4)_0\}$	Proof. For a given $X \in \mathfrak{J}$, consider a space $\mathfrak{X} = \{\alpha X \mid \alpha \in (F_4)_0\}$. Since $(F_4)_0$ is
0902.0431_F01189	61	■ 1.000	$\mathfrak{X}$	$\mathfrak{X}$	compact (Theorem 2.2.5), $\mathfrak{X}$ is also compact. Let $\xi_1^2+\xi_2^2+\xi_3^2$ be the maximal value
0902.0431_F01190	61	■ 1.000	$\xi_1^2+\xi_2^2+\xi_3^2$	$\xi_1^2+\xi_2^2+\xi_3^2$	compact (Theorem 2.2.5), $\mathfrak{X}$ is also compact. Let $\xi_1^2+\xi_2^2+\xi_3^2$ be the maximal value
0902.0431_F01191	61	■ 0.999	$\eta_1^2+\eta_2^2+\eta_3^2$	$\eta_1^2+\eta_2^2+\eta_3^2$	of all $\eta_1^2+\eta_2^2+\eta_3^2$ for $Y=Y(\eta, y) \in \mathfrak{X}$ and let $X_0=X(\xi, x)$ be an element of $\mathfrak{X}$
0902.0431_F01192	61	■ 0.999	$Y=Y(\eta, y) \in \mathfrak{X}$	$Y=Y(\eta, y) \in \mathfrak{X}$	of all $\eta_1^2+\eta_2^2+\eta_3^2$ for $Y=Y(\eta, y) \in \mathfrak{X}$ and let $X_0=X(\xi, x)$ be an element of $\mathfrak{X}$
0902.0431_F01193	61	■ 0.999	$X_0=X(\xi, x)$	$X_0=X(\xi, x)$	of all $\eta_1^2+\eta_2^2+\eta_3^2$ for $Y=Y(\eta, y) \in \mathfrak{X}$ and let $X_0=X(\xi, x)$ be an element of $\mathfrak{X}$
0902.0431_F01194	61	■ 1.000	X_0	X_0	which attains its maximal value. Then X_0 is of diagonal form. Certainly, suppose X_0
0902.0431_F01195	61	■ 0.999	2×3	2×3	is not of diagonal form, for example, the 2×3 entry x_1 of X_0 is non-zero: $x_1 \neq 0$. Let
0902.0431_F01196	61	■ 0.999	x_1	x_1	is not of diagonal form, for example, the 2×3 entry x_1 of X_0 is non-zero: $x_1 \neq 0$. Let
0902.0431_F01197	61	■ 1.000	$a(t)=\frac{x_1}{ x_1 } t, t>0$	$a(t)=\frac{x_1}{ x_1 } t, t>0$	$a(t)=\frac{x_1}{ x_1 } t, t>0$, and construct $\beta_1(a(t)) \in (F_4)_0$ of Lemma 2.8.1. Since $ a(t) = t$
0902.0431_F01198	61	■ 1.000	$\beta_1(a(t)) \in\left(F_4\right)_0$	$\beta_1(a(t)) \in (F_4)_0$	$a(t)=\frac{x_1}{ x_1 } t, t>0$, and construct $\beta_1(a(t)) \in (F_4)_0$ of Lemma 2.8.1. Since $ a(t) = t$
0902.0431_F01199	61	■ 1.000	$ a(t) =t$	$ a(t) = t$	$a(t)=\frac{x_1}{ x_1 } t, t>0$, and construct $\beta_1(a(t)) \in (F_4)_0$ of Lemma 2.8.1. Since $ a(t) = t$
0902.0431_F01200	61	■ 1.000	$\frac{(a(t), x_1)}{ a(t) }= x_1 $	$\frac{(a(t), x_1)}{ a(t) } = x_1 $	and $\frac{(a(t), x_1)}{ a(t) } = x_1 $ for $Y(\eta(t), y(t)) = \beta_1(a(t))X_0 \in \mathfrak{X}$, we have
0902.0431_F01201	61	■ 1.000	$Y(\eta(t), y(t))=\beta_1(a(t)) X_0 \in \mathfrak{X}$	$Y(\eta(t), y(t)) = \beta_1(a(t))X_0 \in \mathfrak{X}$	and $\frac{(a(t), x_1)}{ a(t) } = x_1 $ for $Y(\eta(t), y(t)) = \beta_1(a(t))X_0 \in \mathfrak{X}$, we have

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0902.0431_F01240	63	■ 0.992	$\backslash\mathrm{quad}\backslash\mathrm{left}(F_{\{4\}}\backslash\mathrm{right})^{\{\sigma\}}=\backslash\mathrm{left}(F_{\{4\}}\backslash\mathrm{right})_{\{E_{\{1\}}\}}=\operatornamename{Spin}(9)$	$(F_4)^\sigma = (F_4)_{E_1} = \operatorname{Spin}(9)$	Theorem 2.9.1. $(F_4)^\sigma = (F_4)_{E_1} = \operatorname{Spin}(9)$.
0902.0431_F01241	63	■ 1.000	$\backslash\alpha\in\backslash\mathrm{left}(F_{\{4\}}\backslash\mathrm{right})^{\{\sigma\}}$	$\alpha \in (F_4)^\sigma$	Proof. We shall show that for $\alpha \in (F_4)^\sigma$ we have $\alpha E_1 = E_1$. Let $\alpha \in (F_4)^\sigma$. We
0902.0431_F01242	63	■ 1.000	$\backslash\alpha E_{\{3\}}\in\backslash\mathrm{mathfrak{J}}(2,\backslash\mathrm{mathfrak{C}})$	$\alpha E_3 \in \mathfrak{J}(2, \mathfrak{C})$	Similarly $\alpha E_3 \in \mathfrak{J}(2, \mathfrak{C})$. So $\alpha E_1 = \alpha(E - E_2 - E_3) = E - \alpha E_2 - \alpha E_3$ is of the form
0902.0431_F01243	63	■ 1.000	$\backslash\alpha E_{\{1\}}=\backslash\alpha\backslash\mathrm{left}(E-E_{\{2\}}-E_{\{3\}}\backslash\mathrm{right})=E-\backslash\alpha E_{\{2\}}-\backslash\alpha E_{\{3\}}$	$\alpha E_1 = \alpha(E - E_2 - E_3) = E - \alpha E_2 - \alpha E_3$	Similarly $\alpha E_3 \in \mathfrak{J}(2, \mathfrak{C})$. So $\alpha E_1 = \alpha(E - E_2 - E_3) = E - \alpha E_2 - \alpha E_3$ is of the form
0902.0431_F01244	64	■ 1.000	$\backslash x_{\{2\}}=\backslash x_{\{3\}}=x_{\{1\}}=0$	$\xi_2 = \xi_3 = x_1 = 0$	so that $\xi_2 = \xi_3 = x_1 = 0$, therefore $\alpha E_1 = E_1$, that is, $\alpha \in (F_4)_{E_1}$. Conversely
0902.0431_F01245	64	■ 1.000	$\backslash\mathrm{mathfrak{J}}_{\{\sigma\}}=\backslash\mathrm{left}\{X\in\backslash\mathrm{mathfrak{J}}\mid E_1\circ X=0\}\oplus\backslash\mathrm{mathfrak{C}}_1,\backslash\mathrm{mathfrak{J}}_{-\sigma}=\backslash\mathrm{left}\{X\in\backslash\mathrm{mathfrak{J}}\mid 2E_1\circ X=X\}$	$\mathfrak{J}_\sigma = \{X \in \mathfrak{J} \mid E_1 \circ X = 0\} \oplus \mathfrak{C}_1, \mathfrak{J}_{-\sigma} = \{X \in \mathfrak{J} \mid 2E_1 \circ X = X\}$	$\mathfrak{J}_\sigma = \{X \in \mathfrak{J} \mid E_1 \circ X = 0\} \oplus \mathfrak{C}_1, \mathfrak{J}_{-\sigma} = \{X \in \mathfrak{J} \mid 2E_1 \circ X = X\}$ are invariant under
0902.0431_F01246	64	■ 1.000	$\backslash\alpha\sigma=\sigma\alpha$	$\alpha\sigma = \sigma\alpha$	Thus $\alpha\sigma = \sigma\alpha$, and so $\alpha \in (F_4)^\sigma$. Therefore we have shown that $(F_4)^\sigma = (F_4)_{E_1} \cong$
0902.0431_F01247	64	■ 1.000	$\backslash\mathrm{left}(F_{\{4\}}\backslash\mathrm{right})^{\{\sigma\}}=\backslash\mathrm{left}(F_{\{4\}}\backslash\mathrm{right})_{\{E_{\{1\}}\}}\backslash\mathrm{cong}$	$(F_4)^\sigma = (F_4)_{E_1} \cong$	Thus $\alpha\sigma = \sigma\alpha$, and so $\alpha \in (F_4)^\sigma$. Therefore we have shown that $(F_4)^\sigma = (F_4)_{E_1} \cong$
0902.0431_F01248	64	■ 1.000	$\backslash\alpha\in z\backslash\mathrm{left}(F_{\{4\}}\backslash\mathrm{right})$	$\alpha \in z(F_4)$	Proof. Let $\alpha \in z(F_4)$. From the commutativity with σ : $\sigma\alpha = \alpha\sigma$, we have
0902.0431_F01249	64	■ 1.000	$\backslash\sigma:\backslash\sigma\alpha=\alpha\sigma$	$\sigma : \sigma\alpha = \alpha\sigma$	Proof. Let $\alpha \in z(F_4)$. From the commutativity with σ : $\sigma\alpha = \alpha\sigma$, we have
0902.0431_F01250	64	■ 1.000	$\backslash\alpha\in z\backslash\mathrm{operatorname{Spin}}(9)$	$\alpha \in z(\operatorname{Spin}(9))$	$\alpha \in \operatorname{Spin}(9)$ (Theorem 2.9.1) and so $\alpha \in z(\operatorname{Spin}(9))$. Since the center $z(\operatorname{Spin}(9))$ of
0902.0431_F01251	64	■ 1.000	$z\backslash\mathrm{operatorname{Spin}}(9)$	$z(\operatorname{Spin}(9))$	$\alpha \in \operatorname{Spin}(9)$ (Theorem 2.9.1) and so $\alpha \in z(\operatorname{Spin}(9))$. Since the center $z(\operatorname{Spin}(9))$ of
0902.0431_F01252	64	■ 1.000	$\backslash\sigma\notin z\backslash\mathrm{left}(F_{\{4\}}\backslash\mathrm{right})$	$\sigma \notin z(F_4)$	However, $\sigma \notin z(F_4)$ (Theorem 2.9.1), which implies that $\alpha = 1$.
0902.0431_F01253	64	■ 1.000	$F_{\{4\}}=\backslash\mathrm{left}\{\backslash\alpha\in\backslash\mathrm{operatorname{Iso}}_{\backslash\mathrm{boldsymbol{R}}}\backslash\mathrm{mathfrak{J}}\}\mid\backslash\alpha(X\circ Y)=\alpha X\circ\alpha Y\}$	$F_4 = \{\alpha \in \operatorname{Iso}_{\mathbf{R}}(\mathfrak{J}) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y\}$	Theorem 2.10.2. $F_4 = \{\alpha \in \operatorname{Iso}_{\mathbf{R}}(\mathfrak{J}) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y\}$ is a simply connected
0902.0431_F01254	64	■ 1.000	$\backslash\gamma\in G_2\subset F_4$	$\gamma \in G_2 \subset F_4$	This γ is the same as $\gamma \in G_2 \subset F_4$.
0902.0431_F01255	65	■ 1.000	$\backslash\mathrm{left}(F_{\{4\}}\backslash\mathrm{right})^{\{\gamma\}}$	$(F_4)^\gamma$	We shall study the following subgroup $(F_4)^\gamma$ of F_4 :
0902.0431_F01256	65	■ 1.000	$x_{\{i\}}=m_{\{i\}}+a_{\{i\}}e_{\{4\}}\in\backslash\mathrm{boldsymbol{H}}\oplus\backslash\mathrm{boldsymbol{H}}e_{\{4\}}=\backslash\mathrm{mathfrak{C}}$	$x_i = m_i + a_i e_4 \in \mathbf{H} \oplus \mathbf{H}e_4 = \mathfrak{C}$	where $x_i = m_i + a_i e_4 \in \mathbf{H} \oplus \mathbf{H}e_4 = \mathfrak{C}$. We associate such element $X \in \mathfrak{J}$ with the
0902.0431_F01257	65	■ 1.000	$\backslash\mathrm{mathfrak{J}}(3,\backslash\mathrm{boldsymbol{H}})\oplus\backslash\mathrm{boldsymbol{H}}^3$	$\mathfrak{J}(3, \mathbf{H}) \oplus \mathbf{H}^3$	element of $\mathfrak{J}(3, \mathbf{H}) \oplus \mathbf{H}^3$,
0902.0431_F01258	65	■ 0.998	$\backslash\mathrm{mathfrak{J}}(3,\backslash\mathrm{boldsymbol{H}})$	$\mathfrak{J}(3, \mathbf{H})$	(In $\mathfrak{J}(3, \mathbf{H})$ the multiplication $M \times N$ and the inner product (M, N) are analogously
0902.0431_F01259	65	■ 0.998	$M\times N$	$M \times N$	(In $\mathfrak{J}(3, \mathbf{H})$ the multiplication $M \times N$ and the inner product (M, N) are analogously
0902.0431_F01260	65	■ 0.998	(M,N)	(M, N)	(In $\mathfrak{J}(3, \mathbf{H})$ the multiplication $M \times N$ and the inner product (M, N) are analogously
0902.0431_F01261	65	■ 1.000	$\backslash\mathrm{boldsymbol{H}}^3$	$\mathbf{H}^3$	defined as in $\mathfrak{J}$, and in $\mathbf{H}^3$ the inner product $(\mathbf{a}, \mathbf{b})$ is defined naturally by $\frac{1}{2}(\mathbf{a}^* \mathbf{b} +$
0902.0431_F01262	65	■ 1.000	$\backslash\mathrm{frac}\{1\}{2}\backslash\mathrm{left}(\backslash\mathrm{boldsymbol{a}}^*\backslash\mathrm{boldsymbol{b}}+\backslash\mathrm{right}.$	$\frac{1}{2}(\mathbf{a}^* \mathbf{b} +$	defined as in $\mathfrak{J}$, and in $\mathbf{H}^3$ the inner product $(\mathbf{a}, \mathbf{b})$ is defined naturally by $\frac{1}{2}(\mathbf{a}^* \mathbf{b} +$
0902.0431_F01263	65	■ 0.813	$\backslash\mathrm{left}.\backslash\mathrm{left}.\backslash\mathrm{boldsymbol{b}}^*\backslash\mathrm{boldsymbol{a}}\backslash\mathrm{right})\backslash\mathrm{right})$	$\mathbf{b}^* \mathbf{a}))$	$\mathbf{b}^* \mathbf{a}))$. Since these operations correspond to their respective operations in $\mathfrak{J}$, hereafter,
0902.0431_F01264	65	■ 1.000	$F_{\{4\},\backslash\mathrm{boldsymbol{H}}}$	$F_{4, \mathbf{H}}$	The group $F_{4, \mathbf{H}}$ is defined to be the automorphism group of the Jordan algebra
0902.0431_F01265	65	■ 0.568	$\backslash\mathrm{mathfrak{J}}_{\{\backslash\mathrm{boldsymbol{H}}\}}=\backslash\mathrm{mathfrak{J}}(3,\backslash\mathrm{boldsymbol{H}})$	$\mathfrak{J}_{\mathbf{H}} = \mathfrak{J}(3, \mathbf{H})$	$\mathfrak{J}_{\mathbf{H}} = \mathfrak{J}(3, \mathbf{H})$ (in which multiplications $\circ$ and $\times$ are analogously defined as in $\mathfrak{J}$):
0902.0431_F01266	65	■ 1.000	$\backslash\mathrm{quad}F_{\{4\},\backslash\mathrm{boldsymbol{H}}}\backslash\mathrm{cong}\backslash\mathrm{operatorname{Sp}}(3)/\backslash\mathrm{boldsymbol{Z}}_{\{2\}},\backslash\mathrm{quad}\backslash\mathrm{boldsymbol{Z}}_{\{2\}}=\{E,-E\}$	$F_{4, \mathbf{H}} \cong \operatorname{Sp}(3)/\mathbf{Z}_2, \quad \mathbf{Z}_2 = \{E, -E\}$	Proposition 2.11.1. $F_{4, \mathbf{H}} \cong \operatorname{Sp}(3)/\mathbf{Z}_2, \quad \mathbf{Z}_2 = \{E, -E\}$.
0902.0431_F01267	65	■ 0.964	$\backslash\mathrm{varphi}:\backslash\mathrm{operatorname{Sp}}(3)\backslash\mathrm{rightarrow}F_{\{4\},\backslash\mathrm{boldsymbol{H}}}$	$\varphi : \operatorname{Sp}(3) \rightarrow F_{4, \mathbf{H}}$	Proof. We define a mapping $\varphi : \operatorname{Sp}(3) \rightarrow F_{4, \mathbf{H}}$ by
0902.0431_F01268	65	■ 1.000	$\backslash\alpha\in F_{\{4\},\backslash\mathrm{boldsymbol{H}}}$	$\alpha \in F_{4, \mathbf{H}}$	that φ is onto. For a given $\alpha \in F_{4, \mathbf{H}}$, we consider $\alpha E_i, i = 1, 2, 3$. Since αE_i satisfies
0902.0431_F01269	65	■ 1.000	$\backslash\alpha E_{\{i\}},i=1,2,3$	$\alpha E_i, i = 1, 2, 3$	that φ is onto. For a given $\alpha \in F_{4, \mathbf{H}}$, we consider $\alpha E_i, i = 1, 2, 3$. Since αE_i satisfies
0902.0431_F01270	65	■ 1.000	$\backslash\alpha E_{\{i\}}$	αE_i	that φ is onto. For a given $\alpha \in F_{4, \mathbf{H}}$, we consider $\alpha E_i, i = 1, 2, 3$. Since αE_i satisfies
0902.0431_F01271	65	■ 1.000	$\backslash\alpha E_{\{i\}}\backslash\mathrm{circ}\backslash\alpha E_{\{i\}}=\backslash\alpha E_{\{i\}}$	$\alpha E_i \circ \alpha E_i = \alpha E_i$	$\alpha E_i \circ \alpha E_i = \alpha E_i$ and $\operatorname{tr}(\alpha E_i) = 1$ (that is, $\alpha E_i \in \mathbf{HP}_2 = \{M \in \mathfrak{J}_{\mathbf{H}} \mid M \circ M =$
0902.0431_F01272	65	■ 1.000	$\backslash\mathrm{operatorname{tr}}\backslash\mathrm{left}(\backslash\alpha E_{\{i\}}\backslash\mathrm{right})=1$	$\operatorname{tr}(\alpha E_i) = 1$	$\alpha E_i \circ \alpha E_i = \alpha E_i$ and $\operatorname{tr}(\alpha E_i) = 1$ (that is, $\alpha E_i \in \mathbf{HP}_2 = \{M \in \mathfrak{J}_{\mathbf{H}} \mid M \circ M =$
0902.0431_F01273	65	■ 1.000	$\backslash\alpha E_{\{i\}}\in\backslash\mathrm{boldsymbol{H}}\backslash\mathrm{P}_{\{2\}}=\backslash\mathrm{left}\{M\in\backslash\mathrm{mathfrak{J}}_{\{\backslash\mathrm{boldsymbol{H}}\}}\}\mid M\backslash\mathrm{circ}M=\backslash\mathrm{right}.$	$\alpha E_i \in \mathbf{HP}_2 = \{M \in \mathfrak{J}_{\mathbf{H}} \mid M \circ M =$	$\alpha E_i \circ \alpha E_i = \alpha E_i$ and $\operatorname{tr}(\alpha E_i) = 1$ (that is, $\alpha E_i \in \mathbf{HP}_2 = \{M \in \mathfrak{J}_{\mathbf{H}} \mid M \circ M =$
0902.0431_F01274	66	■ 0.852	$M,\backslash\mathrm{operatorname{tr}}(M)=1\}$	$M, \operatorname{tr}(M) = 1\}$	$M, \operatorname{tr}(M) = 1\}$ (the quaternion projective plane)), there exists $A_i \in \operatorname{Sp}(3)$ such that
0902.0431_F01275	66	■ 0.852	$\backslash A_{\{i\}}\in\backslash\mathrm{operatorname{Sp}}(3)$	$A_i \in \operatorname{Sp}(3)$	$M, \operatorname{tr}(M) = 1\}$ (the quaternion projective plane)), there exists $A_i \in \operatorname{Sp}(3)$ such that

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0902.0431_F01276	66	■ 0.999	$\alpha E_{\{i\}}=A_{\{i\}} E_{\{i\}} A_{\{i\}}^{\{ \}^{\{*\}}}$	$\alpha E_i = A_i E_i A_i^*$	$\alpha E_i = A_i E_i A_i^*$. Let $\mathbf{a}_i = \begin{pmatrix} a_{i1} \\ a_{i2} \\ a_{i3} \end{pmatrix}$ be the i -th column vector of A_i , then we have
0902.0431_F01277	66	■ 0.999	$\boldsymbol{a}_{\{i\}}=\left(\begin{array}{c}a_{\{i\} 1} \\ a_{\{i\} 2} \\ a_{\{i\} 3}\end{array}\right)$	$\mathbf{a}_i = \begin{pmatrix} a_{i1} \\ a_{i2} \\ a_{i3} \end{pmatrix}$	$\alpha E_i = A_i E_i A_i^*$. Let $\mathbf{a}_i = \begin{pmatrix} a_{i1} \\ a_{i2} \\ a_{i3} \end{pmatrix}$ be the i -th column vector of A_i , then we have
0902.0431_F01278	66	■ 0.999	$A_{\{i\}}$	A_i	$\alpha E_i = A_i E_i A_i^*$. Let $\mathbf{a}_i = \begin{pmatrix} a_{i1} \\ a_{i2} \\ a_{i3} \end{pmatrix}$ be the i -th column vector of A_i , then we have
0902.0431_F01279	66	■ 0.999	$\alpha E_{\{i\}}=\left(\begin{array}{ccc}a_{\{i\} 1} \bar{a}_{\{i\} 1} & a_{\{i\} 1} \bar{a}_{\{i\} 2} & a_{\{i\} 1} \bar{a}_{\{i\} 3} \\ a_{\{i\} 2} \bar{a}_{\{i\} 1} & a_{\{i\} 2} \bar{a}_{\{i\} 2} & a_{\{i\} 2} \bar{a}_{\{i\} 3} \\ a_{\{i\} 3} \bar{a}_{\{i\} 1} & a_{\{i\} 3} \bar{a}_{\{i\} 2} & a_{\{i\} 3} \bar{a}_{\{i\} 3}\end{array}\right)$	$\alpha E_i = \begin{pmatrix} a_{i1}\bar{a}_{i1} & a_{i1}\bar{a}_{i2} & a_{i1}\bar{a}_{i3} \\ a_{i2}\bar{a}_{i1} & a_{i2}\bar{a}_{i2} & a_{i2}\bar{a}_{i3} \\ a_{i3}\bar{a}_{i1} & a_{i3}\bar{a}_{i2} & a_{i3}\bar{a}_{i3} \end{pmatrix}$	$\alpha E_i = \begin{pmatrix} a_{i1}\bar{a}_{i1} & a_{i1}\bar{a}_{i2} & a_{i1}\bar{a}_{i3} \\ a_{i2}\bar{a}_{i1} & a_{i2}\bar{a}_{i2} & a_{i2}\bar{a}_{i3} \\ a_{i3}\bar{a}_{i1} & a_{i3}\bar{a}_{i2} & a_{i3}\bar{a}_{i3} \end{pmatrix}$. Construct a matrix $A = (\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3)$, then we
0902.0431_F01280	66	■ 0.999	$A=\left(\boldsymbol{a}_{\{1\}}, \boldsymbol{a}_{\{2\}}, \boldsymbol{a}_{\{3\}}\right)$	$A = (\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3)$	$\alpha E_i = \begin{pmatrix} a_{i1}\bar{a}_{i1} & a_{i1}\bar{a}_{i2} & a_{i1}\bar{a}_{i3} \\ a_{i2}\bar{a}_{i1} & a_{i2}\bar{a}_{i2} & a_{i2}\bar{a}_{i3} \\ a_{i3}\bar{a}_{i1} & a_{i3}\bar{a}_{i2} & a_{i3}\bar{a}_{i3} \end{pmatrix}$. Construct a matrix $A = (\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3)$, then we
0902.0431_F01281	66	■ 1.000	$A A^{\{*\}}=A\left(E_{\{1\}}+E_{\{2\}}+E_{\{3\}}\right) A^{\{*\}}=\alpha E_{\{1\}}+\alpha E_{\{2\}}+\alpha E_{\{3\}}=\alpha E=E$	$AA^* = A(E_1 + E_2 + E_3)A^* = \alpha E_1 + \alpha E_2 + \alpha E_3 = \alpha E = E$	Since $AA^* = A(E_1 + E_2 + E_3)A^* = \alpha E_1 + \alpha E_2 + \alpha E_3 = \alpha E = E$, we have $A \in Sp(3)$.
0902.0431_F01282	66	■ 1.000	$A \in \operatorname{Sp}(3)$	$A \in Sp(3)$	Since $AA^* = A(E_1 + E_2 + E_3)A^* = \alpha E_1 + \alpha E_2 + \alpha E_3 = \alpha E = E$, we have $A \in Sp(3)$.
0902.0431_F01283	66	■ 1.000	$\beta=\varphi(A)^{-1} \alpha$	$\beta = \varphi(A)^{-1}\alpha$	If we let $\beta = \varphi(A)^{-1}\alpha$, then $\beta \in F_{4,\mathbf{H}}$ and satisfies
0902.0431_F01284	66	■ 1.000	$\beta \in F_4, \boldsymbol{H}$	$\beta \in F_{4,\mathbf{H}}$	If we let $\beta = \varphi(A)^{-1}\alpha$, then $\beta \in F_{4,\mathbf{H}}$ and satisfies
0902.0431_F01285	66	■ 1.000	$\beta_1, \beta_2, \beta_3$	$\beta_1, \beta_2, \beta_3$	Analogously as in Theorem 2.7.1, β induces orthogonal transformations $\beta_1, \beta_2, \beta_3$
0902.0431_F01286	66	■ 0.999	$: \boldsymbol{H} \rightarrow \boldsymbol{H}$	$: \mathbf{H} \rightarrow \mathbf{H}$	$: \mathbf{H} \rightarrow \mathbf{H}$ satisfying
0902.0431_F01287	66	■ 0.995	$p=\beta_1 1$	$p = \beta_1 1$	Put $p = \beta_1 1$ and $q = \beta_2 1$, then we have $ p = q = 1$. Letting $m = 1$ and $n = 1$ in
0902.0431_F01288	66	■ 0.995	$q=\beta_2 1$	$q = \beta_2 1$	Put $p = \beta_1 1$ and $q = \beta_2 1$, then we have $ p = q = 1$. Letting $m = 1$ and $n = 1$ in
0902.0431_F01289	66	■ 0.995	$ p = q =1$	$ p = q = 1$	Put $p = \beta_1 1$ and $q = \beta_2 1$, then we have $ p = q = 1$. Letting $m = 1$ and $n = 1$ in
0902.0431_F01290	66	■ 0.995	$m=1$	$m = 1$	Put $p = \beta_1 1$ and $q = \beta_2 1$, then we have $ p = q = 1$. Letting $m = 1$ and $n = 1$ in
0902.0431_F01291	66	■ 0.995	$n=1$	$n = 1$	Put $p = \beta_1 1$ and $q = \beta_2 1$, then we have $ p = q = 1$. Letting $m = 1$ and $n = 1$ in
0902.0431_F01292	66	■ 1.000	$p\left(\beta_2 n\right)=\overline{\beta_3} \bar{n}$	$p(\beta_2 n) = \overline{\beta_3} \bar{n}$	(i), we have $p(\beta_2 n) = \overline{\beta_3} \bar{n}$ and $(\beta_1 m)q = \overline{\beta_3} \bar{m}$, and hence
0902.0431_F01293	66	■ 1.000	$\left(\beta_1 m\right) q=\overline{\beta_3} \bar{m}$	$(\beta_1 m)q = \overline{\beta_3} \bar{m}$	(i), we have $p(\beta_2 n) = \overline{\beta_3} \bar{n}$ and $(\beta_1 m)q = \overline{\beta_3} \bar{m}$, and hence
0902.0431_F01294	66	■ 1.000	$\zeta m=\bar{p}\left(\beta_1 m\right)$	$\zeta m = \bar{p}(\beta_1 m)$	Substitute $\zeta m = \bar{p}(\beta_1 m)$ in (i), then we see that ζ satisfies
0902.0431_F01295	66	■ 1.000	ζ	ζ	Substitute $\zeta m = \bar{p}(\beta_1 m)$ in (i), then we see that ζ satisfies
0902.0431_F01296	66	■ 1.000	$\boldsymbol{H}: \zeta \in \operatorname{Aut}(\boldsymbol{H})$	$\mathbf{H} : \zeta \in \operatorname{Aut}(\mathbf{H})$	that is, ζ is an automorphism of $\mathbf{H}$: $\zeta \in \operatorname{Aut}(\mathbf{H})$. Hence, ζ is expressed by $\zeta m = rm\bar{r}$,
0902.0431_F01297	66	■ 1.000	$\zeta m=r m \bar{r}$	$\zeta m = rm\bar{r}$	that is, ζ is an automorphism of $\mathbf{H}$: $\zeta \in \operatorname{Aut}(\mathbf{H})$. Hence, ζ is expressed by $\zeta m = rm\bar{r}$,
0902.0431_F01298	66	■ 0.996	$r \in \operatorname{Sp}(1)$	$r \in Sp(1)$	using $r \in Sp(1)$ (Proposition 108 of Yokota [58]). Consequently
0902.0431_F01299	66	■ 0.999	$B=\left(\begin{array}{ccc}\bar{q} r & 0 & 0 \\ 0 & p r & 0 \\ 0 & 0 & r\end{array}\right)$	$B = \begin{pmatrix} \bar{q}r & 0 & 0 \\ 0 & pr & 0 \\ 0 & 0 & r \end{pmatrix}$	Construct a matrix $B = \begin{pmatrix} \bar{q}r & 0 & 0 \\ 0 & pr & 0 \\ 0 & 0 & r \end{pmatrix}$, then $B \in Sp(3)$ and we have
0902.0431_F01300	66	■ 0.999	$B \in \operatorname{Sp}(3)$	$B \in Sp(3)$	Construct a matrix $B = \begin{pmatrix} \bar{q}r & 0 & 0 \\ 0 & pr & 0 \\ 0 & 0 & r \end{pmatrix}$, then $B \in Sp(3)$ and we have
0902.0431_F01301	66	■ 1.000	$\beta=\varphi(B)$	$\beta = \varphi(B)$	that is, $\beta = \varphi(B)$. Hence
0902.0431_F01302	66	■ 1.000	$\operatorname{Ker} \varphi=\{E,-E\}$	$\operatorname{Ker} \varphi = \{E, -E\}$	Therefore φ is onto. $\operatorname{Ker} \varphi = \{E, -E\}$ can be easily obtained. Thus we have the
0902.0431_F01303	66	■ 0.997	$\operatorname{Sp}(3) / \boldsymbol{Z}_2 \cong F_4, \boldsymbol{H}$	$Sp(3)/\mathbf{Z}_2 \cong F_{4,\mathbf{H}}$	isomorphism $Sp(3)/\mathbf{Z}_2 \cong F_{4,\mathbf{H}}$.

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0902.0431_F01304	66	■ 0.301	$\left(F_4\right)^{\gamma} \cong(\operatorname{Sp}(1) \times \operatorname{Sp}(3)) / \boldsymbol{Z}_2, \boldsymbol{Z}_2=\{(1, E),(-1,-E)\}$	$(F_4)^{\gamma} \cong(\operatorname{Sp}(1) \times \operatorname{Sp}(3)) / \boldsymbol{Z}_2, \boldsymbol{Z}_2=\{(1, E),(-1,-E)\}$	Theorem 2.11.2. $(F_4)^{\gamma} \cong(\operatorname{Sp}(1) \times \operatorname{Sp}(3)) / \boldsymbol{Z}_2, \boldsymbol{Z}_2=\{(1, E),(-1,-E)\}$.
0902.0431_F01305	67	■ 0.978	$\varphi: \operatorname{Sp}(1) \times \operatorname{Sp}(3) \rightarrow \left(F_4\right)^{\gamma}$	$\varphi: \operatorname{Sp}(1) \times \operatorname{Sp}(3) \rightarrow (F_4)^{\gamma}$	Proof. We define a mapping $\varphi: \operatorname{Sp}(1) \times \operatorname{Sp}(3) \rightarrow (F_4)^{\gamma}$ by
0902.0431_F01306	67	■ 0.867	$\varphi(p, A) \in \left(F_4\right)^{\gamma}$	$\varphi(p, A) \in (F_4)^{\gamma}$	We first show that $\varphi(p, A) \in (F_4)^{\gamma}$. For $p \in \operatorname{Sp}(1), A \in \operatorname{Sp}(3), M, N \in \mathfrak{J}_{\boldsymbol{H}}$ and
0902.0431_F01307	67	■ 0.867	$p \in \operatorname{Sp}(1), A \in \operatorname{Sp}(3), M, N \in \mathfrak{J}_{\boldsymbol{H}}$	$p \in \operatorname{Sp}(1), A \in \operatorname{Sp}(3), M, N \in \mathfrak{J}_{\boldsymbol{H}}$	We first show that $\varphi(p, A) \in (F_4)^{\gamma}$. For $p \in \operatorname{Sp}(1), A \in \operatorname{Sp}(3), M, N \in \mathfrak{J}_{\boldsymbol{H}}$ and
0902.0431_F01308	67	■ 1.000	$\boldsymbol{a}, \boldsymbol{b} \in \boldsymbol{H}^3$	$\boldsymbol{a}, \boldsymbol{b} \in \boldsymbol{H}^3$	$\boldsymbol{a}, \boldsymbol{b} \in \boldsymbol{H}^3$, we have
0902.0431_F01309	67	■ 1.000	$\varphi(p, A)$	$\varphi(p, A)$	From which, we see that $\varphi(p, A)$ satisfies
0902.0431_F01310	67	■ 1.000	$\varphi(p, A) \in F_4$	$\varphi(p, A) \in F_4$	hence $\varphi(p, A) \in F_4$. Clearly $\gamma \varphi(p, A)=\varphi(p, A) \gamma$, so that $\varphi(p, A) \in (F_4)^{\gamma}$. Certainly
0902.0431_F01311	67	■ 1.000	$\gamma \varphi(p, A)=\varphi(p, A) \gamma$	$\gamma \varphi(p, A)=\varphi(p, A) \gamma$	hence $\varphi(p, A) \in F_4$. Clearly $\gamma \varphi(p, A)=\varphi(p, A) \gamma$, so that $\varphi(p, A) \in (F_4)^{\gamma}$. Certainly
0902.0431_F01312	67	■ 1.000	$\alpha \in \left(F_4\right)^{\gamma}$	$\alpha \in (F_4)^{\gamma}$	φ is a homomorphism. We shall show that φ is onto. Let $\alpha \in (F_4)^{\gamma}$. Since the
0902.0431_F01313	67	■ 1.000	$\alpha'=\alpha \mid \mathfrak{J}_{\boldsymbol{H}}$	$\alpha'=\alpha \mid \mathfrak{J}_{\boldsymbol{H}}$	restriction $\alpha'=\alpha \mid \mathfrak{J}_{\boldsymbol{H}}$ of α to $\mathfrak{J}_{\boldsymbol{H}}=\{X \in \mathfrak{J} \mid \gamma X=X\}$ belongs to $F_{4, \boldsymbol{H}}$, there exists
0902.0431_F01314	67	■ 1.000	$\mathfrak{J}_{\boldsymbol{H}}=\{X \in \mathfrak{J} \mid \gamma X=X\}$	$\mathfrak{J}_{\boldsymbol{H}}=\{X \in \mathfrak{J} \mid \gamma X=X\}$	restriction $\alpha'=\alpha \mid \mathfrak{J}_{\boldsymbol{H}}$ of α to $\mathfrak{J}_{\boldsymbol{H}}=\{X \in \mathfrak{J} \mid \gamma X=X\}$ belongs to $F_{4, \boldsymbol{H}}$, there exists
0902.0431_F01315	67	■ 0.975	$\beta=\varphi(1, A)^{-1} \alpha$	$\beta=\varphi(1, A)^{-1} \alpha$	(Proposition 2.11.1). Let $\beta=\varphi(1, A)^{-1} \alpha$, then $\beta \mid \mathfrak{J}_{\boldsymbol{H}}=1$. Therefore $\beta \in G_2$.
0902.0431_F01316	67	■ 0.975	$\beta \mid \mathfrak{J}_{\boldsymbol{H}}=1$	$\beta \mid \mathfrak{J}_{\boldsymbol{H}}=1$	(Proposition 2.11.1). Let $\beta=\varphi(1, A)^{-1} \alpha$, then $\beta \mid \mathfrak{J}_{\boldsymbol{H}}=1$. Therefore $\beta \in G_2$.
0902.0431_F01317	67	■ 0.975	$\beta \in G_2$	$\beta \in G_2$	(Proposition 2.11.1). Let $\beta=\varphi(1, A)^{-1} \alpha$, then $\beta \mid \mathfrak{J}_{\boldsymbol{H}}=1$. Therefore $\beta \in G_2$.
0902.0431_F01318	67	■ 1.000	$\beta \in \operatorname{Spin}(7)$	$\beta \in \operatorname{Spin}(7)$	Certainly, since $\beta \in \operatorname{Spin}(7)$, by letting $\beta=\left(\beta_1, \beta, \kappa \beta\right)$, we see that $\left(\beta_1 x\right)(\beta y)=$
0902.0431_F01319	67	■ 1.000	$\beta=\left(\beta_1, \beta, \kappa \beta\right)$	$\beta=\left(\beta_1, \beta, \kappa \beta\right)$	Certainly, since $\beta \in \operatorname{Spin}(7)$, by letting $\beta=\left(\beta_1, \beta, \kappa \beta\right)$, we see that $\left(\beta_1 x\right)(\beta y)=$
0902.0431_F01320	67	■ 1.000	$\left(\beta_1 x\right)(\beta y)=$	$\left(\beta_1 x\right)(\beta y)=$	Certainly, since $\beta \in \operatorname{Spin}(7)$, by letting $\beta=\left(\beta_1, \beta, \kappa \beta\right)$, we see that $\left(\beta_1 x\right)(\beta y)=$
0902.0431_F01321	67	■ 1.000	$\beta(x y)$	$\beta(x y)$	$\beta(x y)$. Now let $y=1$, then we see $\beta_1=\beta$. Since $\beta \in\left(G_2\right)^{\gamma}$ and $\beta \mid \boldsymbol{H}=1$, there
0902.0431_F01322	67	■ 1.000	$\beta_1=\beta$	$\beta_1=\beta$	$\beta(x y)$. Now let $y=1$, then we see $\beta_1=\beta$. Since $\beta \in\left(G_2\right)^{\gamma}$ and $\beta \mid \boldsymbol{H}=1$, there
0902.0431_F01323	67	■ 1.000	$\beta=\varphi(p, E)$	$\beta=\varphi(p, E)$	so that $\beta=\varphi(p, E)$. Hence we have $\alpha=\varphi(1, A) \beta=\varphi(1, A) \varphi(p, E)=\varphi(p, A)$.
0902.0431_F01324	67	■ 1.000	$\alpha=\varphi(1, A) \beta=\varphi(1, A) \varphi(p, E)=\varphi(p, A)$	$\alpha=\varphi(1, A) \beta=\varphi(1, A) \varphi(p, E)=\varphi(p, A)$	so that $\beta=\varphi(p, E)$. Hence we have $\alpha=\varphi(1, A) \beta=\varphi(1, A) \varphi(p, E)=\varphi(p, A)$.
0902.0431_F01325	67	■ 0.999	$\operatorname{Ker} \varphi=\{(1, E),(-1,-E)\}=\boldsymbol{Z}_2$	$\operatorname{Ker} \varphi=\{(1, E),(-1,-E)\}=\boldsymbol{Z}_2$	Therefore φ is onto. $\operatorname{Ker} \varphi=\{(1, E),(-1,-E)\}=\boldsymbol{Z}_2$ can be easily obtained. Thus
0902.0431_F01326	67	■ 0.740	$(\operatorname{Sp}(1) \times \operatorname{Sp}(3)) / \boldsymbol{Z}_2 \cong\left(F_4\right)^{\gamma}$	$(\operatorname{Sp}(1) \times \operatorname{Sp}(3)) / \boldsymbol{Z}_2 \cong\left(F_4\right)^{\gamma}$	we have the isomorphism $(\operatorname{Sp}(1) \times \operatorname{Sp}(3)) / \boldsymbol{Z}_2 \cong\left(F_4\right)^{\gamma}$.
0902.0431_F01327	67	■ 1.000	$\varphi: \operatorname{Sp}(1) \times \operatorname{Sp}(3) \rightarrow F_4$	$\varphi: \operatorname{Sp}(1) \times \operatorname{Sp}(3) \rightarrow F_4$	γ of the simply connected Lie group F_4 , the fact that $\varphi: \operatorname{Sp}(1) \times \operatorname{Sp}(3) \rightarrow F_4$ is onto
0902.0431_F01328	67	■ 1.000	$\left(\mathfrak{f}_4\right)^{\gamma}$	$\left(\mathfrak{f}_4\right)^{\gamma}$	forms an $\boldsymbol{R}$ -basis of $\left(\mathfrak{f}_4\right)^{\gamma}$. So $\dim\left(\left(\mathfrak{f}_4\right)^{\gamma}\right)=6 \times 2+4 \times 3=24=3+21=\dim(\mathfrak{sp}(1) \oplus$
0902.0431_F01329	67	■ 1.000	$\dim\left(\left(\mathfrak{f}_4\right)^{\gamma}\right)=6 \times 2+4 \times 3=24=3+21=\dim(\mathfrak{sp}(1) \oplus$	$\dim\left(\left(\mathfrak{f}_4\right)^{\gamma}\right)=6 \times 2+4 \times 3=24=3+21=\dim(\mathfrak{sp}(1) \oplus$	forms an $\boldsymbol{R}$ -basis of $\left(\mathfrak{f}_4\right)^{\gamma}$. So $\dim\left(\left(\mathfrak{f}_4\right)^{\gamma}\right)=6 \times 2+4 \times 3=24=3+21=\dim(\mathfrak{sp}(1) \oplus$
0902.0431_F01330	67	■ 0.508	$\mathfrak{sp}(3))$	$\mathfrak{sp}(3))$	$\mathfrak{sp}(3))$. Hence φ is onto.
0902.0431_F01331	68	■ 1.000	$w \in G_2 \subset F_4$	$w \in G_2 \subset F_4$	This w is the same as $w \in G_2 \subset F_4$.
0902.0431_F01332	68	■ 1.000	$\left(F_4\right)^w$	$\left(F_4\right)^w$	We shall study the following subgroup $\left(F_4\right)^w$ of F_4 :
0902.0431_F01333	68	■ 1.000	$\mathfrak{J}(3, \boldsymbol{C}) \oplus M(3, \boldsymbol{C})$	$\mathfrak{J}(3, \boldsymbol{C}) \oplus M(3, \boldsymbol{C})$	of $\mathfrak{J}(3, \boldsymbol{C}) \oplus M(3, \boldsymbol{C})$. In $M(3, \boldsymbol{C})$, we define the exterior product $M \times N$ by
0902.0431_F01334	68	■ 1.000	$M(3, \boldsymbol{C})$	$M(3, \boldsymbol{C})$	of $\mathfrak{J}(3, \boldsymbol{C}) \oplus M(3, \boldsymbol{C})$. In $M(3, \boldsymbol{C})$, we define the exterior product $M \times N$ by
0902.0431_F01335	68	■ 1.000	$M=\left(\boldsymbol{m}_1, \boldsymbol{m}_2, \boldsymbol{m}_3\right), N=\left(\boldsymbol{n}_1, \boldsymbol{n}_2, \boldsymbol{n}_3\right) \in M(3, \boldsymbol{C})$	$M=\left(\boldsymbol{m}_1, \boldsymbol{m}_2, \boldsymbol{m}_3\right), N=\left(\boldsymbol{n}_1, \boldsymbol{n}_2, \boldsymbol{n}_3\right) \in M(3, \boldsymbol{C})$	where $M=\left(\boldsymbol{m}_1, \boldsymbol{m}_2, \boldsymbol{m}_3\right), N=\left(\boldsymbol{n}_1, \boldsymbol{n}_2, \boldsymbol{n}_3\right) \in M(3, \boldsymbol{C})$. Then, for $P, A \in M(3, \boldsymbol{C})$
0902.0431_F01336	68	■ 1.000	$P, A \in M(3, \boldsymbol{C})$	$P, A \in M(3, \boldsymbol{C})$	where $M=\left(\boldsymbol{m}_1, \boldsymbol{m}_2, \boldsymbol{m}_3\right), N=\left(\boldsymbol{n}_1, \boldsymbol{n}_2, \boldsymbol{n}_3\right) \in M(3, \boldsymbol{C})$. Then, for $P, A \in M(3, \boldsymbol{C})$
0902.0431_F01337	68	■ 1.000	$M, N \in M(3, \boldsymbol{C})$	$M, N \in M(3, \boldsymbol{C})$	and $M, N \in M(3, \boldsymbol{C})$, we have
0902.0431_F01338	68	■ 1.000	$\tilde{P}$	$\tilde{P}$	where $\tilde{P}$ and $\tilde{A}$ are the adjoint matrices of P and A , respectively. Further, in $M(3, \boldsymbol{C})$,
0902.0431_F01339	68	■ 1.000	P	P	where $\tilde{P}$ and $\tilde{A}$ are the adjoint matrices of P and A , respectively. Further, in $M(3, \boldsymbol{C})$,

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0902.0431_F01340	68	■ 0.998	$M=\left(m_{ij}\right), N=\left(n_{ij}\right) \in M(3, \boldsymbol{C})$	$M=\left(m_{ij}\right), N=\left(n_{ij}\right) \in M(3, C)$	where $M=\left(m_{ij}\right), N=\left(n_{ij}\right) \in M(3, C)$.
0902.0431_F01341	69	■ 0.999	$\mathfrak{J}(3, \boldsymbol{C})$	$\mathfrak{J}(3, C)$	(in $\mathfrak{J}(3, C)$, multiplications $\circ$ and $\times$ are defined as in $\mathfrak{J}$). Since these operations
0902.0431_F01342	69	■ 1.000	$F_{4, \boldsymbol{C}}$	$F_{4, C}$	The group $F_{4, C}$ is defined to be the automorphism group of the Jordan algebra
0902.0431_F01343	69	■ 0.881	$\mathfrak{J}_{\boldsymbol{C}}=\mathfrak{J}(3, \boldsymbol{C})$:	$\mathfrak{J}_C=\mathfrak{J}(3, C)$:	$\mathfrak{J}_C=\mathfrak{J}(3, C)$:
0902.0431_F01344	69	■ 0.856	$\boldsymbol{Z}_2=\{1, \epsilon\}$	$Z_2=\{1, \epsilon\}$	The group $Z_2=\{1, \epsilon\}$ acts on the group $SU(3)$ by
0902.0431_F01345	69	■ 0.983	$S U(3) \cdot \boldsymbol{Z}_2$	$SU(3) \cdot Z_2$	and the group $SU(3) \cdot Z_2$ be the semi-direct product of groups $SU(3)$ and Z_2 under
0902.0431_F01346	69	■ 0.983	$\boldsymbol{Z}_2$	Z_2	and the group $SU(3) \cdot Z_2$ be the semi-direct product of groups $SU(3)$ and Z_2 under
0902.0431_F01347	69	■ 0.642	$F_{4, \boldsymbol{C}} \cong \left(S U(3) / \boldsymbol{Z}_3\right) \cdot \boldsymbol{Z}_2, \quad \boldsymbol{Z}_3=\left\{E, \omega_1 E, \omega_1^2 E\right\}, \omega_1=\left\{E, \omega_1 E, \omega_1^2 E\right\} \rightleftharpoons \left\{E, \omega_1 E, \omega_1^2 E\right\}$	$F_{4, C} \cong (SU(3) / Z_3) \cdot Z_2, \quad Z_3=\left\{E, \omega_1 E, \omega_1^2 E\right\}, \omega_1=$	Proposition 2.12.1. $F_{4, C} \cong (SU(3) / Z_3) \cdot Z_2, \quad Z_3=\left\{E, \omega_1 E, \omega_1^2 E\right\}, \omega_1=$
0902.0431_F01348	69	■ 0.996	$-\frac{1}{2}+\frac{\sqrt{3}}{2} e_1$	$-\frac{1}{2}+\frac{\sqrt{3}}{2} e_1$	$-\frac{1}{2}+\frac{\sqrt{3}}{2} e_1$.
0902.0431_F01349	69	■ 1.000	$\varphi: S U(3) \cdot \boldsymbol{Z}_2 \rightarrow F_{4, \boldsymbol{C}}$	$\varphi: SU(3) \cdot Z_2 \rightarrow F_{4, C}$	Proof. We define a mapping $\varphi: SU(3) \cdot Z_2 \rightarrow F_{4, C}$ by
0902.0431_F01350	69	■ 0.968	$C, S U(3)$	$C, SU(3)$	proof that φ is onto is the same as that of Proposition 2.11.1 (use $C, SU(3)$, instead
0902.0431_F01351	69	■ 0.947	$\boldsymbol{H}, \operatorname{Sp}(3)$	$H, Sp(3)$	of $H, Sp(3)$, and we only need to modify a little the last part. Since $\zeta \in \operatorname{Aut}(C)$ can
0902.0431_F01352	69	■ 0.947	$\zeta \in \operatorname{Aut}(\boldsymbol{C})$	$\zeta \in \operatorname{Aut}(C)$	of $H, Sp(3)$, and we only need to modify a little the last part. Since $\zeta \in \operatorname{Aut}(C)$ can
0902.0431_F01353	69	■ 1.000	$\zeta=1$	$\zeta=1$	be either $\zeta=1$ or $\zeta=\epsilon$ (where $\epsilon x=\bar{x}, x \in C$), we have
0902.0431_F01354	69	■ 1.000	$\zeta=\epsilon$	$\zeta=\epsilon$	be either $\zeta=1$ or $\zeta=\epsilon$ (where $\epsilon x=\bar{x}, x \in C$), we have
0902.0431_F01355	69	■ 1.000	$\epsilon x=\bar{x}, x \in \boldsymbol{C}$	$\epsilon x=\bar{x}, x \in C$	be either $\zeta=1$ or $\zeta=\epsilon$ (where $\epsilon x=\bar{x}, x \in C$), we have
0902.0431_F01356	69	■ 1.000	$r \in \boldsymbol{C}$	$r \in C$	Now, choose $r \in C$ such that $r^{-3}=\bar{q} p$ and construct a matrix $B=\left(\begin{array}{ccc} \bar{q} r & 0 & 0 \\ 0 & p r & 0 \\ 0 & 0 & r \end{array}\right)$.
0902.0431_F01357	69	■ 1.000	$r^{-3}=\bar{q} p$	$r^{-3}=\bar{q} p$	Now, choose $r \in C$ such that $r^{-3}=\bar{q} p$ and construct a matrix $B=\left(\begin{array}{ccc} \bar{q} r & 0 & 0 \\ 0 & p r & 0 \\ 0 & 0 & r \end{array}\right)$.
0902.0431_F01358	69	■ 1.000	$B \in S U(3)$	$B \in SU(3)$	Then, $B \in SU(3)$ and we have
0902.0431_F01359	69	■ 0.817	$\left(F_4\right)^w \cong(S U(3) \times S U(3)) / \boldsymbol{Z}_3, \quad \boldsymbol{Z}_3=\{(E, E),\left(\omega_1 E, \omega_1 E\right)\}$	$(F_4)^w \cong (SU(3) \times SU(3)) / Z_3, \quad Z_3=\{(E, E), (\omega_1 E, \omega_1 E)\}$	Theorem 2.12.2. $(F_4)^w \cong (SU(3) \times SU(3)) / Z_3, \quad Z_3=\{(E, E), (\omega_1 E, \omega_1 E)\}$,
0902.0431_F01360	69	■ 0.674	$\left(\omega_1^2 E, \omega_1^2 E\right)\}, \omega_1=-\frac{1}{2}+\frac{\sqrt{3}}{2} e_1$	$(\omega_1^2 E, \omega_1^2 E)\}, \omega_1=-\frac{1}{2}+\frac{\sqrt{3}}{2} e_1$	$(\omega_1^2 E, \omega_1^2 E)\}, \omega_1=-\frac{1}{2}+\frac{\sqrt{3}}{2} e_1$.
0902.0431_F01361	69	■ 0.817	$\varphi: S U(3) \times S U(3) \rightarrow\left(F_4\right)^w$	$\varphi: SU(3) \times SU(3) \rightarrow (F_4)^w$	Proof. We define a mapping $\varphi: SU(3) \times SU(3) \rightarrow (F_4)^w$ by
0902.0431_F01362	70	■ 1.000	$\varphi(P, A) \in\left(F_4\right)^w$	$\varphi(P, A) \in (F_4)^w$	We first show that $\varphi(P, A) \in (F_4)^w$. For $P, A \in SU(3)$ and $X+M, Y+N \in$
0902.0431_F01363	70	■ 1.000	$P, A \in S U(3)$	$P, A \in SU(3)$	We first show that $\varphi(P, A) \in (F_4)^w$. For $P, A \in SU(3)$ and $X+M, Y+N \in$
0902.0431_F01364	70	■ 1.000	$X+M, Y+N \in$	$X+M, Y+N \in$	We first show that $\varphi(P, A) \in (F_4)^w$. For $P, A \in SU(3)$ and $X+M, Y+N \in$
0902.0431_F01365	70	■ 1.000	$\varphi(P, A)$	$\varphi(P, A)$	Using them, we see that $\varphi(P, A)$ satisfies
0902.0431_F01366	70	■ 1.000	$\varphi(P, A) \in F_4$	$\varphi(P, A) \in F_4$	hence $\varphi(P, A) \in F_4$. Clearly $w \varphi(P, A)=\varphi(P, A) w$, so that $\varphi(P, A) \in (F_4)^w$. Cer-
0902.0431_F01367	70	■ 1.000	$w \varphi(P, A)=\varphi(P, A) w$	$w \varphi(P, A)=\varphi(P, A) w$	hence $\varphi(P, A) \in F_4$. Clearly $w \varphi(P, A)=\varphi(P, A) w$, so that $\varphi(P, A) \in (F_4)^w$. Cer-
0902.0431_F01368	70	■ 1.000	$\alpha \in\left(F_4\right)^w$	$\alpha \in (F_4)^w$	certainly φ is a homomorphism. We shall show that φ is onto. Let $\alpha \in (F_4)^w$. Since the
0902.0431_F01369	70	■ 1.000	$\alpha^{\prime}=\alpha \mid \mathfrak{J}_{\boldsymbol{C}}$	$\alpha^{\prime}=\alpha \mid \mathfrak{J}_C$	restriction $\alpha^{\prime}=\alpha \mid \mathfrak{J}_C$ of α to $\mathfrak{J}_C=\{X \in \mathfrak{J} \mid w X=X\}$ belongs to $F_{4, C}$, there exists
0902.0431_F01370	70	■ 1.000	$\mathfrak{J}_{\boldsymbol{C}}=\{X \in \mathfrak{J} \mid w X=X\}$	$\mathfrak{J}_C=\{X \in \mathfrak{J} \mid w X=X\}$	restriction $\alpha^{\prime}=\alpha \mid \mathfrak{J}_C$ of α to $\mathfrak{J}_C=\{X \in \mathfrak{J} \mid w X=X\}$ belongs to $F_{4, C}$, there exists
0902.0431_F01371	70	■ 0.603	$\beta=\varphi(E, A)^{-1} \alpha$	$\beta=\varphi(E, A)^{-1} \alpha$	(Proposition 2.12.1). In the former case, let $\beta=\varphi(E, A)^{-1} \alpha$, then $\beta \mid \mathfrak{J}_C=1$, and so
0902.0431_F01372	70	■ 0.603	$\beta \mid \mathfrak{J}_{\boldsymbol{C}}=1$	$\beta \mid \mathfrak{J}_C=1$	(Proposition 2.12.1). In the former case, let $\beta=\varphi(E, A)^{-1} \alpha$, then $\beta \mid \mathfrak{J}_C=1$, and so
0902.0431_F01373	70	■ 1.000	$\beta \in\left(G_2\right)_{e_1}=\left(G_2\right)^w=S U(3)$	$\beta \in\left(G_2\right)_{e_1}=\left(G_2\right)^w=S U(3)$	$\beta \in G_2$. Moreover $\beta \in\left(G_2\right)_{e_1}=\left(G_2\right)^w=S U(3)$ (Theorem 1.9.4), and hence, there

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_F01374	70	■ 1.000	$P \in SU(3)$	$P \in SU(3)$	exists $P \in SU(3)$ such that
0902.0431_F01375	70	■ 1.000	$\gamma_1: \mathfrak{J} \rightarrow \mathfrak{J}$	$\gamma_1: \mathfrak{J} \rightarrow \mathfrak{J}$	In the latter case, consider the mapping $\gamma_1: \mathfrak{J} \rightarrow \mathfrak{J}$ given by $\gamma_1(X + M) = \overline{X} + \overline{M}$,
0902.0431_F01376	70	■ 1.000	$\gamma_1(X+M)=\bar{X}+\bar{M}$	$\gamma_1(X + M) = \bar{X} + \bar{M}$	In the latter case, consider the mapping $\gamma_1: \mathfrak{J} \rightarrow \mathfrak{J}$ given by $\gamma_1(X + M) = \overline{X} + \overline{M}$,
0902.0431_F01377	70	■ 1.000	$X+M \in \mathfrak{J}$	$X + M \in \mathfrak{J}$	$\overline{X + M} \in \mathfrak{J}$ and remember that $\gamma_1 \in G_2 \subset F_4$. Let $\beta = \alpha^{-1}\varphi(E, A)\gamma_1$. Then $\beta \in F_4$
0902.0431_F01378	70	■ 1.000	$\gamma_1 \in G_2 \subset F_4$	$\gamma_1 \in G_2 \subset F_4$	$X + M \in \mathfrak{J}$ and remember that $\gamma_1 \in G_2 \subset F_4$. Let $\beta = \alpha^{-1}\varphi(E, A)\gamma_1$. Then $\beta \in F_4$
0902.0431_F01379	70	■ 1.000	$\beta=\alpha^{-1}\varphi(E, A)\gamma_1$	$\beta = \alpha^{-1}\varphi(E, A)\gamma_1$	$X + M \in \mathfrak{J}$ and remember that $\gamma_1 \in G_2 \subset F_4$. Let $\beta = \alpha^{-1}\varphi(E, A)\gamma_1$. Then $\beta \in F_4$
0902.0431_F01380	70	■ 1.000	$\beta \in F_4$	$\beta \in F_4$	$X + M \in \mathfrak{J}$ and remember that $\gamma_1 \in G_2 \subset F_4$. Let $\beta = \alpha^{-1}\varphi(E, A)\gamma_1$. Then $\beta \in F_4$
0902.0431_F01381	70	■ 0.662	$\beta \widetilde{\mathfrak{J}}_C=1$	$\beta \widetilde{\mathfrak{J}}_C = 1$	and $\beta \mathfrak{J}_C = 1$, which shows that $\beta \in (G_2)_{e_1} = (G_2)^w$ (Theorem 1.9.4) $\subset (F_4)^w$. Since
0902.0431_F01382	70	■ 0.662	$\beta \in (G_2)_{e_1} = (G_2)^w$	$\beta \in (G_2)_{e_1} = (G_2)^w$	and $\beta \mathfrak{J}_C = 1$, which shows that $\beta \in (G_2)_{e_1} = (G_2)^w$ (Theorem 1.9.4) $\subset (F_4)^w$. Since
0902.0431_F01383	70	■ 0.662	$\subset (F_4)^w$	$\subset (F_4)^w$	and $\beta \mathfrak{J}_C = 1$, which shows that $\beta \in (G_2)_{e_1} = (G_2)^w$ (Theorem 1.9.4) $\subset (F_4)^w$. Since
0902.0431_F01384	70	■ 1.000	$\alpha, \varphi(E, A) \in (F_4)^w$	$\alpha, \varphi(E, A) \in (F_4)^w$	$\alpha, \varphi(E, A) \in (F_4)^w$, we have $\gamma_1 \in (F_4)^w$, so that $\gamma_1 \in (G_2)^w$ which is a contradiction
0902.0431_F01385	70	■ 1.000	$\gamma_1 \in (F_4)^w$	$\gamma_1 \in (F_4)^w$	$\alpha, \varphi(E, A) \in (F_4)^w$, we have $\gamma_1 \in (F_4)^w$, so that $\gamma_1 \in (G_2)^w$ which is a contradiction
0902.0431_F01386	70	■ 1.000	$\gamma_1 \in (G_2)^w$	$\gamma_1 \in (G_2)^w$	$\alpha, \varphi(E, A) \in (F_4)^w$, we have $\gamma_1 \in (F_4)^w$, so that $\gamma_1 \in (G_2)^w$ which is a contradiction
0902.0431_F01387	70	■ 0.997	$\text{Ker } \varphi = \{(E, E), (\omega_1 E, \omega_1 E), (\omega_1^2 E, \omega_1^2 E)\} = \mathbf{Z}_3$	$\text{Ker } \varphi = \{(E, E), (\omega_1 E, \omega_1 E), (\omega_1^2 E, \omega_1^2 E)\} = \mathbf{Z}_3$	$\text{Ker } \varphi = \{(E, E), (\omega_1 E, \omega_1 E), (\omega_1^2 E, \omega_1^2 E)\} = \mathbf{Z}_3$ can be easily obtained. Thus we
0902.0431_F01388	70	■ 1.000	$(SU(3) \times SU(3)) / \mathbf{Z}_3 \cong (F_4)^w$	$(SU(3) \times SU(3)) / \mathbf{Z}_3 \cong (F_4)^w$	have the isomorphism $(SU(3) \times SU(3)) / \mathbf{Z}_3 \cong (F_4)^w$.
0902.0431_F01389	71	■ 1.000	$(\mathfrak{f}_4)^w$	$(\mathfrak{f}_4)^w$	forms an $\mathbf{R}$ -basis of $(\mathfrak{f}_4)^w$. So $\dim(\mathfrak{f}_4)^w = 10 + 6 = 16 = 8 + 8 = \dim(\mathfrak{su}(3) + \mathfrak{su}(3))$.
0902.0431_F01390	71	■ 1.000	$\dim(\mathfrak{f}_4)^w = 10 + 6 = 16 = 8 + 8 = \dim(\mathfrak{su}(3) + \mathfrak{su}(3))$	$\dim(\mathfrak{f}_4)^w = 10 + 6 = 16 = 8 + 8 = \dim(\mathfrak{su}(3) + \mathfrak{su}(3))$	forms an $\mathbf{R}$ -basis of $(\mathfrak{f}_4)^w$. So $\dim(\mathfrak{f}_4)^w = 10 + 6 = 16 = 8 + 8 = \dim(\mathfrak{su}(3) + \mathfrak{su}(3))$.
0902.0431_F01391	71	■ 0.977	$(SU(3) \times SU(3)) / \mathbf{Z}_3 \cdot \mathbf{Z}_2$	$(SU(3) \times SU(3)) / \mathbf{Z}_3 \cdot \mathbf{Z}_2$	$(SU(3) \times SU(3)) / \mathbf{Z}_3 \cdot \mathbf{Z}_2$, which is the semi-direct product of the groups $(SU(3) \times$
0902.0431_F01392	71	■ 0.977	$(SU(3) \times$	$(SU(3) \times$	$(SU(3) \times SU(3)) / \mathbf{Z}_3 \cdot \mathbf{Z}_2$, which is the semi-direct product of the groups $(SU(3) \times$
0902.0431_F01393	71	■ 0.993	$\mathbf{Z}_2 = \{1, \gamma_1\}$	$\mathbf{Z}_2 = \{1, \gamma_1\}$	$SU(3)) / \mathbf{Z}_3$ and $\mathbf{Z}_2$ (the action of $\mathbf{Z}_2 = \{1, \gamma_1\}$ on the group $(SU(3) \times SU(3))$ is
0902.0431_F01394	71	■ 0.993	$(SU(3) \times SU(3))$	$(SU(3) \times SU(3))$	$SU(3)) / \mathbf{Z}_3$ and $\mathbf{Z}_2$ (the action of $\mathbf{Z}_2 = \{1, \gamma_1\}$ on the group $(SU(3) \times SU(3))$ is
0902.0431_F01395	71	■ 1.000	$\gamma_1(P, A) = (\bar{P}, \bar{A})$	$\gamma_1(P, A) = (\bar{P}, \bar{A})$	$\gamma_1(P, A) = (\bar{P}, \bar{A})$.
0902.0431_F01396	71	■ 1.000	$\langle X, Y \rangle$	$\langle X, Y \rangle$	We define a positive definite Hermitian inner product $\langle X, Y \rangle$ in $\mathfrak{J}^C$ by
0902.0431_F01397	71	■ 1.000	$\alpha \in \text{Hom}_C(\mathfrak{J}^C)$	$\alpha \in \text{Hom}_C(\mathfrak{J}^C)$	For $\alpha \in \text{Hom}_C(\mathfrak{J}^C)$, we denote the complex conjugate transpose of α with respect to
0902.0431_F01398	71	■ 1.000	$\alpha^*: \langle \alpha^* X, Y \rangle = \langle X, \alpha Y \rangle$	$\alpha^*: \langle \alpha^* X, Y \rangle = \langle X, \alpha Y \rangle$	$\langle X, Y \rangle$ by $\alpha^*: \langle \alpha^* X, Y \rangle = \langle X, \alpha Y \rangle$.
0902.0431_F01399	71	■ 1.000	$\alpha \in F_4^C$	$\alpha \in F_4^C$	Lemma 2.13.1. (1) For $\alpha \in F_4^C$, we have $\alpha^* = \tau\alpha^{-1}\tau \in F_4^C$.
0902.0431_F01400	71	■ 1.000	$\alpha^* = \tau\alpha^{-1}\tau \in F_4^C$	$\alpha^* = \tau\alpha^{-1}\tau \in F_4^C$	Lemma 2.13.1. (1) For $\alpha \in F_4^C$, we have $\alpha^* = \tau\alpha^{-1}\tau \in F_4^C$.
0902.0431_F01401	71	■ 0.999	$\alpha^C: \mathfrak{J}^C \rightarrow \mathfrak{J}^C$	$\alpha^C: \mathfrak{J}^C \rightarrow \mathfrak{J}^C$	(2) For any $\alpha \in F_4$, its complexified mapping $\alpha^C: \mathfrak{J}^C \rightarrow \mathfrak{J}^C$ belongs to F_4^C :
0902.0431_F01402	71	■ 0.998	$\alpha^C \in F_4^C$	$\alpha^C \in F_4^C$	$\alpha^C \in F_4^C$. Identifying α with α^C , we regard F_4 as a subgroup of F_4^C : $F_4 \subset F_4^C$.
0902.0431_F01403	71	■ 0.998	$F_4^C: F_4 \subset F_4^C$	$F_4^C: F_4 \subset F_4^C$	$\alpha^C \in F_4^C$. Identifying α with α^C , we regard F_4 as a subgroup of F_4^C : $F_4 \subset F_4^C$.
0902.0431_F01404	71	■ 1.000	$\langle \alpha^* X, Y \rangle = \langle X, \alpha Y \rangle = (\tau X, \alpha Y) = (\alpha^{-1}\tau X, Y) = \langle \tau\alpha^{-1}\tau X, Y \rangle$	$\langle \alpha^* X, Y \rangle = \langle X, \alpha Y \rangle = (\tau X, \alpha Y) = (\alpha^{-1}\tau X, Y) = \langle \tau\alpha^{-1}\tau X, Y \rangle$	Proof. (1) $\langle \alpha^* X, Y \rangle = \langle X, \alpha Y \rangle = (\tau X, \alpha Y) = (\alpha^{-1}\tau X, Y) = \langle \tau\alpha^{-1}\tau X, Y \rangle$ for
0902.0431_F01405	71	■ 1.000	$X, Y \in \mathfrak{J}^C$	$X, Y \in \mathfrak{J}^C$	all $X, Y \in \mathfrak{J}^C$. Hence $\alpha^* = \tau\alpha^{-1}\tau \in F_4^C$.
0902.0431_F01406	71	■ 0.999	$\tau\alpha X = \alpha\tau X = \alpha X$	$\tau\alpha X = \alpha\tau X = \alpha X$	(2) Let $\alpha \in F_4^C$ satisfy $\tau\alpha = \alpha\tau$. Then, since $\tau\alpha X = \alpha\tau X = \alpha X$, we have
0902.0431_F01407	71	■ 1.000	$\alpha X \in \mathfrak{J}$	$\alpha X \in \mathfrak{J}$	$\alpha X \in \mathfrak{J}$ for $X \in \mathfrak{J}$. Hence α induces an $\mathbf{R}$ -transformation α' of $\mathfrak{J}$ and $\alpha' \in F_4$, further
0902.0431_F01408	71	■ 1.000	$\alpha' \in F_4$	$\alpha' \in F_4$	$\alpha X \in \mathfrak{J}$ for $X \in \mathfrak{J}$. Hence α induces an $\mathbf{R}$ -transformation α' of $\mathfrak{J}$ and $\alpha' \in F_4$, further
0902.0431_F01409	72	■ 0.999	$\text{Iso}_C(\mathfrak{J}^C) = GL(27, C)$	$\text{Iso}_C(\mathfrak{J}^C) = GL(27, C)$	Proof. Evidently F_4^C is an algebraic subgroup of $\text{Iso}_C(\mathfrak{J}^C) = GL(27, C)$. If
0902.0431_F01410	72	■ 1.000	$\alpha^* \in F_4^C$	$\alpha^* \in F_4^C$	$\alpha \in F_4^C$, then $\alpha^* \in F_4^C$ (Lemma 2.13.1.(1)). Hence from Chevalley's lemma, we

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_F01411	72	■ 1.000	$\mathrm{U}\left(\frac{\mathfrak{J}}{\mathfrak{C}}\right)=\left\{\alpha\in\mathrm{Iso}\left(\frac{\mathfrak{J}}{\mathfrak{C}}\right)\mid\langle\alpha X,\alpha Y\rangle=\langle X,Y\rangle\right\}$	$U\left(\mathfrak{J}^C\right)=\left\{\alpha\in\mathrm{Iso}_C\left(\mathfrak{J}^C\right)\mid\langle\alpha X,\alpha Y\rangle=\langle X,Y\rangle\right\}$	where $U(\mathfrak{J}^C) = \{\alpha \in \mathrm{Iso}_C(\mathfrak{J}^C) \mid \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\}$ and $d = \dim F_4^C - \dim F_4 =$
0902.0431_F01412	72	■ 1.000	$d=\operatorname{dim} F_{4\{ \}^C}-\operatorname{dim} F_{4\{ \}}=$	$d = \dim F_4^C - \dim F_4 =$	where $U(\mathfrak{J}^C) = \{\alpha \in \mathrm{Iso}_C(\mathfrak{J}^C) \mid \langle \alpha X, \alpha Y \rangle = \langle X, Y \rangle\}$ and $d = \dim F_4^C - \dim F_4 =$
0902.0431_F01413	72	■ 1.000	$2\times 52-52=52$	$2\times 52-52=52$	$2\times 52-52=52$. Since F_4 is simply connected (Theorem 2.10.2), F_4^C is also simply
0902.0431_F01414	72	■ 1.000	$I_1=\operatorname{diag}(-1,1,1)\in M(3,\boldsymbol{R})$	$I_1 = \operatorname{diag}(-1, 1, 1) \in M(3, \boldsymbol{R})$	where $I_1 = \operatorname{diag}(-1, 1, 1) \in M(3, \boldsymbol{R})$. We define the Jordan multiplication $X \circ Y$ in
0902.0431_F01415	72	■ 1.000	$\frac{\mathfrak{J}}{\mathfrak{C}}\left(3,\frac{\mathfrak{C}}{\mathfrak{C}^{\prime}}\right)$	$\mathfrak{J}(3, \mathfrak{C}')$	$\mathfrak{J}(3, \mathfrak{C}')$ and $\mathfrak{J}(1, 2, \mathfrak{C})$ respectively by
0902.0431_F01416	72	■ 1.000	$\frac{\mathfrak{J}}{\mathfrak{C}}(1,2,\frac{\mathfrak{C}}{\mathfrak{C}})$	$\mathfrak{J}(1, 2, \mathfrak{C})$	$\mathfrak{J}(3, \mathfrak{C}')$ and $\mathfrak{J}(1, 2, \mathfrak{C})$ respectively by
0902.0431_F01417	72	■ 0.998	$\frac{\mathfrak{J}}{\mathfrak{C}}(1$	$\mathfrak{J}(1$	$F_{4(-20)}$ respectively the automorphism groups of the Jordan algebras $\mathfrak{J}(3, \mathfrak{C}')$ and $\mathfrak{J}(1,$
0902.0431_F01418	72	■ 1.000	$2,\frac{\mathfrak{C}}{\mathfrak{C}})$	$2, \mathfrak{C})$	$2, \mathfrak{C})$:
0902.0431_F01419	72	■ 1.000	$z\left(F_{4(4)}\right)$	$z\left(F_{4(4)}\right)$	Theorem 2.14.2. The centers $z(F_{4(4)})$ and $z(F_{4(-20)})$ are trivial:
0902.0431_F01420	72	■ 1.000	$z\left(F_{4(-20)}\right)$	$z\left(F_{4(-20)}\right)$	Theorem 2.14.2. The centers $z(F_{4(4)})$ and $z(F_{4(-20)})$ are trivial:
0902.0431_F01421	73	■ 1.000	λ	λ	If we define an involutive automorphism λ of the group E_6^C by
0902.0431_F01422	73	■ 1.000	$\phi\in\mathfrak{e}_6^C$	$\phi \in \mathfrak{e}_6^C$	(2) Any element $\phi \in \mathfrak{e}_6^C$ is uniquely expressed by
0902.0431_F01423	74	■ 0.539	$\phi\in\mathfrak{e}_6^C$	$\phi \in \mathfrak{e}_6^C$	(2) For $\phi \in \mathfrak{e}_6^C$, by letting $T = \phi E$, we obtain $T \in \mathfrak{J}^C$, and $\operatorname{tr}(T) = 0$. Certainly
0902.0431_F01424	74	■ 0.539	$T=\phi E$	$T = \phi E$	(2) For $\phi \in \mathfrak{e}_6^C$, by letting $T = \phi E$, we obtain $T \in \mathfrak{J}^C$, and $\operatorname{tr}(T) = 0$. Certainly
0902.0431_F01425	74	■ 0.539	$T\in\mathfrak{J}^C$	$T \in \mathfrak{J}^C$	(2) For $\phi \in \mathfrak{e}_6^C$, by letting $T = \phi E$, we obtain $T \in \mathfrak{J}^C$, and $\operatorname{tr}(T) = 0$. Certainly
0902.0431_F01426	74	■ 0.539	$\operatorname{tr}(T)=0$	$\operatorname{tr}(T) = 0$	(2) For $\phi \in \mathfrak{e}_6^C$, by letting $T = \phi E$, we obtain $T \in \mathfrak{J}^C$, and $\operatorname{tr}(T) = 0$. Certainly
0902.0431_F01427	74	■ 1.000	$\operatorname{tr}(T)=(T,E,E)=(\phi E,E,E)=0$	$\operatorname{tr}(T) = (T, E, E) = (\phi E, E, E) = 0$	$\operatorname{tr}(T) = (T, E, E) = (\phi E, E, E) = 0$. If we put $\delta = \phi - \tilde{T}$, then $\delta \in \mathfrak{e}_6^C$. Moreover
0902.0431_F01428	74	■ 1.000	$\delta=\phi-\widetilde{T}$	$\delta = \phi - \tilde{T}$	$\operatorname{tr}(T) = (T, E, E) = (\phi E, E, E) = 0$. If we put $\delta = \phi - \tilde{T}$, then $\delta \in \mathfrak{e}_6^C$. Moreover
0902.0431_F01429	74	■ 1.000	$\delta\in\mathfrak{e}_6^C$	$\delta \in \mathfrak{e}_6^C$	$\operatorname{tr}(T) = (T, E, E) = (\phi E, E, E) = 0$. If we put $\delta = \phi - \tilde{T}$, then $\delta \in \mathfrak{e}_6^C$. Moreover
0902.0431_F01430	74	■ 0.999	$\delta E=\phi E-\widetilde{T}E=T-T=0$	$\delta E = \phi E - \tilde{T}E = T - T = 0$	$\delta \in \mathfrak{f}_4^C$, because $\delta E = \phi E - \tilde{T}E = T - T = 0$. Hence we have $\phi = \delta + \tilde{T}$. To prove
0902.0431_F01431	74	■ 0.999	$\phi=\delta+\widetilde{T}$	$\phi = \delta + \tilde{T}$	$\delta \in \mathfrak{f}_4^C$, because $\delta E = \phi E - \tilde{T}E = T - T = 0$. Hence we have $\phi = \delta + \tilde{T}$. To prove
0902.0431_F01432	74	■ 1.000	$T=0$	$T = 0$	Certainly, let apply it on E , then we have $T = 0$, so that $\delta = 0$. Finally, we have
0902.0431_F01433	74	■ 1.000	$\delta=0$	$\delta = 0$	Certainly, let apply it on E , then we have $T = 0$, so that $\delta = 0$. Finally, we have
0902.0431_F01434	74	■ 1.000	$\operatorname{dim}_C\left(\frac{\mathfrak{e}_6}{\mathfrak{C}}\right)=52+26=78$	$\dim_C\left(\mathfrak{e}_6^C\right)=52+26=78$	$\dim_C(\mathfrak{e}_6^C) = 52 + 26 = 78$ from the expression above. Therefore the theorem is
0902.0431_F01435	74	■ 0.642	$[\phi_1,\phi_2]$	$[\phi_1, \phi_2]$	Theorem 3.2.2. The Lie bracket $[\phi_1, \phi_2]$ in $\mathfrak{e}_6^C$ is given by
0902.0431_F01436	74	■ 1.000	$\phi_i=\delta_i+\widetilde{T}_i,\delta_i\in\mathfrak{f}_4^C,T_i\in\mathfrak{J}_0^C$	$\phi_i = \delta_i + \tilde{T}_i, \delta_i \in \mathfrak{f}_4^C, T_i \in \mathfrak{J}_0^C$	where $\phi_i = \delta_i + \tilde{T}_i, \delta_i \in \mathfrak{f}_4^C, T_i \in \mathfrak{J}_0^C$.
0902.0431_F01437	74	■ 1.000	$[\delta,\widetilde{T}]=\widetilde{\delta T}$	$[\delta, \tilde{T}] = \widetilde{\delta T}$	Proof. It is sufficient to show that $[\delta, \tilde{T}] = \widetilde{\delta T}$ for $\delta \in \mathfrak{f}_4^C, T \in \mathfrak{J}_0^C$. Now,
0902.0431_F01438	74	■ 1.000	$\delta\in\mathfrak{f}_4^C,T\in\mathfrak{J}_0^C$	$\delta \in \mathfrak{f}_4^C, T \in \mathfrak{J}_0^C$	Proof. It is sufficient to show that $[\delta, \tilde{T}] = \widetilde{\delta T}$ for $\delta \in \mathfrak{f}_4^C, T \in \mathfrak{J}_0^C$. Now,
0902.0431_F01439	74	■ 0.997	$\phi=\delta+\widetilde{T}\in\mathfrak{e}_6^C,\delta\in\mathfrak{f}_4^C,T\in\mathfrak{J}_0^C$	$\phi = \delta + \tilde{T} \in \mathfrak{e}_6^C, \delta \in \mathfrak{f}_4^C, T \in \mathfrak{J}_0^C$	Lemma 3.2.3. For $\phi = \delta + \tilde{T} \in \mathfrak{e}_6^C, \delta \in \mathfrak{f}_4^C, T \in \mathfrak{J}_0^C$, we have
0902.0431_F01440	74	■ 1.000	$-{}^t\phi\in\mathfrak{e}_6^C$	$-{}^t\phi \in \mathfrak{e}_6^C$	In particular, $-{}^t\phi \in \mathfrak{e}_6^C$.
0902.0431_F01441	74	■ 0.992	$\left(-{}^t\phi X,Y\right)=-(X,\phi Y)=-(X,\delta Y+\widetilde{T}Y)=-(X,\delta Y)-(X,T\circ Y)$	$\left(-{}^t\phi X, Y\right) = -(X, \phi Y) = -(X, \delta Y + \tilde{T}Y) = -(X, \delta Y) - (X, T \circ Y)$	Proof. $\left(-{}^t\phi X, Y\right) = -(X, \phi Y) = -(X, \delta Y + \tilde{T}Y) = -(X, \delta Y) - (X, T \circ Y)$
0902.0431_F01442	74	■ 0.997	$-{}^t\phi=\delta-\widetilde{T}\in\mathfrak{e}_6^C$	$-{}^t\phi = \delta - \tilde{T} \in \mathfrak{e}_6^C$	Therefore $-{}^t\phi = \delta - \tilde{T} \in \mathfrak{e}_6^C$.
0902.0431_F01443	74	■ 1.000	$\phi\in\mathfrak{e}_6$	$\phi \in \mathfrak{e}_6$	(2) Any element $\phi \in \mathfrak{e}_6$ is uniquely expressed by
0902.0431_F01444	74	■ 0.955	$\tau\lambda(\phi)\tau=\phi$	$\tau\lambda(\phi)\tau = \phi$	(2) For $\phi \in \mathfrak{e}_6^C$, the condition $\phi \in \mathfrak{e}_6$ is equivalent to $\tau\lambda(\phi)\tau = \phi$. Now, if ϕ is
0902.0431_F01445	74	■ 0.955	ϕ	ϕ	(2) For $\phi \in \mathfrak{e}_6^C$, the condition $\phi \in \mathfrak{e}_6$ is equivalent to $\tau\lambda(\phi)\tau = \phi$. Now, if ϕ is

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_F01446	74	■ 0.999	$\phi=\delta+\widetilde{T^{\prime}}, \delta \in \mathfrak{f}_4^C, T^{\prime} \in \mathfrak{J}_0^C$	$\phi = \delta + \widetilde{T'}, \delta \in \mathfrak{f}_4^C, T' \in \mathfrak{J}_0^C$	of the form $\boxed{\phi = \delta + T', \delta \in \mathfrak{f}_4^C, T' \in \mathfrak{J}_0^C}$ (Theorem 3.2.1.(2)), then $\tau\lambda(\phi)\tau = \phi$ is
0902.0431_F01447	75	■ 1.000	$\tau \delta \tau - \widetilde{\tau T^{\prime}} = \delta + \widetilde{T^{\prime}}$	$\tau\delta\tau - \widetilde{\tau T'} = \delta + \widetilde{T'}$	$\boxed{\tau\delta\tau - \tau T' = \delta + T'}$ (Lemma 3.2.3), that is, $\tau\delta\tau = \delta$ and $\tau T' = -T'$. Hence $\delta \in \mathfrak{f}_4$,
0902.0431_F01448	75	■ 1.000	$\tau \delta \tau = \delta$	$\tau\delta\tau = \delta$	$\tau\delta\tau - \widetilde{\tau T'} = \delta + \widetilde{T'}$ (Lemma 3.2.3), that is, $\tau\delta\tau = \delta$ and $\tau T' = -T'$. Hence $\delta \in \mathfrak{f}_4$,
0902.0431_F01449	75	■ 1.000	$\tau T^{\prime} = -T^{\prime}$	$\tau T' = -T'$	$\tau\delta\tau - \widetilde{\tau T'} = \delta + \widetilde{T'}$ (Lemma 3.2.3), that is, $\tau\delta\tau = \delta$ and $\tau T' = -T'$. Hence $\delta \in \mathfrak{f}_4$,
0902.0431_F01450	75	■ 1.000	$T^{\prime}$	T'	and T' is of the form $T' = iT, T \in \mathfrak{J}_0$.
0902.0431_F01451	75	■ 1.000	$T^{\prime}=i T, T \in \mathfrak{J}_0$	$T' = iT, T \in \mathfrak{J}_0$	and T' is of the form $\boxed{T' = iT, T \in \mathfrak{J}_0}$.
0902.0431_F01452	75	■ 1.000	ϕ^*	ϕ^*	Proof. For $\phi \in \mathfrak{e}_6^C$, the conjugate transposed mapping ϕ^* of ϕ with respect to
0902.0431_F01453	75	■ 0.997	$\phi^*=\tau^t \phi \tau \in \mathfrak{e}_6^C$	$\phi^* = \tau^t \phi \tau \in \mathfrak{e}_6^C$	the inner product $\langle X, Y \rangle$ of $\mathfrak{J}^C$ is $\boxed{\phi^* = \tau^t \phi \tau \in \mathfrak{e}_6^C}$, and for $\phi \in \mathfrak{e}_6^C$, ϕ belongs to $\mathfrak{e}_6$
0902.0431_F01454	75	■ 0.997	$\phi \in \mathfrak{e}_6^C, \phi$	$\phi \in \mathfrak{e}_6^C, \phi$	the inner product $\langle X, Y \rangle$ of $\mathfrak{J}^C$ is $\phi^* = \tau^t \phi \tau \in \mathfrak{e}_6^C$, and for $\phi \in \mathfrak{e}_6^C$, ϕ belongs to $\mathfrak{e}_6$
0902.0431_F01455	75	■ 0.999	$\phi^*=-\phi$	$\phi^* = -\phi$	if and only if $\boxed{\phi^* = -\phi}$. Now, any element $\phi \in \mathfrak{e}_6^C$ can be uniquely expressed as
0902.0431_F01456	75	■ 0.997	$p: \mathfrak{e}_6^C \rightarrow \mathfrak{f}_4^C$	$p: \mathfrak{e}_6^C \rightarrow \mathfrak{f}_4^C$	Let $p: \mathfrak{e}_6^C \rightarrow \mathfrak{f}_4^C$ and $q: \mathfrak{e}_6^C \rightarrow \widetilde{\mathfrak{J}}_0^C$ be projections of $\mathfrak{e}_6^C = \mathfrak{f}_4^C \oplus \widetilde{\mathfrak{J}}_0^C$. Now, let $\mathfrak{a}$ be
0902.0431_F01457	75	■ 0.997	$q: \mathfrak{e}_6^C \rightarrow \widetilde{\mathfrak{J}}_0^C$	$q: \mathfrak{e}_6^C \rightarrow \widetilde{\mathfrak{J}}_0^C$	Let $p: \mathfrak{e}_6^C \rightarrow \mathfrak{f}_4^C$ and $q: \mathfrak{e}_6^C \rightarrow \widetilde{\mathfrak{J}}_0^C$ be projections of $\mathfrak{e}_6^C = \mathfrak{f}_4^C \oplus \widetilde{\mathfrak{J}}_0^C$. Now, let $\mathfrak{a}$ be
0902.0431_F01458	75	■ 0.997	$\mathfrak{e}_6^C = \mathfrak{f}_4^C \oplus \widetilde{\mathfrak{J}}_0^C$	$\mathfrak{e}_6^C = \mathfrak{f}_4^C \oplus \widetilde{\mathfrak{J}}_0^C$	Let $p: \mathfrak{e}_6^C \rightarrow \mathfrak{f}_4^C$ and $q: \mathfrak{e}_6^C \rightarrow \widetilde{\mathfrak{J}}_0^C$ be projections of $\boxed{\mathfrak{e}_6^C = \mathfrak{f}_4^C \oplus \widetilde{\mathfrak{J}}_0^C}$. Now, let $\mathfrak{a}$ be
0902.0431_F01459	75	■ 1.000	$\delta \in p(\mathfrak{a})$	$\delta \in p(\mathfrak{a})$	a non-zero ideal of $\mathfrak{e}_6^C$. Then $p(\mathfrak{a})$ is an ideal of $\mathfrak{f}_4^C$. Indeed, if $\delta \in p(\mathfrak{a})$, then there
0902.0431_F01460	75	■ 1.000	$T \in \mathfrak{J}_0^C$	$T \in \mathfrak{J}_0^C$	exists $\boxed{T \in \mathfrak{J}_0^C}$ such that $\delta + \widetilde{T} \in \mathfrak{a}$. For any $\delta_1 \in \mathfrak{f}_4^C$, we have
0902.0431_F01461	75	■ 1.000	$\delta + \widetilde{T} \in \mathfrak{a}$	$\delta + \widetilde{T} \in \mathfrak{a}$	exists $T \in \mathfrak{J}_0^C$ such that $\boxed{\delta + \widetilde{T} \in \mathfrak{a}}$. For any $\delta_1 \in \mathfrak{f}_4^C$, we have
0902.0431_F01462	75	■ 1.000	$\delta_1 \in \mathfrak{f}_4^C$	$\delta_1 \in \mathfrak{f}_4^C$	exists $T \in \mathfrak{J}_0^C$ such that $\delta + \widetilde{T} \in \mathfrak{a}$. For any $\delta_1 \in \mathfrak{f}_4^C$, we have
0902.0431_F01463	75	■ 1.000	$[\delta_1, \delta] \in \mathfrak{a}$	$[\delta_1, \delta] \in \mathfrak{a}$	hence $\boxed{[\delta_1, \delta] \in \mathfrak{a}}$.
0902.0431_F01464	75	■ 1.000	$\mathfrak{f}_4^C \cap \mathfrak{a} \neq \{0\}$	$\mathfrak{f}_4^C \cap \mathfrak{a} \neq \{0\}$	We shall show that either $\mathfrak{f}_4^C \cap \mathfrak{a} \neq \{0\}$ or $\widetilde{\mathfrak{J}}_0^C \cap \mathfrak{a} \neq \{0\}$. Assume that $\mathfrak{f}_4^C \cap \mathfrak{a} = \{0\}$
0902.0431_F01465	75	■ 1.000	$\widetilde{\mathfrak{J}}_0^C \cap \mathfrak{a} \neq \{0\}$	$\widetilde{\mathfrak{J}}_0^C \cap \mathfrak{a} \neq \{0\}$	We shall show that either $\mathfrak{f}_4^C \cap \mathfrak{a} \neq \{0\}$ or $\widetilde{\mathfrak{J}}_0^C \cap \mathfrak{a} \neq \{0\}$. Assume that $\mathfrak{f}_4^C \cap \mathfrak{a} = \{0\}$
0902.0431_F01466	75	■ 1.000	$\mathfrak{f}_4^C \cap \mathfrak{a} = \{0\}$	$\mathfrak{f}_4^C \cap \mathfrak{a} = \{0\}$	We shall show that either $\mathfrak{f}_4^C \cap \mathfrak{a} \neq \{0\}$ or $\widetilde{\mathfrak{J}}_0^C \cap \mathfrak{a} \neq \{0\}$. Assume that $\mathfrak{f}_4^C \cap \mathfrak{a} = \{0\}$
0902.0431_F01467	75	■ 0.876	$\widetilde{\mathfrak{J}}_0^C \cap \mathfrak{a} = \{0\}$	$\widetilde{\mathfrak{J}}_0^C \cap \mathfrak{a} = \{0\}$	and $\widetilde{\mathfrak{J}}_0^C \cap \mathfrak{a} = \{0\}$. Then the mapping $p _{\mathfrak{a}}: \mathfrak{a} \rightarrow \mathfrak{f}_4^C$ is injective because $\widetilde{\mathfrak{J}}_0^C \cap \mathfrak{a} = \{0\}$.
0902.0431_F01468	75	■ 0.876	$p _{\mathfrak{a}}: \mathfrak{a} \rightarrow \mathfrak{f}_4^C$	$p _{\mathfrak{a}}: \mathfrak{a} \rightarrow \mathfrak{f}_4^C$	and $\widetilde{\mathfrak{J}}_0^C \cap \mathfrak{a} = \{0\}$. Then the mapping $\boxed{p _{\mathfrak{a}}: \mathfrak{a} \rightarrow \mathfrak{f}_4^C}$ is injective because $\widetilde{\mathfrak{J}}_0^C \cap \mathfrak{a} = \{0\}$.
0902.0431_F01469	75	■ 1.000	$p(\mathfrak{a}) = \mathfrak{f}_4^C$	$p(\mathfrak{a}) = \mathfrak{f}_4^C$	Since $p(\mathfrak{a})$ is a non-zero ideal of $\mathfrak{f}_4^C$ and $\mathfrak{f}_4^C$ is simple, we have $p(\mathfrak{a}) = \mathfrak{f}_4^C$. Hence
0902.0431_F01470	75	■ 1.000	$\dim_C \mathfrak{a} = \dim_C p(\mathfrak{a}) = \dim_C \mathfrak{f}_4^C = 52$	$\dim_C \mathfrak{a} = \dim_C p(\mathfrak{a}) = \dim_C \mathfrak{f}_4^C = 52$	$\boxed{\dim_C \mathfrak{a} = \dim_C p(\mathfrak{a}) = \dim_C \mathfrak{f}_4^C = 52}$. On the other hand, since $\mathfrak{f}_4^C \cap \mathfrak{a} = \{0\}$,
0902.0431_F01471	75	■ 0.673	$q _{\mathfrak{a}}: \mathfrak{a} \rightarrow \widetilde{\mathfrak{J}}_0^C$	$q _{\mathfrak{a}}: \mathfrak{a} \rightarrow \widetilde{\mathfrak{J}}_0^C$	$q _{\mathfrak{a}}: \mathfrak{a} \rightarrow \widetilde{\mathfrak{J}}_0^C$ is also injective, we have $\dim_C \mathfrak{a} \leq \dim_C \widetilde{\mathfrak{J}}_0^C = \dim_C \mathfrak{J}_0^C = 26$. This
0902.0431_F01472	75	■ 0.673	$\dim_C \mathfrak{a} \leq \dim_C \widetilde{\mathfrak{J}}_0^C = \dim_C \mathfrak{J}_0^C = 26$	$\dim_C \mathfrak{a} \leq \dim_C \widetilde{\mathfrak{J}}_0^C = \dim_C \mathfrak{J}_0^C = 26$	$q _{\mathfrak{a}}: \mathfrak{a} \rightarrow \widetilde{\mathfrak{J}}_0^C$ is also injective, we have $\boxed{\dim_C \mathfrak{a} \leq \dim_C \widetilde{\mathfrak{J}}_0^C = \dim_C \mathfrak{J}_0^C = 26}$. This
0902.0431_F01473	75	■ 1.000	$\mathfrak{f}_4^C \cap \mathfrak{a} = \mathfrak{f}_4^C$	$\mathfrak{f}_4^C \cap \mathfrak{a} = \mathfrak{f}_4^C$	(1) Case $\mathfrak{f}_4^C \cap \mathfrak{a} \neq \{0\}$. From the simplicity of $\mathfrak{f}_4^C$, we have $\mathfrak{f}_4^C \cap \mathfrak{a} = \mathfrak{f}_4^C$, hence
0902.0431_F01474	75	■ 1.000	$\mathfrak{a} \supset \mathfrak{f}_4^C$	$\mathfrak{a} \supset \mathfrak{f}_4^C$	$\boxed{\mathfrak{a} \supset \mathfrak{f}_4^C}$. On the other hand, we have
0902.0431_F01475	75	■ 0.896	$\mathfrak{a} \supset \mathfrak{f}_4^C \oplus \widetilde{\mathfrak{J}}_0^C = \mathfrak{e}_6^C$	$\mathfrak{a} \supset \mathfrak{f}_4^C \oplus \widetilde{\mathfrak{J}}_0^C = \mathfrak{e}_6^C$	Consequently $\boxed{\mathfrak{a} \supset \mathfrak{f}_4^C \oplus \widetilde{\mathfrak{J}}_0^C = \mathfrak{e}_6^C}$.
0902.0431_F01476	75	■ 0.953	$\widetilde{\mathfrak{J}}_0^C \cap \mathfrak{a} \neq \{0\}$	$\widetilde{\mathfrak{J}}_0^C \cap \mathfrak{a} \neq \{0\}$	(2) Case $\widetilde{\mathfrak{J}}_0^C \cap \mathfrak{a} \neq \{0\}$. Let $\tilde{A}$ ($A \in \mathfrak{J}_0^C$) be a non-zero element of $\widetilde{\mathfrak{J}}_0^C \cap \mathfrak{a} \subset \mathfrak{a}$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_F01477	75	■ 0.953	$\widetilde{A} \in \mathfrak{J}_0^C$	$\widetilde{A} (A \in \mathfrak{J}_0^C)$	(2) Case $\widetilde{\mathfrak{J}}_0^C \cap \mathfrak{a} \neq \{0\}$. Let $\widetilde{A} (A \in \mathfrak{J}_0^C)$ be a non-zero element of $\widetilde{\mathfrak{J}}_0^C \cap \mathfrak{a} \subset \mathfrak{a}$.
0902.0431_F01478	75	■ 0.953	$\widetilde{\mathfrak{J}}_0^C \cap \mathfrak{a} \subset \mathfrak{a}$	$\widetilde{\mathfrak{J}}_0^C \cap \mathfrak{a} \subset \mathfrak{a}$	(2) Case $\widetilde{\mathfrak{J}}_0^C \cap \mathfrak{a} \neq \{0\}$. Let $\widetilde{A} (A \in \mathfrak{J}_0^C)$ be a non-zero element of $\widetilde{\mathfrak{J}}_0^C \cap \mathfrak{a} \subset \mathfrak{a}$.
0902.0431_F01479	75	■ 0.999	$0 \neq [\widetilde{A}, \widetilde{B}] \in \mathfrak{f}_4^C \cap \mathfrak{a}$	$0 \neq [\widetilde{A}, \widetilde{B}] \in \mathfrak{f}_4^C \cap \mathfrak{a}$	Choose $B \in \mathfrak{J}_0^C$ such that $[\widetilde{A}, \widetilde{B}] \neq 0$ (Lemma 2.5.4), then $0 \neq [\widetilde{A}, \widetilde{B}] \in \mathfrak{f}_4^C \cap \mathfrak{a}$.
0902.0431_F01480	76	■ 1.000	$\mathfrak{a} = \mathfrak{e}_6^C$	$\mathfrak{a} = \mathfrak{e}_6^C$	Therefore we have $\mathfrak{a} = \mathfrak{e}_6^C$.
0902.0431_F01481	76	■ 0.965	$\mathfrak{e}_6^C \mathfrak{J}^C = \left\{ \sum_i \phi_i A_i \mid \phi_i \in \mathfrak{e}_6^C, A_i \in \mathfrak{J}^C \right\} = \mathfrak{J}^C$	$\mathfrak{e}_6^C \mathfrak{J}^C = \left\{ \sum_i \phi_i A_i \mid \phi_i \in \mathfrak{e}_6^C, A_i \in \mathfrak{J}^C \right\} = \mathfrak{J}^C$	(3) $\mathfrak{e}_6^C \mathfrak{J}^C = \left\{ \sum_i \phi_i A_i \mid \phi_i \in \mathfrak{e}_6^C, A_i \in \mathfrak{J}^C \right\} = \mathfrak{J}^C$.
0902.0431_F01482	76	■ 1.000	$X = X(\xi, x)$	$X = X(\xi, x)$	Proof. (1) Let $\mathfrak{a}$ be a non-zero ideal of $\mathfrak{J}^C$ and $X = X(\xi, x)$ a non-zero element
0902.0431_F01483	76	■ 1.000	$\xi_1 \neq 0$	$\xi_1 \neq 0$	(i) Case $\xi_1 \neq 0$. From $\xi_1 E_1 = (2X \circ E_1 - X) \circ E_1 \in \mathfrak{a}$, we have $E_1 \in \mathfrak{a}$.
0902.0431_F01484	76	■ 1.000	$\xi_1 E_1 = (2X \circ E_1 - X) \circ E_1 \in \mathfrak{a}$	$\xi_1 E_1 = (2X \circ E_1 - X) \circ E_1 \in \mathfrak{a}$	(i) Case $\xi_1 \neq 0$. From $\xi_1 E_1 = (2X \circ E_1 - X) \circ E_1 \in \mathfrak{a}$, we have $E_1 \in \mathfrak{a}$.
0902.0431_F01485	76	■ 1.000	$E_1 \in \mathfrak{a}$	$E_1 \in \mathfrak{a}$	(i) Case $\xi_1 \neq 0$. From $\xi_1 E_1 = (2X \circ E_1 - X) \circ E_1 \in \mathfrak{a}$, we have $E_1 \in \mathfrak{a}$.
0902.0431_F01486	76	■ 1.000	$F_2(1) = 2E_1 \circ F_2(1) \in \mathfrak{a}$	$F_2(1) = 2E_1 \circ F_2(1) \in \mathfrak{a}$	Next, from $F_2(1) = 2E_1 \circ F_2(1) \in \mathfrak{a}$ and $E_1 + E_3 = F_2(1) \circ F_2(1) \in \mathfrak{a}$, we have
0902.0431_F01487	76	■ 1.000	$E_1 + E_3 = F_2(1) \circ F_2(1) \in \mathfrak{a}$	$E_1 + E_3 = F_2(1) \circ F_2(1) \in \mathfrak{a}$	Next, from $F_2(1) = 2E_1 \circ F_2(1) \in \mathfrak{a}$ and $E_1 + E_3 = F_2(1) \circ F_2(1) \in \mathfrak{a}$, we have
0902.0431_F01488	76	■ 1.000	$E_3 = (E_1 + E_3) - E_1 \in \mathfrak{a}$	$E_3 = (E_1 + E_3) - E_1 \in \mathfrak{a}$	$E_3 = (E_1 + E_3) - E_1 \in \mathfrak{a}$. Similarly $E_2 \in \mathfrak{a}$, so that $E = E_1 + E_2 + E_3 \in \mathfrak{a}$. Now,
0902.0431_F01489	76	■ 1.000	$E_2 \in \mathfrak{a}$	$E_2 \in \mathfrak{a}$	$E_3 = (E_1 + E_3) - E_1 \in \mathfrak{a}$. Similarly $E_2 \in \mathfrak{a}$, so that $E = E_1 + E_2 + E_3 \in \mathfrak{a}$. Now,
0902.0431_F01490	76	■ 1.000	$E = E_1 + E_2 + E_3 \in \mathfrak{a}$	$E = E_1 + E_2 + E_3 \in \mathfrak{a}$	$E_3 = (E_1 + E_3) - E_1 \in \mathfrak{a}$. Similarly $E_2 \in \mathfrak{a}$, so that $E = E_1 + E_2 + E_3 \in \mathfrak{a}$. Now,
0902.0431_F01491	76	■ 1.000	$X \in \mathfrak{J}^C, X = E \circ X \in \mathfrak{a}$	$X \in \mathfrak{J}^C, X = E \circ X \in \mathfrak{a}$	for any $X \in \mathfrak{J}^C, X = E \circ X \in \mathfrak{a}$, and so $\mathfrak{a} = \mathfrak{J}^C$. In the case $\xi_2 \neq 0$ or $\xi_3 \neq 0$, the
0902.0431_F01492	76	■ 1.000	$\mathfrak{a} = \mathfrak{J}^C$	$\mathfrak{a} = \mathfrak{J}^C$	for any $X \in \mathfrak{J}^C, X = E \circ X \in \mathfrak{a}$, and so $\mathfrak{a} = \mathfrak{J}^C$. In the case $\xi_2 \neq 0$ or $\xi_3 \neq 0$, the
0902.0431_F01493	76	■ 1.000	$\xi_2 \neq 0$	$\xi_2 \neq 0$	for any $X \in \mathfrak{J}^C, X = E \circ X \in \mathfrak{a}$, and so $\mathfrak{a} = \mathfrak{J}^C$. In the case $\xi_2 \neq 0$ or $\xi_3 \neq 0$, the
0902.0431_F01494	76	■ 1.000	$\xi_3 \neq 0$	$\xi_3 \neq 0$	for any $X \in \mathfrak{J}^C, X = E \circ X \in \mathfrak{a}$, and so $\mathfrak{a} = \mathfrak{J}^C$. In the case $\xi_2 \neq 0$ or $\xi_3 \neq 0$, the
0902.0431_F01495	76	■ 1.000	$\xi_1 = \xi_2 = \xi_3 = 0, x_1 \neq 0$	$\xi_1 = \xi_2 = \xi_3 = 0, x_1 \neq 0$	(ii) Case $\xi_1 = \xi_2 = \xi_3 = 0, x_1 \neq 0$. We have $F_1(x_1) = 4(X \circ E_2) \circ E_3 \in \mathfrak{a}$. Choose
0902.0431_F01496	76	■ 1.000	$F_1(x_1) = 4(X \circ E_2) \circ E_3 \in \mathfrak{a}$	$F_1(x_1) = 4(X \circ E_2) \circ E_3 \in \mathfrak{a}$	(ii) Case $\xi_1 = \xi_2 = \xi_3 = 0, x_1 \neq 0$. We have $F_1(x_1) = 4(X \circ E_2) \circ E_3 \in \mathfrak{a}$. Choose
0902.0431_F01497	76	■ 1.000	$a \in \mathfrak{e}^C$	$a \in \mathfrak{e}^C$	$a \in \mathfrak{e}^C$ such that $(x_1, a) = 1$, then $F_1(x_1) \circ F_1(a) = (x_1, a)(E_2 + E_3) = E_2 + E_3 \in \mathfrak{a}$.
0902.0431_F01498	76	■ 1.000	$(x_1, a) = 1$	$(x_1, a) = 1$	$a \in \mathfrak{e}^C$ such that $(x_1, a) = 1$, then $F_1(x_1) \circ F_1(a) = (x_1, a)(E_2 + E_3) = E_2 + E_3 \in \mathfrak{a}$.
0902.0431_F01499	76	■ 1.000	$F_1(x_1) \circ F_1(a) = (x_1, a)(E_2 + E_3) = E_2 + E_3 \in \mathfrak{a}$	$F_1(x_1) \circ F_1(a) = (x_1, a)(E_2 + E_3) = E_2 + E_3 \in \mathfrak{a}$	$a \in \mathfrak{e}^C$ such that $(x_1, a) = 1$, then $F_1(x_1) \circ F_1(a) = (x_1, a)(E_2 + E_3) = E_2 + E_3 \in \mathfrak{a}$.
0902.0431_F01500	76	■ 1.000	$X \in W$	$X \in W$	$X \in W$, we have
0902.0431_F01501	76	■ 0.598	$\mathfrak{J}^C((1))$	$\mathfrak{J}^C((1))$	Hence, W is an ideal of $\mathfrak{J}^C$. From the simplicity of $\mathfrak{J}^C$ ((1) above), we have $W = \mathfrak{J}^C$.
0902.0431_F01502	76	■ 0.598	$W = \mathfrak{J}^C$	$W = \mathfrak{J}^C$	Hence, W is an ideal of $\mathfrak{J}^C$. From the simplicity of $\mathfrak{J}^C$ ((1) above), we have $W = \mathfrak{J}^C$.
0902.0431_F01503	76	■ 1.000	$\mathfrak{e}_6^C \mathfrak{J}^C$	$\mathfrak{e}_6^C \mathfrak{J}^C$	(3) $\mathfrak{e}_6^C \mathfrak{J}^C$ is an $\mathfrak{e}_6^C$ -invariant C -submodule of $\mathfrak{J}^C$. Hence from the irreducibility
0902.0431_F01504	76	■ 0.925	$\mathfrak{e}_6^C \mathfrak{J}^C = \mathfrak{J}^C$	$\mathfrak{e}_6^C \mathfrak{J}^C = \mathfrak{J}^C$	of $\mathfrak{J}^C$ ((2) above), we have $\mathfrak{e}_6^C \mathfrak{J}^C = \mathfrak{J}^C$.
0902.0431_F01505	76	■ 0.998	$A \vee B \in \mathfrak{e}_6^C$	$A \vee B \in \mathfrak{e}_6^C$	Definition. For $A, B \in \mathfrak{J}^C$, we define an element $A \vee B \in \mathfrak{e}_6^C$ by
0902.0431_F01506	76	■ 1.000	$X \circ (X \times X) = (\det X)E, X \in \mathfrak{J}^C$	$X \circ (X \times X) = (\det X)E, X \in \mathfrak{J}^C$	Proof. Consider Hamilton-Cayley formula $X \circ (X \times X) = (\det X)E, X \in \mathfrak{J}^C$,
0902.0431_F01507	76	■ 1.000	$\lambda A + \mu B + \nu X$	$\lambda A + \mu B + \nu X$	If we put $\lambda A + \mu B + \nu X$ in place of X , then taking the coefficient of $\lambda \mu \nu$, we have
0902.0431_F01508	77	■ 1.000	$E_i \vee E_j = 0, i \neq j$	$E_i \vee E_j = 0, i \neq j$	Lemma 3.4.2. (1) $E_i \vee E_j = 0, i \neq j$.
0902.0431_F01509	77	■ 1.000	$E_1 \vee E_1 = \frac{1}{3}(2E_1 - E_2 - E_3)^\sim$	$E_1 \vee E_1 = \frac{1}{3}(2E_1 - E_2 - E_3)^\sim$	(2) $E_1 \vee E_1 = \frac{1}{3}(2E_1 - E_2 - E_3)^\sim$, more explicitly, we have
0902.0431_F01510	77	■ 0.988	$\phi \in \text{Hom}_C(\mathfrak{J}^C)$	$\phi \in \text{Hom}_C(\mathfrak{J}^C)$	For $\phi \in \text{Hom}_C(\mathfrak{J}^C)$, we often denote $-^t \phi$ by ϕ' :
0902.0431_F01511	77	■ 0.988	$-^t \phi$	$-^t \phi$	For $\phi \in \text{Hom}_C(\mathfrak{J}^C)$, we often denote $-^t \phi$ by ϕ' :
0902.0431_F01512	77	■ 0.988	ϕ'	ϕ'	For $\phi \in \text{Hom}_C(\mathfrak{J}^C)$, we often denote $-^t \phi$ by ϕ' :

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_F01513	78	■ 1.000	$(A \vee B)^{\prime} = \left([\widetilde{A}, \widetilde{B}] + \left(A \circ B - \frac{1}{3}(A, B)E \right)^{\sim} \right)^{\prime} = [\widetilde{A}, \widetilde{B}] - \left(A \circ B - \frac{1}{3}(A, B)E \right)^{\sim}$	$(A \vee B)^{\prime} = \left([\widetilde{A}, \widetilde{B}] + \left(A \circ B - \frac{1}{3}(A, B)E \right)^{\sim} \right)^{\prime} = [\widetilde{A}, \widetilde{B}] - \left(A \circ B - \frac{1}{3}(A, B)E \right)^{\sim}$	(2) $[A \vee B]^{\prime} = \left([\widetilde{A}, \widetilde{B}] + \left(A \circ B - \frac{1}{3}(A, B)E \right)^{\sim} \right)^{\prime} = [\widetilde{A}, \widetilde{B}] - \left(A \circ B - \frac{1}{3}(A, B)E \right)^{\sim}$
0902.0431_F01514	78	■ 0.999	(	(	(Lemma 3.2.3) $= -[\widetilde{B}, \widetilde{A}] - \left(A \circ B - \frac{1}{3}(A, B)E \right)^{\sim} = -B \vee A$.
0902.0431_F01515	78	■ 0.999	$) = -[\widetilde{B}, \widetilde{A}] - \left(A \circ B - \frac{1}{3}(A, B)E \right)^{\sim} = -B \vee A$	$) = -[\widetilde{B}, \widetilde{A}] - \left(A \circ B - \frac{1}{3}(A, B)E \right)^{\sim} = -B \vee A$	(Lemma 3.2.3) $= -[\widetilde{B}, \widetilde{A}] - \left(A \circ B - \frac{1}{3}(A, B)E \right)^{\sim} = -B \vee A$.
0902.0431_F01516	78	■ 0.869	$\phi = \sum_i \left(A_i \vee B_i \right), A_i, B_i \in \mathfrak{J}^C$	$\phi = \sum_i (A_i \vee B_i), A_i, B_i \in \mathfrak{J}^C$	(2) Any element $\phi \in \mathfrak{e}_6^C$ is expressed by $\phi = \sum_i (A_i \vee B_i), A_i, B_i \in \mathfrak{J}^C$.
0902.0431_F01517	78	■ 1.000	$[\phi, A \vee B]X = \phi(A \vee B)X - (A \vee B)\phi X$	$[\phi, A \vee B]X = \phi(A \vee B)X - (A \vee B)\phi X$	Proof. (1) $[\phi, A \vee B]X = \phi(A \vee B)X - (A \vee B)\phi X$
0902.0431_F01518	78	■ 0.988	$\mathfrak{a} = \left\{ \sum_i \left(A_i \vee B_i \right) \mid A_i, B_i \in \mathfrak{J}^C \right\}$	$\mathfrak{a} = \left\{ \sum_i (A_i \vee B_i) \mid A_i, B_i \in \mathfrak{J}^C \right\}$	(2) We can see from (1) that $\mathfrak{a} = \left\{ \sum_i (A_i \vee B_i) \mid A_i, B_i \in \mathfrak{J}^C \right\}$ is an ideal of $\mathfrak{e}_6^C$.
0902.0431_F01519	78	■ 1.000	$\left(\phi_1, \phi_2 \right)_6$	$(\phi_1, \phi_2)_6$	Definition. We define a symmetric inner product $(\phi_1, \phi_2)_6$ in $\mathfrak{e}_6^C$ by
0902.0431_F01520	78	■ 0.749	$\phi \in \mathfrak{e}_6^C, A, B \in \mathfrak{J}^C$	$\phi \in \mathfrak{e}_6^C, A, B \in \mathfrak{J}^C$	(2) For $\phi \in \mathfrak{e}_6^C, A, B \in \mathfrak{J}^C$, we have
0902.0431_F01521	78	■ 0.993	$\phi = \delta + \widetilde{T}, \phi_i = \delta_i + \widetilde{T}_i, \delta, \delta_i \in \mathfrak{f}_4^C, T, T_i \in \mathfrak{J}_0^C$	$\phi = \delta + \widetilde{T}, \phi_i = \delta_i + \widetilde{T}_i, \delta, \delta_i \in \mathfrak{f}_4^C, T, T_i \in \mathfrak{J}_0^C$	Proof. (1) For $\phi = \delta + \widetilde{T}, \phi_i = \delta_i + \widetilde{T}_i, \delta, \delta_i \in \mathfrak{f}_4^C, T, T_i \in \mathfrak{J}_0^C$, we have
0902.0431_F01522	79	■ 1.000	$\phi = \delta + \widetilde{T}, \delta \in \mathfrak{f}_4^C, T \in \mathfrak{J}_0^C$	$\phi = \delta + \widetilde{T}, \delta \in \mathfrak{f}_4^C, T \in \mathfrak{J}_0^C$	(2) For $\phi = \delta + \widetilde{T}, \delta \in \mathfrak{f}_4^C, T \in \mathfrak{J}_0^C$, we have
0902.0431_F01523	79	■ 1.000	B_6	B_6	Theorem 3.5.3. The Killing form B_6 of the Lie algebra $\mathfrak{e}_6^C$ is given by
0902.0431_F01524	79	■ 1.000	$\phi = \phi_1 = \phi_2 = (E_1 - E_2)^{\sim}$	$\phi = \phi_1 = \phi_2 = (E_1 - E_2)^{\sim}$	To determine these k, k' , let $\phi = \phi_1 = \phi_2 = (E_1 - E_2)^{\sim}$. Then we have
0902.0431_F01525	79	■ 1.000	$(\operatorname{ad} \phi)^2$	$(\operatorname{ad} \phi)^2$	On the other hand, $(\operatorname{ad} \phi)^2$ is calculated as follows.
0902.0431_F01526	80	■ 0.877	$= 0$	$= 0$	the others $= 0$.
0902.0431_F01527	80	■ 0.998	$k = 12$	$k = 12$	Therefore $k = 12$. Next, we will calculate $\operatorname{tr}(\phi\phi)$ as follows.
0902.0431_F01528	80	■ 0.998	$\operatorname{tr}(\phi\phi)$	$\operatorname{tr}(\phi\phi)$	Therefore $k = 12$. Next, we will calculate $\operatorname{tr}(\phi\phi)$ as follows.
0902.0431_F01529	80	■ 1.000	$k^{\prime} = 4$	$k^{\prime} = 4$	Therefore $k^{\prime} = 4$.
0902.0431_F01530	80	■ 1.000	$A \vee (A \times A) = 0, A \in \mathfrak{J}^C$	$A \vee (A \times A) = 0, A \in \mathfrak{J}^C$	(1) $A \vee (A \times A) = 0, A \in \mathfrak{J}^C$,
0902.0431_F01531	80	■ 0.999	$A \in \mathfrak{J}^C, A \neq 0$	$A \in \mathfrak{J}^C, A \neq 0$	(2) For $A \in \mathfrak{J}^C, A \neq 0$, there exists $B \in \mathfrak{J}^C$ such that $A \vee B \neq 0$.
0902.0431_F01532	80	■ 0.999	$B \in \mathfrak{J}^C$	$B \in \mathfrak{J}^C$	(2) For $A \in \mathfrak{J}^C, A \neq 0$, there exists $B \in \mathfrak{J}^C$ such that $A \vee B \neq 0$.
0902.0431_F01533	80	■ 0.999	$A \vee B \neq 0$	$A \vee B \neq 0$	(2) For $A \in \mathfrak{J}^C, A \neq 0$, there exists $B \in \mathfrak{J}^C$ such that $A \vee B \neq 0$.
0902.0431_F01534	80	■ 0.990	$(\phi, (A \times A) \vee A)_6 = (\phi(A \times A), A)$	$(\phi, (A \times A) \vee A)_6 = (\phi(A \times A), A)$	Proof. (1) $(\phi, (A \times A) \vee A)_6 = (\phi(A \times A), A)$ (Lemma 3.5.1.(2))
0902.0431_F01535	80	■ 0.867	$(\phi, (A \times A) \vee A)_6 = 0$	$(\phi, (A \times A) \vee A)_6 = 0$	Hence $(\phi, (A \times A) \vee A)_6 = 0$ for all $\phi \in \mathfrak{e}_6^C$, so that $(A \times A) \vee A = 0$, that is,
0902.0431_F01536	80	■ 0.867	$(A \times A) \vee A = 0$	$(A \times A) \vee A = 0$	Hence $(\phi, (A \times A) \vee A)_6 = 0$ for all $\phi \in \mathfrak{e}_6^C$, so that $(A \times A) \vee A = 0$, that is,
0902.0431_F01537	80	■ 1.000	$A \vee (A \times A) = 0$	$A \vee (A \times A) = 0$	$A \vee (A \times A) = 0$ (Lemma 3.4.3.(2)). If we put $\lambda A + \mu B + \nu C$ in the place of A , then
0902.0431_F01538	80	■ 1.000	$\lambda A + \mu B + \nu C$	$\lambda A + \mu B + \nu C$	$A \vee (A \times A) = 0$ (Lemma 3.4.3.(2)). If we put $\lambda A + \mu B + \nu C$ in the place of A , then
0902.0431_F01539	80	■ 1.000	$A \vee B = 0$	$A \vee B = 0$	(2) Assume that $A \vee B = 0$, that is $B \vee A = 0$ (Lemma 3.4.3.(2)) for all $B \in \mathfrak{J}^C$.
0902.0431_F01540	80	■ 1.000	$B \vee A = 0$	$B \vee A = 0$	(2) Assume that $A \vee B = 0$, that is $B \vee A = 0$ (Lemma 3.4.3.(2)) for all $B \in \mathfrak{J}^C$.
0902.0431_F01541	80	■ 1.000	$\phi \in \mathfrak{e}_6^C, 0 = (\phi, B \vee A)_6 = (\phi B, A)$	$\phi \in \mathfrak{e}_6^C, 0 = (\phi, B \vee A)_6 = (\phi B, A)$	Then for any $\phi \in \mathfrak{e}_6^C, 0 = (\phi, B \vee A)_6 = (\phi B, A)$ (Lemma 3.5.1.(2)). Since $\mathfrak{e}_6^C \mathfrak{J}^C =$
0902.0431_F01542	80	■ 1.000	$\mathfrak{e}_6^C \mathfrak{J}^C =$	$\mathfrak{e}_6^C \mathfrak{J}^C =$	Then for any $\phi \in \mathfrak{e}_6^C, 0 = (\phi, B \vee A)_6 = (\phi B, A)$ (Lemma 3.5.1.(2)). Since $\mathfrak{e}_6^C \mathfrak{J}^C =$
0902.0431_F01543	80	■ 1.000	$\left(\mathfrak{J}^C, A \right) = 0$	$(\mathfrak{J}^C, A) = 0$	$\mathfrak{J}^C$ (Proposition 3.3.2.(3)), we have $(\mathfrak{J}^C, A) = 0$, so that $A = 0$.
0902.0431_F01544	81	■ 1.000	$\left(\mathfrak{M}^r \right)^C$	$(\mathfrak{M}^r)^C$	and let $(\mathfrak{M}^r)^C$ be its complexification. In $(\mathfrak{M}^r)^C$, we define a multiplication $X \circ Y$
0902.0431_F01545	81	■ 1.000	$\delta = (\delta_1, \delta_2, \delta_3) \in \mathfrak{d}_4^C$	$\delta = (\delta_1, \delta_2, \delta_3) \in \mathfrak{d}_4^C$	where $X^* = {}^t \overline{X}$. For $\delta = (\delta_1, \delta_2, \delta_3) \in \mathfrak{d}_4^C$ satisfying the principle of triality $(\delta_1 x) y +$
0902.0431_F01546	81	■ 1.000	$(\delta_1 x) y +$	$(\delta_1 x) y +$	where $X^* = {}^t \overline{X}$. For $\delta = (\delta_1, \delta_2, \delta_3) \in \mathfrak{d}_4^C$ satisfying the principle of triality $(\delta_1 x) y +$

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0902.0431_F01547	81	■ 0.998	$x\left(\overline{\delta_2 y}\right)=\overline{\delta_3(x y)}, x, y \in \mathfrak{C}$	$x\left(\delta_2 y\right)=\overline{\delta_3(x y)}, x, y \in \mathfrak{C}$	$x\left(\delta_2 y\right)=\overline{\delta_3(x y)}, x, y \in \mathfrak{C}$, we define a C -linear mapping $\delta: (\mathfrak{M}^r)^C \rightarrow (\mathfrak{M}^r)^C$ by
0902.0431_F01548	81	■ 0.998	$\delta: \left(\mathfrak{M}^r\right)^C \rightarrow \left(\mathfrak{M}^r\right)^C$	$\delta: (\mathfrak{M}^r)^C \rightarrow (\mathfrak{M}^r)^C$	$x\left(\delta_2 y\right)=\overline{\delta_3(x y)}, x, y \in \mathfrak{C}$, we define a C -linear mapping $\delta: (\mathfrak{M}^r)^C \rightarrow (\mathfrak{M}^r)^C$ by
0902.0431_F01549	81	■ 1.000	$\delta: \mathfrak{J}^C \rightarrow \mathfrak{J}^C$	$\delta: \mathfrak{J}^C \rightarrow \mathfrak{J}^C$	Observe that this mapping δ is an extension of $\delta: \mathfrak{J}^C \rightarrow \mathfrak{J}^C$.
0902.0431_F01550	81	■ 1.000	$\delta \in \mathfrak{d}_4^C$	$\delta \in \mathfrak{d}_4^C$	Lemma 3.6.1. For $\delta \in \mathfrak{d}_4^C$, we have
0902.0431_F01551	81	■ 0.958	(i, i)	(i, i)	The (i, i) -element of $\delta X \circ Y + Y \circ \delta Y$ (note that this is contained in C) is equal to
0902.0431_F01552	81	■ 0.958	$\delta X \circ Y + Y \circ \delta Y$	$\delta X \circ Y + Y \circ \delta Y$	The (i, i) -element of $\delta X \circ Y + Y \circ \delta Y$ (note that this is contained in C) is equal to
0902.0431_F01553	81	■ 1.000	$\delta X \circ Y + Y \circ \delta Y (i \neq j)$	$\delta X \circ Y + Y \circ \delta Y (i \neq j)$	The (i, j) -element of $\delta X \circ Y + Y \circ \delta Y (i \neq j)$ is equal to
0902.0431_F01554	81	■ 1.000	i, j, k	i, j, k	(If i, j, k are all distinct, from the principle of triality, we obtain
0902.0431_F01555	81	■ 0.999	$i = k \neq j$	$i = k \neq j$	Even if $i = k \neq j$ or $i \neq k = j$, considering the fact that $\sigma_{ii} = 0$, we see that the
0902.0431_F01556	81	■ 0.999	$i \neq k = j$	$i \neq k = j$	Even if $i = k \neq j$ or $i \neq k = j$, considering the fact that $\sigma_{ii} = 0$, we see that the
0902.0431_F01557	81	■ 0.999	$\sigma_{ii} = 0$	$\sigma_{ii} = 0$	Even if $i = k \neq j$ or $i \neq k = j$, considering the fact that $\sigma_{ii} = 0$, we see that the
0902.0431_F01558	81	■ 1.000	$x_{ii} \in C$	$x_{ii} \in C$	formula above is also valid, since $x_{ii} \in C$)
0902.0431_F01559	81	■ 1.000	$T \in M\left(3, \mathfrak{C}\right)$	$T \in M\left(3, \mathfrak{C}\right)$	For $T \in M(3, \mathfrak{C})$, we define a C -linear mapping $\tilde{T}: \mathfrak{J}^C \rightarrow \mathfrak{J}^C$ by
0902.0431_F01560	81	■ 1.000	$\tilde{T}: \mathfrak{J}^C \rightarrow \mathfrak{J}^C$	$\tilde{T}: \mathfrak{J}^C \rightarrow \mathfrak{J}^C$	For $T \in M(3, \mathfrak{C})$, we define a C -linear mapping $\tilde{T}: \mathfrak{J}^C \rightarrow \mathfrak{J}^C$ by
0902.0431_F01561	82	■ 0.995	$T \in M\left(3, \mathfrak{C}\right), \operatorname{tr}(T)=0$	$T \in M\left(3, \mathfrak{C}\right), \operatorname{tr}(T)=0$	Proposition 3.6.2. For $T \in M(3, \mathfrak{C}), \operatorname{tr}(T)=0$, we have $\tilde{T} \in \mathfrak{e}_6^C$.
0902.0431_F01562	82	■ 0.995	$\tilde{T} \in \mathfrak{e}_6^C$	$\tilde{T} \in \mathfrak{e}_6^C$	Proposition 3.6.2. For $T \in M(3, \mathfrak{C}), \operatorname{tr}(T)=0$, we have $\tilde{T} \in \mathfrak{e}_6^C$.
0902.0431_F01563	82	■ 0.920	$T=T_1+T_2, T_1=\frac{T+T^*}{2}, T_2=\frac{T-T^*}{2}$	$T=T_1+T_2, T_1=\frac{T+T^*}{2}, T_2=\frac{T-T^*}{2}$	Proof. We decompose $T=T_1+T_2, T_1=\frac{T+T^*}{2}, T_2=\frac{T-T^*}{2}$. Then $\tilde{T}_1 \in \mathfrak{e}_6^C$
0902.0431_F01564	82	■ 0.920	$\tilde{T}_1 \in \mathfrak{e}_6^C$	$\tilde{T}_1 \in \mathfrak{e}_6^C$	Proof. We decompose $T=T_1+T_2, T_1=\frac{T+T^*}{2}, T_2=\frac{T-T^*}{2}$. Then $\tilde{T}_1 \in \mathfrak{e}_6^C$
0902.0431_F01565	82	■ 1.000	$\tilde{T}_2 \in \mathfrak{f}_4^C$	$\tilde{T}_2 \in \mathfrak{f}_4^C$	(Lemma 2.4.1.(1)) and $\tilde{T}_2 \in \mathfrak{f}_4^C$ (Proposition 2.3.6) $\subset \mathfrak{e}_6^C$. Therefore $\tilde{T} = \tilde{T}_1 + \tilde{T}_2 \in$
0902.0431_F01566	82	■ 1.000	$\subset \mathfrak{e}_6^C$	$\subset \mathfrak{e}_6^C$	(Lemma 2.4.1.(1)) and $\tilde{T}_2 \in \mathfrak{f}_4^C$ (Proposition 2.3.6) $\subset \mathfrak{e}_6^C$. Therefore $\tilde{T} = \tilde{T}_1 + \tilde{T}_2 \in$
0902.0431_F01567	82	■ 1.000	$\tilde{T} = \tilde{T}_1 + \tilde{T}_2 \in$	$\tilde{T} = \tilde{T}_1 + \tilde{T}_2 \in$	(Lemma 2.4.1.(1)) and $\tilde{T}_2 \in \mathfrak{f}_4^C$ (Proposition 2.3.6) $\subset \mathfrak{e}_6^C$. Therefore $\tilde{T} = \tilde{T}_1 + \tilde{T}_2 \in$
0902.0431_F01568	82	■ 0.777	$\delta \in \mathfrak{d}_4^C$	$\delta \in \mathfrak{d}_4^C$	Lemma 3.6.3. (1) For $\delta \in \mathfrak{d}_4^C$ and $R \in (\mathfrak{M}^r)^C, \operatorname{tr}(R)=0$, we have
0902.0431_F01569	82	■ 0.777	$R \in (\mathfrak{M}^r)^C, \operatorname{tr}(R)=0$	$R \in (\mathfrak{M}^r)^C, \operatorname{tr}(R)=0$	Lemma 3.6.3. (1) For $\delta \in \mathfrak{d}_4^C$ and $R \in (\mathfrak{M}^r)^C, \operatorname{tr}(R)=0$, we have
0902.0431_F01570	82	■ 1.000	$H \in M(3, C), \operatorname{tr}(H)=0$	$H \in M(3, C), \operatorname{tr}(H)=0$	(2) For $H \in M(3, C), \operatorname{tr}(H)=0$ and $T \in M(3, \mathfrak{C}), \operatorname{tr}(T)=0$, we have
0902.0431_F01571	82	■ 0.999	$(\widetilde{\delta R}) X = \delta R \circ X = \delta(R \circ X) - R \circ \delta X$	$(\widetilde{\delta R}) X = \delta R \circ X = \delta(R \circ X) - R \circ \delta X$	Proof. (1) $(\widetilde{\delta R}) X = \delta R \circ X = \delta(R \circ X) - R \circ \delta X$ (Lemma 3.6.1) $= \delta(\widetilde{R} X) -$
0902.0431_F01572	82	■ 0.999	$= \delta(\widetilde{R} X) -$	$= \delta(\widetilde{R} X) -$	Proof. (1) $(\widetilde{\delta R}) X = \delta R \circ X = \delta(R \circ X) - R \circ \delta X$ (Lemma 3.6.1) $= \delta(\widetilde{R} X) -$
0902.0431_F01573	82	■ 0.999	$\widetilde{R}(\delta X) = [\delta, \widetilde{R}] X, X \in \mathfrak{J}^C$	$\widetilde{R}(\delta X) = [\delta, \widetilde{R}] X, X \in \mathfrak{J}^C$	$\widetilde{R}(\delta X) = [\delta, \widetilde{R}] X, X \in \mathfrak{J}^C$. Hence, $\widetilde{\delta R} = [\delta, \widetilde{R}]$.
0902.0431_F01574	82	■ 0.999	$\widetilde{\delta R} = [\delta, \widetilde{R}]$	$\widetilde{\delta R} = [\delta, \widetilde{R}]$	$\widetilde{R}(\delta X) = [\delta, \widetilde{R}] X, X \in \mathfrak{J}^C$. Hence, $\widetilde{\delta R} = [\delta, \widetilde{R}]$.
0902.0431_F01575	82	■ 1.000	$[\widetilde{H}, \widetilde{T}] X = \widetilde{H} \widetilde{T} X - \widetilde{T} \widetilde{H} X$	$[\widetilde{H}, \widetilde{T}] X = \widetilde{H} \widetilde{T} X - \widetilde{T} \widetilde{H} X$	(2) $[\widetilde{H}, \widetilde{T}] X = \widetilde{H} \widetilde{T} X - \widetilde{T} \widetilde{H} X$
0902.0431_F01576	82	■ 1.000	$H \in M(3, C)$	$H \in M(3, C)$	(since $H \in M(3, C)$, products of matrices above are associative)
0902.0431_F01577	83	■ 1.000	$\mu_1 + \mu_2 + \mu_3 = 0$	$\mu_1 + \mu_2 + \mu_3 = 0$	with $\mu_1 + \mu_2 + \mu_3 = 0$.
0902.0431_F01578	83	■ 1.000	$\pm \lambda_k \pm \lambda_l$	$\pm \lambda_k \pm \lambda_l$	I The roots $\pm \lambda_k \pm \lambda_l$ of $\mathfrak{d}_4^C (\subset \mathfrak{f}_4^C \subset \mathfrak{e}_6^C)$ are also roots of $\mathfrak{e}_6^C$. Indeed, let α
0902.0431_F01579	83	■ 1.000	$\mathfrak{d}_4^C (\subset \mathfrak{f}_4^C \subset \mathfrak{e}_6^C)$	$\mathfrak{d}_4^C (\subset \mathfrak{f}_4^C \subset \mathfrak{e}_6^C)$	I The roots $\pm \lambda_k \pm \lambda_l$ of $\mathfrak{d}_4^C (\subset \mathfrak{f}_4^C \subset \mathfrak{e}_6^C)$ are also roots of $\mathfrak{e}_6^C$. Indeed, let α
0902.0431_F01580	83	■ 1.000	$S \in \delta_4^C \subset \mathfrak{e}_6^C$	$S \in \delta_4^C \subset \mathfrak{e}_6^C$	be a root of $\mathfrak{d}_4^C$ and $S \in \delta_4^C \subset \mathfrak{e}_6^C$ be an associated root vector. Then
0902.0431_F01581	83	■ 1.000	$a E_{kl} \in M\left(3, \mathfrak{C}\right)$	$a E_{kl} \in M\left(3, \mathfrak{C}\right)$	II We denote $a E_{kl} \in M(3, \mathfrak{C})$ by $F_{kl}(a): F_{kl}(a) = a E_{kl}, a \in \mathfrak{C}, k \neq l$. For
0902.0431_F01582	83	■ 1.000	$F_{kl}(a): F_{kl}(a) = a E_{kl}, a \in \mathfrak{C}, k \neq l$	$F_{kl}(a): F_{kl}(a) = a E_{kl}, a \in \mathfrak{C}, k \neq l$	II We denote $a E_{kl} \in M(3, \mathfrak{C})$ by $F_{kl}(a): F_{kl}(a) = a E_{kl}, a \in \mathfrak{C}, k \neq l$. For

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0902.0431_F01583	83	■ 1.000	$F_{23}(a)=\left(\begin{array}{lll}\theta & \theta & \theta \\ \theta & \theta & a \\ \theta & \theta & \theta\end{array}\right)$	$F_{23}(a)=\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & a \\ 0 & 0 & 0 \end{pmatrix}$	example, $F_{23}(a)=\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & a \\ 0 & 0 & 0 \end{pmatrix}$. Then, we have
0902.0431_F01584	83	■ 1.000	$a=e_{\{k\}+i} e_{\{4+k\}}$	$a=e_k+ie_{4+k}$	where $a=e_k+ie_{4+k}$ (since $h_\delta a=h_\delta(e_k+ie_{4+k})=\lambda_k(e_k+ie_{4+k})=\lambda_k a$, and
0902.0431_F01585	83	■ 1.000	$h_{\{\delta\}} a=h_{\{\delta\}}\left(e_{\{k\}+i} e_{\{4+k\}}\right)=\lambda_{\{k\}}\left(e_{\{k\}+i} e_{\{4+k\}}\right)=\lambda_{\{k\}} a$	$h_\delta a=h_\delta(e_k+ie_{4+k})=\lambda_k(e_k+ie_{4+k})=\lambda_k a$	where $a=e_k+ie_{4+k}$ (since $h_\delta a=h_\delta(e_k+ie_{4+k})=\lambda_k(e_k+ie_{4+k})=\lambda_k a$, and
0902.0431_F01586	83	■ 0.997	$\left[H, F_{23}(a)\right]=H F_{23}(a)-F_{23}(a) H=\left(\mu_2-\mu_3\right) F_{23}(a)$	$[H, F_{23}(a)]=H F_{23}(a)-F_{23}(a) H=(\mu_2-\mu_3) F_{23}(a))$	$[H, F_{23}(a)]=H F_{23}(a)-F_{23}(a) H=(\mu_2-\mu_3) F_{23}(a))$. Hence $\lambda_k+\frac{1}{2}\left(\mu_2-\mu_3\right)$ is a
0902.0431_F01587	83	■ 0.997	$\lambda_{\{k\}}+\frac{1}{2}\left(\mu_2-\mu_3\right)$	$\lambda_k+\frac{1}{2}\left(\mu_2-\mu_3\right)$	$[H, F_{23}(a)]=H F_{23}(a)-F_{23}(a) H=(\mu_2-\mu_3) F_{23}(a))$. Hence $\lambda_k+\frac{1}{2}\left(\mu_2-\mu_3\right)$ is a
0902.0431_F01588	84	■ 1.000	$\widetilde{F}_{23}\left(e_{\{k\}+i} e_{\{4+k\}}\right)$	$\widetilde{F}_{23}\left(e_k+i e_{4+k}\right)$	root of $\mathfrak{e}_6^C$ and $\widetilde{F}_{23}\left(e_k+i e_{4+k}\right)$ is its root vector. Similarly $-\lambda_k+\frac{1}{2}\left(\mu_2-\mu_3\right)$ is a
0902.0431_F01589	84	■ 1.000	$-\lambda_{\{k\}}+\frac{1}{2}\left(\mu_2-\mu_3\right)$	$-\lambda_k+\frac{1}{2}\left(\mu_2-\mu_3\right)$	root of $\mathfrak{e}_6^C$ and $\widetilde{F}_{23}\left(e_k+i e_{4+k}\right)$ is its root vector. Similarly $-\lambda_k+\frac{1}{2}\left(\mu_2-\mu_3\right)$ is a
0902.0431_F01590	84	■ 1.000	$\widetilde{F}_{23}\left(e_{\{k\}}-i e_{\{4+k\}}\right)$	$\widetilde{F}_{23}\left(e_k-i e_{4+k}\right)$	root of $\mathfrak{e}_6^C$ and $\widetilde{F}_{23}\left(e_k-i e_{4+k}\right)$ is its root vector. Again, from
0902.0431_F01591	84	■ 0.999	$-\lambda_{\{k\}}+\frac{1}{2}\left(\mu_3-\mu_2\right)$	$-\lambda_k+\frac{1}{2}\left(\mu_3-\mu_2\right)$	we see that $-\lambda_k+\frac{1}{2}\left(\mu_3-\mu_2\right)$ is a root of $\mathfrak{e}_6^C$. Similarly $\lambda_k+\frac{1}{2}\left(\mu_3-\mu_2\right)$ is a root of
0902.0431_F01592	84	■ 0.999	$\lambda_{\{k\}}+\frac{1}{2}\left(\mu_3-\mu_2\right)$	$\lambda_k+\frac{1}{2}\left(\mu_3-\mu_2\right)$	we see that $-\lambda_k+\frac{1}{2}\left(\mu_3-\mu_2\right)$ is a root of $\mathfrak{e}_6^C$. Similarly $\lambda_k+\frac{1}{2}\left(\mu_3-\mu_2\right)$ is a root of
0902.0431_F01593	84	■ 0.999	$\widetilde{F}_{32}\left(e_{\{k\}}-i e_{\{4+k\}}\right)$	$\widetilde{F}_{32}\left(e_k-i e_{4+k}\right)$	$\mathfrak{e}_6^C$ and $\widetilde{F}_{32}\left(e_k-i e_{4+k}\right)$ is its root vector. Next, using the relation
0902.0431_F01594	85	■ 1.000	$n_{\{1\}} n_{\{2\}} \cdots n_{\{6\}}$	$n_1 n_2 \cdots n_6$	Proof. In the following, the notation $n_1 n_2 \cdots n_6$ denotes the root $n_1 \alpha_1+n_2 \alpha_2+$
0902.0431_F01595	85	■ 1.000	$n_{\{1\}} \alpha_{\{1\}}+n_{\{2\}} \alpha_{\{2\}}+$	$n_1 \alpha_1+n_2 \alpha_2+$	Proof. In the following, the notation $n_1 n_2 \cdots n_6$ denotes the root $n_1 \alpha_1+n_2 \alpha_2+$
0902.0431_F01596	85	■ 0.578	$\cdots+n_{\{6\}} \alpha_{\{6\}}$	$\cdots+n_6 \alpha_6$	$\cdots+n_6 \alpha_6$. Now, all positive roots of $\mathfrak{e}_6^C$ are represented by
0902.0431_F01597	86	■ 1.000	$\Pi=\left\{\alpha_{\{1\}}, \alpha_{\{2\}}, \cdots, \alpha_{\{6\}}\right\}$	$\Pi=\left\{\alpha_1, \alpha_2, \cdots, \alpha_6\right\}$	Hence $\Pi=\left\{\alpha_1, \alpha_2, \cdots, \alpha_6\right\}$ is a fundamental root system of $\mathfrak{e}_6^C$. The real part $\mathfrak{h}_{\mathbf{R}}$
0902.0431_F01598	86	■ 0.909	$h=\sum_{k=0}^3 \lambda_{\{k\}} H_{\{k\}}+\left(\sum_{j=1}^3 \mu_{\{j\}} E_{\{j\}}\right)^{\sim}, h^{\prime}=\sum_{k=0}^3 \lambda_{\{k\}}^{\prime} H_{\{k\}}+\left(\sum_{j=1}^3 \mu_{\{j\}}^{\prime} E_{\{j\}}\right)^{\sim} \in \mathfrak{h}_{\mathbf{R}}$	$h=\sum_{k=0}^3 \lambda_k H_k+\left(\sum_{j=1}^3 \mu_j E_j\right)^{\sim}, h'=\sum_{k=0}^3 \lambda_k' H_k+\left(\sum_{j=1}^3 \mu_j' E_j\right)^{\sim} \in \mathfrak{h}_{\mathbf{R}}$	for $h=\sum_{k=0}^3 \lambda_k H_k+\left(\sum_{j=1}^3 \mu_j E_j\right)^{\sim}, h'=\sum_{k=0}^3 \lambda_k' H_k+\left(\sum_{j=1}^3 \mu_j' E_j\right)^{\sim} \in \mathfrak{h}_{\mathbf{R}}$. Indeed,
0902.0431_F01599	86	■ 1.000	$\alpha_{\{i\}}\left(B_{\{6\}}\left(H_{\{\alpha\}}, H\right)=\alpha(H), H \in \mathfrak{h}_{\mathbf{R}}\right)$	$\alpha_i\left(B_6\left(H_{\alpha}, H\right)=\alpha(H), H \in \mathfrak{h}_{\mathbf{R}}\right)$	Now, the canonical elements $H_{\alpha_i} \in \mathfrak{h}_{\mathbf{R}}$ corresponding to $\alpha_i\left(B_6\left(H_{\alpha}, H\right)=\alpha(H), H \in \mathfrak{h}_{\mathbf{R}}\right)$
0902.0431_F01600	87	■ 1.000	$\alpha_{\{1\}}, \alpha_{\{2\}}, \cdots, \alpha_{\{6\}}$	$\alpha_1, \alpha_2, \cdots, \alpha_6$	induced by the Killing form B_6 between $\alpha_1, \alpha_2, \cdots, \alpha_6$ and $-\mu$ are given by
0902.0431_F01601	87	■ 1.000	$T \oplus D_{\{5\}}$	$T \oplus D_5$	(1) The first one is a subalgebra of type $T \oplus D_5$ which is obtained as the fixed
0902.0431_F01602	87	■ 1.000	$C_{\{1\}} \oplus A_{\{5\}}$	$C_1 \oplus A_5$	(2) The second one is a subalgebra of type $C_1 \oplus A_5$ which is obtained as the fixed
0902.0431_F01603	87	■ 1.000	$A_{\{2\}} \oplus A_{\{2\}} \oplus A_{\{2\}}$	$A_2 \oplus A_2 \oplus A_2$	(3) The third one is a subalgebra of type $A_2 \oplus A_2 \oplus A_2$ which is obtained as the
0902.0431_F01604	87	■ 1.000	$\tau, \tau \gamma$	$\tau, \tau \gamma$	The Lie algebra $\mathfrak{e}_6$ has furthermore two outer involutions $\tau, \tau \gamma$. The subalgebra
0902.0431_F01605	87	■ 1.000	$C_{\{4\}}$	C_4	points of $\tau \gamma$ is type C_4 .
0902.0431_F01606	87	■ 1.000	$\left(E_{\{6\}}\right)^{\tau}$	$\left(E_6\right)^{\tau}$	We shall study the following subgroup $\left(E_6\right)^{\tau}$ of E_6 :
0902.0431_F01607	87	■ 1.000	$\alpha \in E_{\{6\}}$	$\alpha \in E_6$	If $\alpha \in E_6$ satisfies $\tau \alpha=\alpha \tau$, then $\langle\alpha X, \alpha Y\rangle=\langle\tau \alpha X, \alpha Y\rangle=\langle\alpha \tau X, \alpha Y\rangle=\langle\tau X, Y\rangle=$
0902.0431_F01608	87	■ 1.000	$\langle\alpha X, \alpha Y\rangle=\langle\tau \alpha X, \alpha Y\rangle=\langle\alpha \tau X, \alpha Y\rangle=\langle\tau X, Y\rangle=$	$\langle\alpha X, \alpha Y\rangle=\langle\tau \alpha X, \alpha Y\rangle=\langle\alpha \tau X, \alpha Y\rangle=\langle\tau X, Y\rangle=$	If $\alpha \in E_6$ satisfies $\tau \alpha=\alpha \tau$, then $\langle\alpha X, \alpha Y\rangle=\langle\tau \alpha X, \alpha Y\rangle=\langle\alpha \tau X, \alpha Y\rangle=\langle\tau X, Y\rangle=$
0902.0431_F01609	87	■ 1.000	$\langle\alpha X, \alpha Y\rangle=\langle X, Y\rangle$	$\langle\alpha X, \alpha Y\rangle=\langle X, Y\rangle$	$\langle X, Y\rangle$ and vice versa. Further, for $\alpha \in E_6$, the conditions $\langle\alpha X, \alpha Y\rangle=\langle X, Y\rangle$ and
0902.0431_F01610	87	■ 0.999	$\left(E_{\{6\}}\right)^{\tau}=\left(E_{\{6\}}\right)_E \cong F_4$	$\left(E_6\right)^{\tau}=\left(E_6\right)_E \cong F_4$	Theorem 3.7.1. $\left(E_6\right)^{\tau}=\left(E_6\right)_E \cong F_4$.
0902.0431_F01611	88	■ 1.000	$\left(E_{\{6\}}\right)^{\tau} \cong F_4$	$\left(E_6\right)^{\tau} \cong F_4$	Proof. We shall show $\left(E_6\right)^{\tau} \cong F_4$. Let $\alpha \in\left(E_6\right)^{\tau}$. Then, for $X \in \mathfrak{J}$ we have
0902.0431_F01612	88	■ 1.000	$\alpha \in\left(E_6\right)^{\tau}$	$\alpha \in\left(E_6\right)^{\tau}$	Proof. We shall show $\left(E_6\right)^{\tau} \cong F_4$. Let $\alpha \in\left(E_6\right)^{\tau}$. Then, for $X \in \mathfrak{J}$ we have
0902.0431_F01613	88	■ 1.000	$\alpha \mid \mathfrak{J}$	$\alpha \mid \mathfrak{J}$	$\mathfrak{J}$. Therefore, the restriction $\alpha \mid \mathfrak{J}$ of α to $\mathfrak{J}$ belongs to the group

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0902.0431_F01614	88	■ 1.000	$\alpha^{\mathcal{C}}: \mathfrak{J}^{\mathcal{C}} \rightarrow \mathfrak{J}^{\mathcal{C}}, \alpha^{\mathcal{C}}(X_1 + iX_2) = \alpha X_1 +$ $\mathfrak{J}^{\mathcal{C}}, \alpha^{\mathcal{C}}(X_1 + iX_2) = \alpha X_1 +$ $\mathfrak{J}^{\mathcal{C}}, \alpha^{\mathcal{C}}(X_1 + iX_2) = \alpha X_1 +$	$\alpha^{\mathcal{C}}: \mathfrak{J}^{\mathcal{C}} \rightarrow \mathfrak{J}^{\mathcal{C}}, \alpha^{\mathcal{C}}(X_1 + iX_2) = \alpha X_1 +$	Conversely, for $\alpha \in F_4$, its complexification $\alpha^{\mathcal{C}}: \mathfrak{J}^{\mathcal{C}} \rightarrow \mathfrak{J}^{\mathcal{C}}, \alpha^{\mathcal{C}}(X_1 + iX_2) = \alpha X_1 +$
0902.0431_F01615	88	■ 1.000	$i \alpha X_2$	$i\alpha X_2$	$i\alpha X_2$ belongs to $(E_6)^{\tau}$. Therefore the correspondence $F_4 \ni \alpha \rightarrow \alpha^{\mathcal{C}} \in (E_6)^{\tau}$ gives an
0902.0431_F01616	88	■ 1.000	$F_4 \ni \alpha \rightarrow \alpha^{\mathcal{C}} \in (E_6)^{\tau}$	$F_4 \ni \alpha \rightarrow \alpha^{\mathcal{C}} \in (E_6)^{\tau}$	$i\alpha X_2$ belongs to $(E_6)^{\tau}$. Therefore the correspondence $F_4 \ni \alpha \rightarrow \alpha^{\mathcal{C}} \in (E_6)^{\tau}$ gives an
0902.0431_F01617	88	■ 0.959	$(E_6)_0$	$(E_6)_0$	We denote by $(E_6)_0$ the connected component of E_6 containing the identity 1.
0902.0431_F01618	88	■ 1.000	$\alpha_{12}(t): \mathfrak{J}^{\mathcal{C}} \rightarrow \mathfrak{J}^{\mathcal{C}}$	$\alpha_{12}(t): \mathfrak{J}^{\mathcal{C}} \rightarrow \mathfrak{J}^{\mathcal{C}}$	Lemma 3.8.1. (1) For $t \in \mathbf{R}$, if we define a mapping $\alpha_{12}(t): \mathfrak{J}^{\mathcal{C}} \rightarrow \mathfrak{J}^{\mathcal{C}}$ by
0902.0431_F01619	88	■ 1.000	$\alpha_{12}(t) \in (E_6)_0$	$\alpha_{12}(t) \in (E_6)_0$	then $\alpha_{12}(t) \in (E_6)_0$. Similarly we can define $\alpha_{13}(t), \alpha_{23}(t) \in (E_6)_0$.
0902.0431_F01620	88	■ 1.000	$\alpha_{13}(t), \alpha_{23}(t) \in (E_6)_0$	$\alpha_{13}(t), \alpha_{23}(t) \in (E_6)_0$	then $\alpha_{12}(t) \in (E_6)_0$. Similarly we can define $\alpha_{13}(t), \alpha_{23}(t) \in (E_6)_0$.
0902.0431_F01621	88	■ 1.000	$\alpha_1(a): \mathfrak{J}^{\mathcal{C}} \rightarrow \mathfrak{J}^{\mathcal{C}}$	$\alpha_1(a): \mathfrak{J}^{\mathcal{C}} \rightarrow \mathfrak{J}^{\mathcal{C}}$	(2) For $a \in \mathfrak{C}$, if we define a mapping $\alpha_1(a): \mathfrak{J}^{\mathcal{C}} \rightarrow \mathfrak{J}^{\mathcal{C}}$ by $\alpha_1(a)X(\xi, x) = Y(\eta, y)$,
0902.0431_F01622	88	■ 1.000	$\alpha_1(a)X(\xi, x) = Y(\eta, y)$	$\alpha_1(a)X(\xi, x) = Y(\eta, y)$	(2) For $a \in \mathfrak{C}$, if we define a mapping $\alpha_1(a): \mathfrak{J}^{\mathcal{C}} \rightarrow \mathfrak{J}^{\mathcal{C}}$ by $\alpha_1(a)X(\xi, x) = Y(\eta, y)$,
0902.0431_F01623	88	■ 1.000	$\alpha_1(a) \in (E_6)_0$	$\alpha_1(a) \in (E_6)_0$	(if $a = 0$, then $\frac{\sin a }{ a }$ means 1), then $\alpha_1(a) \in (E_6)_0$.
0902.0431_F01624	88	■ 0.998	$E_1 - E_2 \in \mathfrak{J}_0$	$E_1 - E_2 \in \mathfrak{J}_0$	Proof. (1) For $E_1 - E_2 \in \mathfrak{J}_0$, we have $i(E_1 - E_2)^{\sim} \in \mathfrak{e}_6$ (Theorem 3.2.4.(2)) and
0902.0431_F01625	88	■ 0.998	$i(E_1 - E_2)^{\sim} \in \mathfrak{e}_6$	$i(E_1 - E_2)^{\sim} \in \mathfrak{e}_6$	Proof. (1) For $E_1 - E_2 \in \mathfrak{J}_0$, we have $i(E_1 - E_2)^{\sim} \in \mathfrak{e}_6$ (Theorem 3.2.4.(2)) and
0902.0431_F01626	88	■ 1.000	$\alpha_{12}(t) = \exp it(E_1 - E_2)^{\sim}$	$\alpha_{12}(t) = \exp it(E_1 - E_2)^{\sim}$	$\alpha_{12}(t) = \exp it(E_1 - E_2)^{\sim}$. Hence $\alpha_{12}(t) \in (E_6)_0$.
0902.0431_F01627	88	■ 1.000	$F_1(a) \in \mathfrak{J}_0$	$F_1(a) \in \mathfrak{J}_0$	(2) For $F_1(a) \in \mathfrak{J}_0$, we have $i\tilde{F}_1(a) \in \mathfrak{e}_6$ (Theorem 3.2.4.(2)) and $\alpha_1(a) =$
0902.0431_F01628	88	■ 1.000	$i\tilde{F}_1(a) \in \mathfrak{e}_6$	$i\tilde{F}_1(a) \in \mathfrak{e}_6$	(2) For $F_1(a) \in \mathfrak{J}_0$, we have $i\tilde{F}_1(a) \in \mathfrak{e}_6$ (Theorem 3.2.4.(2)) and $\alpha_1(a) =$
0902.0431_F01629	88	■ 1.000	$\alpha_1(a) =$	$\alpha_1(a) =$	(2) For $F_1(a) \in \mathfrak{J}_0$, we have $i\tilde{F}_1(a) \in \mathfrak{e}_6$ (Theorem 3.2.4.(2)) and $\alpha_1(a) =$
0902.0431_F01630	88	■ 1.000	$\exp i\tilde{F}_1(a)$	$\exp i\tilde{F}_1(a)$	$\exp iF_1(a)$. Hence $\alpha_1(a) \in (E_6)_0$.
0902.0431_F01631	88	■ 1.000	$X \in \mathfrak{J}^{\mathcal{C}}$	$X \in \mathfrak{J}^{\mathcal{C}}$	Proposition 3.8.2. Any element $X \in \mathfrak{J}^{\mathcal{C}}$ can be transformed to a diagonal form
0902.0431_F01632	88	■ 1.000	$\alpha \in (E_6)_0$	$\alpha \in (E_6)_0$	by some element $\alpha \in (E_6)_0$:
0902.0431_F01633	89	■ 1.000	$ \xi $	$ \xi $	Proof. (Hereafter, we use the notation $ \xi $ instead of $\sqrt{\xi(\tau\xi)}$ for $\xi \in \mathbf{R}^{\mathcal{C}} = C$).
0902.0431_F01634	89	■ 1.000	$\sqrt{\xi(\tau\xi)}$	$\sqrt{\xi(\tau\xi)}$	Proof. (Hereafter, we use the notation $ \xi $ instead of $\sqrt{\xi(\tau\xi)}$ for $\xi \in \mathbf{R}^{\mathcal{C}} = C$).
0902.0431_F01635	89	■ 1.000	$\xi \in \mathbf{R}^{\mathcal{C}} = C$	$\xi \in \mathbf{R}^{\mathcal{C}} = C$	Proof. (Hereafter, we use the notation $ \xi $ instead of $\sqrt{\xi(\tau\xi)}$ for $\xi \in \mathbf{R}^{\mathcal{C}} = C$).
0902.0431_F01636	89	■ 1.000	$\mathfrak{X} = \{\alpha X \mid \alpha \in (E_6)_0\}$	$\mathfrak{X} = \{\alpha X \mid \alpha \in (E_6)_0\}$	For a given $X \in \mathfrak{J}^{\mathcal{C}}$, consider a space $\mathfrak{X} = \{\alpha X \mid \alpha \in (E_6)_0\}$. Since $(E_6)_0$ is compact
0902.0431_F01637	89	■ 1.000	$ \xi_1 ^2 + \xi_2 ^2 + \xi_3 ^2$	$ \xi_1 ^2 + \xi_2 ^2 + \xi_3 ^2$	(Theorem 3.1.1), $\mathfrak{X}$ is also compact. Let $ \xi_1 ^2 + \xi_2 ^2 + \xi_3 ^2$ be the maximal value of
0902.0431_F01638	89	■ 1.000	$ \eta_1 ^2 + \eta_2 ^2 + \eta_3 ^2$	$ \eta_1 ^2 + \eta_2 ^2 + \eta_3 ^2$	all $ \eta_1 ^2 + \eta_2 ^2 + \eta_3 ^2$ for $Y = Y(\eta, y) \in \mathfrak{X}$ and let $X_0 = X(\xi, x)$ be an element of
0902.0431_F01639	89	■ 1.000	ξ_2, ξ_3	ξ_2, ξ_3	It is sufficient to prove in the case that ξ_2, ξ_3 are real numbers (otherwise we can
0902.0431_F01640	89	■ 1.000	$\alpha_{12}(t_1)$	$\alpha_{12}(t_1)$	apply some $\alpha_{12}(t_1)$ and $\alpha_{13}(t_2)$ of Lemma 3.8.1.(1)).
0902.0431_F01641	89	■ 1.000	$\alpha_{13}(t_2)$	$\alpha_{13}(t_2)$	apply some $\alpha_{12}(t_1)$ and $\alpha_{13}(t_2)$ of Lemma 3.8.1.(1)).
0902.0431_F01642	89	■ 1.000	$q \neq 0$	$q \neq 0$	(1) Case $q \neq 0$. Let $a(t) = \frac{q}{ q }t$, $t > 0$ and construct $\alpha_1(a(t)) \in (E_6)_0$ of
0902.0431_F01643	89	■ 1.000	$a(t) = \frac{q}{ q }t, t > 0$	$a(t) = \frac{q}{ q }t, t > 0$	(1) Case $q \neq 0$. Let $a(t) = \frac{q}{ q }t$, $t > 0$ and construct $\alpha_1(a(t)) \in (E_6)_0$ of
0902.0431_F01644	89	■ 1.000	$\alpha_1(a(t)) \in (E_6)_0$	$\alpha_1(a(t)) \in (E_6)_0$	(1) Case $q \neq 0$. Let $a(t) = \frac{q}{ q }t$, $t > 0$ and construct $\alpha_1(a(t)) \in (E_6)_0$ of
0902.0431_F01645	89	■ 1.000	$i \frac{(a(t), x_1)}{ a(t) } = i\nu - q $	$i \frac{(a(t), x_1)}{ a(t) } = i\nu - q $	Lemma 3.8.1.(2). Since $ a(t) = t$ and $i \frac{(a(t), x_1)}{ a(t) } = i\nu - q $, where $\nu = \left(\frac{q}{ q }, p\right)$,
0902.0431_F01646	89	■ 1.000	$\nu = \left(\frac{q}{ q }, p\right)$	$\nu = \left(\frac{q}{ q }, p\right)$	Lemma 3.8.1.(2). Since $ a(t) = t$ and $i \frac{(a(t), x_1)}{ a(t) } = i\nu - q $, where $\nu = \left(\frac{q}{ q }, p\right)$,
0902.0431_F01647	89	■ 1.000	$Y(\eta(t), y(t)) = \alpha_1(a(t))X_0 \in \mathfrak{X}$	$Y(\eta(t), y(t)) = \alpha_1(a(t))X_0 \in \mathfrak{X}$	for $Y(\eta(t), y(t)) = \alpha_1(a(t))X_0 \in \mathfrak{X}$, we have
0902.0431_F01648	89	■ 1.000	$q = 0$	$q = 0$	of $ \xi_1 ^2 + \xi_2 ^2 + \xi_3 ^2$. Hence $q = 0$.

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0902.0431_F01649	89	■ 1.000	$p \neq 0$	$p \neq 0$	(2) Case $p \neq 0$. Let $a(t) = \frac{p}{ p }t$, $t > 0$ and construct $\beta_1(a(t)) \in F_4 \subset (E_6)_0$ of
0902.0431_F01650	89	■ 1.000	$a(t)=\frac{p}{ p }t, t>0$	$a(t) = \frac{p}{ p }t, t > 0$	(2) Case $p \neq 0$. Let $a(t) = \frac{p}{ p }t, t > 0$ and construct $\beta_1(a(t)) \in F_4 \subset (E_6)_0$ of
0902.0431_F01651	89	■ 1.000	$\beta_1(a(t)) \in F_4 \subset (E_6)_0$	$\beta_1(a(t)) \in F_4 \subset (E_6)_0$	(2) Case $p \neq 0$. Let $a(t) = \frac{p}{ p }t$, $t > 0$ and construct $\beta_1(a(t)) \in F_4 \subset (E_6)_0$ of
0902.0431_F01652	89	■ 1.000	$ \eta_1(t) ^2 +$	$ \eta_1(t) ^2 +$	Lemma 2.8.1. For $Y(\eta(t), y(t)) = \beta_1(a(t))X_0 \in \mathfrak{X}$, the maximal value of $ \eta_1(t) ^2 +$
0902.0431_F01653	89	■ 1.000	$ \eta_2(t) ^2 + \eta_3(t) ^2$	$ \eta_2(t) ^2 + \eta_3(t) ^2$	$ \eta_2(t) ^2 + \eta_3(t) ^2$ is $ \xi_1 ^2 + \xi_2 ^2 + \xi_3 ^2 + 2 p ^2$ (the calculation is the same as Propo-
0902.0431_F01654	89	■ 1.000	$ \xi_1 ^2 + \xi_2 ^2 + \xi_3 ^2 + 2 p ^2$	$ \xi_1 ^2 + \xi_2 ^2 + \xi_3 ^2 + 2 p ^2$	$ \eta_2(t) ^2 + \eta_3(t) ^2$ is $ \xi_1 ^2 + \xi_2 ^2 + \xi_3 ^2 + 2 p ^2$ (the calculation is the same as Propo-
0902.0431_F01655	89	■ 0.999	$\alpha_2(a), \alpha_3(a) \in (E_6)_0$	$\alpha_2(a), \alpha_3(a) \in (E_6)_0$	$\alpha_2(a), \alpha_3(a) \in (E_6)_0$ analogous to $\alpha_1(a)$ of Lemma 3.8.1.(2). Hence X_0 is of diagonal
0902.0431_F01656	89	■ 0.999	$\alpha_1(a)$	$\alpha_1(a)$	$\alpha_2(a), \alpha_3(a) \in (E_6)_0$ analogous to $\alpha_1(a)$ of Lemma 3.8.1.(2). Hence X_0 is of diagonal
0902.0431_F01657	90	■ 0.999	$E \in IV$	$E \in IV$	The space EIV , called the symmetric space of type EIV , is defined by
0902.0431_F01658	90	■ 0.999	$E_6/F_4 \simeq EIV$	$E_6/F_4 \simeq EIV$	Theorem 3.8.3. $E_6/F_4 \simeq EIV$.
0902.0431_F01659	90	■ 1.000	$X \in EIV$	$X \in EIV$	Proof. For $\alpha \in E_6$ and $X \in EIV$, we have $\alpha X \in EIV$. Hence E_6 acts on EIV .
0902.0431_F01660	90	■ 1.000	$\alpha X \in EIV$	$\alpha X \in EIV$	Proof. For $\alpha \in E_6$ and $X \in EIV$, we have $\alpha X \in EIV$. Hence E_6 acts on EIV .
0902.0431_F01661	90	■ 1.000	$E \in EIV$	$E \in EIV$	sufficient to show that any element $X \in EIV$ can be transformed to $E \in EIV$ by
0902.0431_F01662	90	■ 1.000	$X \in EIV \subset \mathfrak{J}^C$	$X \in EIV \subset \mathfrak{J}^C$	some $\alpha \in (E_6)_0$. Now, $X \in EIV \subset \mathfrak{J}^C$ can be transformed to a diagonal form by
0902.0431_F01663	90	■ 0.709	$\xi_1 = \xi_2 = \xi_3 = 1$	$\xi_1 = \xi_2 = \xi_3 = 1$	This implies that $\xi_1 = \xi_2 = \xi_3 = 1$. Certainly, from $0 \leq (\xi_2 - \xi_3)^2 = \xi_2^2 + \xi_3^2 - 2\xi_2\xi_3 =$
0902.0431_F01664	90	■ 0.709	$0 \leq (\xi_2 - \xi_3)^2 = \xi_2^2 + \xi_3^2 - 2\xi_2\xi_3 =$	$0 \leq (\xi_2 - \xi_3)^2 = \xi_2^2 + \xi_3^2 - 2\xi_2\xi_3 =$	This implies that $\xi_1 = \xi_2 = \xi_3 = 1$. Certainly, from $0 \leq (\xi_2 - \xi_3)^2 = \xi_2^2 + \xi_3^2 - 2\xi_2\xi_3 =$
0902.0431_F01665	90	■ 0.993	$3 - \xi_1^2 - \frac{2}{\xi_1} = -\frac{\xi_1^3 - 3\xi_1 + 2}{\xi_1} = -\frac{(\xi_1 - 1)^2(\xi_1 + 2)}{\xi_1} \leq 0$	$3 - \xi_1^2 - \frac{2}{\xi_1} = -\frac{\xi_1^3 - 3\xi_1 + 2}{\xi_1} = -\frac{(\xi_1 - 1)^2(\xi_1 + 2)}{\xi_1} \leq 0$	$3 - \xi_1^2 - \frac{2}{\xi_1} = -\frac{\xi_1^3 - 3\xi_1 + 2}{\xi_1} = -\frac{(\xi_1 - 1)^2(\xi_1 + 2)}{\xi_1} \leq 0$, we have $\xi_1 = 1$. Similarly
0902.0431_F01666	90	■ 0.993	$\xi_1 = 1$	$\xi_1 = 1$	$3 - \xi_1^2 - \frac{2}{\xi_1} = -\frac{\xi_1^3 - 3\xi_1 + 2}{\xi_1} = -\frac{(\xi_1 - 1)^2(\xi_1 + 2)}{\xi_1} \leq 0$, we have $\xi_1 = 1$. Similarly
0902.0431_F01667	90	■ 1.000	$\xi_2 = \xi_3 = 1$	$\xi_2 = \xi_3 = 1$	$\xi_2 = \xi_3 = 1$ are obtained. Hence $\alpha X = E$, which shows the transitivity of $(E_6)_0$.
0902.0431_F01668	90	■ 1.000	$\alpha X = E$	$\alpha X = E$	$\xi_2 = \xi_3 = 1$ are obtained. Hence $\alpha X = E$, which shows the transitivity of $(E_6)_0$.
0902.0431_F01669	90	■ 0.999	$EIV = (E_6)_0 E, EIV$	$EIV = (E_6)_0 E, EIV$	Since we have $EIV = (E_6)_0 E, EIV$ is connected. Now, the group E_6 acts transitively
0902.0431_F01670	90	■ 1.000	$E_6/F_4 \simeq EIV$	$E_6/F_4 \simeq EIV$	we have the homeomorphism $E_6/F_4 \simeq EIV$. Finally, the connectedness of E_6 follows
0902.0431_F01671	90	■ 1.000	$\alpha \in z(E_6)$	$\alpha \in z(E_6)$	Proof. Let $\alpha \in z(E_6)$. From the commutativity with $\beta \in F_4 \subset E_6$, we have
0902.0431_F01672	90	■ 1.000	$\beta \in F_4 \subset E_6$	$\beta \in F_4 \subset E_6$	Proof. Let $\alpha \in z(E_6)$. From the commutativity with $\beta \in F_4 \subset E_6$, we have
0902.0431_F01673	90	■ 1.000	$\beta \alpha E = \alpha \beta E = \alpha E$	$\beta \alpha E = \alpha \beta E = \alpha E$	$\beta \alpha E = \alpha \beta E = \alpha E$. Let denote $\alpha E = Y = Y(\eta, y) \in \mathfrak{J}^C$, then we have
0902.0431_F01674	90	■ 1.000	$\alpha E = Y = Y(\eta, y) \in \mathfrak{J}^C$	$\alpha E = Y = Y(\eta, y) \in \mathfrak{J}^C$	$\beta \alpha E = \alpha \beta E = \alpha E$. Let denote $\alpha E = Y = Y(\eta, y) \in \mathfrak{J}^C$, then we have
0902.0431_F01675	91	■ 1.000	$T = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$	$T = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$	where $T = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$ and $\begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \in SO(3)$. Then we
0902.0431_F01676	91	■ 1.000	$\begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \in SO(3)$	$\begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \in SO(3)$	where $T = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$ and $\begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \in SO(3)$. Then we
0902.0431_F01677	91	■ 1.000	$y_1 = y_2 = y_3 = 0$	$y_1 = y_2 = y_3 = 0$	have $y_1 = y_2 = y_3 = 0$ and $\eta_1 = \eta_2 = \eta_3 (= \omega)$, that is,
0902.0431_F01678	91	■ 1.000	$\eta_1 = \eta_2 = \eta_3 (= \omega)$	$\eta_1 = \eta_2 = \eta_3 (= \omega)$	have $y_1 = y_2 = y_3 = 0$ and $\eta_1 = \eta_2 = \eta_3 (= \omega)$, that is,
0902.0431_F01679	91	■ 1.000	$\omega^3 = \det(\alpha E) = \det E = 1$	$\omega^3 = \det(\alpha E) = \det E = 1$	and $\omega^3 = \det(\alpha E) = \det E = 1$. Since it is easy to verify that $\omega 1 \in z(E_6)$, we
0902.0431_F01680	91	■ 1.000	$\omega 1 \in z(E_6)$	$\omega 1 \in z(E_6)$	and $\omega^3 = \det(\alpha E) = \det E = 1$. Since it is easy to verify that $\omega 1 \in z(E_6)$, we
0902.0431_F01681	91	■ 1.000	$\omega^{-1} \alpha \in z(E_6)$	$\omega^{-1} \alpha \in z(E_6)$	have $\omega^{-1} \alpha \in z(E_6)$ and $\omega^{-1} \alpha E = E$, and so $\omega^{-1} \alpha \in z(F_4)$ (Theorem 3.7.1). Since
0902.0431_F01682	91	■ 1.000	$\omega^{-1} \alpha E = E$	$\omega^{-1} \alpha E = E$	have $\omega^{-1} \alpha \in z(E_6)$ and $\omega^{-1} \alpha E = E$, and so $\omega^{-1} \alpha \in z(F_4)$ (Theorem 3.7.1). Since
0902.0431_F01683	91	■ 1.000	$\omega^{-1} \alpha \in z(F_4)$	$\omega^{-1} \alpha \in z(F_4)$	have $\omega^{-1} \alpha \in z(E_6)$ and $\omega^{-1} \alpha E = E$, and so $\omega^{-1} \alpha \in z(F_4)$ (Theorem 3.7.1). Since

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0902.0431_F01684	91	■ 1.000	$z\left(F_4\right)=\{1\}$	$z\left(F_4\right)=\{1\}$	$z(F_4) = \{1\}$ (Theorem 2.10.1), we have $\omega^{-1}\alpha = 1$, that is, $\alpha = \omega 1$. This completes
0902.0431_F01685	91	■ 1.000	$\omega ^{-1}\alpha =1$	$\omega ^{-1}\alpha =1$	$z(F_4) = \{1\}$ (Theorem 2.10.1), we have $\omega^{-1}\alpha = 1$, that is, $\alpha = \omega 1$. This completes
0902.0431_F01686	91	■ 1.000	$\alpha =\omega \mathbf{1}$	$\alpha =\omega 1$	$z(F_4) = \{1\}$ (Theorem 2.10.1), we have $\omega^{-1}\alpha = 1$, that is, $\alpha = \omega 1$. This completes
0902.0431_F01687	91	■ 1.000	$E_6=\left\{\alpha \in \operatorname{Iso}_C\left(\mathfrak{J}^C\right) \mid \det (\alpha X)=\det X,\langle \alpha X,\alpha Y\rangle =\langle X,Y\rangle \right\}$	$E_6 = \left\{\alpha \in \operatorname{Iso}_C\left(\mathfrak{J}^C\right) \mid \det (\alpha X)=\det X,\langle \alpha X,\alpha Y\rangle =\langle X,Y\rangle \right\}$	Theorem 3.9.2. $E_6 = \left\{\alpha \in \operatorname{Iso}_C\left(\mathfrak{J}^C\right) \mid \det (\alpha X)=\det X,\langle \alpha X,\alpha Y\rangle =\langle X,Y\rangle \right\}$ is
0902.0431_F01688	91	■ 1.000	$\sigma : \mathfrak{J}^C \rightarrow \mathfrak{J}^C$	$\sigma : \mathfrak{J}^C \rightarrow \mathfrak{J}^C$	Let the C -linear mapping $\sigma : \mathfrak{J}^C \rightarrow \mathfrak{J}^C$ be the complexification of $\sigma \in F_4$ of
0902.0431_F01689	91	■ 1.000	$\sigma \in E_6$	$\sigma \in E_6$	Section 2.9. Then $\sigma \in E_6$ and $\sigma^2 = 1$.
0902.0431_F01690	91	■ 1.000	$\left(E_6\right)^{\sigma}$	$\left(E_6\right)^{\sigma}$	We shall study the following subgroup $\left(E_6\right)^{\sigma}$ of E_6 :
0902.0431_F01691	91	■ 1.000	$\left(\mathfrak{J}^C\right)_{\sigma}$	$\left(\mathfrak{J}^C\right)_{\sigma}$	To this end, we consider the C -subspaces $\left(\mathfrak{J}^C\right)_{\sigma}$ and $\left(\mathfrak{J}^C\right)_{-\sigma}$ of $\mathfrak{J}^C$, which are the
0902.0431_F01692	91	■ 1.000	$\left(\mathfrak{J}^C\right)_{-\sigma}$	$\left(\mathfrak{J}^C\right)_{-\sigma}$	To this end, we consider the C -subspaces $\left(\mathfrak{J}^C\right)_{\sigma}$ and $\left(\mathfrak{J}^C\right)_{-\sigma}$ of $\mathfrak{J}^C$, which are the
0902.0431_F01693	91	■ 0.672	$\mathfrak{E}_1^C=\left\{\xi E_1 \mid \xi \in C\right\}$	$\mathfrak{E}_1^C = \left\{\xi E_1 \mid \xi \in C\right\}$	where $\mathfrak{E}_1^C = \left\{\xi E_1 \mid \xi \in C\right\}$. Then $\mathfrak{J}^C = \left(\mathfrak{J}^C\right)_{\sigma} \oplus \left(\mathfrak{J}^C\right)_{-\sigma}$ (which is the complexification
0902.0431_F01694	91	■ 0.672	$\mathfrak{J}^C=\left(\mathfrak{J}^C\right)_{\sigma} \oplus\left(\mathfrak{J}^C\right)_{-\sigma}$	$\mathfrak{J}^C = \left(\mathfrak{J}^C\right)_{\sigma} \oplus \left(\mathfrak{J}^C\right)_{-\sigma}$	where $\mathfrak{E}_1^C = \left\{\xi E_1 \mid \xi \in C\right\}$. Then $\mathfrak{J}^C = \left(\mathfrak{J}^C\right)_{\sigma} \oplus \left(\mathfrak{J}^C\right)_{-\sigma}$ (which is the complexification
0902.0431_F01695	91	■ 0.999	$\left(\mathfrak{J}^C\right)_{\sigma},\left(\mathfrak{J}^C\right)_{-\sigma}$	$\left(\mathfrak{J}^C\right)_{\sigma},\left(\mathfrak{J}^C\right)_{-\sigma}$	of $\mathfrak{J} = \mathfrak{J}_{\sigma} \oplus \mathfrak{J}_{-\sigma}$ in Section 2.9), and $\left(\mathfrak{J}^C\right)_{\sigma},\left(\mathfrak{J}^C\right)_{-\sigma}$ are invariant under the action of
0902.0431_F01696	91	■ 1.000	$\alpha \in\left(E_6\right)^{\sigma}$	$\alpha \in\left(E_6\right)^{\sigma}$	Lemma 3.10.1. For $\alpha \in\left(E_6\right)^{\sigma}$, there exists $\xi \in C$ such that
0902.0431_F01697	91	■ 1.000	$\xi \in C$	$\xi \in C$	Lemma 3.10.1. For $\alpha \in\left(E_6\right)^{\sigma}$, there exists $\xi \in C$ such that
0902.0431_F01698	92	■ 1.000	$\alpha E_1 \in \mathfrak{J}\left(2,\mathfrak{C}^C\right)$	$\alpha E_1 \in \mathfrak{J}\left(2,\mathfrak{C}^C\right)$	Suppose that $\alpha E_1 \in \mathfrak{J}(2, \mathfrak{C}^C)$. Then $\alpha E = \alpha E_1 + \alpha E_2 + \alpha E_3 \in \mathfrak{J}(2, \mathfrak{C}^C)$, so we can
0902.0431_F01699	92	■ 1.000	$\alpha E=\alpha E_1+\alpha E_2+\alpha E_3 \in \mathfrak{J}\left(2,\mathfrak{C}^C\right)$	$\alpha E = \alpha E_1 + \alpha E_2 + \alpha E_3 \in \mathfrak{J}\left(2,\mathfrak{C}^C\right)$	Suppose that $\alpha E_1 \in \mathfrak{J}(2, \mathfrak{C}^C)$. Then $\alpha E = \alpha E_1 + \alpha E_2 + \alpha E_3 \in \mathfrak{J}(2, \mathfrak{C}^C)$ so we can
0902.0431_F01700	92	■ 1.000	$\alpha E=\xi_2 E_2+\xi_3 E_3+F_1\left(x_1\right), \xi_2, \xi_3 \in C, x_1 \in \mathfrak{C}^C$	$\alpha E = \xi_2 E_2 + \xi_3 E_3 + F_1\left(x_1\right), \xi_2, \xi_3 \in C, x_1 \in \mathfrak{C}^C$	put $\alpha E = \xi_2 E_2 + \xi_3 E_3 + F_1(x_1), \xi_2, \xi_3 \in C, x_1 \in \mathfrak{C}^C$. Hence
0902.0431_F01701	92	■ 1.000	$\alpha E=0$	$\alpha E = 0$	This implies that $\xi_2 = \xi_3 = x_1 = 0$. Hence $\alpha E = 0$, which is a contradiction.
0902.0431_F01702	92	■ 1.000	αE_1	αE_1	Therefore αE_1 is of the form
0902.0431_F01703	92	■ 0.999	$\alpha E_1 \times \alpha E_1=\tau \alpha \tau\left(E_1 \times E_1\right)=0$	$\alpha E_1 \times \alpha E_1 = \tau \alpha \tau\left(E_1 \times E_1\right) = 0$	From $\alpha E_1 \times \alpha E_1 = \tau \alpha \tau\left(E_1 \times E_1\right) = 0$, we have
0902.0431_F01704	92	■ 1.000	$(\tau \xi) \xi=1$	$(\tau \xi) \xi = 1$	we have $(\tau \xi) \xi = 1$.
0902.0431_F01705	92	■ 1.000	$\left(E_6\right)_{E_1}$	$\left(E_6\right)_{E_1}$	In order to investigate the group $\left(E_6\right)^{\sigma}$, we consider the following subgroup $\left(E_6\right)_{E_1}$
0902.0431_F01706	92	■ 1.000	$\left(E_6\right)^{\sigma}:\left(E_6\right)_{E_1} \subset\left(E_6\right)^{\sigma}$	$\left(E_6\right)^{\sigma}:\left(E_6\right)_{E_1} \subset\left(E_6\right)^{\sigma}$	Lemma 3.10.2. $\left(E_6\right)_{E_1}$ is a subgroup of $\left(E_6\right)^{\sigma}$: $\left(E_6\right)_{E_1} \subset\left(E_6\right)^{\sigma}$.
0902.0431_F01707	92	■ 0.926	$\left(\mathfrak{J}^C\right)_{\sigma}=\left\{X \in \mathfrak{J}^C \mid 4 E_1 \times\left(E_1 \times X\right)=X\right\} \oplus \mathfrak{E}_1^C$	$\left(\mathfrak{J}^C\right)_{\sigma} = \left\{X \in \mathfrak{J}^C \mid 4 E_1 \times\left(E_1 \times X\right)=X\right\} \oplus \mathfrak{E}_1^C$	Proof. Since $\left(\mathfrak{J}^C\right)_{\sigma} = \left\{X \in \mathfrak{J}^C \mid 4 E_1 \times\left(E_1 \times X\right)=X\right\} \oplus \mathfrak{E}_1^C$ and $\left(\mathfrak{J}^C\right)_{-\sigma} =$
0902.0431_F01708	92	■ 0.926	$\left(\mathfrak{J}^C\right)_{-\sigma} =$	$\left(\mathfrak{J}^C\right)_{-\sigma} =$	Proof. Since $\left(\mathfrak{J}^C\right)_{\sigma} = \left\{X \in \mathfrak{J}^C \mid 4 E_1 \times\left(E_1 \times X\right)=X\right\} \oplus \mathfrak{E}_1^C$ and $\left(\mathfrak{J}^C\right)_{-\sigma} =$
0902.0431_F01709	92	■ 1.000	$\left\{X \in \mathfrak{J}^C \mid E_1 \times X=0,\left\langle E_1, X\right\rangle =0\right\}$	$\left\{X \in \mathfrak{J}^C \mid E_1 \times X=0,\left\langle E_1, X\right\rangle =0\right\}$	$\left\{X \in \mathfrak{J}^C \mid E_1 \times X=0,\left\langle E_1, X\right\rangle =0\right\}$, these spaces are seen to be invariant under the
0902.0431_F01710	92	■ 1.000	$\alpha \in\left(E_6\right)_{E_1}$	$\alpha \in\left(E_6\right)_{E_1}$	action of $\left(E_6\right)_{E_1}$. Hence for $\alpha \in\left(E_6\right)_{E_1}$, we have $\sigma \alpha = \alpha \sigma$ (the proof is the same as
0902.0431_F01711	92	■ 1.000	$\sigma \alpha = \alpha \sigma$	$\sigma \alpha = \alpha \sigma$	action of $\left(E_6\right)_{E_1}$. Hence for $\alpha \in\left(E_6\right)_{E_1}$, we have $\sigma \alpha = \alpha \sigma$ (the proof is the same as
0902.0431_F01712	93	■ 1.000	V^{10}	V^{10}	We define a 10 dimensional $\mathbf{R}$ -vector space V^{10} by
0902.0431_F01713	93	■ 0.835	$\left(E_6\right)_{E_1} / \operatorname{Spin}(9) \simeq S^9$	$\left(E_6\right)_{E_1} / \operatorname{Spin}(9) \simeq S^9$	Proposition 3.10.3. $\left(E_6\right)_{E_1} / \operatorname{Spin}(9) \simeq S^9$.
0902.0431_F01714	93	■ 1.000	$S^9=\left\{X \in V^{10} \mid\langle X, X\rangle =2\right\}$	$S^9 = \left\{X \in V^{10} \mid\langle X, X\rangle =2\right\}$	Proof. $S^9 = \left\{X \in V^{10} \mid\langle X, X\rangle =2\right\}$ is a 9 dimensional sphere. For $\alpha \in\left(E_6\right)_{E_1}$
0902.0431_F01715	93	■ 1.000	$X \in S^9$	$X \in S^9$	and $X \in S^9$, we have $\alpha X \in S^9$. Indeed,
0902.0431_F01716	93	■ 1.000	$\alpha X \in S^9$	$\alpha X \in S^9$	and $X \in S^9$, we have $\alpha X \in S^9$. Indeed,
0902.0431_F01717	93	■ 1.000	S^9	S^9	Hence the group $\left(E_6\right)_{E_1}$ acts on S^9 . We shall prove that the action is transitive. To

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9902.0431_F01718	93	■ 1.000	$i\left(E_2+E_3\right) \in S^9$	$i\left(E_2+E_3\right) \in S^9$	$i\left(E_2+E_3\right) \in S^9$ by some $\alpha \in \left(E_6\right)_{E_1}$. Now, for a given $X \in S^9$, we can choose $\alpha_{23}\left(t_0\right)$
9902.0431_F01719	93	■ 1.000	$\alpha_{23}\left(t_0\right)$	$\alpha_{23}\left(t_0\right)$	$i\left(E_2+E_3\right) \in S^9$ by some $\alpha \in \left(E_6\right)_{E_1}$. Now, for a given $X \in S^9$, we can choose $\alpha_{23}\left(t_0\right)$
9902.0431_F01720	93	■ 1.000	$V^9=\left\{X \in V^{10} \mid \tau X=X\right\}$	$V^9=\left\{X \in V^{10} \mid \tau X=X\right\}$	where $V^9=\left\{X \in V^{10} \mid \tau X=X\right\}$. (Note that $\alpha_{23}\left(t_0\right) \in \left(E_6\right)_{E_1}$ because $\alpha_{23}\left(t_0\right) E_1=$
9902.0431_F01721	93	■ 1.000	$\alpha_{23}\left(t_0\right) \in \left(E_6\right)_{E_1}$	$\alpha_{23}\left(t_0\right) \in \left(E_6\right)_{E_1}$	where $V^9=\left\{X \in V^{10} \mid \tau X=X\right\}$. (Note that $\alpha_{23}\left(t_0\right) \in \left(E_6\right)_{E_1}$ because $\alpha_{23}\left(t_0\right) E_1=$
9902.0431_F01722	93	■ 1.000	$\alpha_{23}\left(t_0\right) E_1=$	$\alpha_{23}\left(t_0\right) E_1=$	where $V^9=\left\{X \in V^{10} \mid \tau X=X\right\}$. (Note that $\alpha_{23}\left(t_0\right) \in \left(E_6\right)_{E_1}$ because $\alpha_{23}\left(t_0\right) E_1=$
9902.0431_F01723	93	■ 0.769	$\operatorname{Spin}(9)=\left(F_4\right)_{E_1} \subset \left(E_6\right)_{E_1}$	$\operatorname{Spin}(9)=\left(F_4\right)_{E_1} \subset \left(E_6\right)_{E_1}$	E_1). Since the group $\operatorname{Spin}(9)=\left(F_4\right)_{E_1} \subset \left(E_6\right)_{E_1}$ acts transitively on S^8 (Proposition
9902.0431_F01724	93	■ 1.000	$\beta \in \operatorname{Spin}(9)$	$\beta \in \operatorname{Spin}(9)$	2.7.3), there exists $\beta \in \operatorname{Spin}(9)$ such that
9902.0431_F01725	93	■ 1.000	$\alpha_{23}(\pi / 2) \in \left(E_6\right)_{E_1}$	$\alpha_{23}(\pi / 2) \in \left(E_6\right)_{E_1}$	By applying $\alpha_{23}(\pi / 2) \in \left(E_6\right)_{E_1}$ of Lemma 3.8.1.(1), we get
9902.0431_F01726	93	■ 1.000	$\alpha\left(i\left(E_2+E_3\right)\right)=i\left(E_2+E_3\right)$	$\alpha\left(i\left(E_2+E_3\right)\right)=i\left(E_2+E_3\right)$	$\operatorname{Spin}(9)$. Indeed, if $\alpha \in \left(E_6\right)_{E_1}$ satisfies $\alpha\left(i\left(E_2+E_3\right)\right)=i\left(E_2+E_3\right)$, then $\alpha E=\alpha E_1+$
9902.0431_F01727	93	■ 1.000	$\alpha E=\alpha E_1+$	$\alpha E=\alpha E_1+$	$\operatorname{Spin}(9)$. Indeed, if $\alpha \in \left(E_6\right)_{E_1}$ satisfies $\alpha\left(i\left(E_2+E_3\right)\right)=i\left(E_2+E_3\right)$, then $\alpha E=\alpha E_1+$
9902.0431_F01728	93	■ 1.000	$\alpha\left(E_2+E_3\right)=E_1+\left(E_2+E_3\right)=E$	$\alpha\left(E_2+E_3\right)=E_1+\left(E_2+E_3\right)=E$	$\alpha\left(E_2+E_3\right)=E_1+\left(E_2+E_3\right)=E$, so that $\alpha \in F_4$ and hence $\alpha \in \left(F_4\right)_{E_1}=\operatorname{Spin}(9)$.
9902.0431_F01729	93	■ 1.000	$\alpha \in \left(F_4\right)_{E_1}=\operatorname{Spin}(9)$	$\alpha \in \left(F_4\right)_{E_1}=\operatorname{Spin}(9)$	$\alpha\left(E_2+E_3\right)=E_1+\left(E_2+E_3\right)=E$, so that $\alpha \in F_4$ and hence $\alpha \in \left(F_4\right)_{E_1}=\operatorname{Spin}(9)$.
9902.0431_F01730	93	■ 0.994	$\left(E_6\right)_{E_1} / \operatorname{Spin}(9) \simeq S^9$	$\left(E_6\right)_{E_1} / \operatorname{Spin}(9) \simeq S^9$	homeomorphism $\left(E_6\right)_{E_1} / \operatorname{Spin}(9) \simeq S^9$.
9902.0431_F01731	93	■ 1.000	$\left(E_6\right)_{E_1} \cong \operatorname{Spin}(10)$	$\left(E_6\right)_{E_1} \cong \operatorname{Spin}(10)$	Theorem 3.10.4. $\left(E_6\right)_{E_1} \cong \operatorname{Spin}(10)$.
9902.0431_F01732	93	■ 1.000	$p(\alpha)=\alpha \mid V^{10}$	$p(\alpha)=\alpha \mid V^{10}$	by $p(\alpha)=\alpha \mid V^{10}$. The restriction p' of $p:\left(E_6\right)_{E_1} \rightarrow S O(10)$ to $\left(F_4\right)_{E_1}$ coincides
9902.0431_F01733	93	■ 1.000	$p:\left(E_6\right)_{E_1} \rightarrow S O(10)$	$p:\left(E_6\right)_{E_1} \rightarrow S O(10)$	by $p(\alpha)=\alpha \mid V^{10}$. The restriction p' of $p:\left(E_6\right)_{E_1} \rightarrow S O(10)$ to $\left(F_4\right)_{E_1}$ coincides
9902.0431_F01734	93	■ 0.991	$p': \operatorname{Spin}(9) \rightarrow S O(9)$	$p': \operatorname{Spin}(9) \rightarrow S O(9)$	with the homomorphism $p': \operatorname{Spin}(9) \rightarrow S O(9)$ of Theorem 2.7.4. In particular,
9902.0431_F01735	94	■ 1.000	$\operatorname{Ker} p=$	$\operatorname{Ker} p=$	we see that $p:\left(E_6\right)_{E_1} \rightarrow S O(10)$ is onto by the five lemma. We also have $\operatorname{Ker} p=$
9902.0431_F01736	94	■ 1.000	$\{1, \sigma\}$	$\{1, \sigma\}$	$\{1, \sigma\}$. Indeed, $\alpha \in \operatorname{Ker} p$ leaves E_2-E_3 and $i\left(E_2+E_3\right)$ invariant, so that $\alpha E_i=E_i$
9902.0431_F01737	94	■ 1.000	$i\left(E_2+E_3\right)$	$i\left(E_2+E_3\right)$	$\{1, \sigma\}$. Indeed, $\alpha \in \operatorname{Ker} p$ leaves E_2-E_3 and $i\left(E_2+E_3\right)$ invariant, so that $\alpha E_i=E_i$
9902.0431_F01738	94	■ 1.000	$\alpha E_i=E_i$	$\alpha E_i=E_i$	$\{1, \sigma\}$. Indeed, $\alpha \in \operatorname{Ker} p$ leaves E_2-E_3 and $i\left(E_2+E_3\right)$ invariant, so that $\alpha E_i=E_i$
9902.0431_F01739	94	■ 1.000	$\alpha \in \operatorname{Ker} p'$	$\alpha \in \operatorname{Ker} p'$	for $i=1,2,3$. Hence we have $\alpha \in \operatorname{Spin}(8)$ and so $\alpha \in \operatorname{Ker} p'$. Therefore $\alpha=1$ or σ
9902.0431_F01740	94	■ 1.000	$\operatorname{Spin}(10)$	$\operatorname{Spin}(10)$	Therefore the group $\left(E_6\right)_{E_1}$ is isomorphic to the group $\operatorname{Spin}(10)$ as the universal
9902.0431_F01741	94	■ 0.999	$S O(10)$	$S O(10)$	covering group of $S O(10)$.
9902.0431_F01742	94	■ 1.000	$\theta \in C, \theta \neq 0$	$\theta \in C, \theta \neq 0$	For $\theta \in C, \theta \neq 0$, we define a C -linear mapping $\phi(\theta): \mathfrak{J}^C \rightarrow \mathfrak{J}^C$ by
9902.0431_F01743	94	■ 1.000	$\phi(\theta): \mathfrak{J}^C \rightarrow \mathfrak{J}^C$	$\phi(\theta): \mathfrak{J}^C \rightarrow \mathfrak{J}^C$	For $\theta \in C, \theta \neq 0$, we define a C -linear mapping $\phi(\theta): \mathfrak{J}^C \rightarrow \mathfrak{J}^C$ by
9902.0431_F01744	94	■ 1.000	$\alpha_{12}(t), \alpha_{13}(t)$	$\alpha_{12}(t), \alpha_{13}(t)$	(where $\alpha_{12}(t), \alpha_{13}(t)$ are mappings defined in Lemma 3.8.1) which is isomorphic to
9902.0431_F01745	94	■ 1.000	$U(1)=\{\theta \in C \mid(\tau \theta) \theta=1\}$	$U(1)=\{\theta \in C \mid(\tau \theta) \theta=1\}$	the usual unitary group $U(1)=\{\theta \in C \mid(\tau \theta) \theta=1\}$.
9902.0431_F01746	94	■ 0.936	$\mathfrak{J}^C=\mathfrak{E}_1^C \oplus \mathfrak{J}(2, \mathfrak{C})^C \oplus\left(\mathfrak{J}^C\right)_{-\sigma}$	$\mathfrak{J}^C=\mathfrak{E}_1^C \oplus \mathfrak{J}(2, \mathfrak{C})^C \oplus\left(\mathfrak{J}^C\right)_{-\sigma}$	Proof. We consider the decomposition $\mathfrak{J}^C=\mathfrak{E}_1^C \oplus \mathfrak{J}(2, \mathfrak{C})^C \oplus\left(\mathfrak{J}^C\right)_{-\sigma}$, where
9902.0431_F01747	94	■ 1.000	$\phi(\theta) \in U(1)$	$\phi(\theta) \in U(1)$	The restrictions of $\phi(\theta) \in U(1)$ to these spaces are all constant mappings:
9902.0431_F01748	94	■ 0.984	$\beta \in \operatorname{Spin}(10)$	$\beta \in \operatorname{Spin}(10)$	On the other hand, $\beta \in \operatorname{Spin}(10)$ also induces C -linear transformations of these
9902.0431_F01749	94	■ 0.500	$\phi(\theta)$	$\phi(\theta)$	spaces. From this, the commutativity of $\phi(\theta)$ and $\beta: \phi(\theta) \beta=\beta \phi(\theta)$ follows.
9902.0431_F01750	94	■ 0.500	$\beta: \phi(\theta) \beta=\beta \phi(\theta)$	$\beta: \phi(\theta) \beta=\beta \phi(\theta)$	spaces. From this, the commutativity of $\phi(\theta)$ and $\beta: \phi(\theta) \beta=\beta \phi(\theta)$ follows.
9902.0431_F01751	94	■ 0.887	$\left(E_6\right)^\sigma \cong(U(1) \times \operatorname{Spin}(10)) / \mathbf{Z}_4, \quad \mathbf{Z}_4=\{(1, \phi(1)),(-1, \phi(-1))\}$	$\left(E_6\right)^\sigma \cong(U(1) \times \operatorname{Spin}(10)) / \mathbf{Z}_4, \quad \mathbf{Z}_4=\{(1, \phi(1)),(-1, \phi(-1))\}$	Theorem 3.10.7. $\left(E_6\right)^\sigma \cong(U(1) \times \operatorname{Spin}(10)) / \mathbf{Z}_4, \quad \mathbf{Z}_4=\{(1, \phi(1)),(-1, \phi(-1))\}$,
9902.0431_F01752	94	■ 0.907	$(i, \phi(-i)),(-i, \phi(i))\}$	$(i, \phi(-i)),(-i, \phi(i))\}$	$(i, \phi(-i)),(-i, \phi(i))\}$.
9902.0431_F01753	95	■ 0.969	$\varphi: U(1) \times \operatorname{Spin}(10) \rightarrow\left(E_6\right)^\sigma$	$\varphi: U(1) \times \operatorname{Spin}(10) \rightarrow\left(E_6\right)^\sigma$	Proof. We define a mapping $\varphi: U(1) \times \operatorname{Spin}(10) \rightarrow\left(E_6\right)^\sigma$ by
9902.0431_F01754	95	■ 1.000	$\theta \in C$	$\theta \in C$	is a homomorphism. We shall show that φ is onto. For $\alpha \in\left(E_6\right)^\sigma$, there exists $\theta \in C$,
9902.0431_F01755	95	■ 1.000	$(\tau \theta) \theta=1$	$(\tau \theta) \theta=1$	$(\tau \theta) \theta=1$ satisfying

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0902.0431_F01756	95	■ 0.992	$\backslash\mathrm{beta}=\backslash\mathrm{phi}(\backslash\mathrm{theta})^{\wedge}\{-1\} \backslash\mathrm{alpha}$	$\beta = \phi(\theta)^{-1}\alpha$	(Lemma 3.10.1). Let $\beta = \phi(\theta)^{-1}\alpha$, then $\beta E_1 = E_1$, so $\beta \in Spin(10)$ (Theorem
0902.0431_F01757	95	■ 0.992	$\backslash\mathrm{beta} \ E_{\{1\}}=E_{\{1\}}$	$\beta E_1 = E_1$	(Lemma 3.10.1). Let $\beta = \phi(\theta)^{-1}\alpha$, then $\beta E_1 = E_1$, so $\beta \in Spin(10)$ (Theorem
0902.0431_F01758	95	■ 1.000	$\backslash\mathrm{alpha}=\backslash\mathrm{phi}(\backslash\mathrm{theta}) \ \backslash\mathrm{beta}=\backslash\mathrm{varphi}(\backslash\mathrm{theta}, \ \backslash\mathrm{beta})$	$\alpha = \phi(\theta)\beta = \varphi(\theta, \beta)$	3.10.4). Hence, we have $\alpha = \phi(\theta)\beta = \varphi(\theta, \beta)$ and therefore φ is onto. It is easily seen
0902.0431_F01759	95	■ 1.000	$(U(1) \ \backslash\mathrm{times} \ \mathrm{operatorname{Spin}}(10)) \ / \ \backslash\mathrm{boldsymbol{Z}}_{\{4\}} \ \backslash\mathrm{cong} \backslash\mathrm{left}(E_{\{6\}}\backslash\mathrm{right})^{\wedge}\{\backslash\mathrm{sigma}\}$	$(U(1) \times \mathrm{Spin}(10))/\mathbf{Z}_4 \cong (E_6)^\sigma$	Thus we have the isomorphism $U(1) \times Spin(10))/\mathbf{Z}_4 \cong (E_6)^\sigma$.
0902.0431_F01760	95	■ 1.000	$\backslash\mathrm{gamma}: \ \backslash\mathrm{mathfrak{J}}^{\wedge}\{C\} \ \backslash\mathrm{rightarrow} \ \backslash\mathrm{mathfrak{J}}^{\wedge}\{C\}$	$\gamma : \mathfrak{J}^C \rightarrow \mathfrak{J}^C$	Let the C -linear mapping $\gamma : \mathfrak{J}^C \rightarrow \mathfrak{J}^C$ be the complexification of $\gamma \in G_2 \subset F_4$.
0902.0431_F01761	95	■ 1.000	$\backslash\mathrm{gamma} \ \backslash\mathrm{in} \ E_{\{6\}}$	$\gamma \in E_6$	Then $\gamma \in E_6$ and $\gamma^2 = 1$.
0902.0431_F01762	95	■ 1.000	$\backslash\mathrm{gamma}^{\wedge}\{2\}=1$	$\gamma^2 = 1$	Then $\gamma \in E_6$ and $\gamma^2 = 1$.
0902.0431_F01763	95	■ 1.000	$\backslash\mathrm{left}(E_{\{6\}}\backslash\mathrm{right})^{\wedge}\{\backslash\mathrm{gamma}\}$	$(E_6)^\gamma$	We shall study the following subgroup $(E_6)^\gamma$ of E_6 :
0902.0431_F01764	95	■ 1.000	$\backslash\mathrm{mathfrak{J}}=\backslash\mathrm{mathfrak{J}}(3, \ \backslash\mathrm{boldsymbol{H}}) \ \backslash\mathrm{oplus} \ \backslash\mathrm{boldsymbol{H}}^{\wedge}\{3\}$	$\mathfrak{J} = \mathfrak{J}(3, \mathbf{H}) \oplus \mathbf{H}^3$	which is the complexification of the decomposition $\mathfrak{J} = \mathfrak{J}(3, \mathbf{H}) \oplus \mathbf{H}^3$. As usual we
0902.0431_F01765	95	■ 1.000	$\backslash\{X \ \backslash\mathrm{in} \ \backslash\mathrm{mathfrak{J}}(3, \ \backslash\mathrm{boldsymbol{H}}) \} \ \backslash\mathrm{mid} \ \mathrm{operatorname{tr}}(X)=0 \backslash\}$	$\{X \in \mathfrak{J}(3, \mathbf{H}) \mid \mathrm{tr}(X) = 0\}$	denote $\mathfrak{J}(3, \mathbf{H})$ and $\{X \in \mathfrak{J}(3, \mathbf{H}) \mid \mathrm{tr}(X) = 0\}$ by $\mathfrak{J}_H$ and $(\mathfrak{J}_H)_0$, respectively.
0902.0431_F01766	95	■ 1.000	$\backslash\mathrm{mathfrak{J}}_{\backslash\mathrm{boldsymbol{H}}}$	$\mathfrak{J}_H$	denote $\mathfrak{J}(3, \mathbf{H})$ and $\{X \in \mathfrak{J}(3, \mathbf{H}) \mid \mathrm{tr}(X) = 0\}$ by $\mathfrak{J}_H$ and $(\mathfrak{J}_H)_0$, respectively.
0902.0431_F01767	95	■ 1.000	$\backslash\mathrm{left}(\backslash\mathrm{mathfrak{J}}_{\backslash\mathrm{boldsymbol{H}}}\backslash\mathrm{right})_{\{0\}}$	$(\mathfrak{J}_H)_0$	denote $\mathfrak{J}(3, \mathbf{H})$ and $\{X \in \mathfrak{J}(3, \mathbf{H}) \mid \mathrm{tr}(X) = 0\}$ by $\mathfrak{J}_H$ and $(\mathfrak{J}_H)_0$, respectively.
0902.0431_F01768	95	■ 1.000	$\backslash\mathrm{boldsymbol{C}}=\backslash\mathrm{left}\backslash\mathrm{x+y} \ e_{\{1\}} \ \backslash\mathrm{mid} \ \mathrm{x}, \ \mathrm{y} \ \backslash\mathrm{in} \ \backslash\mathrm{boldsymbol{R}}\backslash\mathrm{right}\} \ \backslash\mathrm{subset} \ \backslash\mathrm{mathfrak{C}}$	$\mathbf{C} = \{x + ye_1 \mid x, y \in \mathbf{R}\} \subset \mathfrak{C}$	We consider the embedding $\mathbf{C} = \{x + ye_1 \mid x, y \in \mathbf{R}\} \subset \mathfrak{C}$ and, for an element
0902.0431_F01769	95	■ 1.000	$\mathrm{a}=\mathrm{x+y} \ e_{\{1\}} \ \backslash\mathrm{in} \ \backslash\mathrm{boldsymbol{C}}$	$a = x + ye_1 \in \mathbf{C}$	$a = x + ye_1 \in \mathbf{C}$, we denote by a' the element $x + yi \in C$. Now, we define an $\mathbf{R}$ -linear
0902.0431_F01770	95	■ 1.000	$\mathrm{a}^{\wedge}\{\backslash\mathrm{prime}\}$	a'	$a = x + ye_1 \in \mathbf{C}$, we denote by a' the element $x + yi \in C$. Now, we define an $\mathbf{R}$ -linear
0902.0431_F01771	95	■ 1.000	$\mathrm{x+y} \ i \ \backslash\mathrm{in} \ C$	$x + yi \in C$	$a = x + ye_1 \in \mathbf{C}$, we denote by a' the element $x + yi \in C$. Now, we define an $\mathbf{R}$ -linear
0902.0431_F01772	95	■ 1.000	$\mathrm{k}: \ \backslash\mathrm{boldsymbol{H}} \ \backslash\mathrm{rightarrow} \ M(2, \ C)$	$k : \mathbf{H} \rightarrow M(2, C)$	mapping $k : \mathbf{H} \rightarrow M(2, C)$ by
0902.0431_F01773	96	■ 1.000	$\mathrm{M}^{\wedge}\{*\}=\{ \ }^{\wedge}\{t\} \ \backslash\mathrm{bar}\{M\}, \ M \ \backslash\mathrm{in} \ M(3, \ \backslash\mathrm{boldsymbol{H}})$	$M^* = {}^t\bar{M}, M \in M(3, \mathbf{H})$	and $M^* = {}^t\bar{M}, M \in M(3, \mathbf{H})$. Then the following four properties hold.
0902.0431_F01774	96	■ 1.000	$\mathrm{k}(M \ N)=\mathrm{k}(M) \ \mathrm{k}(N)$	$k(MN) = k(M)k(N)$	(1) $k(MN) = k(M)k(N)$,
0902.0431_F01775	96	■ 0.874	$\backslash\mathrm{quad} \ \backslash\mathrm{tau}^{\wedge}\{t\}(\mathrm{k}(M))=\mathrm{k}\backslash\mathrm{left}(\mathrm{M}^{\wedge}\{*\}\backslash\mathrm{right})$	$\tau^t(k(M)) = k(M^*)$	(2) $\tau^t(k(M)) = k(M^*)$,
0902.0431_F01776	96	■ 0.999	$\mathrm{J}(\mathrm{k}(M))=(\backslash\mathrm{tau}(\mathrm{k}(M))) \ \mathrm{J}$	$J(k(M)) = (\tau(k(M)))J$	(3) $J(k(M)) = (\tau(k(M)))J$,
0902.0431_F01777	96	■ 0.999	$\mathrm{k}(\backslash\mathrm{boldsymbol{a}} \ M)=\mathrm{k}(\backslash\mathrm{boldsymbol{a}}) \ \mathrm{k}(M)$	$k(\mathbf{a}M) = k(\mathbf{a})k(M)$	(4) $k(\mathbf{a}M) = k(\mathbf{a})k(M)$,
0902.0431_F01778	96	■ 1.000	$\mathrm{k}: \ M(3, \ \backslash\mathrm{boldsymbol{H}}) \ \backslash\mathrm{rightarrow} \ M(6, \ C)$	$k : M(3, \mathbf{H}) \rightarrow M(6, C)$	The $\mathbf{R}$ -linear mappings $k : M(3, \mathbf{H}) \rightarrow M(6, C)$ and $k : \mathbf{H}^3 \rightarrow M(2, 6, C)$ are ex-
0902.0431_F01779	96	■ 1.000	$\mathrm{k}: \ \backslash\mathrm{boldsymbol{H}}^{\wedge}\{3\} \ \backslash\mathrm{rightarrow} \ M(2, 6, \ C)$	$k : \mathbf{H}^3 \rightarrow M(2, 6, C)$	The $\mathbf{R}$ -linear mappings $k : M(3, \mathbf{H}) \rightarrow M(6, C)$ and $k : \mathbf{H}^3 \rightarrow M(2, 6, C)$ are ex-
0902.0431_F01780	96	■ 1.000	$\mathrm{k}: \ M(3, \ \backslash\mathrm{boldsymbol{H}})^{\wedge}\{C\} \ \backslash\mathrm{rightarrow} \ M(6, \ C)$	$k : M(3, \mathbf{H})^C \rightarrow M(6, C)$	tended to C -linear mappings $k : M(3, \mathbf{H})^C \rightarrow M(6, C)$ and $k : (\mathbf{H}^3)^C \rightarrow M(2, 6, C)$
0902.0431_F01781	96	■ 1.000	$\mathrm{k}: \backslash\mathrm{left}(\backslash\mathrm{boldsymbol{H}}^{\wedge}\{3\}\backslash\mathrm{right})^{\wedge}\{C\} \ \backslash\mathrm{rightarrow} \ M(2, 6, \ C)$	$k : (\mathbf{H}^3)^C \rightarrow M(2, 6, C)$	tended to C -linear mappings $k : M(3, \mathbf{H})^C \rightarrow M(6, C)$ and $k : (\mathbf{H}^3)^C \rightarrow M(2, 6, C)$
0902.0431_F01782	96	■ 0.539	$(1) \ \backslash\mathrm{sim}(4)$	$(1) \sim (4)$	It is not difficult to see that they satisfy the four properties (1) $\sim$ (4) above.
0902.0431_F01783	96	■ 1.000	$\backslash\mathrm{mathfrak{S}}(6, \ C)$	$\mathfrak{S}(6, C)$	We define a C -vector space $\mathfrak{S}(6, C)$ by
0902.0431_F01784	96	■ 1.000	$\mathrm{k}_{\{J\}}: \ \backslash\mathrm{mathfrak{J}}(3, \ \backslash\mathrm{boldsymbol{H}})^{\wedge}\{C\} \ \backslash\mathrm{rightarrow} \ \backslash\mathrm{mathfrak{S}}(6, \ C)$	$k_J : \mathfrak{J}(3, \mathbf{H})^C \rightarrow \mathfrak{S}(6, C)$	and a C -linear mapping $k_J : \mathfrak{J}(3, \mathbf{H})^C \rightarrow \mathfrak{S}(6, C)$ by
0902.0431_F01785	96	■ 0.999	$\mathrm{k}_{\{J\}}$	k_J	Then k_J is well-defined. Indeed, for $M = M_1 + iM_2 \in \mathfrak{J}(3, \mathbf{H})^C$, we have
0902.0431_F01786	96	■ 0.999	$\mathrm{M}=\mathrm{M}_{\{1\}}+\mathrm{i} \ \mathrm{M}_{\{2\}} \ \backslash\mathrm{in} \ \backslash\mathrm{mathfrak{J}}(3, \ \backslash\mathrm{boldsymbol{H}})^{\wedge}\{C\}$	$M = M_1 + iM_2 \in \mathfrak{J}(3, \mathbf{H})^C$	Then k_J is well-defined. Indeed, for $M = M_1 + iM_2 \in \mathfrak{J}(3, \mathbf{H})^C$, we have
0902.0431_F01787	96	■ 0.985	$\backslash\mathrm{langle} \ S, \ T\backslash\mathrm{rangle}$	$\langle S, T \rangle$	Finally, we define Hermitian inner products $\langle S, T \rangle$ in $\mathfrak{S}(6, C)$ and $\langle P, Q \rangle$ in $M(2, 6,$
0902.0431_F01788	96	■ 0.985	$\backslash\mathrm{langle} \ P, \ Q\backslash\mathrm{rangle}$	$\langle P, Q \rangle$	Finally, we define Hermitian inner products $\langle S, T \rangle$ in $\mathfrak{S}(6, C)$ and $\langle P, Q \rangle$ in $M(2, 6,$
0902.0431_F01789	96	■ 0.985	$M(2, 6$	$M(2, 6$	Finally, we define Hermitian inner products $\langle S, T \rangle$ in $\mathfrak{S}(6, C)$ and $\langle P, Q \rangle$ in $M(2, 6,$
0902.0431_F01790	96	■ 1.000	$\mathrm{k}: \ M(3, \ \backslash\mathrm{boldsymbol{H}})^{\wedge}\{C\} \ \backslash\mathrm{rightarrow} \ M(6, \ C), \ \mathrm{k}: \backslash\mathrm{left}(\backslash\mathrm{boldsymbol{H}}^{\wedge}\{3\}\backslash\mathrm{right})^{\wedge}\{C\} \ \backslash\mathrm{rightarrow} \ M(2, 6, \ C)$	$k : M(3, \mathbf{H})^C \rightarrow M(6, C), k : (\mathbf{H}^3)^C \rightarrow M(2, 6, C)$	Lemma 3.11.1. (1) $k : M(3, \mathbf{H})^C \rightarrow M(6, C), k : (\mathbf{H}^3)^C \rightarrow M(2, 6, C)$ and
0902.0431_F01791	97	■ 1.000	$\mathrm{a}^{\wedge}\{\backslash\mathrm{prime}\}=\mathrm{b}^{\wedge}\{\backslash\mathrm{prime}\}=\mathrm{c}^{\wedge}\{\backslash\mathrm{prime}\}=\mathrm{d}^{\wedge}\{\backslash\mathrm{prime}\}=0$	$a' = b' = c' = d' = 0$	From which, we have $a' = b' = c' = d' = 0$ and so $a = b = c = d = 0$. Now, k is a
0902.0431_F01792	97	■ 1.000	$\mathrm{a}=\mathrm{b}=\mathrm{c}=\mathrm{d}=0$	$a = b = c = d = 0$	From which, we have $a' = b' = c' = d' = 0$ and so $a = b = c = d = 0$. Now, k is a
0902.0431_F01793	97	■ 0.998	$\mathrm{operatorname{dim}}_{\{C\}}\backslash\mathrm{left}(M(3, \ \backslash\mathrm{boldsymbol{H}})^{\wedge}\{C\} \ \backslash\mathrm{right})=36=\mathrm{operatorname{dim}}_{\{C\}}(M(6, \ C))$	$\dim_C(M(3, \mathbf{H})^C) = 36 = \dim_C(M(6, C))$	C -linear isomorphism, because $\dim_C(M(3, \mathbf{H})^C) = 36 = \dim_C(M(6, C))$. The other
0902.0431_F01794	97	■ 1.000	$\mathrm{m}_{\{k\}}=\mathrm{a}_{\{k\}}+\mathrm{b}_{\{k\}} \ e_{\{2\}}$	$m_k = a_k + b_k e_2$	(2) For $m_k = a_k + b_k e_2$ and $n_k = c_k + d_k e_2, a_k, b_k, c_k, d_k \in \mathbf{C}, k = 1, 2$, we have,
0902.0431_F01795	97	■ 1.000	$\mathrm{n}_{\{k\}}=\mathrm{c}_{\{k\}}+\mathrm{d}_{\{k\}} \ e_{\{2\}}, \ \mathrm{a}_{\{k\}}, \ \mathrm{b}_{\{k\}}, \ \mathrm{c}_{\{k\}}, \ \mathrm{d}_{\{k\}} \ \backslash\mathrm{in} \ \backslash\mathrm{boldsymbol{C}}, \ \mathrm{k}=1, 2$	$n_k = c_k + d_k e_2, a_k, b_k, c_k, d_k \in \mathbf{C}, k = 1, 2$	(2) For $m_k = a_k + b_k e_2$ and $n_k = c_k + d_k e_2, a_k, b_k, c_k, d_k \in \mathbf{C}, k = 1, 2$, we have,

Identifier	Page	Conf.	LaTeX source	Rendered	Image
9902.0431_F01796	97	■ 1.000	$\operatorname{det}\left(k_{\mathcal{J}}(M)\right)=\operatorname{det}\left(k(M)\right)$	$\det(k_J(M)) = \det(k(M))$	(3) Note that $\det(k_J(M)) = \det(k(M))$ and $\det M \in C$. Since we know that $\det S$
9902.0431_F01797	97	■ 1.000	$\operatorname{det} M \in C$	$\det M \in C$	(3) Note that $\det(k_J(M)) = \det(k(M))$ and $\det M \in C$. Since we know that $\det S$
9902.0431_F01798	97	■ 1.000	$\operatorname{det} S$	$\det S$	(3) Note that $\det(k_J(M)) = \det(k(M))$ and $\det M \in C$. Since we know that $\det S$
9902.0431_F01799	97	■ 1.000	s_{ij}	s_{ij}	with entries s_{ij} in S , we can easily see that
9902.0431_F01800	97	■ 1.000	$k_{\mathcal{J}}(M)$	$k_J(M)$	Now, since $k_J(M)$ is skew-symmetric, using the above result, we have
9902.0431_F01801	97	■ 1.000	$E_{6, \boldsymbol{H}}$	$E_{6, \boldsymbol{H}}$	We now consider a group $E_{6, \boldsymbol{H}}$ defined by replacing $\mathfrak{C}$ by $\boldsymbol{H}$ in the definition of
9902.0431_F01802	98	■ 0.924	$\operatorname{Sp}(3) / \boldsymbol{Z}_2$	$\operatorname{Sp}(3) / \boldsymbol{Z}_2$	(see Lemma 2.2.4). Moreover, it is isomorphic to the group $\operatorname{Sp}(3) / \boldsymbol{Z}_2$ (Proposition
9902.0431_F01803	98	■ 0.999	$\mathfrak{e}_{6, \boldsymbol{H}}$	$\mathfrak{e}_{6, \boldsymbol{H}}$	2.11.1). The Lie algebra $\mathfrak{e}_{6, \boldsymbol{H}}$ of the group $E_{6, \boldsymbol{H}}$ is given by
9902.0431_F01804	98	■ 1.000	$\mathfrak{f}_{4, \boldsymbol{H}} = \{\delta \in \mathfrak{e}_{6, \boldsymbol{H}} \mid \delta E = 0\}$	$\mathfrak{f}_{4, \boldsymbol{H}} = \{\delta \in \mathfrak{e}_{6, \boldsymbol{H}} \mid \delta E = 0\}$	Where $\mathfrak{f}_{4, \boldsymbol{H}} = \{\delta \in \mathfrak{e}_{6, \boldsymbol{H}} \mid \delta E = 0\}$ is the Lie algebra of the group $F_{4, \boldsymbol{H}}$ and is
9902.0431_F01805	98	■ 1.000	$\mathfrak{sp}(3)$	$\mathfrak{sp}(3)$	isomorphic to the Lie algebra $\mathfrak{sp}(3)$ by the mapping $\varphi_* : \mathfrak{sp}(3) \rightarrow \mathfrak{f}_{4, \boldsymbol{H}}$ given by
9902.0431_F01806	98	■ 1.000	$\varphi_* : \mathfrak{sp}(3) \rightarrow \mathfrak{f}_{4, \boldsymbol{H}}$	$\varphi_* : \mathfrak{sp}(3) \rightarrow \mathfrak{f}_{4, \boldsymbol{H}}$	isomorphic to the Lie algebra $\mathfrak{sp}(3)$ by the mapping $\varphi_* : \mathfrak{sp}(3) \rightarrow \mathfrak{f}_{4, \boldsymbol{H}}$ given by
9902.0431_F01807	98	■ 0.999	$\dim \mathfrak{e}_{6, \boldsymbol{H}} = 21 + 15 = 35$	$\dim \mathfrak{e}_{6, \boldsymbol{H}} = 21 + 15 = 35$	(Proposition 2.11.1) and we see that $\dim \mathfrak{e}_{6, \boldsymbol{H}} = 21 + 15 = 35$. As in Theorem 3.8.3,
9902.0431_F01808	98	■ 0.913	$E_{6, \boldsymbol{H}} \cong SU(6) / \boldsymbol{Z}_2, \quad \boldsymbol{Z}_2 = \{E, -E\}$	$E_{6, \boldsymbol{H}} \cong SU(6) / \boldsymbol{Z}_2, \quad \boldsymbol{Z}_2 = \{E, -E\}$	Proposition 3.11.3. $E_{6, \boldsymbol{H}} \cong SU(6) / \boldsymbol{Z}_2, \quad \boldsymbol{Z}_2 = \{E, -E\}$.
9902.0431_F01809	98	■ 1.000	$SU(6) = \{A \in M(6, C) \mid (\tau^t A) A = E, \det A = 1\}$	$SU(6) = \{A \in M(6, C) \mid (\tau^t A) A = E, \det A = 1\}$	Proof. Let $SU(6) = \{A \in M(6, C) \mid (\tau^t A) A = E, \det A = 1\}$ and define a
9902.0431_F01810	98	■ 0.963	$\varphi : SU(6) \rightarrow E_{6, \boldsymbol{H}}$	$\varphi : SU(6) \rightarrow E_{6, \boldsymbol{H}}$	mapping $\varphi : SU(6) \rightarrow E_{6, \boldsymbol{H}}$ by
9902.0431_F01811	98	■ 1.000	$\varphi(A) \in E_{6, \boldsymbol{H}}$	$\varphi(A) \in E_{6, \boldsymbol{H}}$	We first have to prove that $\varphi(A) \in E_{6, \boldsymbol{H}}$. Indeed, we have $(\det(\varphi(A)M))^2 =$
9902.0431_F01812	98	■ 1.000	$(\det(\varphi(A)M))^2 =$	$(\det(\varphi(A)M))^2 =$	We first have to prove that $\varphi(A) \in E_{6, \boldsymbol{H}}$. Indeed, we have $(\det(\varphi(A)M))^2 =$
9902.0431_F01813	98	■ 0.933	$\det(k_J(\varphi(A)M)) ($	$\det(k_J(\varphi(A)M)) ($	$\det(k_J(\varphi(A)M))$ (Lemma 3.11.1) $= \det(A(k_J(M))^t A) = \det(k_J(M)) = (\det M)^2$
9902.0431_F01814	98	■ 0.933	$) = \det(A(k_J(M))^t A) = \det(k_J(M)) = (\det M)^2$	$) = \det(A(k_J(M))^t A) = \det(k_J(M)) = (\det M)^2$	$\det(k_J(\varphi(A)M))$ (Lemma 3.11.1) $= \det(A(k_J(M))^t A) = \det(k_J(M)) = (\det M)^2$
9902.0431_F01815	98	■ 1.000	$\det(\varphi(A)M) = \pm \det M$	$\det(\varphi(A)M) = \pm \det M$	(Lemma 3.11.1), and so $\det(\varphi(A)M) = \pm \det M$. On the other hand, since $E_{6, \boldsymbol{H}}$ is
9902.0431_F01816	98	■ 1.000	$\det(\varphi(A)M)$	$\det(\varphi(A)M)$	connected (Lemma 3.11.2), the sign of $\det(\varphi(A)M)$ is constant, that is, independent
9902.0431_F01817	98	■ 1.000	$\det M \neq 0$	$\det M \neq 0$	of A (assuming that $\det M \neq 0$). Therefore
9902.0431_F01818	99	■ 0.996	$\operatorname{Ker} \varphi = \{E, -E\} = \boldsymbol{Z}_2$	$\operatorname{Ker} \varphi = \{E, -E\} = \boldsymbol{Z}_2$	Evidently φ is a homomorphism. That $\operatorname{Ker} \varphi = \{E, -E\} = \boldsymbol{Z}_2$ can be easily obtained.
9902.0431_F01819	99	■ 0.999	$\dim SU(6) = 35 = \dim E_{6, \boldsymbol{H}}$	$\dim SU(6) = 35 = \dim E_{6, \boldsymbol{H}}$	Since the group $E_{6, \boldsymbol{H}}$ is connected and $\dim SU(6) = 35 = \dim E_{6, \boldsymbol{H}}$ (Lemma 3.11.2),
9902.0431_F01820	99	■ 1.000	$SU(6) / \boldsymbol{Z}_2 \cong E_{6, \boldsymbol{H}}$	$SU(6) / \boldsymbol{Z}_2 \cong E_{6, \boldsymbol{H}}$	φ is onto. Thus we have the isomorphism $SU(6) / \boldsymbol{Z}_2 \cong E_{6, \boldsymbol{H}}$.
9902.0431_F01821	99	■ 0.470	$(E_6)^\gamma \cong (Sp(1) \times SU(6)) / \boldsymbol{Z}_2, \quad \boldsymbol{Z}_2 = \{(1, E), (-1, -E)\}$	$(E_6)^\gamma \cong (Sp(1) \times SU(6)) / \boldsymbol{Z}_2, \quad \boldsymbol{Z}_2 = \{(1, E), (-1, -E)\}$	Theorem 3.11.4. $(E_6)^\gamma \cong (Sp(1) \times SU(6)) / \boldsymbol{Z}_2, \quad \boldsymbol{Z}_2 = \{(1, E), (-1, -E)\}$.
9902.0431_F01822	99	■ 0.993	$\varphi : Sp(1) \times SU(6) \rightarrow (E_6)^\gamma$	$\varphi : Sp(1) \times SU(6) \rightarrow (E_6)^\gamma$	Proof. We define a mapping $\varphi : Sp(1) \times SU(6) \rightarrow (E_6)^\gamma$ by
9902.0431_F01823	99	■ 1.000	$\varphi(p, A) \in (E_6)^\gamma$	$\varphi(p, A) \in (E_6)^\gamma$	We first need to prove that $\varphi(p, A) \in (E_6)^\gamma$.
9902.0431_F01824	99	■ 0.925	${}^t \varphi(p, A)^{-1} = \tau \varphi(p, A) \tau$	${}^t \varphi(p, A)^{-1} = \tau \varphi(p, A) \tau$	Claim 1. ${}^t \varphi(p, A)^{-1} = \tau \varphi(p, A) \tau$.
9902.0431_F01825	99	■ 1.000	$\tau^t \varphi(p, A) \tau = \varphi(\bar{p}, \tau^t A)$	$\tau^t \varphi(p, A) \tau = \varphi(\bar{p}, \tau^t A)$	and so we have ${}^t \varphi(p, A) \tau = \varphi(\bar{p}, \tau^t A)$, which implies that ${}^t \varphi(p, A)^{-1} = \tau \varphi(p, A) \tau$.
9902.0431_F01826	99	■ 0.833	$\varphi(p, A) \in (E_6)^\gamma$	$\varphi(p, A) \in (E_6)^\gamma$	Claim 2. $\varphi(p, A) \in (E_6)^\gamma$.
9902.0431_F01827	99	■ 1.000	$\alpha = \varphi(p, A)$	$\alpha = \varphi(p, A)$	Proof. We will first show that $\alpha = \varphi(p, A)$ satisfies
9902.0431_F01828	99	■ 1.000	$\alpha M \times \alpha N = {}^t \alpha^{-1} (M \times N)$	$\alpha M \times \alpha N = {}^t \alpha^{-1} (M \times N)$	The equality $\alpha M \times \alpha N = {}^t \alpha^{-1} (M \times N)$ is evident from $\det(\alpha M) = \det M$ (Proposition
9902.0431_F01829	99	■ 1.000	$\det(\alpha M) = \det M$	$\det(\alpha M) = \det M$	The equality $\alpha M \times \alpha N = {}^t \alpha^{-1} (M \times N)$ is evident from $\det(\alpha M) = \det M$ (Proposition
9902.0431_F01830	100	■ 1.000	$\varphi(p, A) \in E_6$	$\varphi(p, A) \in E_6$	We have therefore shown that $\varphi(p, A) \in E_6$. Clearly, $\gamma \varphi(p, A) = \varphi(p, A) \gamma$, so that
9902.0431_F01831	100	■ 1.000	$\alpha \in (E_6)^\gamma$	$\alpha \in (E_6)^\gamma$	We shall now show that φ is onto. Let $\alpha \in (E_6)^\gamma$. Since the restriction $\alpha' =$
9902.0431_F01832	100	■ 1.000	$\alpha' =$	$\alpha' =$	We shall now show that φ is onto. Let $\alpha \in (E_6)^\gamma$. Since the restriction $\alpha' =$

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0902.0431_F01833	100	■ 1.000	$\alpha \mid \left(\mathfrak{J}_{\boldsymbol{H}}\right)^{\mathrm{C}}$	$\alpha \mid (\mathfrak{J}_{\boldsymbol{H}})^C$	$\alpha \mid (\mathfrak{J}_{\boldsymbol{H}})^C$ of α to $(\mathfrak{J}_{\boldsymbol{H}})^C$ belongs to $E_{6,\boldsymbol{H}}$, there exists $A \in SU(6)$ such that $\alpha' = \varphi(A)$
0902.0431_F01834	100	■ 1.000	$\left(\mathfrak{J}_{\boldsymbol{H}}\right)^{\mathrm{C}}$	$(\mathfrak{J}_{\boldsymbol{H}})^C$	$\alpha \mid (\mathfrak{J}_{\boldsymbol{H}})^C$ of α to $(\mathfrak{J}_{\boldsymbol{H}})^C$ belongs to $E_{6,\boldsymbol{H}}$, there exists $A \in SU(6)$ such that $\alpha' = \varphi(A)$
0902.0431_F01835	100	■ 1.000	$A \in \mathrm{S} \, \mathrm{U}(6)$	$A \in SU(6)$	$\alpha \mid (\mathfrak{J}_{\boldsymbol{H}})^C$ of α to $(\mathfrak{J}_{\boldsymbol{H}})^C$ belongs to $E_{6,\boldsymbol{H}}$, there exists $A \in SU(6)$ such that $\alpha' = \varphi(A)$
0902.0431_F01836	100	■ 1.000	$\alpha^{\prime}=\varphi(A)$	$\alpha' = \varphi(A)$	$\alpha \mid (\mathfrak{J}_{\boldsymbol{H}})^C$ of α to $(\mathfrak{J}_{\boldsymbol{H}})^C$ belongs to $E_{6,\boldsymbol{H}}$, there exists $A \in SU(6)$ such that $\alpha' = \varphi(A)$
0902.0431_F01837	100	■ 1.000	$\beta \mid \left(\mathfrak{J}_{\boldsymbol{H}}\right)^{\mathrm{C}}=1$	$\beta \mid (\mathfrak{J}_{\boldsymbol{H}})^C = 1$	(Proposition 3.11.3). If we put $\beta = \varphi(1, A)^{-1}\alpha$, then $\beta \mid (\mathfrak{J}_{\boldsymbol{H}})^C = 1$. Hence, by the
0902.0431_F01838	100	■ 0.972	$p \in \mathrm{S} \, \mathrm{p}(1)$	$p \in Sp(1)$	same argument as in Theorem 2.11.2, there exists $p \in Sp(1)$ such that $\beta = \varphi(p, E)$,
0902.0431_F01839	100	■ 0.822	$(\mathrm{S} \, \mathrm{p}(1) \times \mathrm{S} \, \mathrm{U}(6)) / \boldsymbol{Z}_2 \cong \left(E_6\right)^{\gamma}$	$(Sp(1) \times SU(6)) / \boldsymbol{Z}_2 \cong (E_6)^\gamma$	we have the isomorphism $[Sp(1) \times SU(6)] / \boldsymbol{Z}_2 \cong (E_6)^\gamma$.
0902.0431_F01840	100	■ 1.000	$\left(E_6\right)^{\tau \gamma}$	$(E_6)^{\tau\gamma}$	we shall study the following subgroup $(E_6)^{\tau\gamma}$ of E_6 :
0902.0431_F01841	100	■ 1.000	$\left(\mathfrak{J}^{\mathrm{C}}\right)_{\tau \gamma}$	$(\mathfrak{J}^C)_{\tau\gamma}$	For this end, we consider $\boldsymbol{R}$ -vector subspaces $(\mathfrak{J}^C)_{\tau\gamma}$ and $(\mathfrak{J}^C)_{-\tau\gamma}$ of $\mathfrak{J}^C$, which are
0902.0431_F01842	100	■ 1.000	$\left(\mathfrak{J}^{\mathrm{C}}\right)_{-\tau \gamma}$	$(\mathfrak{J}^C)_{-\tau\gamma}$	For this end, we consider $\boldsymbol{R}$ -vector subspaces $(\mathfrak{J}^C)_{\tau\gamma}$ and $(\mathfrak{J}^C)_{-\tau\gamma}$ of $\mathfrak{J}^C$, which are
0902.0431_F01843	101	■ 0.993	$\left(\mathfrak{J}^{\mathrm{C}}\right)_{\tau \gamma}: \mathfrak{J}^{\mathrm{C}}=\left(\left(\mathfrak{J}^{\mathrm{C}}\right)_{\tau \gamma}\right)^{\mathrm{C}}$	$(\mathfrak{J}^C)_{\tau\gamma} : \mathfrak{J}^C = \left((\mathfrak{J}^C)_{\tau\gamma}\right)^C$	In particular, $\mathfrak{J}^C$ is the complexification of $[\mathfrak{J}^C]_{\tau\gamma} : \mathfrak{J}^C = ((\mathfrak{J}^C)_{\tau\gamma})^C$.
0902.0431_F01844	101	■ 0.997	$P \circ Q$	$P \circ Q$	we define the Jordan multiplication $P \circ Q$ and an inner product (P, Q) respectively
0902.0431_F01845	101	■ 0.997	$(P, \, Q)$	(P, Q)	we define the Jordan multiplication $P \circ Q$ and an inner product (P, Q) respectively
0902.0431_F01846	101	■ 0.984	$\operatorname{Sp}(4)$	$\mathrm{Sp}(4)$	The group $Sp(4)$ acts on $\mathfrak{J}(4, \boldsymbol{H})$ by the mapping $\mu : Sp(4) \times \mathfrak{J}(4, \boldsymbol{H}) \rightarrow \mathfrak{J}(4, \boldsymbol{H})$,
0902.0431_F01847	101	■ 0.984	$\mathfrak{J}(4, \boldsymbol{H})$	$\mathfrak{J}(4, \boldsymbol{H})$	The group $Sp(4)$ acts on $\mathfrak{J}(4, \boldsymbol{H})$ by the mapping $\mu : Sp(4) \times \mathfrak{J}(4, \boldsymbol{H}) \rightarrow \mathfrak{J}(4, \boldsymbol{H})$,
0902.0431_F01848	101	■ 0.984	$\mu: \operatorname{Sp}(4) \times \mathfrak{J}(4, \boldsymbol{H}) \rightarrow \mathfrak{J}(4, \boldsymbol{H})$	$\mu : \mathrm{Sp}(4) \times \mathfrak{J}(4, \boldsymbol{H}) \rightarrow \mathfrak{J}(4, \boldsymbol{H})$	The group $Sp(4)$ acts on $\mathfrak{J}(4, \boldsymbol{H})$ by the mapping $\mu : Sp(4) \times \mathfrak{J}(4, \boldsymbol{H}) \rightarrow \mathfrak{J}(4, \boldsymbol{H})$,
0902.0431_F01849	101	■ 1.000	$\mu(A, P)=A \, P \, A^{*}$	$\mu(A, P) = APA^*$	$[\mu(A, P) = APA^*]$. Then we have
0902.0431_F01850	101	■ 0.999	$\boldsymbol{H} P_3$	$\boldsymbol{H} P_3$	The quaternion projective space $\boldsymbol{H} P_3$ is defined by
0902.0431_F01851	101	■ 1.000	$\mathfrak{J}(4, \boldsymbol{H})^{\mathrm{C}}$	$\mathfrak{J}(4, \boldsymbol{H})^C$	Finally, in the complexification $\mathfrak{J}(4, \boldsymbol{H})^C$ of $\mathfrak{J}(4, \boldsymbol{H})$, we extend naturally the Jordan
0902.0431_F01852	102	■ 1.000	$\mathfrak{J}(4, \boldsymbol{H})_{\{0\}}$	$\mathfrak{J}(4, \boldsymbol{H})_0$	We denote by $\mathfrak{J}(4, \boldsymbol{H})_0$ the space $\{P \in \mathfrak{J}(4, \boldsymbol{H}) \mid \mathrm{tr}(P) = 0\}$ and by $\mathfrak{J}(4, \boldsymbol{H})_0^C$ its
0902.0431_F01853	102	■ 1.000	$\{P \in \mathfrak{J}(4, \boldsymbol{H}) \mid \operatorname{tr}(P)=0\}$	$\{P \in \mathfrak{J}(4, \boldsymbol{H}) \mid \mathrm{tr}(P) = 0\}$	We denote by $\mathfrak{J}(4, \boldsymbol{H})_0$ the space $\{P \in \mathfrak{J}(4, \boldsymbol{H}) \mid \mathrm{tr}(P) = 0\}$ and by $\mathfrak{J}(4, \boldsymbol{H})_0^C$ its
0902.0431_F01854	102	■ 1.000	$\mathfrak{J}(4, \boldsymbol{H})_{\{0\}}^{\mathrm{C}}$	$\mathfrak{J}(4, \boldsymbol{H})_0^C$	We denote by $\mathfrak{J}(4, \boldsymbol{H})_0$ the space $\{P \in \mathfrak{J}(4, \boldsymbol{H}) \mid \mathrm{tr}(P) = 0\}$ and by $\mathfrak{J}(4, \boldsymbol{H})_0^C$ its
0902.0431_F01855	102	■ 1.000	$g: \mathfrak{J}^{\mathrm{C}} \rightarrow \mathfrak{J}(4, \boldsymbol{H})_{\{0\}}^{\mathrm{C}}$	$g : \mathfrak{J}^C \rightarrow \mathfrak{J}(4, \boldsymbol{H})_0^C$	Definition. We define a C -linear mapping $[g : \mathfrak{J}^C \rightarrow \mathfrak{J}(4, \boldsymbol{H})_0^C]$ by
0902.0431_F01856	102	■ 1.000	g	g	The restriction of g to $(\mathfrak{J}^C)_{\tau\gamma}$ is given by
0902.0431_F01857	102	■ 1.000	$g: \left(\mathfrak{J}^{\mathrm{C}}\right)_{\tau \gamma} \rightarrow \mathfrak{J}(4, \boldsymbol{H})_{\{0\}}$	$g : (\mathfrak{J}^C)_{\tau\gamma} \rightarrow \mathfrak{J}(4, \boldsymbol{H})_0$	The restriction of g to $(\mathfrak{J}^C)_{\tau\gamma}$ induces an $\boldsymbol{R}$ -linear isomorphism $[g : (\mathfrak{J}^C)_{\tau\gamma} \rightarrow \mathfrak{J}(4, \boldsymbol{H})_0]$.
0902.0431_F01858	102	■ 1.000	$\dim_{\mathrm{C}} \mathfrak{J}^{\mathrm{C}}=27=\dim_{\mathrm{C}} \mathfrak{J}(4, \boldsymbol{H})_{\{0\}}^{\mathrm{C}}, g$	$\dim_C \mathfrak{J}^C = 27 = \dim_C \mathfrak{J}(4, \boldsymbol{H})_0^C, g$	$[\dim_C \mathfrak{J}^C = 27 = \dim_C \mathfrak{J}(4, \boldsymbol{H})_0^C, g]$ is a C -linear isomorphism. Now, for $X = M + \boldsymbol{a}$,
0902.0431_F01859	102	■ 1.000	$X=M+\boldsymbol{a}$	$X = M + \boldsymbol{a}$	$\dim_C \mathfrak{J}^C = 27 = \dim_C \mathfrak{J}(4, \boldsymbol{H})_0^C, g$ is a C -linear isomorphism. Now, for $X = M + \boldsymbol{a}$,
0902.0431_F01860	102	■ 1.000	$Y=N+\boldsymbol{b} \in \left(\mathfrak{J}_{\boldsymbol{H}}\right)^{\mathrm{C}} \oplus\left(\boldsymbol{H}^3\right)^{\mathrm{C}}=\mathfrak{J}^{\mathrm{C}}$	$Y = N + \boldsymbol{b} \in (\mathfrak{J}_{\boldsymbol{H}})^C \oplus (\boldsymbol{H}^3)^C = \mathfrak{J}^C$	$[\boldsymbol{Y} = N + \boldsymbol{b} \in (\mathfrak{J}_{\boldsymbol{H}})^C \oplus (\boldsymbol{H}^3)^C = \mathfrak{J}^C]$, we have
0902.0431_F01861	103	■ 1.000	$g \, X \circ g \, Y=g(\gamma(X \times Y))+\frac{1}{4}(\gamma X, Y) E$	$gX \circ gY = g(\gamma(X \times Y)) + \frac{1}{4}(\gamma X, Y)E$	Thus the first equality $[gX \circ gY = g(\gamma(X \times Y)) + \frac{1}{4}(\gamma X, Y)E]$ is proved. Taking the
0902.0431_F01862	103	■ 1.000	$(\gamma X, Y)=\langle X, Y\rangle$	$(\gamma X, Y) = \langle X, Y \rangle$	Finally, if we note that $[\gamma X, Y] = \langle X, Y \rangle$ for $X, Y \in (\mathfrak{J}^C)_{\tau\gamma}$, then we can easily show
0902.0431_F01863	103	■ 1.000	$X, Y \in\left(\mathfrak{J}^{\mathrm{C}}\right)_{\tau \gamma}$	$X, Y \in (\mathfrak{J}^C)_{\tau\gamma}$	Finally, if we note that $(\gamma X, Y) = \langle X, Y \rangle$ for $[X, Y \in (\mathfrak{J}^C)_{\tau\gamma}]$, then we can easily show
0902.0431_F01864	103	■ 0.952	$\operatorname{Sp}(4) / \boldsymbol{Z}_2 \cong \left(E_6\right)^{\tau \gamma}, \quad \boldsymbol{Z}_2=\{E,-E\}$	$(E_6)^{\tau\gamma} \cong \mathrm{Sp}(4) / \boldsymbol{Z}_2, \quad \boldsymbol{Z}_2 = \{E, -E\}$	Theorem 3.12.2. $[E_6]^{\tau\gamma} \cong Sp(4) / \boldsymbol{Z}_2, \quad \boldsymbol{Z}_2 = \{E, -E\}]$.
0902.0431_F01865	103	■ 0.995	$\varphi: \operatorname{Sp}(4) \rightarrow\left(E_6\right)^{\tau \gamma}$	$\varphi : \mathrm{Sp}(4) \rightarrow (E_6)^{\tau\gamma}$	Proof. We define a mapping $[\varphi : Sp(4) \rightarrow (E_6)^{\tau\gamma}]$ by

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0902.0431_F01866	103	■ 1.000	$\varphi(A) \in \left(E_6\right)^{\tau \gamma}$	$\varphi(A) \in (E_6)^{\tau \gamma}$	We first have to prove that $\varphi(A) \in (E_6)^{\tau \gamma}$. Let $Z = \varphi(A)X$ and use Lemma 3.12.1,
0902.0431_F01867	103	■ 1.000	$Z=\varphi(A) X$	$Z = \varphi(A)X$	We first have to prove that $\varphi(A) \in (E_6)^{\tau \gamma}$. Let $Z = \varphi(A)X$ and use Lemma 3.12.1,
0902.0431_F01868	103	■ 1.000	$\varphi(A) \in E_6$	$\varphi(A) \in E_6$	Hence $\varphi(A) \in E_6$. To prove $\varphi(A) \in (E_6)^{\tau \gamma}$, namely, $\tau \gamma \varphi(A) = \varphi(A) \tau \gamma$, it is sufficient
0902.0431_F01869	103	■ 1.000	$\tau \gamma \varphi(A)=\varphi(A) \tau \gamma$	$\tau \gamma \varphi(A) = \varphi(A) \tau \gamma$	Hence $\varphi(A) \in E_6$. To prove $\varphi(A) \in (E_6)^{\tau \gamma}$, namely, $\tau \gamma \varphi(A) = \varphi(A) \tau \gamma$, it is sufficient
0902.0431_F01870	103	■ 1.000	$\mathfrak{J}^C=\left(\left(\mathfrak{J}^C\right)^{\tau \gamma}\right)^C$	$\mathfrak{J}^C = \left(\left(\mathfrak{J}^C\right)^{\tau \gamma}\right)^C$	since $\mathfrak{J}^C = \left(\left(\mathfrak{J}^C\right)^{\tau \gamma}\right)^C$. However this is evident. Indeed, if $X \in \left(\mathfrak{J}^C\right)^{\tau \gamma}$, then $gX \in$
0902.0431_F01871	103	■ 1.000	$X \in \left(\mathfrak{J}^C\right)^{\tau \gamma}$	$X \in \left(\mathfrak{J}^C\right)^{\tau \gamma}$	since $\mathfrak{J}^C = \left(\left(\mathfrak{J}^C\right)^{\tau \gamma}\right)^C$. However this is evident. Indeed, if $X \in \left(\mathfrak{J}^C\right)^{\tau \gamma}$, then $gX \in$
0902.0431_F01872	103	■ 1.000	$g X \in$	$gX \in$	since $\mathfrak{J}^C = \left(\left(\mathfrak{J}^C\right)^{\tau \gamma}\right)^C$. However this is evident. Indeed, if $X \in \left(\mathfrak{J}^C\right)^{\tau \gamma}$, then $gX \in$
0902.0431_F01873	103	■ 1.000	$\varphi(A) X \in \left(\mathfrak{J}^C\right)^{\tau \gamma}$	$\varphi(A)X \in \left(\mathfrak{J}^C\right)^{\tau \gamma}$	$\mathfrak{J}(4, \boldsymbol{H})_0$, so that $\varphi(A)X \in \left(\mathfrak{J}^C\right)^{\tau \gamma}$. Hence $\tau \gamma \varphi(A) \tau \gamma X = \tau \gamma \varphi(A)X = \varphi(A)X$.
0902.0431_F01874	103	■ 1.000	$\tau \gamma \varphi(A) \tau \gamma X=\tau \gamma \varphi(A) X=\varphi(A) X$	$\tau \gamma \varphi(A) \tau \gamma X = \tau \gamma \varphi(A)X = \varphi(A)X$	$\mathfrak{J}(4, \boldsymbol{H})_0$, so that $\varphi(A)X \in \left(\mathfrak{J}^C\right)^{\tau \gamma}$. Hence $\tau \gamma \varphi(A) \tau \gamma X = \tau \gamma \varphi(A)X = \varphi(A)X$.
0902.0431_F01875	103	■ 1.000	$\alpha \in \left(E_6\right)^{\tau \gamma}$	$\alpha \in (E_6)^{\tau \gamma}$	Evidently φ is a homomorphism. We shall show that φ is onto. For $\alpha \in (E_6)^{\tau \gamma}$, we
0902.0431_F01876	104	■ 1.000	$P \in \mathfrak{J}(4, \boldsymbol{H})$	$P \in \mathfrak{J}(4, \boldsymbol{H})$	then we have $P \in \mathfrak{J}(4, \boldsymbol{H})$ and $P^2 = P$, $\text{tr}(P) = 1$, that is, $P \in \boldsymbol{HP}^3$. Indeed,
0902.0431_F01877	104	■ 1.000	$P^2=P, \operatorname{tr}(P)=1$	$P^2 = P, \text{tr}(P) = 1$	then we have $P \in \mathfrak{J}(4, \boldsymbol{H})$ and $P^2 = P$, $\text{tr}(P) = 1$, that is, $P \in \boldsymbol{HP}^3$. Indeed,
0902.0431_F01878	104	■ 1.000	$P \in \boldsymbol{HP}^3$	$P \in \boldsymbol{HP}^3$	then we have $P \in \mathfrak{J}(4, \boldsymbol{H})$ and $P^2 = P$, $\text{tr}(P) = 1$, that is, $P \in \boldsymbol{HP}^3$. Indeed,
0902.0431_F01879	104	■ 0.999	$A \in \operatorname{Sp}(4)$	$A \in \text{Sp}(4)$	Hence there exists $A \in \text{Sp}(4)$ such that
0902.0431_F01880	104	■ 1.000	$g E=2 E_1-\frac{1}{2} E$	$gE = 2E_1 - \frac{1}{2}E$	Since $gE = 2E_1 - \frac{1}{2}E$, we have
0902.0431_F01881	104	■ 1.000	$\beta E=E$	$\beta E = E$	Putting $\beta = \varphi(A)^{-1} \alpha$, we have $\beta E = E$, and so $\beta \in F_4$ (Theorem 3.7.1). Also β
0902.0431_F01882	104	■ 1.000	$\tau \gamma \beta=\beta \tau \gamma$	$\tau \gamma \beta = \beta \tau \gamma$	satisfies $\tau \gamma \beta = \beta \tau \gamma$ and $\tau \beta = \beta \tau$ (Theorem 3.7.1), hence $\gamma \beta = \beta \gamma$, and therefore
0902.0431_F01883	104	■ 1.000	$\tau \beta=\beta \tau$	$\tau \beta = \beta \tau$	satisfies $\tau \gamma \beta = \beta \tau \gamma$ and $\tau \beta = \beta \tau$ (Theorem 3.7.1), hence $\gamma \beta = \beta \gamma$, and therefore
0902.0431_F01884	104	■ 1.000	$\gamma \beta=\beta \gamma$	$\gamma \beta = \beta \gamma$	satisfies $\tau \gamma \beta = \beta \tau \gamma$ and $\tau \beta = \beta \tau$ (Theorem 3.7.1), hence $\gamma \beta = \beta \gamma$, and therefore
0902.0431_F01885	104	■ 0.979	$\beta \in\left(F_4\right)^{\gamma}$	$\beta \in \left(F_4\right)^{\gamma}$	$\beta \in \left(F_4\right)^{\gamma}$. From Theorem 2.11.2, there exist $p \in \text{Sp}(1)$ and $D \in \text{Sp}(3)$ such that
0902.0431_F01886	104	■ 0.979	$D \in \operatorname{Sp}(3)$	$D \in \text{Sp}(3)$	$\beta \in \left(F_4\right)^{\gamma}$. From Theorem 2.11.2, there exist $p \in \text{Sp}(1)$ and $D \in \text{Sp}(3)$ such that
0902.0431_F01887	104	■ 0.911	$B=\left(\begin{array}{cc} p & 0 \\ 0 & D \end{array}\right)$	$B = \left(\begin{array}{cc} p & 0 \\ 0 & D \end{array}\right)$	Let $B = \left(\begin{array}{cc} p & 0 \\ 0 & D \end{array}\right)$, then $B \in \text{Sp}(4)$ and we have
0902.0431_F01888	104	■ 0.911	$B \in \operatorname{Sp}(4)$	$B \in \text{Sp}(4)$	Let $B = \left(\begin{array}{cc} p & 0 \\ 0 & D \end{array}\right)$, then $B \in \text{Sp}(4)$ and we have
0902.0431_F01889	104	■ 1.000	$M+\boldsymbol{a} \in\left(\mathfrak{J}_{\boldsymbol{H}}\right)^C \oplus\left(\boldsymbol{H}^3\right)^C=\mathfrak{J}^C$	$M + \boldsymbol{a} \in \left(\mathfrak{J}_{\boldsymbol{H}}\right)^C \oplus \left(\boldsymbol{H}^3\right)^C = \mathfrak{J}^C$	Certainly, for $M + \boldsymbol{a} \in \left(\mathfrak{J}_{\boldsymbol{H}}\right)^C \oplus \left(\boldsymbol{H}^3\right)^C = \mathfrak{J}^C$,
0902.0431_F01890	105	■ 0.806	$\operatorname{Sp}(4) / \boldsymbol{Z}_2 \cong\left(E_6\right)^{\tau \gamma}$	$\text{Sp}(4) / \boldsymbol{Z}_2 \cong (E_6)^{\tau \gamma}$	isomorphism $\text{Sp}(4) / \boldsymbol{Z}_2 \cong (E_6)^{\tau \gamma}$.
0902.0431_F01891	105	■ 1.000	$(S U(3) \times S U(3) \times S U(3)) / \boldsymbol{Z}_3$	$(SU(3) \times SU(3) \times SU(3)) / \boldsymbol{Z}_3$	3.13. Automorphism w of order 3 and subgroup $(SU(3) \times SU(3) \times SU(3)) / \boldsymbol{Z}_3$
0902.0431_F01892	105	■ 1.000	$w: \mathfrak{J}^C \rightarrow \mathfrak{J}^C$	$w : \mathfrak{J}^C \rightarrow \mathfrak{J}^C$	Let the C -linear mapping $w : \mathfrak{J}^C \rightarrow \mathfrak{J}^C$ be the complexification of $w \in G_2 \subset F_4$
0902.0431_F01893	105	■ 1.000	$w \in E_6$	$w \in E_6$	of Section 2.12. Then $w \in E_6$ and $w^3 = 1$.
0902.0431_F01894	105	■ 1.000	$\left(E_6\right)^w$	$(E_6)^w$	We shall study the following subgroup $(E_6)^w$ of E_6 :
0902.0431_F01895	105	■ 1.000	$\{X \in \mathfrak{J}(3, \boldsymbol{C}) \mid \operatorname{tr}(X)=0\}$	$\{X \in \mathfrak{J}(3, \boldsymbol{C}) \mid \text{tr}(X) = 0\}$	For convenience, we denote $\mathfrak{J}(3, \boldsymbol{C})$ and $\{X \in \mathfrak{J}(3, \boldsymbol{C}) \mid \text{tr}(X) = 0\}$ by $\mathfrak{J}_{\boldsymbol{C}}$ and $(\mathfrak{J}_{\boldsymbol{C}})_0$,
0902.0431_F01896	105	■ 1.000	$\mathfrak{J}_{\boldsymbol{C}}$	$\mathfrak{J}_{\boldsymbol{C}}$	For convenience, we denote $\mathfrak{J}(3, \boldsymbol{C})$ and $\{X \in \mathfrak{J}(3, \boldsymbol{C}) \mid \text{tr}(X) = 0\}$ by $\mathfrak{J}_{\boldsymbol{C}}$ and $(\mathfrak{J}_{\boldsymbol{C}})_0$,
0902.0431_F01897	105	■ 1.000	$\left(\mathfrak{J}_{\boldsymbol{C}}\right)_0$	$(\mathfrak{J}_{\boldsymbol{C}})_0$	For convenience, we denote $\mathfrak{J}(3, \boldsymbol{C})$ and $\{X \in \mathfrak{J}(3, \boldsymbol{C}) \mid \text{tr}(X) = 0\}$ by $\mathfrak{J}_{\boldsymbol{C}}$ and $(\mathfrak{J}_{\boldsymbol{C}})_0$,
0902.0431_F01898	105	■ 1.000	$E_{6, \boldsymbol{C}}$	$E_{6, \boldsymbol{C}}$	The group $E_{6, \boldsymbol{C}}$ is defined to be obtained by replacing $\mathfrak{J}^C$ with $(\mathfrak{J}_{\boldsymbol{C}})^C$ in the
0902.0431_F01899	105	■ 1.000	$\left(\mathfrak{J}_{\boldsymbol{C}}\right)^C$	$(\mathfrak{J}_{\boldsymbol{C}})^C$	The group $E_{6, \boldsymbol{C}}$ is defined to be obtained by replacing $\mathfrak{J}^C$ with $(\mathfrak{J}_{\boldsymbol{C}})^C$ in the
0902.0431_F01900	105	■ 1.000	$F_{4, \boldsymbol{C}}=\left\{\alpha \in \operatorname{Iso}_{\boldsymbol{R}}\left(\mathfrak{J}_{\boldsymbol{C}}\right) \mid \alpha(X \circ Y)=\alpha X \circ \alpha Y\right\}$	$F_{4, \boldsymbol{C}} = \{\alpha \in \text{Iso}_{\boldsymbol{R}}(\mathfrak{J}_{\boldsymbol{C}}) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y\}$	which is also defined by the group $F_{4, \boldsymbol{C}} = \{\alpha \in \text{Iso}_{\boldsymbol{R}}(\mathfrak{J}_{\boldsymbol{C}}) \mid \alpha(X \circ Y) = \alpha X \circ \alpha Y\}$,
0902.0431_F01901	105	■ 0.998	$\left(S U(3) / \boldsymbol{Z}_3\right) \cdot \boldsymbol{Z}_2$	$(SU(3) / \boldsymbol{Z}_3) \cdot \boldsymbol{Z}_2$	moreover, it is isomorphic to the group $(SU(3) / \boldsymbol{Z}_3) \cdot \boldsymbol{Z}_2$ (Proposition 2.12.1). The

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0902.0431_F01902	105	■ 1.000	$\mathrm{e}_{6,C}$	$\mathfrak{e}_{6,C}$	Lie algebra $\mathfrak{e}_{6,C}$ of the group $E_{6,C}$ is
0902.0431_F01903	105	■ 1.000	$\mathfrak{f}_{4,C} = \{\delta \in \mathfrak{e}_{6,C} \mid \delta E = 0\}$	$\mathfrak{f}_{4,C} = \{\delta \in \mathfrak{e}_{6,C} \mid \delta E = 0\}$	(Theorem 3.2.4), where $\mathfrak{f}_{4,C} = \{\delta \in \mathfrak{e}_{6,C} \mid \delta E = 0\}$ is the Lie algebra of the group
0902.0431_F01904	106	■ 1.000	$\pi_0(F_{4,C}) \rightarrow \pi_0(E_{6,C}) \rightarrow \pi_0(EIV_C)$	$\pi_0(F_{4,C}) \rightarrow \pi_0(E_{6,C}) \rightarrow \pi_0(EIV_C)$	Proof. From the exact sequence $\pi_0(F_{4,C}) \rightarrow \pi_0(E_{6,C}) \rightarrow \pi_0(EIV_C)$, that is,
0902.0431_F01905	106	■ 1.000	$Z_2 \rightarrow \pi_0(E_{6,C}) \rightarrow 0$	$Z_2 \rightarrow \pi_0(E_{6,C}) \rightarrow 0$	$Z_2 \rightarrow \pi_0(E_{6,C}) \rightarrow 0$ (Proposition 2.12.1), we see that $\pi_0(E_{6,C})$ is 0 or Z_2 .
0902.0431_F01906	106	■ 1.000	$\pi_0(E_{6,C})$	$\pi_0(E_{6,C})$	$Z_2 \rightarrow \pi_0(E_{6,C}) \rightarrow 0$ (Proposition 2.12.1), we see that $\pi_0(E_{6,C})$ is 0 or Z_2 .
0902.0431_F01907	106	■ 1.000	$h: C \oplus C \rightarrow C^C$	$h: C \oplus C \rightarrow C^C$	We define mappings $h: C \oplus C \rightarrow C^C$ and $h: M(3, C) \oplus M(3, C) \rightarrow M(3, C)^C$
0902.0431_F01908	106	■ 1.000	$h: M(3, C) \oplus M(3, C) \rightarrow M(3, C)^C$	$h: M(3, C) \oplus M(3, C) \rightarrow M(3, C)^C$	We define mappings $h: C \oplus C \rightarrow C^C$ and $h: M(3, C) \oplus M(3, C) \rightarrow M(3, C)^C$
0902.0431_F01909	106	■ 1.000	$h: M(3, C) \oplus M(3, C) \rightarrow$	$h: M(3, C) \oplus M(3, C) \rightarrow$	Lemma 3.13.2. The mappings $h: C \oplus C \rightarrow C^C$ and $h: M(3, C) \oplus M(3, C) \rightarrow$
0902.0431_F01910	106	■ 1.000	$M(3, C)^C$	$M(3, C)^C$	$M(3, C)^C$ satisfy the following four conditions.
0902.0431_F01911	106	■ 0.999	$h(a, b)h(a', b') = h(aa', bb'), h(A, B)h(A', B') = h(AA', BB')$	$h(a, b)h(a', b') = h(aa', bb'), h(A, B)h(A', B') = h(AA', BB')$	(2) $h(a, b)h(a', b') = h(aa', bb'), h(A, B)h(A', B') = h(AA', BB')$.
0902.0431_F01912	106	■ 0.989	$\tau h(a, b) = h(b, a), \overline{h(a, b)} = h(\bar{b}, \bar{a})$	$\tau h(a, b) = h(b, a), \overline{h(a, b)} = h(\bar{b}, \bar{a})$	(3) $\tau h(a, b) = h(b, a), h(a, b) = h(\bar{b}, \bar{a})$.
0902.0431_F01913	106	■ 1.000	$\det(h(A, B)) = h(\det A, \det B)$	$\det(h(A, B)) = h(\det A, \det B)$	(4) $\det(h(A, B)) = h(\det A, \det B)$.
0902.0431_F01914	106	■ 1.000	$\iota^2 = \iota, \bar{\iota}^2 = \bar{\iota}, \iota + \bar{\iota} = 1$	$\iota^2 = \iota, \bar{\iota}^2 = \bar{\iota}, \iota + \bar{\iota} = 1$	Proof. It is easy to prove, noting that $\iota^2 = \iota, \bar{\iota}^2 = \bar{\iota}, \iota + \bar{\iota} = 1$.
0902.0431_F01915	106	■ 0.956	$\mathfrak{e}_{6,C} \cong \mathfrak{su}(3) \oplus \mathfrak{su}(3)$	$\mathfrak{e}_{6,C} \cong \mathfrak{su}(3) \oplus \mathfrak{su}(3)$	Lemma 3.13.3. $\mathfrak{e}_{6,C} \cong \mathfrak{su}(3) \oplus \mathfrak{su}(3)$.
0902.0431_F01916	106	■ 1.000	$\phi_C: \mathfrak{su}(3) \oplus \mathfrak{su}(3) \rightarrow \mathfrak{e}_{6,C}$	$\phi_C: \mathfrak{su}(3) \oplus \mathfrak{su}(3) \rightarrow \mathfrak{e}_{6,C}$	Proof. The mapping $\phi_C: \mathfrak{su}(3) \oplus \mathfrak{su}(3) \rightarrow \mathfrak{e}_{6,C}$,
0902.0431_F01917	106	■ 0.878	$SU(3) \times SU(3)$	$SU(3) \times SU(3)$	We define an action of the group $Z_2 = \{1, \epsilon\}$ on $SU(3) \times SU(3)$ by
0902.0431_F01918	106	■ 0.990	$(SU(3) \times SU(3)) \cdot Z_2$	$(SU(3) \times SU(3)) \cdot Z_2$	and let $(SU(3) \times SU(3)) \cdot Z_2$ be the semi-direct product of the groups $SU(3) \times SU(3)$
0902.0431_F01919	106	■ 0.990	$SU(3) \times SU(3)$	$SU(3) \times SU(3)$	and let $(SU(3) \times SU(3)) \cdot Z_2$ be the semi-direct product of the groups $SU(3) \times SU(3)$
0902.0431_F01920	107	■ 0.874	$E_{6,C} \cong ((SU(3) \times SU(3))/Z_3) \cdot Z_2, Z_3 = \{(E, E), (\omega_1 E$	$E_{6,C} \cong ((SU(3) \times SU(3))/Z_3) \cdot Z_2, Z_3 = \{(E, E), (\omega_1 E$	Proposition 3.13.4. $E_{6,C} \cong ((SU(3) \times SU(3))/Z_3) \cdot Z_2, Z_3 = \{(E, E), (\omega_1 E,$
0902.0431_F01921	107	■ 0.763	$(\omega_1 E), (\omega_1^2 E, \omega_1^2 E)\}, \omega_1 = -\frac{1}{2} + \frac{\sqrt{3}}{2} e_1$	$(\omega_1 E), (\omega_1^2 E, \omega_1^2 E)\}, \omega_1 = -\frac{1}{2} + \frac{\sqrt{3}}{2} e_1$	$(\omega_1 E), (\omega_1^2 E, \omega_1^2 E)\}, \omega_1 = -\frac{1}{2} + \frac{\sqrt{3}}{2} e_1$.
0902.0431_F01922	107	■ 1.000	$\varphi: (SU(3) \times SU(3)) \cdot Z_2 \rightarrow E_{6,C}$	$\varphi: (SU(3) \times SU(3)) \cdot Z_2 \rightarrow E_{6,C}$	Proof. We define a mapping $\varphi: (SU(3) \times SU(3)) \cdot Z_2 \rightarrow E_{6,C}$ by
0902.0431_F01923	107	■ 0.975	$\alpha = \varphi((A, B), 1) \in E_{6,C}$	$\alpha = \varphi((A, B), 1) \in E_{6,C}$	First we have to show that $\alpha = \varphi((A, B), 1) \in E_{6,C}$. Using $\det(h(A, B)) = h(\det A,$
0902.0431_F01924	107	■ 0.975	$\det(h(A, B)) = h(\det A$	$\det(h(A, B)) = h(\det A$	First we have to show that $\alpha = \varphi((A, B), 1) \in E_{6,C}$. Using $\det(h(A, B)) = h(\det A,$
0902.0431_F01925	107	■ 0.862	$\det B)(\text{Lemma 3.13.2.(4)}) = h(1, 1) = 1$	$\det B)(\text{Lemma 3.13.2.(4)}) = h(1, 1) = 1$	$\det B)(\text{Lemma 3.13.2.(4)}) = h(1, 1) = 1$ and $\tau h(A, B)^* h(A, B) = h(A^*, B^*) h(A, B)$
0902.0431_F01926	107	■ 0.862	$\tau h(A, B)^* h(A, B) = h(A^*, B^*) h(A, B)$	$\tau h(A, B)^* h(A, B) = h(A^*, B^*) h(A, B)$	$\det B)(\text{Lemma 3.13.2.(4)}) = h(1, 1) = 1$ and $\tau h(A, B)^* h(A, B) = h(A^*, B^*) h(A, B)$
0902.0431_F01927	107	■ 0.963	$= h(A^* A, B^* B) = h(E, E) = E$	$= h(A^* A, B^* B) = h(E, E) = E$	(Lemma 3.13.2.(3)) $= h(A^* A, B^* B) = h(E, E) = E$, we have
0902.0431_F01928	107	■ 1.000	$\alpha \in E_{6,C}$	$\alpha \in E_{6,C}$	Hence $\alpha \in E_{6,C}$. Since $\varphi((E, E), \epsilon) = \epsilon \in G_{2,C} (= \text{Aut}(C)) \subset F_{4,C} \subset E_{6,C}$, we also

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0902.0431_F01929	107	■ 1.000	$\varphi((E, E), \epsilon) = \epsilon \in G_{2,C} (= \text{Aut}(C)) \subset F_{4,C} \subset E_{6,C}$		Hence $\alpha \in E_{6,C}$. Since $\varphi((E, E), \epsilon) = \epsilon \in G_{2,C} (= \text{Aut}(C)) \subset F_{4,C} \subset E_{6,C}$, we also
0902.0431_F01930	107	■ 0.997	$\varphi((A, B), \epsilon) = \varphi((A, B), 1) \varphi((E, E), \epsilon) \in E_{6,C}$		have $\varphi((A, B), \epsilon) = \varphi((A, B), 1) \varphi((E, E), \epsilon) \in E_{6,C}$. Next, we shall show that φ is a
0902.0431_F01931	107	■ 1.000	$\text{Ker } \varphi = \{(E, E), (\omega_1 E, \omega_1 E), (\omega_1^2 E, \omega_1^2 E)\} \times 1 = \mathbf{Z}_3 \times 1$		That $\text{Ker } \varphi = \{(E, E), (\omega_1 E, \omega_1 E), (\omega_1^2 E, \omega_1^2 E)\} \times 1 = \mathbf{Z}_3 \times 1$ can be easily ob-
0902.0431_F01932	107	■ 1.000	$\text{Ker } \varphi$	$\text{Ker } \varphi$	tained. In particular, $\text{Ker } \varphi$ is discrete. Hence φ induces an injective homomorphism
0902.0431_F01933	107	■ 0.725	$\dim(\mathfrak{su}(3) \oplus \mathfrak{su}(3)) = \dim(\mathfrak{e}_{6,C})$		In particular, $\dim(\mathfrak{su}(3) \oplus \mathfrak{su}(3)) = \dim(\mathfrak{e}_{6,C})$, so φ_* is an isomorphism (φ_* coincides
0902.0431_F01934	107	■ 0.725	$\left(\varphi_*\right)$	$(\varphi_*$	In particular, $\dim(\mathfrak{su}(3) \oplus \mathfrak{su}(3)) = \dim(\mathfrak{e}_{6,C})$, so φ_* is an isomorphism (φ_* coincides
0902.0431_F01935	107	■ 0.924	ϕ_C	ϕ_C	with ϕ_C of Lemma 3.13.3). Hence, φ induces the surjection $\varphi : SU(3) \times SU(3) \rightarrow$
0902.0431_F01936	107	■ 0.924	$\varphi : SU(3) \times SU(3) \rightarrow$	$\varphi : SU(3) \times SU(3) \rightarrow$	with ϕ_C of Lemma 3.13.3). Hence, φ induces the surjection $\varphi : SU(3) \times SU(3) \rightarrow$
0902.0431_F01937	107	■ 1.000	$(E_{6,C})_0$	$(E_{6,C})_0$	$(E_{6,C})_0$ (which denotes the connected component of $E_{6,C}$ containing the identity 1).
0902.0431_F01938	107	■ 0.993	$\epsilon = \varphi((E, E), \epsilon) \notin (E_{6,C})_0$	$\epsilon = \varphi((E, E), \epsilon) \notin (E_{6,C})_0$	However $\epsilon = \varphi((E, E), \epsilon) \notin (E_{6,C})_0$. Certainly, for any $A, B \in SU(3)$,
0902.0431_F01939	107	■ 0.993	$A, B \in SU(3)$	$A, B \in SU(3)$	However $\epsilon = \varphi((E, E), \epsilon) \notin (E_{6,C})_0$. Certainly, for any $A, B \in SU(3)$,
0902.0431_F01940	107	■ 1.000	$((SU(3) \times SU(3))/\mathbf{Z}_3) \cdot \mathbf{Z}_2 \cong E_{6,C}$	$((SU(3) \times SU(3))/\mathbf{Z}_3) \cdot \mathbf{Z}_2 \cong E_{6,C}$	$((SU(3) \times SU(3))/\mathbf{Z}_3) \cdot \mathbf{Z}_2 \cong E_{6,C}$.
0902.0431_F01941	108	■ 0.588	$(E_6)^w \cong (SU(3) \times SU(3) \times SU(3))/\mathbf{Z}_3, \quad \mathbf{Z}_3 = \{(E, E, E)$	$(E_6)^w \cong (SU(3) \times SU(3) \times SU(3))/\mathbf{Z}_3, \quad \mathbf{Z}_3 = \{(E, E, E)$	Theorem 3.13.5. $(E_6)^w \cong (SU(3) \times SU(3) \times SU(3))/\mathbf{Z}_3, \quad \mathbf{Z}_3 = \{(E, E, E)$,
0902.0431_F01942	108	■ 0.929	$(\omega_1 E, \omega_1 E, \omega_1 E), (\omega_1^2 E, \omega_1^2 E, \omega_1^2 E)\}, \omega_1 = -\frac{1}{2} + \frac{\sqrt{3}}{2} e_1$	$(\omega_1 E, \omega_1 E, \omega_1 E), (\omega_1^2 E, \omega_1^2 E, \omega_1^2 E)\}, \omega_1 = -\frac{1}{2} + \frac{\sqrt{3}}{2} e_1$	$(\omega_1 E, \omega_1 E, \omega_1 E), (\omega_1^2 E, \omega_1^2 E, \omega_1^2 E)\}, \omega_1 = -\frac{1}{2} + \frac{\sqrt{3}}{2} e_1$.
0902.0431_F01943	108	■ 0.917	$\varphi : SU(3) \times SU(3) \times SU(3) \rightarrow (E_6)^w$	$\varphi : SU(3) \times SU(3) \times SU(3) \rightarrow (E_6)^w$	Proof. We define a mapping $\varphi : SU(3) \times SU(3) \times SU(3) \rightarrow (E_6)^w$ by
0902.0431_F01944	108	■ 1.000	$\alpha = \varphi(P, A, B) \in (E_6)^w$	$\alpha = \varphi(P, A, B) \in (E_6)^w$	We first have to show that $\alpha = \varphi(P, A, B) \in (E_6)^w$, that is, α leaves
0902.0431_F01945	108	■ 1.000	$X + M, Y + N \in (\mathfrak{J}_C)^C \oplus M(3, C)^C = \mathfrak{J}^C$	$X + M, Y + N \in (\mathfrak{J}_C)^C \oplus M(3, C)^C = \mathfrak{J}^C$	Now, for $X + M, Y + N \in (\mathfrak{J}_C)^C \oplus M(3, C)^C = \mathfrak{J}^C$, the relations
0902.0431_F01946	108	■ 1.000	$w\alpha = \alpha w$	$w\alpha = \alpha w$	and so $\alpha \in E_6$. Clearly, $w\alpha = \alpha w$, so that $\alpha \in (E_6)^w$. Evidently φ is a homomor-
0902.0431_F01947	108	■ 1.000	$\alpha \in (E_6)^w$	$\alpha \in (E_6)^w$	and so $\alpha \in E_6$. Clearly, $w\alpha = \alpha w$, so that $\alpha \in (E_6)^w$. Evidently φ is a homomor-
0902.0431_F01948	108	■ 1.000	$(\mathfrak{J}^C)_w = \{X \in \mathfrak{J}^C \mid wX = X\} = (\mathfrak{J}_C)^C$	$(\mathfrak{J}^C)_w = \{X \in \mathfrak{J}^C \mid wX = X\} = (\mathfrak{J}_C)^C$	to $(\mathfrak{J}^C)_w = \{X \in \mathfrak{J}^C \mid wX = X\} = (\mathfrak{J}_C)^C$ belongs to $E_{6,C}$: $\alpha' \in E_{6,C}$. Hence, there
0902.0431_F01949	108	■ 1.000	$E_{6,C} : \alpha' \in E_{6,C}$	$E_{6,C} : \alpha' \in E_{6,C}$	to $(\mathfrak{J}^C)_w = \{X \in \mathfrak{J}^C \mid wX = X\} = (\mathfrak{J}_C)^C$ belongs to $E_{6,C}$: $\alpha' \in E_{6,C}$. Hence, there
0902.0431_F01950	108	■ 1.000	$\beta = \varphi(E, A, B)^{-1} \alpha$	$\beta = \varphi(E, A, B)^{-1} \alpha$	(Proposition 3.13.4). In the former case, let $\beta = \varphi(E, A, B)^{-1} \alpha$, then $\beta (\mathfrak{J}_C)^C = 1$.
0902.0431_F01951	108	■ 1.000	$\beta (\mathfrak{J}_C)^C = 1$	$\beta (\mathfrak{J}_C)^C = 1$	(Proposition 3.13.4). In the former case, let $\beta = \varphi(E, A, B)^{-1} \alpha$, then $\beta (\mathfrak{J}_C)^C = 1$.
0902.0431_F01952	109	■ 1.000	$\gamma_1 : \mathfrak{J}^C \rightarrow \mathfrak{J}^C, \gamma_1(X + M) = \bar{X} + \bar{M}, X +$	$\gamma_1 : \mathfrak{J}^C \rightarrow \mathfrak{J}^C, \gamma_1(X + M) = \bar{X} + \bar{M}, X +$	In the latter case, consider the mapping $\gamma_1 : \mathfrak{J}^C \rightarrow \mathfrak{J}^C, \gamma_1(X + M) = \bar{X} + \bar{M}, X +$
0902.0431_F01953	109	■ 1.000	$M \in (\mathfrak{J}_C)^C \oplus M(3, C)^C = \mathfrak{J}^C$	$M \in (\mathfrak{J}_C)^C \oplus M(3, C)^C = \mathfrak{J}^C$	$M \in (\mathfrak{J}_C)^C \oplus M(3, C)^C = \mathfrak{J}^C$ and remember that $\gamma_1 \in G_2 \subset F_4 \subset E_6$. Let
0902.0431_F01954	109	■ 1.000	$\gamma_1 \in G_2 \subset F_4 \subset E_6$	$\gamma_1 \in G_2 \subset F_4 \subset E_6$	$M \in (\mathfrak{J}_C)^C \oplus M(3, C)^C = \mathfrak{J}^C$ and remember that $\gamma_1 \in G_2 \subset F_4 \subset E_6$. Let
0902.0431_F01955	109	■ 1.000	$\beta = \alpha^{-1} \varphi(E, A, B) \gamma_1$	$\beta = \alpha^{-1} \varphi(E, A, B) \gamma_1$	$\beta = \alpha^{-1} \varphi(E, A, B) \gamma_1$, then $\beta \in E_6$ and $\beta (\mathfrak{J}_C)^C = 1$. Hence $\beta \in (G_2)_{e_1} = (G_2)^w$
0902.0431_F01956	109	■ 1.000	$\beta \in E_6$	$\beta \in E_6$	$\beta = \alpha^{-1} \varphi(E, A, B) \gamma_1$, then $\beta \in E_6$ and $\beta (\mathfrak{J}_C)^C = 1$. Hence $\beta \in (G_2)_{e_1} = (G_2)^w$

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0902.0431_F01957	109	■1.000	$\subset\left(E_6\right)^w$	$\subset\left(E_6\right)^w$	(Theorem 1.9.4) $\subset\left(E_6\right)^w$. Since α and $\varphi(E, A, B) \in\left(E_6\right)^w$, we have $\gamma_1 \in\left(E_6\right)^w$, so
0902.0431_F01958	109	■1.000	$\varphi(E, A, B) \in\left(E_6\right)^w$	$\varphi(E, A, B) \in\left(E_6\right)^w$	(Theorem 1.9.4) $\subset\left(E_6\right)^w$. Since α and $\varphi(E, A, B) \in\left(E_6\right)^w$, we have $\gamma_1 \in\left(E_6\right)^w$, so
0902.0431_F01959	109	■1.000	$\gamma_1 \in\left(E_6\right)^w$	$\gamma_1 \in\left(E_6\right)^w$	(Theorem 1.9.4) $\subset\left(E_6\right)^w$. Since α and $\varphi(E, A, B) \in\left(E_6\right)^w$, we have $\gamma_1 \in\left(E_6\right)^w$, so
0902.0431_F01960	109	■0.837	$\operatorname{Ker} \varphi=\left\{\left(E, E, E\right),\left(\omega_1 E, \omega_1 E, \omega_1 E\right),\left(\omega_1^2 E, \omega_1^2 E, \omega_1^2 E\right)\right\}=\mathbf{Z}_3$	$\operatorname{Ker} \varphi=\left\{\left(E, E, E\right),\left(\omega_1 E, \omega_1 E, \omega_1 E\right),\left(\omega_1^2 E, \omega_1^2 E, \omega_1^2 E\right)\right\}=\mathbf{Z}_3$	is onto. That $\operatorname{Ker} \varphi=\left\{\left(E, E, E\right),\left(\omega_1 E, \omega_1 E, \omega_1 E\right),\left(\omega_1^2 E, \omega_1^2 E, \omega_1^2 E\right)\right\}=\mathbf{Z}_3$ can
0902.0431_F01961	109	■0.996	$\left(S U(3) \times S U(3) \times S U(3)\right) /\boldsymbol{\mathrm{Z}}_3 \cong$	$\left(S U(3) \times S U(3) \times S U(3)\right) /\mathbf{Z}_3 \cong$	be easily obtained. Thus we have the isomorphism $\left(S U(3) \times S U(3) \times S U(3)\right) /\mathbf{Z}_3 \cong$
0902.0431_F01962	109	■0.998	$S U(3) \times S U(3) \times S U(3) \rightarrow\left(E_6\right)^w$	$S U(3) \times S U(3) \times S U(3) \rightarrow\left(E_6\right)^w$	$S U(3) \times S U(3) \times S U(3) \rightarrow\left(E_6\right)^w$ is onto can be proved as follows. The elements
0902.0431_F01963	109	■0.997	$\left(\mathfrak{e}_6\right)^w$	$\left(\mathfrak{e}_6\right)^w$	forms an $\mathbf{R}$ -basis of $\left(\mathfrak{e}_6\right)^w$. So $\dim\left(\left(\mathfrak{e}_6\right)^w\right)=16+8=24=8+8+8=\dim(\mathfrak{su}(3) \oplus$
0902.0431_F01964	109	■0.997	$\operatorname{dim}\left(\left(\mathfrak{e}_6\right)^w\right)=16+8=24=8+8+8=\operatorname{dim}(\mathfrak{su}(3) \oplus$	$\operatorname{dim}\left(\left(\mathfrak{e}_6\right)^w\right)=16+8=24=8+8+8=\dim(\mathfrak{su}(3) \oplus$	forms an $\mathbf{R}$ -basis of $\left(\mathfrak{e}_6\right)^w$. So $\dim\left(\left(\mathfrak{e}_6\right)^w\right)=16+8=24=8+8+8=\dim(\mathfrak{su}(3) \oplus$
0902.0431_F01965	109	■0.971	$\mathfrak{su}(3) \oplus \mathfrak{su}(3))$	$\mathfrak{su}(3) \oplus \mathfrak{su}(3))$	$\mathfrak{su}(3) \oplus \mathfrak{su}(3)$, Hence φ is onto.
0902.0431_F01966	109	■1.000	$\left(\left(S U(3) \times S U(3) \times S U(3)\right) /\boldsymbol{\mathrm{Z}}_3\right) \cdot \mathbf{Z}_2$	$\left(\left(S U(3) \times S U(3) \times S U(3)\right) /\mathbf{Z}_3\right) \cdot \mathbf{Z}_2$	$\left(\left(S U(3) \times S U(3) \times S U(3)\right) /\mathbf{Z}_3\right) \cdot \mathbf{Z}_2$ which is the semi-direct product of the groups
0902.0431_F01967	109	■1.000	$S U(3) \times S U(3) \times S U(3)$	$S U(3) \times S U(3) \times S U(3)$	$S U(3) \times S U(3) \times S U(3)$ is $\gamma_1(P, A, B)=\left(\overline{P}, \overline{B}, \overline{A}\right)$.
0902.0431_F01968	109	■1.000	$\gamma_1(P, A, B)=\left(\bar{P}, \bar{B}, \bar{A}\right)$	$\gamma_1(P, A, B)=\left(\bar{P}, \bar{B}, \bar{A}\right)$	$S U(3) \times S U(3) \times S U(3)$ is $\gamma_1(P, A, B)=\left(\bar{P}, \bar{B}, \bar{A}\right)$.
0902.0431_F01969	109	■0.999	α^*	α^*	conjugate transposed mapping α^* of $\alpha \in E_6^C$ with respect to the inner product
0902.0431_F01970	109	■0.999	$\alpha \in E_6^C$	$\alpha \in E_6^C$	conjugate transposed mapping α^* of $\alpha \in E_6^C$ with respect to the inner product
0902.0431_F01971	109	■1.000	$\alpha^*=\tau^t \alpha \tau \in E_6^C$	$\alpha^*=\tau^t \alpha \tau \in E_6^C$	$\langle X, Y\rangle$ is $\alpha^*=\tau^t \alpha \tau \in E_6^C$. Hence, from Chevalley's lemma, we have
0902.0431_F01972	110	■1.000	$\langle X, Y\rangle_\gamma$	$\langle X, Y\rangle_\gamma$	We define Hermitian inner products $\langle X, Y\rangle_\gamma$ and $\langle X, Y\rangle_\sigma$ in $\mathfrak{J}(3, \mathfrak{C}^C)$ respectively
0902.0431_F01973	110	■1.000	$\langle X, Y\rangle_\sigma$	$\langle X, Y\rangle_\sigma$	We define Hermitian inner products $\langle X, Y\rangle_\gamma$ and $\langle X, Y\rangle_\sigma$ in $\mathfrak{J}(3, \mathfrak{C}^C)$ respectively
0902.0431_F01974	110	■1.000	$\mathfrak{J}(3, \mathfrak{C}^C)$	$\mathfrak{J}(3, \mathfrak{C}^C)$	We define Hermitian inner products $\langle X, Y\rangle_\gamma$ and $\langle X, Y\rangle_\sigma$ in $\mathfrak{J}(3, \mathfrak{C}^C)$ respectively
0902.0431_F01975	111	■0.906	$\left(\begin{array}{l} X \\ Y \\ \xi \\ \eta \end{array}\right)$	$\left(\begin{array}{l} X \\ Y \\ \xi \\ \eta \end{array}\right)$	$\mathfrak{P}^C$ is a 56 dimensional C -vector space. An element $\left(\begin{array}{l} X \\ Y \\ \xi \\ \eta \end{array}\right)$ of $\mathfrak{P}^C$ is often denoted by
0902.0431_F01976	111	■0.996	(X, Y, ξ, η)	(X, Y, ξ, η)	(X, Y, ξ, η) or sometimes by $\dot{X}+Y+\dot{\xi}+\eta$. In $\mathfrak{P}^C$, we define an inner product (P, Q) , a
0902.0431_F01977	111	■0.996	$\dot{X}+Y+\dot{\xi}+\eta$	$\dot{X}+Y+\dot{\xi}+\eta$	(X, Y, ξ, η) or sometimes by $\dot{X}+Y+\dot{\xi}+\eta$. In $\mathfrak{P}^C$, we define an inner product (P, Q) , a
0902.0431_F01978	111	■1.000	$\{P, Q\}$	$\{P, Q\}$	$\{P, Q\}$ respectively by
0902.0431_F01979	111	■1.000	$P=(X, Y, \xi, \eta), Q=(Z, W, \zeta, \omega) \in \mathfrak{P}^C$	$P=(X, Y, \xi, \eta), Q=(Z, W, \zeta, \omega) \in \mathfrak{P}^C$	where $P=(X, Y, \xi, \eta), Q=(Z, W, \zeta, \omega) \in \mathfrak{P}^C$.
0902.0431_F01980	111	■0.977	$\phi \in \mathfrak{e}_6^C, A, B \in \mathfrak{J}^C, \nu \in C$	$\phi \in \mathfrak{e}_6^C, A, B \in \mathfrak{J}^C, \nu \in C$	For $\phi \in \mathfrak{e}_6^C, A, B \in \mathfrak{J}^C, \nu \in C$, we define a C -linear mapping $\Phi(\phi, A, B, \nu):$
0902.0431_F01981	111	■0.977	$\Phi(\phi, A, B, \nu)$	$\Phi(\phi, A, B, \nu)$	For $\phi \in \mathfrak{e}_6^C, A, B \in \mathfrak{J}^C, \nu \in C$, we define a C -linear mapping $\Phi(\phi, A, B, \nu):$
0902.0431_F01982	111	■1.000	$\mathfrak{P}^C \rightarrow \mathfrak{P}^C$	$\mathfrak{P}^C \rightarrow \mathfrak{P}^C$	$\mathfrak{P}^C \rightarrow \mathfrak{P}^C$ by
0902.0431_F01983	111	■1.000	$P \times Q: \mathfrak{P}^C \rightarrow \mathfrak{P}^C$	$P \times Q: \mathfrak{P}^C \rightarrow \mathfrak{P}^C$	mapping $P \times Q: \mathfrak{P}^C \rightarrow \mathfrak{P}^C$ by
0902.0431_F01984	112	■1.000	$P, Q, R \in \mathfrak{P}^C$	$P, Q, R \in \mathfrak{P}^C$	Lemma 4.1.1. For $P, Q, R \in \mathfrak{P}^C$, we have
0902.0431_F01985	112	■1.000	$P \times Q=Q \times P$	$P \times Q=Q \times P$	(1) $P \times Q=Q \times P$.
0902.0431_F01986	112	■1.000	$(P \times Q) P-(P \times P) Q+\frac{3}{8}\{P, Q\} P=0$	$(P \times Q) P-(P \times P) Q+\frac{3}{8}\{P, Q\} P=0$	(2) $(P \times Q) P-(P \times P) Q+\frac{3}{8}\{P, Q\} P=0$.
0902.0431_F01987	112	■1.000	$(P \times R) Q-(Q \times R) P+\frac{1}{8}\{Q, R\} P-\frac{1}{8}\{P, R\} Q-\frac{1}{4}\{P, Q\} R=0$	$(P \times R) Q-(Q \times R) P+\frac{1}{8}\{Q, R\} P-\frac{1}{8}\{P, R\} Q-\frac{1}{4}\{P, Q\} R=0$	(3) $(P \times R) Q-(Q \times R) P+\frac{1}{8}\{Q, R\} P-\frac{1}{8}\{P, R\} Q-\frac{1}{4}\{P, Q\} R=0$.
0902.0431_F01988	112	■0.910	$=\cdots$	$=\cdots$	$=\cdots$ (using the formula $X \vee Y$ of Lemma 3.4.1), etc.) $\cdots$

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0902.0431_F01989	112	■ 0.910	$X \vee Y$	$X \vee Y$	$= \cdots$ (using the formula $X \vee Y$ of Lemma 3.4.1), etc.) $\cdots$
0902.0431_F01990	112	■ 1.000	$P+R$	$P + R$	(3) In the equality (2), put $\overline{P+R}$ in the place of P , then we have
0902.0431_F01991	112	■ 1.000	Q	Q	Exchanging P with Q , we see that
0902.0431_F01992	112	■ 0.786	$((\mathrm{i})-(\mathrm{ii}))\,\div 3$	$((\mathrm{i})-(\mathrm{ii}))\div 3$	Taking $((\mathrm{i})-(\mathrm{ii}))\div 3$, we have
0902.0431_F01993	112	■ 1.000	$\mathfrak{M}^C$	$\mathfrak{M}^C$	We define a space $\mathfrak{M}^C$, called the complex Freudenthal manifold, by
0902.0431_F01994	112	■ 1.000	$\xi \neq 0, \eta \neq 0$	$\xi \neq 0, \eta \neq 0$	Lemma 4.1.2. <i>The following elements (assuming $\xi \neq 0, \eta \neq 0$) of $\mathfrak{P}^C$</i>
0902.0431_F01995	113	■ 1.000	$X \vee (X \times X) = 0$	$X \vee (X \times X) = 0$	Proof. This is clear from $\overline{X \vee (X \times X)} = \overline{0}$ (Lemma 3.5.4.(1)) and $(X \times X) \times$
0902.0431_F01996	113	■ 1.000	$(X \times X) \times$	$(X \times X) \times$	Proof. This is clear from $X \vee (X \times X) = 0$ (Lemma 3.5.4.(1)) and $(X \times X) \times$
0902.0431_F01997	113	■ 1.000	$(X \times X) = (\operatorname{det} X) X$	$(X \times X) = (\det X)X$	$\overline{(X \times X)} = (\det X) \overline{X}$ (Lemma 2.1.1.(3)).
0902.0431_F01998	113	■ 0.999	$\alpha \in E_7^C$	$\alpha \in E_7^C$	Proposition 4.2.2. <i>For $\alpha \in E_7^C$ (and so, for $\alpha \in E_7$), we have</i>
0902.0431_F01999	113	■ 0.999	$\alpha \in E_7$	$\alpha \in E_7$	Proposition 4.2.2. <i>For $\alpha \in E_7^C$ (and so, for $\alpha \in E_7$), we have</i>
0902.0431_F02000	113	■ 1.000	$\alpha \mathfrak{M}^C \subset \mathfrak{M}^C$	$\alpha \mathfrak{M}^C \subset \mathfrak{M}^C$	Proof. (1) It is sufficient to prove that $\alpha \mathfrak{M}^C \subset \mathfrak{M}^C$. Now, for $P \in \mathfrak{M}^C$, we have
0902.0431_F02001	113	■ 1.000	$P \in \mathfrak{M}^C$	$P \in \mathfrak{M}^C$	Proof. (1) It is sufficient to prove that $\alpha \mathfrak{M}^C \subset \mathfrak{M}^C$. Now, for $P \in \mathfrak{M}^C$, we have
0902.0431_F02002	113	■ 1.000	$\alpha P \times \alpha P = \alpha(P \times P) \alpha^{-1} = \alpha 0 \alpha^{-1} = 0$	$\alpha P \times \alpha P = \alpha(P \times P) \alpha^{-1} = \alpha 0 \alpha^{-1} = \overline{0}$. Hence $\alpha P \in \mathfrak{M}^C$.	
0902.0431_F02003	113	■ 1.000	$\alpha P \in \mathfrak{M}^C$	$\alpha P \in \mathfrak{M}^C$	$\alpha P \times \alpha P = \alpha(P \times P) \alpha^{-1} = \alpha 0 \alpha^{-1} = 0$. Hence $\alpha P \in \mathfrak{M}^C$.
0902.0431_F02004	113	■ 1.000	$\{\alpha P, \alpha Q\} = \{P, Q\}$	$\{\alpha P, \alpha Q\} = \{P, Q\}$	It follows $\overline{\{\alpha P, \alpha Q\}} = \overline{\{P, Q\}}$.
0902.0431_F02005	113	■ 0.974	$[\Phi_1, \Phi_2]$	$[\Phi_1, \Phi_2]$	The Lie bracket $[\Phi_1, \Phi_2]$ in $\mathfrak{e}_7^C$ is given by
0902.0431_F02006	114	■ 1.000	$\Phi_i \in \mathfrak{e}_7^C$	$\Phi_i \in \mathfrak{e}_7^C$	check the form of the Lie bracket in $\mathfrak{e}_7^C$. For $\overline{\Phi_i} \in \mathfrak{e}_7^C$ and $P \in \mathfrak{P}^C$, we have
0902.0431_F02007	114	■ 1.000	$P \in \mathfrak{P}^C$	$P \in \mathfrak{P}^C$	check the form of the Lie bracket in $\mathfrak{e}_7^C$. For $\Phi_i \in \mathfrak{e}_7^C$ and $P \in \mathfrak{P}^C$, we have
0902.0431_F02008	114	■ 1.000	Φ	Φ	This Φ is that of the theorem.
0902.0431_F02009	114	■ 0.994	$\Phi \in \operatorname{Hom}_C(\mathfrak{P}^C)$	$\Phi \in \operatorname{Hom}_C(\mathfrak{P}^C)$	if $\overline{\Phi} \in \operatorname{Hom}_C(\mathfrak{P}^C)$ belongs to $\mathfrak{e}_7^C$, then Φ satisfies
0902.0431_F02010	114	■ 0.994	$\Phi \in \mathfrak{e}_7^C$	$\Phi \in \mathfrak{e}_7^C$	Since $\overline{\Phi} \in \mathfrak{e}_7^C$ is a C -linear mapping of $\mathfrak{P}^C = \mathfrak{J}^C \oplus \mathfrak{J}^C \oplus C \oplus C$, Φ is of the form
0902.0431_F02011	114	■ 0.994	$\mathfrak{P}^C = \mathfrak{J}^C \oplus \mathfrak{J}^C \oplus C \oplus C, \Phi$	$\mathfrak{P}^C = \mathfrak{J}^C \oplus \mathfrak{J}^C \oplus C \oplus C, \Phi$	Since $\Phi \in \mathfrak{e}_7^C$ is a C -linear mapping of $\overline{\mathfrak{P}^C} = \overline{\mathfrak{J}^C} \oplus \overline{\mathfrak{J}^C} \oplus C \oplus C$, Φ is of the form
0902.0431_F02012	114	■ 1.000	$0 \neq r \in C$	$0 \neq r \in C$	For $\overline{0} \neq \overline{r} \in \overline{C}$, we define a C -linear isomorphism $f_r : \mathfrak{P}^C \rightarrow \mathfrak{P}^C$ by
0902.0431_F02013	114	■ 1.000	$f_r : \mathfrak{P}^C \rightarrow \mathfrak{P}^C$	$f_r : \mathfrak{P}^C \rightarrow \mathfrak{P}^C$	For $0 \neq r \in C$, we define a C -linear isomorphism $\overline{f_r} : \overline{\mathfrak{P}^C} \rightarrow \overline{\mathfrak{P}^C}$ by
0902.0431_F02014	114	■ 1.000	f_r	f_r	then it is easy to see that f_r satisfies
0902.0431_F02015	114	■ 0.985	$f_r \alpha f_r^{-1} \in E_7^C$	$f_r \alpha f_r^{-1} \in E_7^C$	Hence we see that for $\alpha \in E_7^C$ we have $\overline{f_r \alpha f_r^{-1}} \in \overline{E_7^C}$. Therefore, for $\Phi \in \mathfrak{e}_7^C$, we
0902.0431_F02016	115	■ 1.000	$\Phi_{-3}(0, 0, 1, 0) \times (0, 0, 1, 0) = 0$	$\Phi_{-3}(0, 0, 1, 0) \times (0, 0, 1, 0) = 0$	The relation $\overline{\Phi_{-3}(0, 0, 1, 0)} \times \overline{(0, 0, 1, 0)} = \overline{0}$ implies $\kappa = 0$, hence $\Phi_{-3} = 0$. Similarly
0902.0431_F02017	115	■ 1.000	$\kappa = 0$	$\kappa = 0$	The relation $\Phi_{-3}(0, 0, 1, 0) \times (0, 0, 1, 0) = 0$ implies $\kappa = 0$, hence $\Phi_{-3} = 0$. Similarly
0902.0431_F02018	115	■ 1.000	$\Phi_{-3} = 0$	$\Phi_{-3} = 0$	The relation $\Phi_{-3}(0, 0, 1, 0) \times (0, 0, 1, 0) = 0$ implies $\kappa = 0$, hence $\Phi_{-3} = 0$. Similarly
0902.0431_F02019	115	■ 1.000	$\Phi_3 = 0$	$\Phi_3 = 0$	$\Phi_3 = 0$. The relation $\Phi_{-2}(0, 0, 1, 0) \times (0, 0, 1, 0) = 0$ implies $C = 0$. Moreover, the
0902.0431_F02020	115	■ 1.000	$\Phi_{-2}(0, 0, 1, 0) \times (0, 0, 1, 0) = 0$	$\Phi_{-2}(0, 0, 1, 0) \times (0, 0, 1, 0) = 0$	$\Phi_3 = 0$. The relation $\overline{\Phi_{-2}(0, 0, 1, 0)} \times \overline{(0, 0, 1, 0)} = \overline{0}$ implies $C = 0$. Moreover, the
0902.0431_F02021	115	■ 1.000	$C = 0$	$C = 0$	$\Phi_3 = 0$. The relation $\Phi_{-2}(0, 0, 1, 0) \times (0, 0, 1, 0) = 0$ implies $C = 0$. Moreover, the
0902.0431_F02022	115	■ 0.997	$\Phi_{-2} P \times P = 0$	$\Phi_{-2} P \times P = 0$	relation $\Phi_{-2} P \times P = 0$ for $P = (Y \times Y, Y, 1, \det Y) \in \mathfrak{M}^C$, that is, $(0, 0, 0, d(Y)) \times$
0902.0431_F02023	115	■ 0.997	$P = (Y \times Y, Y, 1, \operatorname{det} Y) \in \mathfrak{M}^C$	$P = (Y \times Y, Y, 1, \det Y) \in \mathfrak{M}^C$	relation $\Phi_{-2} P \times P = 0$ for $\overline{P} = (Y \times Y, Y, 1, \det Y) \in \overline{\mathfrak{M}^C}$, that is, $(0, 0, 0, d(Y)) \times$
0902.0431_F02024	115	■ 0.997	$(0, 0, 0, d(Y)) \times$	$(0, 0, 0, d(Y)) \times$	relation $\Phi_{-2} P \times P = 0$ for $P = (Y \times Y, Y, 1, \det Y) \in \mathfrak{M}^C$, that is, $\overline{(0, 0, 0, d(Y))} \times$
0902.0431_F02025	115	■ 1.000	$(Y \times Y, Y, 1, \operatorname{det} Y) = 0$	$(Y \times Y, Y, 1, \det Y) = 0$	$\overline{(Y \times Y, Y, 1, \det Y)} = \overline{0}$ implies $d = 0$. Hence $\Phi_{-2} = 0$. Similarly $\Phi_2 = 0$. The relation
0902.0431_F02026	115	■ 1.000	$d = 0$	$d = 0$	$(Y \times Y, Y, 1, \det Y) = 0$ implies $d = 0$. Hence $\Phi_{-2} = 0$. Similarly $\Phi_2 = 0$. The relation
0902.0431_F02027	115	■ 1.000	$\Phi_{-2} = 0$	$\Phi_{-2} = 0$	$(Y \times Y, Y, 1, \det Y) = 0$ implies $d = 0$. Hence $\overline{\Phi_{-2}} = \overline{0}$. Similarly $\Phi_2 = 0$. The relation
0902.0431_F02028	115	■ 1.000	$\Phi_2 = 0$	$\Phi_2 = 0$	$(Y \times Y, Y, 1, \det Y) = 0$ implies $d = 0$. Hence $\Phi_{-2} = 0$. Similarly $\Phi_2 = 0$. The relation
0902.0431_F02029	115	■ 0.997	$\Phi_{-1} P \times P = 0$	$\Phi_{-1} P \times P = 0$	$\Phi_{-1} P \times P = 0$ for $P = (Y \times Y, Y, 1, \det Y) \in \mathfrak{M}^C$, that is, $(l(Y), B, 0, b(Y \times Y)) \times$
0902.0431_F02030	115	■ 0.997	$(l(Y), B, 0, b(Y \times Y)) \times$	$(l(Y), B, 0, b(Y \times Y)) \times$	$\Phi_{-1} P \times P = 0$ for $P = (Y \times Y, Y, 1, \det Y) \in \mathfrak{M}^C$, that is, $\overline{(l(Y), B, 0, b(Y \times Y))} \times$
0902.0431_F02031	115	■ 1.000	$P = (X, X \times X, \operatorname{det} X, 1) \in \mathfrak{M}^C$	$P = (X, X \times X, \det X, 1) \in \mathfrak{M}^C$	Next, the relation $\Phi_{-1} P \times P = 0$ for $P = (X, X \times X, \det X, 1) \in \mathfrak{M}^C$, that is,

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_F02069	118	■ 1.000	$=\{\tau \lambda P, Q\}$	$=\{\tau \lambda P, Q\}$	$=\{\tau \lambda P, Q\}$ (Proposition 4.2.2.(2)) $=\langle P, Q \rangle$ (Lemma 4.3.3), $P, Q \in \mathfrak{P}^C$. Hence,
0902.0431_F02070	118	■ 1.000	$=\langle P, Q \rangle$	$=\langle P, Q \rangle$	$=\{\tau \lambda P, Q\}$ (Proposition 4.2.2.(2)) $=\langle P, Q \rangle$ (Lemma 4.3.3), $P, Q \in \mathfrak{P}^C$. Hence,
0902.0431_F02071	119	■ 1.000	$\tau \Phi(\phi, A$	$\tau \Phi(\phi, A$	Therefore the theorem follows from Theorem 4.3.1, Lemma 4.3.3.(5) and $\tau \Phi(\phi, A,$
0902.0431_F02072	119	■ 0.817	$B, \nu) \tau = \Phi(\tau \phi \tau, \tau A, \tau B, \tau \nu)$	$B, \nu) \tau = \Phi(\tau \phi \tau, \tau A, \tau B, \tau \nu)$	$\tau B, \nu) \tau = \Phi(\tau \phi \tau, \tau A, \tau B, \tau \nu)$.
0902.0431_F02073	119	■ 0.998	Φ^*	Φ^*	Proof. For $\Phi \in \mathfrak{e}_7^C$, the conjugate transposed mapping Φ^* of Φ with respect to
0902.0431_F02074	119	■ 0.993	$\Phi^* = \tau \lambda \Phi \lambda \tau \in \mathfrak{e}_7^C$	$\Phi^* = \tau \lambda \Phi \lambda \tau \in \mathfrak{e}_7^C$	the inner product $\langle P, Q \rangle$ of $\mathfrak{P}^C$ is $\Phi^* = \tau \lambda \Phi \lambda \tau \in \mathfrak{e}_7^C$, and for $\Phi \in \mathfrak{e}_7^C$, Φ belongs to
0902.0431_F02075	119	■ 0.993	$\Phi \in \mathfrak{e}_7^C, \Phi$	$\Phi \in \mathfrak{e}_7^C, \Phi$	the inner product $\langle P, Q \rangle$ of $\mathfrak{P}^C$ is $\Phi^* = \tau \lambda \Phi \lambda \tau \in \mathfrak{e}_7^C$, and for $\Phi \in \mathfrak{e}_7^C$, Φ belongs to
0902.0431_F02076	119	■ 0.822	$\Phi^* = -\Phi$	$\Phi^* = -\Phi$	$\mathfrak{e}_7$ if and only if $\Phi^* = -\Phi$. Now, any element $\Phi \in \mathfrak{e}_7^C$ is represented as
0902.0431_F02077	119	■ 0.999	$\mathfrak{e}_6^C = \{\Phi(\phi, 0, 0, 0) \in \mathfrak{e}_7^C \mid \phi \in \mathfrak{e}_6^C\}$	$\mathfrak{e}_6^C = \{\Phi(\phi, 0, 0, 0) \in \mathfrak{e}_7^C \mid \phi \in \mathfrak{e}_6^C\}$	where $\mathfrak{e}_6^C = \{\Phi(\phi, 0, 0, 0) \in \mathfrak{e}_7^C \mid \phi \in \mathfrak{e}_6^C\}$ and $\mathfrak{N}^C = \{\Phi(0, A, B, \nu) \in \mathfrak{e}_7^C \mid A, B \in$
0902.0431_F02078	119	■ 0.999	$\mathfrak{N}^C = \{\Phi(0, A, B, \nu) \in \mathfrak{e}_7^C \mid A, B \in$	$\mathfrak{N}^C = \{\Phi(0, A, B, \nu) \in \mathfrak{e}_7^C \mid A, B \in$	where $\mathfrak{e}_6^C = \{\Phi(\phi, 0, 0, 0) \in \mathfrak{e}_7^C \mid \phi \in \mathfrak{e}_6^C\}$ and $\mathfrak{N}^C = \{\Phi(0, A, B, \nu) \in \mathfrak{e}_7^C \mid A, B \in$
0902.0431_F02079	119	■ 1.000	$\mathfrak{J}^C, \nu \in C\}$	$\mathfrak{J}^C, \nu \in C\}$	$\mathfrak{J}^C, \nu \in C\}$. Let $p : \mathfrak{e}_7^C \rightarrow \mathfrak{e}_6^C$ and $q : \mathfrak{e}_7^C \rightarrow \mathfrak{N}^C$ be projections of $\mathfrak{e}_7^C = \mathfrak{e}_6^C \oplus \mathfrak{N}^C$.
0902.0431_F02080	119	■ 1.000	$p : \mathfrak{e}_7^C \rightarrow \mathfrak{e}_6^C$	$p : \mathfrak{e}_7^C \rightarrow \mathfrak{e}_6^C$	$\mathfrak{J}^C, \nu \in C\}$. Let $p : \mathfrak{e}_7^C \rightarrow \mathfrak{e}_6^C$ and $q : \mathfrak{e}_7^C \rightarrow \mathfrak{N}^C$ be projections of $\mathfrak{e}_7^C = \mathfrak{e}_6^C \oplus \mathfrak{N}^C$.
0902.0431_F02081	119	■ 1.000	$q : \mathfrak{e}_7^C \rightarrow \mathfrak{N}^C$	$q : \mathfrak{e}_7^C \rightarrow \mathfrak{N}^C$	$\mathfrak{J}^C, \nu \in C\}$. Let $p : \mathfrak{e}_7^C \rightarrow \mathfrak{e}_6^C$ and $q : \mathfrak{e}_7^C \rightarrow \mathfrak{N}^C$ be projections of $\mathfrak{e}_7^C = \mathfrak{e}_6^C \oplus \mathfrak{N}^C$.
0902.0431_F02082	119	■ 1.000	$\mathfrak{e}_7^C = \mathfrak{e}_6^C \oplus \mathfrak{N}^C$	$\mathfrak{e}_7^C = \mathfrak{e}_6^C \oplus \mathfrak{N}^C$	$\mathfrak{J}^C, \nu \in C\}$. Let $p : \mathfrak{e}_7^C \rightarrow \mathfrak{e}_6^C$ and $q : \mathfrak{e}_7^C \rightarrow \mathfrak{N}^C$ be projections of $\mathfrak{e}_7^C = \mathfrak{e}_6^C \oplus \mathfrak{N}^C$.
0902.0431_F02083	119	■ 1.000	$\phi \in p(\mathfrak{a})$	$\phi \in p(\mathfrak{a})$	Now, let $\mathfrak{a}$ be a non-zero ideal of $\mathfrak{e}_7^C$. Then $p(\mathfrak{a})$ is an ideal of $\mathfrak{e}_6^C$. Indeed, if $\phi \in p(\mathfrak{a})$,
0902.0431_F02084	119	■ 1.000	$\Phi(0, A, B, \nu) \in \mathfrak{N}^C$	$\Phi(0, A, B, \nu) \in \mathfrak{N}^C$	then there exists $\Phi(0, A, B, \nu) \in \mathfrak{N}^C$ such that $\Phi(\phi, A, B, \nu) \in \mathfrak{a}$. For any $\phi_1 \in \mathfrak{e}_6^C$,
0902.0431_F02085	119	■ 1.000	$\Phi(\phi, A, B, \nu) \in \mathfrak{a}$	$\Phi(\phi, A, B, \nu) \in \mathfrak{a}$	then there exists $\Phi(0, A, B, \nu) \in \mathfrak{N}^C$ such that $\Phi(\phi, A, B, \nu) \in \mathfrak{a}$. For any $\phi_1 \in \mathfrak{e}_6^C$,
0902.0431_F02086	119	■ 1.000	$\phi_1 \in \mathfrak{e}_6^C$	$\phi_1 \in \mathfrak{e}_6^C$	then there exists $\Phi(0, A, B, \nu) \in \mathfrak{N}^C$ such that $\Phi(\phi, A, B, \nu) \in \mathfrak{a}$. For any $\phi_1 \in \mathfrak{e}_6^C$,
0902.0431_F02087	119	■ 1.000	$[\phi_1, \phi] \in p(\mathfrak{a})$	$[\phi_1, \phi] \in p(\mathfrak{a})$	(Theorem 4.3.1), hence $[\phi_1, \phi] \in p(\mathfrak{a})$.
0902.0431_F02088	119	■ 1.000	$\mathfrak{e}_6^C \cap \mathfrak{a} \neq \{0\}$	$\mathfrak{e}_6^C \cap \mathfrak{a} \neq \{0\}$	We shall show that either $\mathfrak{e}_6^C \cap \mathfrak{a} \neq \{0\}$ or $\mathfrak{N}^C \cap \mathfrak{a} \neq \{0\}$. Assume that $\mathfrak{e}_6^C \cap \mathfrak{a} =$
0902.0431_F02089	119	■ 1.000	$\mathfrak{N}^C \cap \mathfrak{a} \neq \{0\}$	$\mathfrak{N}^C \cap \mathfrak{a} \neq \{0\}$	We shall show that either $\mathfrak{e}_6^C \cap \mathfrak{a} \neq \{0\}$ or $\mathfrak{N}^C \cap \mathfrak{a} \neq \{0\}$. Assume that $\mathfrak{e}_6^C \cap \mathfrak{a} =$
0902.0431_F02090	119	■ 1.000	$\mathfrak{e}_6^C \cap \mathfrak{a} =$	$\mathfrak{e}_6^C \cap \mathfrak{a} =$	We shall show that either $\mathfrak{e}_6^C \cap \mathfrak{a} \neq \{0\}$ or $\mathfrak{N}^C \cap \mathfrak{a} \neq \{0\}$. Assume that $\mathfrak{e}_6^C \cap \mathfrak{a} =$
0902.0431_F02091	119	■ 1.000	$\{0\}$	$\{0\}$	$\{0\}$ and $\mathfrak{N}^C \cap \mathfrak{a} = \{0\}$. Then the mapping $p _{\mathfrak{a}} : \mathfrak{a} \rightarrow \mathfrak{e}_6^C$ is injective because $\mathfrak{N}^C \cap \mathfrak{a} =$
0902.0431_F02092	119	■ 1.000	$\mathfrak{N}^C \cap \mathfrak{a} = \{0\}$	$\mathfrak{N}^C \cap \mathfrak{a} = \{0\}$	$\{0\}$ and $\mathfrak{N}^C \cap \mathfrak{a} = \{0\}$. Then the mapping $p _{\mathfrak{a}} : \mathfrak{a} \rightarrow \mathfrak{e}_6^C$ is injective because $\mathfrak{N}^C \cap \mathfrak{a} =$
0902.0431_F02093	119	■ 1.000	$p _{\mathfrak{a}} : \mathfrak{a} \rightarrow \mathfrak{e}_6^C$	$p _{\mathfrak{a}} : \mathfrak{a} \rightarrow \mathfrak{e}_6^C$	$\{0\}$ and $\mathfrak{N}^C \cap \mathfrak{a} = \{0\}$. Then the mapping $p _{\mathfrak{a}} : \mathfrak{a} \rightarrow \mathfrak{e}_6^C$ is injective because $\mathfrak{N}^C \cap \mathfrak{a} =$
0902.0431_F02094	119	■ 1.000	$\mathfrak{N}^C \cap \mathfrak{a} =$	$\mathfrak{N}^C \cap \mathfrak{a} =$	$\{0\}$ and $\mathfrak{N}^C \cap \mathfrak{a} = \{0\}$. Then the mapping $p _{\mathfrak{a}} : \mathfrak{a} \rightarrow \mathfrak{e}_6^C$ is injective because $\mathfrak{N}^C \cap \mathfrak{a} =$
0902.0431_F02095	119	■ 0.999	$p(\mathfrak{a}) = \mathfrak{e}_6^C$	$p(\mathfrak{a}) = \mathfrak{e}_6^C$	$\{0\}$. Since $p(\mathfrak{a})$ is a non-zero ideal of $\mathfrak{e}_6^C$ and $\mathfrak{e}_6^C$ is simple, we have $p(\mathfrak{a}) = \mathfrak{e}_6^C$. Hence
0902.0431_F02096	119	■ 0.996	$\dim_C(\mathfrak{a}) = \dim_C(p(\mathfrak{a})) = \dim_C(\mathfrak{e}_6^C) = 78$	$\dim_C(\mathfrak{a}) = \dim_C(p(\mathfrak{a})) = \dim_C(\mathfrak{e}_6^C) = 78$	$\dim_C(\mathfrak{a}) = \dim_C(p(\mathfrak{a})) = \dim_C(\mathfrak{e}_6^C) = 78$. On the other hand, since $\mathfrak{e}_6^C \cap \mathfrak{a} = \{0\}$,
0902.0431_F02097	119	■ 0.996	$\mathfrak{e}_6^C \cap \mathfrak{a} = \{0\}$	$\mathfrak{e}_6^C \cap \mathfrak{a} = \{0\}$	$\dim_C(\mathfrak{a}) = \dim_C(p(\mathfrak{a})) = \dim_C(\mathfrak{e}_6^C) = 78$. On the other hand, since $\mathfrak{e}_6^C \cap \mathfrak{a} = \{0\}$,
0902.0431_F02098	119	■ 1.000	$q _{\mathfrak{a}} : \mathfrak{a} \rightarrow \mathfrak{N}^C$	$q _{\mathfrak{a}} : \mathfrak{a} \rightarrow \mathfrak{N}^C$	$q _{\mathfrak{a}} : \mathfrak{a} \rightarrow \mathfrak{N}^C$ is also injective. Hence we have $\dim_C(\mathfrak{a}) \leq \dim_C(\mathfrak{N}^C) = 27+27+1 = 55$.
0902.0431_F02099	119	■ 1.000	$\dim_C(\mathfrak{a}) \leq \dim_C(\mathfrak{N}^C) = 27 + 27 + 1 = 55$	$\dim_C(\mathfrak{a}) \leq \dim_C(\mathfrak{N}^C) = 27 + 27 + 1 = 55$	$q _{\mathfrak{a}} : \mathfrak{a} \rightarrow \mathfrak{N}^C$ is also injective. Hence we have $\dim_C(\mathfrak{a}) \leq \dim_C(\mathfrak{N}^C) = 27+27+1 = 55$.
0902.0431_F02100	119	■ 1.000	$\mathfrak{e}_6^C \cap \mathfrak{a} = \mathfrak{e}_6^C$	$\mathfrak{e}_6^C \cap \mathfrak{a} = \mathfrak{e}_6^C$	(1) Case $\mathfrak{e}_6^C \cap \mathfrak{a} \neq \{0\}$. From the simplicity of $\mathfrak{e}_6^C$, we have $\mathfrak{e}_6^C \cap \mathfrak{a} = \mathfrak{e}_6^C$, hence
0902.0431_F02101	119	■ 1.000	$\mathfrak{a} \supset \mathfrak{e}_6^C$	$\mathfrak{a} \supset \mathfrak{e}_6^C$	$\mathfrak{a} \supset \mathfrak{e}_6^C$. On the other hand, we have
0902.0431_F02102	119	■ 1.000	$\Phi(0, 0, \mathfrak{J}^C, 0) \subset \mathfrak{a}$	$\Phi(0, 0, \mathfrak{J}^C, 0) \subset \mathfrak{a}$	Similarly we have $\Phi(0, 0, \mathfrak{J}^C, 0) \subset \mathfrak{a}$. Moreover, from

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0902.0431_F02103	120	0.602	$\Phi(0,0,0,1) \in \mathfrak{a}$	$\Phi(0,0,0,1) \in \mathfrak{a}$	we have $\Phi(0,0,0,1) \in \mathfrak{a}$, and so $\mathfrak{a} \supset \mathfrak{N}^C$. Hence $\mathfrak{a} \supset \mathfrak{e}_6^C \oplus \mathfrak{N}^C = \mathfrak{e}_7^C$.
0902.0431_F02104	120	0.602	$\mathfrak{a} \supset \mathfrak{N}^C$	$\mathfrak{a} \supset \mathfrak{N}^C$	we have $\Phi(0,0,0,1) \in \mathfrak{a}$, and so $\mathfrak{a} \supset \mathfrak{N}^C$. Hence $\mathfrak{a} \supset \mathfrak{e}_6^C \oplus \mathfrak{N}^C = \mathfrak{e}_7^C$.
0902.0431_F02105	120	0.602	$\mathfrak{a} \supset \mathfrak{e}_6^C \oplus \mathfrak{N}^C = \mathfrak{e}_7^C$	$\mathfrak{a} \supset \mathfrak{e}_6^C \oplus \mathfrak{N}^C = \mathfrak{e}_7^C$	we have $\Phi(0,0,0,1) \in \mathfrak{a}$, and so $\mathfrak{a} \supset \mathfrak{N}^C$. Hence $\mathfrak{a} \supset \mathfrak{e}_6^C \oplus \mathfrak{N}^C = \mathfrak{e}_7^C$.
0902.0431_F02106	120	1.000	$\Phi(0, A, B, \nu)$	$\Phi(0, A, B, \nu)$	(2) Case $\mathfrak{N}^C \cap \mathfrak{a} \neq \{0\}$. Let $\Phi(0, A, B, \nu)$ be a non-zero element of $\mathfrak{N}^C \cap \mathfrak{a}$.
0902.0431_F02107	120	1.000	$\mathfrak{N}^C \cap \mathfrak{a}$	$\mathfrak{N}^C \cap \mathfrak{a}$	(2) Case $\mathfrak{N}^C \cap \mathfrak{a} \neq \{0\}$. Let $\Phi(0, A, B, \nu)$ be a non-zero element of $\mathfrak{N}^C \cap \mathfrak{a}$.
0902.0431_F02108	120	1.000	$\Phi(0, A, B, \nu), A \neq 0$	$\Phi(0, A, B, \nu), A \neq 0$	(i) Case $\Phi(0, A, B, \nu), A \neq 0$. Choose $B_1 \in \mathfrak{J}^C$ such that $A \vee B_1 \neq 0$ (Lemma
0902.0431_F02109	120	1.000	$B_1 \in \mathfrak{J}^C$	$B_1 \in \mathfrak{J}^C$	(i) Case $\Phi(0, A, B, \nu), A \neq 0$. Choose $B_1 \in \mathfrak{J}^C$ such that $A \vee B_1 \neq 0$ (Lemma
0902.0431_F02110	120	1.000	$A \vee B_1 \neq 0$	$A \vee B_1 \neq 0$	(i) Case $\Phi(0, A, B, \nu), A \neq 0$. Choose $B_1 \in \mathfrak{J}^C$ such that $A \vee B_1 \neq 0$ (Lemma
0902.0431_F02111	120	1.000	$[A \vee B_1, \phi] \neq 0$	$[A \vee B_1, \phi] \neq 0$	3.5.4.(2)), and choose $\phi \in \mathfrak{e}_6^C$ such that $[A \vee B_1, \phi] \neq 0$ (since $\mathfrak{e}_6^C$ is simple, such a
0902.0431_F02112	120	1.000	$\Phi(0, A, B, \nu), B \neq 0$	$\Phi(0, A, B, \nu), B \neq 0$	(ii) Case $\Phi(0, A, B, \nu), B \neq 0$. This case is also reduced to the case (1) in a
0902.0431_F02113	120	1.000	$\Phi(0,0,0,\nu), \nu \neq 0$	$\Phi(0,0,0,\nu), \nu \neq 0$	(iii) Case $\Phi(0,0,0,\nu), \nu \neq 0$. If we choose $0 \neq A \in \mathfrak{J}^C$, then we have
0902.0431_F02114	120	1.000	$0 \neq A \in \mathfrak{J}^C$	$0 \neq A \in \mathfrak{J}^C$	(iii) Case $\Phi(0,0,0,\nu), \nu \neq 0$. If we choose $0 \neq A \in \mathfrak{J}^C$, then we have
0902.0431_F02115	120	1.000	$\mathfrak{a} = \mathfrak{e}_7^C$	$\mathfrak{a} = \mathfrak{e}_7^C$	Thus we have $\mathfrak{a} = \mathfrak{e}_7^C$.
0902.0431_F02116	120	0.990	$\mathfrak{e}_7^C \mathfrak{P}^C = \left\{ \sum_k \Phi_k P_k \mid \Phi_k \in \mathfrak{e}_7^C, P_k \in \mathfrak{P}^C \right\} = \mathfrak{P}^C$	$\mathfrak{e}_7^C \mathfrak{P}^C = \left\{ \sum_k \Phi_k P_k \mid \Phi_k \in \mathfrak{e}_7^C, P_k \in \mathfrak{P}^C \right\} = \mathfrak{P}^C$	(2) $\mathfrak{e}_7^C \mathfrak{P}^C = \left\{ \sum_k \Phi_k P_k \mid \Phi_k \in \mathfrak{e}_7^C, P_k \in \mathfrak{P}^C \right\} = \mathfrak{P}^C$.
0902.0431_F02117	120	1.000	$(0,0,0,1) \in W$	$(0,0,0,1) \in W$	that if $(0,0,0,1) \in W$, then we have $W = \mathfrak{P}^C$. Indeed,
0902.0431_F02118	120	1.000	$W = \mathfrak{P}^C$	$W = \mathfrak{P}^C$	that if $(0,0,0,1) \in W$, then we have $W = \mathfrak{P}^C$. Indeed,
0902.0431_F02119	120	1.000	$P = (X, Y, \xi, \eta)$	$P = (X, Y, \xi, \eta)$	Hence we have $W = \mathfrak{P}^C$. Now, let $P = (X, Y, \xi, \eta)$ be a non-zero element of W .
0902.0431_F02120	120	1.000	$W \ni P = (X, Y, \xi, \eta), X \neq 0$	$W \ni P = (X, Y, \xi, \eta), X \neq 0$	(i) Case $W \ni P = (X, Y, \xi, \eta), X \neq 0$. Then we have
0902.0431_F02121	120	0.991	$((\mathrm{a}) - (\mathrm{b})) \div 2, ((\mathrm{a}) - (\mathrm{c})) \div 8$	$((\mathrm{a}) - (\mathrm{b})) \div 2, ((\mathrm{a}) - (\mathrm{c})) \div 8$	Taking $((\mathrm{a}) - (\mathrm{b})) \div 2, ((\mathrm{a}) - (\mathrm{c})) \div 8$, we have $(X, 0, -\xi, 2\eta) \in W, (0, 0, \xi, \eta) \in W$, re-
0902.0431_F02122	120	0.991	$(X, 0, -\xi, 2\eta) \in W, (0, 0, \xi, \eta) \in W$	$(X, 0, -\xi, 2\eta) \in W, (0, 0, \xi, \eta) \in W$	Taking $((\mathrm{a}) - (\mathrm{b})) \div 2, ((\mathrm{a}) - (\mathrm{c})) \div 8$, we have $(X, 0, -\xi, 2\eta) \in W, (0, 0, \xi, \eta) \in W$, re-
0902.0431_F02123	120	1.000	$(X, 0, 0, 3\eta) \in W$	$(X, 0, 0, 3\eta) \in W$	spectively. Consequently $(X, 0, 0, 3\eta) \in W$. Next, if we choose $X_1 \in \mathfrak{J}^C$ such that
0902.0431_F02124	120	1.000	$X_1 \in \mathfrak{J}^C$	$X_1 \in \mathfrak{J}^C$	spectively. Consequently $(X, 0, 0, 3\eta) \in W$. Next, if we choose $X_1 \in \mathfrak{J}^C$ such that
0902.0431_F02125	120	1.000	$(X_1, X) \neq 0$	$(X_1, X) \neq 0$	$(X_1, X) \neq 0$, from
0902.0431_F02126	121	1.000	$P = (0, Y, \xi, \eta), Y \neq 0$	$P = (0, Y, \xi, \eta), Y \neq 0$	(ii) Case $P = (0, Y, \xi, \eta), Y \neq 0$. If we choose $B \in \mathfrak{J}^C$ such that $B \times Y \neq 0$, from
0902.0431_F02127	121	1.000	$B \times Y \neq 0$	$B \times Y \neq 0$	(ii) Case $P = (0, Y, \xi, \eta), Y \neq 0$. If we choose $B \in \mathfrak{J}^C$ such that $B \times Y \neq 0$, from
0902.0431_F02128	121	1.000	$P = (0, 0, \xi, \eta), \xi \neq 0$	$P = (0, 0, \xi, \eta), \xi \neq 0$	(iii) Case $P = (0, 0, \xi, \eta), \xi \neq 0$. For $0 \neq B \in \mathfrak{J}^C$, we have
0902.0431_F02129	121	1.000	$0 \neq B \in \mathfrak{J}^C$	$0 \neq B \in \mathfrak{J}^C$	(iii) Case $P = (0, 0, \xi, \eta), \xi \neq 0$. For $0 \neq B \in \mathfrak{J}^C$, we have
0902.0431_F02130	121	0.979	$\mathfrak{e}_7^C \mathfrak{P}^C$	$\mathfrak{e}_7^C \mathfrak{P}^C$	(2) Since $\mathfrak{e}_7^C \mathfrak{P}^C$ is an $\mathfrak{e}_7^C$ -invariant C -submodule of $\mathfrak{P}^C$, $\mathfrak{e}_7^C \mathfrak{P}^C = \mathfrak{P}^C$ follows
0902.0431_F02131	121	0.979	$\mathfrak{P}^C, \mathfrak{e}_7^C \mathfrak{P}^C = \mathfrak{P}^C$	$\mathfrak{P}^C, \mathfrak{e}_7^C \mathfrak{P}^C = \mathfrak{P}^C$	(2) Since $\mathfrak{e}_7^C \mathfrak{P}^C$ is an $\mathfrak{e}_7^C$ -invariant C -submodule of $\mathfrak{P}^C$, $\mathfrak{e}_7^C \mathfrak{P}^C = \mathfrak{P}^C$ follows
0902.0431_F02132	121	0.859	$\Phi = \sum_i (P_i \times Q_i), P_i, Q_i \in$	$\Phi = \sum_i (P_i \times Q_i), P_i, Q_i \in$	Lemma 4.4.3. Any element $\Phi \in \mathfrak{e}_7^C$ is expressed by $\Phi = \sum_i (P_i \times Q_i), P_i, Q_i \in$
0902.0431_F02133	121	1.000	$[\Phi, P \times Q] = \Phi P \times Q + P \times \Phi Q$	$[\Phi, P \times Q] = \Phi P \times Q + P \times \Phi Q$	Proof. Since $[\Phi, P \times Q] = \Phi P \times Q + P \times \Phi Q$ (Proposition 4.3.2), $\mathfrak{a} = \left\{ \sum_i (P_i \times$
0902.0431_F02134	121	1.000	$\mathfrak{a} = \left\{ \sum_i (P_i \times$	$\mathfrak{a} = \left\{ \sum_i (P_i \times$	Proof. Since $[\Phi, P \times Q] = \Phi P \times Q + P \times \Phi Q$ (Proposition 4.3.2), $\mathfrak{a} = \left\{ \sum_i (P_i \times$
0902.0431_F02135	121	0.999	$Q_i) \mid P_i, Q_i \in \mathfrak{P}^C \right\}$	$Q_i) \mid P_i, Q_i \in \mathfrak{P}^C \}$	$Q_i) \mid P_i, Q_i \in \mathfrak{P}^C \}$ is an ideal of $\mathfrak{e}_7^C$. From the simplicity of $\mathfrak{e}_7^C$ (Theorem 4.4.1), we
0902.0431_F02136	121	0.971	$(\Phi_1, \Phi_2)_7$	$(\Phi_1, \Phi_2)_7$	Definition. We define a symmetric inner product $(\Phi_1, \Phi_2)_7$ in $\mathfrak{e}_7^C$ by
0902.0431_F02137	121	0.959	$\Phi_i = \Phi(\phi_i, A_i, B_i, \nu_i) \in \mathfrak{e}_7^C$	$\Phi_i = \Phi(\phi_i, A_i, B_i, \nu_i) \in \mathfrak{e}_7^C$	where $\Phi_i = \Phi(\phi_i, A_i, B_i, \nu_i) \in \mathfrak{e}_7^C$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_F02138	121	0.761	$\Phi \in \mathfrak{e}_7^C, P, Q \in \mathfrak{P}^C$	$\Phi \in \mathfrak{e}_7^C, P, Q \in \mathfrak{P}^C$	(2) For $\Phi \in \mathfrak{e}_7^C, P, Q \in \mathfrak{P}^C$, we have
0902.0431_F02139	121	1.000	$[\Phi, \Phi_1], \Phi_2$	$([\Phi, \Phi_1], \Phi_2)_7$	Proof. (1) $[\Phi, \Phi_1, \Phi_2]$
0902.0431_F02140	122	1.000	B_7	B_7	Theorem 4.5.2. The Killing form B_7 of the Lie algebra $\mathfrak{e}_7^C$ is given by
0902.0431_F02141	122	0.682	$\Phi_i = \Phi(\phi_i, A_i, B_i, \nu_i) \in \mathfrak{e}_7^C$	$\Phi_i = \Phi(\phi_i, A_i, B_i, \nu_i) \in \mathfrak{e}_7^C$	where $\Phi_i = \Phi(\phi_i, A_i, B_i, \nu_i) \in \mathfrak{e}_7^C$ and B_6 is the Killing form of $\mathfrak{e}_6^C$.
0902.0431_F02142	122	1.000	$\Phi_0 = \Phi_1 = \Phi_2 = \Phi(0, 0, 0, 1)$	$\Phi_0 = \Phi_1 = \Phi_2 = \Phi(0, 0, 0, 1)$	To determine these k, k' , let $\Phi_0 = \Phi_1 = \Phi_2 = \Phi(0, 0, 0, 1)$. Then we have
0902.0431_F02143	123	1.000	$k = -9$	$k = -9$	Therefore $k = -9$. Next, from
0902.0431_F02144	123	1.000	$P \in \mathfrak{P}^C, P \neq 0$	$P \in \mathfrak{P}^C, P \neq 0$	Lemma 4.5.3. For $P \in \mathfrak{P}^C, P \neq 0$, there exists $Q \in \mathfrak{P}^C$ such that $P \times Q \neq 0$.
0902.0431_F02145	123	1.000	$Q \in \mathfrak{P}^C$	$Q \in \mathfrak{P}^C$	Lemma 4.5.3. For $P \in \mathfrak{P}^C, P \neq 0$, there exists $Q \in \mathfrak{P}^C$ such that $P \times Q \neq 0$.
0902.0431_F02146	123	1.000	$P \times Q \neq 0$	$P \times Q \neq 0$	Lemma 4.5.3. For $P \in \mathfrak{P}^C, P \neq 0$, there exists $Q \in \mathfrak{P}^C$ such that $P \times Q \neq 0$.
0902.0431_F02147	123	1.000	$P \times Q = 0$	$P \times Q = 0$	Proof. Assume that $P \times Q = 0$ for all $Q \in \mathfrak{P}^C$. Then for any $\Phi \in \mathfrak{e}_7^C$,
0902.0431_F02148	123	1.000	$0 = (\Phi, P \times Q)_7 = (\Phi, Q \times P)_7 = \{\Phi Q, P\}$	$0 = (\Phi, P \times Q)_7 = (\Phi, Q \times P)_7 = \{\Phi Q, P\}$	$0 = (\Phi, P \times Q)_7 = (\Phi, Q \times P)_7 = \{\Phi Q, P\}$ (Lemma 4.5.1.(2)). Since $\mathfrak{e}_7^C \mathfrak{P}^C = \mathfrak{P}^C$
0902.0431_F02149	123	1.000	$\mathfrak{e}_7^C \mathfrak{P}^C = \mathfrak{P}^C$	$\mathfrak{e}_7^C \mathfrak{P}^C = \mathfrak{P}^C$	$0 = (\Phi, P \times Q)_7 = (\Phi, Q \times P)_7 = \{\Phi Q, P\}$ (Lemma 4.5.1.(2)). Since $\mathfrak{e}_7^C \mathfrak{P}^C = \mathfrak{P}^C$
0902.0431_F02150	123	1.000	$\{\mathfrak{P}^C, P\} = 0$	$\{\mathfrak{P}^C, P\} = 0$	(Proposition 4.4.2.(2)), we have $\{\mathfrak{P}^C, P\} = 0$, so that $P = 0$.
0902.0431_F02151	123	1.000	$P = 0$	$P = 0$	(Proposition 4.4.2.(2)), we have $\{\mathfrak{P}^C, P\} = 0$, so that $P = 0$.
0902.0431_F02152	124	0.844	$\mathfrak{e}_7^C$	$\mathfrak{e}_7^C$	(where $H_k = -iG_{k4+k}$), then $\mathfrak{h}$ is an abelian subalgebra of $\mathfrak{e}_7^C$ (it will be a Cartan sub-
0902.0431_F02153	124	1.000	$h_\delta = \sum_{k=0}^3 \lambda_k H_k, H = \sum_{j=1}^3 \mu_j E_j$	$h_\delta = \sum_{k=0}^3 \lambda_k H_k, H = \sum_{j=1}^3 \mu_j E_j$	algebra of $\mathfrak{e}_7^C$). In the following calculations, we put $h_\delta = \sum_{k=0}^3 \lambda_k H_k, H = \sum_{j=1}^3 \mu_j E_j$.
0902.0431_F02154	124	1.000	$S \in \mathfrak{e}_6^C \subset \mathfrak{e}_7^C$	$S \in \mathfrak{e}_6^C \subset \mathfrak{e}_7^C$	$S \in \mathfrak{e}_6^C \subset \mathfrak{e}_7^C$ be a root vector belonging to α . Then
0902.0431_F02155	124	0.999	$\pm \left(\mu_j + \frac{2}{3} \nu \right), 0 \leq j \leq 3$	$\pm \left(\mu_j + \frac{2}{3} \nu \right), 0 \leq j \leq 3$	Hence $\pm \left(\mu_j + \frac{2}{3} \nu \right), 0 \leq j \leq 3$, are roots of $\mathfrak{e}_7^C$.
0902.0431_F02156	125	1.000	$\pm \lambda_k - \frac{1}{2} \mu_1 + \frac{2}{3} \nu, 0 \leq k \leq 3$	$\pm \lambda_k - \frac{1}{2} \mu_1 + \frac{2}{3} \nu, 0 \leq k \leq 3$	Hence $\pm \lambda_k - \frac{1}{2} \mu_1 + \frac{2}{3} \nu, 0 \leq k \leq 3$, are roots of $\mathfrak{e}_7^C$. The remainders of roots can be
0902.0431_F02157	125	0.965	$\nu h_\delta, \kappa \pi h_\delta$	$\nu h_\delta, \kappa \pi h_\delta$	similarly found ($\nu h_\delta, \kappa \pi h_\delta$ in Theorem 3.6.4 will be used).
0902.0431_F02158	125	1.000	$n_1 n_2 \cdots n_7$	$n_1 n_2 \cdots n_7$	Proof. In the following, the notation $n_1 \alpha_1 + n_2 \alpha_2 +$
0902.0431_F02159	125	1.000	$\cdots + n_7 \alpha_7$	$\cdots + n_7 \alpha_7$	$\cdots + n_7 \alpha_7$. Now, all positive roots of $\mathfrak{e}_7^C$ are represented by
0902.0431_F02160	128	1.000	$\Pi = \{\alpha_1, \alpha_2, \dots, \alpha_7\}$	$\Pi = \{\alpha_1, \alpha_2, \dots, \alpha_7\}$	Hence $\Pi = \{\alpha_1, \alpha_2, \dots, \alpha_7\}$ is a fundamental root system of $\mathfrak{e}_7^C$. The real part $\mathfrak{h}_R$
0902.0431_F02161	128	0.394	$h = \Phi \left(\sum_{k=0}^3 \lambda_k H_k + \left(\sum_{j=1}^3 \mu_j E_j \right)^\sim, 0, 0, \nu \right), h' = \Phi \left(\sum_{k=0}^3 \lambda_k' H_k + \left(\sum_{j=1}^3 \mu_j' E_j \right)^\sim, 0, \right)$	$h = \Phi \left(\sum_{k=0}^3 \lambda_k H_k + \left(\sum_{j=1}^3 \mu_j E_j \right)^\sim, 0, 0, \nu \right), h' = \Phi \left(\sum_{k=0}^3 \lambda_k' H_k + \left(\sum_{j=1}^3 \mu_j' E_j \right)^\sim, 0, \right)$	for $h = \Phi \left(\sum_{k=0}^3 \lambda_k H_k + \left(\sum_{j=1}^3 \mu_j E_j \right)^\sim, 0, 0, \nu \right), h' = \Phi \left(\sum_{k=0}^3 \lambda_k' H_k + \left(\sum_{j=1}^3 \mu_j' E_j \right)^\sim, 0, \right)$,
0902.0431_F02162	128	1.000	$0, \nu' \in \mathfrak{h}_R$	$0, \nu' \in \mathfrak{h}_R$	$0, \nu' \in \mathfrak{h}_R$. Indeed, from Theorem 4.5.2, we have
0902.0431_F02163	128	1.000	$\alpha_i (B_7(H_\alpha, H) = \alpha(H), H \in$	$\alpha_i (B_7(H_\alpha, H) = \alpha(H), H \in$	Now, the canonical elements $H_{\alpha_i} \in \mathfrak{h}_R$ corresponding to $\alpha_i (B_7(H_\alpha, H) = \alpha(H), H \in$
0902.0431_F02164	128	0.508	$\mathfrak{h}$	$\mathfrak{h}$	$\mathfrak{h}$) are determined as follows.
0902.0431_F02165	128	1.000	$\alpha_1, \alpha_2, \dots, \alpha_7$	$\alpha_1, \alpha_2, \dots, \alpha_7$	induced by the Killing form B_7 between $\alpha_1, \alpha_2, \dots, \alpha_7$ and $-\mu$ are given by
0902.0431_F02166	129	1.000	$T \oplus E_6$	$T \oplus E_6$	(1) The first one is a subalgebra of type $T \oplus E_6$ which is obtained as the fixed
0902.0431_F02167	129	1.000	$A_1 \oplus D_6$	$A_1 \oplus D_6$	(2) The second one is a subalgebra of type $A_1 \oplus D_6$ which is obtained as the fixed
0902.0431_F02168	129	1.000	A_7	A_7	(3) The third one is a subalgebra of type A_7 which is obtained as the fixed points
0902.0431_F02169	129	1.000	$\lambda \gamma$	$\lambda \gamma$	of an involution $\lambda \gamma$ of $\mathfrak{e}_7$.
0902.0431_F02170	129	1.000	$A_2 \oplus A_5$	$A_2 \oplus A_5$	(4) The fourth one is a subalgebra of type $A_2 \oplus A_5$ which is obtained as the fixed

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0902.0431_F02171	129	■ 1.000	$\left(E_7\right)_{\{(\theta, 0, 1, \theta)\}}$	$(E_7)_{(0,0,1,0)}$	We shall study the following subgroup $(E_7)_{(0,0,1,0)}$ of E_7 :
0902.0431_F02172	129	■ 1.000	$\alpha(\theta, 0, 1, \theta)=(\theta, 0, 1, \theta)$	$\alpha(0, 0, 1, 0)=(0, 0, 1, 0)$	Lemma 4.7.1. If $\alpha \in E_7$ satisfies $\alpha(0, 0, 1, 0)=(0, 0, 1, 0)$, then α also satisfies
0902.0431_F02173	129	■ 1.000	$\alpha(\theta, 0, \theta, 1)=(\theta, 0, \theta, 1)$	$\alpha(0, 0, 0, 1)=(0, 0, 0, 1)$	$\alpha(0, 0, 0, 1)=(0, 0, 0, 1)$, and conversely.
0902.0431_F02174	129	■ 0.986	$\alpha \in E_6=\left\{\alpha \in \operatorname{Iso}_C\left(\mathfrak{J}^C\right) \mid \det (\alpha X)=\det X,\langle \alpha X\right.$ $\left.\left(\frac{1}{\alpha X}\right) \mid \operatorname{det}(\alpha X)=\operatorname{det} X,\langle \alpha X\right.$	$\alpha \in E_6=\left\{\alpha \in \operatorname{Iso}_C\left(\mathfrak{J}^C\right) \mid \det (\alpha X)=\det X,\langle \alpha X$	Proof. We associate an element $\alpha \in E_6=\left\{\alpha \in \operatorname{Iso}_C\left(\mathfrak{J}^C\right) \mid \det (\alpha X)=\det X,\langle \alpha X,$
0902.0431_F02175	129	■ 0.996	$\alpha Y \rangle=\langle X, Y \rangle$	$\alpha Y\rangle=\langle X, Y\rangle\}$	$\langle \alpha Y\rangle=\langle X, Y\rangle$ with the element
0902.0431_F02176	130	■ 1.000	$\widetilde{\alpha} \in E_7$	$\tilde{\alpha} \in E_7$	We first have to prove that $\tilde{\alpha} \in E_7$. For $P=(X, Y, \xi, \eta), Q=(Z, W, \zeta, \omega) \in \mathfrak{P}^C$, we
0902.0431_F02177	130	■ 0.999	$\widetilde{\alpha} P, \widetilde{\alpha} Q\rangle=\langle P, Q\rangle$	$\langle \tilde{\alpha} P, \tilde{\alpha} Q\rangle=\langle P, Q\rangle$	and $\langle \tilde{\alpha} P, \tilde{\alpha} Q\rangle=\langle P, Q\rangle$ is evident. Hence $\tilde{\alpha} \in E_7$, moreover $\tilde{\alpha} \in (E_7)_{(0,0,1,0)}$.
0902.0431_F02178	130	■ 0.999	$\widetilde{\alpha} \in\left(E_7\right)_{\{(\theta, 0, 1, \theta)\}}$	$\tilde{\alpha} \in (E_7)_{(0,0,1,0)}$	and $\langle \tilde{\alpha} P, \tilde{\alpha} Q\rangle=\langle P, Q\rangle$ is evident. Hence $\tilde{\alpha} \in E_7$, moreover $\tilde{\alpha} \in (E_7)_{(0,0,1,0)}$.
0902.0431_F02179	130	■ 1.000	$\alpha(\theta, 0, \theta, 1)=$	$\alpha(0, 0, 0, 1)=$	Conversely, suppose that $\alpha \in E_7$ satisfies $\alpha(0, 0, 1, 0)=(0, 0, 1, 0)$ and $\alpha(0, 0, 0, 1)=$
0902.0431_F02180	130	■ 1.000	$\theta \neq \eta \in C$	$0 \neq \eta \in C$	holds for all $0 \neq \eta \in C$. Comparing the coefficients of η of both sides, we have $\delta=0$.
0902.0431_F02181	130	■ 1.000	η	η	holds for all $0 \neq \eta \in C$. Comparing the coefficients of η of both sides, we have $\delta=0$.
0902.0431_F02182	130	■ 1.000	$\alpha\left(\frac{1}{\xi}(Y \times Y), Y, \xi, \frac{1}{\xi^2} \det Y\right) \in \mathfrak{M}^C$	$\alpha\left(\frac{1}{\xi}(Y \times Y), Y, \xi, \frac{1}{\xi^2} \det Y\right) \in \mathfrak{M}^C$	Similarly, from $\alpha\left(\frac{1}{\xi}(Y \times Y), Y, \xi, \frac{1}{\xi^2} \det Y\right) \in \mathfrak{M}^C$, we have $\epsilon=0$. Furthermore, since
0902.0431_F02183	130	■ 1.000	$\epsilon=0$	$\epsilon=0$	Similarly, from $\alpha\left(\frac{1}{\xi}(Y \times Y), Y, \xi, \frac{1}{\xi^2} \det Y\right) \in \mathfrak{M}^C$, we have $\epsilon=0$. Furthermore, since
0902.0431_F02184	131	■ 1.000	$\beta \in E_6^C$	$\beta \in E_6^C$	which implies that $\beta \in E_6^C$. The equality $\langle \alpha \dot{X}, \alpha \dot{Y}\rangle=\langle \dot{X}, \dot{Y}\rangle$ implies $\langle \beta X, \beta Y\rangle=$
0902.0431_F02185	131	■ 1.000	$\langle \alpha \dot{X}, \alpha \dot{Y}\rangle=\langle \dot{X}, \dot{Y}\rangle$	$\langle \alpha \dot{X}, \alpha \dot{Y}\rangle=\langle \dot{X}, \dot{Y}\rangle$	which implies that $\beta \in E_6^C$. The equality $\langle \alpha \dot{X}, \alpha \dot{Y}\rangle=\langle \dot{X}, \dot{Y}\rangle$ implies $\langle \beta X, \beta Y\rangle=$
0902.0431_F02186	131	■ 1.000	$\langle \beta X, \beta Y\rangle=$	$\langle \beta X, \beta Y\rangle=$	which implies that $\beta \in E_6^C$. The equality $\langle \alpha \dot{X}, \alpha \dot{Y}\rangle=\langle \dot{X}, \dot{Y}\rangle$ implies $\langle \beta X, \beta Y\rangle=$
0902.0431_F02187	131	■ 1.000	$\beta_1=\tau \beta \tau$	$\beta_1=\tau \beta \tau$	we obtain $\beta_1=\tau \beta \tau$. Indeed, putting $X \times X$ instead of X , we have
0902.0431_F02188	131	■ 1.000	$X \times X$	$X \times X$	we obtain $\beta_1=\tau \beta \tau$. Indeed, putting $X \times X$ instead of X , we have
0902.0431_F02189	131	■ 1.000	$\beta_1 X=\tau \beta \tau X$	$\beta_1 X=\tau \beta \tau X$	If $\det X \neq 0$, then we have $\beta_1 X=\tau \beta \tau X$, Since β_1 and $\tau \beta \tau$ are linear mappings (of
0902.0431_F02190	131	■ 1.000	$\tau \beta \tau$	$\tau \beta \tau$	If $\det X \neq 0$, then we have $\beta_1 X=\tau \beta \tau X$, Since β_1 and $\tau \beta \tau$ are linear mappings (of
0902.0431_F02191	131	■ 1.000	$\varphi_1(\theta): \mathfrak{P}^C \rightarrow \mathfrak{P}^C$	$\varphi_1(\theta): \mathfrak{P}^C \rightarrow \mathfrak{P}^C$	For $\theta \in C, \theta \neq 0$, we define a C -linear transformation $\varphi_1(\theta): \mathfrak{P}^C \rightarrow \mathfrak{P}^C$ by
0902.0431_F02192	131	■ 1.000	$\varphi_1(\theta) \in E_7^C$	$\varphi_1(\theta) \in E_7^C$	Then $\varphi_1(\theta) \in E_7^C$.
0902.0431_F02193	131	■ 1.000	$U(1)=\{\theta \in$	$U(1)=\{\theta \in$	as a subargroup. This subgroup is isomorphic to the usual unitary group $U(1)=\{\theta \in$
0902.0431_F02194	131	■ 1.000	$C \mid(\tau \theta) \theta=1\}$	$C \mid(\tau \theta) \theta=1\}$	$C \mid(\tau \theta) \theta=1\}$.
0902.0431_F02195	131	■ 1.000	$\varphi_1(\theta) \in E_7$	$\varphi_1(\theta) \in E_7$	Proof. It is easy to check that $\varphi_1(\theta) \in E_7$.
0902.0431_F02196	131	■ 1.000	$(E_7)_0$	$(E_7)_0$	We denote by $(E_7)_0$ the connected component of E_7 containing the identity 1.
0902.0431_F02197	131	■ 1.000	$a \in C$	$a \in C$	Lemma 4.8.1. For $a \in C$, if we define a mapping $\alpha_i(a): \mathfrak{P}^C \rightarrow \mathfrak{P}^C, i=1,2,3$
0902.0431_F02198	131	■ 1.000	$\alpha_i(a): \mathfrak{P}^C \rightarrow \mathfrak{P}^C, i=1,2,3$	$\alpha_i(a): \mathfrak{P}^C \rightarrow \mathfrak{P}^C, i=1,2,3$	Lemma 4.8.1. For $a \in C$, if we define a mapping $\alpha_i(a): \mathfrak{P}^C \rightarrow \mathfrak{P}^C, i=1,2,3$
0902.0431_F02199	131	■ 1.000	$\alpha_i(a) \in (E_7)_0$	$\alpha_i(a) \in (E_7)_0$	$\left(\text { if } a=0, \text { then } \frac{\sin a }{ a } \text { means } 1\right)$, then $\alpha_i(a) \in (E_7)_0$, where $p_i: \mathfrak{J}^C \rightarrow \mathfrak{J}^C$ is the
0902.0431_F02200	131	■ 1.000	$p_i: \mathfrak{J}^C \rightarrow \mathfrak{J}^C$	$p_i: \mathfrak{J}^C \rightarrow \mathfrak{J}^C$	$\left(\text { if } a=0, \text { then } \frac{\sin a }{ a } \text { means } 1\right)$, then $\alpha_i(a) \in (E_7)_0$, where $p_i: \mathfrak{J}^C \rightarrow \mathfrak{J}^C$ is the
0902.0431_F02201	132	■ 0.998	δ_{ij}	δ_{ij}	where δ_{ij} is the Kronecker delta symble. The mappings $\alpha_1(a_1), \alpha_2(a_2), \alpha(a_3),\left(a_i \in C\right)$
0902.0431_F02202	132	■ 0.998	$\alpha_1\left(a_1\right), \alpha_2\left(a_2\right), \alpha\left(a_3\right),\left(a_i \in C\right)$	$\alpha_1\left(a_1\right), \alpha_2\left(a_2\right), \alpha\left(a_3\right),\left(a_i \in C\right)$	where δ_{ij} is the Kronecker delta symble. The mappings $\alpha_1(a_1), \alpha_2(a_2), \alpha(a_3),\left(a_i \in C\right)$
0902.0431_F02203	132	■ 1.000	$\alpha_i(a)=\exp \Phi_i(a)$	$\alpha_i(a)=\exp \Phi_i(a)$	(Theorem 4.3.4), we have $\alpha_i(a)=\exp \Phi_i(a)$. Hence $\alpha_i(a) \in (E_7)_0$. The relation
0902.0431_F02204	132	■ 1.000	$\left[\Phi_i\left(a_i\right), \Phi_j\left(a_j\right)\right]=0$	$\left[\Phi_i\left(a_i\right), \Phi_j\left(a_j\right)\right]=0$	$\left[\Phi_i\left(a_i\right), \Phi_j\left(a_j\right)\right]=0$ shows that $\alpha_i\left(a_i\right)$ and $\alpha_j\left(a_j\right)$ are commutative.
0902.0431_F02205	132	■ 1.000	$\alpha_i\left(a_i\right)$	$\alpha_i\left(a_i\right)$	$\left[\Phi_i\left(a_i\right), \Phi_j\left(a_j\right)\right]=0$ shows that $\alpha_i\left(a_i\right)$ and $\alpha_j\left(a_j\right)$ are commutative.
0902.0431_F02206	132	■ 1.000	$\alpha_j\left(a_j\right)$	$\alpha_j\left(a_j\right)$	$\left[\Phi_i\left(a_i\right), \Phi_j\left(a_j\right)\right]=0$ shows that $\alpha_i\left(a_i\right)$ and $\alpha_j\left(a_j\right)$ are commutative.
0902.0431_F02207	132	■ 1.000	$P \in \mathfrak{M}^C, P \neq 0$	$P \in \mathfrak{M}^C, P \neq 0$	Proposition 4.8.2. Any element $P \in \mathfrak{M}^C, P \neq 0$ can be transformed to a

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0902.0431_F02208	132	■1.000	$\alpha \in \mathrm{in}\left(E_7\right)_{\{0\}}$	$\alpha \in (E_7)_0$	diagonal form by some element $\alpha \in (E_7)_0$:
0902.0431_F02209	132	■1.000	$P=(X, Y, \xi, \eta) \in \mathfrak{M}^C$	$P=(X, Y, \xi, \eta) \in \mathfrak{M}^C$	Proof. Let $P=(X, Y, \xi, \eta) \in \mathfrak{M}^C$. We shall first show that P can be transformed
0902.0431_F02210	132	■1.000	$\xi \neq 0$	$\xi \neq 0$	to a diagonal form with $\xi \neq 0$.
0902.0431_F02211	132	■1.000	$P=(X, Y, \xi, \eta), \xi \neq 0$	$P=(X, Y, \xi, \eta), \xi \neq 0$	(1) Case $P=(X, Y, \xi, \eta), \xi \neq 0$. In this case, $X=\frac{1}{\xi}(Y \times Y)$. Choose $\beta \in E_6$
0902.0431_F02212	132	■1.000	$X=\frac{1}{\xi}(Y \times Y)$	$X=\frac{1}{\xi}(Y \times Y)$	(1) Case $P=(X, Y, \xi, \eta), \xi \neq 0$. In this case, $X=\frac{1}{\xi}(Y \times Y)$. Choose $\beta \in E_6$
0902.0431_F02213	132	■1.000	$\tau \beta \tau Y$	$\tau \beta \tau Y$	such that $\tau \beta \tau Y$ is diagonal (Proposition 3.8.2), then
0902.0431_F02214	132	■1.000	$P=(X, Y, 0, \eta), Y \neq 0$	$P=(X, Y, 0, \eta), Y \neq 0$	(2) Case $P=(X, Y, 0, \eta), Y \neq 0$. Choose $\beta \in E_6$ so that
0902.0431_F02215	132	■0.887	$\tau \beta \tau Y \neq 0$	$\tau \beta \tau Y \neq 0$	(Proposition 3.8.2). Since $\tau \beta \tau Y \neq 0$, some η_i is non-zero: $\eta_i \neq 0$. Applying
0902.0431_F02216	132	■0.887	η_i	η_i	(Proposition 3.8.2). Since $\tau \beta \tau Y \neq 0$, some η_i is non-zero: $\eta_i \neq 0$. Applying
0902.0431_F02217	132	■0.887	$\eta_i \neq 0$	$\eta_i \neq 0$	(Proposition 3.8.2). Since $\tau \beta \tau Y \neq 0$, some η_i is non-zero: $\eta_i \neq 0$. Applying
0902.0431_F02218	132	■1.000	$\alpha_i(-\pi / 2) \in \mathrm{in}\left(E_7\right)_{\{0\}}$	$\alpha_i(-\pi / 2) \in (E_7)_0$	$\alpha_i(-\pi / 2) \in (E_7)_0$ of Lemma 4.8.1 on βP , we get
0902.0431_F02219	132	■1.000	βP	βP	$\alpha_i(-\pi / 2) \in (E_7)_0$ of Lemma 4.8.1 on βP , we get
0902.0431_F02220	132	■1.000	$P=(X, 0, 0, \eta), X \neq 0$	$P=(X, 0, 0, \eta), X \neq 0$	(3) Case $P=(X, 0, 0, \eta), X \neq 0$. Choose $\beta \in E_6$ so that $\beta X=\xi_1 E_1+\xi_2 E_2+\xi_3 E_3$,
0902.0431_F02221	132	■1.000	$\beta X=\xi_1 E_1+\xi_2 E_2+\xi_3 E_3$	$\beta X=\xi_1 E_1+\xi_2 E_2+\xi_3 E_3$	(3) Case $P=(X, 0, 0, \eta), X \neq 0$. Choose $\beta \in E_6$ so that $\beta X=\xi_1 E_1+\xi_2 E_2+\xi_3 E_3$,
0902.0431_F02222	132	■0.943	$\xi_i \in C$	$\xi_i \in C$	$\xi_i \in C$ (Proposition 3.8.2). Since $\beta X \neq 0$, some ξ_i is non-zero: $\xi_i \neq 0$. Then
0902.0431_F02223	132	■0.943	$\beta X \neq 0$	$\beta X \neq 0$	$\xi_i \in C$ (Proposition 3.8.2). Since $\beta X \neq 0$, some ξ_i is non-zero: $\xi_i \neq 0$. Then
0902.0431_F02224	132	■0.943	ξ_i	ξ_i	$\xi_i \in C$ (Proposition 3.8.2). Since $\beta X \neq 0$, some ξ_i is non-zero: $\xi_i \neq 0$. Then
0902.0431_F02225	132	■0.943	$\xi_i \neq 0$	$\xi_i \neq 0$	$\xi_i \in C$ (Proposition 3.8.2). Since $\beta X \neq 0$, some ξ_i is non-zero: $\xi_i \neq 0$. Then
0902.0431_F02226	133	■1.000	$P=(0, 0, 0, \eta), \eta \neq 0$	$P=(0, 0, 0, \eta), \eta \neq 0$	(4) Case $P=(0, 0, 0, \eta), \eta \neq 0$. Then
0902.0431_F02227	133	■1.000	$\phi_1(\theta) \in U(1) \subset \mathrm{in}\left(E_7\right)_{\{0\}}$	$\phi_1(\theta) \in U(1) \subset (E_7)_0$	Furthermore, by applying some $\phi_1(\theta) \in U(1) \subset (E_7)_0$ of Theorem 4.7.3 on it, then ξ
0902.0431_F02228	133	■1.000	ξ	ξ	Furthermore, by applying some $\phi_1(\theta) \in U(1) \subset (E_7)_0$ of Theorem 4.7.3 on it, then ξ
0902.0431_F02229	133	■1.000	$\xi>0$	$\xi>0$	becomes $\xi>0$. Thus the proof of Proposition 4.8.2 is completed.
0902.0431_F02230	133	■1.000	$\mathfrak{M}_1$	$\mathfrak{M}_1$	We define a space $\mathfrak{M}_1$, called the compact Freudenthal manifold, by
0902.0431_F02231	133	■1.000	$P \in \mathfrak{M}_1$	$P \in \mathfrak{M}_1$	Proof. For $\alpha \in E_7$ and $P \in \mathfrak{M}_1$, we have $\alpha P \in \mathfrak{M}_1$. Hence E_7 acts on $\mathfrak{M}_1$. We
0902.0431_F02232	133	■1.000	$\alpha P \in \mathfrak{M}_1$	$\alpha P \in \mathfrak{M}_1$	Proof. For $\alpha \in E_7$ and $P \in \mathfrak{M}_1$, we have $\alpha P \in \mathfrak{M}_1$. Hence E_7 acts on $\mathfrak{M}_1$. We
0902.0431_F02233	133	■0.998	$(0, 0, 1, 0) \in \mathfrak{M}_1$	$(0, 0, 1, 0) \in \mathfrak{M}_1$	to show that any element $P \in \mathfrak{M}_1$ can be transformed to $(0, 0, 1, 0) \in \mathfrak{M}_1$ by some
0902.0431_F02234	133	■1.000	$\langle \alpha P, \alpha P \rangle = \langle P, P \rangle = 1$	$\langle \alpha P, \alpha P \rangle = \langle P, P \rangle = 1$	(Proposition 4.8.2). From the condition $\langle \alpha P, \alpha P \rangle = \langle P, P \rangle = 1$, we have
0902.0431_F02235	133	■1.000	$r_1, r_2, r_3 \in \boldsymbol{R}, 0 \leq r_i < \frac{\pi}{2}$	$r_1, r_2, r_3 \in \boldsymbol{R}, 0 \leq r_i < \frac{\pi}{2}$	Choose $r_1, r_2, r_3 \in \boldsymbol{R}, 0 \leq r_i < \frac{\pi}{2}$, such that
0902.0431_F02236	134	■0.952	$\eta_i=0$	$\eta_i=0$	(if $\eta_i=0$, then a_i means 0), we have
0902.0431_F02237	134	■1.000	αP	αP	Therefore, we see that αP is of the form
0902.0431_F02238	134	■1.000	$=\alpha_3(a_3) \alpha_2(a_2) \alpha_1(a_1)(0, 0, 1, 0)$	$=\alpha_3(a_3) \alpha_2(a_2) \alpha_1(a_1)(0, 0, 1, 0)$	$=\alpha_3(a_3) \alpha_2(a_2) \alpha_1(a_1)(0, 0, 1, 0)$,
0902.0431_F02239	134	■1.000	$\mathfrak{M}_1=(E_7)_0(0, 0, 1, 0), \mathfrak{M}_1$	$\mathfrak{M}_1=(E_7)_0(0, 0, 1, 0), \mathfrak{M}_1$	This shows the transitivity of $(E_7)_0$. Since we have $\mathfrak{M}_1=(E_7)_0(0, 0, 1, 0)$, $\mathfrak{M}_1$ is
0902.0431_F02240	134	■1.000	$E_7 / E_6 \simeq \mathfrak{M}_1$	$E_7 / E_6 \simeq \mathfrak{M}_1$	$E_7 / E_6 \simeq \mathfrak{M}_1$. Finally, the connectedness of E_7 follows from the connectedness of $\mathfrak{M}_1$
0902.0431_F02241	134	■1.000	$\alpha \in z\left(E_7\right)$	$\alpha \in z\left(E_7\right)$	Proof. Let $\alpha \in z\left(E_7\right)$. From the commutativity with $\beta \in E_6 \subset E_7$, we have
0902.0431_F02242	134	■1.000	$\beta \in E_6 \subset E_7$	$\beta \in E_6 \subset E_7$	Proof. Let $\alpha \in z\left(E_7\right)$. From the commutativity with $\beta \in E_6 \subset E_7$, we have
0902.0431_F02243	134	■1.000	$\beta \alpha(0, 0, 1, 0)=\alpha \beta(0, 0, 1, 0)=\alpha(0, 0, 1, 0)$	$\beta \alpha(0, 0, 1, 0)=\alpha \beta(0, 0, 1, 0)=\alpha(0, 0, 1, 0)$	$\beta \alpha(0, 0, 1, 0)=\alpha \beta(0, 0, 1, 0)=\alpha(0, 0, 1, 0)$. If we denote $\alpha(0, 0, 1, 0)=(X, Y, \xi, \eta) \in$
0902.0431_F02244	134	■1.000	$\alpha(0, 0, 1, 0)=(X, Y, \xi, \eta) \in$	$\alpha(0, 0, 1, 0)=(X, Y, \xi, \eta) \in$	$\beta \alpha(0, 0, 1, 0)=\alpha \beta(0, 0, 1, 0)=\alpha(0, 0, 1, 0)$. If we denote $\alpha(0, 0, 1, 0)=(X, Y, \xi, \eta) \in$
0902.0431_F02245	134	■1.000	$(\beta X, \tau \beta \tau Y, \xi, \eta)=(X, Y, \xi, \eta)$	$(\beta X, \tau \beta \tau Y, \xi, \eta)=(X, Y, \xi, \eta)$	$\mathfrak{P}^C$, then from $(\beta X, \tau \beta \tau Y, \xi, \eta)=(X, Y, \xi, \eta)$, we have
0902.0431_F02246	135	■1.000	$X=Y=0$	$X=Y=0$	Hence $X=Y=0$. Therefore, $\alpha(0, 0, 1, 0)$ is of the form
0902.0431_F02247	135	■1.000	$\alpha(0, 0, 1, 0)$	$\alpha(0, 0, 1, 0)$	Hence $X=Y=0$. Therefore, $\alpha(0, 0, 1, 0)$ is of the form
0902.0431_F02248	135	■1.000	$\alpha(0, 0, 1, 0) \in \mathfrak{M}^C$	$\alpha(0, 0, 1, 0) \in \mathfrak{M}^C$	From the condition $\alpha(0, 0, 1, 0) \in \mathfrak{M}^C$, we have $\xi \eta=0$. Suppose $\xi=0$, then
0902.0431_F02249	135	■1.000	$\xi \eta=0$	$\xi \eta=0$	From the condition $\alpha(0, 0, 1, 0) \in \mathfrak{M}^C$, we have $\xi \eta=0$. Suppose $\xi=0$, then
0902.0431_F02250	135	■1.000	$\xi=0$	$\xi=0$	From the condition $\alpha(0, 0, 1, 0) \in \mathfrak{M}^C$, we have $\xi \eta=0$. Suppose $\xi=0$, then
0902.0431_F02251	135	■1.000	$\alpha(0, 0, 1, 0)=(0, 0, 0, \eta), \eta \neq 0$	$\alpha(0, 0, 1, 0)=(0, 0, 0, \eta), \eta \neq 0$	$\alpha(0, 0, 1, 0)=(0, 0, 0, \eta), \eta \neq 0$. Also from the commutativity with $\varphi_1(\theta) \in U(1) \subset E_7$

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0902.0431_F02252	135	■ 1.000	$\varphi_1(\theta) \in U(1) \subset E_7$	$\varphi_1(\theta) \in U(1) \subset E_7$	$\alpha(0,0,1,0) = (0,0,0,\eta), \eta \neq 0$. Also from the commutativity with $\varphi_1(\theta) \in U(1) \subset E_7$
0902.0431_F02253	135	■ 1.000	$\theta^{\{-3\}} \eta = \theta^{\{3\}} \eta$	$\theta^{-3} \eta = \theta^3 \eta$	and so $\theta^{-3} \eta = \theta^3 \eta$ for all θ . But this is a contradiction. Hence $\xi \neq 0, \eta = 0$,
0902.0431_F02254	135	■ 1.000	θ	θ	and so $\theta^{-3} \eta = \theta^3 \eta$ for all θ . But this is a contradiction. Hence $\xi \neq 0, \eta = 0$,
0902.0431_F02255	135	■ 1.000	$\xi \neq 0, \eta = 0$	$\xi \neq 0, \eta = 0$	and so $\theta^{-3} \eta = \theta^3 \eta$ for all θ . But this is a contradiction. Hence $\xi \neq 0, \eta = 0$,
0902.0431_F02256	135	■ 1.000	$\alpha(0,0,1,0) = (0,0, \xi, 0)$	$\alpha(0,0,1,0) = (0,0,\xi,0)$	that is, $\alpha(0,0,1,0) = (0,0,\xi,0)$. Similarly we have $\alpha(0,0,0,1) = (0,0,0,\zeta)$. Since
0902.0431_F02257	135	■ 1.000	$\alpha(0,0,0,1) = (0,0,0, \zeta)$	$\alpha(0,0,0,1) = (0,0,0,\zeta)$	that is, $\alpha(0,0,1,0) = (0,0,\xi,0)$. Similarly we have $\alpha(0,0,0,1) = (0,0,0,\zeta)$. Since
0902.0431_F02258	135	■ 1.000	$\{\alpha(0,0,1,0), \alpha(0,0,1,0)\} = 1$	$\{\alpha(0,0,1,0), \alpha(0,0,1,0)\} = 1$	$\{\alpha(0,0,1,0), \alpha(0,0,1,0)\} = 1$, we have $\xi\zeta = 1$, and therefore
0902.0431_F02259	135	■ 1.000	$\xi\zeta = 1$	$\xi\zeta = 1$	$\{\alpha(0,0,1,0), \alpha(0,0,1,0)\} = 1$, we have $\xi\zeta = 1$, and therefore
0902.0431_F02260	135	■ 1.000	$\xi = \xi^{-1}$	$\xi = \xi^{-1}$	Hence $\xi = \xi^{-1}$, that is, $\xi = \pm 1$. In the case $\xi = 1$, we have $\alpha \in E_6$ (Theorem 4.7.2),
0902.0431_F02261	135	■ 1.000	$\xi = \pm 1$	$\xi = \pm 1$	Hence $\xi = \xi^{-1}$, that is, $\xi = \pm 1$. In the case $\xi = 1$, we have $\alpha \in E_6$ (Theorem 4.7.2),
0902.0431_F02262	135	■ 1.000	$\xi = 1$	$\xi = 1$	Hence $\xi = \xi^{-1}$, that is, $\xi = \pm 1$. In the case $\xi = 1$, we have $\alpha \in E_6$ (Theorem 4.7.2),
0902.0431_F02263	135	■ 1.000	$\alpha \in z(E_6) = \{1, \omega 1, \omega^2 1\}$	$\alpha \in z(E_6) = \{1, \omega 1, \omega^2 1\}$	so that $\alpha \in z(E_6) = \{1, \omega 1, \omega^2 1\}$ (Theorem 3.9.1), that is,
0902.0431_F02264	135	■ 1.000	$\omega' = \omega'^{-1}$	$\omega' = \omega'^{-1}$	for all $X \in \mathfrak{J}^C$, which shows that $\omega' = \omega'^{-1}$, hence $\omega' = 1$. Therefore $\alpha = 1$. In the
0902.0431_F02265	135	■ 1.000	$\omega' = 1$	$\omega' = 1$	for all $X \in \mathfrak{J}^C$, which shows that $\omega' = \omega'^{-1}$, hence $\omega' = 1$. Therefore $\alpha = 1$. In the
0902.0431_F02266	135	■ 1.000	$\xi = -1$	$\xi = -1$	case $\xi = -1$, we have $-\alpha \in z(E_6)$, so that by the similar argument as above we have
0902.0431_F02267	135	■ 1.000	$-\alpha \in z(E_6)$	$-\alpha \in z(E_6)$	case $\xi = -1$, we have $-\alpha \in z(E_6)$, so that by the similar argument as above we have
0902.0431_F02268	135	■ 1.000	$-\alpha = 1$	$-\alpha = 1$	$-\alpha = 1$. Thus we have $z(E_7) = \{1, -1\}$.
0902.0431_F02269	135	■ 1.000	$z(E_7) = \{1, -1\}$	$z(E_7) = \{1, -1\}$	$-\alpha = 1$. Thus we have $z(E_7) = \{1, -1\}$.
0902.0431_F02270	135	■ 1.000	$E_7 = \{\alpha \in \text{Iso}_C(\mathfrak{P}^C) \mid \alpha(P \times Q)\alpha^{-1} = \alpha P \times \alpha Q, \langle \alpha P, \alpha Q \rangle =$	$E_7 = \{\alpha \in \text{Iso}_C(\mathfrak{P}^C) \mid \alpha(P \times Q)\alpha^{-1} = \alpha P \times \alpha Q, \langle \alpha P, \alpha Q \rangle =$	Theorem 4.9.2. $E_7 = \{\alpha \in \text{Iso}_C(\mathfrak{P}^C) \mid \alpha(P \times Q)\alpha^{-1} = \alpha P \times \alpha Q, \langle \alpha P, \alpha Q \rangle =$
0902.0431_F02271	135	■ 0.983	$\langle P, Q \rangle$	$\langle P, Q \rangle$	$\langle P, Q \rangle$ is a simply connected compact Lie group of type E_7 .
0902.0431_F02272	135	■ 0.857	$\text{subgroup}(U(1) \times E_6) / \mathbf{Z}_3$	$\text{subgroup}(U(1) \times E_6) / \mathbf{Z}_3$	4.10. Involution ι and subgroup $(U(1) \times E_6) / \mathbf{Z}_3$ of E_7
0902.0431_F02273	136	■ 1.000	$\iota = \varphi_1(i) \in U(1) \subset E_7, \iota^2 = -1 \in z(E_7)$	$\iota = \varphi_1(i) \in U(1) \subset E_7, \iota^2 = -1 \in z(E_7)$	Then, $\iota = \varphi_1(i) \in U(1) \subset E_7, \iota^2 = -1 \in z(E_7)$ (Theorem 4.9.1) and so $\iota^4 = 1$.
0902.0431_F02274	136	■ 1.000	$\iota^4 = 1$	$\iota^4 = 1$	Then, $\iota = \varphi_1(i) \in U(1) \subset E_7, \iota^2 = -1 \in z(E_7)$ (Theorem 4.9.1) and so $\iota^4 = 1$.
0902.0431_F02275	136	■ 1.000	$\delta = \alpha_1 \left(\frac{i\pi}{4} \right) \alpha_2 \left(\frac{i\pi}{4} \right) \alpha_3 \left(\frac{i\pi}{4} \right)$	$\delta = \alpha_1 \left(\frac{i\pi}{4} \right) \alpha_2 \left(\frac{i\pi}{4} \right) \alpha_3 \left(\frac{i\pi}{4} \right)$	Proof. For $\delta = \alpha_1 \left(\frac{i\pi}{4} \right) \alpha_2 \left(\frac{i\pi}{4} \right) \alpha_3 \left(\frac{i\pi}{4} \right)$ (Lemma 4.8.1), we have $\iota = \delta^{-1} \lambda \delta$.
0902.0431_F02276	136	■ 1.000	$\iota = \delta^{-1} \lambda \delta$	$\iota = \delta^{-1} \lambda \delta$	Proof. For $\delta = \alpha_1 \left(\frac{i\pi}{4} \right) \alpha_2 \left(\frac{i\pi}{4} \right) \alpha_3 \left(\frac{i\pi}{4} \right)$ (Lemma 4.8.1), we have $\iota = \delta^{-1} \lambda \delta$.
0902.0431_F02277	136	■ 0.562	$\widetilde{\iota} : E_7 \rightarrow E_7$	$\widetilde{\iota} : E_7 \rightarrow E_7$	ι induces an involutive automorphism $\widetilde{\iota} : E_7 \rightarrow E_7$ by
0902.0431_F02278	136	■ 1.000	$(E_7)^\iota$	$(E_7)^\iota$	We shall now study the following subgroup $(E_7)^\iota$ of E_7 :
0902.0431_F02279	136	■ 1.000	$(E_7)^\iota \cong (U(1) \times E_6) / \mathbf{Z}_3, \quad \mathbf{Z}_3 = \{(1,1), (\omega, \omega 1), (\omega^2, \omega^2 1)\}$	$(E_7)^\iota \cong (U(1) \times E_6) / \mathbf{Z}_3, \quad \mathbf{Z}_3 = \{(1,1), (\omega, \omega 1), (\omega^2, \omega^2 1)\}$	Theorem 4.10.2 $(E_7)^\iota \cong (U(1) \times E_6) / \mathbf{Z}_3, \quad \mathbf{Z}_3 = \{(1,1), (\omega, \omega 1), (\omega^2, \omega^2 1)\}$,
0902.0431_F02280	136	■ 0.999	$\omega = -\frac{1}{2} + \frac{\sqrt{3}}{2}i \in C$	$\omega = -\frac{1}{2} + \frac{\sqrt{3}}{2}i \in C$	$\omega = -\frac{1}{2} + \frac{\sqrt{3}}{2}i \in C$.
0902.0431_F02281	136	■ 1.000	$\varphi : U(1) \times E_6 \rightarrow (E_7)^\iota$	$\varphi : U(1) \times E_6 \rightarrow (E_7)^\iota$	Proof. We define a mapping $\varphi : U(1) \times E_6 \rightarrow (E_7)^\iota$ by
0902.0431_F02282	136	■ 1.000	$\varphi(\theta, \beta) \in (E_7)^\iota$	$\varphi(\theta, \beta) \in (E_7)^\iota$	Evidently $\varphi(\theta, \beta) \in (E_7)^\iota$. Since $\varphi_1(\theta)$ and β are commutative, φ is a homomorphism.
0902.0431_F02283	136	■ 1.000	$\varphi_1(\theta)$	$\varphi_1(\theta)$	Evidently $\varphi(\theta, \beta) \in (E_7)^\iota$. Since $\varphi_1(\theta)$ and β are commutative, φ is a homomorphism.
0902.0431_F02284	136	■ 1.000	$\alpha \in (E_7)^\iota$	$\alpha \in (E_7)^\iota$	We shall prove that φ is onto. Let $\alpha \in (E_7)^\iota$. From $\iota\alpha = \alpha\iota$, α is seen to be of the
0902.0431_F02285	136	■ 1.000	$\iota\alpha = \alpha\iota, \alpha$	$\iota\alpha = \alpha\iota, \alpha$	We shall prove that φ is onto. Let $\alpha \in (E_7)^\iota$. From $\iota\alpha = \alpha\iota$, α is seen to be of the
0902.0431_F02286	136	■ 1.000	$\alpha(0,0,1,0), \alpha(0,0,0,1) \in \mathfrak{M}^C$	$\alpha(0,0,1,0), \alpha(0,0,0,1) \in \mathfrak{M}^C$	The condition $\alpha(0,0,1,0), \alpha(0,0,0,1) \in \mathfrak{M}^C$ implies that
0902.0431_F02287	136	■ 1.000	$M = N = 0$	$M = N = 0$	We shall first show that $M = N = 0$. Suppose that $M \neq 0, \mu = 0$. Then, the
0902.0431_F02288	136	■ 1.000	$M \neq 0, \mu = 0$	$M \neq 0, \mu = 0$	We shall first show that $M = N = 0$. Suppose that $M \neq 0, \mu = 0$. Then, the

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0902.0431_F02289	136	■ 1.000	$\backslash\{\alpha(0,0,1,0), \alpha(0,0,0,1)\}=\backslash\{(0,0,1,0), (0,0,0,1)\}=1$	$\{\alpha(0,0,1,0), \alpha(0,0,0,1)\} = \{(0,0,1,0), (0,0,0,1)\} = 1$	condition $\backslash\{\alpha(0,0,1,0), \alpha(0,0,0,1)\} = \{(0,0,1,0), (0,0,0,1)\} = 1$ implies that
0902.0431_F02290	136	■ 1.000	$N \neq 0, \nu=0$	$N \neq 0, \nu = 0$	Hence we have $N \neq 0, \nu = 0$. From
0902.0431_F02291	137	■ 1.000	$a(X) \operatorname{det} X=a(X)^2 b(X \times X)$	$a(X) \det X = a(X)^2 b(X \times X)$	Hence we have $a(X) \det X = a(X)^2 b(X \times X)$. Furthermore we have
0902.0431_F02292	137	■ 1.000	$\mu=0$	$\mu = 0$	Indeed, from $\mu = 0$, we deduce that $a \neq 0$. Since $a : \mathfrak{J}^C \rightarrow C$ is a linear form, the
0902.0431_F02293	137	■ 1.000	$a \neq 0$	$a \neq 0$	Indeed, from $\mu = 0$, we deduce that $a \neq 0$. Since $a : \mathfrak{J}^C \rightarrow C$ is a linear form, the
0902.0431_F02294	137	■ 1.000	$a: \mathfrak{J}^C \rightarrow C$	$a : \mathfrak{J}^C \rightarrow C$	Indeed, from $\mu = 0$, we deduce that $a \neq 0$. Since $a : \mathfrak{J}^C \rightarrow C$ is a linear form, the
0902.0431_F02295	137	■ 0.993	$\backslash\left\{X \in \mathfrak{J}^C \mid a(X) \neq 0\right\}$	$\{X \in \mathfrak{J}^C \mid a(X) \neq 0\}$	set $\{X \in \mathfrak{J}^C \mid a(X) \neq 0\}$ is dense in $\mathfrak{J}^C$ and the correspondence $\det X$ and $b(X \times X)$
0902.0431_F02296	137	■ 0.993	$b(X \times X)$	$b(X \times X)$	set $\{X \in \mathfrak{J}^C \mid a(X) \neq 0\}$ is dense in $\mathfrak{J}^C$ and the correspondence $\det X$ and $b(X \times X)$
0902.0431_F02297	137	■ 1.000	$a(X)=0$	$a(X) = 0$	is continuous with respect to X , (v) is also valid for $X \in \mathfrak{J}^C$ such that $a(X) = 0$.
0902.0431_F02298	137	■ 1.000	$M=0$	$M = 0$	that $M = 0$. Similarly we can prove that $N = 0$. Therefore $\alpha(0,0,1,0) = (0,0,\mu,0)$,
0902.0431_F02299	137	■ 1.000	$N=0$	$N = 0$	that $M = 0$. Similarly we can prove that $N = 0$. Therefore $\alpha(0,0,1,0) = (0,0,\mu,0)$,
0902.0431_F02300	137	■ 1.000	$\alpha(0,0,1,0)=(0,0, \mu, 0)$	$\alpha(0,0,1,0) = (0,0,\mu,0)$	that $M = 0$. Similarly we can prove that $N = 0$. Therefore $\alpha(0,0,1,0) = (0,0,\mu,0)$,
0902.0431_F02301	137	■ 0.627	$\alpha(0,0,0,1)=(0,0,0, \nu)$	$\alpha(0,0,0,1) = (0,0,0,\nu)$	$\alpha(0,0,0,1) = (0,0,0,\nu)$. From the condition $\{\alpha\dot{1}, \alpha\dot{1}\} = 1, \langle \alpha\dot{1}, \alpha\dot{1} \rangle = 1$, we deduce
0902.0431_F02302	137	■ 0.627	$\backslash\{\alpha \mathrm{i}, \alpha \underline{1}\}=1, \backslash\angle \alpha \mathrm{i}, \alpha \mathrm{i} \backslash\angle=1$	$\{\alpha\dot{1}, \alpha\dot{1}\} = 1, \langle \alpha\dot{1}, \alpha\dot{1} \rangle = 1$	$\alpha(0,0,0,1) = (0,0,0,\nu)$. From the condition $\{\alpha\dot{1}, \alpha\dot{1}\} = 1, \langle \alpha\dot{1}, \alpha\dot{1} \rangle = 1$, we deduce
0902.0431_F02303	137	■ 1.000	$\theta^3=\mu$	$\theta^3 = \mu$	If we choose $\theta \in C$ such that $\theta^3 = \mu$ and let $\beta = \varphi_1(\theta)^{-1}\alpha$, then $\beta(0,0,1,0) =$
0902.0431_F02304	137	■ 1.000	$\beta=\varphi_1(\theta)^{-1} \alpha$	$\beta = \varphi_1(\theta)^{-1}\alpha$	If we choose $\theta \in C$ such that $\theta^3 = \mu$ and let $\beta = \varphi_1(\theta)^{-1}\alpha$, then $\beta(0,0,1,0) =$
0902.0431_F02305	137	■ 1.000	$\beta(0,0,1,0)=$	$\beta(0,0,1,0) =$	If we choose $\theta \in C$ such that $\theta^3 = \mu$ and let $\beta = \varphi_1(\theta)^{-1}\alpha$, then $\beta(0,0,1,0) =$
0902.0431_F02306	137	■ 1.000	$(0,0,1,0), \beta(0,0,0,1)=(0,0,0,1)$	$(0,0,1,0), \beta(0,0,0,1) = (0,0,0,1)$	$\beta(0,0,0,1) = (0,0,0,1)$. Hence, $\beta \in E_6$ (Theorem 4.7.2) and we have
0902.0431_F02307	138	■ 0.966	$\operatorname{Ker} \varphi=\left\{(1,1),\left(\omega,\omega^2\right)\right\}=\mathbf{Z}_3$	$\operatorname{Ker} \varphi = \{(1,1), (\omega, \omega^2), (\omega^2, \omega^2 1)\} = \mathbf{Z}_3$	This shows φ is onto. That $\operatorname{Ker} \varphi = \{(1,1), (\omega, \omega^2), (\omega^2, \omega^2 1)\} = \mathbf{Z}_3$ is easily ob-
0902.0431_F02308	138	■ 0.999	$\left(U(1) \times E_6\right) / \mathbf{Z}_3 \cong\left(E_7\right)^t$	$(U(1) \times E_6) / \mathbf{Z}_3 \cong (E_7)^t$	tained. Thus we have the isomorphism $(U(1) \times E_6) / \mathbf{Z}_3 \cong (E_7)^t$.
0902.0431_F02309	138	■ 1.000	$\varphi: U(1) \times E_6 \rightarrow\left(E_6\right)^t$	$\varphi : U(1) \times E_6 \rightarrow (E_6)^t$	the simply connected Lie group E_7 . Hence, to show that $\varphi : U(1) \times E_6 \rightarrow (E_6)^t$ is
0902.0431_F02310	138	■ 1.000	$\varphi_*: \mathfrak{u}(1) \oplus \mathfrak{e}_6 \rightarrow\left(\mathfrak{e}_7\right)^t$	$\varphi_* : \mathfrak{u}(1) \oplus \mathfrak{e}_6 \rightarrow (\mathfrak{e}_7)^t$	onto, it is sufficient to show that $\varphi_* : \mathfrak{u}(1) \oplus \mathfrak{e}_6 \rightarrow (\mathfrak{e}_7)^t$ is onto, which is easily shown.
0902.0431_F02312	138	■ 1.000	$\sigma \in F_4 \subset E_6 \subset E_7$	$\sigma \in F_4 \subset E_6 \subset E_7$	$\sigma \in F_4$ regarding as $\sigma \in F_4 \subset E_6 \subset E_7$.
0902.0431_F02313	138	■ 1.000	$\left(E_7\right)^{\sigma}$	$(E_7)^\sigma$	We shall now study the following subgroup $(E_7)^\sigma$ of E_7 :
0902.0431_F02314	138	■ 1.000	$\kappa, \mu: \mathfrak{P}^C \rightarrow \mathfrak{P}^C$	$\kappa, \mu : \mathfrak{P}^C \rightarrow \mathfrak{P}^C$	To this end, we define two C -linear mappings $\kappa, \mu : \mathfrak{P}^C \rightarrow \mathfrak{P}^C$ by
0902.0431_F02315	138	■ 1.000	μ	μ	The explicit forms of κ and μ are given by
0902.0431_F02316	138	■ 0.997	$\quad \kappa \mu=\mu \kappa, \quad \left\{\begin{array}{l} \kappa \sigma=\sigma \kappa \\ \mu \sigma=\sigma \mu, \end{array}\right. \quad \left\{\begin{array}{l} \kappa \lambda=-\lambda \kappa \\ \mu \lambda=-\lambda \mu . \end{array}\right.$	$\kappa \mu = \mu \kappa, \quad \left\{\begin{array}{l} \kappa \sigma = \sigma \kappa \\ \mu \sigma = \sigma \mu, \end{array}\right. \quad \left\{\begin{array}{l} \kappa \lambda = -\lambda \kappa \\ \mu \lambda = -\lambda \mu . \end{array}\right.$	Lemma 4.11.1. (1) $\kappa \mu = \mu \kappa, \quad \left\{\begin{array}{l} \kappa \sigma = \sigma \kappa \\ \mu \sigma = \sigma \mu, \end{array}\right. \quad \left\{\begin{array}{l} \kappa \lambda = -\lambda \kappa \\ \mu \lambda = -\lambda \mu . \end{array}\right.$
0902.0431_F02317	139	■ 1.000	$\kappa \alpha=\alpha \kappa$	$\kappa \alpha = \alpha \kappa$	(2) If $\alpha \in E_7$ satisfies $\kappa \alpha = \alpha \kappa$, then α also satisfies $\sigma \alpha = \alpha \sigma$.
0902.0431_F02318	139	■ 1.000	$\sigma=\exp \pi i \kappa$	$\sigma = \exp \pi i \kappa$	(2) Since $\sigma = \exp \pi i \kappa$, we have $\sigma \alpha = (\exp \pi i \kappa) \alpha = \alpha (\exp \pi i \kappa) = \alpha \sigma$.
0902.0431_F02319	139	■ 1.000	$\sigma \alpha=(\exp \pi i \kappa) \alpha=\alpha(\exp \pi i \kappa)=\alpha \sigma$	$\sigma \alpha = (\exp \pi i \kappa) \alpha = \alpha (\exp \pi i \kappa) = \alpha \sigma$	(2) Since $\sigma = \exp \pi i \kappa$, we have $\sigma \alpha = (\exp \pi i \kappa) \alpha = \alpha (\exp \pi i \kappa) = \alpha \sigma$.
0902.0431_F02320	139	■ 1.000	$\left(E_7\right)^{\kappa, \mu}$	$(E_7)^{\kappa, \mu}$	We shall first study the following subgroups $(E_7)^{\kappa, \mu}$ and $((E_7)^{\kappa, \mu})_{(0, E_1, 0, 1)}$ of E_7 :
0902.0431_F02321	139	■ 1.000	$\left(\left(E_7\right)^{\kappa, \mu}\right)_{(0, E_1, 0, 1)}$	$((E_7)^{\kappa, \mu})_{(0, E_1, 0, 1)}$	We shall first study the following subgroups $(E_7)^{\kappa, \mu}$ and $((E_7)^{\kappa, \mu})_{(0, E_1, 0, 1)}$ of E_7 :
0902.0431_F02322	139	■ 1.000	$\left(\mathfrak{e}_7\right)^{\sigma},\left(\mathfrak{e}_7\right)^{\kappa, \mu},\left(\left(\mathfrak{e}_7\right)^{\kappa, \mu}\right)_{(0, E_1, 0, 1)}$	$(\mathfrak{e}_7)^\sigma, (\mathfrak{e}_7)^{\kappa, \mu}, ((\mathfrak{e}_7)^{\kappa, \mu})_{(0, E_1, 0, 1)}$	Proposition 4.11.2. The Lie algebras $(\mathfrak{e}_7)^\sigma, (\mathfrak{e}_7)^{\kappa, \mu}, ((\mathfrak{e}_7)^{\kappa, \mu})_{(0, E_1, 0, 1)}$ of the

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0902.0431_F02323	139	■ 1.000	$\left(E_7\right)^{\sigma}, \left(E_7\right)^{\kappa, \mu}, \left(\left(E_7\right)^{\kappa, \mu}\right)_{(0, E_1, 0, 1)}$		groups $\left(E_7\right)^{\sigma}, \left(E_7\right)^{\kappa, \mu}, \left(\left(E_7\right)^{\kappa, \mu}\right)_{(0, E_1, 0, 1)}$ are respectively given by
0902.0431_F02324	139	■ 1.000	$\Phi=\Phi(\phi, A,-\tau A, \nu) \in \mathfrak{e}_7$	$\Phi=\Phi(\phi, A,-\tau A, \nu) \in \mathfrak{e}_7$	(2) Suppose that $\Phi=\Phi(\phi, A,-\tau A, \nu) \in \mathfrak{e}_7$ satisfies $\kappa \Phi=\Phi \kappa$ and $\mu \Phi=\Phi \mu$.
0902.0431_F02325	139	■ 1.000	$\kappa \Phi=\Phi \kappa$	$\kappa \Phi=\Phi \kappa$	(2) Suppose that $\Phi=\Phi(\phi, A,-\tau A, \nu) \in \mathfrak{e}_7$ satisfies $\kappa \Phi=\Phi \kappa$ and $\mu \Phi=\Phi \mu$.
0902.0431_F02326	139	■ 1.000	$\mu \Phi=\Phi \mu$	$\mu \Phi=\Phi \mu$	(2) Suppose that $\Phi=\Phi(\phi, A,-\tau A, \nu) \in \mathfrak{e}_7$ satisfies $\kappa \Phi=\Phi \kappa$ and $\mu \Phi=\Phi \mu$.
0902.0431_F02327	139	■ 1.000	$\kappa \Phi P=\Phi \kappa P, P=(X, Y, \xi, \eta) \in \mathfrak{P}^C$	$\kappa \Phi P=\Phi \kappa P, P=(X, Y, \xi, \eta) \in \mathfrak{P}^C$	Comparing the ξ -term of $\kappa \Phi P=\Phi \kappa P, P=(X, Y, \xi, \eta) \in \mathfrak{P}^C$, we have
0902.0431_F02328	139	■ 0.915	$Y=E_1$	$Y=E_1$	Let $Y=E_1$, then we have $(A, E_1)=0$. Next, comparing the η -term of $\mu \Phi=\Phi \mu$, we
0902.0431_F02329	139	■ 0.915	$\left(A, E_1\right)=0$	$(A, E_1)=0$	Let $Y=E_1$, then we have $(A, E_1)=0$. Next, comparing the η -term of $\mu \Phi=\Phi \mu$, we
0902.0431_F02330	139	■ 1.000	$\phi \in\left(\mathfrak{e}_6\right)^{\sigma}$	$\phi \in\left(\mathfrak{e}_6\right)^{\sigma}$	Since $\phi \in\left(\mathfrak{e}_6\right)^{\sigma}$, we can set $\phi E_1=k E_1, k \in i \boldsymbol{R}$ (Lemma 3.10.1). Hence let $X=E_1$ in
0902.0431_F02331	139	■ 1.000	$\phi E_1=k E_1, k \in i \boldsymbol{R}$	$\phi E_1=k E_1, k \in i \boldsymbol{R}$	Since $\phi \in\left(\mathfrak{e}_6\right)^{\sigma}$, we can set $\phi E_1=k E_1, k \in i \boldsymbol{R}$ (Lemma 3.10.1). Hence let $X=E_1$ in
0902.0431_F02332	139	■ 1.000	$X=E_1$	$X=E_1$	Since $\phi \in\left(\mathfrak{e}_6\right)^{\sigma}$, we can set $\phi E_1=k E_1, k \in i \boldsymbol{R}$ (Lemma 3.10.1). Hence let $X=E_1$ in
0902.0431_F02333	139	■ 1.000	$k=-\frac{2}{3} \nu$	$k=-\frac{2}{3} \nu$	(i), then we have $k=-\frac{2}{3} \nu$. Conversely, if $\Phi=\Phi(\phi, A,-\tau A, \nu) \in \mathfrak{e}_7$ has the condition
0902.0431_F02334	139	■ 0.976	$A \in\left(\mathfrak{J}^C\right)_{\sigma}$	$A \in\left(\mathfrak{J}^C\right)_{\sigma}$	(1) For $A \in\left(\mathfrak{J}^C\right)_{\sigma}$, we have $\kappa_1(A \times X)=\kappa_1 A \times \kappa_1 X, X \in \mathfrak{J}^C$.
0902.0431_F02335	139	■ 0.976	$\kappa_1(A \times X)=\kappa_1 A \times \kappa_1 X, X \in \mathfrak{J}^C$	$\kappa_1(A \times X)=\kappa_1 A \times \kappa_1 X, X \in \mathfrak{J}^C$	(1) For $A \in\left(\mathfrak{J}^C\right)_{\sigma}$, we have $\kappa_1(A \times X)=\kappa_1 A \times \kappa_1 X, X \in \mathfrak{J}^C$.
0902.0431_F02336	139	■ 1.000	$\kappa_1 \phi=\phi \kappa_1$	$\kappa_1 \phi=\phi \kappa_1$	(2) For $\phi \in\left(\mathfrak{e}_6\right)^{\sigma}$, we have $\kappa_1 \phi=\phi \kappa_1$.
0902.0431_F02337	139	■ 1.000	$A \in\left(\mathfrak{J}^C\right)_{\sigma},\left(E_1, A\right)=0$	$A \in\left(\mathfrak{J}^C\right)_{\sigma},\left(E_1, A\right)=0$	(3) For $A \in\left(\mathfrak{J}^C\right)_{\sigma},\left(E_1, A\right)=0$, we have $\kappa_1 A=-A$ and
0902.0431_F02338	139	■ 1.000	$\kappa_1 A=-A$	$\kappa_1 A=-A$	(3) For $A \in\left(\mathfrak{J}^C\right)_{\sigma},\left(E_1, A\right)=0$, we have $\kappa_1 A=-A$ and
0902.0431_F02339	140	■ 1.000	$\Phi=\Phi(\phi, A,-\tau A, \nu) \in\left(\mathfrak{e}_7\right)^{\kappa, \mu}$	$\Phi=\Phi(\phi, A,-\tau A, \nu) \in\left(\mathfrak{e}_7\right)^{\kappa, \mu}$	If $\Phi=\Phi(\phi, A,-\tau A, \nu) \in\left(\mathfrak{e}_7\right)^{\kappa, \mu}$ satisfies $\Phi((0, E_1, 0, 1))=0$, then
0902.0431_F02340	140	■ 1.000	$\Phi((0, E_1, 0, 1))=0$	$\Phi((0, E_1, 0, 1))=0$	If $\Phi=\Phi(\phi, A,-\tau A, \nu) \in\left(\mathfrak{e}_7\right)^{\kappa, \mu}$ satisfies $\Phi((0, E_1, 0, 1))=0$, then
0902.0431_F02341	140	■ 1.000	$\nu=\tau \nu$	$\nu=\tau \nu$	so that $\nu=\tau \nu$. Together with $\tau \nu=-\nu$, we have $\nu=0$ and $\phi E_1=0$, furthermore,
0902.0431_F02342	140	■ 1.000	$\tau \nu=-\nu$	$\tau \nu=-\nu$	so that $\nu=\tau \nu$. Together with $\tau \nu=-\nu$, we have $\nu=0$ and $\phi E_1=0$, furthermore,
0902.0431_F02343	140	■ 1.000	$\nu=0$	$\nu=0$	so that $\nu=\tau \nu$. Together with $\tau \nu=-\nu$, we have $\nu=0$ and $\phi E_1=0$, furthermore,
0902.0431_F02344	140	■ 1.000	$\phi E_1=0$	$\phi E_1=0$	so that $\nu=\tau \nu$. Together with $\tau \nu=-\nu$, we have $\nu=0$ and $\phi E_1=0$, furthermore,
0902.0431_F02345	140	■ 1.000	$2 A \times E_1=\tau A$	$2 A \times E_1=\tau A$	we have $2 A \times E_1=\tau A$ (in this case, $A \in\left(\mathfrak{J}^C\right)_{\sigma}$ and $(E_1, A)=0$ automatically hold).
0902.0431_F02346	140	■ 1.000	$\left(E_1, A\right)=0$	$(E_1, A)=0$	we have $2 A \times E_1=\tau A$ (in this case, $A \in\left(\mathfrak{J}^C\right)_{\sigma}$ and $\left(E_1, A\right)=0$ automatically hold).
0902.0431_F02347	140	■ 1.000	$\nu \in i \boldsymbol{R}$	$\nu \in i \boldsymbol{R}$	For $\nu \in i \boldsymbol{R}$, we define a C -linear mapping $\phi(\nu): \mathfrak{J}^C \rightarrow \mathfrak{J}^C$ by
0902.0431_F02348	140	■ 1.000	$\phi(\nu): \mathfrak{J}^C \rightarrow \mathfrak{J}^C$	$\phi(\nu): \mathfrak{J}^C \rightarrow \mathfrak{J}^C$	For $\nu \in i \boldsymbol{R}$, we define a C -linear mapping $\phi(\nu): \mathfrak{J}^C \rightarrow \mathfrak{J}^C$ by
0902.0431_F02349	140	■ 1.000	$\phi(\nu) \in\left(\mathfrak{e}_6\right)^{\sigma}$	$\phi(\nu) \in\left(\mathfrak{e}_6\right)^{\sigma}$	(Lemma 3.4.2.(2)). Then $\phi(\nu) \in\left(\mathfrak{e}_6\right)^{\sigma}$.
0902.0431_F02350	140	■ 0.999	$\mathfrak{a}_1=\left\{\Phi(\phi(\nu), a E_1,-\tau a E_1, \nu) \mid a \in C, \nu \in i \boldsymbol{R}\right\}$	$\mathfrak{a}_1=\left\{\Phi(\phi(\nu), a E_1,-\tau a E_1, \nu) \mid a \in C, \nu \in i \boldsymbol{R}\right\}$	Proposition 4.11.4. (1) $\mathfrak{a}_1=\left\{\Phi(\phi(\nu), a E_1,-\tau a E_1, \nu) \mid a \in C, \nu \in i \boldsymbol{R}\right\}$
0902.0431_F02351	140	■ 1.000	$\left(\mathfrak{e}_7\right)^{\sigma}$	$\left(\mathfrak{e}_7\right)^{\sigma}$	is a Lie subalgebra of $\left(\mathfrak{e}_7\right)^{\sigma}$ and isomorphic to the Lie algebra $\mathfrak{su}(2)$.
0902.0431_F02352	140	■ 1.000	$\mathfrak{su}(2)$	$\mathfrak{su}(2)$	is a Lie subalgebra of $\left(\mathfrak{e}_7\right)^{\sigma}$ and isomorphic to the Lie algebra $\mathfrak{su}(2)$.
0902.0431_F02353	140	■ 1.000	$\mathfrak{a}_1$	$\mathfrak{a}_1$	(2) The Lie algebra $\left(\mathfrak{e}_7\right)^{\sigma}$ is isomorphic to the direct sum of Lie algebras $\mathfrak{a}_1$ and
0902.0431_F02354	140	■ 1.000	$\left(\mathfrak{e}_7\right)^{\kappa, \mu}$	$\left(\mathfrak{e}_7\right)^{\kappa, \mu}$	$\left(\mathfrak{e}_7\right)^{\kappa, \mu}$:
0902.0431_F02355	140	■ 1.000	$\varphi_*: \mathfrak{a}_1 \rightarrow \mathfrak{su}(2)=\left\{D \in M(2, C) \mid \tau\left({}^t D\right)=-D\right\}$	$\varphi_*: \mathfrak{a}_1 \rightarrow \mathfrak{su}(2)=\left\{D \in M(2, C) \mid \tau\left({}^t D\right)=-D\right\}$	Proof. (1) The mapping $\varphi_*: \mathfrak{a}_1 \rightarrow \mathfrak{su}(2)=\left\{D \in M(2, C) \mid \tau\left({}^t D\right)=-D\right\}$ defined
0902.0431_F02356	140	■ 0.993	$\nu'=\frac{1}{3} \nu+\frac{1}{2}\left(E_1, \phi E_1\right), a=\left(E_1, A\right)$	$\nu'=\frac{1}{3} \nu+\frac{1}{2}\left(E_1, \phi E_1\right), a=\left(E_1, A\right)$	where $\nu'=\frac{1}{3} \nu+\frac{1}{2}\left(E_1, \phi E_1\right), a=\left(E_1, A\right)$, gives an isomorphism as Lie algebras.
0902.0431_F02357	141	■ 1.000	$\alpha_i(a) \in E_7, i=2,3$	$\alpha_i(a) \in E_7, i=2,3$	where $\alpha_i(a) \in E_7, i=2,3$ are defined in Lemma 4.8.1.
0902.0431_F02358	141	■ 1.000	$\Phi(0,-\tau a E_i, a E_i, 0) \in\left(\mathfrak{e}_7\right)^{\kappa, \mu}$	$\Phi(0,-\tau a E_i, a E_i, 0) \in\left(\mathfrak{e}_7\right)^{\kappa, \mu}$	Proof. Since $\Phi(0,-\tau a E_i, a E_i, 0) \in\left(\mathfrak{e}_7\right)^{\kappa, \mu}$, we have $\alpha_i(a)=\exp \Phi(0,-\tau a E_i,$
0902.0431_F02359	141	■ 1.000	$\alpha_i(a)=\exp \Phi(0,-\tau a E_i$	$\alpha_i(a)=\exp \Phi(0,-\tau a E_i$	Proof. Since $\Phi(0,-\tau a E_i, a E_i, 0) \in\left(\mathfrak{e}_7\right)^{\kappa, \mu}$, we have $\alpha_i(a)=\exp \Phi(0,-\tau a E_i,$

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0902.0431_F02360	141	■ 0.999	$\backslash\mathrm{left.a\ E_{\{i\}},\ 0\right)}\ \backslash\mathrm{in}\backslash\mathrm{left}(E_{\{7\}\right)}^{\backslash\mathrm{kappa},\ \backslash\mathrm{mu}\},\ i=2,3$	$aE_i, 0) \in (E_7)^{\kappa,\mu}, i = 2, 3$	$aE_i, 0) \in (E_7)^{\kappa,\mu}, i = 2, 3$. Since $\alpha_2(a)$ and $\alpha_3(\tau a)$ are commutative, we have
0902.0431_F02361	141	■ 0.999	$\backslash\mathrm{alpha}_{\{2\}}(a)$	$\alpha_2(a)$	$aE_i, 0) \in (E_7)^{\kappa,\mu}, i = 2, 3$. Since $\alpha_2(a)$ and $\alpha_3(\tau a)$ are commutative, we have
0902.0431_F02362	141	■ 0.999	$\backslash\mathrm{alpha}_{\{3\}}(\backslash\mathrm{tau}\ a)$	$\alpha_3(\tau a)$	$aE_i, 0) \in (E_7)^{\kappa,\mu}, i = 2, 3$. Since $\alpha_2(a)$ and $\alpha_3(\tau a)$ are commutative, we have
0902.0431_F02363	141	■ 1.000	$\backslash\mathrm{Phi}\backslash\mathrm{left}(0,-\backslash\mathrm{tau}\ a\ E_{\{2\}}-a\ E_{\{3\}},\ a\ E_{\{2\}}+\backslash\mathrm{tau}\ a\ E_{\{3\}},\ 0\right)}\ \backslash\mathrm{in}\backslash\mathrm{left}(\backslash\mathrm{left}(\backslash\mathrm{mathfrak{e}}_{\{7\}\right)}^{\backslash\mathrm{kappa},\ \backslash\mathrm{mu}\}\backslash\mathrm{right})_{\backslash\mathrm{left}(0,\ E_{\{1\}},\ 0,1\right)}\}$	$\Phi(0, -\tau aE_2 - aE_3, aE_2 + \tau aE_3, 0) \in ((\mathfrak{e}_7)^{\kappa,\mu})_{(0,E_1,0,1)}$	Since $\Phi(0, -\tau aE_2 - aE_3, aE_2 + \tau aE_3, 0) \in ((\mathfrak{e}_7)^{\kappa,\mu})_{(0,E_1,0,1)}$, we have $\alpha_{23}(a) \in$
0902.0431_F02364	141	■ 1.000	$\backslash\mathrm{alpha}_{\{23\}}(a)\ \backslash\mathrm{in}$	$\alpha_{23}(a) \in$	Since $\Phi(0, -\tau aE_2 - aE_3, aE_2 + \tau aE_3, 0) \in ((\mathfrak{e}_7)^{\kappa,\mu})_{(0,E_1,0,1)}$, we have $\alpha_{23}(a) \in$
0902.0431_F02365	141	■ 1.000	$\backslash\mathrm{left}(\backslash\mathrm{left}(E_{\{7\}\right)}^{\backslash\mathrm{kappa},\ \backslash\mathrm{nu}\}\backslash\mathrm{right})_{\backslash\mathrm{left}(0,\ E_{\{1\}},\ 0,1\right)}\}$	$((E_7)^{\kappa,\nu})_{(0,E_1,0,1)}$	$((E_7)^{\kappa,\nu})_{(0,E_1,0,1)}$.
0902.0431_F02366	141	■ 1.000	$\backslash\mathrm{alpha}\ \backslash\mathrm{in}\backslash\mathrm{left}(\backslash\mathrm{left}(E_{\{7\}\right)}^{\backslash\mathrm{kappa},\ \backslash\mathrm{mu}\}\backslash\mathrm{right})_{\backslash\mathrm{left}(0,\ E_{\{1\}},\ 0,1\right)}\}$	$\alpha \in ((E_7)^{\kappa,\mu})_{(0,E_1,0,1)}$	Lemma 4.11.6. For $\alpha \in ((E_7)^{\kappa,\mu})_{(0,E_1,0,1)}$, we have
0902.0431_F02367	141	■ 1.000	$\backslash\mathrm{alpha}\ \backslash\mathrm{in}\backslash\mathrm{left}(E_{\{7\}\right)}^{\backslash\mathrm{kappa},\ \backslash\mathrm{mu}\}$	$\alpha \in (E_7)^{\kappa,\mu}$	Proof. If $\alpha \in (E_7)^{\kappa,\mu}$ satisfies $\alpha(0, E_1, 0, 1) = (0, E_1, 0, 1)$ and $\alpha(0, -E_1, 0, 1) =$
0902.0431_F02368	141	■ 1.000	$\backslash\mathrm{alpha}\backslash\mathrm{left}(0,\ E_{\{1\}},\ 0,1\right)}=\backslash\mathrm{left}(0,\ E_{\{1\}},\ 0,1\right)}$	$\alpha(0, E_1, 0, 1) = (0, E_1, 0, 1)$	Proof. If $\alpha \in (E_7)^{\kappa,\mu}$ satisfies $\alpha(0, E_1, 0, 1) = (0, E_1, 0, 1)$ and $\alpha(0, -E_1, 0, 1) =$
0902.0431_F02369	141	■ 1.000	$\backslash\mathrm{alpha}\backslash\mathrm{left}(0,-E_{\{1\}},\ 0,1\right)}=$	$\alpha(0, -E_1, 0, 1) =$	Proof. If $\alpha \in (E_7)^{\kappa,\mu}$ satisfies $\alpha(0, E_1, 0, 1) = (0, E_1, 0, 1)$ and $\alpha(0, -E_1, 0, 1) =$
0902.0431_F02370	141	■ 0.989	$\backslash\mathrm{left}(0,-E_{\{1\}},\ 0,1\right)}$	$(0, -E_1, 0, 1)$	$(0, -E_1, 0, 1)$, then we have $\alpha(0, 0, 0, 1) = (0, 0, 0, 1)$ and $\alpha(0, E_1, 0, 0) = (0, E_1, 0, 0)$,
0902.0431_F02371	141	■ 0.989	$\backslash\mathrm{alpha}\backslash\mathrm{left}(0,\ E_{\{1\}},\ 0,0\right)}=\backslash\mathrm{left}(0,\ E_{\{1\}},\ 0,0\right)}$	$\alpha(0, E_1, 0, 0) = (0, E_1, 0, 0)$	$(0, -E_1, 0, 1)$, then we have $\alpha(0, 0, 0, 1) = (0, 0, 0, 1)$ and $\alpha(0, E_1, 0, 0) = (0, E_1, 0, 0)$,
0902.0431_F02372	141	■ 1.000	$\backslash\mathrm{alpha}(0,0,1,0)=\backslash\mathrm{alpha}\ \backslash\mathrm{mu}\backslash\mathrm{left}(0,\ E_{\{1\}},\ 0,0\right)}=\backslash\mathrm{mu}\ \backslash\mathrm{alpha}\backslash\mathrm{left}(0,\ E_{\{1\}},\ 0,0\right)}=\backslash\mathrm{mu}\backslash\mathrm{left}(0,\ E_{\{1\}},\ 0,0\right)}=$	$\alpha(0, 0, 1, 0) = \alpha\mu(0, E_1, 0, 0) = \mu\alpha(0, E_1, 0, 0) = \mu(0, E_1, 0, 0) =$	which imply that $\alpha(0, 0, 1, 0) = \alpha\mu(0, E_1, 0, 0) = \mu\alpha(0, E_1, 0, 0) = \mu(0, E_1, 0, 0) =$
0902.0431_F02373	141	■ 0.957	$\backslash\mathrm{alpha}(0,0,1,0)=$	$\alpha(0, 0, 1, 0) =$	$(0, 0, 1, 0)$. The converse can be similarly proved. If $\alpha \in E_7$ satisfies $\alpha(0, 0, 1, 0) =$
0902.0431_F02374	141	■ 0.966	$(0,0,1,0)$	$(0, 0, 1, 0)$	$(0, 0, 1, 0)$, then $\alpha \in E_6$ (Theorem 4.7.2), and from the condition $\alpha E_1 = E_1$, we obtain
0902.0431_F02375	141	■ 1.000	$\backslash\mathrm{alpha}\ \backslash\mathrm{in}\ \backslash\mathrm{operatorname{Spin}}(10)$	$\alpha \in \mathrm{Spin}(10)$	$\alpha \in \mathrm{Spin}(10)$, (Theorem 3.10.4). The converse also holds.
0902.0431_F02376	141	■ 1.000	$V^{\wedge 11}$	V^{11}	We define an 11 dimensional $\mathbf{R}$ -vector space V^{11} by
0902.0431_F02377	142	■ 1.000	$(P,\ P)_{\backslash\mathrm{mu}\}$	$(P, P)_\mu$	with the norm $(P, P)_\mu$ given by
0902.0431_F02378	142	■ 0.985	$\backslash\mathrm{quad}\backslash\mathrm{left}(\backslash\mathrm{left}(E_{\{7\}\right)}^{\backslash\mathrm{kappa},\ \backslash\mathrm{mu}\}\backslash\mathrm{right})_{\backslash\mathrm{left}(0,\ E_{\{1\}},\ 0,1\right)}\} / \backslash\mathrm{operatorname{Spin}}(10)\ \backslash\mathrm{simeq}\ S^{\wedge 10}$	$((E_7)^{\kappa,\mu})_{(0,E_1,0,1)} / \mathrm{Spin}(10) \simeq S^{10}$	Proposition 4.11.7. $((E_7)^{\kappa,\mu})_{(0,E_1,0,1)} / \mathrm{Spin}(10) \simeq S^{10}$.
0902.0431_F02379	142	■ 1.000	$S^{\wedge 10}=\backslash\mathrm{left}\{P\ \backslash\mathrm{in}\ V^{\wedge 11}\ \backslash\mathrm{mid}(P,\ P)_{\backslash\mathrm{mu}\}=1\right)\}$	$S^{10} = \{P \in V^{11} \mid (P, P)_\mu = 1\}$	Proof. $S^{10} = \{P \in V^{11} \mid (P, P)_\mu = 1\}$ is a 10 dimensional sphere. For $\alpha \in$
0902.0431_F02380	142	■ 1.000	$\backslash\mathrm{alpha}\ \backslash\mathrm{in}$	$\alpha \in$	Proof. $S^{10} = \{P \in V^{11} \mid (P, P)_\mu = 1\}$ is a 10 dimensional sphere. For $\alpha \in$
0902.0431_F02381	142	■ 1.000	$P\ \backslash\mathrm{in}\ S^{\wedge 10}$	$P \in S^{10}$	$((E_7)^{\kappa,\mu})_{(0,E_1,0,1)}$ and $P \in S^{10}$, we have $\alpha P \in S^{10}$ (Proposition 4.2.2, Lemma 4.3.3).
0902.0431_F02382	142	■ 1.000	$\backslash\mathrm{alpha}\ P\ \backslash\mathrm{in}\ S^{\wedge 10}$	$\alpha P \in S^{10}$	$((E_7)^{\kappa,\mu})_{(0,E_1,0,1)}$ and $P \in S^{10}$, we have $\alpha P \in S^{10}$ (Proposition 4.2.2, Lemma 4.3.3).
0902.0431_F02383	142	■ 1.000	$S^{\wedge 10}$	S^{10}	Hence the group $((E_7)^{\kappa,\mu})_{(0,E_1,0,1)}$ acts on S^{10} . We shall prove that this action is
0902.0431_F02384	142	■ 1.000	$\backslash\mathrm{left}(0,-i\ E_{\{1\}},\ 0,\ i\right)}\ \backslash\mathrm{in}\ S^{\wedge 10}$	$(0, -iE_1, 0, i) \in S^{10}$	transformed to $(0, -iE_1, 0, i) \in S^{10}$ by some $\alpha \in ((E_7)^{\kappa,\mu})_{(0,E_1,0,1)}$. Now, for a given
0902.0431_F02385	142	■ 1.000	$a\ \backslash\mathrm{in}\ \backslash\mathrm{boldsymbol{R}},\ 0\ \leq a < \backslash\mathrm{frac}\{\backslash\mathrm{pi}\}\{4\}$	$a \in \mathbf{R}, 0 \leq a < \frac{\pi}{4}$	choose $a \in \mathbf{R}, 0 \leq a < \frac{\pi}{4}$, such that
0902.0431_F02386	142	■ 1.000	$\backslash\mathrm{tau}\ \backslash\mathrm{xi}-\backslash\mathrm{xi}=0$	$\tau\xi - \xi = 0$	(If $\tau\xi - \xi = 0$, then we choose $a = \frac{\pi}{4}$). Applying $\alpha_{23}(a)$ of Lemma 4.11.5 on P , then
0902.0431_F02387	142	■ 1.000	$a=\backslash\mathrm{frac}\{\backslash\mathrm{pi}\}\{4\}$	$a = \frac{\pi}{4}$	(If $\tau\xi - \xi = 0$, then we choose $a = \frac{\pi}{4}$). Applying $\alpha_{23}(a)$ of Lemma 4.11.5 on P , then
0902.0431_F02388	142	■ 1.000	$\backslash\mathrm{alpha}_{\{23\}}(a)$	$\alpha_{23}(a)$	(If $\tau\xi - \xi = 0$, then we choose $a = \frac{\pi}{4}$). Applying $\alpha_{23}(a)$ of Lemma 4.11.5 on P , then
0902.0431_F02389	142	■ 1.000	$\backslash\mathrm{alpha}_{\{23\}}(a)\ P$	$\alpha_{23}(a)P$	the η -part of $\alpha_{23}(a)P$ becomes
0902.0431_F02390	142	■ 1.000	$\backslash\mathrm{beta}\ \backslash\mathrm{in}\ \backslash\mathrm{operatorname{Spin}}(10)=\backslash\mathrm{left}(E_{\{6\}\right)}_{\{E_{\{1\}}\}}\ \backslash\mathrm{subset}\backslash\mathrm{left}(\backslash\mathrm{left}(E_{\{7\}\right)}^{\backslash\mathrm{kappa},\ \backslash\mathrm{mu}\}\backslash\mathrm{right})_{\backslash\mathrm{left}(0,\ E_{\{1\}},\ 0,1\right)}\}$	$\beta \in \mathrm{Spin}(10) = (E_6)_{E_1} \subset ((E_7)^{\kappa,\mu})_{(0,E_1,0,1)}$	$\beta \in \mathrm{Spin}(10) = (E_6)_{E_1} \subset ((E_7)^{\kappa,\mu})_{(0,E_1,0,1)}$ such that
0902.0431_F02391	142	■ 1.000	$\backslash\mathrm{alpha}_{\{23\}}(-\pi/4)\ \backslash\mathrm{in}\backslash\mathrm{left}(\backslash\mathrm{left}(E_{\{7\}\right)}^{\backslash\mathrm{kappa},\ \backslash\mathrm{mu}\}\backslash\mathrm{right})_{\backslash\mathrm{left}(0,\ E_{\{1\}},\ 0,1\right)}\}$	$\alpha_{23}(-\pi/4) \in ((E_7)^{\kappa,\mu})_{(0,E_1,0,1)}$	Again, applying $\alpha_{23}(-\pi/4) \in ((E_7)^{\kappa,\mu})_{(0,E_1,0,1)}$ of Lemma 4.11.5 on the above, we
0902.0431_F02392	142	■ 0.999	$\backslash\mathrm{left}(0-i\ E_{\{1\}},\ 0,\ i\right)}$	$(0 - iE_1, 0, i)$	$((E_7)^{\kappa,\mu})_{(0,E_1,0,1)}$ at $(0 - iE_1, 0, i)$ is $\mathrm{Spin}(10)$ (Lemma 4.11.6). Thus we have the
0902.0431_F02393	142	■ 0.999	$\backslash\mathrm{left}(\backslash\mathrm{left}(E_{\{7\}\right)}^{\backslash\mathrm{kappa},\ \backslash\mathrm{mu}\}\backslash\mathrm{right})_{\backslash\mathrm{left}(0,\ E_{\{1\}},\ 0,1\right)}\} / \backslash\mathrm{operatorname{Spin}}(10)\ \backslash\mathrm{simeq}\ S^{\wedge 10}$	$((E_7)^{\kappa,\mu})_{(0,E_1,0,1)} / \mathrm{Spin}(10) \simeq S^{10}$	homeomorphism $((E_7)^{\kappa,\mu})_{(0,E_1,0,1)} / \mathrm{Spin}(10) \simeq S^{10}$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_F02394	142	0.506	$\backslash\mathrm{quad}\backslash\mathrm{left}(\backslash\mathrm{left}(E_7\backslash\mathrm{right})^{\backslash\mathrm{kappa}},\backslash\mathrm{mu}\backslash\mathrm{right})_{\backslash\mathrm{left}(0,E_1,0,1)}\backslash\mathrm{cong}\backslash\mathrm{operatorname{Spin}}(11)$	$((E_7)^{\kappa,\mu})_{(0,E_1,0,1)} \cong \mathrm{Spin}(11)$	Theorem 4.11.8. $(E_7)^{\kappa,\mu}_{(0,E_1,0,1)} \cong \mathrm{Spin}(11)$.
0902.0431_F02395	143	1.000	$\backslash\mathrm{left}(\backslash\mathrm{left}(E_7\backslash\mathrm{right})^{\backslash\mathrm{kappa}},\backslash\mathrm{mu}\backslash\mathrm{right})_{\backslash\mathrm{left}(0,E_1,0,1)}\backslash\mathrm{rightarrow}S_0(11)=S_0\backslash\mathrm{left}(V^{11}\backslash\mathrm{right})$	$((E_7)^{\kappa,\mu})_{(0,E_1,0,1)} \rightarrow SO(11) = SO(V^{11})$	$((E_7)^{\kappa,\mu})_{(0,E_1,0,1)} \rightarrow SO(11) = SO(V^{11})$ by $p(\alpha) = \alpha V^{11}$. The restriction p' of p
0902.0431_F02396	143	1.000	$p(\alpha)=\alpha\mid V^{11}$	$p(\alpha) = \alpha V^{11}$	$((E_7)^{\kappa,\mu})_{(0,E_1,0,1)} \rightarrow SO(11) = SO(V^{11})$ by $p(\alpha) = \alpha V^{11}$. The restriction p' of p
0902.0431_F02397	143	0.995	$p^{\backslash\mathrm{prime}}:\backslash\mathrm{operatorname{Spin}}(10)\backslash\mathrm{rightarrow}S_0(10)=S_0\backslash\mathrm{left}(V^{10}\backslash\mathrm{right})$	$p' : \mathrm{Spin}(10) \rightarrow SO(10) = SO(V^{10})$	to $(E_6)_{E_1}$ coincides with the homomorphism $p' : \mathrm{Spin}(10) \rightarrow SO(10) = SO(V^{10})$
0902.0431_F02398	143	1.000	$V^{10}=\backslash\mathrm{left}\backslash\{P\in V^{11}\mid P=(X,0,0,0)\backslash\mathrm{right}\}$	$V^{10} = \{P \in V^{11} \mid P = (X, 0, 0, 0)\}$	(where $V^{10} = \{P \in V^{11} \mid P = (X, 0, 0, 0)\}$). In particular, $p' : \mathrm{Spin}(10) \rightarrow SO(10)$ is
0902.0431_F02399	143	1.000	$p^{\backslash\mathrm{prime}}:\backslash\mathrm{operatorname{Spin}}(10)\backslash\mathrm{rightarrow}S_0(10)$	$p' : \mathrm{Spin}(10) \rightarrow SO(10)$	(where $V^{10} = \{P \in V^{11} \mid P = (X, 0, 0, 0)\}$). In particular, $p' : \mathrm{Spin}(10) \rightarrow SO(10)$ is
0902.0431_F02400	143	1.000	$p:\backslash\mathrm{left}(\backslash\mathrm{left}(E_7\backslash\mathrm{right})^{\backslash\mathrm{kappa}},\backslash\mathrm{mu}\backslash\mathrm{right})_{\backslash\mathrm{left}(0,E_1,0,1)}\backslash\mathrm{rightarrow}S_0(11)$	$p : ((E_7)^{\kappa,\mu})_{(0,E_1,0,1)} \rightarrow SO(11)$	we see that $p : ((E_7)^{\kappa,\mu})_{(0,E_1,0,1)} \rightarrow SO(11)$ is onto by the five lemma. Using the five
0902.0431_F02401	143	0.989	$\backslash\mathrm{operatorname{Ker}}p$	$\mathrm{Ker} p$	lemme again, we see that $\mathrm{Ker} p$ coincides with $\mathrm{Ker} p'$. Hence $\mathrm{Ker} p = \{1, \sigma\}$ (Theorem
0902.0431_F02402	143	0.989	$\backslash\mathrm{operatorname{Ker}}p^{\backslash\mathrm{prime}}$	$\mathrm{Ker} p'$	lemme again, we see that $\mathrm{Ker} p$ coincides with $\mathrm{Ker} p'$. Hence $\mathrm{Ker} p = \{1, \sigma\}$ (Theorem
0902.0431_F02403	143	1.000	$\backslash\mathrm{operatorname{Spin}}(11)$	$\mathrm{Spin}(11)$	Therefore the group $((E_7)^{\kappa,\mu})_{(0,E_1,0,1)}$ is isomorphic to the group $\mathrm{Spin}(11)$ as the
0902.0431_F02404	143	1.000	$S_0(11)$	$SO(11)$	universal covering group of $SO(11)$.
0902.0431_F02405	143	1.000	$\alpha(t):\backslash\mathrm{mathfrak{P}}^C\backslash\mathrm{rightarrow}\backslash\mathrm{mathfrak{P}}^C$	$\alpha(t) : \mathfrak{P}^C \rightarrow \mathfrak{P}^C$	Lemma 4.11.9. For $t \in \mathbf{R}$, we define a mapping $\alpha(t) : \mathfrak{P}^C \rightarrow \mathfrak{P}^C$ by
0902.0431_F02406	143	1.000	$\alpha(t)\in\backslash\mathrm{left}(\backslash\mathrm{left}(E_7\backslash\mathrm{right})^{\backslash\mathrm{kappa}},\backslash\mathrm{mu}\backslash\mathrm{right})_{\backslash\mathrm{left}(0,E_1,0,1)}$	$\alpha(t) \in (E_7)^{\kappa,\mu}$	then $\alpha(t) \in (E_7)^{\kappa,\mu}$.
0902.0431_F02407	143	1.000	$\nu=i\,t\in i\,\mathbf{R}$	$\nu = it \in i\mathbf{R}$	Proof. For $\nu = it \in i\mathbf{R}$, let $\phi(\nu) = 2\nu E_1 \vee E_1 \in (\mathfrak{e}_6)^\sigma$. Then, $\Phi(\phi(\nu), 0, 0, -2\nu) \in$
0902.0431_F02408	143	1.000	$\phi(\nu)=2\nu E_1\vee E_1\in\backslash\mathrm{left}(\backslash\mathrm{mathfrak{e}}_6\backslash\mathrm{right})^{\backslash\mathrm{sigma}}$	$\phi(\nu) = 2\nu E_1 \vee E_1 \in (\mathfrak{e}_6)^\sigma$	Proof. For $\nu = it \in i\mathbf{R}$, let $\phi(\nu) = 2\nu E_1 \vee E_1 \in (\mathfrak{e}_6)^\sigma$. Then, $\Phi(\phi(\nu), 0, 0, -2\nu) \in$
0902.0431_F02409	143	1.000	$\Phi(\phi(\nu),0,0,-2\nu)\in\backslash\mathrm{left}(\backslash\mathrm{mathfrak{e}}_7\backslash\mathrm{right})^{\backslash\mathrm{kappa}},\backslash\mathrm{nu}\backslash\mathrm{right})_{\backslash\mathrm{left}(0,E_1,0,1)}$	$\Phi(\phi(\nu), 0, 0, -2\nu) \in$	Proof. For $\nu = it \in i\mathbf{R}$, let $\phi(\nu) = 2\nu E_1 \vee E_1 \in (\mathfrak{e}_6)^\sigma$. Then, $\Phi(\phi(\nu), 0, 0, -2\nu) \in$
0902.0431_F02410	143	0.999	$\backslash\mathrm{left}(\backslash\mathrm{mathfrak{e}}_7\backslash\mathrm{right})^{\backslash\mathrm{kappa}},\backslash\mathrm{nu}\backslash\mathrm{right})_{\backslash\mathrm{left}(0,E_1,0,1)}$	$(\mathfrak{e}_7)^{\kappa,\nu}$	$(\mathfrak{e}_7)^{\kappa,\nu}$ (Proposition 4.11.2) and $\alpha(t) = \exp \Phi(\phi(\nu), 0, 0, -2\nu)$. Hence we have $\alpha(t) \in$
0902.0431_F02411	143	0.999	$\alpha(t)=\exp\Phi(\phi(\nu),0,0,-2\nu)$	$\alpha(t) = \exp \Phi(\phi(\nu), 0, 0, -2\nu)$	$(\mathfrak{e}_7)^{\kappa,\nu}$ (Proposition 4.11.2) and $\alpha(t) = \exp \Phi(\phi(\nu), 0, 0, -2\nu)$. Hence we have $\alpha(t) \in$
0902.0431_F02412	143	0.999	$\alpha(t)\in\backslash\mathrm{left}(\backslash\mathrm{mathfrak{e}}_7\backslash\mathrm{right})^{\backslash\mathrm{kappa}},\backslash\mathrm{nu}\backslash\mathrm{right})_{\backslash\mathrm{left}(0,E_1,0,1)}$	$\alpha(t) \in$	$(\mathfrak{e}_7)^{\kappa,\nu}$ (Proposition 4.11.2) and $\alpha(t) = \exp \Phi(\phi(\nu), 0, 0, -2\nu)$. Hence we have $\alpha(t) \in$
0902.0431_F02413	143	1.000	V^{12}	V^{12}	We define a 12 dimensional $\mathbf{R}$ -vector space V^{12} by
0902.0431_F02414	143	1.000	$\backslash\mathrm{left}(E_7\backslash\mathrm{right})^{\backslash\mathrm{kappa}},\backslash\mathrm{mu}\backslash\mathrm{right})_{\backslash\mathrm{left}(0,E_1,0,1)}\backslash\mathrm{simeq}S^{11}$	$(E_7)^{\kappa,\mu} / \mathrm{Spin}(11) \simeq S^{11}$	Proposition 4.11.10. $(E_7)^{\kappa,\mu} / \mathrm{Spin}(11) \simeq S^{11}$.
0902.0431_F02415	144	1.000	$S^{11}=\backslash\mathrm{left}\backslash\{P\in V^{12}\mid(P,P)_\mu=1\backslash\mathrm{right}\}$	$S^{11} = \{P \in V^{12} \mid (P, P)_\mu = 1\}$	Proof. $S^{11} = \{P \in V^{12} \mid (P, P)_\mu = 1\}$ is an 11 dimensional sphere. For $\alpha \in$
0902.0431_F02416	144	1.000	$P\in S^{11}$	$P \in S^{11}$	$(E_7)^{\kappa,\mu}$ and $P \in S^{11}$, we have $\alpha P \in S^{11}$ (Proposition 4.2.2, Lemma 4.3.3). Hence
0902.0431_F02417	144	1.000	$\alpha P\in S^{11}$	$\alpha P \in S^{11}$	$(E_7)^{\kappa,\mu}$ and $P \in S^{11}$, we have $\alpha P \in S^{11}$ (Proposition 4.2.2, Lemma 4.3.3). Hence
0902.0431_F02418	144	1.000	S^{11}	S^{11}	the group $(E_7)^{\kappa,\mu}$ acts on S^{11} . We shall prove that this action is transitive. To
0902.0431_F02419	144	1.000	$\backslash\mathrm{left}(0,E_1,0,1)\in S^{11}$	$(0, E_1, 0, 1) \in S^{11}$	$(0, E_1, 0, 1) \in S^{11}$ by some $\alpha \in (E_7)^{\kappa,\mu}$. Now, for a given
0902.0431_F02420	144	1.000	$e^{-2it}\eta\in i\,\mathbf{R}$	$e^{-2it}\eta \in i\mathbf{R}$	we choose $t \in \mathbf{R}$ such that $e^{-2it}\eta \in i\mathbf{R}$. Applying $\alpha(t)$ of Lemma 4.11.9 on P , we get
0902.0431_F02421	144	1.000	$\alpha(t)$	$\alpha(t)$	we choose $t \in \mathbf{R}$ such that $e^{-2it}\eta \in i\mathbf{R}$. Applying $\alpha(t)$ of Lemma 4.11.9 on P , we get
0902.0431_F02422	144	1.000	$\beta\in\backslash\mathrm{Spin}}(11)=((E_7)^{\kappa,\mu})_{\backslash\mathrm{left}(0,E_1,0,1)}$	$\beta \in \mathrm{Spin}(11) = ((E_7)^{\kappa,\mu})_{(0,E_1,0,1)}$	$\beta \in \mathrm{Spin}(11) = ((E_7)^{\kappa,\mu})_{(0,E_1,0,1)}$ such that
0902.0431_F02423	144	1.000	$\alpha(-\pi/4)\in(E_7)^{\kappa,\mu}$	$\alpha(-\pi/4) \in (E_7)^{\kappa,\mu}$	If we further apply $\alpha(-\pi/4) \in (E_7)^{\kappa,\mu}$ of Lemma 4.11.9 on the above, then we have
0902.0431_F02424	144	1.000	$\backslash\mathrm{left}(0,E_1,0,1)\in S^{11}$	$(0, E_1, 0, 1) \in S^{11}$	This shows the transitivity of $(E_7)^{\kappa,\mu}$. The isotropy subgroup of $(E_7)^{\kappa,\mu}$ at $(0, E_1, 0, 1)$
0902.0431_F02425	144	0.993	$\backslash\mathrm{left}(E_7\backslash\mathrm{right})^{\backslash\mathrm{kappa}},\backslash\mathrm{mu}\backslash\mathrm{right})_{\backslash\mathrm{left}(0,E_1,0,1)}\backslash\mathrm{simeq}S^{11}$	$(E_7)^{\kappa,\mu} / \mathrm{Spin}(11)$	is $\mathrm{Spin}(11)$ (Theorem 4.11.8). Thus we have the homeomorphism $(E_7)^{\kappa,\mu} / \mathrm{Spin}(11)$
0902.0431_F02426	144	1.000	$\backslash\mathrm{simeq}S^{11}$	$\simeq S^{11}$	$\simeq S^{11}$.
0902.0431_F02427	144	1.000	$\backslash\mathrm{quad}\backslash\mathrm{left}(E_7\backslash\mathrm{right})^{\backslash\mathrm{kappa}},\backslash\mathrm{mu}\backslash\mathrm{right})_{\backslash\mathrm{left}(0,E_1,0,1)}\backslash\mathrm{cong}\backslash\mathrm{operatorname{Spin}}(12)$	$(E_7)^{\kappa,\mu} \cong \mathrm{Spin}(12)$	Theorem 4.11.11. $(E_7)^{\kappa,\mu} \cong \mathrm{Spin}(12)$.
0902.0431_F02428	144	1.000	$p(\alpha)=\alpha\mid V^{12}$	$p(\alpha) = \alpha V^{12}$	by $p(\alpha) = \alpha V^{12}$. The restriction p' of p to $(E_6)^{\kappa,\mu}$ coincides with the homomorphism
0902.0431_F02429	144	1.000	$\backslash\mathrm{left}(E_6\backslash\mathrm{right})^{\backslash\mathrm{kappa}},\backslash\mathrm{mu}\backslash\mathrm{right})_{\backslash\mathrm{left}(0,E_1,0,1)}$	$(E_6)^{\kappa,\mu}$	by $p(\alpha) = \alpha V^{12}$. The restriction p' of p to $(E_6)^{\kappa,\mu}$ coincides with the homomorphism
0902.0431_F02430	144	0.920	$p^{\backslash\mathrm{prime}}:\backslash\mathrm{operatorname{Spin}}(11)\backslash\mathrm{rightarrow}S_0(11)$	$p' : \mathrm{Spin}(11) \rightarrow SO(11)$	$p' : \mathrm{Spin}(11) \rightarrow SO(11)$ of Theorem 4.11.8. In particular, $p' : \mathrm{Spin}(11) \rightarrow SO(11)$ is
0902.0431_F02431	144	1.000	$p:\backslash\mathrm{left}(E_7\backslash\mathrm{right})^{\backslash\mathrm{kappa}},\backslash\mathrm{mu}\backslash\mathrm{right})_{\backslash\mathrm{left}(0,E_1,0,1)}\backslash\mathrm{rightarrow}S_0(12)$	$p : (E_7)^{\kappa,\mu} \rightarrow SO(12)$	we see that $p : (E_7)^{\kappa,\mu} \rightarrow SO(12)$ is onto by the five lemma. Using the five lemme
0902.0431_F02432	145	0.988	$\backslash\mathrm{operatorname{Spin}}(12)$	$\mathrm{Spin}(12)$	Therefore the group $(E_7)^{\kappa,\mu}$ is isomorphic to the group $\mathrm{Spin}(12)$ as the universal
0902.0431_F02433	145	1.000	$S_0(12)$	$SO(12)$	covering group of $SO(12)$.
0902.0431_F02434	145	0.484	$z(\backslash\mathrm{left}(\backslash\mathrm{left}(E_7\backslash\mathrm{right})^{\backslash\mathrm{kappa}},\backslash\mathrm{mu}\backslash\mathrm{right})_{\backslash\mathrm{left}(0,E_1,0,1)}\backslash\mathrm{right})_{\backslash\mathrm{left}(0,E_1,0,1)}$	$z(\backslash\mathrm{left}(\backslash\mathrm{left}(E_7\backslash\mathrm{right})^{\backslash\mathrm{kappa}},\backslash\mathrm{mu}\backslash\mathrm{right})_{\backslash\mathrm{left}(0,E_1,0,1)}\backslash\mathrm{right})_{\backslash\mathrm{left}(0,E_1,0,1)}$	Theorem 4.11.12. The center $z(\mathrm{Spin}(12))$ of $\mathrm{Spin}(12)$ is

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_F02435	145	■ 0.979	$SU(2)=\left\{\left.A\in M(2,C)\right ^t(\tau A)A=\right.$	$SU(2)=\left\{A\in M(2,C)\right ^t(\tau A)A=$	$which\ is\ isomorphic\ to\ the\ special\ unitary\ group\ SU(2)=\{A\in M(2,C)\, ^t(\tau A)A=$
0902.0431_F02436	145	■ 0.988	$E,\operatorname{det}A=1\}$	$E,\det A=1\}$	$\boxed{E,\det A=1\}$. Here, for $A\in SU(2)$, a mapping $\varphi_2(A):\mathfrak{P}^C\rightarrow\mathfrak{P}^C$ is defined by
0902.0431_F02437	145	■ 0.988	$A\in SU(2)$	$A\in SU(2)$	$E,\det A=1\}$. Here, for $A\in SU(2)$, a mapping $\varphi_2(A):\mathfrak{P}^C\rightarrow\mathfrak{P}^C$ is defined by
0902.0431_F02438	145	■ 0.988	$\varphi_2(A):\mathfrak{P}^C\rightarrow\mathfrak{P}^C$	$\varphi_2(A):\mathfrak{P}^C\rightarrow\mathfrak{P}^C$	$E,\det A=1\}$. Here, for $A\in SU(2)$, a mapping $\varphi_2(A):\mathfrak{P}^C\rightarrow\mathfrak{P}^C$ is defined by
0902.0431_F02439	145	■ 0.998	$\Phi(\phi(\nu),aE_1,-\tau aE_1,\nu)\in\mathfrak{a}_1(\phi(\nu)=2\nu E_1\vee E_1,\nu\in i\boldsymbol{R},a\in$	$\Phi(\phi(\nu),aE_1,-\tau aE_1,\nu)\in\mathfrak{a}_1(\phi(\nu)=2\nu E_1\vee E_1,\nu\in i\boldsymbol{R},a\in$	Proof. The action of $\boxed{\Phi(\phi(\nu),aE_1,-\tau aE_1,\nu)\in\mathfrak{a}_1(\phi(\nu)=2\nu E_1\vee E_1,\nu\in i\boldsymbol{R},a\in$
0902.0431_F02440	145	■ 0.998	$A=\exp\left(\begin{array}{cc}\nu&a\\-\tau a&-\nu\end{array}\right)\in SU(2)$	$A=\exp\left(\begin{array}{cc}\nu&a\\-\tau a&-\nu\end{array}\right)\in SU(2)$	Therefore, for $\boxed{A=\exp\left(\begin{array}{cc}\nu&a\\-\tau a&-\nu\end{array}\right)\in SU(2)}$, we have
0902.0431_F02441	146	■ 0.707	$\left(E_7\right)^{\sigma} \cong(SU(2) \times \operatorname{Spin}(12)) / \boldsymbol{Z}_2, \boldsymbol{Z}_2=\{(E, 1),(-E,-\sigma)\}$	$(E_7)^\sigma\cong(SU(2)\times\operatorname{Spin}(12))/\boldsymbol{Z}_2,\boldsymbol{Z}_2=\{(E,1),(-E,-\sigma)\}$	Theorem 4.11.15. $\boxed{(E_7)^\sigma\cong(SU(2)\times\operatorname{Spin}(12))/\boldsymbol{Z}_2,\boldsymbol{Z}_2=\{(E,1),(-E,-\sigma)\}}$.
0902.0431_F02442	146	■ 0.947	$\varphi:SU(2)\times\operatorname{Spin}(12)\rightarrow(E_7)^\sigma$	$\varphi:SU(2)\times\operatorname{Spin}(12)\rightarrow(E_7)^\sigma$	Proof. We define a mapping $\boxed{\varphi:SU(2)\times\operatorname{Spin}(12)\rightarrow(E_7)^\sigma}$ by
0902.0431_F02443	146	■ 0.768	$SU(2)$	$SU(2)$	Since the Lie algebras $\mathfrak{a}_1$ and $(\mathfrak{e}_7)^{\kappa,\mu}$ of $SU(2)$ and $\operatorname{Spin}(12)$ are elementwise commu-
0902.0431_F02444	146	■ 0.840	$\varphi_2(A)\in SU(2)$	$\varphi_2(A)\in SU(2)$	tative (Proposition 4.11.4.(2)), $\varphi_2(A)\in SU(2)$ and $\beta\in\operatorname{Spin}(12)$ are commutative
0902.0431_F02445	146	■ 0.840	$\beta\in\operatorname{Spin}(12)$	$\beta\in\operatorname{Spin}(12)$	tative (Proposition 4.11.4.(2)), $\varphi_2(A)\in SU(2)$ and $\beta\in\operatorname{Spin}(12)$ are commutative
0902.0431_F02446	146	■ 1.000	$\varphi_2(A)\beta=\beta\varphi_2(A)$	$\varphi_2(A)\beta=\beta\varphi_2(A)$	: $\varphi_2(A)\beta=\beta\varphi_2(A)$. Hence φ is a homomorphism. We shall show that φ is onto.
0902.0431_F02447	146	■ 1.000	$\varphi_*:\mathfrak{a}_1\oplus(\mathfrak{e}_7)^{\kappa,\mu}\rightarrow(\mathfrak{e}_7)^\sigma$	$\varphi_*:\mathfrak{a}_1\oplus(\mathfrak{e}_7)^{\kappa,\mu}\rightarrow(\mathfrak{e}_7)^\sigma$	show that its differential mapping $\boxed{\varphi_*:\mathfrak{a}_1\oplus(\mathfrak{e}_7)^{\kappa,\mu}\rightarrow(\mathfrak{e}_7)^\sigma}$ is onto. However this
0902.0431_F02448	146	■ 1.000	$\operatorname{Ker}\varphi=\{(E,1),(-E,\varphi_2(-E))\}=$	$\operatorname{Ker}\varphi=\{(E,1),(-E,\varphi_2(-E))\}=$	has been already shown in Proposition 4.11.4.(2). $\boxed{\operatorname{Ker}\varphi=\{(E,1),(-E,\varphi_2(-E))\}=}$
0902.0431_F02449	146	■ 1.000	$\{(E,1),(-E,-\sigma)\}=\boldsymbol{Z}_2$	$\{(E,1),(-E,-\sigma)\}=\boldsymbol{Z}_2$	$\boxed{\{(E,1),(-E,-\sigma)\}=\boldsymbol{Z}_2}$ (Theorem 4.11.13) is easily obtained. Thus we have the
0902.0431_F02450	146	■ 0.792	$(SU(2)\times\operatorname{Spin}(12))/\boldsymbol{Z}_2\cong(E_7)^\sigma$	$(SU(2)\times\operatorname{Spin}(12))/\boldsymbol{Z}_2\cong(E_7)^\sigma$	isomorphism $\boxed{SU(2)\times\operatorname{Spin}(12))/\boldsymbol{Z}_2\cong(E_7)^\sigma}$.
0902.0431_F02451	146	■ 1.000	$X\in(\mathfrak{J}^C)_\sigma$	$X\in(\mathfrak{J}^C)_\sigma$	Claim 1. Any element $\boxed{X\in(\mathfrak{J}^C)_\sigma}$ can be transformed to a diagonal form by some
0902.0431_F02452	146	■ 1.000	$\xi_2\geq 0,\xi_3\geq 0$	$\xi_2\geq 0,\xi_3\geq 0$	Moreover we can choose $\alpha\in(E_6)^\sigma$ so that $\xi_2\geq 0,\xi_3\geq 0$.
0902.0431_F02453	146	■ 1.000	$i(E_1-E_2)^\sim,i(E_1-E_3)^\sim,i\widetilde{F}_1(a),\widetilde{A}_1(a)(a\in\mathfrak{C})\in(\mathfrak{e}_6)^\sigma$	$i(E_1-E_2)^\sim,i(E_1-E_3)^\sim,i\widetilde{F}_1(a),\widetilde{A}_1(a)(a\in\mathfrak{C})\in(\mathfrak{e}_6)^\sigma$	Proof. Recall that $\boxed{i(E_1-E_2)^\sim,i(E_1-E_3)^\sim,i\widetilde{F}_1(a),\widetilde{A}_1(a)(a\in\mathfrak{C})\in(\mathfrak{e}_6)^\sigma}$, then
0902.0431_F02454	146	■ 1.000	$P\in(\mathfrak{M}^C)_\sigma=\{P\in\mathfrak{M}^C\mid\sigma P=P\}$	$P\in(\mathfrak{M}^C)_\sigma=\{P\in\mathfrak{M}^C\mid\sigma P=P\}$	Claim 2. Any element $\boxed{P\in(\mathfrak{M}^C)_\sigma=\{P\in\mathfrak{M}^C\mid\sigma P=P\}}$ can be transformed to
0902.0431_F02455	146	■ 1.000	$\alpha\in((E_7)^\sigma)_0$	$\alpha\in((E_7)^\sigma)_0$	a diagonal form by some $\boxed{\alpha\in((E_7)^\sigma)_0}$ (the connected component of $(E_7)^\sigma$ containing
0902.0431_F02456	146	■ 1.000	$\Phi(0,-\tau aE_i,aE_i,0)\in(\mathfrak{e}_7)^\sigma,i=1,2,3$	$\Phi(0,-\tau aE_i,aE_i,0)\in(\mathfrak{e}_7)^\sigma,i=1,2,3$	Proof. Recall $\boxed{\Phi(0,-\tau aE_i,aE_i,0)\in(\mathfrak{e}_7)^\sigma,i=1,2,3}$, then we can prove analo-
0902.0431_F02457	146	■ 0.926	$(E_7)^\sigma/(E_6)^\sigma\simeq(\mathfrak{M}_1)_\sigma=\{P\in\mathfrak{M}_1\mid\sigma P=P\}$	$(E_7)^\sigma/(E_6)^\sigma\simeq(\mathfrak{M}_1)_\sigma=\{P\in\mathfrak{M}_1\mid\sigma P=P\}$	Claim 3. $\boxed{(E_7)^\sigma/(E_6)^\sigma\simeq(\mathfrak{M}_1)_\sigma=\{P\in\mathfrak{M}_1\mid\sigma P=P\}}$.
0902.0431_F02458	146	■ 1.000	$\alpha_i(a)$	$\alpha_i(a)$	Proof. Remark that $\alpha_i(a)$ of Lemma 4.8.1 belongs to $((E_7)^\sigma)_0$, then this claim is
0902.0431_F02459	146	■ 1.000	$((E_7)^\sigma)_0$	$((E_7)^\sigma)_0$	Proof. Remark that $\alpha_i(a)$ of Lemma 4.8.1 belongs to $\boxed{((E_7)^\sigma)_0}$, then this claim is
0902.0431_F02460	146	■ 0.909	$(E_6)^\sigma\cong(U(1)\times\operatorname{Spin}(10))/\boldsymbol{Z}_4($	$(E_6)^\sigma\cong(U(1)\times\operatorname{Spin}(10))/\boldsymbol{Z}_4($	connectedness of $\boxed{(E_6)^\sigma\cong(U(1)\times\operatorname{Spin}(10))/\boldsymbol{Z}_4}$ (Theorem 3.10.7) and $(\mathfrak{M}_1)_\sigma$.
0902.0431_F02461	146	■ 0.909	$(\mathfrak{M}_1)_\sigma$	$(\mathfrak{M}_1)_\sigma$	connectedness of $(E_6)^\sigma\cong(U(1)\times\operatorname{Spin}(10))/\boldsymbol{Z}_4$ (Theorem 3.10.7) and $(\mathfrak{M}_1)_\sigma$.
0902.0431_F02462	147	■ 0.998	$\gamma\in G_2\subset F_4\subset E_6\subset E_7$	$\gamma\in G_2\subset F_4\subset E_6\subset E_7$	where γ is the same as $\boxed{\gamma\in G_2\subset F_4\subset E_6\subset E_7}$.
0902.0431_F02463	147	■ 1.000	$(E_7)^{\tau\gamma}$	$(E_7)^{\tau\gamma}$	We shall study the following subgroup $(E_7)^{\tau\gamma}$ of E_7 :

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0902.0431_F02464	147	■ 1.000	$\left(\mathfrak{P}^C\right)_{\tau\gamma}, \left(\mathfrak{P}^C\right)_{-\tau\gamma}$	$(\mathfrak{P}^C)_{\tau\gamma}, (\mathfrak{P}^C)_{-\tau\gamma}$	To this end, we consider $\mathbf{R}$ -vector subspaces $(\mathfrak{P}^C)_{\tau\gamma}, (\mathfrak{P}^C)_{-\tau\gamma}$ of $\mathfrak{P}^C$, which are
0902.0431_F02465	147	■ 0.999	$\left(\mathfrak{P}^C\right)_{\tau\gamma}: \mathfrak{P}^C = \left(\left(\mathfrak{P}^C\right)_{\tau\gamma}\right)^C$	$(\mathfrak{P}^C)_{\tau\gamma}: \mathfrak{P}^C = \left((\mathfrak{P}^C)_{\tau\gamma}\right)^C$	In paticular, $\mathfrak{P}^C$ is the complexification of $(\mathfrak{P}^C)_{\tau\gamma}: \mathfrak{P}^C = ((\mathfrak{P}^C)_{\tau\gamma})^C$.
0902.0431_F02466	147	■ 1.000	$k: M(4, \mathbf{H}) \rightarrow$	$k: M(4, \mathbf{H}) \rightarrow$	Analogous to Section 3.11, we can define the $\mathbf{R}$ -linear mapping $\mathfrak{k}: M(4, \mathbf{H}) \rightarrow$
0902.0431_F02467	147	■ 0.989	$M(8, \mathbf{C})$	$M(8, \mathbf{C})$	$M(8, \mathbf{C})$,
0902.0431_F02468	147	■ 1.000	$B \in \mathfrak{su}(8)$	$B \in \mathfrak{su}(8)$	Lemma 4.12.1. Any element $B \in \mathfrak{su}(8)$ is uniquely expressed by
0902.0431_F02469	147	■ 0.931	$D_1 = \frac{B - J\bar{B}J}{2}, T_1 = \frac{B + J\bar{B}J}{2e_1} \in M(8, \mathbf{C})$	$D_1 = \frac{B - J\bar{B}J}{2}, T_1 = \frac{B + J\bar{B}J}{2e_1} \in M(8, \mathbf{C})$	Proof. For $B \in \mathfrak{su}(8)$, let $D_1 = \frac{B - J\bar{B}J}{2}, T_1 = \frac{B + J\bar{B}J}{2e_1} \in M(8, \mathbf{C})$, then we
0902.0431_F02470	147	■ 1.000	$D = k^{-1}(D_1), T = k^{-1}(T_1) \in M(4, \mathbf{H})$	$D = k^{-1}(D_1), T = k^{-1}(T_1) \in M(4, \mathbf{H})$	Then, $D = k^{-1}(D_1), T = k^{-1}(T_1) \in M(4, \mathbf{H})$ are the required elements. To prove
0902.0431_F02471	148	■ 1.000	$J D_1 + e_1 J T_1 = 0$	$J D_1 + e_1 J T_1 = 0$	Certainly, from the condition, we have $J D_1 + e_1 J T_1 = 0$, so $\bar{D}_1 J + e_1 \bar{T}_1 J = 0$, and
0902.0431_F02472	148	■ 1.000	$\bar{D}_1 J + e_1 \bar{T}_1 J = 0$	$\bar{D}_1 J + e_1 \bar{T}_1 J = 0$	Certainly, from the condition, we have $J D_1 + e_1 J T_1 = 0$, so $\bar{D}_1 J + e_1 \bar{T}_1 J = 0$, and
0902.0431_F02473	148	■ 1.000	$D_1 J - e_1 T_1 J = 0$	$D_1 J - e_1 T_1 J = 0$	so that $D_1 J - e_1 T_1 J = 0$, that is, $D_1 - e_1 T_1 = 0$. Together with the first equation,
0902.0431_F02474	148	■ 1.000	$D_1 - e_1 T_1 = 0$	$D_1 - e_1 T_1 = 0$	so that $D_1 J - e_1 T_1 J = 0$, that is, $D_1 - e_1 T_1 = 0$. Together with the first equation,
0902.0431_F02475	148	■ 0.998	$D_1 = T_1 = 0$	$D_1 = T_1 = 0$	we have $D_1 = T_1 = 0$.
0902.0431_F02476	148	■ 1.000	$g: \mathfrak{J}^C \rightarrow \mathfrak{J}(4, \mathbf{H})^C, g(M + \mathbf{a}) =$	$g: \mathfrak{J}^C \rightarrow \mathfrak{J}(4, \mathbf{H})^C, g(M + \mathbf{a}) =$	After this, we will use the $\mathbf{C}$ -linear mapping $q: \mathfrak{J}^C \rightarrow \mathfrak{J}(4, \mathbf{H})^C, q(M + \mathbf{a}) =$
0902.0431_F02477	148	■ 0.998	$\left(\begin{array}{cc} \frac{1}{2} \operatorname{tr}(M) & ia \\ ia^* & M - \frac{1}{2} \operatorname{tr}(M) E \end{array}\right)$	$\left(\begin{array}{cc} \frac{1}{2} \operatorname{tr}(M) & ia \\ ia^* & M - \frac{1}{2} \operatorname{tr}(M) E \end{array}\right)$	$\left(\begin{array}{cc} \frac{1}{2} \operatorname{tr}(M) & ia \\ ia^* & M - \frac{1}{2} \operatorname{tr}(M) E \end{array}\right)$, the homomorphism $\varphi: Sp(4) \rightarrow (E_6)^{\tau\gamma}$, $\varphi(A)X =$
0902.0431_F02478	148	■ 0.998	$\varphi: Sp(4) \rightarrow (E_6)^{\tau\gamma}, \varphi(A)X =$	$\varphi: Sp(4) \rightarrow (E_6)^{\tau\gamma}, \varphi(A)X =$	$\left(\begin{array}{cc} \frac{1}{2} \operatorname{tr}(M) & ia \\ ia^* & M - \frac{1}{2} \operatorname{tr}(M) E \end{array}\right)$, the homomorphism $\varphi: Sp(4) \rightarrow (E_6)^{\tau\gamma}$, $\varphi(A)X =$
0902.0431_F02479	148	■ 1.000	$g^{-1}(A(gX)A^*), X \in \mathfrak{J}^C$	$g^{-1}(A(gX)A^*), X \in \mathfrak{J}^C$	$g^{-1}(A(gX)A^*), X \in \mathfrak{J}^C$ and its differential mapping $\varphi_*: \mathfrak{sp}(4) \rightarrow (\mathfrak{e}_6)^{\tau\gamma}$, $\varphi_*(D)X =$
0902.0431_F02480	148	■ 1.000	$\varphi_*: \mathfrak{sp}(4) \rightarrow (\mathfrak{e}_6)^{\tau\gamma}, \varphi_*(D)X =$	$\varphi_*: \mathfrak{sp}(4) \rightarrow (\mathfrak{e}_6)^{\tau\gamma}, \varphi_*(D)X =$	$g^{-1}(A(gX)A^*), X \in \mathfrak{J}^C$ and its differential mapping $\varphi_*: \mathfrak{sp}(4) \rightarrow (\mathfrak{e}_6)^{\tau\gamma}$, $\varphi_*(D)X =$
0902.0431_F02481	148	■ 1.000	$g^{-1}(D(gX) + (gX)D^*), X \in \mathfrak{J}^C$	$g^{-1}(D(gX) + (gX)D^*), X \in \mathfrak{J}^C$	$q^{-1}(D(qX) + (qX)D^*), X \in \mathfrak{J}^C$ which are defined in Section 3.12.
0902.0431_F02482	148	■ 1.000	$(\mathfrak{e}_7)^{\tau\gamma}$	$(\mathfrak{e}_7)^{\tau\gamma}$	Proposition 4.12.2. The Lie algebra $(\mathfrak{e}_7)^{\tau\gamma}$ of the group $(E_7)^{\tau\gamma}$ is
0902.0431_F02483	148	■ 1.000	$\mathfrak{S}(8, \mathbf{C})$	$\mathfrak{S}(8, \mathbf{C})$	Analogous to Section 3.11, we define a $\mathbf{C}$ -vector space $\mathfrak{S}(8, \mathbf{C})$ by
0902.0431_F02484	148	■ 1.000	$k_J: \mathfrak{J}(4, \mathbf{H})^C \rightarrow \mathfrak{S}(8, \mathbf{C})^C$	$k_J: \mathfrak{J}(4, \mathbf{H})^C \rightarrow \mathfrak{S}(8, \mathbf{C})^C$	and a $\mathbf{C}$ -linear mapping $\mathfrak{k}_J: \mathfrak{J}(4, \mathbf{H})^C \rightarrow \mathfrak{S}(8, \mathbf{C})^C$ by
0902.0431_F02485	148	■ 0.999	$J = \operatorname{diag}(J, J, J, J) \in M(8, \mathbf{C}), J = \left(\begin{array}{cc} 0 & 1 \\ -1 & 0 \end{array}\right)$	$J = \operatorname{diag}(J, J, J, J) \in M(8, \mathbf{C}), J = \left(\begin{array}{cc} 0 & 1 \\ -1 & 0 \end{array}\right)$	where $J = \operatorname{diag}(J, J, J, J) \in M(8, \mathbf{C}), J = \left(\begin{array}{cc} 0 & 1 \\ -1 & 0 \end{array}\right)$.
0902.0431_F02486	148	■ 0.658	$\chi: \mathfrak{P}^C \rightarrow \mathfrak{S}(8, \mathbf{C})^C$	$\chi: \mathfrak{P}^C \rightarrow \mathfrak{S}(8, \mathbf{C})^C$	Definition. We define a $\mathbf{C}$ -linear isomorphism $\chi: \mathfrak{P}^C \rightarrow \mathfrak{S}(8, \mathbf{C})^C$ by
0902.0431_F02487	148	■ 1.000	$(\mathfrak{e}_7)^{\tau\gamma} \cong \mathfrak{su}(8)$	$(\mathfrak{e}_7)^{\tau\gamma} \cong \mathfrak{su}(8)$	Proposition 4.12.3. $(\mathfrak{e}_7)^{\tau\gamma} \cong \mathfrak{su}(8)$.
0902.0431_F02488	148	■ 0.822	$\varphi_*: \mathfrak{su}(8) \rightarrow (\mathfrak{e}_7)^{\tau\gamma}$	$\varphi_*: \mathfrak{su}(8) \rightarrow (\mathfrak{e}_7)^{\tau\gamma}$	This isomorphism is given by the mapping $\varphi_*: \mathfrak{su}(8) \rightarrow (\mathfrak{e}_7)^{\tau\gamma}$,
0902.0431_F02489	149	■ 1.000	$\varphi_*(B) \in (\mathfrak{e}_7)^{\tau\gamma}$	$\varphi_*(B) \in (\mathfrak{e}_7)^{\tau\gamma}$	Proof. We first prove that for $B \in \mathfrak{su}(8)$ we have $\varphi_*(B) \in (\mathfrak{e}_7)^{\tau\gamma}$.
0902.0431_F02490	149	■ 1.000	$B = k(D), D \in \mathfrak{sp}(4)$	$B = k(D), D \in \mathfrak{sp}(4)$	(i) For $B = k(D), D \in \mathfrak{sp}(4)$, we have
0902.0431_F02491	149	■ 0.815	$\varphi_*(C)X = g^{-1}(C(gX) + (gX)C^*)$	$\varphi_*(C)X = g^{-1}(C(gX) + (gX)C^*)$	(recall $\varphi_*(C)X = g^{-1}(C(gX) + (gX)C^*)$)

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0902.0431_F02492	149	■ 1.000	$\varphi_{*}(k(D))=\Phi(\varphi_{*}(D),\,0,0,0)\in(\mathfrak{e}_7)^{\tau\gamma}$	$\varphi_{*}(k(D)) = \Phi(\varphi_{*}(D), 0, 0, 0) \in (\mathfrak{e}_7)^{\tau\gamma}$	Hence, we have $\varphi_{*}(k(D)) = \Phi(\varphi_{*}(D), 0, 0, 0) \in (\mathfrak{e}_7)^{\tau\gamma}$.
0902.0431_F02493	149	■ 0.995	$B=e_1k(T),\,T\in\mathfrak{J}(4,\boldsymbol{H})_0$	$B = e_1k(T), T \in \mathfrak{J}(4, \boldsymbol{H})_0$	(ii) For $B = e_1k(T), T \in \mathfrak{J}(4, \boldsymbol{H})_0$ denote $T = gA, A \in (\mathfrak{J}^C)_{\tau\gamma}$
0902.0431_F02494	149	■ 0.995	$T=gA,\,A\in(\mathfrak{J}^C)_{\tau\gamma}$	$T = gA, A \in (\mathfrak{J}^C)_{\tau\gamma}$	(ii) For $B = e_1k(T), T \in \mathfrak{J}(4, \boldsymbol{H})_0$ (denote $T = gA, A \in (\mathfrak{J}^C)_{\tau\gamma}$)
0902.0431_F02495	149	■ 1.000	$\varphi_{*}\left(e_1k(T)\right)=\Phi(0,\,A,-\gamma A,\,0)\in(\mathfrak{e}_7)^{\tau\gamma}$	$\varphi_{*}(e_1k(T)) = \Phi(0, A, -\gamma A, 0) \in (\mathfrak{e}_7)^{\tau\gamma}$	Hence we have $\varphi_{*}(e_1k(T)) = \Phi(0, A, -\gamma A, 0) \in (\mathfrak{e}_7)^{\tau\gamma}$.
0902.0431_F02496	150	■ 0.997	$\varphi:\mathfrak{su}(8)\rightarrow(\mathfrak{e}_7)^{\tau\gamma}$	$\varphi : \mathfrak{su}(8) \rightarrow (\mathfrak{e}_7)^{\tau\gamma}$	Consequently we see that the mapping $\varphi : \mathfrak{su}(8) \rightarrow (\mathfrak{e}_7)^{\tau\gamma}$ is well-defined. We can
0902.0431_F02497	150	■ 0.998	$\left(E_7\right)^{\tau\gamma} \cong S U(8) / \boldsymbol{Z}_2,\quad \boldsymbol{Z}_2=\{E,-E\}$	$(E_7)^{\tau\gamma} \cong SU(8)/\boldsymbol{Z}_2, \quad \boldsymbol{Z}_2 = \{E, -E\}$	Theorem 4.12.5. $(E_7)^{\tau\gamma} \cong SU(8)/\boldsymbol{Z}_2, \quad \boldsymbol{Z}_2 = \{E, -E\}$.
0902.0431_F02498	150	■ 1.000	$\varphi:S U(8) \rightarrow\left(E_7\right)^{\tau\gamma}$	$\varphi : SU(8) \rightarrow (E_7)^{\tau\gamma}$	Proof. We define a mapping $\varphi : SU(8) \rightarrow (E_7)^{\tau\gamma}$ by
0902.0431_F02499	150	■ 0.615	$\varphi(A) \in\left(E_7\right)^{\tau\gamma}$	$\varphi(A) \in (E_7)^{\tau\gamma}$	We first prove $\varphi(A) \in (E_7)^{\tau\gamma}$. To prove this, for the differential mapping $\varphi_{*} : \mathfrak{su}(8) \rightarrow$
0902.0431_F02500	150	■ 0.615	$\varphi_{*}: \mathfrak{su}(8) \rightarrow$	$\varphi_{*} : \mathfrak{su}(8) \rightarrow$	We first prove $\varphi(A) \in (E_7)^{\tau\gamma}$. To prove this, for the differential mapping $\varphi_{*} : \mathfrak{su}(8) \rightarrow$
0902.0431_F02501	150	■ 1.000	$\varphi_{*}(D) \in\left(\mathfrak{e}_7\right)^{\tau\gamma}$	$\varphi_{*}(D) \in (\mathfrak{e}_7)^{\tau\gamma}$	it is sufficient to show that φ_{*} is well-defined, that is, $\varphi_{*}(D) \in (\mathfrak{e}_7)^{\tau\gamma}$. However
0902.0431_F02502	150	■ 0.983	$\varphi_{*}: \mathfrak{su}(8) \rightarrow\left(\mathfrak{e}_7\right)^{\tau\gamma}$	$\varphi_{*} : \mathfrak{su}(8) \rightarrow (\mathfrak{e}_7)^{\tau\gamma}$	homomorphism. Since $\varphi_{*} : \mathfrak{su}(8) \rightarrow (\mathfrak{e}_7)^{\tau\gamma}$ is onto and $(E_7)^{\tau\gamma}$ is connected (Lemma
0902.0431_F02503	150	■ 1.000	$S U(8) / \boldsymbol{Z}_2 \cong\left(E_7\right)^{\tau\gamma}$	$SU(8)/\boldsymbol{Z}_2 \cong (E_7)^{\tau\gamma}$	Thus we have the isomorphism $SU(8)/\boldsymbol{Z}_2 \cong (E_7)^{\tau\gamma}$.
0902.0431_F02504	150	■ 1.000	$\varphi:S U(8) \rightarrow$	$\varphi : SU(8) \rightarrow$	Remark. Without using Lemma 4.12.4, the fact that the mapping $\varphi : SU(8) \rightarrow$
0902.0431_F02505	150	■ 1.000	$a \in \boldsymbol{R}, \alpha_i(a)$	$a \in \boldsymbol{R}, \alpha_i(a)$	Claim 1. For $a \in \boldsymbol{R}, \alpha_i(a)$ of Lemma 4.8.1 belongs to $\varphi(SU(8))$.
0902.0431_F02506	150	■ 1.000	$\varphi(S U(8))$	$\varphi(SU(8))$	Claim 1. For $a \in \boldsymbol{R}, \alpha_i(a)$ of Lemma 4.8.1 belongs to $\varphi(SU(8))$.
0902.0431_F02507	150	■ 0.582	$\alpha_i(a)=\exp \left(\Phi\left(0,-a E_i, a E_i, 0\right)\right) \in \exp \varphi_{*}(\mathfrak{su}(8))$	$\alpha_i(a) = \exp(\Phi(0, -aE_i, aE_i, 0)) \in \exp \varphi_{*}(\mathfrak{su}(8))$	Proof. $\alpha_i(a) = \exp(\Phi(0, -aE_i, aE_i, 0)) \in \exp \varphi_{*}(\mathfrak{su}(8))$ (Proposition 4.12.3) =
0902.0431_F02508	150	■ 0.991	$\varphi(\exp (\mathfrak{su}(8))) \in \varphi(S U(8))$	$\varphi(\exp(\mathfrak{su}(8))) \in \varphi(SU(8))$	$\varphi(\exp(\mathfrak{su}(8))) \in \varphi(SU(8))$.
0902.0431_F02509	150	■ 1.000	$P \in\left(\mathfrak{M}^C\right)_{\tau \gamma}=\left\{P \in \mathfrak{M}^C \mid \tau \gamma P=P\right\}$	$P \in (\mathfrak{M}^C)_{\tau\gamma} = \{P \in \mathfrak{M}^C \mid \tau\gamma P = P\}$	Claim 2. Any element $P \in (\mathfrak{M}^C)_{\tau\gamma} = \{P \in \mathfrak{M}^C \mid \tau\gamma P = P\}$ can be transformed
0902.0431_F02510	150	■ 1.000	$\alpha \in \varphi(S U(8))$	$\alpha \in \varphi(SU(8))$	to a diagonal form by some $\alpha \in \varphi(SU(8))$:
0902.0431_F02511	150	■ 1.000	$P=(X, Y, \xi, \eta) \in\left(\mathfrak{M}^C\right)_{\tau \gamma}$	$P = (X, Y, \xi, \eta) \in (\mathfrak{M}^C)_{\tau\gamma}$	Proof. Let $P = (X, Y, \xi, \eta) \in (\mathfrak{M}^C)_{\tau\gamma}$. If $\xi \neq 0$. Then
0902.0431_F02512	150	■ 1.000	$Y \in\left(\mathfrak{J}^C\right)_{\tau \gamma}$	$Y \in (\mathfrak{J}^C)_{\tau\gamma}$	Since $Y \in (\mathfrak{J}^C)_{\tau\gamma}$, we have $\gamma Y \in (\mathfrak{J}^C)_{\tau\gamma}$, so that $g(\gamma Y) \in \mathfrak{J}(4, \boldsymbol{H})_0$. Hence, there
0902.0431_F02513	150	■ 1.000	$\gamma Y \in\left(\mathfrak{J}^C\right)_{\tau \gamma}$	$\gamma Y \in (\mathfrak{J}^C)_{\tau\gamma}$	Since $Y \in (\mathfrak{J}^C)_{\tau\gamma}$, we have $\gamma Y \in (\mathfrak{J}^C)_{\tau\gamma}$, so that $g(\gamma Y) \in \mathfrak{J}(4, \boldsymbol{H})_0$. Hence, there
0902.0431_F02514	150	■ 1.000	$g(\gamma Y) \in \mathfrak{J}(4, \boldsymbol{H})_0$	$g(\gamma Y) \in \mathfrak{J}(4, \boldsymbol{H})_0$	Since $Y \in (\mathfrak{J}^C)_{\tau\gamma}$, we have $\gamma Y \in (\mathfrak{J}^C)_{\tau\gamma}$, so that $g(\gamma Y) \in \mathfrak{J}(4, \boldsymbol{H})_0$. Hence, there
0902.0431_F02515	150	■ 0.998	$D \in \operatorname{Sp}(4)$	$D \in \operatorname{Sp}(4)$	exists $D \in \operatorname{Sp}(4)$ such that
0902.0431_F02516	151	■ 1.000	$\xi<0$	$\xi < 0$	If $\xi < 0$, apply $\alpha_1(\pi)$ of Claim 1 on it, then ξ becomes $\xi > 0$.
0902.0431_F02517	151	■ 1.000	$\alpha_1(\pi)$	$\alpha_1(\pi)$	If $\xi < 0$, apply $\alpha_1(\pi)$ of Claim 1 on it, then ξ becomes $\xi > 0$.
0902.0431_F02518	151	■ 1.000	$\alpha \in\left(E_7\right)^{\tau \gamma}$	$\alpha \in (E_7)^{\tau\gamma}$	Claims 1, 2. For a given $\alpha \in (E_7)^{\tau\gamma}$, consider the element $P = \alpha \dot{1} \in (\mathfrak{M}^C)_{\tau\gamma}$. We
0902.0431_F02519	151	■ 1.000	$P=\alpha \dot{1} \in\left(\mathfrak{M}^C\right)_{\tau \gamma}$	$P = \alpha \dot{1} \in (\mathfrak{M}^C)_{\tau\gamma}$	Claims 1, 2. For a given $\alpha \in (E_7)^{\tau\gamma}$, consider the element $P = \alpha \dot{1} \in (\mathfrak{M}^C)_{\tau\gamma}$. We
0902.0431_F02520	151	■ 0.997	$\beta \in \varphi(S U(8))$	$\beta \in \varphi(SU(8))$	first transform P to a diagonal form (Claim 2) by some $\beta \in \varphi(SU(8))$, and we have,
0902.0431_F02521	151	■ 0.973	$a_i=\frac{\eta_i}{\left \eta_i\right } r_i\left(\eta_i\right.$	$a_i = \frac{\eta_i}{ \eta_i } r_i(\eta_i$	where $a_i = \frac{\eta_i}{ \eta_i } r_i$ (η_i is a diagonal element of Y). Since $\eta_i \in \boldsymbol{R}$, we have $\alpha_i(a_i) \in$
0902.0431_F02522	151	■ 0.973	$\left.\eta_i\right) \in \boldsymbol{R}$	$\eta_i \in \boldsymbol{R}$	where $a_i = \frac{\eta_i}{ \eta_i } r_i$ (η_i is a diagonal element of Y). Since $\eta_i \in \boldsymbol{R}$, we have $\alpha_i(a_i) \in$
0902.0431_F02523	151	■ 0.973	$\alpha_i\left(a_i\right) \in$	$\alpha_i(a_i) \in$	where $a_i = \frac{\eta_i}{ \eta_i } r_i$ (η_i is a diagonal element of Y). Since $\eta_i \in \boldsymbol{R}$, we have $\alpha_i(a_i) \in$

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0902.0431_F02524	151	■ 0.997	$\widetilde{\alpha}=\alpha_1\left(a_1\right)^{-1} \alpha_2\left(a_2\right)^{-1} \alpha_3\left(a_3\right)^{-1} \beta \alpha$	$\tilde{\alpha}=\alpha_1\left(a_1\right)^{-1} \alpha_2\left(a_2\right)^{-1} \alpha_3\left(a_3\right)^{-1} \beta \alpha$	$\varphi(SU(8))$ (Claim 1). If we put $\tilde{\alpha}=\alpha_1\left(a_1\right)^{-1} \alpha_2\left(a_2\right)^{-1} \alpha_3\left(a_3\right)^{-1} \beta \alpha$, then, $\tilde{\alpha} \in E_6$
0902.0431_F02525	151	■ 0.997	$\widetilde{\alpha} \in E_6$	$\tilde{\alpha} \in E_6$	$\varphi(SU(8))$ (Claim 1). If we put $\tilde{\alpha}=\alpha_1\left(a_1\right)^{-1} \alpha_2\left(a_2\right)^{-1} \alpha_3\left(a_3\right)^{-1} \beta \alpha$, then, $\tilde{\alpha} \in E_6$
0902.0431_F02526	151	■ 0.995	$\tau \gamma \widetilde{\alpha}=\widetilde{\alpha} \tau \gamma$	$\tau \gamma \tilde{\alpha}=\tilde{\alpha} \tau \gamma$	(Theorem 4.7.2) and $\tilde{\alpha}$ satisfies $\tau \gamma \tilde{\alpha}=\tilde{\alpha} \tau \gamma$. Hence $\tilde{\alpha} \in\left(E_6\right)^{\tau \gamma}=\varphi(S p(4))$ (Theorem
0902.0431_F02527	151	■ 0.995	$\widetilde{\alpha} \in\left(E_6\right)^{\tau \gamma}=\varphi(S p(4))$	$\tilde{\alpha} \in\left(E_6\right)^{\tau \gamma}=\varphi(S p(4))$	(Theorem 4.7.2) and $\tilde{\alpha}$ satisfies $\tau \gamma \tilde{\alpha}=\tilde{\alpha} \tau \gamma$. Hence $\tilde{\alpha} \in\left(E_6\right)^{\tau \gamma}=\varphi(S p(4))$ (Theorem
0902.0431_F02528	151	■ 0.997	$\subset \varphi(S U(8))$	$\subset \varphi(S U(8))$	3.12.2) $\subset \varphi(S U(8))$, Therefore $\alpha=\beta^{-1} \alpha_3\left(a_3\right) \alpha_2\left(a_2\right) \alpha_1\left(a_1\right) \tilde{\alpha} \in \varphi(S U(8))$. This shows
0902.0431_F02529	151	■ 0.997	$\alpha=\beta^{-1} \alpha_3\left(a_3\right) \alpha_2\left(a_2\right) \alpha_1\left(a_1\right) \tilde{\alpha} \in \varphi(S U(8))$	$\alpha=\beta^{-1} \alpha_3\left(a_3\right) \alpha_2\left(a_2\right) \alpha_1\left(a_1\right) \tilde{\alpha} \in \varphi(S U(8))$	3.12.2) $\subset \varphi(S U(8))$, Therefore $\alpha=\beta^{-1} \alpha_3\left(a_3\right) \alpha_2\left(a_2\right) \alpha_1\left(a_1\right) \tilde{\alpha} \in \varphi(S U(8))$. This shows
0902.0431_F02530	151	■ 1.000	$w \in G_2 \subset F_4 \subset E_6 \subset E_7$	$w \in G_2 \subset F_4 \subset E_6 \subset E_7$	This w is the same as $w \in G_2 \subset F_4 \subset E_6 \subset E_7$.
0902.0431_F02531	151	■ 1.000	$\left(E_7\right)^w$	$\left(E_7\right)^w$	We shall study the following subgroup $\left(E_7\right)^w$ of E_7 :
0902.0431_F02532	151	■ 1.000	$E_{7, \mathcal{C}}$	$E_{7, \mathcal{C}}$	We consider the group $E_{7, \mathcal{C}}$ replaced with $\mathcal{C}$ in the place $\mathfrak{C}$ in the definition of
0902.0431_F02533	152	■ 0.999	$\left((S U(3) \times S U(3)) / \mathbb{Z}_3\right) \cdot \mathbb{Z}_2$	$\left((S U(3) \times S U(3)) / \mathbb{Z}_3\right) \cdot \mathbb{Z}_2$	which is isomorphic to the group $\left((S U(3) \times S U(3)) / \mathbb{Z}_3\right) \cdot \mathbb{Z}_2$ (Proposition 3.13.4).
0902.0431_F02534	152	■ 1.000	$\mathfrak{e}_{7, \mathcal{C}}$	$\mathfrak{e}_{7, \mathcal{C}}$	The Lie algebra $\mathfrak{e}_{7, \mathcal{C}}$ of the group $E_{7, \mathcal{C}}$ is given by
0902.0431_F02535	152	■ 1.000	$\pi_0\left(E_{6, \mathcal{C}}\right) \rightarrow \pi_0\left(E_{7, \mathcal{C}}\right) \rightarrow \pi_0\left(\left(\mathfrak{M}_{\mathcal{C}}\right)_1\right)$	$\pi_0\left(E_{6, \mathcal{C}}\right) \rightarrow \pi_0\left(E_{7, \mathcal{C}}\right) \rightarrow \pi_0\left(\left(\mathfrak{M}_{\mathcal{C}}\right)_1\right)$	Proof. From the exact sequence $\pi_0\left(E_{6, \mathcal{C}}\right) \rightarrow \pi_0\left(E_{7, \mathcal{C}}\right) \rightarrow \pi_0\left(\left(\mathfrak{M}_{\mathcal{C}}\right)_1\right)$, that is,
0902.0431_F02536	152	■ 1.000	$\mathbb{Z}_2 \rightarrow \pi_0\left(E_{7, \mathcal{C}}\right) \rightarrow 0$	$\mathbb{Z}_2 \rightarrow \pi_0\left(E_{7, \mathcal{C}}\right) \rightarrow 0$	$\mathbb{Z}_2 \rightarrow \pi_0\left(E_{7, \mathcal{C}}\right) \rightarrow 0$ (Proposition 3.13.4), we see that $\pi_0\left(E_{7, \mathcal{C}}\right)$ is 0 or $\mathbb{Z}_2$.
0902.0431_F02537	152	■ 1.000	$\pi_0\left(E_{7, \mathcal{C}}\right)$	$\pi_0\left(E_{7, \mathcal{C}}\right)$	$\mathbb{Z}_2 \rightarrow \pi_0\left(E_{7, \mathcal{C}}\right) \rightarrow 0$ (Proposition 3.13.4), we see that $\pi_0\left(E_{7, \mathcal{C}}\right)$ is 0 or $\mathbb{Z}_2$.
0902.0431_F02538	152	■ 0.989	$h': \mathcal{C} \rightarrow \mathcal{C}$	$h': \mathcal{C} \rightarrow \mathcal{C}$	Let $h': \mathcal{C} \rightarrow \mathcal{C}$ be the $\mathcal{R}$ -linear isomorphism defined by
0902.0431_F02539	152	■ 0.996	V, W	V, W	Now, let V, W be $\mathcal{C}$ - and $\mathcal{C}$ -vector spaces, respectively. A linear mapping $f: V \rightarrow$
0902.0431_F02540	152	■ 0.996	$f: V \rightarrow$	$f: V \rightarrow$	Now, let V, W be $\mathcal{C}$ - and $\mathcal{C}$ -vector spaces, respectively. A linear mapping $f: V \rightarrow$
0902.0431_F02541	152	■ 0.415	$\mathcal{C}-\mathcal{C}$	$\mathcal{C}-\mathcal{C}$	Similarly, a $\mathcal{C}$ - $\mathcal{C}$ -linear mapping $g: W \rightarrow V$ is defined.
0902.0431_F02542	152	■ 0.415	$g: W \rightarrow V$	$g: W \rightarrow V$	Similarly, a $\mathcal{C}$ - $\mathcal{C}$ -linear mapping $g: W \rightarrow V$ is defined.
0902.0431_F02543	152	■ 0.886	$h': \mathcal{C}^{\mathcal{C}} \rightarrow \mathcal{C}$	$h': \mathcal{C}^{\mathcal{C}} \rightarrow \mathcal{C}$	Definition. Let $h': \mathcal{C}^{\mathcal{C}} \rightarrow \mathcal{C}$ be a $\mathcal{C}$ - $\mathcal{C}$ -linear mapping defined by
0902.0431_F02544	152	■ 0.886	$\mathcal{C}-\mathcal{C}$	$\mathcal{C}-\mathcal{C}$	Definition. Let $h': \mathcal{C}^{\mathcal{C}} \rightarrow \mathcal{C}$ be a $\mathcal{C}$ - $\mathcal{C}$ -linear mapping defined by
0902.0431_F02545	152	■ 0.999	$\Lambda^3\left(\mathcal{C}^6\right)$	$\Lambda^3\left(\mathcal{C}^6\right)$	Now, let $\Lambda^3\left(\mathcal{C}^6\right)$ be the third exterior product of $\mathcal{C}$ -vector space $\mathcal{C}^6$ and we define
0902.0431_F02546	152	■ 0.999	$\mathcal{C}^6$	$\mathcal{C}^6$	Now, let $\Lambda^3\left(\mathcal{C}^6\right)$ be the third exterior product of $\mathcal{C}$ -vector space $\mathcal{C}^6$ and we define
0902.0431_F02547	152	■ 0.818	$f:\left(\mathfrak{P}_{\mathcal{C}}\right)^{\mathcal{C}} \rightarrow \Lambda^3\left(\mathcal{C}^6\right)$	$f:\left(\mathfrak{P}_{\mathcal{C}}\right)^{\mathcal{C}} \rightarrow \Lambda^3\left(\mathcal{C}^6\right)$	a $\mathcal{C}$ - $\mathcal{C}$ -linear isomorphism $f:\left(\mathfrak{P}_{\mathcal{C}}\right)^{\mathcal{C}} \rightarrow \Lambda^3\left(\mathcal{C}^6\right)$ by
0902.0431_F02548	152	■ 1.000	$\left\{e_1, e_2, \cdots, e_6\right\}$	$\left\{e_1, e_2, \cdots, e_6\right\}$	$\left\{e_1, e_2, \cdots, e_6\right\}$ is the canonical basis of $\mathcal{C}^6$ and $x_{ijk} \in \mathcal{C}$ are skew-symmetric tensor:
0902.0431_F02549	152	■ 1.000	$x_{ijk} \in \mathcal{C}$	$x_{ijk} \in \mathcal{C}$	$\left\{e_1, e_2, \cdots, e_6\right\}$ is the canonical basis of $\mathcal{C}^6$ and $x_{ijk} \in \mathcal{C}$ are skew-symmetric tensor:
0902.0431_F02550	152	■ 0.984	$x_{i' j' k'}=\operatorname{sgn}\left(\begin{array}{ccc} i & j & k \\ i' & j' & k' \end{array}\right) x_{ijk}$	$x_{i' j' k'}=\operatorname{sgn}\left(\begin{array}{ccc} i & j & k \\ i' & j' & k' \end{array}\right) x_{ijk}$	$x_{i' j' k'}=\operatorname{sgn}\left(\begin{array}{ccc} i & j & k \\ i' & j' & k' \end{array}\right) x_{ijk}$, where
0902.0431_F02551	153	■ 1.000	$f^{-1}: \Lambda^3\left(\mathcal{C}^6\right) \rightarrow\left(\mathfrak{P}_{\mathcal{C}}\right)^{\mathcal{C}}$	$f^{-1}: \Lambda^3\left(\mathcal{C}^6\right) \rightarrow\left(\mathfrak{P}_{\mathcal{C}}\right)^{\mathcal{C}}$	The inverse mapping $f^{-1}: \Lambda^3\left(\mathcal{C}^6\right) \rightarrow\left(\mathfrak{P}_{\mathcal{C}}\right)^{\mathcal{C}}$ of f is given by
0902.0431_F02552	153	■ 0.979	$h: \mathcal{C} \oplus \mathcal{C} \rightarrow \mathcal{C}^{\mathcal{C}}, h: \mathcal{C} \rightarrow \mathcal{C}$	$h: \mathcal{C} \oplus \mathcal{C} \rightarrow \mathcal{C}^{\mathcal{C}}, h: \mathcal{C} \rightarrow \mathcal{C}$	where $h: \mathcal{C} \oplus \mathcal{C} \rightarrow \mathcal{C}^{\mathcal{C}}, h: \mathcal{C} \rightarrow \mathcal{C}$ are $\mathcal{C}$ - $\mathcal{C}$ -linear mappings defined respectively by
0902.0431_F02553	153	■ 1.000	$S U(6)$	$S U(6)$	The group $S U(6)$ acts naturally on $\Lambda^3\left(\mathcal{C}^6\right)$, that is, the action of $A \in S U(6)$ on
0902.0431_F02554	153	■ 1.000	$a \wedge b \wedge c \in \Lambda^3\left(\mathcal{C}^6\right)$	$a \wedge b \wedge c \in \Lambda^3\left(\mathcal{C}^6\right)$	$a \wedge b \wedge c \in \Lambda^3\left(\mathcal{C}^6\right)$ is defined by

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0902.0431_F02555	153	■ 1.000	$D \in \mathfrak{su}(6)$	$D \in \mathfrak{su}(6)$	Hence, the action of $D \in \mathfrak{su}(6)$ on $\Lambda^3(C^6)$ is given by
0902.0431_F02556	154	■ 1.000	$\mathfrak{su}(6)$	$\mathfrak{su}(6)$	(2) The Lie algebra $\mathfrak{e}_{7,C}$ is isomorphic to the Lie algebra $\mathfrak{su}(6)$ as Lie algebras:
0902.0431_F02557	154	■ 0.342	$\varphi_C : \mathfrak{su}(6) \rightarrow \mathfrak{e}_{7,C}$	$\varphi_C : \mathfrak{su}(6) \rightarrow \mathfrak{e}_{7,C}$	This isomorphism is given by the mapping $\varphi_C : \mathfrak{su}(6) \rightarrow \mathfrak{e}_{7,C}$,
0902.0431_F02558	154	■ 1.000	$\phi_C(B, C)X = h(B, C)X + Xh(B, C)^*$	$\phi_C(B, C)X = h(B, C)X + Xh(B, C)^*$	where $\phi_C : \mathfrak{su}(3) \oplus \mathfrak{su}(3) \rightarrow \mathfrak{e}_{6,C}$ is defined by $\phi_C(B, C)X = h(B, C)X + Xh(B, C)^*$,
0902.0431_F02559	154	■ 1.000	$X \in (\mathfrak{J}_C)^C$	$X \in (\mathfrak{J}_C)^C$	$X \in (\mathfrak{J}_C)^C$ (Lemma 3.13.3).
0902.0431_F02560	154	■ 0.980	$D = \begin{pmatrix} B' & L \\ -L^* & C' \end{pmatrix} \in \mathfrak{su}(6), B', C' \in \mathfrak{u}(3), L \in M(3, C)$	$D = \begin{pmatrix} B' & L \\ -L^* & C' \end{pmatrix} \in \mathfrak{su}(6), B', C' \in \mathfrak{u}(3), L \in M(3, C)$	Proof. (1) For $D = \begin{pmatrix} B' & L \\ -L^* & C' \end{pmatrix} \in \mathfrak{su}(6), B', C' \in \mathfrak{u}(3), L \in M(3, C)$, we let
0902.0431_F02561	154	■ 1.000	$E, A_{ij} \in M(3, C)$	$E, A_{ij} \in M(3, C)$	where $E, A_{ij} \in M(3, C)$, and let $SU(6) \cdot Z_2$ be the semi-direct product of the groups
0902.0431_F02562	154	■ 1.000	$SU(6) \cdot Z_2$	$SU(6) \cdot Z_2$	where $E, A_{ij} \in M(3, C)$, and let $SU(6) \cdot Z_2$ be the semi-direct product of the groups
0902.0431_F02563	154	■ 0.977	$E_7 \cong (SU(6)/Z_3) \cdot Z_2, Z_3 = \{E, \omega_1 E, \omega_1^2 E\}, \omega_1 =$	$E_{7,C} \cong (SU(6)/Z_3) \cdot Z_2, Z_3 = \{E, \omega_1 E, \omega_1^2 E\}, \omega_1 =$	Proposition 4.13.3. $E_{7,C} \cong (SU(6)/Z_3) \cdot Z_2, Z_3 = \{E, \omega_1 E, \omega_1^2 E\}, \omega_1 =$
0902.0431_F02564	154	■ 1.000	$\psi : SU(6) \cdot Z_2 \rightarrow E_{7,C}$	$\psi : SU(6) \cdot Z_2 \rightarrow E_{7,C}$	Proof. We define a mapping $\psi : SU(6) \cdot Z_2 \rightarrow E_{7,C}$ by
0902.0431_F02565	154	■ 1.000	$\psi(A, 1) \in E_{7,C}$	$\psi(A, 1) \in E_{7,C}$	We first have to show that $\psi(A, 1) \in E_{7,C}$. To prove this, it is sufficient to show that
0902.0431_F02566	154	■ 1.000	$\psi_* : \mathfrak{su}(6) \rightarrow \mathfrak{e}_{7,C}$	$\psi_* : \mathfrak{su}(6) \rightarrow \mathfrak{e}_{7,C}$	the differential mapping $\psi_* : \mathfrak{su}(6) \rightarrow \mathfrak{e}_{7,C}$ of ψ :
0902.0431_F02567	154	■ 1.000	ψ	ψ	the differential mapping $\psi_* : \mathfrak{su}(6) \rightarrow \mathfrak{e}_{7,C}$ of ψ :
0902.0431_F02568	154	■ 0.998	$\psi_C : \mathfrak{su}(6) \rightarrow \mathfrak{e}_{7,C}$	$\psi_C : \mathfrak{su}(6) \rightarrow \mathfrak{e}_{7,C}$	coincides with the mapping $\psi_C : \mathfrak{su}(6) \rightarrow \mathfrak{e}_{7,C}$ of Lemma 4.13.2. We put
0902.0431_F02569	155	■ 1.000	$\bar{b}_{ii} = -b_{ii}, \bar{c}_{ii} = -c_{ii}, b_{11} + b_{22} + b_{33} = c_{11} + c_{22} + c_{33} = 0, \bar{\nu} = -\nu$	$\bar{b}_{ii} = -b_{ii}, \bar{c}_{ii} = -c_{ii}, b_{11} + b_{22} + b_{33} = c_{11} + c_{22} + c_{33} = 0, \bar{\nu} = -\nu$	where $\bar{b}_{ii} = -b_{ii}, \bar{c}_{ii} = -c_{ii}, b_{11} + b_{22} + b_{33} = c_{11} + c_{22} + c_{33} = 0, \bar{\nu} = -\nu$.
0902.0431_F02570	155	■ 1.000	$P = (0, 0, 1, 0), \psi_*(D)P$	$P = (0, 0, 1, 0), \psi_*(D)P$	(1) For $P = (0, 0, 1, 0), \psi_*(D)P$ is calculated as follows.
0902.0431_F02571	155	■ 1.000	$P = (E_1, 0, 0, 0), \psi_*(D)P$	$P = (E_1, 0, 0, 0), \psi_*(D)P$	(2) For $P = (E_1, 0, 0, 0), \psi_*(D)P$ is calculated as follows.
0902.0431_F02572	156	■ 1.000	$P = (F_1(1), 0, 0, 0), \psi_*(D)P$	$P = (F_1(1), 0, 0, 0), \psi_*(D)P$	(3) For $P = (F_1(1), 0, 0, 0), \psi_*(D)P$ is calculated as follows.
0902.0431_F02573	157	■ 0.977	$\mathfrak{P}_C^C$	$\mathfrak{P}_C^C$	(4) For the other generator of $\mathfrak{P}_C^C$, that is, $P = (X, 0, 0, 0), (0, X, 0, 0)$ where
0902.0431_F02574	157	■ 0.977	$P = (X, 0, 0, 0), (0, X, 0, 0)$	$P = (X, 0, 0, 0), (0, X, 0, 0)$	(4) For the other generator of $\mathfrak{P}_C^C$, that is, $P = (X, 0, 0, 0), (0, X, 0, 0)$ where
0902.0431_F02575	157	■ 1.000	$X = E_i, F_i(1), F_i(e_1), i = 1, 2, 3$	$X = E_i, F_i(1), F_i(e_1), i = 1, 2, 3$	$X = E_i, F_i(1), F_i(e_1), i = 1, 2, 3$ and $P = (0, 0, 0, 1)$, we have also
0902.0431_F02576	157	■ 1.000	$P = (0, 0, 0, 1)$	$P = (0, 0, 0, 1)$	$X = E_i, F_i(1), F_i(e_1), i = 1, 2, 3$ and $P = (0, 0, 0, 1)$, we have also
0902.0431_F02577	157	■ 0.999	$\psi(E, \epsilon)P = \bar{P}$	$\psi(E, \epsilon)P = \bar{P}$	Thus we see that $\psi(A, 1) \in E_{7,C}$ for $A \in SU(6)$. Since $\psi(E, \epsilon)P = \bar{P}$, we see
0902.0431_F02578	157	■ 1.000	$\psi(E, \epsilon) = \epsilon \in G_{2,C} (= \text{Aut}(C)) \subset F_{4,C} \subset E_{6,C} \subset E_{7,C}$	$\psi(E, \epsilon) = \epsilon \in G_{2,C} (= \text{Aut}(C)) \subset F_{4,C} \subset E_{6,C} \subset E_{7,C}$	$\psi(E, \epsilon) = \epsilon \in G_{2,C} (= \text{Aut}(C)) \subset F_{4,C} \subset E_{6,C} \subset E_{7,C}$. We shall show that
0902.0431_F02579	157	■ 0.997	$\psi(A, \epsilon) = \varphi(A, 1)\varphi(E, \epsilon) \in E_{7,C}$	$\psi(A, \epsilon) = \varphi(A, 1)\varphi(E, \epsilon) \in E_{7,C}$	Thus, for $A \in SU(6)$, we have $\psi(A, \epsilon) = \varphi(A, 1)\varphi(E, \epsilon) \in E_{7,C}$. Since ψ induces a
0902.0431_F02580	157	■ 0.789	$\psi_* : \mathfrak{su}(6) \rightarrow \mathfrak{e}_{7,C}, \psi : SU(6) \rightarrow (E_{7,C})_0$	$\psi_* : \mathfrak{su}(6) \rightarrow \mathfrak{e}_{7,C}, \psi : SU(6) \rightarrow (E_{7,C})_0$	surjection $\psi_* : \mathfrak{su}(6) \rightarrow \mathfrak{e}_{7,C}, \psi : SU(6) \rightarrow (E_{7,C})_0$ (which is the connected component
0902.0431_F02581	157	■ 0.997	$\epsilon = \psi(E, \epsilon) \notin (E_{7,C})_0$	$\epsilon = \psi(E, \epsilon) \notin (E_{7,C})_0$	of $E_{7,C}$ containing the identity 1) is onto. However $\epsilon = \psi(E, \epsilon) \notin (E_{7,C})_0$. Indeed,
0902.0431_F02582	157	■ 0.817	$E_{7,C}$	$E_{7,C}$	does not hold. Therefore $E_{7,C}$ has just two connected components (see Lemma 4.13.1).
0902.0431_F02583	157	■ 1.000	$\text{Ker } \psi = \{E, \omega_1 E, \omega_1^2 E\} \times 1 = Z_3 \times 1$	$\text{Ker } \psi = \{E, \omega_1 E, \omega_1^2 E\} \times 1 = Z_3 \times 1$	Hence $\psi : SU(6) \cdot Z_2 \rightarrow E_{7,C}$ is onto. $\text{Ker } \psi = \{E, \omega_1 E, \omega_1^2 E\} \times 1 = Z_3 \times 1$ easily
0902.0431_F02584	157	■ 1.000	$(SU(6)/Z_3) \cdot Z_2 \cong E_{7,C}$	$(SU(6)/Z_3) \cdot Z_2 \cong E_{7,C}$	obtained. Thus we have the isomorphism $(SU(6)/Z_3) \cdot Z_2 \cong E_{7,C}$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_F02585	157	■ 1.000	$\left(\mathfrak{P}_C\right)^C \oplus \left(M(3, C)^C \oplus M(3, C)^C\right)$	$(\mathfrak{P}_C)^C \oplus (M(3, C)^C \oplus M(3, C)^C)$	We identify $\mathfrak{P}_C^C \oplus (M(3, C)^C \oplus M(3, C)^C)$ with $\mathfrak{P}^C$ (using the identification
0902.0431_F02586	157	■ 1.000	$\mathfrak{J}_C \oplus M(3, C)^C$	$\mathfrak{J}_C^C \oplus M(3, C)^C$	with $\mathfrak{J}^C$ in Section 3.13) by
0902.0431_F02587	158	■ 0.822	$\mu: M(6, C) \rightarrow M(3, C)^C \oplus M(3, C)^C$	$\mu: M(6, C) \rightarrow M(3, C)^C \oplus M(3, C)^C$	Further, we define a C - C -linear mapping $u: M(6, C) \rightarrow M(3, C)^C \oplus M(3, C)^C$ by
0902.0431_F02588	158	■ 1.000	$M_{ij} \in M(3, C)$	$M_{ij} \in M(3, C)$	where $M_{ij} \in M(3, C)$. The inverse mapping $\mu^{-1}: M(3, C)^C \oplus M(3, C)^C \rightarrow M(6, C)$
0902.0431_F02589	158	■ 1.000	$\mu^{-1}: M(3, C)^C \oplus M(3, C)^C \rightarrow M(6, C)$	$\mu^{-1}: M(3, C)^C \oplus M(3, C)^C \rightarrow M(6, C)$	where $M_{ij} \in M(3, C)$. The inverse mapping $u^{-1}: M(3, C)^C \oplus M(3, C)^C \rightarrow M(6, C)$
0902.0431_F02590	158	■ 1.000	$\widetilde{M} \in M(6, C)$	$\widetilde{M} \in M(6, C)$	Lemma 4.13.4. For $D \in \mathfrak{su}(6)$ and $\widetilde{M} \in M(6, C)$, we have
0902.0431_F02591	159	■ 0.615	$\phi_C(B, C)M = M\tau h(B, C)^* = -Mh(B, C)$	$\phi_C(B, C)M = M\tau h(B, C)^* = -Mh(B, C)$	(using $\phi_C(B, C)M = M\tau h(B, C)^* = -Mh(B, C)$ and $-2\tau h(L) \times N = N\tau h(L)$ etc.)
0902.0431_F02592	159	■ 0.615	$-2\tau h(L) \times N = N\tau h(L)$	$-2\tau h(L) \times N = N\tau h(L)$	(using $\phi_C(B, C)M = M\tau h(B, C)^* = -Mh(B, C)$ and $-2\tau h(L) \times N = N\tau h(L)$ etc.)
0902.0431_F02593	159	■ 0.797	$\xrightarrow{\mu^{-1}} \dots$	$\xrightarrow{\mu^{-1}} \dots$	$\xrightarrow{\mu^{-1}} \dots$ by simple calculations $\dots$
0902.0431_F02594	159	■ 0.797	$\dots$	$\dots$	$\xrightarrow{\mu^{-1}} \dots$ by simple calculations $\dots$
0902.0431_F02595	159	■ 0.809	$f: \mathfrak{P}^C \rightarrow \Lambda^3(C^6) \oplus M(6, C)$	$f: \mathfrak{P}^C \rightarrow \Lambda^3(C^6) \oplus M(6, C)$	Definition. We define a C - C -linear isomorphism $f: \mathfrak{P}^C \rightarrow \Lambda^3(C^6) \oplus M(6, C)$
0902.0431_F02596	159	■ 1.000	$SU(3) \times SU(6)$	$SU(3) \times SU(6)$	The group $SU(3) \times SU(6)$ acts on $\Lambda^3(C^6) \oplus M(6, C)$ by
0902.0431_F02597	159	■ 1.000	$\Lambda^3(C^6) \oplus M(6, C)$	$\Lambda^3(C^6) \oplus M(6, C)$	The group $SU(3) \times SU(6)$ acts on $\Lambda^3(C^6) \oplus M(6, C)$ by
0902.0431_F02598	159	■ 1.000	$Q\widetilde{M}$	$Q\widetilde{M}$	where $Q\widetilde{M}$ means $\begin{pmatrix} Q & 0 \\ 0 & Q \end{pmatrix} \begin{pmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{pmatrix} = \begin{pmatrix} QM_{11} & QM_{12} \\ QM_{21} & QM_{22} \end{pmatrix}, M_{ij} \in M(3, C)$.
0902.0431_F02599	159	■ 1.000	$\left(\begin{array}{cc} Q & 0 \\ 0 & Q \end{array}\right) \left(\begin{array}{cc} M_{11} & M_{12} \\ M_{21} & M_{22} \end{array}\right) = \left(\begin{array}{cc} QM_{11} & QM_{12} \\ QM_{21} & QM_{22} \end{array}\right), M_{ij} \in M(3, C)$	$\left(\begin{array}{cc} Q & 0 \\ 0 & Q \end{array}\right) \left(\begin{array}{cc} M_{11} & M_{12} \\ M_{21} & M_{22} \end{array}\right) = \left(\begin{array}{cc} QM_{11} & QM_{12} \\ QM_{21} & QM_{22} \end{array}\right), M_{ij} \in M(3, C)$	where $Q\widetilde{M}$ means $\begin{pmatrix} Q & 0 \\ 0 & Q \end{pmatrix} \begin{pmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{pmatrix} = \begin{pmatrix} QM_{11} & QM_{12} \\ QM_{21} & QM_{22} \end{pmatrix}, M_{ij} \in M(3, C)$.
0902.0431_F02600	159	■ 0.918	$(E_7)^w \cong (SU(3) \times SU(6))/Z_3, \quad Z_3 = \{(E, E), (\omega_1 E, \omega_1 E)\}$	$(E_7)^w \cong (SU(3) \times SU(6))/Z_3, \quad Z_3 = \{(E, E), (\omega_1 E, \omega_1 E)\}$	Theorem 4.13.5. $(E_7)^w \cong (SU(3) \times SU(6))/Z_3, \quad Z_3 = \{(E, E), (\omega_1 E, \omega_1 E)\}$,
0902.0431_F02601	159	■ 1.000	$\psi: SU(3) \times SU(6) \rightarrow (E_7)^w$	$\psi: SU(3) \times SU(6) \rightarrow (E_7)^w$	Proof. We defined a mapping $\psi: SU(3) \times SU(6) \rightarrow (E_7)^w$ by
0902.0431_F02602	159	■ 1.000	$\psi(Q, A) \in E_7$	$\psi(Q, A) \in E_7$	We first have to prove that $\psi(Q, A) \in E_7$. To prove this, since $\varphi(Q, E) \in (E_6)^w \subset$
0902.0431_F02603	159	■ 1.000	$\varphi(Q, E) \in (E_6)^w \subset$	$\varphi(Q, E) \in (E_6)^w \subset$	We first have to prove that $\psi(Q, A) \in E_7$. To prove this, since $\varphi(Q, E) \in (E_6)^w \subset$
0902.0431_F02604	159	■ 1.000	$\psi(E, A) \in E_7$	$\psi(E, A) \in E_7$	$(E_7)^w$, it suffices to show that $\psi(E, A) \in E_7$. Moreover, it is sufficient to show that,
0902.0431_F02605	159	■ 0.910	$\psi_*: \mathfrak{su}(3) \oplus \mathfrak{su}(6) \rightarrow \mathfrak{e}_7$	$\psi_*: \mathfrak{su}(3) \oplus \mathfrak{su}(6) \rightarrow \mathfrak{e}_7$	for the differential mapping $\psi_*: \mathfrak{su}(3) \oplus \mathfrak{su}(6) \rightarrow \mathfrak{e}_7$ of ψ , $\psi_*(0, D)$ coincides with
0902.0431_F02606	159	■ 0.910	$\psi, \psi_*(0, D)$	$\psi, \psi_*(0, D)$	for the differential mapping $\psi_*: \mathfrak{su}(3) \oplus \mathfrak{su}(6) \rightarrow \mathfrak{e}_7$ of ψ , $\psi_*(0, D)$ coincides with
0902.0431_F02607	159	■ 1.000	$(\mathfrak{P}^C)_w = \{P \in \mathfrak{P}^C \mid wP = P\} = (\mathfrak{P}_C)^C$	$(\mathfrak{P}^C)_w = \{P \in \mathfrak{P}^C \mid wP = P\} = (\mathfrak{P}_C)^C$	$(\mathfrak{P}^C)_w = \{P \in \mathfrak{P}^C \mid wP = P\} = (\mathfrak{P}_C)^C$, obviously we have $w\psi(Q, A) = \psi(Q, A)w$,
0902.0431_F02608	159	■ 1.000	$w\psi(Q, A) = \psi(Q, A)w$	$w\psi(Q, A) = \psi(Q, A)w$	$(\mathfrak{P}^C)_w = \{P \in \mathfrak{P}^C \mid wP = P\} = (\mathfrak{P}_C)^C$, obviously we have $w\psi(Q, A) = \psi(Q, A)w$,
0902.0431_F02609	159	■ 1.000	$\psi(Q, A) \in (E_7)^w$	$\psi(Q, A) \in (E_7)^w$	hence $\psi(Q, A) \in (E_7)^w$. Evidently ψ is a homomorphism. We shall show that ψ is
0902.0431_F02610	159	■ 1.000	$\alpha \in (E_7)^w$	$\alpha \in (E_7)^w$	onto. Let $\alpha \in (E_7)^w$. Since the restriction α' to $(\mathfrak{P}^C)_w = (\mathfrak{P}_C)^C$ of α belongs to
0902.0431_F02611	159	■ 1.000	$(\mathfrak{P}^C)_w = (\mathfrak{P}_C)^C$	$(\mathfrak{P}^C)_w = (\mathfrak{P}_C)^C$	onto. Let $\alpha \in (E_7)^w$. Since the restriction α' to $(\mathfrak{P}^C)_w = (\mathfrak{P}_C)^C$ of α belongs to
0902.0431_F02612	159	■ 0.999	$\beta = \psi(E, A)^{-1}\alpha$	$\beta = \psi(E, A)^{-1}\alpha$	(Proposition 4.13.3). In the former case, let $\beta = \psi(E, A)^{-1}\alpha$, then $\beta (\mathfrak{P}_C)^C = 1$,
0902.0431_F02613	159	■ 0.999	$\beta (\mathfrak{P}_C)^C = 1$	$\beta (\mathfrak{P}_C)^C = 1$	(Proposition 4.13.3). In the former case, let $\beta = \psi(E, A)^{-1}\alpha$, then $\beta (\mathfrak{P}_C)^C = 1$,

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0902.0431_F02614	159	■ 0.999	$\backslash beta \ \backslash in \backslash left(G_{\{2\}}\backslash right)^{\{w\}}=S \ U(3)$	$\beta \in (G_2)^w = SU(3)$	hence $\beta \in G_2$. Furthermore, $\beta \in (G_2)^w = SU(3)$ (Theorem 1.9.4), so there exists
0902.0431_F02615	159	■ 1.000	$Q \ \backslash in \ S \ U(3)$	$Q \in SU(3)$	$Q \in SU(3)$ such that
0902.0431_F02616	160	■ 1.000	$\backslash gamma_{\{1\}} \ \backslash in \ G_{\{2\}} \ \backslash subset \ F_{\{4\}} \ \backslash subset \ E_{\{6\}} \ \backslash subset \ E_{\{7\}}$	$\gamma_1 \in G_2 \subset F_4 \subset E_6 \subset E_7$	Then, $\gamma_1 \in G_2 \subset F_4 \subset E_6 \subset E_7$. From the same argument as Section 1, we have $\gamma_1 \alpha \in$
0902.0431_F02617	160	■ 1.000	$\backslash gamma_{\{1\}} \ \backslash alpha \ \backslash in$	$\gamma_1 \alpha \in$	Then, $\gamma_1 \in G_2 \subset F_4 \subset E_6 \subset E_7$. From the same argument as Section 1, we have $\gamma_1 \alpha \in$
0902.0431_F02618	160	■ 0.998	$\backslash gamma_{\{1\}} \ \backslash in \backslash left(G_{\{2\}}\backslash right)^{\{w\}}=S \ U(3)$	$\gamma_1 \in (G_2)^w = SU(3)$	$(E_7)^w$, hence $\gamma_1 \in (G_2)^w = SU(3)$. However this is a contradiction (Theorem 1.9.4).
0902.0431_F02619	160	■ 1.000	$\backslash operatorname{Ker} \ \backslash psi=\backslash left\{(E, \ E), \backslash left(\backslash omega_{\{1\}} \ E, \ \backslash omega_{\{1\}} \ E\backslash right), \backslash left(\backslash omega_{\{1\}} \{ \ }^{\{2\}} \ E, \ \backslash omega_{\{1\}} \{ \ }^{\{2\}} \ E\backslash right)\backslash right\}=\backslash boldsymbol{Z}_{\{3\}}$	$\text{Ker } \psi = \{(E, E), (\omega_1 E, \omega_1 E), (\omega_1^2 E, \omega_1^2 E)\} = \mathbf{Z}_3$	Therefore that ψ is onto is shown. $\text{Ker} \psi = \{(E, E), (\omega_1 E, \omega_1 E), (\omega_1^2 E, \omega_1^2 E)\} = \mathbf{Z}_3$
0902.0431_F02620	160	■ 0.999	$(S \ U(3) \ \backslash times \ S \ U(6)) \ / \ \backslash boldsymbol{Z}_{\{3\}} \ \backslash cong \ \backslash left(E_{\{7\}}\backslash right)^{\{w\}}$	$(SU(3) \times SU(6))/\mathbf{Z}_3 \cong (E_7)^w$	is easily obtained. Thus we have the isomorphism $SU(3) \times SU(6))/\mathbf{Z}_3 \cong (E_7)^w$.
0902.0431_F02621	160	■ 1.000	$\backslash left((S \ U(3) \ \backslash times \ S \ U(6)) \ / \ \backslash boldsymbol{Z}_{\{3\}}\backslash right) \ \backslash cdot \ \backslash boldsymbol{Z}_{\{2\}}$	$((SU(3) \times SU(6))/\mathbf{Z}_3) \cdot \mathbf{Z}_2$	product $(SU(3) \times SU(6))/\mathbf{Z}_3 \cdot \mathbf{Z}_2$ (the action of the group $\mathbf{Z}_2 = \{1, \gamma\}$ to the group
0902.0431_F02622	160	■ 1.000	$\backslash boldsymbol{Z}_{\{2\}}=\{1, \ \backslash gamma\backslash }$	$\mathbf{Z}_2 = \{1, \gamma\}$	product $((SU(3) \times SU(6))/\mathbf{Z}_3) \cdot \mathbf{Z}_2$ (the action of the group $\mathbf{Z}_2 = \{1, \gamma\}$ to the group
0902.0431_F02623	160	■ 0.953	$\backslash left.\backslash gamma(Q, \ A)=\backslash left(\backslash bar{Q}, \ \overline{\backslash operatorname{Ad}\backslash left(J_{\{3\}}\backslash right) \ A}\backslash right)\backslash right)$	$\gamma(Q, A) = \left(\bar{Q}, \text{Ad}(J_3) \overline{A}\right)$	$SU(3) \times SU(6)$ is $\gamma(Q, A) = (\bar{Q}, \text{Ad}(J_3) \overline{A})$.
0902.0431_F02624	160	■ 1.000	$\backslash left(\backslash mathfrak{e}_{\{7\}}\backslash right)^{\{w\}}$	$(\mathfrak{e}_7)^w$	forms an $\mathbf{R}$ -basis of $(\mathfrak{e}_7)^w$. So, $\dim(\mathfrak{e}_7)^w = 10 + 14 + 6 \times 3 + 1 = 43 = 8 + 35 =$
0902.0431_F02625	160	■ 1.000	$\backslash operatorname{dim}\backslash left(\backslash mathfrak{e}_{\{7\}} \ \backslash right)^{\{w\}}=10+14+6 \ \backslash times \ 3+1=43=8+35=$	$\dim(\mathfrak{e}_7)^w = 10 + 14 + 6 \times 3 + 1 = 43 = 8 + 35 =$	forms an $\mathbf{R}$ -basis of $(\mathfrak{e}_7)^w$. So, $\dim(\mathfrak{e}_7)^w = 10 + 14 + 6 \times 3 + 1 = 43 = 8 + 35 =$
0902.0431_F02626	160	■ 0.995	$\backslash operatorname{dim}(\backslash mathfrak{su}(3) \ \backslash oplus \ \backslash mathfrak{su}(6))$	$\dim(\mathfrak{su}(3) \oplus \mathfrak{su}(6))$	$\dim(\mathfrak{su}(3) \oplus \mathfrak{su}(6))$. Hence φ is onto.
0902.0431_F02627	160	■ 1.000	$\backslash operatorname{Iso}_{\{C\}}\backslash left(\backslash mathfrak{P}^{\{C\}}\backslash right)=G \ L(78, \ C)$	$\text{Iso}_C(\mathfrak{P}^C) = GL(78, C)$	Proof. Evidently E_7^C is an algebraic subgroup of $\text{Iso}_C(\mathfrak{P}^C) = GL(78, C)$. If
0902.0431_F02628	160	■ 1.000	$\backslash langle \ X, \ Y\rangle\!:\backslash langle\backslash alpha \ X, \ Y\rangle\!=\backslash left \ \backslash langle \ X, \ \backslash alpha^*\ Y\rangle\!\backslash right\langle \backslash langle \ X, \ Y\rangle\!:\langle \alpha X, Y\rangle = \langle X, \alpha^* Y\rangle$	$\langle X, Y\rangle : \langle \alpha X, Y\rangle = \langle X, \alpha^* Y\rangle$	$\langle X, Y\rangle : \langle \alpha X, Y\rangle = \langle X, \alpha^* Y\rangle$ is $\alpha^* = \tau \lambda \alpha^{-1} \lambda^{-1} \tau \in E_7^C$. Hence, from Chevalley's
0902.0431_F02629	160	■ 1.000	$\backslash alpha^*=\backslash tau \ \backslash lambda \ \backslash alpha^{\{-1\}} \ \backslash lambda^{\{-1\}} \ \backslash tau \ \backslash in \ E_{\{7\}} \{ \ }^{\{C\}}$	$\alpha^* = \tau \lambda \alpha^{-1} \lambda^{-1} \tau \in E_7^C$	$\langle X, Y\rangle : \langle \alpha X, Y\rangle = \langle X, \alpha^* Y\rangle$ is $\alpha^* = \tau \lambda \alpha^{-1} \lambda^{-1} \tau \in E_7^C$. Hence, from Chevalley's
0902.0431_F02630	161	■ 0.993	$P, \ Q \ \backslash in \ \backslash mathfrak{P}$	$P, Q \in \mathfrak{P}$	For $P, Q \in \mathfrak{P}$ or $\mathfrak{P}'$, we define an $\mathbf{R}$ -linear mapping $P \times Q : \mathfrak{P} \rightarrow \mathfrak{P}$ or $\mathfrak{P}' \rightarrow \mathfrak{P}'$ as
0902.0431_F02631	161	■ 0.993	$\backslash mathfrak{P}^{\backslash prime}$	$\mathfrak{P}'$	For $P, Q \in \mathfrak{P}$ or $\mathfrak{P}'$, we define an $\mathbf{R}$ -linear mapping $P \times Q : \mathfrak{P} \rightarrow \mathfrak{P}$ or $\mathfrak{P}' \rightarrow \mathfrak{P}'$ as
0902.0431_F02632	161	■ 0.993	$P \ \backslash times \ Q: \ \backslash mathfrak{P} \ \backslash rightharpoonup \ \backslash mathfrak{P}$	$P \times Q : \mathfrak{P} \rightarrow \mathfrak{P}$	For $P, Q \in \mathfrak{P}$ or $\mathfrak{P}'$, we define an $\mathbf{R}$ -linear mapping $P \times Q : \mathfrak{P} \rightarrow \mathfrak{P}$ or $\mathfrak{P}' \rightarrow \mathfrak{P}'$ as
0902.0431_F02633	161	■ 0.993	$\backslash mathfrak{P}^{\backslash prime} \ \backslash rightharpoonup \ \backslash mathfrak{P}^{\backslash prime}$	$\mathfrak{P}' \rightarrow \mathfrak{P}'$	For $P, Q \in \mathfrak{P}$ or $\mathfrak{P}'$, we define an $\mathbf{R}$ -linear mapping $P \times Q : \mathfrak{P} \rightarrow \mathfrak{P}$ or $\mathfrak{P}' \rightarrow \mathfrak{P}'$ as
0902.0431_F02634	161	■ 1.000	$\backslash langle \ P, \ Q\rangle\!_{\{\backslash sigma\}}$	$\langle P, Q\rangle_\sigma$	similar to Section 4.1. And we define a Hermitian inner product $\langle P, Q\rangle_\sigma$ in $\mathfrak{P}^C$ by
0902.0431_F02635	162	■ 1.000	$133+56 \ \backslash times \ 2+3=248$	$133 + 56 \times 2 + 3 = 248$	Theorem 5.1.1. In a $133+56 \times 2+3=248$ dimensional C -vector space
0902.0431_F02636	162	■ 1.000	$\backslash left[R_{\{1\}}, \ R_{\{2\}}\backslash right]$	$[R_1, R_2]$	if we define a Lie bracket $[R_1, R_2]$ by
0902.0431_F02637	163	■ 1.000	$\backslash mathfrak{K}^{\{C\}}=\backslash mathfrak{P}^{\{C\}} \ \backslash oplus \ \backslash mathfrak{P}^{\{C\}} \ \backslash oplus \ C \ \backslash oplus \ C \ \backslash oplus \ C$	$\mathfrak{K}^C = \mathfrak{P}^C \oplus \mathfrak{P}^C \oplus C \oplus C \oplus C$	where $\mathfrak{K}^C = \mathfrak{P}^C \oplus \mathfrak{P}^C \oplus C \oplus C \oplus C$. Let $p : \mathfrak{e}_8^C \rightarrow \mathfrak{e}_7^C$ and $q : \mathfrak{e}_8^C \rightarrow \mathfrak{K}^C$ be
0902.0431_F02638	163	■ 1.000	$p: \ \backslash mathfrak{e}_{\{8\}} \{ \ }^{\{C\}} \ \backslash rightharpoonup \ \backslash mathfrak{e}_{\{7\}} \{ \ }^{\{C\}}$	$p : \mathfrak{e}_8^C \rightarrow \mathfrak{e}_7^C$	where $\mathfrak{K}^C = \mathfrak{P}^C \oplus \mathfrak{P}^C \oplus C \oplus C \oplus C$. Let $p : \mathfrak{e}_8^C \rightarrow \mathfrak{e}_7^C$ and $q : \mathfrak{e}_8^C \rightarrow \mathfrak{K}^C$ be
0902.0431_F02639	163	■ 1.000	$q: \ \backslash mathfrak{e}_{\{8\}} \{ \ }^{\{C\}} \ \backslash rightharpoonup \ \backslash mathfrak{K}^{\{C\}}$	$q : \mathfrak{e}_8^C \rightarrow \mathfrak{K}^C$	where $\mathfrak{K}^C = \mathfrak{P}^C \oplus \mathfrak{P}^C \oplus C \oplus C \oplus C$. Let $p : \mathfrak{e}_8^C \rightarrow \mathfrak{e}_7^C$ and $q : \mathfrak{e}_8^C \rightarrow \mathfrak{K}^C$ be
0902.0431_F02640	163	■ 1.000	$\backslash mathfrak{e}_{\{8\}} \{ \ }^{\{C\}}=\backslash mathfrak{e}_{\{7\}} \{ \ }^{\{C\}} \ \backslash oplus \ \backslash mathfrak{K}^{\{C\}}$	$\mathfrak{e}_8^C = \mathfrak{e}_7^C \oplus \mathfrak{K}^C$	projections of $\mathfrak{e}_8^C = \mathfrak{e}_7^C \oplus \mathfrak{K}^C$. Now, let $\mathfrak{a}$ be a non-zero ideal of $\mathfrak{e}_8^C$. Then $p(\mathfrak{a})$ is
0902.0431_F02641	163	■ 1.000	$\backslash Phi \ \backslash in \ p(\backslash mathfrak{a})$	$\Phi \in p(\mathfrak{a})$	an ideal of $\mathfrak{e}_7^C$. Indeed, if $\Phi \in p(\mathfrak{a})$, then there exists $(0, P, Q, r, s, t) \in \mathfrak{K}^C$ such that
0902.0431_F02642	163	■ 1.000	$(\emptyset, \ P, \ Q, \ r, \ s, \ t) \ \backslash in \ \backslash mathfrak{K}^{\{C\}}$	$(0, P, Q, r, s, t) \in \mathfrak{K}^C$	an ideal of $\mathfrak{e}_7^C$. Indeed, if $\Phi \in p(\mathfrak{a})$, then there exists $(0, P, Q, r, s, t) \in \mathfrak{K}^C$ such that
0902.0431_F02643	163	■ 0.970	$(\backslash Phi, \ P, \ Q, \ r, \ s, \ t) \ \backslash in \ \backslash mathfrak{a}$	$(\Phi, P, Q, r, s, t) \in \mathfrak{a}$	$(\Phi, P, Q, r, s, t) \in \mathfrak{a}$. For any $\Phi_1 \in \mathfrak{e}_7^C$, we have
0902.0431_F02644	163	■ 0.970	$\backslash Phi_{\{1\}} \ \backslash in \ \backslash mathfrak{e}_{\{7\}} \{ \ }^{\{C\}}$	$\Phi_1 \in \mathfrak{e}_7^C$	$(\Phi, P, Q, r, s, t) \in \mathfrak{a}$. For any $\Phi_1 \in \mathfrak{e}_7^C$, we have
0902.0431_F02645	163	■ 1.000	$\backslash left[\backslash Phi_{\{1\}}, \ \backslash Phi\backslash right] \ \backslash in \ p(\backslash mathfrak{a})$	$[\Phi_1, \Phi] \in p(\mathfrak{a})$	hence $[\Phi_1, \Phi] \in p(\mathfrak{a})$.
0902.0431_F02646	163	■ 1.000	$\backslash mathfrak{e}_{\{7\}} \{ \ }^{\{C\}} \ \backslash cap \ \backslash mathfrak{a} \ \backslash neq \backslash \{0\}$	$\mathfrak{e}_7^C \cap \mathfrak{a} \neq \{0\}$	We shall show that either $\mathfrak{e}_7^C \cap \mathfrak{a} \neq \{0\}$ or $\mathfrak{K}^C \cap \mathfrak{a} \neq \{0\}$. Assume that $\mathfrak{e}_7^C \cap \mathfrak{a} = \{0\}$
0902.0431_F02647	163	■ 1.000	$\backslash mathfrak{K}^{\{C\}} \ \backslash cap \ \backslash mathfrak{a} \ \backslash neq \backslash \{0\}$	$\mathfrak{K}^C \cap \mathfrak{a} \neq \{0\}$	We shall show that either $\mathfrak{e}_7^C \cap \mathfrak{a} \neq \{0\}$ or $\mathfrak{K}^C \cap \mathfrak{a} \neq \{0\}$. Assume that $\mathfrak{e}_7^C \cap \mathfrak{a} = \{0\}$
0902.0431_F02648	163	■ 1.000	$\backslash mathfrak{e}_{\{7\}} \{ \ }^{\{C\}} \ \backslash cap \ \backslash mathfrak{a}=\backslash \{0\}$	$\mathfrak{e}_7^C \cap \mathfrak{a} = \{0\}$	We shall show that either $\mathfrak{e}_7^C \cap \mathfrak{a} \neq \{0\}$ or $\mathfrak{K}^C \cap \mathfrak{a} \neq \{0\}$. Assume that $\mathfrak{e}_7^C \cap \mathfrak{a} = \{0\}$
0902.0431_F02649	163	■ 1.000	$\backslash mathfrak{K}^{\{C\}} \ \backslash cap \ \backslash mathfrak{a}=\backslash \{0\}$	$\mathfrak{K}^C \cap \mathfrak{a} = \{0\}$	and $\mathfrak{K}^C \cap \mathfrak{a} = \{0\}$. Then the mapping $p _{\mathfrak{a}} : \mathfrak{a} \rightarrow \mathfrak{e}_7^C$ is injective because $\mathfrak{K}^C \cap \mathfrak{a} = \{0\}$.
0902.0431_F02650	163	■ 1.000	$p \ \backslash mid \ \backslash mathfrak{a}: \ \backslash mathfrak{a} \ \backslash rightharpoonup \ \backslash mathfrak{e}_{\{7\}} \{ \ }^{\{C\}}$	$p \mid \mathfrak{a} : \mathfrak{a} \rightarrow \mathfrak{e}_7^C$	and $\mathfrak{K}^C \cap \mathfrak{a} = \{0\}$. Then the mapping $p _{\mathfrak{a}} : \mathfrak{a} \rightarrow \mathfrak{e}_7^C$ is injective because $\mathfrak{K}^C \cap \mathfrak{a} = \{0\}$.

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0902.0431_F02651	163	■ 1.000	$p(\frac{a}{e})=\frac{e}{7}\{ \}^C$	$p(\mathfrak{a}) = \mathfrak{e}_7^C$	Since $p(\mathfrak{a})$ is a non-zero ideal of $\mathfrak{e}_7^C$ and $\mathfrak{e}_7^C$ is simple, we have $p(\mathfrak{a}) = \mathfrak{e}_7^C$. Hence
0902.0431_F02652	163	■ 1.000	$\operatorname{dim}_C(\frac{a}{e})=\operatorname{dim}_C(p(\frac{a}{e}))=\operatorname{dim}_C(\frac{e}{7}\{ \}^C\left(\frac{e}{7}\{ \}^C\right)=133$	$\dim_C(\mathfrak{a}) = \dim_C(p(\mathfrak{a})) = \dim_C(\mathfrak{e}_7^C) = 133$	$\dim_C(\mathfrak{a}) = \dim_C(p(\mathfrak{a})) = \dim_C(\mathfrak{e}_7^C) = 133$ On the other hand, since $\mathfrak{e}_7^C \cap \mathfrak{a} = \{0\}$,
0902.0431_F02653	163	■ 1.000	$q \mid \frac{a}{K}: \frac{a}{K} \rightarrow \frac{a}{K}$	$q \mid \mathfrak{a} : \mathfrak{a} \rightarrow \mathfrak{K}^C$	$q \mid \mathfrak{a} : \mathfrak{a} \rightarrow \mathfrak{K}^C$ is also injective. Hence we have $\dim_C(\mathfrak{a}) \leq \dim_C(\mathfrak{K}^C) = 56 \times 2 + 3 = 115$.
0902.0431_F02654	163	■ 1.000	$\operatorname{dim}_C(\frac{a}{K}) \leq \operatorname{dim}_C(\frac{a}{K}) \leq \operatorname{dim}_C(\frac{K}{K})=56 \times 2+3=115$	$\dim_C(\mathfrak{a}) \leq \dim_C(\mathfrak{K}^C) = 56 \times 2 + 3 = 115$	$q \mid \mathfrak{a} : \mathfrak{a} \rightarrow \mathfrak{K}^C$ is also injective. Hence we have $\dim_C(\mathfrak{a}) \leq \dim_C(\mathfrak{K}^C) = 56 \times 2 + 3 = 115$.
0902.0431_F02655	163	■ 1.000	$\frac{e}{7}\{ \}^C \cap \frac{a}{e} = \frac{e}{7}\{ \}^C$	$\mathfrak{e}_7^C \cap \mathfrak{a} = \mathfrak{e}_7^C$	(1) Case $\mathfrak{e}_7^C \cap \mathfrak{a} \neq \{0\}$. From the simplicity of $\mathfrak{e}_7^C$, we have $\mathfrak{e}_7^C \cap \mathfrak{a} = \mathfrak{e}_7^C$, hence
0902.0431_F02656	163	■ 1.000	$\frac{a}{K} \supset \frac{e}{7}\{ \}^C$	$\mathfrak{a} \supset \mathfrak{e}_7^C$	$\mathfrak{a} \supset \mathfrak{e}_7^C$. On the other hand, we have
0902.0431_F02657	163	■ 1.000	$\frac{a}{K} \supset \frac{e}{7}\{ \}^C \oplus \frac{K}{K} = \frac{e}{8}\{ \}^C$	$\mathfrak{a} \supset \mathfrak{e}_7^C \oplus \mathfrak{K}^C = \mathfrak{e}_8^C$	Therefore, $\mathfrak{a} \supset \mathfrak{e}_7^C \oplus \mathfrak{K}^C = \mathfrak{e}_8^C$ which implies $\mathfrak{a} = \mathfrak{e}_8^C$.
0902.0431_F02658	163	■ 1.000	$\frac{a}{K} = \frac{e}{8}\{ \}^C$	$\mathfrak{a} = \mathfrak{e}_8^C$	Therefore, $\mathfrak{a} \supset \mathfrak{e}_7^C \oplus \mathfrak{K}^C = \mathfrak{e}_8^C$ which implies $\mathfrak{a} = \mathfrak{e}_8^C$.
0902.0431_F02659	163	■ 1.000	$R=(\theta, P, Q, r, s, t)$	$R = (0, P, Q, r, s, t)$	(2) Case $\mathfrak{K}^C \cap \mathfrak{a} \neq \{0\}$. Let $R = (0, P, Q, r, s, t)$ be a non-zero element of $\mathfrak{K}^C \cap \mathfrak{a} \subset \mathfrak{a}$.
0902.0431_F02660	163	■ 1.000	$\frac{K}{K} \cap \frac{a}{K} \subset \frac{a}{K}$	$\mathfrak{K}^C \cap \mathfrak{a} \subset \mathfrak{a}$	(2) Case $\mathfrak{K}^C \cap \mathfrak{a} \neq \{0\}$. Let $R = (0, P, Q, r, s, t)$ be a non-zero element of $\mathfrak{K}^C \cap \mathfrak{a} \subset \mathfrak{a}$.
0902.0431_F02661	163	■ 1.000	$R=(\theta, P, Q, r, s, t), P \neq 0$	$R = (0, P, Q, r, s, t), P \neq 0$	(i) Case $R = (0, P, Q, r, s, t), P \neq 0$. We have
0902.0431_F02662	163	■ 0.996	$P_1 \in \frac{P}{P}\{ \}^C$	$P_1 \in \mathfrak{P}^C$	We choose $P_1 \in \mathfrak{P}^C$ so that $P \times P_1 \neq 0$ (Lemma 4.5.3) and choose $\Phi \in \mathfrak{e}_7^C$ so that
0902.0431_F02663	163	■ 0.996	$P \times P_1 \neq 0$	$P \times P_1 \neq 0$	We choose $P_1 \in \mathfrak{P}^C$ so that $P \times P_1 \neq 0$ (Lemma 4.5.3) and choose $\Phi \in \mathfrak{e}_7^C$ so that
0902.0431_F02664	163	■ 0.911	$\left[\Phi, P \times P_1\right] \neq 0$	$[\Phi, P \times P_1] \neq 0$	$[\Phi, P \times P_1] \neq 0$. (Since $\mathfrak{e}_7^C$ simple, the center of $\mathfrak{e}_7^C$ consists only of 0, so such Φ
0902.0431_F02665	164	■ 1.000	$R=(\theta, P, Q, r, s, t), Q \neq 0$	$R = (0, P, Q, r, s, t), Q \neq 0$	(ii) Case $R = (0, P, Q, r, s, t), Q \neq 0$. The argument is similar to (i).
0902.0431_F02666	164	■ 1.000	$R=(\theta, 0, 0, r, s, t), r \neq 0$	$R = (0, 0, 0, r, s, t), r \neq 0$	(iii) Case $R = (0, 0, 0, r, s, t), r \neq 0$. For $0 \neq P \in \mathfrak{P}^C$, we have
0902.0431_F02667	164	■ 1.000	$\theta \neq P \in \frac{P}{P}\{ \}^C$	$0 \neq P \in \mathfrak{P}^C$	(iii) Case $R = (0, 0, 0, r, s, t), r \neq 0$. For $0 \neq P \in \mathfrak{P}^C$, we have
0902.0431_F02668	164	■ 1.000	$R=(\theta, 0, 0, 0, s, t), s \neq 0$	$R = (0, 0, 0, 0, s, t), s \neq 0$	(iv) Case $R = (0, 0, 0, 0, s, t), s \neq 0$. We have
0902.0431_F02669	164	■ 1.000	$R=(\theta, 0, 0, 0, 0, t), t \neq 0$	$R = (0, 0, 0, 0, 0, t), t \neq 0$	(v) Case $R = (0, 0, 0, 0, 0, t), t \neq 0$. The argument is similar to (iv).
0902.0431_F02670	164	■ 0.942	$R \in \frac{e}{8}\{ \}^C$	$R \in \mathfrak{e}_8^C$	For $R \in \mathfrak{e}_8^C$ we denote $\operatorname{ad} R : \mathfrak{e}_8^C \rightarrow \mathfrak{e}_8^C$, that is, $\operatorname{ad} R(R_1) = [R, R_1]$, by $\Theta(R) =$
0902.0431_F02671	164	■ 0.942	$\operatorname{ad} R: \frac{e}{8}\{ \}^C \rightarrow \frac{e}{8}\{ \}^C$	$\operatorname{ad} R : \mathfrak{e}_8^C \rightarrow \mathfrak{e}_8^C$	For $R \in \mathfrak{e}_8^C$ we denote $\operatorname{ad} R : \mathfrak{e}_8^C \rightarrow \mathfrak{e}_8^C$, that is, $\operatorname{ad} R(R_1) = [R, R_1]$, by $\Theta(R) =$
0902.0431_F02672	164	■ 0.942	$\operatorname{ad} R\left(R_1\right)=\left[R, R_1\right]$	$\operatorname{ad} R(R_1) = [R, R_1]$	For $R \in \mathfrak{e}_8^C$ we denote $\operatorname{ad} R : \mathfrak{e}_8^C \rightarrow \mathfrak{e}_8^C$, that is, $\operatorname{ad} R(R_1) = [R, R_1]$, by $\Theta(R) =$
0902.0431_F02673	164	■ 0.942	$\Theta(R)=$	$\Theta(R) =$	For $R \in \mathfrak{e}_8^C$ we denote $\operatorname{ad} R : \mathfrak{e}_8^C \rightarrow \mathfrak{e}_8^C$, that is, $\operatorname{ad} R(R_1) = [R, R_1]$, by $\Theta(R) =$
0902.0431_F02674	164	■ 0.999	$\operatorname{ad} R$	$\operatorname{ad} R$	$\operatorname{ad} R$. Since $\mathfrak{e}_8^C$ is simple (Theorem 5.2.1), we obtain an isomorphism of Lie algebras
0902.0431_F02675	164	■ 0.996	$\Theta(R)$	$\Theta(R)$	by assigning $\Theta(R)$ to R . Moreover, $\mathfrak{e}_8^C$ is isomorphic to the algebra
0902.0431_F02676	164	■ 0.996	R	R	by assigning $\Theta(R)$ to R . Moreover, $\mathfrak{e}_8^C$ is isomorphic to the algebra
0902.0431_F02677	164	■ 0.993	$\frac{e}{8}\{ \}^C \cong \operatorname{Der}\left(\frac{e}{8}\{ \}^C\right)$	$\mathfrak{e}_8^C \cong \operatorname{Der}(\mathfrak{e}_8^C)$	Hereafter we often denote $\Theta(R)$ by R identifying $\mathfrak{e}_8^C \cong \operatorname{Der}(\mathfrak{e}_8^C)$.
0902.0431_F02678	164	■ 0.994	$\left(R_1, R_2\right)_8$	$(R_1, R_2)_8$	Definition. We define a symmetric inner product $(R_1, R_2)_8$ in $\mathfrak{e}_8^C$ by
0902.0431_F02679	164	■ 0.766	$R_i=\left(\Phi_i, P_i, Q_i, r_i, s_i, t_i\right) \in \frac{e}{8}\{ \}^C$	$R_i = (\Phi_i, P_i, Q_i, r_i, s_i, t_i) \in \mathfrak{e}_8^C$	where $R_i = (\Phi_i, P_i, Q_i, r_i, s_i, t_i) \in \mathfrak{e}_8^C$.
0902.0431_F02680	164	■ 1.000	$\left(\left[R, R_1\right], R_2\right)_8$	$([R, R_1], R_2)_8$	Proof. $([R, R_1], R_2)_8$
0902.0431_F02681	165	■ 1.000	B_8	B_8	Theorem 5.3.2. The Killing form B_8 of the Lie algebra $\mathfrak{e}_8^C$ is given by
0902.0431_F02682	165	■ 0.737	$R_i=\left(\Phi_i, P_i, Q_i, r_i, s_i, t_i\right) \in \frac{e}{8}\{ \}^C$	$R_i = (\Phi_i, P_i, Q_i, r_i, s_i, t_i) \in \mathfrak{e}_8^C$	where $R_i = (\Phi_i, P_i, Q_i, r_i, s_i, t_i) \in \mathfrak{e}_8^C$ and B_7 is the Killing form of $\mathfrak{e}_7^C$.
0902.0431_F02683	165	■ 1.000	$R_1=R_2=(0,0,0,1,0,0)=1$	$R_1 = R_2 = (0, 0, 0, 1, 0, 0) = 1$	To determine k , let $R_1 = R_2 = (0, 0, 0, 1, 0, 0) = 1$. Then, we have
0902.0431_F02684	165	■ 1.000	$k=-15$	$k = -15$	Therefore $k = -15$. Thus we have $B_8(R_1, R_2) = -15(R_1, R_2)_8$.
0902.0431_F02685	165	■ 1.000	$B_8\left(R_1, R_2\right)=-15\left(R_1, R_2\right)_8$	$B_8(R_1, R_2) = -15(R_1, R_2)_8$	Therefore $k = -15$. Thus we have $B_8(R_1, R_2) = -15(R_1, R_2)_8$.
0902.0431_F02686	166	■ 1.000	$\operatorname{Inn}\left(\frac{e}{8}\{ \}^C\right)$	$\operatorname{Inn}(\mathfrak{e}_8^C)$	Proof. Denote by $\operatorname{Inn}(\mathfrak{e}_8^C)$ the subgroup generated by the inner automorphisms

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0902.0431_F02687	166	■ 1.000	$\exp(\Theta(R)), R \in \mathfrak{e}_8^C$	$\exp(\Theta(R)), R \in \mathfrak{e}_8^C$	$\exp(\Theta(R)), R \in \mathfrak{e}_8^C$ in the automorphism group $\text{Aut}(\mathfrak{e}_8^C) = E_8^C$ of the Lie algebra
0902.0431_F02688	166	■ 1.000	$\operatorname{Aut}(\mathfrak{e}_8^C) = E_8^C$	$\text{Aut}(\mathfrak{e}_8^C) = E_8^C$	$\exp(\Theta(R)), R \in \mathfrak{e}_8^C$ in the automorphism group $\text{Aut}(\mathfrak{e}_8^C) = E_8^C$ of the Lie algebra
0902.0431_F02689	166	■ 0.988	$\operatorname{Aut}(\mathfrak{e}_8^C) / \operatorname{Inn}(\mathfrak{e}_8^C) = \{1\}$	$\text{Aut}(\mathfrak{e}_8^C) / \text{Inn}(\mathfrak{e}_8^C) = \{1\}$	$\mathfrak{e}_8^C$. It is known that $\text{Aut}(\mathfrak{e}_8^C) / \text{Inn}(\mathfrak{e}_8^C) = \{1\}$ holds for the C -algebra of E_8 type
0902.0431_F02690	166	■ 1.000	$R \times R: \mathfrak{e}_8^C \rightarrow \mathfrak{e}_8^C$	$R \times R: \mathfrak{e}_8^C \rightarrow \mathfrak{e}_8^C$	lowing fact. For $R \in \mathfrak{e}_8^C$, we define a C -linear mapping $\boxed{R \times R: \mathfrak{e}_8^C \rightarrow \mathfrak{e}_8^C}$ by
0902.0431_F02691	166	■ 1.000	$\mathfrak{W}^C$	$\mathfrak{W}^C$	and we define a space $\mathfrak{W}^C$ by
0902.0431_F02692	166	■ 0.999	$\left(E_8^C\right)_{1-} = \{\alpha \in E_8^C \mid \alpha 1_- = 1_-\} = \exp(\Phi(0, 0, \mathfrak{P}^C, 0, 0, C)) E_7$	$(E_8^C)_{1-} = \{\alpha \in E_8^C \mid \alpha 1_- = 1_-\} = \exp(\Phi(0, 0, \mathfrak{P}^C, 0, 0, C)) E_7$	where $\boxed{(E_8^C)_{1-} = \{\alpha \in E_8^C \mid \alpha 1_- = 1_-\} = \exp(\Phi(0, 0, \mathfrak{P}^C, 0, 0, C)) E_7}$. The connect-
0902.0431_F02693	166	■ 0.997	$\left(E_8^C\right)_{1-}$	$(E_8^C)_{1-}$	edness of E_8^C follows from the connectedness of $(E_8^C)_{1-}$ and $\mathfrak{W}^C$. (See Imai and
0902.0431_F02694	166	■ 1.000	$\operatorname{Der}(\mathfrak{e}_8^C) \cong \mathfrak{e}_8^C$	$\text{Der}(\mathfrak{e}_8^C) \cong \mathfrak{e}_8^C$	The Lie algebra of the group E_8^C is $\text{Der}(\mathfrak{e}_8^C) \cong \mathfrak{e}_8^C$, and therefore we have shown
0902.0431_F02695	166	■ 1.000	λ, λ'	λ, λ'	We define C -linear transformations λ, λ' of $\mathfrak{e}_8^C$ respectively by
0902.0431_F02696	166	■ 0.973	λ'	λ'	mappings λ and λ' preserve the Lie bracket in $\mathfrak{e}_8^C$, that is, $\lambda, \lambda' \in \text{Aut}(\mathfrak{e}_8^C) = E_8^C$.
0902.0431_F02697	166	■ 0.973	$\lambda, \lambda' \in \text{Aut}(\mathfrak{e}_8^C) = E_8^C$	$\lambda, \lambda' \in \text{Aut}(\mathfrak{e}_8^C) = E_8^C$	mappings λ and λ' preserve the Lie bracket in $\mathfrak{e}_8^C$, that is, $\boxed{\lambda, \lambda' \in \text{Aut}(\mathfrak{e}_8^C) = E_8^C}$.
0902.0431_F02698	167	■ 0.994	$\langle R_1, R_2 \rangle$	$\langle R_1, R_2 \rangle$	Definition. We define a Hermitian inner product $\boxed{\langle R_1, R_2 \rangle}$ in $\mathfrak{e}_8^C$ by
0902.0431_F02699	167	■ 0.772	$R_i = (\Phi_i, P_i, Q_i, r_i, s_i, t_i) \in \mathfrak{e}_8^C, i = 1, 2$	$R_i = (\Phi_i, P_i, Q_i, r_i, s_i, t_i) \in \mathfrak{e}_8^C, i = 1, 2$	Proof. Let $\boxed{R_i = (\Phi_i, P_i, Q_i, r_i, s_i, t_i) \in \mathfrak{e}_8^C, i = 1, 2}$. Since $\tau \tilde{\lambda} R_1 = (\tau \lambda \Phi_1 \lambda^{-1} \tau,$
0902.0431_F02700	167	■ 0.772	$\tau \tilde{\lambda} R_1 = (\tau \lambda \Phi_1 \lambda^{-1} \tau,$	$\tau \tilde{\lambda} R_1 = (\tau \lambda \Phi_1 \lambda^{-1} \tau,$	Proof. Let $R_i = (\Phi_i, P_i, Q_i, r_i, s_i, t_i) \in \mathfrak{e}_8^C, i = 1, 2$. Since $\boxed{\tau \tilde{\lambda} R_1 = (\tau \lambda \Phi_1 \lambda^{-1} \tau,$
0902.0431_F02701	167	■ 0.853	$\tau \lambda Q_1, -\tau \lambda P_1, -\tau r_1, -\tau t_1, -\tau s_1$	$\tau \lambda Q_1, -\tau \lambda P_1, -\tau r_1, -\tau t_1, -\tau s_1$	$\boxed{\tau \lambda Q_1, -\tau \lambda P_1, -\tau r_1, -\tau t_1, -\tau s_1}$), we have, by Theorem 5.3.2, that
0902.0431_F02702	167	■ 1.000	$(\tau \lambda \Phi_1 \lambda^{-1} \tau, \Phi_2)_7$	$(\tau \lambda \Phi_1 \lambda^{-1} \tau, \Phi_2)_7$	Hence, it is sufficient to show that $\boxed{\tau \lambda \Phi_1 \lambda^{-1} \tau, \Phi_2}_7$ is positive definite. Let $\Phi_i =$
0902.0431_F02703	167	■ 1.000	$\Phi_i =$	$\Phi_i =$	Hence, it is sufficient to show that $(\tau \lambda \Phi_1 \lambda^{-1} \tau, \Phi_2)_7$ is positive definite. Let $\Phi_i =$
0902.0431_F02704	167	■ 1.000	$\Phi(\phi_i, A_i, B_i, \nu_i), i = 1, 2$	$\Phi(\phi_i, A_i, B_i, \nu_i), i = 1, 2$	$\boxed{\Phi(\phi_i, A_i, B_i, \nu_i), i = 1, 2}$. Since $\tau \lambda \Phi_1 \lambda^{-1} \tau = \Phi(-\tau^t \phi_1 \tau, -\tau B_1, -\tau A_1, -\tau \nu_1)$, we have
0902.0431_F02705	167	■ 1.000	$\tau \lambda \Phi_1 \lambda^{-1} \tau = \Phi(-\tau^t \phi_1 \tau, -\tau B_1, -\tau A_1, -\tau \nu_1)$	$\tau \lambda \Phi_1 \lambda^{-1} \tau = \Phi(-\tau^t \phi_1 \tau, -\tau B_1, -\tau A_1, -\tau \nu_1)$	$\Phi(\phi_i, A_i, B_i, \nu_i), i = 1, 2$. Since $\boxed{\tau \lambda \Phi_1 \lambda^{-1} \tau = \Phi(-\tau^t \phi_1 \tau, -\tau B_1, -\tau A_1, -\tau \nu_1)}$, we have
0902.0431_F02706	167	■ 1.000	$(\tau^t \phi_1 \tau, \phi_2)_6$	$(\tau^t \phi_1 \tau, \phi_2)_6$	Therefore, it is enough to show that $\boxed{\tau^t \phi_1 \tau, \phi_2}_6$ is positive definite. Let $\phi_i = \delta_i + \tilde{T}_i \in$
0902.0431_F02707	167	■ 1.000	$\phi_i = \delta_i + \tilde{T}_i \in$	$\phi_i = \delta_i + \tilde{T}_i \in$	Therefore, it is enough to show that $(\tau^t \phi_1 \tau, \phi_2)_6$ is positive definite. Let $\phi_i = \delta_i + \tilde{T}_i \in$
0902.0431_F02708	167	■ 0.996	$\mathfrak{e}_6^C, \delta_i \in \mathfrak{f}_4^C, \tilde{T}_i \in \mathfrak{J}_0^C, i = 1, 2$	$\mathfrak{e}_6^C, \delta_i \in \mathfrak{f}_4^C, \tilde{T}_i \in \mathfrak{J}_0^C, i = 1, 2$	$\boxed{\mathfrak{e}_6^C, \delta_i \in \mathfrak{f}_4^C, \tilde{T}_i \in \mathfrak{J}_0^C, i = 1, 2}$. Since $\tau^t \phi_1 \tau = -\tau \delta_1 \tau + \tau \tilde{T}_1$, we have
0902.0431_F02709	167	■ 0.996	$\tau^t \phi_1 \tau = -\tau \delta_1 \tau + \tau \tilde{T}_1$	$\tau^t \phi_1 \tau = -\tau \delta_1 \tau + \tau \tilde{T}_1$	$\mathfrak{e}_6^C, \delta_i \in \mathfrak{f}_4^C, \tilde{T}_i \in \mathfrak{J}_0^C, i = 1, 2$. Since $\boxed{\tau^t \phi_1 \tau = -\tau \delta_1 \tau + \tau \tilde{T}_1}$, we have
0902.0431_F02710	167	■ 1.000	$-(\tau \delta_1 \tau, \delta_2)_4$	$-(\tau \delta_1 \tau, \delta_2)_4$	Consequently, it is sufficient to show that $\boxed{-(\tau \delta_1 \tau, \delta_2)_4}$ is positive definite, which can
0902.0431_F02711	168	■ 1.000	$\mathfrak{e}_8$	$\mathfrak{e}_8$	Theorem 5.5.3. The Lie algebra $\mathfrak{e}_8$ of the group E_8 is
0902.0431_F02712	168	■ 1.000	$R = (\Phi, P, Q, r, s, t) \in \mathfrak{e}_8^C$	$R = (\Phi, P, Q, r, s, t) \in \mathfrak{e}_8^C$	Proof. For $\boxed{R = (\Phi, P, Q, r, s, t) \in \mathfrak{e}_8^C}$, since
0902.0431_F02713	168	■ 1.000	$\tau \tilde{\lambda} R = R$	$\tau \tilde{\lambda} R = R$	the condition $\tau \tilde{\lambda} R = R$ is equivalent to $\tau \lambda \Phi = \Phi \lambda \tau, Q = -\tau \lambda P, \tau r = -r, t = -\tau s,$
0902.0431_F02714	168	■ 1.000	$\tau \lambda \Phi = \Phi \lambda \tau, Q = -\tau \lambda P, \tau r = -r, t = -\tau s$	$\tau \lambda \Phi = \Phi \lambda \tau, Q = -\tau \lambda P, \tau r = -r, t = -\tau s$	the condition $\tau \tilde{\lambda} R = R$ is equivalent to $\boxed{\tau \lambda \Phi = \Phi \lambda \tau, Q = -\tau \lambda P, \tau r = -r, t = -\tau s}$,
0902.0431_F02715	168	■ 1.000	R^*	R^*	Proof. For $R \in \mathfrak{e}_8^C$, the conjugate transposed mapping R^* of R with respect to
0902.0431_F02716	168	■ 0.975	$R^* = \tau \tilde{\lambda} R \tilde{\lambda} \tau \in \mathfrak{e}_8^C$	$R^* = \tau \tilde{\lambda} R \tilde{\lambda} \tau \in \mathfrak{e}_8^C$	the inner product $\langle R_1, R_2 \rangle$ of $\mathfrak{e}_8^C$ is $\boxed{R^* = \tau \tilde{\lambda} R \tilde{\lambda} \tau \in \mathfrak{e}_8^C}$, and for $R \in \mathfrak{e}_8^C$, R belongs
0902.0431_F02717	168	■ 0.975	$R \in \mathfrak{e}_8^C, R$	$R \in \mathfrak{e}_8^C, R$	the inner product $\langle R_1, R_2 \rangle$ of $\mathfrak{e}_8^C$ is $R^* = \tau \tilde{\lambda} R \tilde{\lambda} \tau \in \mathfrak{e}_8^C$, and for $R \in \mathfrak{e}_8^C$, R belongs

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0902.0431_F02718	168	■ 1.000	$R^{\ast}=-R$	$R^{\ast}=-R$	to $\mathfrak{e}_8$ if and only if $\boxed{R^{\ast}=-R}$. Now, any element $R \in \mathfrak{e}_8^C$ is represented by
0902.0431_F02719	168	■ 1.000	$R \in \mathfrak{e}_8$	$R \in \mathfrak{e}_8$	Analogously as in $\mathfrak{e}_8^C$, for $R \in \mathfrak{e}_8$, we identify R with $\Theta(R)$ and regard $\mathfrak{e}_8 \cong \Theta(\mathfrak{e}_8)$.
0902.0431_F02720	168	■ 1.000	$\mathfrak{e}_8 \cong \Theta(\mathfrak{e}_8)$	$\mathfrak{e}_8 \cong \Theta(\mathfrak{e}_8)$	Analogously as in $\mathfrak{e}_8^C$, for $R \in \mathfrak{e}_8$, we identify R with $\Theta(R)$ and regard $\mathfrak{e}_8 \cong \Theta(\mathfrak{e}_8)$.
0902.0431_F02721	168	■ 1.000	$\operatorname{Iso}_C(\mathfrak{e}_8^C) = GL(248, C)$	$\operatorname{Iso}_C(\mathfrak{e}_8^C) = GL(248, C)$	Proof. Evidently E_8^C is an algebraic subgroup of $\boxed{\operatorname{Iso}_C(\mathfrak{e}_8^C) = GL(248, C)}$. For
0902.0431_F02722	168	■ 1.000	$\alpha \in E_8^C$	$\alpha \in E_8^C$	$\alpha \in E_8^C$, the conjugate transposed mapping α^* of α with respect to the inner product
0902.0431_F02723	168	■ 1.000	$\alpha^* = \tau \tilde{\lambda} \alpha^{-1} \tilde{\lambda} \tau \in E_8^C$	$\alpha^* = \tau \tilde{\lambda} \alpha^{-1} \tilde{\lambda} \tau \in E_8^C$	$\langle R_1, R_2 \rangle$ is $\boxed{\alpha^* = \tau \lambda \alpha^{-1} \lambda \tau \in E_8^C}$. Therefore, by Chevalley's lemma, we have
0902.0431_F02724	170	■ 0.905	$\mathfrak{h}_7$	$\mathfrak{h}_7$	Proof. Let $\mathfrak{h}_7$ be the Cartan subalgebra of $\mathfrak{e}_7^C$ given in Theorem 4.6.1, then
0902.0431_F02725	170	■ 1.000	$Y=0, \xi=\eta=0$	$Y=0, \xi=\eta=0$	By putting $\boxed{Y=0, \xi=\eta=0}$, we obtain
0902.0431_F02726	170	■ 1.000	$X=F_2(a), F_3(a)$	$X=F_2(a), F_3(a)$	We can also obtain roots by letting $\boxed{X=F_2(a), F_3(a)}$.
0902.0431_F02727	170	■ 1.000	$X=0, \xi=\eta=0$	$X=0, \xi=\eta=0$	By putting $\boxed{X=0, \xi=\eta=0}$ and further $Y=E_k, Y=F_i(a)$, we can again roots.
0902.0431_F02728	170	■ 1.000	$Y=E_k, Y=F_i(a)$	$Y=E_k, Y=F_i(a)$	By putting $X=0, \xi=\eta=0$ and further $\boxed{Y=E_k, Y=F_i(a)}$, we can again roots.
0902.0431_F02729	170	■ 1.000	$X=Y=0, \xi=1, \eta=0$	$X=Y=0, \xi=1, \eta=0$	By putting $\boxed{X=Y=0, \xi=1, \eta=0}$, we can again the root $\nu+r$. Similarly, we can
0902.0431_F02730	170	■ 1.000	$\nu+r$	$\nu+r$	By putting $X=Y=0, \xi=1, \eta=0$, we can again the root $\nu+r$. Similarly, we can
0902.0431_F02731	170	■ 1.000	$-\nu+r$	$-\nu+r$	obtain the root $\boxed{-\nu+r}$.
0902.0431_F02732	171	■ 1.000	$2r$	$2r$	we obtain the roots $2r$ and $-2r$.
0902.0431_F02733	171	■ 1.000	$-2r$	$-2r$	we obtain the roots $2r$ and $-2r$.
0902.0431_F02734	171	■ 1.000	$n_1 n_2 \cdots n_8$	$n_1 n_2 \cdots n_8$	Proof. In the following, the notation $\boxed{\alpha_1 \alpha_2 \cdots \alpha_8}$ denotes the root $n_1 \alpha_1 + n_2 \alpha_2 +$
0902.0431_F02735	171	■ 0.998	$\cdots + n_8 \alpha_8$	$\cdots + n_8 \alpha_8$	$\cdots + n_8 \alpha_8$. Now, all positive roots of $\mathfrak{e}_8^C$ are expressed by
0902.0431_F02765	176	■ 0.998	$\Pi = \{\alpha_1, \alpha_2, \cdots, \alpha_8\}$	$\Pi = \{\alpha_1, \alpha_2, \cdots, \alpha_8\}$	Hence $\boxed{\Pi = \{\alpha_1, \alpha_2, \cdots, \alpha_8\}}$ is a fundamental root system of $\mathfrak{e}_8^C$. The real part $\mathfrak{h}_R$
0902.0431_F02766	176	■ 0.925	$\Phi(h) = \left\{ \Phi \left(\sum_{k=0}^3 \lambda_k H_k + \left(\sum_{j=1}^3 \mu_j E_j \right)^{\sim}, 0, 0, \nu \right) \in (\mathfrak{h}_7)_R \right.$	$\Phi(h) = \left\{ \Phi \left(\sum_{k=0}^3 \lambda_k H_k + \left(\sum_{j=1}^3 \mu_j E_j \right)^{\sim}, 0, 0, \nu \right) \in (\mathfrak{h}_7)_R \right.$	(where $\boxed{\Phi(h) = \left\{ \Phi \left(\sum_{k=0}^3 \lambda_k H_k + \left(\sum_{j=1}^3 \mu_j E_j \right)^{\sim}, 0, 0, \nu \right) \in (\mathfrak{h}_7)_R \right.}$ (Theorem 4.6.2)). The
0902.0431_F02767	176	■ 1.000	$\tilde{h} = (\Phi(h), 0, 0, r, 0), \tilde{h}' = (\Phi(h'), 0, 0, r', 0) \in \mathfrak{h}$	$\tilde{h} = (\Phi(h), 0, 0, r, 0), \tilde{h}' = (\Phi(h'), 0, 0, r', 0) \in \mathfrak{h}$	for $\boxed{\tilde{h} = (\Phi(h), 0, 0, r, 0), \tilde{h}' = (\Phi(h'), 0, 0, r', 0) \in \mathfrak{h}}$, where $\Phi(h) = \Phi \left(\sum_{k=0}^3 \lambda_k H_k +$
0902.0431_F02768	176	■ 1.000	$\Phi(h) = \Phi \left(\sum_{k=0}^3 \lambda_k H_k +$	$\Phi(h) = \Phi \left(\sum_{k=0}^3 \lambda_k H_k +$	for $\tilde{h} = (\Phi(h), 0, 0, r, 0), \tilde{h}' = (\Phi(h'), 0, 0, r', 0) \in \mathfrak{h}$, where $\boxed{\Phi(h) = \Phi \left(\sum_{k=0}^3 \lambda_k H_k +$
0902.0431_F02769	176	■ 0.449	$\left(\sum_{j=1}^3 \mu_j E_j \right)^{\sim}, 0, 0, \nu \right), \Phi(h') = \Phi \left(\sum_{k=0}^3 \lambda'_k H_k + \left(\sum_{j=1}^3 \mu'_j E_j \right)^{\sim}, 0, 0, \nu' \right) \in (\mathfrak{h}_7)_R$	$\left(\sum_{j=1}^3 \mu_j E_j \right)^{\sim}, 0, 0, \nu \right), \Phi(h') = \Phi \left(\sum_{k=0}^3 \lambda'_k H_k + \left(\sum_{j=1}^3 \mu'_j E_j \right)^{\sim}, 0, 0, \nu' \right) \in (\mathfrak{h}_7)_R$	$\boxed{\left(\sum_{j=1}^3 \mu_j E_j \right)^{\sim}, 0, 0, \nu \right), \Phi(h') = \Phi \left(\sum_{k=0}^3 \lambda'_k H_k + \left(\sum_{j=1}^3 \mu'_j E_j \right)^{\sim}, 0, 0, \nu' \right) \in (\mathfrak{h}_7)_R}$. Indeed,
0902.0431_F02770	176	■ 0.991	$\alpha_i(B_8(H_\alpha, H)) = \alpha(H), H \in$	$\alpha_i(B_8(H_\alpha, H)) = \alpha(H), H \in$	Now, the canonical elements $H_{\alpha_i} \in \mathfrak{h}_R$ associated with $\boxed{\alpha_i(B_8(H_\alpha, H)) = \alpha(H), H \in}$
0902.0431_F02771	177	■ 0.997	$\alpha_1, \alpha_2, \cdots, \alpha_8$	$\alpha_1, \alpha_2, \cdots, \alpha_8$	induced by the Killing form B_8 between $\boxed{\alpha_1, \alpha_2, \cdots, \alpha_8}$ and $-\mu$ are given by
0902.0431_F02772	177	■ 1.000	$A_1 \oplus E_7$	$A_1 \oplus E_7$	(1) The first one is a subalgebra of type $A_1 \oplus E_7$ which is obtained as the fixed
0902.0431_F02773	177	■ 1.000	D_8	D_8	(2) The second one is a subalgebra of type D_8 which is obtained as the fixed points
0902.0431_F02774	177	■ 1.000	$A_2 \oplus E_6$	$A_2 \oplus E_6$	(3) The third one is a subalgebra of type $A_2 \oplus E_6$ which is obtained as the fixed
0902.0431_F02775	177	■ 1.000	A_8	A_8	(4) The fourth one is a subalgebra of type A_8 which is obtained as the fixed points
0902.0431_F02776	177	■ 1.000	$A_4 \oplus A_4$	$A_4 \oplus A_4$	(5) The fifth one is a subalgebra of type $\boxed{A_4 \oplus A_4}$ which is obtained as the fixed
0902.0431_F02777	178	■ 0.999	$(E_8^C)_{1,1^-,1_-}$	$(E_8^C)_{1,1^-,1_-}$	We shall first study the following subgroup $\boxed{(E_8^C)_{1,1^-,1_-}}$ of E_8^C :

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0902.0431_F02778	178	0.683	$\text{\quad}\left(E_{\{8\}}\right)_{1,1^{\{-}\},\,1_{\{-}\}}\text{\cong}E_{\{7\}}^{\{C\}}$	$(E_8^C)_{1,1^-,1_-} \cong E_7^C$	Proposition 5.7.1. $(E_8^C)_{1,1^-,1_-} \cong E_7^Q$.
0902.0431_F02779	178	1.000	$\beta \in E_{\{7\}}^{\{C\}}$	$\beta \in E_7^C$	Proof. For $\beta \in E_7^C$, we define a C -linear mapping $\tilde{\beta} : \mathfrak{e}_8^C \rightarrow \mathfrak{e}_8^C$ by
0902.0431_F02780	178	1.000	$\widetilde{\beta} : \mathfrak{e}_{\{8\}}^{\{C\}} \rightarrow \mathfrak{e}_{\{8\}}^{\{C\}}$	$\tilde{\beta} : \mathfrak{e}_8^C \rightarrow \mathfrak{e}_8^C$	Proof. For $\beta \in E_7^C$, we define a C -linear mapping $\tilde{\beta} : \mathfrak{e}_8^C \rightarrow \mathfrak{e}_8^Q$ by
0902.0431_F02781	178	0.992	$(\operatorname{Ad} \beta) \Phi = \beta \Phi \beta^{-1}, \Phi \in \mathfrak{e}_7^C$	$(\operatorname{Ad} \beta) \Phi = \beta \Phi \beta^{-1}, \Phi \in \mathfrak{e}_7^C$	where $(\operatorname{Ad} \beta) \Phi = \beta \Phi \beta^{-1}, \Phi \in \mathfrak{e}_7^C$. It is easy to check that $\tilde{\beta} \in (E_8^C)_{1,1^-,1_-}$. Con-
0902.0431_F02782	178	0.992	$\widetilde{\beta} \in \left(E_{\{8\}}\right)_{1,1^{\{-}\},\,1_{\{-}\}}$	$\tilde{\beta} \in (E_8^C)_{1,1^-,1_-}$	where $(\operatorname{Ad} \beta) \Phi = \beta \Phi \beta^{-1}, \Phi \in \mathfrak{e}_7^C$. It is easy to check that $\tilde{\beta} \in (E_8^C)_{1,1^-,1_-}$. Con-
0902.0431_F02783	178	1.000	$\alpha 1 = 1, \alpha 1^{\{-}\} = 1^{\{-}\}$	$\alpha 1 = 1, \alpha 1^- = 1^-$	versely, if $\alpha \in E_8^C$ satisfies $\alpha 1 = 1, \alpha 1^- = 1^-$ and $\alpha 1_- = 1_-$, then α is of the
0902.0431_F02784	178	1.000	$\alpha 1_{\{-}\} = 1_{\{-}\}$	$\alpha 1_- = 1_-$	versely, if $\alpha \in E_8^C$ satisfies $\alpha 1 = 1, \alpha 1^- = 1^-$ and $\alpha 1_- = 1_-$, then α is of the
0902.0431_F02785	178	1.000	$[\alpha \Phi, 1] = [\alpha \Phi, \alpha 1] = \alpha [\Phi, 1] = 0$	$[\alpha \Phi, 1] = [\alpha \Phi, \alpha 1] = \alpha [\Phi, 1] = 0$	From the relation $[\alpha \Phi, 1] = [\alpha \Phi, \alpha 1] = \alpha [\Phi, 1] = 0$, that is,
0902.0431_F02786	178	1.000	$\beta_{21} = \beta_{31} = 0$	$\beta_{21} = \beta_{31} = 0$	we obtain $\beta_{21} = \beta_{31} = 0$ and $a_2 = a_3 = 0$. Furthermore, from $[\alpha \Phi, 1^-] = [\alpha \Phi, \alpha 1^-] =$
0902.0431_F02787	178	1.000	$a_{\{2\}} = a_{\{3\}} = 0$	$a_2 = a_3 = 0$	we obtain $\beta_{21} = \beta_{31} = 0$ and $a_2 = a_3 = 0$. Furthermore, from $[\alpha \Phi, 1^-] = [\alpha \Phi, \alpha 1^-] =$
0902.0431_F02788	178	1.000	$\left[\alpha \Phi, 1^{\{-}\}\right] = \left[\alpha \Phi, \alpha 1^{\{-}\}\right] =$	$[\alpha \Phi, 1^-] = [\alpha \Phi, \alpha 1^-] =$	we obtain $\beta_{21} = \beta_{31} = 0$ and $a_2 = a_3 = 0$. Furthermore, from $[\alpha \Phi, 1^-] = [\alpha \Phi, \alpha 1^-] =$
0902.0431_F02789	178	1.000	$\alpha \left[\Phi, 1^{\{-}\}\right] = 0$	$\alpha [\Phi, 1^-] = 0$	$[\alpha \Phi, 1^-] = 0$, that is,
0902.0431_F02790	178	1.000	$a_{\{1\}} = 0$	$a_1 = 0$	we obtain $a_1 = 0$. Using the fact that $[\alpha P^-, 1] = [\alpha P^-, \alpha 1] = \alpha [P^-, 1] = -\alpha P^-$,
0902.0431_F02791	178	1.000	$\left[\alpha P^-, 1\right] = \left[\alpha P^-, \alpha 1\right] = \alpha \left[P^-, 1\right] = -\alpha P^-$	$[\alpha P^-, 1] = [\alpha P^-, \alpha 1] = \alpha [P^-, 1] = -\alpha P^-$	we obtain $a_1 = 0$. Using the fact that $[\alpha P^-, 1] = [\alpha P^-, \alpha 1] = \alpha [P^-, 1] = -\alpha P^-$,
0902.0431_F02792	178	1.000	$\beta_{12} = \beta_{32} = 0$	$\beta_{12} = \beta_{32} = 0$	we obtain $\beta_{12} = \beta_{32} = 0$ and $b_1 = b_2 = b_3 = 0$. Similarly, from $[\alpha Q_-, 1] =$
0902.0431_F02793	178	1.000	$b_{\{1\}} = b_{\{2\}} = b_{\{3\}} = 0$	$b_1 = b_2 = b_3 = 0$	we obtain $\beta_{12} = \beta_{32} = 0$ and $b_1 = b_2 = b_3 = 0$. Similarly, from $[\alpha Q_-, 1] =$
0902.0431_F02794	178	1.000	$\left[\alpha Q_-, 1\right] =$	$[\alpha Q_-, 1] =$	we obtain $\beta_{12} = \beta_{32} = 0$ and $b_1 = b_2 = b_3 = 0$. Similarly, from $[\alpha Q_-, 1] =$
0902.0431_F02795	178	1.000	$\left[\alpha Q_-, \alpha 1\right] = \alpha \left[Q_-, 1\right] = \alpha Q_-$	$[\alpha Q_-, \alpha 1] = \alpha [Q_-, 1] = \alpha Q_-$	$[\alpha Q_-, \alpha 1] = \alpha [Q_-, 1] = \alpha Q_-$, we obtain $\beta_{13} = \beta_{23} = 0$ and $c_1 = c_2 = c_3 = 0$.
0902.0431_F02796	178	1.000	$\beta_{13} = \beta_{23} = 0$	$\beta_{13} = \beta_{23} = 0$	$[\alpha Q_-, \alpha 1] = \alpha [Q_-, 1] = \alpha Q_-$, we obtain $\beta_{13} = \beta_{23} = 0$ and $c_1 = c_2 = c_3 = 0$.
0902.0431_F02797	178	1.000	$c_{\{1\}} = c_{\{2\}} = c_{\{3\}} = 0$	$c_1 = c_2 = c_3 = 0$	$[\alpha Q_-, \alpha 1] = \alpha [Q_-, 1] = \alpha Q_-$, we obtain $\beta_{13} = \beta_{23} = 0$ and $c_1 = c_2 = c_3 = 0$.
0902.0431_F02798	179	1.000	$\left[(0, P, 0, 0, 0, 0), (0, 0, Q, 0, 0, 0)\right] = \left(P \times Q, 0, 0, -\frac{1}{8}\{P, Q\}, 0, 0\right)$	$[(0, P, 0, 0, 0, 0), (0, 0, Q, 0, 0, 0)] = \left(P \times Q, 0, 0, -\frac{1}{8}\{P, Q\}, 0, 0\right)$	By applying α on $[(0, P, 0, 0, 0, 0), (0, 0, Q, 0, 0, 0)] = \left(P \times Q, 0, 0, -\frac{1}{8}\{P, Q\}, 0, 0\right)$, we
0902.0431_F02799	179	1.000	$\left[\left(\left(0, \beta_2 P, 0, 0, 0, 0\right), \left(0, 0, \beta_3 Q, 0, 0, 0\right)\right), \left(0, 0, \beta_1(P \times Q), 0, 0, -\frac{1}{8}\{P, Q\}, 0, 0\right)\right] = \left(\beta_1(P \times Q), 0, 0, -\frac{1}{8}\{P, Q\}, 0, 0\right)$	$[(0, \beta_2 P, 0, 0, 0, 0), (0, 0, \beta_3 Q, 0, 0, 0)] = \left(\beta_1(P \times Q), 0, 0, -\frac{1}{8}\{P, Q\}, 0, 0\right)$	since $[(0, \beta_2 P, 0, 0, 0, 0), (0, 0, \beta_3 Q, 0, 0, 0)] = \left(\beta_1(P \times Q), 0, 0, -\frac{1}{8}\{P, Q\}, 0, 0\right)$. Again,
0902.0431_F02800	179	1.000	$\left[(0, P, 0, 0, 0, 0), (0, Q, 0, 0, 0, 0)\right] = \left(0, 0, 0, \frac{1}{4}\{P, Q\}, 0, 0\right)$	$[(0, P, 0, 0, 0, 0), (0, Q, 0, 0, 0, 0)] = \left(0, 0, 0, \frac{1}{4}\{P, Q\}, 0, 0\right)$	by applying α on $[(0, P, 0, 0, 0, 0), (0, Q, 0, 0, 0, 0)] = \left(0, 0, 0, \frac{1}{4}\{P, Q\}, 0, 0\right)$, we obtain
0902.0431_F02801	179	1.000	$\left[(\Phi, 0, 0, 0, 0, 0), (0, P, 0, 0, 0, 0)\right] = (0, \Phi P, 0, 0, 0, 0)$	$[(\Phi, 0, 0, 0, 0, 0), (0, P, 0, 0, 0, 0)] = (0, \Phi P, 0, 0, 0, 0)$	Further, by applying α on $[(\Phi, 0, 0, 0, 0, 0), (0, P, 0, 0, 0, 0)] = (0, \Phi P, 0, 0, 0, 0)$, we ob-
0902.0431_F02802	179	1.000	$\beta_2 = \beta_3$	$\beta_2 = \beta_3$	hence $\beta_2 = \beta_3$, which we denote by β . If we put $\beta^{-1}P$ in (iii) in the place of P , we
0902.0431_F02803	179	1.000	$\beta^{-1}P$	$\beta^{-1}P$	hence $\beta_2 = \beta_3$, which we denote by β . If we put $\beta^{-1}P$ in (iii) in the place of P , we
0902.0431_F02804	179	1.000	$v : \mathfrak{e}_{\{8\}}^{\{C\}} \rightarrow \mathfrak{e}_{\{8\}}^{\{C\}}$	$v : \mathfrak{e}_8^C \rightarrow \mathfrak{e}_8^C$	Definition. We define a C -linear mapping $v : \mathfrak{e}_8^C \rightarrow \mathfrak{e}_8^Q$ by
0902.0431_F02805	179	0.992	$v \in E_{\{8\}}$	$v \in E_8$	Then $v \in E_8$ and $v^2 = 1$. Note that v is the central element -1 of E_7 regarding as
0902.0431_F02806	179	0.992	$v^2 = 1$	$v^2 = 1$	Then $v \in E_8$ and $v^2 = 1$. Note that v is the central element -1 of E_7 regarding as
0902.0431_F02807	179	1.000	$\left(E_{\{8\}}\right)^v$	$(E_8)^v$	We shall study the following subgroup $(E_8)^v$ of E_8 :
0902.0431_F02808	180	1.000	$\alpha \in E_{\{8\}}$	$\alpha \in E_8$	Lemma 5.7.2. If $\alpha \in E_8$ satisfies $\alpha 1_- = 1_-$, then it also satisfies $\alpha 1 = 1$ and
0902.0431_F02809	180	1.000	$\alpha 1^{\{-}\} = 1^{\{-}\}$	$\alpha 1^- = 1^-$	$\alpha 1^- = 1^-$.
0902.0431_F02810	180	0.685	$\alpha 1 = (\Phi, P, Q, r, s, t)$	$\alpha 1 = (\Phi, P, Q, r, s, t)$	Proof. Let $\alpha 1 = (\Phi, P, Q, r, s, t)$. From the relation $[\alpha 1, 1_-] = [\alpha 1, \alpha 1_-] =$
0902.0431_F02811	180	0.685	$\left[\alpha 1_1, 1_-\right] = \left[\alpha 1, \alpha 1_-\right] =$	$[\alpha 1_1 1_-] = [\alpha 1, \alpha 1_-] =$	Proof. Let $\alpha 1 = (\Phi, P, Q, r, s, t)$. From the relation $[\alpha 1, 1_-] = [\alpha 1, \alpha 1_-] =$
0902.0431_F02812	180	1.000	$\alpha \left[1, 1_-\right] = -2 \alpha 1_- = -2 1_-$	$\alpha [1, 1_-] = -2 \alpha 1_- = -2 1_-$	$[\alpha 1, 1_-] = -2 \alpha 1_- = -2 1_-$. we have
0902.0431_F02813	180	1.000	$P = 0, s = 0, r = 1$	$P = 0, s = 0, r = 1$	from which we obtain $P = 0, s = 0, r = 1$. Further, $\langle \alpha 1, \alpha 1 \rangle = \langle 1, 1 \rangle = 8$, that is,

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0902.0431_F02814	180	■ 1.000	$\backslash \langle \alpha 1, \alpha 1 \rangle = \langle \alpha 1, \alpha 1 \rangle = 8$	$\langle \alpha 1, \alpha 1 \rangle = \langle 1, 1 \rangle = 8$	from which we obtain $P = 0, s = 0, r = 1$. Further, $\langle \alpha 1, \alpha 1 \rangle = \langle 1, 1 \rangle = 8$, that is,
0902.0431_F02815	180	■ 1.000	$\backslash \langle \Phi, \Phi \rangle + \langle Q, Q \rangle + 8 + 4(\tau t)t = 8$	$\langle \Phi, \Phi \rangle + \langle Q, Q \rangle + 8 + 4(\tau t)t = 8$	$\langle \Phi, \Phi \rangle + \langle Q, Q \rangle + 8 + 4(\tau t)t = 8$, which implies that $\Phi = 0, Q = 0, t = 0$. Therefore
0902.0431_F02816	180	■ 1.000	$\Phi = 0, Q = 0, t = 0$	$\Phi = 0, Q = 0, t = 0$	$\langle \Phi, \Phi \rangle + \langle Q, Q \rangle + 8 + 4(\tau t)t = 8$, which implies that $\Phi = 0, Q = 0, t = 0$. Therefore
0902.0431_F02817	180	■ 1.000	$\backslash \left[1^-, 1_- \right] = 1$	$[1^-, 1_-] = 1$	$\alpha 1 = 1$. By applying α on $[1^-, 1_-] = 1$, we obtain $\alpha 1^- = 1^-$ by a similar method to
0902.0431_F02818	180	■ 0.295	$\backslash \left(E_8 \right)_1$	$(E_8)_1$	$(E_8)_{1_-}$ of E_8 :
0902.0431_F02819	180	■ 0.848	$\backslash \text{quad} \backslash \left(E_8 \right)_1 \cong E_7$	$(E_8)_{1_-} \cong E_7$	Theorem 5.7.3. $(E_8)_{1_-} \cong E_7$.
0902.0431_F02820	180	■ 0.967	$\backslash \left(E_8 \right)_1 = \left(E_8 \right)_{1, 1^-, 1_-}$	$(E_8)_{1_-} = (E_8)_{1, 1^-, 1_-}$	Proof. $(E_8)_{1_-} = (E_8)_{1, 1^-, 1_-}$ (Lemma 5.7.2)
0902.0431_F02821	180	■ 1.000	$\mathfrak{W}_1$	$\mathfrak{W}_1$	Remark. We define a space $\mathfrak{W}_1$ by
0902.0431_F02822	180	■ 0.963	$SU(2) = \{ A \in M(2, C) \mid (\tau^t A) A = E, \det A = 1 \}$	$SU(2) = \{ A \in M(2, C) \mid (\tau^t A) A = E, \det A = 1 \}$	which is isomorphic to the group $SU(2) = \{ A \in M(2, C) \mid (\tau^t A) A = E, \det A = 1 \}$,
0902.0431_F02823	180	■ 1.000	$A = \begin{pmatrix} a & -\tau b \\ b & \tau a \end{pmatrix} \in SU(2), \varphi_3(A) : \mathfrak{e}_8^C \rightarrow \mathfrak{e}_8^C$	$A = \begin{pmatrix} a & -\tau b \\ b & \tau a \end{pmatrix} \in SU(2), \varphi_3(A) : \mathfrak{e}_8^C \rightarrow \mathfrak{e}_8^C$	where, for $A = \begin{pmatrix} a & -\tau b \\ b & \tau a \end{pmatrix} \in SU(2), \varphi_3(A) : \mathfrak{e}_8^C \rightarrow \mathfrak{e}_8^C$ is defined by
0902.0431_F02824	181	■ 1.000	$A = \begin{pmatrix} a & -\tau b \\ b & \tau a \end{pmatrix} \in SU(2), \varphi_3(A) : \mathfrak{e}_8^C \rightarrow \mathfrak{e}_8^C$	$A = \begin{pmatrix} a & -\tau b \\ b & \tau a \end{pmatrix} \in SU(2), \varphi_3(A) : \mathfrak{e}_8^C \rightarrow \mathfrak{e}_8^C$	Proof. For $A = \begin{pmatrix} a & -\tau b \\ b & \tau a \end{pmatrix} \in SU(2), \varphi_3(A) : \mathfrak{e}_8^C \rightarrow \mathfrak{e}_8^C$, we have $\phi(A) =$
0902.0431_F02825	181	■ 1.000	$\phi(A) =$	$\phi(A) =$	Proof. For $A = \begin{pmatrix} a & -\tau b \\ b & \tau a \end{pmatrix} \in SU(2), \varphi_3(A) : \mathfrak{e}_8^C \rightarrow \mathfrak{e}_8^C$, we have $\phi(A) =$
0902.0431_F02826	181	■ 0.998	$\exp(\Theta(0, 0, 0, i\nu, \rho, -\tau\rho)) \in (E_8)^v$	$\exp(\Theta(0, 0, 0, i\nu, \rho, -\tau\rho)) \in (E_8)^v$	$\exp(\Theta(0, 0, 0, i\nu, \rho, -\tau\rho)) \in (E_8)^v$.
0902.0431_F02827	181	■ 0.995	$\left(E_8 \right)^v \cong (SU(2) \times E_7) / \mathbf{Z}_2, \mathbf{Z}_2 = \{ (E, 1), (-E, -1) \}$	$(E_8)^v \cong (SU(2) \times E_7) / \mathbf{Z}_2, \mathbf{Z}_2 = \{ (E, 1), (-E, -1) \}$	Theorem 5.7.6. $(E_8)^v \cong (SU(2) \times E_7) / \mathbf{Z}_2, \mathbf{Z}_2 = \{ (E, 1), (-E, -1) \}$.
0902.0431_F02828	181	■ 0.935	$\varphi : SU(2) \times E_7 \rightarrow (E_8)^v$	$\varphi : SU(2) \times E_7 \rightarrow (E_8)^v$	Proof. We define a mapping $\varphi : SU(2) \times E_7 \rightarrow (E_8)^v$ by
0902.0431_F02829	181	■ 1.000	$\varphi_3(A) \in \varphi_3(SU(A))$	$\varphi_3(A) \in \varphi_3(SU(A))$	Since $\varphi_3(A) \in \varphi_3(SU(A))$ and $\beta \in E_7$ commute, φ is a homomorphism. Since $(E_8)^v$
0902.0431_F02830	181	■ 1.000	$\beta \in E_7$	$\beta \in E_7$	Since $\varphi_3(A) \in \varphi_3(SU(A))$ and $\beta \in E_7$ commute, φ is a homomorphism. Since $(E_8)^v$
0902.0431_F02831	181	■ 1.000	$\varphi_* : \mathfrak{su}(8) \oplus \mathfrak{e}_7 \rightarrow (\mathfrak{e}_8)^v$	$\varphi_* : \mathfrak{su}(8) \oplus \mathfrak{e}_7 \rightarrow (\mathfrak{e}_8)^v$	differential mapping $\varphi_* : \mathfrak{su}(8) \oplus \mathfrak{e}_7 \rightarrow (\mathfrak{e}_8)^v$ of φ is onto. But which is not difficult to
0902.0431_F02832	181	■ 1.000	$\text{Ker} \varphi = \{ (E, 1), (-E, -1) \} = \mathbf{Z}_2$	$\text{Ker} \varphi = \{ (E, 1), (-E, -1) \} = \mathbf{Z}_2$	$\text{Ker} \varphi = \{ (E, 1), (-E, -1) \} = \mathbf{Z}_2$ is easily obtained. Thus we have the isomorphism
0902.0431_F02833	181	■ 0.999	$(SU(2) \times E_7) / \mathbf{Z}_2 \cong (E_8)^v$	$(SU(2) \times E_7) / \mathbf{Z}_2 \cong (E_8)^v$	$(SU(2) \times E_7) / \mathbf{Z}_2 \cong (E_8)^v$.
0902.0431_F02834	181	■ 0.990	$\tilde{\lambda} \gamma : \mathfrak{e}_8^C \rightarrow \mathfrak{e}_8^C$	$\tilde{\lambda} \gamma : \mathfrak{e}_8^C \rightarrow \mathfrak{e}_8^C$	We define a C -linear mapping $\tilde{\lambda} \gamma : \mathfrak{e}_8^C \rightarrow \mathfrak{e}_8^C$ by
0902.0431_F02835	181	■ 0.730	$\tilde{\lambda} \gamma \in E_8$	$\tilde{\lambda} \gamma \in E_8$	Then $\tilde{\lambda} \gamma \in E_8$ and $(\tilde{\lambda} \gamma)^2 = 1$.
0902.0431_F02836	181	■ 0.730	$(\tilde{\lambda} \gamma)^2 = 1$	$(\tilde{\lambda} \gamma)^2 = 1$	Then $\tilde{\lambda} \gamma \in E_8$ and $(\tilde{\lambda} \gamma)^2 = 1$.
0902.0431_F02837	181	■ 0.999	$(E_8)^{\tilde{\lambda} \gamma}$	$(E_8)^{\tilde{\lambda} \gamma}$	We shall study the following subgroup $(E_8)^{\tilde{\lambda} \gamma}$ of E_8 :
0902.0431_F02838	181	■ 1.000	$l : M(8, C) \rightarrow M(16, \mathbf{R})$	$l : M(8, C) \rightarrow M(16, \mathbf{R})$	We define an $\mathbf{R}$ -linear mapping $l : M(8, C) \rightarrow M(16, \mathbf{R})$ by
0902.0431_F02839	182	■ 1.000	$I, J \in M(16, \mathbf{R})$	$I, J \in M(16, \mathbf{R})$	Further, we define $l : M(16, \mathbf{R})$ by
0902.0431_F02840	182	■ 1.000	$IJ = -JI$	$IJ = -JI$	then $l(IJ) = -l(JI)$, and for $X, Y \in M(8, C)$, we have
0902.0431_F02841	182	■ 1.000	$X, Y \in M(8, C)$	$X, Y \in M(8, C)$	then $IJ = -JI$, and for $X, Y \in M(8, C)$, we have
0902.0431_F02842	182	■ 1.000	$l(XY) = l(X)l(Y)$	$l(XY) = l(X)l(Y)$	(1) $l(XY) = l(X)l(Y)$,
0902.0431_F02843	182	■ 0.965	$l(X) = l(\tau X)I, J l(X) = l(X)J$	$l(X) = l(\tau X)I, J l(X) = l(X)J$	(2) $l(X) = l(\tau X)I, J l(X) = l(X)J$,
0902.0431_F02844	182	■ 1.000	${}^t l(X) = l(\tau^t X)$	${}^t l(X) = l(\tau^t X)$	(3) $l(X) = l(\tau^t X)$
0902.0431_F02845	182	■ 0.989	$l(u(8)) = \{ B \in \mathfrak{so}(16) \mid JB = BJ \}$	$l(u(8)) = \{ B \in \mathfrak{so}(16) \mid JB = BJ \}$	Lemma 5.8.1. (1) $l(u(8)) = \{ B \in \mathfrak{so}(16) \mid JB = BJ \}$.

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0902.0431_F02846	182	■ 1.000	$B \in \mathfrak{so}(16)$	$B \in \mathfrak{so}(16)$	(2) Any element $B \in \mathfrak{so}(16)$ is uniquely expressed by
0902.0431_F02847	182	■ 0.991	$D \in \mathfrak{u}(8)$	$D \in \mathfrak{u}(8)$	Proof. (1) If $D \in \mathfrak{u}(8)$, then we have $Jl(D) = l(D)J$ and ${}^t l(D) = l(\tau {}^t D) = -l(D)$.
0902.0431_F02848	182	■ 0.991	$Jl(D) = l(D)J$	$Jl(D) = l(D)J$	Proof. (1) If $D \in \mathfrak{u}(8)$, then we have $Jl(D) = l(D)J$ and ${}^t l(D) = l(\tau {}^t D) = -l(D)$.
0902.0431_F02849	182	■ 0.991	$\{ \}^{\{t\}} l(D) = l(\tau {}^t D) = -l(D)$	${}^t l(D) = l(\tau {}^t D) = -l(D)$	Proof. (1) If $D \in \mathfrak{u}(8)$, then we have $Jl(D) = l(D)J$ and ${}^t l(D) = l(\tau {}^t D) = -l(D)$.
0902.0431_F02850	182	■ 1.000	$JB = BJ$	$JB = BJ$	Conversely, suppose that $B \in \mathfrak{so}(16)$ satisfies $JB = BJ$. Let $B = l(D), D \in M(8, C)$.
0902.0431_F02851	182	■ 1.000	$B = l(D), D \in M(8, C)$	$B = l(D), D \in M(8, C)$	Conversely, suppose that $B \in \mathfrak{so}(16)$ satisfies $JB = BJ$. Let $B = l(D), D \in M(8, C)$.
0902.0431_F02852	182	■ 1.000	$\tau {}^t D = -D$	$\tau {}^t D = -D$	implies $\tau {}^t D = -D$, that, is, $D \in \mathfrak{u}(8)$. Next, for $S \in \mathfrak{S}(8, C)$, we have
0902.0431_F02853	182	■ 1.000	$S \in \mathfrak{S}(8, C)$	$S \in \mathfrak{S}(8, C)$	implies $\tau {}^t D = -D$, that, is, $D \in \mathfrak{u}(8)$. Next, for $S \in \mathfrak{S}(8, C)$, we have
0902.0431_F02854	182	■ 1.000	$JB = -BJ$	$JB = -BJ$	Conversely, suppose that $B \in \mathfrak{so}(16)$ satisfies $JB = -BJ$. Since the element BI
0902.0431_F02855	182	■ 1.000	BI	BI	Conversely, suppose that $B \in \mathfrak{so}(16)$ satisfies $JB = -BJ$. Since the element BI
0902.0431_F02856	182	■ 1.000	$JBI = BIJ$	$JBI = BIJ$	satisfies $JBI = BIJ$, we let $BI = l(S), S \in M(8, C)$. Then the relation
0902.0431_F02857	182	■ 1.000	$BI = l(S), S \in M(8, C)$	$BI = l(S), S \in M(8, C)$	satisfies $JBI = BIJ$, we let $BI = l(S), S \in M(8, C)$. Then the relation
0902.0431_F02858	182	■ 1.000	$-S = {}^t S$	$-S = {}^t S$	implies $-S = {}^t S$, that is, $S \in \mathfrak{S}(8, C)$.
0902.0431_F02859	182	■ 1.000	$B = \frac{B - JBJ}{2} + \frac{B + JBJ}{2}$	$B = \frac{B - JBJ}{2} + \frac{B + JBJ}{2}$	(2) Let $B = \frac{B - JBJ}{2} + \frac{B + JBJ}{2}$ and use (1) above.
0902.0431_F02860	182	■ 1.000	$\chi : (\mathfrak{P}^C)_{\tau\gamma} \rightarrow$	$\chi : (\mathfrak{P}^C)_{\tau\gamma} \rightarrow$	The following Lemmas 5.8.2 and 5.8.3 are properties of the mappings $\chi : (\mathfrak{P}^C)_{\tau\gamma} \rightarrow$
0902.0431_F02861	182	■ 0.950	$\mathfrak{S}(8, C)$	$\mathfrak{S}(8, C)$	$\mathfrak{S}(8, C)$ and $\varphi_* : \mathfrak{su}(8) \rightarrow (\mathfrak{e}_6)^{\lambda\gamma}$ of Section 4.12, and will be used in the proof of
0902.0431_F02862	182	■ 0.950	$\varphi_* : \mathfrak{su}(8) \rightarrow (\mathfrak{e}_6)^{\lambda\gamma}$	$\varphi_* : \mathfrak{su}(8) \rightarrow (\mathfrak{e}_6)^{\lambda\gamma}$	$\mathfrak{S}(8, C)$ and $\varphi_* : \mathfrak{su}(8) \rightarrow (\mathfrak{e}_6)^{\lambda\gamma}$ of Section 4.12, and will be used in the proof of
0902.0431_F02863	182	■ 0.999	$\varphi_* : \mathfrak{sp}(4) \rightarrow (\mathfrak{e}_6)^{\lambda\gamma}$	$\varphi_* : \mathfrak{sp}(4) \rightarrow (\mathfrak{e}_6)^{\lambda\gamma}$	Lemma 5.8.2 The Lie isomorphism $\varphi_* : \mathfrak{sp}(4) \rightarrow (\mathfrak{e}_6)^{\lambda\gamma}$ defined by $(\varphi_* D)X =$
0902.0431_F02864	182	■ 0.999	$(\varphi_* D)X =$	$(\varphi_* D)X =$	Lemma 5.8.2 The Lie isomorphism $\varphi_* : \mathfrak{sp}(4) \rightarrow (\mathfrak{e}_6)^{\lambda\gamma}$ defined by $(\varphi_* D)X =$
0902.0431_F02865	183	■ 1.000	$[gX_1, gX_2] \in \mathfrak{sp}(4)$	$[gX_1, gX_2] \in \mathfrak{sp}(4)$	Proof. Note that $[gX_1, gX_2] \in \mathfrak{sp}(4)$. Now, since
0902.0431_F02866	183	■ 1.000	$S, S_1, S_2 \in \mathfrak{S}(8, C)$	$S, S_1, S_2 \in \mathfrak{S}(8, C)$	Lemma 5.8.3. For $S, S_1, S_2 \in \mathfrak{S}(8, C)$, we have
0902.0431_F02867	183	■ 0.985	$\lambda\gamma\chi^{-1}(S) = -\chi^{-1}(iS)$	$\lambda\gamma\chi^{-1}(S) = -\chi^{-1}(iS)$	(1) $\lambda\gamma\chi^{-1}(S) = -\chi^{-1}(iS)$.
0902.0431_F02868	183	■ 0.984	$\mathrm{tr}(S_1\tau S_2 - S_2\tau S_1) = 4i\{\chi^{-1}S_1, \chi^{-1}S_2\}$	$\mathrm{tr}(S_1\tau S_2 - S_2\tau S_1) = 4i\{\chi^{-1}S_1, \chi^{-1}S_2\}$	(2) $\mathrm{tr}(S_1\tau S_2 - S_2\tau S_1) = 4i\{\chi^{-1}S_1, \chi^{-1}S_2\}$.
0902.0431_F02869	183	■ 1.000	$\chi^{-1}S = P = (X, Y, \xi, \eta)$	$\chi^{-1}S = P = (X, Y, \xi, \eta)$	Proof. (1) Let $\chi^{-1}S = P = (X, Y, \xi, \eta)$. Then
0902.0431_F02870	183	■ 1.000	$\chi^{-1}S_i = P_i = (X_i, T_i, \xi_i, \eta_i), i = 1, 2$	$\chi^{-1}S_i = P_i = (X_i, T_i, \xi_i, \eta_i), i = 1, 2$	(2),(3) Let $\chi^{-1}S_i = P_i = (X_i, T_i, \xi_i, \eta_i), i = 1, 2$. Noting
0902.0431_F02871	184	■ 0.999	$D = -[gX_1, gX_2] - [g(\gamma Y_1), g(\gamma Y_2)] \in \mathfrak{sp}(4)$	$D = -[gX_1, gX_2] - [g(\gamma Y_1), g(\gamma Y_2)] \in \mathfrak{sp}(4)$	(denote $D = -[gX_1, gX_2] - [g(\gamma Y_1), g(\gamma Y_2)] \in \mathfrak{sp}(4)$,
0902.0431_F02872	185	■ 0.986	$(\mathfrak{e}_8)^{\tilde{\lambda}\gamma}$	$(\mathfrak{e}_8)^{\tilde{\lambda}\gamma}$	Theorem 5.8.4. The Lie algebra $(\mathfrak{e}_8)^{\tilde{\lambda}\gamma}$ of the Lie group $(E_8)^{\tilde{\lambda}\gamma}$ is given by
0902.0431_F02873	185	■ 0.989	$\mathfrak{so}(16)$	$\mathfrak{so}(16)$	and $(\mathfrak{e}_8)^{\tilde{\lambda}\gamma}$ is isomorphic to the Lie algebra $\mathfrak{so}(16)$ by the mapping $\zeta : \mathfrak{so}(16) \rightarrow (\mathfrak{e}_8)^{\tilde{\lambda}\gamma}$
0902.0431_F02874	185	■ 0.989	$\zeta : \mathfrak{so}(16) \rightarrow (\mathfrak{e}_8)^{\tilde{\lambda}\gamma}$	$\zeta : \mathfrak{so}(16) \rightarrow (\mathfrak{e}_8)^{\tilde{\lambda}\gamma}$	and $(\mathfrak{e}_8)^{\tilde{\lambda}\gamma}$ is isomorphic to the Lie algebra $\mathfrak{so}(16)$ by the mapping $\zeta : \mathfrak{so}(16) \rightarrow (\mathfrak{e}_8)^{\tilde{\lambda}\gamma}$
0902.0431_F02875	185	■ 0.937	$D \in \mathfrak{su}(8), S \in \mathfrak{S}(8, C), c \in \mathbf{R}$	$D \in \mathfrak{su}(8), S \in \mathfrak{S}(8, C), c \in \mathbf{R}$	where $D \in \mathfrak{su}(8), S \in \mathfrak{S}(8, C), c \in \mathbf{R}$, and $\varphi_* : \mathfrak{su}(8) \rightarrow (\mathfrak{e}_7)^{\lambda\gamma}, \chi : (\mathfrak{P}^C)_{\mathrm{tau}\gamma} \rightarrow$
0902.0431_F02876	185	■ 0.937	$\varphi_* : \mathfrak{su}(8) \rightarrow (\mathfrak{e}_7)^{\lambda\gamma}, \chi : (\mathfrak{P}^C)_{\mathrm{tau}\gamma} \rightarrow$	$\varphi_* : \mathfrak{su}(8) \rightarrow (\mathfrak{e}_7)^{\lambda\gamma}, \chi : (\mathfrak{P}^C)_{\mathrm{tau}\gamma} \rightarrow$	where $D \in \mathfrak{su}(8), S \in \mathfrak{S}(8, C), c \in \mathbf{R}$, and $\varphi_* : \mathfrak{su}(8) \rightarrow (\mathfrak{e}_7)^{\lambda\gamma}, \chi : (\mathfrak{P}^C)_{\mathrm{tau}\gamma} \rightarrow$
0902.0431_F02877	185	■ 1.000	$(\varphi_* D)\lambda\gamma = \lambda\gamma(\varphi_* D)$	$(\varphi_* D)\lambda\gamma = \lambda\gamma(\varphi_* D)$	Since $(\varphi_* D)\lambda\gamma = \lambda\gamma(\varphi_* D)$ and $(\varphi_* D)\chi^{-1}S = \chi^{-1}(DS + S^t D)$, they are equal.
0902.0431_F02878	185	■ 1.000	$(\varphi_* D)\chi^{-1}S = \chi^{-1}(DS + S^t D)$	$(\varphi_* D)\chi^{-1}S = \chi^{-1}(DS + S^t D)$	Since $(\varphi_* D)\lambda\gamma = \lambda\gamma(\varphi_* D)$ and $(\varphi_* D)\chi^{-1}S = \chi^{-1}(DS + S^t D)$, they are equal.
0902.0431_F02879	186	■ 1.000	$\zeta[l(D), l(icE)] = \zeta[D, icE] = \zeta 0 = 0$	$\zeta[l(D), l(icE)] = \zeta[D, icE] = \zeta 0 = 0$	(3) $\zeta[l(D), l(icE)] = \zeta[D, icE] = \zeta 0 = 0$

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0902.0431_F02880	186	■ 0.986	$\left(E_{\{8\}}\right)^{\widetilde{\lambda}\gamma} \cong Ss(16)$	$(E_8)^{\tilde{\lambda}\gamma} \cong Ss(16)$	Precisely we have $(E_8)^{\lambda\gamma} \cong Ss(16)$, below we will give an outline of the proof.
0902.0431_F02881	187	■ 1.000	$\mathfrak{e}_{\{8(8)\}}$	$\mathfrak{e}_{8(8)}$	we obtain the following decomposition of $\mathfrak{e}_{8(8)}$:
0902.0431_F02882	187	■ 0.820	$\left(\mathfrak{e}_{\{8(8)\}}\right)^{\widetilde{\lambda}\gamma} \cong Ss(16)$	$(\mathfrak{e}_{8(8)})^{\tilde{\lambda}\gamma} \cong Ss(16)$	Since $(\mathfrak{e}_{8(8)})^{\lambda\gamma} \cong Ss(16)$ (Theorem 5.8.4), this is the Cartan decomposition of $\mathfrak{e}_{8(8)}$.
0902.0431_F02883	187	■ 0.318	$\left[\left(\mathfrak{e}_{\{8(8)\}}\right)^{\widetilde{\lambda}\gamma}, \left(\mathfrak{e}_{\{8(8)\}}\right)_{-\widetilde{\lambda}\gamma}\right] \subset \left(\mathfrak{e}_{\{8(8)\}}\right)_{-\widetilde{\lambda}\gamma}$	$\left[\left(\mathfrak{e}_{8(8)}\right)^{\tilde{\lambda}\gamma}, \left(\mathfrak{e}_{8(8)}\right)_{-\tilde{\lambda}\gamma}\right] \subset \left(\mathfrak{e}_{8(8)}\right)_{-\tilde{\lambda}\gamma}$	Since $\left[(\mathfrak{e}_{8(8)})^{\lambda\gamma}, (\mathfrak{e}_{8(8)})_{-\tilde{\lambda}\gamma}\right] \subset (\mathfrak{e}_{8(8)})_{-\tilde{\lambda}\gamma}$, we obtain a representation φ of $(\mathfrak{e}_{8(8)})^{\tilde{\lambda}\gamma}$ to
0902.0431_F02884	187	■ 0.318	$\left(\mathfrak{e}_{\{8(8)\}}\right)^{\widetilde{\lambda}\gamma}$	$(\mathfrak{e}_{8(8)})^{\tilde{\lambda}\gamma}$	Since $\left[(\mathfrak{e}_{8(8)})^{\tilde{\lambda}\gamma}, (\mathfrak{e}_{8(8)})_{-\tilde{\lambda}\gamma}\right] \subset (\mathfrak{e}_{8(8)})_{-\tilde{\lambda}\gamma}$, we obtain a representation φ of $(\mathfrak{e}_{8(8)})^{\tilde{\lambda}\gamma}$ to
0902.0431_F02885	187	■ 0.777	$\left(\mathfrak{e}_{\{8(8)\}}\right)_{-\widetilde{\lambda}\gamma}$	$(\mathfrak{e}_{8(8)})_{-\tilde{\lambda}\gamma}$	$(\mathfrak{e}_{8(8)})_{-\tilde{\lambda}\gamma}$
0902.0431_F02886	187	■ 0.989	φ^C	φ^C	thermore, the complex representation φ^C of φ to $\left((\mathfrak{e}_{8(8)})_{-\tilde{\lambda}\gamma}\right)^C = (\mathfrak{e}_8^C)_{-\tilde{\lambda}\gamma}$ is also
0902.0431_F02887	187	■ 0.989	$\left(\left(\mathfrak{e}_{\{8(8)\}}\right)_{-\widetilde{\lambda}\gamma}\right)^C = (\mathfrak{e}_8^C)_{-\tilde{\lambda}\gamma}$	$\left(\left(\mathfrak{e}_{8(8)}\right)_{-\tilde{\lambda}\gamma}\right)^C = (\mathfrak{e}_8^C)_{-\tilde{\lambda}\gamma}$	thermore, the complex representation φ^C of φ to $\left((\mathfrak{e}_{8(8)})_{-\tilde{\lambda}\gamma}\right)^C = (\mathfrak{e}_8^C)_{-\tilde{\lambda}\gamma}$ is also
0902.0431_F02888	187	■ 0.770	$(\mathfrak{e}_{8(8)})_{-\tilde{\lambda}\gamma}$	$(\mathfrak{e}_{8(8)})_{-\tilde{\lambda}\gamma}$	irreducible, since $(\mathfrak{e}_{8(8)})_{-\tilde{\lambda}\gamma}$ is simple (see (8.8.3) of the same book). The following
0902.0431_F02889	187	■ 0.978	$z\left((E_{\{8\}})^{\widetilde{\lambda}\gamma}\right)$	$z\left((E_8)^{\tilde{\lambda}\gamma}\right)$	Proposition 5.8.6. The center $z((E_8)^{\tilde{\lambda}\gamma})$ of the group $(E_8)^{\tilde{\lambda}\gamma}$ is a group of order
0902.0431_F02890	187	■ 0.989	$\{1, \widetilde{\lambda}\gamma\} \subset z\left((E_{\{8\}})^{\widetilde{\lambda}\gamma}\right)$	$\{1, \tilde{\lambda}\gamma\} \subset z\left((E_8)^{\tilde{\lambda}\gamma}\right)$	Proof. Evidently, $\{1, \tilde{\lambda}\gamma\} \subset z((E_8)^{\tilde{\lambda}\gamma})$. Conversely, let $\alpha \in z((E_8)^{\tilde{\lambda}\gamma})$. Since
0902.0431_F02891	187	■ 0.989	$\alpha \in z\left((E_{\{8\}})^{\widetilde{\lambda}\gamma}\right)$	$\alpha \in z\left((E_8)^{\tilde{\lambda}\gamma}\right)$	Proof. Evidently, $\{1, \tilde{\lambda}\gamma\} \subset z((E_8)^{\tilde{\lambda}\gamma})$. Conversely, let $\alpha \in z((E_8)^{\tilde{\lambda}\gamma})$. Since
0902.0431_F02892	187	■ 0.711	$(E_{\{8\}})^{\widetilde{\lambda}\gamma}$	$(E_8)^{\tilde{\lambda}\gamma}$	the representation of $(E_8)^{\tilde{\lambda}\gamma}$ to $(\mathfrak{e}_8^C)_{-\tilde{\lambda}\gamma}$ irreducible (Lemma 5.8.5), we see, by using
0902.0431_F02893	187	■ 0.711	$(\mathfrak{e}_8^C)_{-\tilde{\lambda}\gamma}$	$(\mathfrak{e}_8^C)_{-\tilde{\lambda}\gamma}$	the representation of $(E_8)^{\tilde{\lambda}\gamma}$ to $(\mathfrak{e}_8^C)_{-\tilde{\lambda}\gamma}$ irreducible (Lemma 5.8.5), we see, by using
0902.0431_F02894	187	■ 0.777	$B_{\{8\}}(R, R')$	$B_8(R, R')$	Since the Killing form $B_8(R, R')$ is invariant under α : $B_8(\alpha R, \alpha R') = B_8(R, R')$, we
0902.0431_F02895	187	■ 0.777	$\alpha : B_{\{8\}}(\alpha R, \alpha R') = B_{\{8\}}(R, R')$	$\alpha : B_8(\alpha R, \alpha R') = B_8(R, R')$	Since the Killing form $B_8(R, R')$ is invariant under α : $B_8(\alpha R, \alpha R') = B_8(R, R')$, we
0902.0431_F02896	188	■ 0.999	$k^2=1$	$k^2=1$	which implies that $k^2=1$. By Theorem 5.8.4, we have $(\mathfrak{e}_8^C)^{\tilde{\lambda}\gamma} \cong \mathfrak{so}(16, C)$ which is
0902.0431_F02897	188	■ 0.999	$(\mathfrak{e}_8^C)^{\tilde{\lambda}\gamma} \cong \mathfrak{so}(16, C)$	$(\mathfrak{e}_8^C)^{\tilde{\lambda}\gamma} \cong \mathfrak{so}(16, C)$	which implies that $k^2=1$. By Theorem 5.8.4, we have $(\mathfrak{e}_8^C)^{\lambda\gamma} \cong \mathfrak{so}(16, C)$ which is
0902.0431_F02898	188	■ 0.995	$(\mathfrak{e}_8^C)^{\tilde{\lambda}\gamma}$	$(\mathfrak{e}_8^C)^{\tilde{\lambda}\gamma}$	simple, and hence we see that $(\mathfrak{e}_8^C)^{\tilde{\lambda}\gamma}$ is generated by $(\mathfrak{e}_8^C)^{-\tilde{\lambda}\gamma}$:
0902.0431_F02899	188	■ 0.995	$(\mathfrak{e}_8^C)^{-\tilde{\lambda}\gamma}$	$(\mathfrak{e}_8^C)^{-\tilde{\lambda}\gamma}$	simple, and hence we see that $(\mathfrak{e}_8^C)^{\tilde{\lambda}\gamma}$ is generated by $(\mathfrak{e}_8^C)^{-\tilde{\lambda}\gamma}$:
0902.0431_F02900	188	■ 0.995	$k^2=1$	$k^2=1$	Consequently, α satisfies $k^2=1$ on $(\mathfrak{e}_8^C)_{-\tilde{\lambda}\gamma}$, that is, the identity mapping. When
0902.0431_F02901	188	■ 0.998	$k=1$	$k=1$	$k=1$, we have $\alpha=1$, when $k=-1$, we have $\alpha=\tilde{\lambda}\gamma$. Thus we have proved the
0902.0431_F02902	188	■ 0.998	$k=-1$	$k=-1$	$k=1$, we have $\alpha=1$, when $k=-1$, we have $\alpha=\tilde{\lambda}\gamma$. Thus we have proved the
0902.0431_F02903	188	■ 0.998	$\alpha=\widetilde{\lambda}\gamma$	$\alpha=\tilde{\lambda}\gamma$	$k=1$, we have $\alpha=1$, when $k=-1$, we have $\alpha=\tilde{\lambda}\gamma$. Thus we have proved the
0902.0431_F02904	188	■ 1.000	ω_7	ω_7	hence the dominant root of dimension 128 is either ω_7 or ω_8 . (Here, $\omega_1, \omega_2, \dots, \omega_8$
0902.0431_F02905	188	■ 1.000	ω_8	ω_8	hence the dominant root of dimension 128 is either ω_7 or ω_8 . (Here, $\omega_1, \omega_2, \dots, \omega_8$
0902.0431_F02906	188	■ 1.000	$\omega_1, \omega_2, \dots, \omega_8$	$\omega_1, \omega_2, \dots, \omega_8$	hence the dominant root of dimension 128 is either ω_7 or ω_8 . (Here, $\omega_1, \omega_2, \dots, \omega_8$
0902.0431_F02907	188	■ 0.986	$Spin(16)$	$Spin(16)$	are fundamental weights). On the other hand, $Spin(16)$ has two complex irreducible
0902.0431_F02908	188	■ 1.000	Δ_{16}^+	Δ_{16}^+	representation Δ_{16}^+ and Δ_{16}^- , called spinor representations. Furthermore, both of
0902.0431_F02909	188	■ 1.000	Δ_{16}^-	Δ_{16}^-	representation Δ_{16}^+ and Δ_{16}^- , called spinor representations. Furthermore, both of
0902.0431_F02910	188	■ 0.982	$SO(16)$	$SO(16)$	Δ_{16}^+ and Δ_{16}^- are not representation of $SO(16)$. Now, by Lemma 5.8.5, $(E_8)^{\tilde{\lambda}\gamma}$ has

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0902.0431_F02911	188	■ 0.934	$\backslash quad \backslash left(E_{\{8\}} \backslash right)^{\backslash \widetilde{\backslash lambda} \backslash gamma} \backslash cong S s(16)$	$(E_8)^{\tilde{\lambda}\gamma} \cong Ss(16)$	Theorem 5.8.7. $(E_8)^{\tilde{\lambda}\gamma} \cong Ss(16)$.
0902.0431_F02912	188	■ 1.000	$\backslash sigma \backslash in F_{\{4\}} \backslash subset E_{\{6\}} \backslash subset E_{\{7\}} \backslash subset E_{\{8\}}$	$\sigma \in F_4 \subset E_6 \subset E_7 \subset E_8$	This is the same as $\overline{\sigma} \in F_4 \subset E_6 \subset E_7 \subset E_8$. We define a subgroup $(E_8)^\sigma$ by
0902.0431_F02913	188	■ 1.000	$\backslash left(E_{\{8\}} \backslash right)^{\backslash \sigma}$	$(E_8)^\sigma$	This is the same as $\sigma \in F_4 \subset E_6 \subset E_7 \subset E_8$. We define a subgroup $(E_8)^\sigma$ by
0902.0431_F02914	188	■ 0.996	$\backslash left(\backslash mathfrak{e}_{\{8\}} \backslash right)^{\backslash \sigma}$	$(\mathfrak{e}_8)^\sigma$	Indeed, we can prove that the Lie algebra $(\mathfrak{e}_8)^\sigma$ of the group $(E_8)^\sigma$ is isomorphic to
0902.0431_F02915	188	■ 0.959	$\backslash mathfrak{s o}(16)=\backslash left\{\backslash left.X \backslash in M(16, \backslash boldsymbol{R}) \backslash right\}^t X=-X \backslash right\}$	$\mathfrak{so}(16) = \left\{ X \in M(16, \mathbf{R}) \mid {}^t X = -X \right\}$	the Lie algebra $\overline{\mathfrak{so}}(16) = \{X \in M(16, \mathbf{R}) \mid {}^t X = -X\}$. Hnece as the same the case the
0902.0431_F02916	188	■ 0.785	$\backslash left(E_{\{8\}} \backslash right)^{\backslash \widetilde{\backslash lambda} \backslash gamma} \backslash cong S s(16) \backslash cong \backslash left(E_{\{8\}} \backslash right)^{\backslash \sigma}$	$(E_8)^{\tilde{\lambda}\gamma} \cong Ss(16) \cong (E_8)^\sigma$	proof of $(E_8)^{\tilde{\lambda}\gamma} \cong Ss(16) \cong (E_8)^\sigma$ is not concretely. M. Gomyo [10] find the group
0902.0431_F02917	189	■ 1.000	$\backslash alpha \backslash in z \backslash left(E_{\{8\}} \backslash right)$	$\alpha \in z(E_8)$	Proof. Let $\alpha \in z(E_8)$. The relation $v\alpha = \alpha v$ implies that $\alpha \in \varphi(SU(2) \times E_7) \cong$
0902.0431_F02918	189	■ 1.000	$v \backslash alpha=\alpha v$	$v\alpha = \alpha v$	Proof. Let $\alpha \in z(E_8)$. The relation $v\alpha = \alpha v$ implies that $\alpha \in \varphi(SU(2) \times E_7) \cong$
0902.0431_F02919	189	■ 1.000	$\backslash alpha \backslash in \backslash varphi \backslash left(S U(2) \backslash times E_{\{7\}} \backslash right) \backslash cong$	$\alpha \in \varphi(SU(2) \times E_7) \cong$	Proof. Let $\alpha \in z(E_8)$. The relation $v\alpha = \alpha v$ implies that $\overline{\alpha} \in \varphi(SU(2) \times E_7) \cong$
0902.0431_F02920	189	■ 0.834	$\backslash alpha \backslash in z \backslash left(\backslash varphi \backslash left(S U(2) \backslash times E_{\{7\}} \backslash right) \backslash right)$	$\alpha \in z(\varphi(SU(2) \times E_7))$	$(SU(2) \times E_7)/\mathbf{Z}_2$ (Theorem 5.7.6), and so $\alpha \in z(\varphi(SU(2) \times E_7))$. Therefore by
0902.0431_F02921	189	■ 1.000	$v \backslash notin z \backslash left(E_{\{8\}} \backslash right)$	$v \notin z(E_8)$	However $v \notin z(E_8)$ (Theorem 5.7.6). Hence $\alpha = 1$.
0902.0431_F02922	189	■ 1.000	$\backslash left(S U(3) \backslash times E_{\{6\}} \backslash right) / \backslash boldsymbol{Z}_{\{3\}}$	$(SU(3) \times E_6)/\mathbf{Z}_3$	5.10. Automorphism w of order 3 and subgroup $[SU(3) \times E_6]/\mathbf{Z}_3$ of E_8
0902.0431_F02923	189	■ 1.000	$\backslash mathfrak{e}_{\{8\}}^C, \backslash left\backslash langle R_{\{1\}}, R_{\{2\}} \backslash right \backslash rangle, \backslash tau \backslash \widetilde{\backslash lambda}, w$	$\mathfrak{e}_8^C, \langle R_1, R_2 \rangle, \tau \tilde{\lambda}, w$	notation as $\overline{\mathfrak{e}}_8^C, \langle R_1, R_2 \rangle, \tau \tilde{\lambda}, w$, even if these are different from those used in the
0902.0431_F02924	189	■ 1.000	$27 \backslash times 3=78$	$27 \times 3 = 78$	We consider a $27 \times 3 = 78$ dimensional C -vector space
0902.0431_F02925	189	■ 1.000	$\backslash left(\backslash mathfrak{J}^{\{C\}} \backslash right)^{\{3\}}$	$(\mathfrak{J}^C)^3$	In $(\mathfrak{J}^C)^3$, we define an inner product $(\mathbf{X}, \mathbf{Y})$, a Hermitian inner product $\langle \mathbf{X}, \mathbf{Y} \rangle$, a
0902.0431_F02926	189	■ 1.000	$(\backslash boldsymbol{X}, \backslash boldsymbol{Y})$	$(\mathbf{X}, \mathbf{Y})$	In $(\mathfrak{J}^C)^3$, we define an inner product $(\mathbf{X}, \mathbf{Y})$, a Hermitian inner product $\langle \mathbf{X}, \mathbf{Y} \rangle$, a
0902.0431_F02927	189	■ 1.000	$\backslash langle \backslash boldsymbol{X}, \backslash boldsymbol{Y} \backslash rangle$	$\langle \mathbf{X}, \mathbf{Y} \rangle$	In $(\mathfrak{J}^C)^3$, we define an inner product $(\mathbf{X}, \mathbf{Y})$, a Hermitian inner product $\langle \mathbf{X}, \mathbf{Y} \rangle$, a
0902.0431_F02928	189	■ 0.996	$\backslash boldsymbol{X} \backslash times \backslash boldsymbol{Y}$	$\mathbf{X} \times \mathbf{Y}$	cross product $\mathbf{X} \times \mathbf{Y}$, an element $\mathbf{X} \cdot \mathbf{Y}$ of $\mathfrak{sl}(3, C)$ and an element $\mathbf{X} \vee \mathbf{Y}$ of $\mathfrak{e}_6^C$
0902.0431_F02929	189	■ 0.996	$\backslash boldsymbol{X} \backslash cdot \backslash boldsymbol{Y}$	$\mathbf{X} \cdot \mathbf{Y}$	cross product $\mathbf{X} \times \mathbf{Y}$, an element $\overline{\mathbf{X} \cdot \mathbf{Y}}$ of $\mathfrak{sl}(3, C)$ and an element $\mathbf{X} \vee \mathbf{Y}$ of $\mathfrak{e}_6^C$
0902.0431_F02930	189	■ 0.996	$\backslash boldsymbol{X} \backslash vee \backslash boldsymbol{Y}$	$\mathbf{X} \vee \mathbf{Y}$	cross product $\mathbf{X} \times \mathbf{Y}$, an element $\mathbf{X} \cdot \mathbf{Y}$ of $\mathfrak{sl}(3, C)$ and an element $\mathbf{X} \vee \mathbf{Y}$ of $\mathfrak{e}_6^C$
0902.0431_F02931	189	■ 0.972	$\backslash boldsymbol{X}=\backslash left(\backslash begin{array}{l} X_{\{1\}} \backslash \backslash X_{\{2\}} \backslash \backslash X_{\{3\}} \backslash end{array} \backslash right), \backslash boldsymbol{Y}=\backslash left(\backslash begin{array}{l} Y_{\{1\}} \backslash \backslash Y_{\{2\}} \backslash \backslash Y_{\{3\}} \backslash end{array} \backslash right) \backslash in \backslash left(\backslash mathfrak{J}^{\{C\}} \backslash right)^{\{3\}}$	$\mathbf{X} = \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix}, \mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ Y_3 \end{pmatrix} \in (\mathfrak{J}^C)^3$	where $\mathbf{X} = \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix}, \mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ Y_3 \end{pmatrix} \in (\mathfrak{J}^C)^3$. Further, for $\phi \in \text{Hom}_C(\mathfrak{J}^C), D = (d_{ij}) \in$
0902.0431_F02932	189	■ 0.972	$\backslash phi \backslash in \backslash operatorname{Hom}_{\{C\}} \backslash left(\backslash mathfrak{J}^{\{C\}} \backslash right), D=\backslash left(d_{\{i j\}} \backslash right) \backslash in$	$\phi \in \text{Hom}_C(\mathfrak{J}^C), D = (d_{ij}) \in$	where $\mathbf{X} = \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix}, \mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ Y_3 \end{pmatrix} \in (\mathfrak{J}^C)^3$. Further, for $\phi \in \text{Hom}_C(\mathfrak{J}^C), D = (d_{ij}) \in$
0902.0431_F02933	189	■ 0.983	$M(3, C)$	$M(3, C)$	$\overline{M(3, C)}$ and $\mathbf{X} = \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix} \in (\mathfrak{J}^C)^3$, elements $\phi \mathbf{X}, D \mathbf{X} \in (\mathfrak{J}^C)^3$ are naturally defined
0902.0431_F02934	189	■ 0.983	$\backslash boldsymbol{X}=\backslash left(\backslash begin{array}{c} X_{\{1\}} \backslash \backslash X_{\{2\}} \backslash \backslash X_{\{3\}} \backslash end{array} \backslash right) \backslash in \backslash left(\backslash mathfrak{J}^{\{C\}} \backslash right)^{\{3\}}$	$\mathbf{X} = \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix} \in (\mathfrak{J}^C)^3$	$M(3, C)$ and $\mathbf{X} = \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix} \in (\mathfrak{J}^C)^3$, elements $\phi \mathbf{X}, D \mathbf{X} \in (\mathfrak{J}^C)^3$ are naturally defined
0902.0431_F02935	189	■ 0.983	$\backslash phi \backslash boldsymbol{X}, D \backslash boldsymbol{X} \backslash in \backslash left(\backslash mathfrak{J}^{\{C\}} \backslash right)^{\{3\}}$	$\phi \mathbf{X}, D \mathbf{X} \in (\mathfrak{J}^C)^3$	$M(3, C)$ and $\mathbf{X} = \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix} \in (\mathfrak{J}^C)^3$, elements $\overline{\phi \mathbf{X}}, D \mathbf{X} \in (\mathfrak{J}^C)^3$ are naturally defined
0902.0431_F02936	190	■ 0.999	$8+78+27 \backslash times 3+27 \backslash times 3=248$	$8 + 78 + 27 \times 3 + 27 \times 3 = 248$	Theorem 5.10.1. In an $\overline{8 + 78 + 27 \times 3 + 27 \times 3 = 248}$ dimensional C -vector
0902.0431_F02937	190	■ 0.902	$\backslash \widetilde{\backslash mathfrak{e}}_{\{8\}}^{\{C\}}=\backslash mathfrak{e}_{\{7\}}^{\{C\}} \backslash oplus \backslash mathfrak{P}^{\{C\}} \backslash oplus \backslash mathfrak{P}^{\{C\}} \backslash oplus C \backslash oplus C \backslash oplus C$	$\tilde{\mathfrak{e}}_8^C = \mathfrak{e}_7^C \oplus \mathfrak{P}^C \oplus \mathfrak{P}^C \oplus C \oplus C \oplus C$	Proof. Let $\tilde{\mathfrak{e}}_8^C = \mathfrak{e}_7^C \oplus \mathfrak{P}^C \oplus \mathfrak{P}^C \oplus C \oplus C \oplus C$ be the C -Lie algebra constructed
0902.0431_F02938	190	■ 0.900	$f: \backslash \widetilde{\backslash mathfrak{e}}_{\{8\}}^{\{C\}} \rightarrow \backslash mathfrak{e}_{\{8\}}^{\{C\}}$	$f: \tilde{\mathfrak{e}}_8^C \rightarrow \mathfrak{e}_8^C$	in Theorem 5.1.1. We define a mapping $f: \tilde{\mathfrak{e}}_8^C \rightarrow \mathfrak{e}_8^C$ by

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0902.0431_F02939	190	0.699	$\widetilde{\mathfrak{e}}_{\{8\}}^{\{ \}^{\{C\}}}\cong \mathfrak{e}_{\{8\}}^{\{ \}^{\{C\}}}$	$\widetilde{\mathfrak{e}}_8^C \cong \mathfrak{e}_8^C$	culations. Thus we have the isomorphism $\widetilde{\mathfrak{e}}_8^C \cong \mathfrak{e}_8^C$.
0902.0431_F02940	190	0.999	$\widetilde{\mathfrak{e}}_{\{8\}}^{\{ \}^{\{C\}}}$	$\widetilde{\mathfrak{e}}_8^C$	Using the Killing form of $\widetilde{\mathfrak{e}}_8^C$ which is obtained in Theorem 5.3.2, we see that the
0902.0431_F02941	190	0.655	$\left(R_i=\left(D_i,\phi_i,\boldsymbol{X}_i,\boldsymbol{Y}_i\right)\in\mathfrak{e}_{\{8\}}^{\{ \}^{\{C\}}}\right)$	$(R_i = (D_i, \phi_i, \boldsymbol{X}_i, \boldsymbol{Y}_i) \in \mathfrak{e}_8^C)$	$(R_i = (D_i, \phi_i, \boldsymbol{X}_i, \boldsymbol{Y}_i) \in \mathfrak{e}_8^C)$, where B_6 is the Killing form of $\mathfrak{e}_6^C$. We define a
0902.0431_F02942	190	1.000	$\tau\widetilde{\lambda}$	$\tau\widetilde{\lambda}$	complex conjugate transformation $\tau\widetilde{\lambda}$ of $\mathfrak{e}_8^C$ by
0902.0431_F02943	191	1.000	$w\in E_{\{8\}}$	$w\in E_8$	where $\omega=-\frac{1}{2}+\frac{\sqrt{3}}{2}i\in C$. Then $w\in E_8$ and $w^3=1$.
0902.0431_F02944	191	1.000	$\left(E_{\{8\}}\right)^w$	$(E_8)^w$	Now, we shall study the following subgroup $(E_8)^w$ of E_8 :
0902.0431_F02945	191	0.545	$\left(E_{\{8\}}\right)^w\cong\left(SU(3)\times E_6\right)/\boldsymbol{Z}_3,\quad\boldsymbol{Z}_3=\left\{\left(E,1\right),\left(\omega E,\omega^21\right),\left(\omega^2E,\omega1\right)\right\}$	$(E_8)^w \cong (SU(3) \times E_6)/\boldsymbol{Z}_3, \quad \boldsymbol{Z}_3 = \{(E, 1), (\omega E, \omega^2 1), (\omega^2 E,$	Theorem 5.10.2. $(E_8)^w \cong (SU(3) \times E_6)/\boldsymbol{Z}_3, \quad \boldsymbol{Z}_3 = \{(E, 1), (\omega E, \omega^2 1), (\omega^2 E,$
0902.0431_F02946	191	0.991	$\omega1)\}$	$\omega1)\}$	$\omega1)\}$.
0902.0431_F02947	191	1.000	$\varphi_1:SU(3)\rightarrow\left(E_{\{8\}}\right)^w$	$\varphi_1:SU(3)\rightarrow(E_8)^w$	Proof. We first define a mapping $\varphi_1:SU(3)\rightarrow(E_8)^w$ by
0902.0431_F02948	191	0.990	$\varphi_1(A)\in\left(E_{\{8\}}\right)^w$	$\varphi_1(A)\in(E_8)^w$	We have to prove that $\varphi_1(A)\in(E_8)^w$. Indeed, since the action of $D_1=(D_1,0,0,0)\in$
0902.0431_F02949	191	0.990	$D_{\{1\}}=\left(D_{\{1\}},0,0,0\right)\in$	$D_1=(D_1,0,0,0)\in$	We have to prove that $\varphi_1(A)\in(E_8)^w$. Indeed, since the action of $D_1=(D_1,0,0,0)\in$
0902.0431_F02950	191	0.994	$\mathfrak{su}(3)\subset\mathfrak{sl}(3,C)\subset\mathfrak{e}_8^C$	$\mathfrak{su}(3)\subset\mathfrak{sl}(3,C)\subset\mathfrak{e}_8^C$	$\mathfrak{su}(3)\subset\mathfrak{sl}(3,C)\subset\mathfrak{e}_8^C$ is given by
0902.0431_F02951	191	0.985	$A=\exp D_{\{1\}}$	$A=\exp D_1$	for $A=\exp D_1$, we have $\varphi_1(A)=\exp(\mathrm{ad}(D_1))\in\mathrm{Aut}(\mathfrak{e}_8^C)$. And from
0902.0431_F02952	191	0.985	$\varphi_1(A)=\exp\left(\mathrm{ad}\left(\left(D_{\{1\}}\right)\right)\right)\in\mathrm{Aut}\left(\mathfrak{e}_{\{8\}}^{\{ \}^{\{C\}}}\right)$	$\varphi_1(A)=\exp(\mathrm{ad}(D_1))\in\mathrm{Aut}(\mathfrak{e}_8^C)$	for $A=\exp D_1$, we have $\varphi_1(A)=\exp(\mathrm{ad}(D_1))\in\mathrm{Aut}(\mathfrak{e}_8^C)$. And from
0902.0431_F02953	191	1.000	$\varphi_1(A)\in E_{\{8\}}$	$\varphi_1(A)\in E_8$	we see that $\varphi_1(A)\in E_8$. Evidently, $w\varphi_1(A)=\varphi_1(A)w$, hence, $\varphi_1(A)\in(E_8)^w$. Next,
0902.0431_F02954	191	1.000	$w\varphi_1(A)=\varphi_1(A)w$	$w\varphi_1(A)=\varphi_1(A)w$	we see that $\varphi_1(A)\in E_8$. Evidently, $w\varphi_1(A)=\varphi_1(A)w$, hence, $\varphi_1(A)\in(E_8)^w$. Next,
0902.0431_F02955	191	1.000	$\varphi_2:E_6\rightarrow\left(E_{\{8\}}\right)^w$	$\varphi_2:E_6\rightarrow(E_8)^w$	we define a mapping $\varphi_2:E_6\rightarrow(E_8)^w$ by
0902.0431_F02956	191	1.000	$\varphi_2(\alpha)\in\left(E_{\{8\}}\right)^w$	$\varphi_2(\alpha)\in(E_8)^w$	We have to prove that $\varphi_2(\alpha)\in(E_8)^w$. Indeed, since the action of an element $\phi' =$
0902.0431_F02957	191	1.000	$\phi'^{}$	$\phi'^{}$	We have to prove that $\varphi_2(\alpha)\in(E_8)^w$. Indeed, since the action of an element $\phi' =$
0902.0431_F02958	191	1.000	$\left(0,\phi',0,0\right)\in\mathfrak{e}_6\subset\mathfrak{e}_6^C\subset\mathfrak{e}_8^C$	$(0, \phi', 0, 0) \in \mathfrak{e}_6 \subset \mathfrak{e}_6^C \subset \mathfrak{e}_8^C$	$(0, \phi', 0, 0) \in \mathfrak{e}_6 \subset \mathfrak{e}_6^C \subset \mathfrak{e}_8^C$ is given by
0902.0431_F02959	191	0.990	$\alpha=\exp\phi'^{}$	$\alpha=\exp\phi'^{}$	for $\alpha=\exp\phi'^{}$, we have $\varphi_2(\alpha)=\exp(\mathrm{ad}(\phi'))\in\mathrm{Aut}(\mathfrak{e}_8^C)$. And from
0902.0431_F02960	191	0.990	$\varphi_2(\alpha)=\exp\left(\mathrm{ad}\left(\phi'\right)\right)\in\mathrm{Aut}\left(\mathfrak{e}_{\{8\}}^{\{ \}^{\{C\}}}\right)$	$\varphi_2(\alpha)=\exp(\mathrm{ad}(\phi'))\in\mathrm{Aut}(\mathfrak{e}_8^C)$	for $\alpha=\exp\phi'^{}$, we have $\varphi_2(\alpha)=\exp(\mathrm{ad}(\phi'))\in\mathrm{Aut}(\mathfrak{e}_8^C)$. And from
0902.0431_F02961	192	1.000	$\varphi_2(\alpha)\in E_{\{8\}}$	$\varphi_2(\alpha)\in E_8$	we see that $\varphi_2(\alpha)\in E_8$. Evidently, $w\varphi_2(\alpha)=\varphi_2(\alpha)w$, hence, $\varphi_2(\alpha)\in(E_8)^w$. Now,
0902.0431_F02962	192	1.000	$w\varphi_2(\alpha)=\varphi_2(\alpha)w$	$w\varphi_2(\alpha)=\varphi_2(\alpha)w$	we see that $\varphi_2(\alpha)\in E_8$. Evidently, $w\varphi_2(\alpha)=\varphi_2(\alpha)w$, hence, $\varphi_2(\alpha)\in(E_8)^w$. Now,
0902.0431_F02963	192	0.798	$\varphi:SU(3)\times E_6\rightarrow\left(E_{\{8\}}\right)^w$	$\varphi:SU(3)\times E_6\rightarrow(E_8)^w$	we define a mapping $\varphi:SU(3)\times E_6\rightarrow(E_8)^w$ by
0902.0431_F02964	192	1.000	$\varphi_1(A)$	$\varphi_1(A)$	Since $\varphi_1(A)$ and $\varphi_2(\alpha)$ commute, φ is a homomorphism. It is not difficult to show
0902.0431_F02965	192	1.000	$\varphi_2(\alpha)$	$\varphi_2(\alpha)$	Since $\varphi_1(A)$ and $\varphi_2(\alpha)$ commute, φ is a homomorphism. It is not difficult to show
0902.0431_F02966	192	0.999	$\mathrm{Ker}\varphi=\left\{\left(E,1\right),\left(\omega E,\omega^21\right),\left(\omega^2E,\omega1\right)\right\}=\boldsymbol{Z}_3$	$\mathrm{Ker}\varphi = \{(E, 1), (\omega E, \omega^2 1), (\omega^2 E, \omega 1)\} = \boldsymbol{Z}_3$	that $\mathrm{Ker}\varphi = \{(E, 1), (\omega E, \omega^2 1), (\omega^2 E, \omega 1)\} = \boldsymbol{Z}_3$. Finally, since $(E_8)^w$ is connected
0902.0431_F02967	192	0.996	$\left(SU(3)\times E_6\right)/\boldsymbol{Z}_3\cong\left(E_{\{8\}}\right)^w$	$(SU(3) \times E_6)/\boldsymbol{Z}_3 \cong (E_8)^w$	φ is onto. Thus we have the isomorphism $SU(3) \times E_6/\boldsymbol{Z}_3 \cong (E_8)^w$.
0902.0431_F02968	192	1.000	$\Lambda^k\left(C^n\right)$	$\Lambda^k(C^n)$	erties of the exterior C -vector space $\Lambda^k(C^n)$. Let $e_1, \cdots, e_n$ be the canonical C -basis
0902.0431_F02969	192	1.000	$e_1, \cdots, e_n$	$e_1, \cdots, e_n$	erties of the exterior C -vector space $\Lambda^k(C^n)$. Let $e_1, \cdots, e_n$ be the canonical C -basis
0902.0431_F02970	192	0.755	n	n	of n -dimensional C -vector space C^n and $(\boldsymbol{x}, \boldsymbol{y})$ the inner product in C^n satisfying
0902.0431_F02971	192	0.755	C^n	C^n	of n -dimensional C -vector space C^n and $(\boldsymbol{x}, \boldsymbol{y})$ the inner product in C^n satisfying
0902.0431_F02972	192	0.755	$(\boldsymbol{x}, \boldsymbol{y})$	$(\boldsymbol{x}, \boldsymbol{y})$	of n -dimensional C -vector space C^n and $(\boldsymbol{x}, \boldsymbol{y})$ the inner product in C^n satisfying

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0902.0431_F02973	192	■ 1.000	$\left(\boldsymbol{e}_i, \boldsymbol{e}_j\right)=\delta_{i j}$	$(\boldsymbol{e}_i, \boldsymbol{e}_j)=\delta_{i j}$	$\boldsymbol{e}_i, \boldsymbol{e}_j)=\delta_i$. In $\Lambda^k(C^n)$, we define an inner product by
0902.0431_F02974	192	■ 0.985	$\boldsymbol{e}_{i_1}\wedge\cdots\wedge\boldsymbol{e}_{i_k}, i_1<\cdots<i_k$	$\boldsymbol{e}_{i_1}\wedge\cdots\wedge\boldsymbol{e}_{i_k}, i_1<\cdots<i_k$	Then, $\boldsymbol{e}_{i_1}\wedge\cdots\wedge\boldsymbol{e}_{i_k}, i_1<\cdots<i_k$ forms an orthonormal C -basis of $\Lambda^k(C^n)$. For
0902.0431_F02975	192	■ 1.000	$\boldsymbol{u} \in \Lambda^k(C^n)$	$\boldsymbol{u} \in \Lambda^k(C^n)$	$\boldsymbol{u} \in \Lambda^k(C^n)$, we define an element $\ast \boldsymbol{u} \in \Lambda^{n-k}(C^n)$ by
0902.0431_F02976	192	■ 1.000	$\ast \boldsymbol{u} \in \Lambda^{n-k}(C^n)$	$\ast \boldsymbol{u} \in \Lambda^{n-k}(C^n)$	$\boldsymbol{u} \in \Lambda^k(C^n)$, we define an element $\ast \boldsymbol{u} \in \Lambda^{n-k}(C^n)$ by
0902.0431_F02977	192	■ 1.000	$S L(n, C)$	$SL(n, C)$	The group $SL(n, C)$ naturally acts on $\Lambda^k(C^n)$ as
0902.0431_F02978	193	■ 1.000	$\mathfrak{sl}(n, C)$	$\mathfrak{sl}(n, C)$	Hence the Lie algebra $\mathfrak{sl}(n, C)$ acts on $\Lambda^k(C^n)$ as
0902.0431_F02979	193	■ 0.997	$A \in S L(n, C), D \in \mathfrak{sl}(n, C)$	$A \in SL(n, C), D \in \mathfrak{sl}(n, C)$	Lemma 5.11.1. For $A \in SL(n, C), D \in \mathfrak{sl}(n, C)$ and $\boldsymbol{u}, \boldsymbol{v} \in \Lambda^k(C^n)$, we have
0902.0431_F02980	193	■ 0.997	$\boldsymbol{u}, \boldsymbol{v} \in \Lambda^k(C^n)$	$\boldsymbol{u}, \boldsymbol{v} \in \Lambda^k(C^n)$	Lemma 5.11.1. For $A \in SL(n, C), D \in \mathfrak{sl}(n, C)$ and $\boldsymbol{u}, \boldsymbol{v} \in \Lambda^k(C^n)$, we have
0902.0431_F02981	193	■ 0.643	$\left(A \boldsymbol{u}, \boldsymbol{v}\right)=\left(\boldsymbol{u}, A \boldsymbol{v}\right), \quad\left(D \boldsymbol{u}, \boldsymbol{v}\right)+\left(\boldsymbol{u},-D \boldsymbol{v}\right)=0$	$(A \boldsymbol{u}, {}^t A^{-1} \boldsymbol{v})=(\boldsymbol{u}, \boldsymbol{v}), \quad(D \boldsymbol{u}, \boldsymbol{v})+(\boldsymbol{u},-{}^t D \boldsymbol{v})=0$	(1) $(A \boldsymbol{u}, {}^t A^{-1} \boldsymbol{v})=(\boldsymbol{u}, \boldsymbol{v}), \quad(D \boldsymbol{u}, \boldsymbol{v})+(\boldsymbol{u},-{}^t D \boldsymbol{v})=0$.
0902.0431_F02982	193	■ 0.739	$\left(A \boldsymbol{u}\right)=A^{-1}\left(\boldsymbol{u}\right), \quad\left(D \boldsymbol{u}\right)=-D\left(\boldsymbol{u}\right)$	$\ast(A \boldsymbol{u})={}^t A^{-1}(\ast \boldsymbol{u}), \quad \ast(D \boldsymbol{u})=-{}^t D^{-1}(\ast \boldsymbol{u})$	(2) $\ast(A \boldsymbol{u})={}^t A^{-1}(\ast \boldsymbol{u}), \quad \ast(D \boldsymbol{u})=-{}^t D^{-1}(\ast \boldsymbol{u})$.
0902.0431_F02983	193	■ 1.000	$\boldsymbol{u}, \boldsymbol{v} \in \Lambda^k(C^n)(1 \leq k \leq n)$	$\boldsymbol{u}, \boldsymbol{v} \in \Lambda^k(C^n)(1 \leq k \leq n)$	For $\boldsymbol{u}, \boldsymbol{v} \in \Lambda^k(C^n)(1 \leq k \leq n)$, we define a C -linear mapping $\boldsymbol{u} \times \boldsymbol{v}$ of C^n by
0902.0431_F02984	193	■ 1.000	$\boldsymbol{u} \times \boldsymbol{v}$	$\boldsymbol{u} \times \boldsymbol{v}$	For $\boldsymbol{u}, \boldsymbol{v} \in \Lambda^k(C^n)(1 \leq k \leq n)$, we define a C -linear mapping $\boldsymbol{u} \times \boldsymbol{v}$ of C^n by
0902.0431_F02985	193	■ 0.950	$\operatorname{tr}(\boldsymbol{u} \times \boldsymbol{v})=0, \boldsymbol{u} \times \boldsymbol{v}$	$\operatorname{tr}(\boldsymbol{u} \times \boldsymbol{v})=0, \boldsymbol{u} \times \boldsymbol{v}$	Since $\operatorname{tr}(\boldsymbol{u} \times \boldsymbol{v})=0, \boldsymbol{u} \times \boldsymbol{v}$ can be regarded as element of $\mathfrak{sl}(n, C)$ with respect to the
0902.0431_F02986	193	■ 0.950	$\mathfrak{sl}(n, C)$	$\mathfrak{sl}(n, C)$	Since $\operatorname{tr}(\boldsymbol{u} \times \boldsymbol{v})=0, \boldsymbol{u} \times \boldsymbol{v}$ can be regarded as element of $\mathfrak{sl}(n, C)$ with respect to the
0902.0431_F02987	193	■ 0.732	$A(\boldsymbol{u} \times \boldsymbol{v}) A^{-1}=A \boldsymbol{u} \times A \boldsymbol{v},[D, \boldsymbol{u} \times \boldsymbol{v}]=D \boldsymbol{u} \times \boldsymbol{v}+\boldsymbol{u} \times(-D \boldsymbol{v})$	$A(\boldsymbol{u} \times \boldsymbol{v}) A^{-1}=A \boldsymbol{u} \times {}^t A^{-1} \boldsymbol{v},[D, \boldsymbol{u} \times \boldsymbol{v}]=D \boldsymbol{u} \times \boldsymbol{v}+\boldsymbol{u} \times(-{}^t D \boldsymbol{v})$	(1) $A(\boldsymbol{u} \times \boldsymbol{v}) A^{-1}=A \boldsymbol{u} \times {}^t A^{-1} \boldsymbol{v}, \quad[D, \boldsymbol{u} \times \boldsymbol{v}]=D \boldsymbol{u} \times \boldsymbol{v}+\boldsymbol{u} \times(-{}^t D \boldsymbol{v})$.
0902.0431_F02988	193	■ 1.000	$\left(\boldsymbol{u} \times \boldsymbol{v}\right)=\boldsymbol{v} \times \boldsymbol{u}, \quad \tau(\boldsymbol{u} \times \boldsymbol{v})=\tau(\boldsymbol{u}) \times \tau(\boldsymbol{v})$	${}^t(\boldsymbol{u} \times \boldsymbol{v})=\boldsymbol{v} \times \boldsymbol{u}, \quad \tau(\boldsymbol{u} \times \boldsymbol{v})=\tau(\boldsymbol{u}) \times \tau(\boldsymbol{v})$	(2) ${}^t(\boldsymbol{u} \times \boldsymbol{v})=\boldsymbol{v} \times \boldsymbol{u}, \quad \tau(\boldsymbol{u} \times \boldsymbol{v})=\tau(\boldsymbol{u}) \times \tau(\boldsymbol{v})$.
0902.0431_F02989	193	■ 1.000	$\operatorname{tr}(D(\boldsymbol{u} \times \boldsymbol{v}))=(-1)^{n-k}(D \boldsymbol{u}, \boldsymbol{v})$	$\operatorname{tr}(D(\boldsymbol{u} \times \boldsymbol{v}))=(-1)^{n-k}(D \boldsymbol{u}, \boldsymbol{v})$	(3) $\operatorname{tr}(D(\boldsymbol{u} \times \boldsymbol{v}))=(-1)^{n-k}(D \boldsymbol{u}, \boldsymbol{v})$.
0902.0431_F02990	193	■ 1.000	$80+84+84=248$	$80+84+84=248$	Theorem 5.11.3. In an $80+84+84=248$ dimensional C -vector space
0902.0431_F02991	193	■ 1.000	$\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w} \in \Lambda^3(C^9)$	$\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w} \in \Lambda^3(C^9)$	Lemma 5.11.4. For $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w} \in \Lambda^3(C^9)$, we have
0902.0431_F02992	193	■ 0.926	$\boldsymbol{u} \times(\boldsymbol{v} \wedge \boldsymbol{w})+\boldsymbol{v} \times(\boldsymbol{w} \wedge \boldsymbol{u})+\boldsymbol{w} \times(\boldsymbol{u} \wedge \boldsymbol{v})=0$	$\boldsymbol{u} \times(\boldsymbol{v} \wedge \boldsymbol{w})+\boldsymbol{v} \times(\boldsymbol{w} \wedge \boldsymbol{u})+\boldsymbol{w} \times(\boldsymbol{u} \wedge \boldsymbol{v})=0$	(1) $\boldsymbol{u} \times(\boldsymbol{v} \wedge \boldsymbol{w})+\boldsymbol{v} \times(\boldsymbol{w} \wedge \boldsymbol{u})+\boldsymbol{w} \times(\boldsymbol{u} \wedge \boldsymbol{v})=0$,
0902.0431_F02993	193	■ 0.800	$(\boldsymbol{u} \times \boldsymbol{w}) \boldsymbol{v}-(\boldsymbol{v} \times \boldsymbol{w}) \boldsymbol{u}+\left((\boldsymbol{u} \times \boldsymbol{v}) \wedge \boldsymbol{w}\right)=0$	$(\boldsymbol{u} \times \boldsymbol{w}) \boldsymbol{v}-(\boldsymbol{v} \times \boldsymbol{w}) \boldsymbol{u}+\left((\boldsymbol{u} \times \boldsymbol{v}) \wedge \boldsymbol{w}\right)=0$	(2) $\boldsymbol{u} \times \boldsymbol{w}) \boldsymbol{v}-(\boldsymbol{v} \times \boldsymbol{w}) \boldsymbol{u}+\left((\boldsymbol{u} \times \boldsymbol{v}) \wedge \boldsymbol{w}\right)=0$.
0902.0431_F02994	194	■ 0.999	$\boldsymbol{u}=\boldsymbol{u}_1 \wedge \boldsymbol{u}_2 \wedge \boldsymbol{u}_3, \boldsymbol{v}=\boldsymbol{u}_4 \wedge \boldsymbol{u}_5 \wedge \boldsymbol{u}_6$	$\boldsymbol{u}=\boldsymbol{u}_1 \wedge \boldsymbol{u}_2 \wedge \boldsymbol{u}_3, \boldsymbol{v}=\boldsymbol{u}_4 \wedge \boldsymbol{u}_5 \wedge \boldsymbol{u}_6$	Proof. Let $\boldsymbol{u}=\boldsymbol{u}_1 \wedge \boldsymbol{u}_2 \wedge \boldsymbol{u}_3, \boldsymbol{v}=\boldsymbol{u}_4 \wedge \boldsymbol{u}_5 \wedge \boldsymbol{u}_6$ and $\boldsymbol{w}=\boldsymbol{u}_7 \wedge \boldsymbol{u}_8 \wedge \boldsymbol{u}_9$. For
0902.0431_F02995	194	■ 0.999	$\boldsymbol{w}=\boldsymbol{u}_7 \wedge \boldsymbol{u}_8 \wedge \boldsymbol{u}_9$	$\boldsymbol{w}=\boldsymbol{u}_7 \wedge \boldsymbol{u}_8 \wedge \boldsymbol{u}_9$	Proof. Let $\boldsymbol{u}=\boldsymbol{u}_1 \wedge \boldsymbol{u}_2 \wedge \boldsymbol{u}_3, \boldsymbol{v}=\boldsymbol{u}_4 \wedge \boldsymbol{u}_5 \wedge \boldsymbol{u}_6$ and $\boldsymbol{w}=\boldsymbol{u}_7 \wedge \boldsymbol{u}_8 \wedge \boldsymbol{u}_9$. For
0902.0431_F02996	194	■ 0.998	$\boldsymbol{x}, \boldsymbol{y} \in C^9$	$\boldsymbol{x}, \boldsymbol{y} \in C^9$	$\boldsymbol{x}, \boldsymbol{y} \in C^9$, we have

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0902.0431_F02997	194	■ 1.000	$\boldsymbol{x}=\sum_{i=1}^9 x_i \boldsymbol{e}_i, \boldsymbol{u}_j=\sum_{k=1}^9 u_{jk} \boldsymbol{e}_k$	$\boldsymbol{x}=\sum_{i=1}^9 x_i \boldsymbol{e}_i, \boldsymbol{u}_j=\sum_{k=1}^9 u_{jk} \boldsymbol{e}_k$	Denote $\boldsymbol{x}=\sum_{i=1}^9 x_i \boldsymbol{e}_i, \boldsymbol{u}_j=\sum_{k=1}^9 u_{jk} \boldsymbol{e}_k$ and $U=(u_{jk}) \in M(9, C)$. Hence we have
0902.0431_F02998	194	■ 1.000	$U=\left(u_{jk}\right) \in M(9, C)$	$U=(u_{jk}) \in M(9, C)$	Denote $\boldsymbol{x}=\sum_{i=1}^9 x_i \boldsymbol{e}_i, \boldsymbol{u}_j=\sum_{k=1}^9 u_{jk} \boldsymbol{e}_k$ and $U=(u_{jk}) \in M(9, C)$. Hence we have
0902.0431_F02999	194	■ 1.000	$\widetilde{u}_{jk}$	$\widetilde{u}_{jk}$	where $\widetilde{u}_{jk}$ is the cofactor of u_{jk} of the matrix U . Therefore
0902.0431_F03000	194	■ 1.000	u_{jk}	u_{jk}	where $\widetilde{u}_{jk}$ is the cofactor of u_{jk} of the matrix U . Therefore
0902.0431_F03001	194	■ 1.000	U	U	where $\widetilde{u}_{jk}$ is the cofactor of u_{jk} of the matrix U . Therefore
0902.0431_F03002	194	■ 1.000	$\boldsymbol{u}=\boldsymbol{u}_1 \wedge \boldsymbol{u}_2 \wedge \boldsymbol{u}_3$	$\boldsymbol{u}=\boldsymbol{u}_1 \wedge \boldsymbol{u}_2 \wedge \boldsymbol{u}_3$	Thus (1) is proved. Next, let $\boldsymbol{u}=\boldsymbol{u}_1 \wedge \boldsymbol{u}_2 \wedge \boldsymbol{u}_3$ and $\boldsymbol{v}=\boldsymbol{v}_1 \wedge \boldsymbol{v}_2 \wedge \boldsymbol{v}_3$. Using (i), for
0902.0431_F03003	194	■ 1.000	$\boldsymbol{v}=\boldsymbol{v}_1 \wedge \boldsymbol{v}_2 \wedge \boldsymbol{v}_3$	$\boldsymbol{v}=\boldsymbol{v}_1 \wedge \boldsymbol{v}_2 \wedge \boldsymbol{v}_3$	Thus (1) is proved. Next, let $\boldsymbol{u}=\boldsymbol{u}_1 \wedge \boldsymbol{u}_2 \wedge \boldsymbol{u}_3$ and $\boldsymbol{v}=\boldsymbol{v}_1 \wedge \boldsymbol{v}_2 \wedge \boldsymbol{v}_3$. Using (i), for
0902.0431_F03004	194	■ 1.000	$\boldsymbol{a} \in \Lambda^3\left(C^9\right)$	$\boldsymbol{a} \in \Lambda^3\left(C^9\right)$	any $\boldsymbol{a} \in \Lambda^3\left(C^9\right)$, we have
0902.0431_F03005	195	■ 0.997	$\mathfrak{e}_8^C=\mathfrak{sl}(9, C) \oplus \Lambda^3\left(C^9\right) \oplus \Lambda^3\left(C^9\right)$	$\mathfrak{e}_8^C=\mathfrak{sl}(9, C) \oplus \Lambda^3\left(C^9\right) \oplus \Lambda^3\left(C^9\right)$	Theorem 5.11.5. $\mathfrak{e}_8^C=\mathfrak{sl}(9, C) \oplus \Lambda^3\left(C^9\right) \oplus \Lambda^3\left(C^9\right)$ is a simple C -Lie algebra of
0902.0431_F03006	195	■ 1.000	$I=\{i, j, k\}(i < j < k)$	$I=\{i, j, k\}(i < j < k)$	For a subset $I=\{i, j, k\}(i < j < k)$ of $\{1, 2, \cdots, 9\}$, we put
0902.0431_F03007	195	■ 1.000	$\{1, 2, \cdots, 9\}$	$\{1, 2, \cdots, 9\}$	For a subset $I=\{i, j, k\}(i < j < k)$ of $\{1, 2, \cdots, 9\}$, we put
0902.0431_F03008	195	■ 1.000	$\mathfrak{g}=\mathfrak{e}_8^C$	$\mathfrak{g}=\mathfrak{e}_8^C$	Now, let $\mathfrak{a}$ be a non-zero ideal of $\mathfrak{g}=\mathfrak{e}_8^C$.
0902.0431_F03009	195	■ 1.000	$\mathfrak{sl}(9, C) \cap \mathfrak{a}=\{0\}$	$\mathfrak{sl}(9, C) \cap \mathfrak{a}=\{0\}$	(1) Case $\mathfrak{sl}(9, C) \cap \mathfrak{a}=\{0\}$ and $\mathfrak{q} \cap \mathfrak{a}=\{0\}$. Let $p: \mathfrak{g} \rightarrow \mathfrak{sl}(9, C)$ be the projection.
0902.0431_F03010	195	■ 1.000	$\mathfrak{q} \cap \mathfrak{a}=\{0\}$	$\mathfrak{q} \cap \mathfrak{a}=\{0\}$	(1) Case $\mathfrak{sl}(9, C) \cap \mathfrak{a}=\{0\}$ and $\mathfrak{q} \cap \mathfrak{a}=\{0\}$. Let $p: \mathfrak{g} \rightarrow \mathfrak{sl}(9, C)$ be the projection.
0902.0431_F03011	195	■ 1.000	$p: \mathfrak{g} \rightarrow \mathfrak{sl}(9, C)$	$p: \mathfrak{g} \rightarrow \mathfrak{sl}(9, C)$	(1) Case $\mathfrak{sl}(9, C) \cap \mathfrak{a}=\{0\}$ and $\mathfrak{q} \cap \mathfrak{a}=\{0\}$. Let $p: \mathfrak{g} \rightarrow \mathfrak{sl}(9, C)$ be the projection.
0902.0431_F03012	195	■ 1.000	$p(\mathfrak{a})=0$	$p(\mathfrak{a})=0$	If $p(\mathfrak{a})=0$, then $\mathfrak{a}$ is contained in $\mathfrak{q}$, which contradicts $\mathfrak{q} \cap \mathfrak{a}=\{0\}$. Hence, $p(\mathfrak{a})$ is a
0902.0431_F03013	195	■ 1.000	$\mathfrak{q}$	$\mathfrak{q}$	If $p(\mathfrak{a})=0$, then $\mathfrak{a}$ is contained in $\mathfrak{q}$, which contradicts $\mathfrak{q} \cap \mathfrak{a}=\{0\}$. Hence, $p(\mathfrak{a})$ is a
0902.0431_F03014	195	■ 1.000	$\mathfrak{sl}(9, C)$	$\mathfrak{sl}(9, C)$	non-zero ideal of $\mathfrak{sl}(9, C)$, so we have $p(\mathfrak{a})=\mathfrak{sl}(9, C)$. For an element $D=\sum_{i=1}^8 H_i \in$
0902.0431_F03015	195	■ 1.000	$p(\mathfrak{a})=\mathfrak{sl}(9, C)$	$p(\mathfrak{a})=\mathfrak{sl}(9, C)$	non-zero ideal of $\mathfrak{sl}(9, C)$, so we have $p(\mathfrak{a})=\mathfrak{sl}(9, C)$. For an element $D=\sum_{i=1}^8 H_i \in$
0902.0431_F03016	195	■ 1.000	$D=\sum_{i=1}^8 H_i \in$	$D=\sum_{i=1}^8 H_i \in$	non-zero ideal of $\mathfrak{sl}(9, C)$, so we have $p(\mathfrak{a})=\mathfrak{sl}(9, C)$. For an element $D=\sum_{i=1}^8 H_i \in$
0902.0431_F03017	195	■ 0.831	$\mathfrak{sl}(9, C), H_i=E_{ii}-E_{99}$	$\mathfrak{sl}(9, C), H_i=E_{ii}-E_{99}$	$\mathfrak{sl}(9, C), H_i=E_{ii}-E_{99}$, there exists an element $(\boldsymbol{u}, \boldsymbol{v})=\left(\sum_I u_I \boldsymbol{e}_I, \sum_J v_J \boldsymbol{e}_J\right) \in \mathfrak{q}$
0902.0431_F03018	195	■ 0.831	$(\boldsymbol{u}, \boldsymbol{v})=\left(\sum_I u_I \boldsymbol{e}_I, \sum_J v_J \boldsymbol{e}_J\right) \in \mathfrak{q}$	$(\boldsymbol{u}, \boldsymbol{v})=\left(\sum_I u_I \boldsymbol{e}_I, \sum_J v_J \boldsymbol{e}_J\right) \in \mathfrak{q}$	$\mathfrak{sl}(9, C), H_i=E_{ii}-E_{99}$, there exists an element $(\boldsymbol{u}, \boldsymbol{v})=\left(\sum_I u_I \boldsymbol{e}_I, \sum_J v_J \boldsymbol{e}_J\right) \in \mathfrak{q}$
0902.0431_F03019	195	■ 0.998	$(D, \boldsymbol{u}, \boldsymbol{v}) \in \mathfrak{a}$	$(D, \boldsymbol{u}, \boldsymbol{v}) \in \mathfrak{a}$	such that $(D, \boldsymbol{u}, \boldsymbol{v}) \in \mathfrak{a}$. Since $[(D, 0, 0), (X, \boldsymbol{u}, \boldsymbol{v})]=(0, D \boldsymbol{u}, -{}^t D \boldsymbol{v}) \in \mathfrak{q} \cap \mathfrak{a}=\{0\}$, we
0902.0431_F03020	195	■ 0.998	$[(D, 0, 0), (X, \boldsymbol{u}, \boldsymbol{v})]=(0, D \boldsymbol{u}, -{}^t D \boldsymbol{v}) \in \mathfrak{q} \cap \mathfrak{a}=\{0\}$	$[(D, 0, 0), (X, \boldsymbol{u}, \boldsymbol{v})]=(0, D \boldsymbol{u}, -{}^t D \boldsymbol{v}) \in \mathfrak{q} \cap \mathfrak{a}=\{0\}$	such that $(D, \boldsymbol{u}, \boldsymbol{v}) \in \mathfrak{a}$. Since $[(D, 0, 0), (X, \boldsymbol{u}, \boldsymbol{v})]=(0, D \boldsymbol{u}, -{}^t D \boldsymbol{v}) \in \mathfrak{q} \cap \mathfrak{a}=\{0\}$, we
0902.0431_F03021	195	■ 0.882	$u_I=0$	$u_I=0$	i.e., $u_I=0$ and $v_J=0$. Then, $0 \neq (D, \boldsymbol{u}, \boldsymbol{v})=(D, 0, 0) \in \mathfrak{sl}(9, C) \cap \mathfrak{a}=\{0\}$. This is
0902.0431_F03022	195	■ 0.882	$v_J=0$	$v_J=0$	i.e., $u_I=0$ and $v_J=0$. Then, $0 \neq (D, \boldsymbol{u}, \boldsymbol{v})=(D, 0, 0) \in \mathfrak{sl}(9, C) \cap \mathfrak{a}=\{0\}$. This is
0902.0431_F03023	195	■ 0.882	$0 \neq (D, \boldsymbol{u}, \boldsymbol{v})=(D, 0, 0) \in \mathfrak{sl}(9, C) \cap \mathfrak{a}=\{0\}$	$0 \neq (D, \boldsymbol{u}, \boldsymbol{v})=(D, 0, 0) \in \mathfrak{sl}(9, C) \cap \mathfrak{a}=\{0\}$	i.e., $u_I=0$ and $v_J=0$. Then, $0 \neq (D, \boldsymbol{u}, \boldsymbol{v})=(D, 0, 0) \in \mathfrak{sl}(9, C) \cap \mathfrak{a}=\{0\}$. This is
0902.0431_F03024	195	■ 1.000	$\mathfrak{sl}(9, C) \cap \mathfrak{a} \neq \{0\}$	$\mathfrak{sl}(9, C) \cap \mathfrak{a} \neq \{0\}$	(2) Case $\mathfrak{sl}(9, C) \cap \mathfrak{a} \neq \{0\}$. Since $\mathfrak{sl}(9, C) \cap \mathfrak{a}$ is a non-zero ideal of $\mathfrak{sl}(9, C)$, we
0902.0431_F03025	195	■ 1.000	$\mathfrak{sl}(9, C) \cap \mathfrak{a}$	$\mathfrak{sl}(9, C) \cap \mathfrak{a}$	(2) Case $\mathfrak{sl}(9, C) \cap \mathfrak{a} \neq \{0\}$. Since $\mathfrak{sl}(9, C) \cap \mathfrak{a}$ is a non-zero ideal of $\mathfrak{sl}(9, C)$, we
0902.0431_F03026	195	■ 0.999	$\mathfrak{sl}(9, C) \subset \mathfrak{a}$	$\mathfrak{sl}(9, C) \subset \mathfrak{a}$	have $\mathfrak{sl}(9, C) \subset \mathfrak{a}$. For any $\boldsymbol{e}_i \wedge \boldsymbol{e}_j \wedge \boldsymbol{e}_k \in \Lambda^3\left(C^9\right)$, put
0902.0431_F03027	195	■ 0.999	$\boldsymbol{e}_i \wedge \boldsymbol{e}_j \wedge \boldsymbol{e}_k \in \Lambda^3\left(C^9\right)$	$\boldsymbol{e}_i \wedge \boldsymbol{e}_j \wedge \boldsymbol{e}_k \in \Lambda^3\left(C^9\right)$	have $\mathfrak{sl}(9, C) \subset \mathfrak{a}$. For any $\boldsymbol{e}_i \wedge \boldsymbol{e}_j \wedge \boldsymbol{e}_k \in \Lambda^3\left(C^9\right)$ put
0902.0431_F03028	195	■ 1.000	$(D, 0, 0) \in \mathfrak{sl}(9, C) \subset \mathfrak{a}$	$(D, 0, 0) \in \mathfrak{sl}(9, C) \subset \mathfrak{a}$	Since $(D, 0, 0) \in \mathfrak{sl}(9, C) \subset \mathfrak{a}$, we see that
0902.0431_F03029	196	■ 1.000	$\mathfrak{q} \subset \mathfrak{a}$	$\mathfrak{q} \subset \mathfrak{a}$	It follows that $\mathfrak{q} \subset \mathfrak{a}$. Hence we have $\mathfrak{a}=\mathfrak{g}$.
0902.0431_F03030	196	■ 1.000	$\mathfrak{a}=\mathfrak{g}$	$\mathfrak{a}=\mathfrak{g}$	It follows that $\mathfrak{q} \subset \mathfrak{a}$. Hence we have $\mathfrak{a}=\mathfrak{g}$.
0902.0431_F03031	196	■ 1.000	$\mathfrak{q} \cap \mathfrak{a} \neq \{0\}$	$\mathfrak{q} \cap \mathfrak{a} \neq \{0\}$	(3) Case $\mathfrak{q} \cap \mathfrak{a} \neq \{0\}$. Let $R=(0, \boldsymbol{u}, \boldsymbol{v})$ be a non-zero element of $\mathfrak{q} \cap \mathfrak{a}$. In the
0902.0431_F03032	196	■ 1.000	$R=(0, \boldsymbol{u}, \boldsymbol{v})$	$R=(0, \boldsymbol{u}, \boldsymbol{v})$	(3) Case $\mathfrak{q} \cap \mathfrak{a} \neq \{0\}$. Let $R=(0, \boldsymbol{u}, \boldsymbol{v})$ be a non-zero element of $\mathfrak{q} \cap \mathfrak{a}$. In the

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0902.0431_F03033	196	■ 1.000	$\mathfrak{q} \cap \mathfrak{a}$	$\mathfrak{q} \cap \mathfrak{a}$	(3) Case $\mathfrak{q} \cap \mathfrak{a} \neq \{0\}$. Let $R = (0, \mathbf{u}, \mathbf{v})$ be a non-zero element of $\mathfrak{q} \cap \mathfrak{a}$. In the
0902.0431_F03034	196	■ 1.000	$\boldsymbol{u} \neq 0$	$\mathbf{u} \neq 0$	case $\mathbf{u} \neq 0$, we put $\mathbf{u} = \sum_I u_I \mathbf{e}_I$. Without loss of generality, we may assume that
0902.0431_F03035	196	■ 1.000	$\boldsymbol{u}=\sum_{\mathcal{I}} u_{\mathcal{I}} \boldsymbol{e}_{\mathcal{I}}$	$\mathbf{u} = \sum_I u_I \mathbf{e}_I$	case $\mathbf{u} \neq 0$, we put $\mathbf{u} = \sum_I u_I \mathbf{e}_I$. Without loss of generality, we may assume that
0902.0431_F03036	196	■ 1.000	$u_{\{123\}}=1$	$u_{\{123\}} = 1$	$u_{\{123\}} = 1$. Putting $S_{ij} = (E_{ii} - E_{jj}, 0, 0) \in \mathfrak{g}$ and $T = (0, 0, \mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_4) \in \mathfrak{g}$, we
0902.0431_F03037	196	■ 1.000	$S_{\{i\ j\}}=\left(E_{\{i\ i\}}-E_{\{j\ j\}},\ 0,0\right) \in \mathfrak{g}$	$S_{ij} = (E_{ii} - E_{jj}, 0, 0) \in \mathfrak{g}$	$u_{\{123\}} = 1$. Putting $\mathbf{S}_{ii} = (E_{ii} - E_{jj}, 0, 0) \in \mathfrak{g}$ and $T = (0, 0, \mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_4) \in \mathfrak{g}$, we
0902.0431_F03038	196	■ 1.000	$T=\left(0,0,\ \boldsymbol{e}_1 \wedge \boldsymbol{e}_2 \wedge \boldsymbol{e}_4\right) \in \mathfrak{g}$	$T = (0, 0, \mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_4) \in \mathfrak{g}$	$u_{\{123\}} = 1$. Putting $S_{ij} = (E_{ii} - E_{jj}, 0, 0) \in \mathfrak{g}$ and $\mathbf{T} = (0, 0, \mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_4) \in \mathfrak{g}$, we
0902.0431_F03039	196	■ 1.000	$\boldsymbol{v} \neq 0$	$\mathbf{v} \neq 0$	Then we can reduce this case to the case (2). In case $\mathbf{v} \neq 0$, we ca similarly reduce
0902.0431_F03040	196	■ 0.997	$\mathfrak{e}_8^C=\mathfrak{sl}(9,C) \oplus$	$\mathfrak{e}_8^C = \mathfrak{sl}(9, C) \oplus$	Proposition 5.11.6. The Killing form B_8 of the Lie algebra $\mathfrak{e}_8^C = \mathfrak{sl}(9, C) \oplus$
0902.0431_F03041	196	■ 1.000	$\Lambda^3\left(C^9\right) \oplus \Lambda^3\left(C^9\right)$	$\Lambda^3(C^9) \oplus \Lambda^3(C^9)$	$\Lambda^3(C^9) \oplus \Lambda^3(C^9)$ is given by
0902.0431_F03042	196	■ 1.000	B	B	Proof. We consider a symmetric bilinear form B of $\mathfrak{e}_8^C$:
0902.0431_F03043	196	■ 1.000	$B_{\{8\}}\left(R_{\{1\}}, R_{\{2\}}\right)=k\ B\left(R_{\{1\}}, R_{\{2\}}\right)$	$B_8(R_1, R_2) = kB(R_1, R_2)$	simple, there exists $k \in C$ such that $B_8(R_1, R_2) = kB(R_1, R_2)$ for all $R_i \in \mathfrak{e}_8^C$. To
0902.0431_F03044	196	■ 1.000	$R_{\{i\}} \in \mathfrak{e}_{\{8\}}^{\{ \}^{\{C\}}}$	$R_i \in \mathfrak{e}_8^C$	simple, there exists $k \in C$ such that $B_8(R_1, R_2) = kB(R_1, R_2)$ for all $\mathbf{R}_i \in \mathfrak{e}_8^C$. To
0902.0431_F03045	196	■ 1.000	$R=R_{\{1\}}=R_{\{2\}}=\left(E_{\{11\}}-E_{\{22\}},\ 0,0\right) \in \mathfrak{e}_{\{8\}}^{\{ \}^{\{C\}}}$	$R = R_1 = R_2 = (E_{11} - E_{22}, 0, 0) \in \mathfrak{e}_8^C$	determined k , let $\mathbf{R} = R_1 = R_2 = (E_{11} - E_{22}, 0, 0) \in \mathfrak{e}_8^C$. Then we have
0902.0431_F03046	196	■ 1.000	$k=60$	$k = 60$	Therefore $k = 60$.
0902.0431_F03047	197	■ 1.000	$w_{\{3\}} \in E_{\{8\}}$	$w_3 \in E_8$	where $\omega = -\frac{1}{2} + \frac{\sqrt{3}}{2}i \in C$. Then, $w_3 \in E_8$ and $w_3^3 = 1$.
0902.0431_F03048	197	■ 1.000	$w_{\{3\}}^{\{ \}^{\{3\}}}=1$	$w_3^3 = 1$	where $\omega = -\frac{1}{2} + \frac{\sqrt{3}}{2}i \in C$. Then, $w_3 \in E_8$ and $\mathbf{w}_3^3 = \mathbf{1}$.
0902.0431_F03049	197	■ 1.000	$\left(E_{\{8\}}\right)^{w_{\{3\}}}$	$(E_8)^{w_3}$	Now, we study the following subgroup $(E_8)^{w_3}$ of E_8 :
0902.0431_F03050	197	■ 0.907	$\quad \quad \quad \left(E_{\{8\}}\right)^{w_{\{3\}}} \cong S\ U(9) / \boldsymbol{Z}_{\{3\}}, \quad \quad \quad \left(E_{\{8\}}\right)^{w_{\{3\}}} = \left\{E, \omega E, \omega^2 E, \omega^3 E\right\}$	$(E_8)^{w_3} \cong SU(9)/Z_3, \quad Z_3 = \{E, \omega E, \omega^2 E\}$	Theorem 5.11.7. $(E_8)^{w_3} \cong SU(9)/Z_3, \quad Z_3 = \{E, \omega E, \omega^2 E\}$.
0902.0431_F03051	197	■ 1.000	$\varphi: S\ U(9) \rightarrow \left(E_{\{8\}}\right)^{w_{\{3\}}}$	$\varphi: SU(9) \rightarrow (E_8)^{w_3}$	Proof. We define a mapping $\varphi: SU(9) \rightarrow (E_8)^{w_3}$ by
0902.0431_F03052	197	■ 0.903	$\varphi(A) \in \left(E_{\{8\}}\right)^{w_{\{3\}}}$	$\varphi(A) \in (E_8)^{w_3}$	φ is well-defined: $\varphi(A) \in (E_8)^{w_3}$. Indeed, for $A = \exp X, X \in \mathfrak{su}(9)$, we have
0902.0431_F03053	197	■ 0.903	$A=\exp X,\ X \in \mathfrak{su}(9)$	$A = \exp X, X \in \mathfrak{su}(9)$	φ is well-defined: $\varphi(A) \in (E_8)^{w_3}$. Indeed, for $\mathbf{A} = \exp X, X \in \mathfrak{su}(9)$, we have
0902.0431_F03054	197	■ 1.000	$\varphi(A) \in E_{\{8\}}$	$\varphi(A) \in E_8$	Hence $\varphi(A) \in E_8$. Clearly $w_3 \varphi(A) = \varphi(A) w_3$. Therefore $\varphi(A) \in (E_8)^{w_3}$. Obviously
0902.0431_F03055	197	■ 1.000	$w_{\{3\}} \varphi(A)=\varphi(A) w_{\{3\}}$	$w_3 \varphi(A) = \varphi(A) w_3$	Hence $\varphi(A) \in E_8$. Clearly $\mathbf{w}_3 \varphi(A) = \varphi(A) \mathbf{w}_3$. Therefore $\varphi(A) \in (E_8)^{w_3}$. Obviously
0902.0431_F03056	197	■ 0.996	$\left(\mathfrak{e}_{\{8\}}\right)^{w_{\{3\}}}$	$(\mathfrak{e}_8)^{w_3}$	φ is a homomorphism. We shall show that φ is onto. Since the Lie algebra $(\mathfrak{e}_8)^{w_3}$ of
0902.0431_F03057	197	■ 0.998	$\ker \varphi=\left\{E, \omega E, \omega^2 E\right\}=\boldsymbol{Z}_{\{3\}}$	$\ker \varphi = \{E, \omega E, \omega^2 E\} = Z_3$	difficult to see that $\ker \varphi = \{E, \omega E, \omega^2 E\} = Z_3$. Thus we have the isomorphism
0902.0431_F03058	197	■ 0.608	$S\ U(9) / \boldsymbol{Z}_{\{3\}} \cong \left(E_{\{8\}}\right)^{w_{\{3\}}}$	$SU(9)/Z_3 \cong (E_8)^{w_3}$	$SU(9)/Z_3 \cong (E_8)^{w_3}$.
0902.0431_F03059	197	■ 1.000	$48+50 \times 4=248$	$48 + 50 \times 4 = 248$	Theorem 5.12.1. In a $48+50 \times 4=248$ dimensional C -vector space
0902.0431_F03060	198	■ 1.000	$\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \Lambda^1\left(C^5\right)=C^5$	$\mathbf{x}, \mathbf{y}, \mathbf{z} \in \Lambda^1(C^5) = C^5$	Lemma 5.12.2. For $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \Lambda^1(C^5) = C^5$ and $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \Lambda^2(C^5)$, we have
0902.0431_F03061	198	■ 1.000	$\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{c} \in \Lambda^2\left(C^5\right)$	$\mathbf{a}, \mathbf{b}, \mathbf{c} \in \Lambda^2(C^5)$	Lemma 5.12.2. For $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \Lambda^1(C^5) = C^5$ and $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \Lambda^2(C^5)$, we have
0902.0431_F03062	198	■ 0.514	$\quad \quad \quad \boldsymbol{a} \wedge \boldsymbol{b} \wedge \boldsymbol{c}+\boldsymbol{b} \wedge \boldsymbol{c} \wedge \boldsymbol{a}+\boldsymbol{c} \wedge \boldsymbol{a} \wedge \boldsymbol{b}=0$	$\mathbf{a} \wedge \mathbf{b} \wedge \mathbf{c} + \mathbf{b} \wedge \mathbf{c} \wedge \mathbf{a} + \mathbf{c} \wedge \mathbf{a} \wedge \mathbf{b} = 0$	(1) $\mathbf{a} \wedge \mathbf{b} \wedge \mathbf{c} + \mathbf{b} \wedge \mathbf{c} \wedge \mathbf{a} + \mathbf{c} \wedge \mathbf{a} \wedge \mathbf{b} = \mathbf{0}$,
0902.0431_F03063	198	■ 0.928	$\quad \quad \quad \boldsymbol{a} \wedge \boldsymbol{b} \wedge \boldsymbol{c}+\boldsymbol{b} \wedge \boldsymbol{c} \wedge \boldsymbol{a}+\boldsymbol{c} \wedge \boldsymbol{a} \wedge \boldsymbol{b}=0$	$\mathbf{a} \wedge \mathbf{b} \wedge \mathbf{c} + \mathbf{b} \wedge \mathbf{c} \wedge \mathbf{a} + \mathbf{c} \wedge \mathbf{a} \wedge \mathbf{b} = 0$	(2) $\mathbf{a} \wedge \mathbf{b} \wedge \mathbf{c} + \mathbf{b} \wedge \mathbf{c} \wedge \mathbf{a} + \mathbf{c} \wedge \mathbf{a} \wedge \mathbf{b} = \mathbf{0}$,

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0902.0431_F03064	198	0.740	$*(\boldsymbol{x} \wedge \boldsymbol{y}) \wedge \boldsymbol{z} = (\boldsymbol{x}, \boldsymbol{z})\boldsymbol{y} - (\boldsymbol{y}, \boldsymbol{z})\boldsymbol{x}$	$*(\boldsymbol{x} \wedge \boldsymbol{y}) \wedge \boldsymbol{z} = (\boldsymbol{x}, \boldsymbol{z})\boldsymbol{y} - (\boldsymbol{y}, \boldsymbol{z})\boldsymbol{x}$	(3) $*(\boldsymbol{x} \wedge \boldsymbol{y}) \wedge \boldsymbol{z} = (\boldsymbol{x}, \boldsymbol{z})\boldsymbol{y} - (\boldsymbol{y}, \boldsymbol{z})\boldsymbol{x}$,
0902.0431_F03065	198	1.000	$\boldsymbol{x} \wedge *(\boldsymbol{a} \wedge \boldsymbol{y}) + *(\boldsymbol{y} \wedge (\boldsymbol{a} \wedge \boldsymbol{x})) - (\boldsymbol{x}, \boldsymbol{y})\boldsymbol{a} = 0$	$\boldsymbol{x} \wedge *(\boldsymbol{a} \wedge \boldsymbol{y}) + *(\boldsymbol{y} \wedge (\boldsymbol{a} \wedge \boldsymbol{x})) - (\boldsymbol{x}, \boldsymbol{y})\boldsymbol{a} = 0$	(4) $\boldsymbol{x} \wedge *(\boldsymbol{a} \wedge \boldsymbol{y}) + *(\boldsymbol{y} \wedge (\boldsymbol{a} \wedge \boldsymbol{x})) - (\boldsymbol{x}, \boldsymbol{y})\boldsymbol{a} = 0$,
0902.0431_F03066	198	0.658	$\boldsymbol{a} \wedge *(\boldsymbol{b} \wedge \boldsymbol{x}) - *(\boldsymbol{b} \wedge *(\boldsymbol{a} \wedge \boldsymbol{x})) - (\boldsymbol{a}, \boldsymbol{b})\boldsymbol{x} = 0$	$*(\boldsymbol{a} \wedge *(\boldsymbol{b} \wedge \boldsymbol{x})) - *(\boldsymbol{b} \wedge *(\boldsymbol{a} \wedge \boldsymbol{x})) - (\boldsymbol{a}, \boldsymbol{b})\boldsymbol{x} = 0$	(5) $*(\boldsymbol{a} \wedge *(\boldsymbol{b} \wedge \boldsymbol{x})) - *(\boldsymbol{b} \wedge *(\boldsymbol{a} \wedge \boldsymbol{x})) - (\boldsymbol{a}, \boldsymbol{b})\boldsymbol{x} = 0$,
0902.0431_F03067	198	0.974	$\boldsymbol{a} \times *(\boldsymbol{b} \wedge \boldsymbol{x}) + \boldsymbol{b} \times *(\boldsymbol{a} \wedge \boldsymbol{x}) - \boldsymbol{x} \times *(\boldsymbol{a} \wedge \boldsymbol{b}) = 0$	$\boldsymbol{a} \times *(\boldsymbol{b} \wedge \boldsymbol{x}) + \boldsymbol{b} \times *(\boldsymbol{a} \wedge \boldsymbol{x}) - \boldsymbol{x} \times *(\boldsymbol{a} \wedge \boldsymbol{b}) = 0$	(6) $\boldsymbol{a} \times *(\boldsymbol{b} \wedge \boldsymbol{x}) + \boldsymbol{b} \times *(\boldsymbol{a} \wedge \boldsymbol{x}) - \boldsymbol{x} \times *(\boldsymbol{a} \wedge \boldsymbol{b}) = 0$,
0902.0431_F03068	198	0.748	$*(\boldsymbol{a} \wedge \boldsymbol{x}) \times \boldsymbol{y} - *(\boldsymbol{a} \wedge \boldsymbol{y}) \times \boldsymbol{x} + \boldsymbol{a} \times (\boldsymbol{x} \wedge \boldsymbol{y}) = 0$	$*(\boldsymbol{a} \wedge \boldsymbol{x}) \times \boldsymbol{y} - *(\boldsymbol{a} \wedge \boldsymbol{y}) \times \boldsymbol{x} + \boldsymbol{a} \times (\boldsymbol{x} \wedge \boldsymbol{y}) = 0$	(7) $*(\boldsymbol{a} \wedge \boldsymbol{x}) \times \boldsymbol{y} - *(\boldsymbol{a} \wedge \boldsymbol{y}) \times \boldsymbol{x} + \boldsymbol{a} \times (\boldsymbol{x} \wedge \boldsymbol{y}) = 0$,
0902.0431_F03069	198	0.999	$(\boldsymbol{a} \wedge \boldsymbol{b})\boldsymbol{c} = *(\boldsymbol{a} \wedge \boldsymbol{c}) \wedge \boldsymbol{b} - \frac{1}{5}(\boldsymbol{a}, \boldsymbol{b})\boldsymbol{c} - (\boldsymbol{b}, \boldsymbol{c})\boldsymbol{a}$	$(\boldsymbol{a} \wedge \boldsymbol{b})\boldsymbol{c} = *(\boldsymbol{a} \wedge \boldsymbol{c}) \wedge \boldsymbol{b} - \frac{1}{5}(\boldsymbol{a}, \boldsymbol{b})\boldsymbol{c} - (\boldsymbol{b}, \boldsymbol{c})\boldsymbol{a}$	(8) $(\boldsymbol{a} \wedge \boldsymbol{b})\boldsymbol{c} = *(\boldsymbol{a} \wedge \boldsymbol{c}) \wedge \boldsymbol{b} - \frac{1}{5}(\boldsymbol{a}, \boldsymbol{b})\boldsymbol{c} - (\boldsymbol{b}, \boldsymbol{c})\boldsymbol{a}$,
0902.0431_F03070	198	0.643	$(\boldsymbol{x} \times \boldsymbol{y})\boldsymbol{a} = - *(\boldsymbol{y} \wedge *(\boldsymbol{x} \wedge \boldsymbol{a})) + \frac{3}{5}(\boldsymbol{x}, \boldsymbol{y})\boldsymbol{a}$	$(\boldsymbol{x} \times \boldsymbol{y})\boldsymbol{a} = - *(\boldsymbol{y} \wedge *(\boldsymbol{x} \wedge \boldsymbol{a})) + \frac{3}{5}(\boldsymbol{x}, \boldsymbol{y})\boldsymbol{a}$	(9) $(\boldsymbol{x} \times \boldsymbol{y})\boldsymbol{a} = - *(\boldsymbol{y} \wedge *(\boldsymbol{x} \wedge \boldsymbol{a})) + \frac{3}{5}(\boldsymbol{x}, \boldsymbol{y})\boldsymbol{a}$.
0902.0431_F03071	198	0.999	$\boldsymbol{a} = \boldsymbol{a}_1 \wedge \boldsymbol{a}_2, \boldsymbol{b} = \boldsymbol{a}_3 \wedge \boldsymbol{a}_4, \boldsymbol{c} = \boldsymbol{a}_5 \wedge \boldsymbol{a}_6$	$\boldsymbol{a} = \boldsymbol{a}_1 \wedge \boldsymbol{a}_2, \boldsymbol{b} = \boldsymbol{a}_3 \wedge \boldsymbol{a}_4, \boldsymbol{c} = \boldsymbol{a}_5 \wedge \boldsymbol{a}_6$	Proof. (1) Let $\boldsymbol{a} = \boldsymbol{a}_1 \wedge \boldsymbol{a}_2, \boldsymbol{b} = \boldsymbol{a}_3 \wedge \boldsymbol{a}_4, \boldsymbol{c} = \boldsymbol{a}_5 \wedge \boldsymbol{a}_6$ and $\boldsymbol{a}_i = \sum_{j=1}^5 a_{ij} \boldsymbol{e}_j$. Since
0902.0431_F03072	198	0.999	$\boldsymbol{a}_i = \sum_{j=1}^5 a_{ij} \boldsymbol{e}_j$	$\boldsymbol{a}_i = \sum_{j=1}^5 a_{ij} \boldsymbol{e}_j$	Proof. (1) Let $\boldsymbol{a} = \boldsymbol{a}_1 \wedge \boldsymbol{a}_2, \boldsymbol{b} = \boldsymbol{a}_3 \wedge \boldsymbol{a}_4, \boldsymbol{c} = \boldsymbol{a}_5 \wedge \boldsymbol{a}_6$ and $\boldsymbol{a}_i = \sum_{j=1}^5 a_{ij} \boldsymbol{e}_j$. Since
0902.0431_F03073	199	1.000	$\boldsymbol{v} \in \Lambda^1(C^5) = C^5$	$\boldsymbol{v} \in \Lambda^1(C^5) = C^5$	(3) For any $\boldsymbol{v} \in \Lambda^1(C^5) = C^5$, we have
0902.0431_F03074	199	1.000	$\boldsymbol{a} = \boldsymbol{a}_1 \wedge \boldsymbol{a}_2$	$\boldsymbol{a} = \boldsymbol{a}_1 \wedge \boldsymbol{a}_2$	(4),(5) Let $\boldsymbol{a} = \boldsymbol{a}_1 \wedge \boldsymbol{a}_2$. Since
0902.0431_F03075	199	0.999	$\boldsymbol{v}, \boldsymbol{w} \in \Lambda^1(C^5) = C^5$	$\boldsymbol{v}, \boldsymbol{w} \in \Lambda^1(C^5) = C^5$	For $\boldsymbol{v}, \boldsymbol{w} \in \Lambda^1(C^5) = C^5$, we have
0902.0431_F03076	200	1.000	$\boldsymbol{c} = \boldsymbol{c}_1 \wedge \boldsymbol{c}_2$	$\boldsymbol{c} = \boldsymbol{c}_1 \wedge \boldsymbol{c}_2$	(8) Let $\boldsymbol{a} = \boldsymbol{a}_1 \wedge \boldsymbol{a}_2$ and $\boldsymbol{c} = \boldsymbol{c}_1 \wedge \boldsymbol{c}_2$. Since
0902.0431_F03077	201	1.000	$\boldsymbol{d} \in \Lambda^2(C^5)$	$\boldsymbol{d} \in \Lambda^2(C^5)$	On the other hand, for $\boldsymbol{d} \in \Lambda^2(C^5)$, we have
0902.0431_F03078	201	0.591	$\mathfrak{e}_8^C = \mathfrak{g}_0 \oplus \mathfrak{g}_1 \oplus \mathfrak{g}_2 \oplus \mathfrak{g}_{-2} \oplus \mathfrak{g}_{-1}$	$\mathfrak{e}_8^C = \mathfrak{g}_0 \oplus \mathfrak{g}_1 \oplus \mathfrak{g}_2 \oplus \mathfrak{g}_{-2} \oplus \mathfrak{g}_{-1}$	Theorem 5.12.3. The Lie algebra $\mathfrak{e}_8^C = \mathfrak{g}_0 \oplus \mathfrak{g}_1 \oplus \mathfrak{g}_2 \oplus \mathfrak{g}_{-2} \oplus \mathfrak{g}_{-1}$ is a C -simple
0902.0431_F03079	201	1.000	$\mathfrak{g} = \mathfrak{e}_8^C = \mathfrak{g}_{01} \oplus \mathfrak{g}_{12} \oplus \mathfrak{q}$	$\mathfrak{g} = \mathfrak{e}_8^C = \mathfrak{g}_{01} \oplus \mathfrak{g}_{12} \oplus \mathfrak{q}$	$\mathfrak{g} = \mathfrak{e}_8^C = \mathfrak{g}_{01} \oplus \mathfrak{g}_{12} \oplus \mathfrak{q}$, where
0902.0431_F03080	201	0.999	$\mathfrak{g}_{01} \cap \mathfrak{a} = \{0\}, \mathfrak{g}_{02} \cap \mathfrak{a} = \{0\}$	$\mathfrak{g}_{01} \cap \mathfrak{a} = \{0\}, \mathfrak{g}_{02} \cap \mathfrak{a} = \{0\}$	(1) Case $\mathfrak{g}_{01} \cap \mathfrak{a} = \{0\}, \mathfrak{g}_{02} \cap \mathfrak{a} = \{0\}$ and $\mathfrak{q} \cap \mathfrak{a} = \{0\}$. Let $p_i : \mathfrak{g} \rightarrow \mathfrak{g}_{0i}$
0902.0431_F03081	201	0.999	$p_i : \mathfrak{g} \rightarrow \mathfrak{g}_{0i}$	$p_i : \mathfrak{g} \rightarrow \mathfrak{g}_{0i}$	(1) Case $\mathfrak{g}_{01} \cap \mathfrak{a} = \{0\}, \mathfrak{g}_{02} \cap \mathfrak{a} = \{0\}$ and $\mathfrak{q} \cap \mathfrak{a} = \{0\}$. Let $p_i : \mathfrak{g} \rightarrow \mathfrak{g}_{0i}$
0902.0431_F03082	201	1.000	$(i=1,2)$	$(i=1,2)$	$i=1,2$ denote the projection. If $p_1(\mathfrak{a}) = \{0\}$ and $p_2(\mathfrak{a}) = \{0\}$, then $\mathfrak{a}$ is contained

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0902.0431_F03083	201	■ 1.000	$p_{\{1\}}(\frac{a}{\mathfrak{a}})=\{0\}$	$p_1(\mathfrak{a}) = \{0\}$	$(i = 1, 2)$ denote the projection. If $p_1(\mathfrak{a}) = \{0\}$ and $p_2(\mathfrak{a}) = \{0\}$, then $\mathfrak{a}$ is contained
0902.0431_F03084	201	■ 1.000	$p_{\{2\}}(\frac{a}{\mathfrak{a}})=\{0\}$	$p_2(\mathfrak{a}) = \{0\}$	$(i = 1, 2)$ denote the projection. If $p_1(\mathfrak{a}) = \{0\}$ and $p_2(\mathfrak{a}) = \{0\}$, then $\mathfrak{a}$ is contained
0902.0431_F03085	201	■ 1.000	$p_{\{1\}}(\frac{a}{\mathfrak{a}})=\frac{\mathfrak{g}_{\{01\}}}{\mathfrak{g}_{01}}$	$p_1(\mathfrak{a}) = \mathfrak{g}_{01}$	assume that $p_1(\mathfrak{a}) = \mathfrak{g}_{01}$, because $\mathfrak{g}_{01}$ is a simple Lie algebra. For $C = \sum_{i=1}^4 H_i \in$
0902.0431_F03086	201	■ 1.000	$\frac{\mathfrak{g}_{\{01\}}}{\mathfrak{g}_{01}}$	$\mathfrak{g}_{01}$	assume that $p_1(\mathfrak{a}) = \mathfrak{g}_{01}$, because $\mathfrak{g}_{01}$ is a simple Lie algebra. For $C = \sum_{i=1}^4 H_i \in$
0902.0431_F03087	201	■ 1.000	$C=\sum_{i=1}^4 H_{\{i\}} \in$	$C = \sum_{i=1}^4 H_i \in$	assume that $p_1(\mathfrak{a}) = \mathfrak{g}_{01}$, because $\mathfrak{g}_{01}$ is a simple Lie algebra. For $C = \sum_{i=1}^4 H_i \in$
0902.0431_F03088	201	■ 0.531	$H_{\{i\}}=E_{\{i\}}-E_{55}$	$H_i = E_{ii} - E_{55}$	$\mathfrak{sl}(3, C)$ where $H_i = E_{ii} - E_{55}$, there exists $(D, g_1, g_2, g_{-2}, g_{-1}) \in \mathfrak{g}_{01} \oplus \mathfrak{q}$ such that
0902.0431_F03089	201	■ 0.531	$\left(D, g_{\{1\}}, g_{\{2\}}, g_{\{-2\}}, g_{\{-1\}}\right) \in \frac{\mathfrak{g}_{\{01\}}}{\mathfrak{g}_{01}} \oplus \frac{\mathfrak{q}}{\mathfrak{q}}$	$(D, g_1, g_2, g_{-2}, g_{-1}) \in \mathfrak{g}_{01} \oplus \mathfrak{q}$	$\mathfrak{sl}(3, C)$ where $H_i = E_{ii} - E_{55}$, there exists $(D, g_1, g_2, g_{-2}, g_{-1}) \in \mathfrak{g}_{01} \oplus \mathfrak{q}$ such that
0902.0431_F03090	201	■ 1.000	$\left(C, D, g_{\{1\}}, g_{\{2\}}, g_{\{-2\}}, g_{\{-1\}}\right) \in \frac{\mathfrak{a}}{\mathfrak{a}}$	$(C, D, g_1, g_2, g_{-2}, g_{-1}) \in \mathfrak{a}$	$(C, D, g_1, g_2, g_{-2}, g_{-1}) \in \mathfrak{q}$. Since
0902.0431_F03091	201	■ 0.869	$\left[C, g_{\{i\}}\right]=0(i=1,2,-2,-1)$	$[C, g_i] = 0(i = 1, 2, -2, -1)$	we have $[C, g_i] = 0$ ($i = 1, 2, -2, -1$). Since any eigenvalue of $\text{ad}X$ is not 0, we have
0902.0431_F03092	201	■ 0.869	$\text{operatorname{ad}} X$	$\text{ad} X$	we have $[C, g_i] = 0$ ($i = 1, 2, -2, -1$). Since any eigenvalue of $\text{ad}X$ is not 0, we have
0902.0431_F03093	201	■ 1.000	$g_{\{i\}}=0$	$g_i = 0$	$g_i = 0$. Then we have $(C, D) \in \mathfrak{g}_0 \cap \mathfrak{a}$. Since
0902.0431_F03094	201	■ 1.000	$(C, D) \in \frac{\mathfrak{g}_{\{0\}}}{\mathfrak{g}_0} \cap \frac{\mathfrak{a}}{\mathfrak{a}}$	$(C, D) \in \mathfrak{g}_0 \cap \mathfrak{a}$	$g_i = 0$. Then we have $(C, D) \in \mathfrak{g}_0 \cap \mathfrak{q}$. Since
0902.0431_F03095	202	■ 1.000	$\frac{\mathfrak{g}_{\{01\}}}{\mathfrak{g}_{01}} \cap \frac{\mathfrak{a}}{\mathfrak{a}} \neq \{0\}$	$\mathfrak{g}_{01} \cap \mathfrak{a} \neq \{0\}$	we have $\mathfrak{g}_{01} \cap \mathfrak{a} \neq \{0\}$. This is a contradiction.
0902.0431_F03096	202	■ 1.000	$\frac{\mathfrak{g}_{\{02\}}}{\mathfrak{g}_{02}} \cap \frac{\mathfrak{a}}{\mathfrak{a}} \neq \{0\}$	$\mathfrak{g}_{02} \cap \mathfrak{a} \neq \{0\}$	(2) Case $\mathfrak{g}_{01} \cap \mathfrak{a} \neq \{0\}$ or $\mathfrak{g}_{02} \cap \mathfrak{a} \neq \{0\}$. We may assume that $\mathfrak{g}_{01} \cap \mathfrak{a} \neq \{0\}$.
0902.0431_F03097	202	■ 1.000	$\frac{\mathfrak{g}_{\{01\}}}{\mathfrak{g}_{01}} \subset \frac{\mathfrak{a}}{\mathfrak{a}}$	$\mathfrak{g}_{01} \subset \mathfrak{a}$	Since $\mathfrak{g}_{01}$ is simple, we have $\mathfrak{g}_{01} \subset \mathfrak{a}$. Since $[\mathfrak{g}_{01}, \mathfrak{g}_i] = \mathfrak{g}_i$ ($i = 1, 2, -2, -1$), we have
0902.0431_F03098	202	■ 1.000	$\left[\frac{\mathfrak{g}_{\{01\}}}{\mathfrak{g}_{01}}, \frac{\mathfrak{g}_{\{i\}}}{\mathfrak{g}_i}\right]=\frac{\mathfrak{g}_{\{i\}}}{\mathfrak{g}_i}(i=1,2,-2,-1)$	$[\mathfrak{g}_{01}, \mathfrak{g}_i] = \mathfrak{g}_i(i = 1, 2, -2, -1)$	Since $\mathfrak{g}_{01}$ is simple, we have $\mathfrak{g}_{01} \subset \mathfrak{a}$. Since $[\mathfrak{g}_{01}, \mathfrak{g}_i] = \mathfrak{g}_i$ ($i = 1, 2, -2, -1$), we have
0902.0431_F03099	202	■ 1.000	$\frac{\mathfrak{q}}{\mathfrak{q}} \subset \frac{\mathfrak{a}}{\mathfrak{a}}$	$\mathfrak{q} \subset \mathfrak{a}$	$\mathfrak{q} \subset \mathfrak{a}$. Since
0902.0431_F03100	202	■ 1.000	$\frac{\mathfrak{g}_{\{02\}}}{\mathfrak{g}_{02}} \subset \frac{\mathfrak{a}}{\mathfrak{a}}$	$\mathfrak{g}_{02} \subset \mathfrak{a}$	we have $\mathfrak{g}_{02} \cap \mathfrak{a} \neq \{0\}$. It follows that $\mathfrak{g}_{02} \subset \mathfrak{a}$. Hence we have $\mathfrak{a} = \mathfrak{g}$.
0902.0431_F03101	202	■ 1.000	$R=\left(g_{\{1\}}, g_{\{2\}}, g_{\{-2\}}, g_{\{-1\}}\right)\left(g_{\{i\}} \in \frac{\mathfrak{g}_{\{i\}}}{\mathfrak{g}_i}\right)$	$R = (g_1, g_2, g_{-2}, g_{-1}) (g_i \in \mathfrak{g}_i)$	(3) Case $\mathfrak{q} \cap \mathfrak{a} \neq \{0\}$. Let $R = (g_1, g_2, g_{-2}, g_{-1}) (g_i \in \mathfrak{g}_i)$ be a non-zero element
0902.0431_F03102	202	■ 1.000	$g_{\{1\}} \neq 0$	$g_1 \neq 0$	of $\mathfrak{q} \cap \mathfrak{a}$. In the case $g_1 \neq 0$, we put $g_1 = \sum_{i,j < k} g_{ijk} e_i \otimes (e_j \wedge e_k)$. Without loss of
0902.0431_F03103	202	■ 1.000	$g_{\{1\}}=\sum_{i,j < k} g_{ijk} \boldsymbol{e}_{\{i\}} \otimes \left(\boldsymbol{e}_{\{j\}} \wedge \boldsymbol{e}_{\{k\}}\right)$	$g_1 = \sum_{i,j < k} g_{ijk} e_i \otimes (e_j \wedge e_k)$	of $\mathfrak{q} \cap \mathfrak{a}$. In the case $g_1 \neq 0$, we put $g_1 = \sum_{i,j < k} g_{ijk} e_i \otimes (e_j \wedge e_k)$. Without loss of
0902.0431_F03104	202	■ 1.000	$g_{\{112\}} \neq 0$	$g_{112} \neq 0$	generality, we may assume that $g_{112} \neq 0$. Putting $S_{ijkl} = (E_{ii} - E_{jj}, E_{kk} - E_{ll}) \in \mathfrak{g}_0$
0902.0431_F03105	202	■ 1.000	$S_{\{i\ j\ k\ l\}}=\left(E_{\{i\ i\}}-E_{\{j\ j\}}, E_{\{k\ k\}}-E_{\{l\ l\}}\right) \in \frac{\mathfrak{g}_{\{0\}}}{\mathfrak{g}_0}$	$S_{ijkl} = (E_{ii} - E_{jj}, E_{kk} - E_{ll}) \in \mathfrak{g}_0$	generality, we may assume that $g_{112} \neq 0$. Putting $S_{ijkl} = (E_{ii} - E_{jj}, E_{kk} - E_{ll}) \in \mathfrak{g}_0$
0902.0431_F03106	202	■ 1.000	$T=\boldsymbol{e}_{\{2\}} \otimes \boldsymbol{e}_{\{1\}} \wedge \boldsymbol{e}_{\{-1\}}$	$T = e_2 \otimes e_1 \wedge e_2 \in \mathfrak{g}_{-1}$	and $T = e_2 \otimes e_1 \wedge e_2 \in \mathfrak{g}_{-1}$, we have
0902.0431_F03107	202	■ 1.000	$g_{\{i\}} \neq 0(i=2,-2,-1)$	$g_i \neq 0(i = 2, -2, -1)$	Then we can reduce this case to the case (2). In the case $g_i \neq 0$ ($i = 2, -2, -1$), we
0902.0431_F03108	202	■ 0.985	$\frac{\mathfrak{e}_{\{8\}}}{\mathfrak{e}_8} \wedge^C \frac{\mathfrak{s}_{\{l\}}}{\mathfrak{s}_l}(5, C) \oplus$	$\mathfrak{e}_8^C = \mathfrak{sl}(5, C) \oplus$	Proposition 5.12.4. The Killing form B_8 of the Lie algebra $\mathfrak{e}_8^C = \mathfrak{sl}(5, C) \oplus$
0902.0431_F03109	202	■ 0.999	$\frac{\mathfrak{s}_{\{l\}}}{\mathfrak{s}_l} \frac{\mathfrak{l}_{\{l\}}}{\mathfrak{l}_l}(5, C) \oplus \frac{\mathfrak{g}_{\{1\}}}{\mathfrak{g}_1} \oplus \frac{\mathfrak{g}_{\{2\}}}{\mathfrak{g}_2} \oplus \frac{\mathfrak{g}_{\{-2\}}}{\mathfrak{g}_{-2}} \oplus \frac{\mathfrak{g}_{\{-1\}}}{\mathfrak{g}_{-1}}$	$\mathfrak{sl}(5, C) \oplus \mathfrak{g}_1 \oplus \mathfrak{g}_2 \oplus \mathfrak{g}_{-2} \oplus \mathfrak{g}_{-1}$	$\mathfrak{sl}(5, C) \oplus \mathfrak{g}_1 \oplus \mathfrak{g}_2 \oplus \mathfrak{g}_{-2} \oplus \mathfrak{g}_{-1}$ is given by
0902.0431_F03110	202	■ 0.657	$R_{\{i\}}=\left(C_{\{i\}}, D_{\{i\}}, \boldsymbol{x}_{\{i\}} \otimes \boldsymbol{a}_{\{i\}}, \boldsymbol{b}_{\{i\}} \otimes \boldsymbol{y}_{\{i\}}, \boldsymbol{c}_{\{i\}} \otimes \boldsymbol{z}_{\{i\}}, \boldsymbol{w}_{\{i\}} \otimes \boldsymbol{d}_{\{i\}}\right) \in \frac{\mathfrak{e}_{\{8\}}}{\mathfrak{e}_8} \wedge^C$	$R_i = (C_i, D_i, \boldsymbol{x}_i \otimes \boldsymbol{a}_i, \boldsymbol{b}_i \otimes \boldsymbol{y}_i, \boldsymbol{c}_i \otimes \boldsymbol{z}_i, \boldsymbol{w}_i \otimes \boldsymbol{d}_i) \in \mathfrak{e}_8^C$	where $R_i = (C_i, D_i, \boldsymbol{x}_i \otimes \boldsymbol{a}_i, \boldsymbol{b}_i \otimes \boldsymbol{y}_i, \boldsymbol{c}_i \otimes \boldsymbol{z}_i, \boldsymbol{w}_i \otimes \boldsymbol{d}_i) \in \mathfrak{e}_8^C$.
0902.0431_F03111	202	■ 1.000	$R=R_{\{1\}}=R_{\{2\}}=\left(E_{\{11\}}-E_{\{22\}}, 0,0,0,0,0\right) \in \frac{\mathfrak{e}_{\{8\}}}{\mathfrak{e}_8} \wedge^C$	$R = R_1 = R_2 = (E_{11} - E_{22}, 0, 0, 0, 0, 0) \in \mathfrak{e}_8^C$	determined k , let $R = R_1 = R_2 = (E_{11} - E_{22}, 0, 0, 0, 0, 0) \in \mathfrak{e}_8^C$. Then we have
0902.0431_F03112	203	■ 1.000	$\zeta=\exp (2 \pi i / 5) \in C$	$\zeta = \exp(2\pi i/5) \in C$	Let $\zeta = \exp(2\pi i/5) \in C$ and we define a C -linear transformation z_5 of $\mathfrak{e}_8^C$ by
0902.0431_F03113	203	■ 1.000	$z_{\{5\}} \in E_{\{8\}}$	$z_5 \in E_8$	Then $z_5 \in E_8$ and $z_5^5 = 1$.
0902.0431_F03114	203	■ 1.000	$z_{\{5\}}^5=1$	$z_5^5 = 1$	Then $z_5 \in E_8$ and $z_5^5 = 1$.
0902.0431_F03115	203	■ 1.000	$\left(E_{\{8\}}\right)^{z_{\{5\}}}$	$(E_8)^{z_5}$	Now, we study the following subgroup $(E_8)^{z_5}$ of E_8 :
0902.0431_F03116	203	■ 0.998	$\left(E_{\{8\}}\right)^{z_{\{5\}}} \cong(S U(5) \times S U(5)) / \boldsymbol{Z}_{\{5\}}, \boldsymbol{Z}_{\{5\}}=\left\{(E, E),\left(\zeta E, \zeta^2 E\right),\right. \\ \left.\left(\zeta^2 E, E\right)\right\}$	$(E_8)^{z_5} \cong (SU(5) \times SU(5))/\boldsymbol{Z}_5, \boldsymbol{Z}_5 = \{(E, E), (\zeta E, \zeta^2 E),$	Theorem 5.12.5. $(E_8)^{z_5} \cong (SU(5) \times SU(5))/\boldsymbol{Z}_5, \boldsymbol{Z}_5 = \{(E, E), (\zeta E, \zeta^2 E),$

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0902.0431_F03117	203	0.520	$\backslash\mathrm{left}.\backslash\mathrm{left}(\backslash\mathrm{zeta}^{\wedge}\{2\} E, \backslash\mathrm{zeta}^{\wedge}\{4\} E\backslash\mathrm{right}),\backslash\mathrm{left}(\backslash\mathrm{zeta}^{\wedge}\{3\} E, \backslash\mathrm{zeta} E\backslash\mathrm{right}),\backslash\mathrm{left}(\backslash\mathrm{zeta}^{\wedge}\{4\} E, \backslash\mathrm{zeta}^{\wedge}\{3\} E\backslash\mathrm{right})\backslash\mathrm{right}\backslash, \backslash\mathrm{zeta}=\backslash\mathrm{exp}\left(2\backslash\mathrm{pi} i / 5\right)$	$(\zeta^2 E, \zeta^4 E), (\zeta^3 E, \zeta E), (\zeta^4 E, \zeta^3 E)\}, \zeta = \exp(2\pi i/5)$	$(\zeta^2 E, \zeta^4 E), (\zeta^3 E, \zeta E), (\zeta^4 E, \zeta^3 E)\}, \zeta = \exp(2\pi i/5)$.
0902.0431_F03118	203	0.936	$\backslash\mathrm{varphi}_{\{1\}}, \backslash\mathrm{varphi}_{\{2\}}: S U(5) \backslash\mathrm{rightarrow} E_{\{8\}}$	$\varphi_1, \varphi_2: SU(5) \rightarrow E_8$	Proof. @We define mappings $\varphi_1, \varphi_2: SU(5) \rightarrow E_8$ respectively by
0902.0431_F03119	203	1.000	$\backslash\mathrm{varphi}_{\{1\}}$	φ_1	φ_1 and φ_2 are well-defined: $\varphi_1(A), \varphi_2(B) \in E_8$. Indeed, for $Z \in \mathfrak{su}(5)$, we have
0902.0431_F03120	203	1.000	$\backslash\mathrm{varphi}_{\{2\}}$	φ_2	φ_1 and φ_2 are well-defined: $\varphi_1(A), \varphi_2(B) \in E_8$. Indeed, for $Z \in \mathfrak{su}(5)$, we have
0902.0431_F03121	203	1.000	$\backslash\mathrm{varphi}_{\{1\}}(A), \backslash\mathrm{varphi}_{\{2\}}(B) \backslash\mathrm{in} E_{\{8\}}$	$\varphi_1(A), \varphi_2(B) \in E_8$	φ_1 and φ_2 are well-defined: $\varphi_1(A), \varphi_2(B) \in E_8$. Indeed, for $Z \in \mathfrak{su}(5)$, we have
0902.0431_F03122	203	1.000	$Z \backslash\mathrm{in} \backslash\mathrm{mathfrak{s} u}(5)$	$Z \in \mathfrak{su}(5)$	φ_1 and φ_2 are well-defined: $\varphi_1(A), \varphi_2(B) \in E_8$. Indeed, for $Z \in \mathfrak{su}(5)$, we have
0902.0431_F03123	203	1.000	$(Z, 0) \backslash\mathrm{in} \backslash\mathrm{mathfrak{g}}_{\{0\}}$	$(Z, 0) \in \mathfrak{g}_0$	$(Z, 0) \in \mathfrak{g}_0$ and
0902.0431_F03124	203	1.000	$\backslash\mathrm{varphi}_{\{1\}}(A) \backslash\mathrm{in} \backslash\mathrm{operatorname{Aut}}\backslash\mathrm{left}(\backslash\mathrm{mathfrak{e}}_{\{8\}}\{ \}^{\wedge}\{C\}\backslash\mathrm{right})=E_{\{8\}}\{ \}^{\wedge}\{C\}$	$\varphi_1(A) \in \mathrm{Aut}(\mathfrak{e}_8^C) = E_8^C$	Hence $\varphi_1(A) \in \mathrm{Aut}(\mathfrak{e}_8^C) = E_8^C$. Using Lemma 5.11.1, we have
0902.0431_F03125	204	1.000	$\backslash\mathrm{varphi}_{\{2\}}(B) \backslash\mathrm{in} E_{\{8\}}$	$\varphi_2(B) \in E_8$	Therefore $\varphi_1(A) \in E_8$. Similarly $\varphi_2(B) \in E_8$.
0902.0431_F03126	204	0.981	$\backslash\mathrm{varphi}: S U(5) \backslash\mathrm{times} S U(5) \backslash\mathrm{rightarrow} E_{\{8\}}$	$\varphi: SU(5) \times SU(5) \rightarrow E_8$	Now, we define a mapping $\varphi: SU(5) \times SU(5) \rightarrow E_8$ by
0902.0431_F03127	204	1.000	$\backslash\mathrm{varphi}_{\{2\}}(B)$	$\varphi_2(B)$	Since $\varphi_1(A)$ and $\varphi_2(B)$ commute, φ is a homomorphism. We shall show that φ is
0902.0431_F03128	204	1.000	$\backslash\mathrm{left}(\backslash\mathrm{mathfrak{e}}_{\{8\}}\backslash\mathrm{right})^{\wedge}\{z_{\{5\}}\}=\backslash\mathrm{mathfrak{s} u}(5) \backslash\mathrm{oplus} \backslash\mathrm{mathfrak{s} u}(5)$	$(\mathfrak{e}_8)^{z_5} = \mathfrak{su}(5) \oplus \mathfrak{su}(5)$	onto. Since $(\mathfrak{e}_8)^{z_5} = \mathfrak{su}(5) \oplus \mathfrak{su}(5)$, the differential φ_* is onto. It is not difficult to see
0902.0431_F03129	204	1.000	$(S U(5) \backslash\mathrm{times}$	$(SU(5) \times$	Further, since $(E_8)^{z_5}$ is connected, φ is onto. Thus we have the isomorphism $(SU(5) \times$
0902.0431_F03130	204	0.546	$S U(5)) / \backslash\mathrm{boldsymbol{Z}}_{\{5\}} \backslash\mathrm{cong}\backslash\mathrm{left}(E_{\{8\}} \backslash\mathrm{right})^{\wedge}\{z_{\{5\}}\}$	$SU(5))/\mathbf{Z}_5 \cong (E_8)^{z_5}$	$SU(5))/\mathbf{Z}_5 \cong (E_8)^{z_5}$.
0902.0431_F03131	204	0.971	$R_{\{1\}}, R_{\{2\}} \backslash\mathrm{in} \backslash\mathrm{mathfrak{e}}_{\{8(8)\}}$	$R_1, R_2 \in \mathfrak{e}_{8(8)}$	For $R_1, R_2 \in \mathfrak{e}_{8(8)}$ or $\mathfrak{e}_{8(-24)}$, we define a Lie bracket $[R_1, R_2]$ as similar to $\mathfrak{e}_8^C$ of
0902.0431_F03132	204	0.971	$\backslash\mathrm{mathfrak{e}}_{\{8(-24)\}}$	$\mathfrak{e}_{8(-24)}$	For $R_1, R_2 \in \mathfrak{e}_{8(8)}$ or $\mathfrak{e}_{8(-24)}$, we define a Lie bracket $[R_1, R_2]$ as similar to $\mathfrak{e}_8^C$ of
0902.0431_F03133	205	0.574	$E_{\{6\}} /(U(1) \backslash\mathrm{operatorname{Spin}}(10)), E_{\{7\}} /\backslash\mathrm{left}(U(1) E_{\{6\}}\backslash\mathrm{right})$	$E_6/(U(1) \mathrm{Spin}(10)), E_7/(U(1) E_6)$	[1] K. Abe and I. Yokota, Realization of spaces $E_6/(U(1) \mathrm{Spin}(10)), E_7/(U(1) E_6)$,
0902.0431_F03134	205	1.000	$E_{\{8\}} /\backslash\mathrm{left}(U(1) E_{\{7\}}\backslash\mathrm{right})$	$E_8/(U(1) E_7)$	$E_8/(U(1) E_7)$ and their volumes, Tokyo J. Math. 20(1997), 73-86.
0902.0431_F03135	205	1.000	$E_{\{7(-133)\}}$	$E_{7(-133)}$	type $E_{7(-25)}$ and $E_{7(-133)}$, J. Fac. Sci. Shinshu Univ. 15(1980), 47-52.
0902.0431_F03136	205	1.000	$E_{\{8(-248)\}}$	$E_{8(-248)}$	[16] T. Imai and I. Yokota, Simply connected compact simple Lie group $E_{8(-248)}$ of
0902.0431_F03137	206	0.946	$\backslash\mathrm{mathfrak{g}}_{\{0\}}$	$\mathfrak{g}_0$	[17] S. Kaneyuki, On the subalgebra $\mathfrak{g}_0$ and $\mathfrak{g}_{ev}$ of semisimple Lie algebra, J. Math.
0902.0431_F03138	206	0.946	$\backslash\mathrm{mathfrak{g}}_{\{e v\}}$	$\mathfrak{g}_{ev}$	[17] S. Kaneyuki, On the subalgebra $\mathfrak{g}_0$ and $\mathfrak{g}_{ev}$ of semisimple Lie algebra, J. Math.
0902.0431_F03139	206	1.000	$F_{\{4\}}, E_{\{6\}}$	F_4, E_6	ceptional Lie groups F_4, E_6 and E_7 , Tsukuba J. Math. 22(1998), 705-721.
0902.0431_F03140	206	1.000	$\backslash\mathrm{sigma}^{\wedge}\{\backslash\mathrm{prime}\}$	σ'	[25] T. Miyashita, Decomposition of spinor groups by the involution σ' in exceptional
0902.0431_F03141	206	1.000	$\backslash\mathrm{sigma}, \backslash\mathrm{gamma}$	σ, γ	[26] T. Miyashita, Fixed points subgroups by two involutive automorphisms σ, γ
0902.0431_F03142	206	1.000	$\backslash\mathrm{mathfrak{g}}_{\{e v\}}, \backslash\mathrm{mathfrak{g}}_{\{0\}}$	$\mathfrak{g}_{ev}, \mathfrak{g}_0$	and group realizations of $\mathfrak{g}_{ev}, \mathfrak{g}_0$, Part III, $G = E_8$, Japanese J. Math. 26-1(2000),
0902.0431_F03143	206	1.000	$G=E_{\{8\}}$	$G = E_8$	and group realizations of $\mathfrak{g}_{ev}, \mathfrak{g}_0$, Part III, $G = E_8$, Japanese J. Math. 26-1(2000),
0902.0431_F03144	206	1.000	$G^{\wedge}\{\backslash\mathrm{sigma}, \backslash\mathrm{sigma}^{\wedge}\{\backslash\mathrm{prime}\}\}$	$G^{\sigma, \sigma'}$	[28] T. Miyashita and I. Yokota, Fixed points subgroups $G^{\sigma, \sigma'}$ by two involutive
0902.0431_F03145	206	1.000	$\backslash\mathrm{sigma}, \backslash\mathrm{sigma}^{\wedge}\{\backslash\mathrm{prime}\}$	σ, σ'	automorphisms σ, σ' of compact exceptional Lie groups F_4, E_6 and E_7 , Tsukuba
0902.0431_F03146	206	0.990	$\backslash\mathrm{left}(\backslash\mathrm{mathfrak{e}}_{\{8\}}\backslash\mathrm{right})^{\wedge}\{\backslash\mathrm{sigma}, \backslash\mathrm{sigma}^{\wedge}\{\backslash\mathrm{prime}\}\}$	$(\mathfrak{e}_8)^{\sigma, \sigma'}$	[29] T. Miyashita and I. Yokota, An explicit isomorphism between $(\mathfrak{e}_8)^{\sigma, \sigma'}$ and $\mathfrak{so}(8) \oplus$
0902.0431_F03147	206	0.990	$\backslash\mathrm{mathfrak{s} o}\{8\} \backslash\mathrm{oplus}$	$\mathfrak{so}(8) \oplus$	[29] T. Miyashita and I. Yokota, An explicit isomorphism between $(\mathfrak{e}_8)^{\sigma, \sigma'}$ and $\mathfrak{so}(8) \oplus$
0902.0431_F03148	206	0.928	$\backslash\mathrm{mathfrak{s} o}\{8\}$	$\mathfrak{so}(8)$	$\mathfrak{so}(8)$ as Lie algebras, Math. J. Toyama Univ. 24(2001), 151-158.
0902.0431_F03149	206	1.000	$\backslash\mathrm{mathfrak{g}}_{\{e d\}}$	$\mathfrak{g}_{ed}$	$\mathfrak{g}$ and group realizations of $\mathfrak{g}_{ev}, \mathfrak{g}_0$ and $\mathfrak{g}_{ed}$, Part II, $G = E_7$, Case 1, J. Kyoto
0902.0431_F03150	206	1.000	$G=E_{\{7\}}$	$G = E_7$	$\mathfrak{g}$ and group realizations of $\mathfrak{g}_{ev}, \mathfrak{g}_0$ and $\mathfrak{g}_{ed}$, Part II, $G = E_7$, Case 1, J. Kyoto
0902.0431_F03151	207	0.998	$\backslash\mathrm{left}(G_{\{2\}}\backslash\mathrm{right})^{\wedge}\{C\} / S L(3, C) \backslash\mathrm{simeq}\backslash\mathrm{left}(S^{\wedge}\{C\} \backslash\mathrm{right})^{\wedge}\{6\}, G_{\{2(2)\}} / S L(3, \backslash\mathrm{boldsymbol{R}}) \backslash\mathrm{simeq} S_{\{3,4\}}$	$(G_2)^C/SL(3, C) \simeq (S^C)^6, G_{2(2)}/SL(3, \mathbf{R}) \simeq S_{3,4}$	and homeomorphisms $(G_2)^C/SL(3, C) \simeq (S^C)^6, G_{2(2)}/SL(3, \mathbf{R}) \simeq S_{3,4}$, Math.
0902.0431_F03152	207	1.000	$E_{\{6(6)\}}$	$E_{6(6)}$	[37] O. Shukuzawa and I. Yokota, Non-compact simple Lie group $E_{6(6)}$ of type E_6 ,
0902.0431_F03153	207	1.000	$E_{\{6(-14)\}}$	$E_{6(-14)}$	[38] O. Shukuzawa and I. Yokota, Non-compact simple Lie group $E_{6(-14)}$ and $E_{6(2)}$
0902.0431_F03154	207	1.000	$E_{\{6(2)\}}$	$E_{6(2)}$	[38] O. Shukuzawa and I. Yokota, Non-compact simple Lie group $E_{6(-14)}$ and $E_{6(2)}$
0902.0431_F03155	207	1.000	$F_{\{4,1\}}$	$F_{4,1}$	[42] I. Yokota, On a non-compact simple Lie group $F_{4,1}$ of type F_4 , J. Fac. Sci.
0902.0431_F03156	207	1.000	$G_{\{2\}}\{ \}^{\wedge}\{\backslash\mathrm{prime}\}$	G_2'	[43] I. Yokota, Non-compat simple Lie group G_2' of type G_2 , J. Fac. Sci. Shinshu
0902.0431_F03157	207	1.000	$F_{\{4,2\}}$	$F_{4,2}$	[44] I. Yokota, Non-compact simple Lie group $F_{4,2}$ of type F_4 , J. Fac. Sci. Shinshu
0902.0431_F03158	208	1.000	$\{ \}^{\wedge}\{\backslash\mathrm{dagger}\}$	$\dagger$	[46] [†] I. Yokota, Simply connected compact simple Lie group $E_{6(-78)}$ of type E_6 and

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0902.0431_F03159	208	■ 1.000	$E_{6(-78)}$	$E_{6(-78)}$	[46] [†] I. Yokota, Simply connected compact simple Lie group $E_{6(-78)}$ of type E_6 and
0902.0431_F03160	208	■ 1.000	$(S\ U(8)) / \boldsymbol{Z}_2$	$(SU(8))/Z_2$	[47] [*] I. Yokota, Subgroup $SU(8)/Z_2$ of compact simple Lie group E_7 and non-
0902.0431_F03161	208	■ 1.000	$E_{7(7)}$	$E_{7(7)}$	compact simple Lie group $E_{7(7)}$ of type E_7 , Math. J. Okayama Univ. 24(1982),
0902.0431_F03162	208	■ 0.807	$G, \mathrm{I}, G=G_2, F_4, E_6$	$G, \mathrm{I}, G=G_2, F_4, E_6$	ceptional Lie groups $G, \mathrm{I}, G=G_2, F_4, E_6$, J. Fac. Sci. Shinshu Univ. 20(1985),
0902.0431_F03163	208	■ 1.000	$G=G_2, F_4$	$G=G_2, F_4$	G , Part I, $G=G_2, F_4$ and E_6 , Part II, $G=E_7$, Part III, $G=E_8$, Tsukuba J.
0902.0431_F03164	208	■ 0.650	$\mathfrak{g}_{\mathrm{eV}}, \mathfrak{g}_0$	$\mathfrak{g}_{\mathrm{eV}}, \mathfrak{g}_0$	alizations of $\mathfrak{g}_{\mathrm{eV}}, \mathfrak{g}_0$, Part I, $G=G_2, F_4, E_6$, Part II, $G=E_7$, Japan. J. Math.
0902.0431_F03165	208	■ 0.650	$G=G_2, F_4, E_6$	$G=G_2, F_4, E_6$	alizations of $\mathfrak{g}_{\mathrm{eV}}, \mathfrak{g}_0$, Part I, $G=G_2, F_4, E_6$, Part II, $G=E_7$, Japan. J. Math.
0902.0431_F03166	208	■ 0.995	$\mathfrak{g}_{\mathrm{ev}}, \mathfrak{g}_0$	$\mathfrak{g}_{\mathrm{ev}}, \mathfrak{g}_0$	alizations of $\mathfrak{g}_{\mathrm{eV}}, \mathfrak{g}_0$ and $\mathfrak{g}_{ed}$, Part I, $G=G_2, F_4, E_6$, J. Math. Kyoto Univ.
0902.0431_F03167	208	■ 0.999	E_8 / E_7	E_8 / E_7	[53] I. Yokota, T. Imai and O. Yasukura, On the homogeneous space E_8 / E_7 , J. Math.,
0902.0431_F03168	208	■ 0.999	$\left((S\ U(3) \times S\ U(6)) / Z_2\right) \cdot Z_2$	$((SU(3) \times SU(6))/Z_2) \cdot Z_2$	[54] I. Yokota, T. Ishihara and O. Yasukura, Subgroup $(SU(3) \times SU(6))/Z_2 \cdot Z_2$
0902.0431_F03169	208	■ 1.000	$E_{7, \sigma}$	$E_{7, \sigma}$	Lie group E_7 and non-compact simple group $E_{7, \sigma}$ of type $E_{7(-5)}$, Hiroshima Univ.
0902.0431_F03170	208	■ 1.000	$E_{7(-5)}$	$E_{7(-5)}$	Lie group E_7 and non-compact simple group $E_{7, \sigma}$ of type $E_{7(-5)}$, Hiroshima Univ.